The Material Economy

Energy, Minerals, Machines, Intelligence and Power in the Global Economy. A Twelve-Month Course

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Version 0.1, November 5, 2026

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Preface and How to Use This Course

Preface: why this course exists

Bagus, this book was written for you, overnight, on November 4 and 5, 2026. It exists because you keep asking a question that no single school subject answers. You are interested in artificial intelligence (AI), robotics, mathematics, finance and the global financial system, and you want to become the founder and chief executive of a business that does good and makes money. Each of those interests is usually taught in a separate room. Computer science does not explain why a cobalt export ban in the Democratic Republic of the Congo (DRC) changes the price of a robot. Economics classes rarely explain why a lithography machine from one Dutch company decides which countries can build AI. Finance courses rarely start from a smelter that needs a billion dollars before it earns a cent. This course puts all of those rooms into one building.

The driving question of the course is this:

> How do energy, ores, processing, manufacturing, chips, robots, AI, labor, finance and geopolitics fit together in the global economy, and who captures the value along the chain from ore to robot to AI?

Behind that question sits a sharper one that you care about most: could Africa or Southeast Asia build a cheap robot supply chain, from ore in the ground to a finished robot, by the 2040s? The course does not dodge this. Month 9 and Month 12 give a direct answer, with a reasoned judgment you are free to argue with.

The single most important idea you will meet is in the first chapter: **where something is made and where its value is captured are different things.** A country can dig most of the world's cobalt and keep a small share of the money it earns at the end of the chain. Another country can make almost nothing physical and still capture most of the profit, because it owns the design, the brand, the key machine or the currency the deal is priced in. Every month after that is a closer look at one layer of the chain, asking the same thing: what is scarce here, who controls it, and why?

How the twelve months build on each other

The course is built like a stack, from the ground up, with two outer layers wrapped around it.

The first three months give you the tools. **Month 1** teaches the basic machinery of the world economy: gross domestic product (GDP), value added, trade, global value chains, the "smile curve" and how money crosses borders. **Month 2** asks why some countries end up owning the scarce, high-value parts of the chain while others stay stuck exporting raw rock; the answer begins with institutions. **Month 3** looks at the actual playbooks that countries used to industrialize, from Britain to China, and why many copies failed.

The middle six months follow the physical chain, layer by layer. **Month 4** is energy, the base layer that every other step burns. **Month 5** is the critical minerals themselves, and the vital difference between reserves, mine production and refining. **Month 6** is processing and the politics of "downstreaming": can mining countries refine their own ore and keep the value? **Month 7** follows refined materials into factories and global supply chains. **Month 8** zooms into the most concentrated part of the chain, semiconductors and AI compute. **Month 9** takes a robot apart, joint by joint, and traces each part back to its minerals and suppliers.

The last three months step back from the machines. **Month 10** asks what happens to workers and wages when machines that see, move and reason become cheap, and who owns those machines. **Month 11** follows the money that decides which projects get built at all: the dollar system, interest rates, debt, sovereign wealth funds and venture capital. **Month 12** puts everything under the pressure of geopolitics, builds scenarios to 2050, answers the Africa and Southeast Asia question, and ends with a capstone on what a founder should build.

Each month ends with a synthesis that redraws the system with the new layer added. By the end, you should be able to draw the whole system from memory and defend your own view of where the value goes.

What kind of book this is

The book takes positions: when experts disagree, the chapters lay out the sides and then give a view marked "Judgment:", which you should treat as an argument to test, not a fact to memorize. It is global in scope. Indonesia appears as one important case, but so do the Congo, Chile, Morocco, South Korea, Taiwan, the Gulf, Mexico, Norway and many others. And it is written in plain English, with jargon explained on first use.

How to use this course

The rhythm: one month per chapter

The course has twelve chapters called Months. Each Month is designed to take one calendar month: four weeks, at about five hours a week, so about twenty hours per chapter and about 240 hours for the whole course. That is a serious but realistic load next to school, and it leaves room for university preparation and building things.

Every Month has the same shape, so you always know where you are:

- **Month overview** (read first): why the topic matters and how it links to the rest of the course.
- **Learning goals**: six or seven things you should be able to do by the end.
- **Weeks 1 to 4**, each with a core lesson, a case study, a "Key numbers" box and a "Think like a founder" box.
- **Month synthesis**: a diagram or table that fits the month's pieces together.
- **Common misconceptions**, each corrected.
- **Exercises**, a mix of calculations and essays.
- **Reading list**, glossary terms and sources.

A suggested week

A five-hour week might look like this:

- **Session 1 (about 90 minutes):** read the core lesson slowly, with your notebook open. Stop at every worked number example and redo the arithmetic yourself.
- **Session 2 (about 60 minutes):** read the case study, then the Key numbers box. Write three sentences: what happened, why it happened, and who captured the value.
- **Session 3 (about 60 minutes):** read the "Think like a founder" box and write your own version: one business idea that this week's lesson suggests, and one reason it could fail.
- **Session 4 (about 90 minutes):** do the reading or follow one source from the Sources list, and add anything new or contradictory to your notebook.

In Week 4, swap some of that time for the synthesis, misconceptions and exercises. If a week runs long, let it.

How to do the exercises

Each Month has seven or eight exercises. Some are quantitative (for example, working out how much a seven-point difference in interest rates costs a billion-dollar refinery each year, or estimating a robot's bill of materials). Some are essays (for example, judging whether a country's export ban worked).

- **Do the numbers by hand first**, then check them in a spreadsheet or a short Python script. Your mathematics is an advantage here; use it. A model you built yourself teaches far more than a number you read.
- **Write down your assumptions** before you calculate. Most real disagreements in economics are disagreements about assumptions, not arithmetic.
- **For essays, take a side.** Write a one-sentence claim, give the strongest evidence for it, give the strongest argument against it, and end with your own judgment. Aim for 500 to 800 words. This is the same structure the chapters use.
- **Check your work against the model answers, but only after you have tried.** Appendix A answers every exercise in Months 1 to 6 and Appendix B every exercise in Months 7 to 12, with every step of each calculation shown and a short model essay for each essay question. If your number differs, find the first line where your working and the model's part ways.
- **Check at least one number against a live source.** The world moves fast; a figure from 2025 may already be out of date. Finding the newer number, and explaining why it changed, is part of the exercise.

Keep a notebook

Keep one notebook for the whole course, on paper or digital (a single document or a notes app works well). Give it four sections:

- **The map.** Redraw the master map (below) at the end of every month, adding what you learned. By Month 12 it should be yours, not this book's.
- **Numbers to remember.** Around ten figures per month that you can quote from memory, each with its year and source. A founder who knows the real numbers wins arguments and spots bad deals.
- **Questions and disagreements.** Every time you doubt a claim in this book, write it down. Many will turn into good research questions.
- **Founder ideas.** Every business idea the course gives you, however rough, with a note on where it would capture value and what would kill it.

Turn it into your own thesis

The final goal is not to finish the book. It is to write your own argument. Over the year, build toward a thesis of about 5,000 to 10,000 words on one question inside the driving question. Some examples:

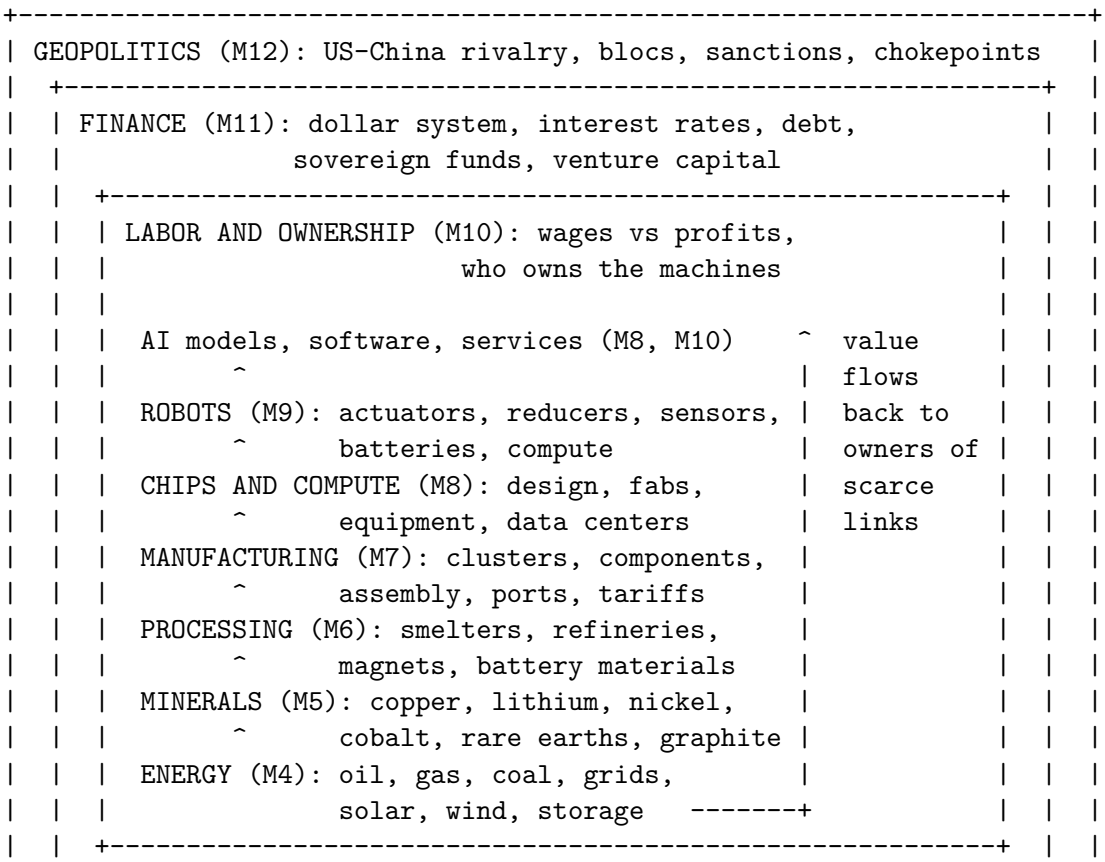
- Which single layer of the robot supply chain could a Southeast Asian or African country realistically own by 2040, and what would it take?
- Is downstreaming a good industrial policy, and under what conditions?
- Who will capture the value of cheap AI models: model makers, chip makers, or the companies that apply them to local problems?

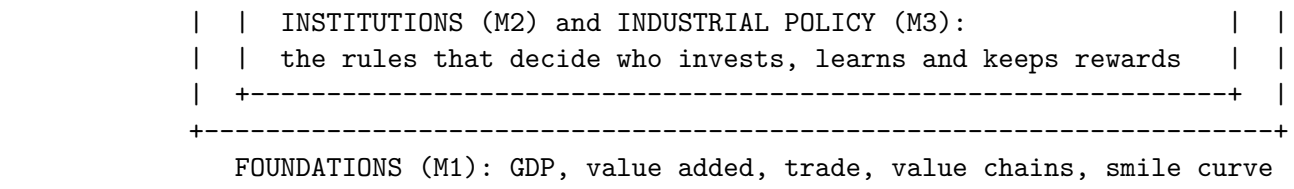
The easiest path is a one-page memo at the end of each month answering: "What did this month change about my answer to the driving question?" Twelve memos make the skeleton of a thesis, and Month 12's capstone gives a framework for the final version. Writing it is close to writing a startup's strategy, and the result is useful for university applications and research programs.

The master map

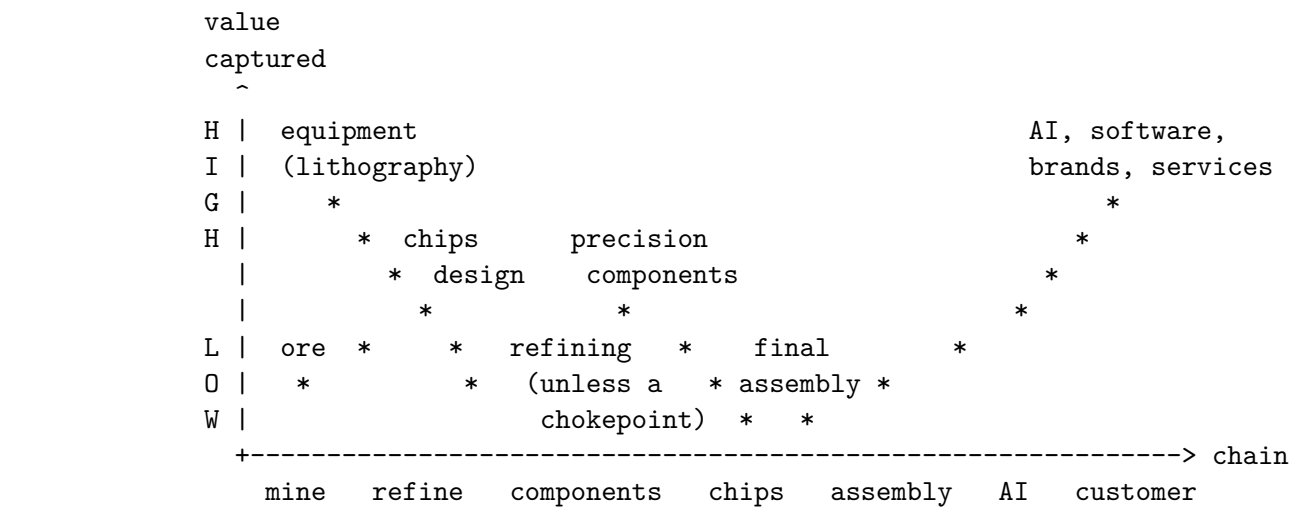
The whole system in one picture

Read this diagram from the bottom up. Each layer depends on the ones beneath it. Institutions and industrial policy sit underneath everything as the rules of the game; finance and geopolitics wrap around everything from the outside. The arrows on the right show money and value flowing back to whoever owns the scarce links.





A second, simpler way to see it is as the "smile curve" laid along the chain. Value tends to be high at the scarce ends (key machines and designs on one side, software, brands, customers and services on the other) and low in the crowded middle:



The exceptions matter as much as the rule. Refining is usually a thin-margin business, but when one country controls nearly all of a step (as China does with rare earth magnets) it holds leverage over everything downstream. Learning to spot those exceptions is the core skill of the course.

The layers at a glance

The table below compresses the map. "Main players" names the countries or firms that dominate the layer in 2025-2026, as described in the chapters. "Where value is captured" names what is scarce in that layer.

Layer	Main countries or firms (2025-2026)	Where value is captured	Month
Foundations: trade and value chains	US, China, EU, Japan; global brands such as Apple	The ends of the smile curve: design, IP, brands, services	1
Institutions and industrial policy	Korea, Taiwan, China as builders; Botswana, Norway as resource cases	Credible rules, state capacity, export discipline	2, 3
Energy	Gulf producers, US, Russia; China for solar, batteries and grids	Low-cost supply and control of routes such as Hormuz	4

Layer	Main countries or firms (2025-2026)	Where value is captured	Month
Minerals (mining)	Chile, DRC, Australia, Indonesia, China, South Africa	Low-cost, high-grade deposits, mostly in price booms	5
Processing and refining	China (most minerals); Indonesia (nickel); Morocco's OCP (phosphate)	Cheap energy plus scale plus know-how; chokepoint leverage	5, 6
Manufacturing and components	China; Japan and Germany (precision); Vietnam, India, Mexico, Morocco (assembly)	Dense supplier clusters and hard-to-replace components	7
Chips and compute	TSMC, ASML, Nvidia, SK hynix, Samsung, Micron; US cloud firms	Tools, leading-edge fabs, accelerators, memory	8
Robots	China (scale; Unitree, others); Japan (reducers, robot arms); Germany, Korea, US	Precision components, software, integration and service	9
AI and labor	Frontier AI labs; owners of capital everywhere	Owners of models, data and machines; workers only where institutions share gains	10
Finance	US Treasury and Federal Reserve; Norway, Gulf and Singapore funds; venture capital	Reserve currency, cheap capital, early equity in winners	11
Geopolitics	US, China; middle powers; holders of chokepoints	Whoever controls a bottleneck, until alternatives exist	12

A month-by-month guide

Month 1: How the Global Economy Works. You learn to measure an economy with GDP and value added, and why the numbers can mislead, through the case of Ireland's "leprechaun economics". You then study why countries trade (comparative advantage and the China shock), how production was sliced into global value chains, and who really captured the value of an iPhone 4. Week 4 covers the balance of payments, exchange rates and capital flows, ending with the 1997-1998 Asian financial crisis. The month closes with a first rough table of who captures value at each link of the chain, which every later month refines.

Month 2: Institutions and Why Nations Get Rich. This month explains the theory of inclusive and extractive institutions behind the 2024 economics Nobel prize, starting from the natural experiment of North and South Korea. It studies the resource curse and how Botswana partly

escaped it, then the rentier states of the Gulf, with Saudi Arabia's Vision 2030 under the strain of the 2026 war. Week 4 goes back into history: how war built European states, how colonial rule shaped African ones, and the long story of the Congo from rubber to cobalt. The conclusion: what is in the ground matters less than who makes and enforces the rules.

Month 3: The Industrialization Playbooks. You follow the actual sequence of policies used by Britain, the United States and Germany, then the developmental states of Japan and South Korea (with POSCO, "the steel mill nobody would fund"), then Taiwan and China (with TSMC and the invention of the chip foundry). Week 4 asks why the same tools failed in much of Latin America and Africa, and studies the return of industrial policy through the US CHIPS Act and the government's stake in Intel. The key lesson is that protection worked only when paired with export discipline.

Month 4: Energy, the Base Layer. This month links energy use to GDP and shows how China built its power system. It treats the oil and gas market, the Organization of the Petroleum Exporting Countries and its allies (OPEC+), and the 2026 Iran war, which disrupted the Strait of Hormuz and produced what the chapter calls the largest oil supply shock in history. Week 3 covers coal, power grids and petrostates; Week 4 covers the collapse in solar and battery costs, with Pakistan's solar rush as the case. The judgment: the energy advantage of the 2040s goes to whoever combines sun or other clean resources with cheap capital, strong grids and industry to use the power.

Month 5: Critical Minerals from Geology to Market. You learn the three-way difference between reserves, mine production and refining, and why they sit in different countries, with copper in Chile and the DRC as the first case. Week 2 covers the battery metals and the lithium boom and bust of 2021-2026; Week 3 covers rare earths, platinum-group metals and China's refining dominance, including the 2025 rare earth export controls. Week 4 explains how prices are made on exchanges like the London Metal Exchange (LME), why cycles are so violent, and the 2022 nickel short squeeze. The rule of the month: the chokepoint captures leverage, but not always profit.

Month 6: Processing, Downstreaming and Resource Nationalism. This month opens the economics of a smelter (mostly a machine for turning energy into metal) and the strange world of near-zero copper smelting fees. It studies Indonesia's nickel ore ban as the largest downstreaming experiment, and its harder bauxite sequel. Week 3 looks at African bargaining power in the DRC, Zambia, Zimbabwe, Botswana and Guinea, with Morocco's state phosphate company OCP as a champion that climbed the chain; Week 4 covers Chilean lithium and the Lobito Corridor rail line. It ends with a scorecard you can apply to any country and mineral: processing is necessary but not sufficient.

Month 7: Manufacturing and Global Supply Chains. You study why factory clusters form and why China's manufacturing dominance reinforces itself, then the "China plus one" shift to Vietnam, Thailand, Malaysia, India and Indonesia, with Vietnam as the success story that still has a value-capture problem. Week 3 covers nearshoring to Mexico and North Africa, with Morocco's car industry as the model and Ethiopia as a warning. Week 4 covers shipping, chokepoints and the trade wars of 2018-2026, ending with the rare earth magnet squeeze of 2025. The thread: value goes to whoever controls the hardest-to-replace step.

Month 8: Semiconductors, Compute and the AI Supply Chain. This month explains how chips are designed and made and why the leading edge lives in Taiwan, with TSMC as the case. Week 2 covers the chipmaking machines and the export-control "chip war", with ASML's extreme ultraviolet (EUV) lithography monopoly as the single point of failure. Week 3 shows why memory and advanced packaging became the bottlenecks of the AI boom, including the 2026 memory squeeze; Week 4 covers data centers, electricity and the economics of AI models, with the data center boom

in Johor, Malaysia. Value concentrates where substitutes are fewest.

Month 9: Robotics, Anatomy and Industrial Economics. You separate the industrial robot arms already working at scale from the humanoids in the news, and see how China became first the world's largest robot market and then a major robot maker. Week 2 takes a robot apart, from motors, reducers and magnets back to the minerals, including the 2025 magnet shock; Week 3 builds bills of materials and cost curves, with Unitree as a profitable humanoid maker. Week 4 maps the players by country and opens the Africa and Southeast Asia question, using Indonesia's nickel and the battery-to-robot link as a case. Value sits in precision components, software and service, not in assembly.

Month 10: AI, Robots and the Labor Economy. This month builds the economics of productivity and why respected economists disagree so widely about how much AI will add, with call centres from Fresno to Manila as the case. Week 2 introduces the task framework, which separates automation from augmentation, and compares what robots did to American and German towns. Week 3 looks at 2023-2026 evidence on wages, inequality, the "missing first rung" for young workers and who owns the machines; Week 4 uses history, growth theory and the policy toolbox, ending with Alaska's oil dividend as a model for a robot dividend. Technology sets what is possible; institutions and ownership decide who benefits.

Month 11: Money, Finance and Sovereign Capital. You learn how the dollar system works and what happened when Russia's reserves were frozen in 2022. Week 2 covers interest rates, capital flows and crises, with Indonesia's 2026 "double squeeze" of a weak currency and index-exclusion risk as a live case. Week 3 covers debt, development finance and Chinese lending, with Zambia's default and slow recovery; Week 4 covers sovereign wealth funds and venture capital, including Indonesia's Danantara fund and how founders raise money. The key idea: the cost of capital is a competitive weapon, and institutions set the risk premium.

Month 12: Geopolitics, Scenarios and Strategy, 2026-2050. This month studies the US-China rivalry and the tools of economic statecraft, with rare earths used as a weapon in 2025-2026, then blocs and chokepoints, with the 2026 Hormuz crisis as a stress test. Week 3 builds scenarios to 2050 and gives the course's direct answer on whether Africa or Southeast Asia can build an ore-to-robot chain by the 2040s. Week 4 is the capstone, asking what a founder should build, with BYD's rise from batteries to an integrated empire as the case, and closes with the whole system on one page and the course in seven sentences.

The appendices

Five appendices follow Month 12. They are reference tools, not extra chapters.

- **Appendix A: Model Answers, Months 1-6.** A worked answer to every exercise in Months 1 to 6. Calculations show each step; essays get a model answer of about 200 to 400 words that takes a clear position, plus a note on what a strong answer must include.
- **Appendix B: Model Answers, Months 7-12.** The same for Months 7 to 12, ending with the capstone exercise of drawing your own system map.
- **Appendix C: The Mathematical Toolkit of the Material Economy.** Fourteen short sections (C.1 to C.14) on the mathematics the Months use in passing: compounding and the rule of 70, present value and the cost of capital, the levelized cost of electricity, learning curves, growth accounting, input-output tables, value-added accounting, market concentration, elasticities and cost curves, debt dynamics, exchange-rate and terms-of-trade shocks, expected

value and scenarios, and common statistical traps, ending with a one-page formula sheet. Each section has a worked example, a short Python snippet and two practice problems with answers.

- **Appendix D: Country Atlas of the Material Economy.** One-page profiles of 22 economies that matter for the chain from ore to robot to AI, from the United States and China to Indonesia, the DRC and Zambia, each with a key-facts table, its role in the chain, its strengths and bottlenecks, what changed in 2025-2026 and a 2040 outlook, plus a comparison table of all 22.
- **Appendix E: Master Glossary and Reading List.** Every glossary term from the twelve Months merged into one alphabetized glossary (about 220 entries, each tagged with the Month that introduces it), each Month's recommended books, reports and articles by title and author, and ten core data sources to keep using after the course.

Each Month also ends with its own reading list, glossary terms and sources, so you can look up a term or a reference without leaving the chapter.

Ten big ideas of the course

1. Where things are made is not where value is captured. This is the first idea and the one that runs through every month. Gross exports, tonnes of ore and factory floor space measure activity, not reward. The iPhone case in Month 1 and the smile curve applied to a robot in Month 7 both show that the physical middle of a chain often keeps a small share, while the owners of designs, key components, brands and customer relationships keep most of it. Whenever you read a headline about a country "becoming a hub", ask what share of the value added stays there and who owns it.

2. Energy is the base layer. Every step from mine to data center converts energy into something else. A smelter is largely a machine for turning electricity into metal, and an AI data center is a machine for turning electricity into computation. Month 4 shows that energy shocks, like the 2026 Hormuz disruption, travel through the entire chain, and that the falling cost of solar and batteries is changing which countries could have cheap power. No country can build the middle of the ore-to-robot chain without cheap, reliable electricity.

3. Owning a resource is not the same as capturing its value. Months 2, 5 and 6 show the DRC mining most of the world's cobalt while much of the value is captured elsewhere, and show Norway and Botswana turning resource income into lasting wealth while Venezuela did not. The difference lies in bargaining power, processing capacity, ownership, and above all institutions. Geology is luck; what a country does with it is a choice.

4. Power concentrates at chokepoints, often in unglamorous places. The narrowest step in a chain controls the whole of it. In 2025-2026 those steps were rare earth refining and magnets (China), extreme ultraviolet lithography machines (ASML), leading-edge chip fabrication (TSMC), high-bandwidth memory, the Strait of Hormuz and the dollar payment system. A chokepoint grants leverage, not always profit: Month 5 notes that rare earths are a tiny share of a car's value yet can stop car production.

5. Institutions are the deepest cause. Month 2 argues that whether a country climbs the chain depends less on what it has than on who makes the rules, who enforces them, and whether ordinary people and firms can invest, learn and keep the rewards. Institutions also show up in money: Month 11 shows that trust, transparency and contract enforcement lower a country's cost of borrowing. Good institutions are built by legitimate local actors over time, with outsiders as partners at most.

6. Industrial policy works only with discipline. Month 3 shows that almost every rich country

protected and subsidized its industries at some point, so "free markets alone" is not the historical record. But protection without pressure produced small, inefficient firms and debt crises across much of Latin America and Africa. What separated Korea, Taiwan and China was export discipline: firms had to win in world markets to keep their support. The same test applies to the US CHIPS Act and to every downstreaming policy today.

7. Systems beat parts, and clusters reinforce themselves. China's lead in manufacturing, batteries and robots is not one advantage but a system: dense supplier clusters, a giant home market, state capital, fast factory building and control of key processing. Month 7 explains why such clusters are self-reinforcing, and Month 9 shows a robot built with a Chinese supply chain costing roughly a third of one built without it. A newcomer cannot copy a whole system quickly; it wins by choosing layers.

8. The cost of capital decides who can build. Smelters, fabs, data centers and robot factories all need huge sums years before they earn anything. Month 11's worked example shows that a seven-point difference in interest rates on a billion-dollar refinery is worth over 50 million dollars a year. Countries and founders who can raise patient, cheap capital, in a currency they can repay, can build layers that others cannot. That is why finance, not just engineering, decides much of the map.

9. Technology decides what is possible; ownership decides who benefits. Month 10 shows that the same robots led to job losses in some American towns and to reallocation in German ones, and that the same oil produced a curse in some places and a citizen dividend in Alaska and Norway. AI and robots are the next great asset class whose income must be divided between workers, founders, states and foreign shareholders. Whether automation raises most people's incomes depends on policy and on who owns the machines.

10. Geopolitics turns interdependence into leverage, so founders should start where value is hard to import. Month 12 shows trade remains positive-sum, but in 2025-2026 every major power used its bottlenecks as weapons. The strategic layers of the chain are splitting into blocs while commercial layers stay more open. For a founder in Africa or Southeast Asia, the course's capstone judgment is to start in the integration and application layer, making robots and AI useful in local conditions, where capital needs are low and local knowledge cannot easily be imported, then earn cheaper capital and move toward scarcer capabilities over time.

A note on sources, accuracy and this version

The date

Everything in this book is written **as of November 2026**, using 2025 and 2026 data wherever possible. Several stories were still unfolding when it was written, including the war involving Iran that began on February 28, 2026, the disruption of shipping through the Strait of Hormuz, US tariff changes after the Supreme Court ruling of February 2026, high US interest rates, and the memory chip squeeze. Treat any figure about those events as a snapshot. When you study a month, check whether the story has moved on.

The [V] and [E] marks

Each month ends with a list of sources, and each source carries one of two marks:

- **[V] means verified:** the writing agent fetched the page and read it.

- **[E] means excerpt only:** the information came from a search result snippet, and the full page was not opened.

In this version, most sources are marked [E] and a minority are [V]. An [E] source is not necessarily wrong, but it is weaker: snippets can drop the qualifiers, dates and conditions around a number. Before you rely on a number for an essay, a pitch or your thesis, open the original, check the figure, year and definition, and mark it [V] in your notebook if it holds. Founders who check their numbers make fewer expensive mistakes.

How the chapters treat numbers and judgments

Every chapter followed the same rules: important numbers carry a year and a source; no statistic, quote or book is invented; uncertain figures are given as "roughly" or a range, or left out. Some tables, such as the shares of robot value in Month 7, are labelled as illustrative teaching estimates rather than measured data. The "Judgment:" passages are reasoned opinions as of November 2026; disagree with them freely, as long as you argue from evidence.

Version 0.1

This is **version 0.1**: a first complete draft, written overnight on November 4 and 5, 2026, by AI writing agents and then reviewed by AI critic agents that fact-checked and deepened the chapters while time allowed. It certainly contains errors, gaps, uneven coverage and some repetition (the 2025 rare earth controls, for example, appear in Months 5, 7, 9 and 12, each from a different angle). Corrections, newer data and your disagreements belong in version 0.2. Keep a list as you read; the best version of this book will be the one you rewrite yourself.

Month 1: How the Global Economy Works

Month overview

This course asks one big question: when a lump of rock leaves the ground in the Democratic Republic of the Congo, Indonesia or Chile, and years later part of it ends up inside a robot arm or an AI data center, who gets paid along the way, and why do some people get paid so much more than others? Before we can follow the rock, we need a map of the ground it travels over. That map is this Month.

Month 1 gives you the basic tools that every later Month uses. In Week 1 you learn how economists measure an economy: gross domestic product (GDP), the idea of "value added", and the traps hidden inside these numbers. In Week 2 you learn why countries trade at all, using the idea of comparative advantage, and you look at the real size and shape of world trade in 2025 and 2026, including the tariff shocks of those years. In Week 3 you learn how production has been sliced into pieces spread across many countries (global value chains), and why the profits are so unevenly spread along those chains (the "smile curve"). In Week 4 you learn how money and capital move between countries: the balance of payments, exchange rates, foreign direct investment, remittances, and what happens when money suddenly runs away, as it did in Asia in 1997 and 1998.

The thread running through the whole Month is the difference between where things are made and where value is captured. A country can make huge quantities of something and still keep little of its value. Another country can make almost nothing physical and keep most of it. Learning to see that difference is the single most useful skill for a founder who wants to build something in the material economy.

Month 2 then asks why some countries end up on the high-value ends of the chain while others stay stuck exporting raw rock, an answer that has much to do with institutions, power and history.

Learning goals

By the end of this Month you should be able to:

- Explain what GDP measures, compute it using the value-added method, and explain the difference between nominal GDP, real GDP, GDP at purchasing power parity (PPP), and gross national income (GNI).
- Use the idea of comparative advantage to show, with numbers, why two countries can both gain from trade even when one is better at everything.

- Describe the size and structure of world trade in 2025-2026, the role of China, and the effect of the 2025-2026 United States tariff shocks.
- Explain what a global value chain is, draw the smile curve, and explain the mechanisms that push profits toward design, brands and scarce technology and away from routine assembly.
- Read a simple balance of payments, explain the main types of cross-border capital flows (foreign direct investment, portfolio flows, bank loans, remittances and aid), and explain why the US dollar dominates.
- Explain how a currency crisis happens, using the 1997-1998 Asian financial crisis as the worked case.
- Apply all of this to a first rough map of value capture along the chain from ore to robot to AI.

Week 1: Measuring an economy: GDP, value added and their traps

Core lesson

What GDP is trying to measure

Gross domestic product (GDP) is the market value of all final goods and services produced inside a country's borders during a period, usually a year. "Final" means goods and services bought by their last user, not things that are used up to make something else. A steel sheet bought by a car factory is not counted on its own, because its value is already inside the price of the car. If we counted both the steel and the car, we would count the steel twice.

There are three ways to measure GDP, and in principle they give the same answer:

- The production (value-added) method: add up, for every firm, the value of what it sells minus the value of the inputs it bought from other firms. That difference is the firm's value added.
- The expenditure method: add up spending on final goods and services: household consumption (C), investment by firms and in housing (I), government purchases (G), and exports minus imports (X minus M). This gives the famous identity GDP = C + I + G + (X - M).
- The income method: add up all the incomes earned in production: wages and salaries, profits, rents and interest, plus taxes on production minus subsidies.

The three methods match because every dollar spent on a final good is, in the end, somebody's income, and the price of every good can be broken into the value added by each firm that touched it.

Value added: the most important idea in this course

The value-added method matters most for us, because the driving question of this course, who captures value from ore to robot to AI, is really a question about value added. Here is a simple worked example. The numbers are invented to show the mechanism, not taken from any real company.

Imagine an industrial robot arm that is finally sold to a factory for 30,000 US dollars. Its life story is:

Stage	Sells output for (US dollars)	Bought inputs for (US dollars)	Value added (US dollars)
Mine digs ore	200	0	200
Smelter and refinery make metals	900	200	700

Stage	Sells output for (US dollars)	Bought inputs for (US dollars)	Value added (US dollars)
Component makers (motors, gear reducers, chips, sensors)	12,000	900	11,100
Robot company designs, assembles, writes software	22,000	12,000	10,000
Distributor and system integrator install it	30,000	22,000	8,000
Total			30,000

Add up the value-added column and you get exactly 30,000 dollars, the final price. Notice what this table already tells you. The mine, which did the dirtiest and often most dangerous work, captured less than 1 percent of the final value. The component makers and the robot company captured about 70 percent between them. If the mine is in one country, the smelter in a second, the component makers in a third and the robot company in a fourth, then each country's GDP grows only by the value added inside its borders. This is why a country that "has the minerals" is not automatically rich from the robot boom. Real numbers for real chains come in Months 5, 6 and 9; for now, hold on to the mechanism.

Value added is also split between workers and owners. Of the 10,000 dollars of value added by the robot company, some is paid as wages (labor income) and the rest is profit, interest and rent (capital income). Who owns the company matters: if a foreign firm owns it, the profits may leave the country. That is the difference between GDP and GNI, which we meet below.

Nominal, real and purchasing power parity

GDP is first measured in a country's own currency at current prices. That is "nominal GDP". It can rise just because prices rose. To see whether the economy actually produced more, statisticians compute "real GDP", which values output at the prices of a fixed base year. When people say "the economy grew 5 percent", they almost always mean real GDP.

To compare countries, we must convert currencies. There are two ways:

- At market exchange rates. This tells you what an economy's output is worth in the global market, for example how many imported machines or chips it could buy. The International Monetary Fund (IMF) projected in its October 2026 World Economic Outlook that the United States would produce about 32.6 trillion dollars of GDP in 2026 and China about 21.5 trillion dollars, out of a world total of about 128 trillion dollars (IMF October 2026 data, as tabulated by Visual Capitalist).
- At purchasing power parity (PPP). PPP adjusts for the fact that many things, especially local services such as haircuts, rent and street food, are much cheaper in poorer countries. A PPP dollar is designed to buy the same basket of goods everywhere. On this measure, China's economy in 2026 is about 44.6 trillion "international dollars", roughly 37 percent larger than the United States. India rises from sixth to third place, and Indonesia rises from 17th (about 1.48 trillion dollars at market rates) to 7th (about 5.48 trillion international dollars) (same source).

Which is right? Both, for different questions. If you want to know living standards, PPP is better. If you want to know how much power a country has to buy foreign machines, pay foreign debts or fund foreign investment, market exchange rates are better. A founder raising money or importing equipment lives in the market-exchange-rate world. A founder selling haircuts lives in the PPP world.

Growth rates and the rule of 70

Small differences in growth rates become huge over time. A quick tool is the "rule of 70": divide 70 by the annual growth rate in percent to get roughly how many years it takes to double. At 7 percent growth an economy doubles in about 10 years. At 2 percent it takes about 35 years. Over a 25-year career, a country growing at 7 percent multiplies its output about 5.4 times (1.07 to the power 25), while a country growing at 2 percent multiplies it about 1.6 times. That gap is the story of East Asia versus much of Latin America after 1980, which Month 3 tells in detail.

Today the world economy is growing slowly. The IMF estimated global growth at 3.5 percent in 2025 and projected 3.1 percent in 2026 and 3.4 percent in 2027. It described two forces pulling in opposite directions: a strong investment boom in artificial intelligence (AI) and technology, and an energy shock from the war involving Iran and the partial closure of the Strait of Hormuz, which pushed global inflation up to a projected 4.7 percent in 2026 (IMF World Economic Outlook, October 2026). The IMF warned that energy-importing economies outside the technology value chain face the hardest outlook. That sentence alone is a summary of this whole course: energy and position in the technology chain now decide a lot of national fortunes.

GDP versus GNI, and why the difference matters for poor and resource-rich countries

GDP counts production inside a country, whoever owns it. Gross national income (GNI) counts the income of a country's residents, wherever it is earned. To go from GDP to GNI, you add income residents earn abroad (for example profits of your companies' foreign factories) and subtract income foreigners take out (for example profits of foreign-owned mines sent home to shareholders).

For many mining countries, GNI is noticeably smaller than GDP because foreign owners take a large share of mining profits home. That is not automatically bad: the foreigners brought the capital and took the risk. But it means that a country's GDP growth from a mining boom can overstate how much richer its own people became. This is one of the first places where the ore-to-robot question shows up in the national accounts.

A worked example: where a mining dollar goes

The gap between GDP and GNI becomes much clearer with numbers. Imagine a copper and cobalt mine in a country we will call Country M. The mine is 80 percent owned by a foreign company and 20 percent by Country M's state mining company. All numbers are invented to show the mechanism, but the structure is realistic.

In one year the mine sells metal worth 1,000 million dollars. To produce it, it buys 350 million dollars of imported goods and services (diesel, explosives, chemicals, spare parts, foreign engineering contractors) and 50 million dollars of inputs from local firms (electricity, trucking, catering). The mine's own value added is therefore 1,000 - 350 - 50 = 600 million dollars. The local suppliers add another 50 million dollars of value inside Country M (assume, for simplicity, that their own inputs are local). So the mine adds about 650 million dollars to Country M's GDP. The 350 million dollars

of imports add to other countries' GDP: the countries that make the trucks, the chemicals and the software.

Now follow the 600 million dollars of the mine's value added to the people who receive it:

Use of the mine's value added	Million dollars	Who receives it
Wages of local workers	60	Residents
Royalty paid to the state (a tax on production)	50	Resident state
Depreciation (wear and tear of machines)	150	Replaces capital, mostly imported
Interest on loans from foreign banks	40	Non-residents
Corporate income tax (30 percent of 300 profit)	90	Resident state
After-tax profit to the state company (20 percent of 210)	42	Resident state
After-tax profit to the foreign owner (80 percent of 210)	168	Non-residents
Total	600	

To move from GDP to GNI, statisticians subtract income paid to non-residents: here the 40 million dollars of interest and the 168 million dollars of profit belonging to the foreign owner (whether it is sent home as dividends or kept in the country as "reinvested earnings", it belongs to a non-resident). So of the 650 million dollars of GDP linked to this mine, only about 442 million dollars, 68 percent, is national income. And of the original 1,000 million dollars of copper and cobalt sold, the income that stays with Country M's people and government (local wages 60, local suppliers 50, royalty 50, tax 90, state dividend 42) is 292 million dollars, about 29 cents per dollar of metal.

That 29 cents is not "unfair" by itself. The foreign owner put up the capital, carried the risk of a dry hole, and brought the technology. But the table shows exactly where a government or a founder can change the split. There are five levers, and every later Month uses them:

- Fiscal terms: royalties and taxes (Month 6). Raising the royalty from 5 to 10 percent adds 50 million dollars of royalty, though the treasury's net gain is smaller (about 35 million dollars) because profit, and so corporate tax, falls by 50 million dollars too; and if higher royalties make the next mine unprofitable, it may not be built at all.
- Ownership: a larger state or local private stake turns non-resident profit into resident income, but it also requires residents to supply capital and accept risk. Botswana's 50:50 joint venture with De Beers is the classic case (Month 2).
- Local content: if local firms could supply 150 million dollars of the 350 million dollars of imported inputs, Country M's GDP would rise by up to 150 million dollars without mining one more tonne. This is often the most realistic entry point for local founders: maintenance, spare parts, trucking, catering, environmental monitoring, software.
- Processing: selling refined metal instead of concentrate or hydroxide raises revenue per tonne, but it also raises imported inputs (energy, reagents, equipment), so the net gain in value added is smaller than the price difference. Month 6 tests when this pays.

- Cost of capital: the 40 million dollars of foreign interest depends on how risky lenders think Country M is. Better institutions lower that number for every project at once (Months 2 and 11).

What do real numbers show? In 2024, GNI was about 89 percent of GDP in Zambia and Mongolia, 95 percent in Chile, 97 percent in the DRC and Indonesia, and 98 percent in Botswana (World Bank, World Development Indicators, author's calculation). Compare Ireland's GNI at 71 percent of GDP in 2025 (CSO, 2026). Two lessons. First, the Zambian and Mongolian gaps are what a large, mostly foreign-owned mining sector looks like in the accounts. Second, read low gaps carefully: Botswana's small gap partly reflects its 50 percent ownership of Debswana, but a small measured gap in a country with weak statistics, such as the DRC, may also reflect how hard it is to record profits that leave through trading companies and transfer pricing (the prices that related companies charge each other). A number is only as good as the institution that measures it.

For the course question, the example gives a first rule of thumb: a country captures value from its ore through wages, local suppliers, taxes and ownership, not through tonnes. A country that exports twice as many tonnes with the same fiscal terms, ownership and supplier base doubles the 29 cents, but it does not change the share.

What GDP misses

GDP is a measure of market production, not of well-being. It misses several things that matter for this course:

- Unpaid household work, such as cooking and childcare, which is mostly done by women and counts for nothing.
- The informal economy: small unregistered businesses, street trade and casual labor. In many developing countries this is a large share of jobs, so GDP figures there are less precise.
- The depletion of natural capital. When a country mines and sells a ton of nickel, the sale adds to GDP, but the country now has one ton less nickel in the ground. A company would record that as using up an asset; GDP records it as pure income. The World Bank's "Changing Wealth of Nations" work tries to correct this by measuring natural, produced and human capital together. For resource exporters this is not a footnote. A country can show strong GDP growth while it is actually running down its wealth, which is one mechanism of the "resource curse" we study in Month 2.
- Pollution and environmental damage, which reduce well-being but are not subtracted.
- Distribution. GDP per person can rise while most people's income stays flat if the gains go to a few.

None of this means GDP is useless. It is the best single measure of how much an economy produces, and it is strongly linked to things people care about, such as health and education. It means you must ask, each time you see a GDP number, "produced by whom, owned by whom, and at what cost to the stock of wealth?"

Case study: Ireland and "leprechaun economics"

In July 2016 Ireland's Central Statistics Office (CSO) published its national accounts for 2015. Real GDP had grown by 26.3 percent in one year, and nominal GDP by 32.4 percent (CSO, National Income and Expenditure 2015); later revisions put the real figure at 24.4 percent, still by far the largest in the European Union (CSO, Measuring Ireland's Progress 2021). No real economy grows

by a quarter in a year without a war ending or an oil field opening. The economist Paul Krugman mocked the number as "leprechaun economics", and the name stuck.

What had happened? A small number of very large multinational companies had moved huge balance sheets, mostly intellectual property such as patents and the rights to use them, into Irish subsidiaries. The CSO itself explained that the jump "was driven by relocations of entire balance sheets to Ireland" (CSO, National Income and Expenditure 2017 FAQ). Once intellectual property sits in an Irish company, the profits it earns from selling products worldwide are recorded as Irish production. Factories did not appear overnight; accountants moved assets for tax reasons.

The mechanism is worth understanding because it is the smile curve showing up in GDP statistics. The most valuable part of a modern product is often not the physical object but the intellectual property: the design, the software, the brand. That property is easy to move on paper, so it tends to be booked wherever taxes are lowest. Ireland's corporate tax rate of 12.5 percent (raised to 15 percent for the largest groups from 2024 under the global minimum tax agreed through the Organisation for Economic Co-operation and Development, OECD) made it an attractive place to book it.

Ireland's response was to invent a better measure. Modified gross national income (GNI*) removes the profits of companies that re-domiciled to Ireland and the depreciation of foreign-owned intellectual property and leased aircraft. The gap is enormous. In 2025 Ireland's GDP was about 602 billion euros but its GNI* was about 334 billion euros, only 55.4 percent of GDP (CSO, Measuring Ireland's Progress 2026). In other words, almost half of what Ireland's GDP calls "Irish production" is, for practical purposes, profit that belongs to foreign shareholders.

Three lessons follow. First, GDP measures where value is booked, not necessarily where it is created or who keeps it. Second, the parts of the value chain that are easiest to move on paper (intellectual property, brands, financing) are also the parts that capture the most profit, which is part of why they are so mobile. Third, for a poor country, attracting a foreign assembly plant raises GDP but may raise national income much less if the profits leave. When you later read that a country "exports 50 billion dollars of electronics", ask how much of that is domestic value added and how much stays at home after profits are paid out.

Key numbers

- World GDP in 2026 (projected, market exchange rates): about 128.0 trillion US dollars (IMF World Economic Outlook database, October 2026, read via the IMF Data Mapper).
- United States GDP 2026 (projected): about 32.6 trillion dollars; China: about 21.5 trillion dollars (IMF World Economic Outlook database, October 2026).
- China GDP at purchasing power parity 2026 (projected): about 44.6 trillion international dollars, about 37 percent above the United States (IMF, October 2026).
- Indonesia GDP 2026 (projected): about 1.48 trillion dollars at market rates, rank 17; about 5.48 trillion international dollars at PPP, rank 7 (IMF, October 2026).
- World GDP in 2025 (estimate, market exchange rates): about 118.5 trillion US dollars (IMF World Economic Outlook database, October 2026).
- World real GDP growth: 3.5 percent in 2025, 3.1 percent projected for 2026, 3.4 percent projected for 2027 (IMF World Economic Outlook database, October 2026).
- Global headline inflation: 4.1 percent in 2025, 4.7 percent projected for 2026, 4.1 percent for 2027 (IMF World Economic Outlook database, October 2026).

- Ireland's real GDP growth in 2015: 26.3 percent as first published in July 2016, since revised by the CSO to 24.4 percent (CSO Ireland, 2016 and Measuring Ireland's Progress 2021).
- Ireland's GNI* as a share of GDP in 2025: 55.4 percent (CSO Ireland, Measuring Ireland's Progress 2026).

Think like a founder

The value-added table is a founder's tool, not just an economist's. For any product you want to build, sketch the chain and ask at which step value added is high and defensible. In the robot example, the mine and the final assembly step are crowded and squeezed; components with real know-how (precision gear reducers, motors, control software) and the customer relationship (the integrator who knows how to make the robot useful in a specific factory) are where margins tend to live. A young founder in Southeast Asia or Africa will rarely start by beating a global robot maker at its core product; a more realistic entry point is a high-value niche close to local customers that outsiders find hard to serve, such as integration, maintenance or software for a local industry. Also note the Ireland lesson in reverse: where you legally book your intellectual property and profits is itself a strategic decision with tax, investor and political consequences. Founders who ignore it pay for it later.

Week 2: Why countries trade: comparative advantage and the real shape of world trade

Core lesson

Absolute advantage and comparative advantage

In 1776 Adam Smith argued that countries should specialize in what they make most cheaply and trade for the rest. That is "absolute advantage". But what if one country is better at making everything? In 1817 David Ricardo showed that trade still pays. The key idea is "comparative advantage": a country should specialize in the goods it gives up the least to produce, measured in terms of other goods it could have made instead. That "what you give up" is called opportunity cost.

Here is a worked example with two invented countries. Country A is rich and technologically advanced. Country B is poorer. The table shows the hours of work needed to make one unit of each good.

	Robot gearbox	Shirt
Country A	10 hours	5 hours
Country B	40 hours	8 hours

Country A is better at both goods (it has an absolute advantage in both). Now compute opportunity costs:

- In A, one gearbox costs 10 hours, which could have made 2 shirts. So one gearbox "costs" 2 shirts.
- In B, one gearbox costs 40 hours, which could have made 5 shirts. So one gearbox "costs" 5 shirts.

Gearboxes are relatively cheaper in A (2 shirts) than in B (5 shirts). Shirts are relatively cheaper in B (one shirt costs 1/5 of a gearbox) than in A (one shirt costs 1/2 of a gearbox). So A has a comparative advantage in gearboxes and B in shirts.

Suppose they trade at a price of 1 gearbox = 3 shirts, somewhere between the two countries' internal costs. Then:

- A spends 10 hours making one gearbox and trades it for 3 shirts. Making those 3 shirts itself would have taken 15 hours. A saves 5 hours.
- B spends 24 hours making 3 shirts and trades them for one gearbox. Making the gearbox itself would have taken 40 hours. B saves 16 hours.

Both gain. B gains even though it is worse at everything, and A gains even though it could make shirts more cheaply than B. That is the power, and the surprise, of comparative advantage. Notice also what this example does not say: it says nothing about whether B should be happy to stay a shirt-maker forever. B's workers earn less per hour because B is less productive overall. Comparative advantage tells you how to make the best of today's productivity. It does not tell you how to raise productivity tomorrow. That is the central question of Months 2 and 3.

Where comparative advantage comes from

Ricardo's model explains trade by differences in technology (productivity). Later theories add more causes:

- Factor endowments. The Heckscher-Ohlin model, developed by two Swedish economists in the early twentieth century, says countries export goods that intensively use the factors they have in abundance. A country with many workers and little capital exports labor-intensive goods (shirts, assembly); a country with much capital and many engineers exports capital- and skill-intensive goods (machine tools, chips); a country with rich ore bodies exports minerals. This is why Chile exports copper and Germany exports machinery.
- Economies of scale. In the late 1970s and 1980s Paul Krugman and others showed that even identical countries can gain from trade if production gets cheaper at large scale. Each country specializes in some varieties and serves the whole world market. Which country ends up making what can depend on history and luck: whoever starts first and grows big gets the lowest costs. This idea, called "new trade theory", explains a lot about chips, aircraft and robots, where the fixed costs of design and factories are huge.
- Firm differences. Marc Melitz showed in the 2000s that only the most productive firms export, so trade shifts workers toward stronger firms.
- Agglomeration. Industries cluster because suppliers, skilled workers and ideas are all nearby. Once a cluster exists, it is hard to copy. China's electronics cluster around Shenzhen and its robotics and battery clusters in the Yangtze River Delta are the most powerful examples today (Month 7).

A final, very practical tool is the "gravity model" of trade: trade between two countries tends to rise with the size of their economies and fall with the distance between them, just as gravity rises with mass and falls with distance. Distance here includes geography, but also language, legal systems and politics. The gravity model predicts trade flows remarkably well, and it explains why most countries trade most with their neighbors.

Dynamic comparative advantage

Comparative advantage is not fixed. South Korea in 1960 had a comparative advantage in wigs, textiles and plywood. By 2000 it had one in memory chips, ships and cars. It got there by investing heavily in education, infrastructure and firms that learned by doing, often with the government pushing and protecting them for a time. Economists call this "dynamic comparative advantage": a country can build the advantages it wants to have, though the process is slow, risky and full of failures. Whether governments should try to do this, and how, is one of the biggest debates in development economics, and Month 3 takes it head on.

The real size of world trade in 2025

According to the World Trade Organization (WTO), the dollar value of world merchandise (goods) exports was 26.26 trillion dollars in 2025, up 7 percent from 2024. Commercial services exports were 9.56 trillion dollars, up 8 percent. Goods and services trade together, measured on a balance-of-payments basis, came to about 34.65 trillion dollars (WTO, Global Trade Outlook and Statistics, March 2026). World GDP at market exchange rates was about 118.5 trillion dollars in 2025 (IMF World Economic Outlook database, October 2026), so trade in goods and services equals roughly 29 percent of world output (34.65 divided by 118.5, with each shipment counted once, on the export side).

In volume terms, goods trade grew 4.6 percent in 2025, far faster than the WTO had forecast, and faster than world GDP. The main reason was a surge in demand for AI-related goods (servers, chips and other equipment), which offset the drag from higher tariffs (WTO, March 2026). For 2026 the WTO's baseline forecast was much slower goods trade growth of about 1.9 percent, before the full force of the energy shock was known.

China remained the world's largest goods exporter in 2025, with exports of 3.77 trillion dollars, about 17.5 percent of world exports when trade inside the European Union is excluded (WTO, Global Trade Outlook and Statistics, March 2026, Appendix Table 2; 14.4 percent if intra-EU trade is counted). China's trade surplus, exports minus imports, reached a record 1.189 trillion dollars in 2025, the first time any country's annual surplus had passed 1 trillion dollars (Reuters, January 14, 2026, citing Chinese customs). Its exports grew 5.5 percent in dollar terms while imports were flat at about 2.58 trillion dollars (South China Morning Post, January 2026, citing customs data).

What trade balances mean, and what they do not

People often treat a trade surplus as "winning" and a deficit as "losing". That is wrong, or at least far too simple. A basic accounting identity explains why. For any economy:

current account balance = national saving minus domestic investment

The current account is mainly the trade balance plus net income from abroad. If a country saves more than it invests at home, it must lend the difference to the rest of the world, and the way it does that is by exporting more than it imports. If a country invests more than it saves, it must borrow from abroad, and it does that by importing more than it exports.

So China's huge surplus is the other face of very high saving (by households, firms and the state) relative to investment and consumption at home, and the US deficit reflects low US saving and the world's strong demand for dollar assets. Matthew Klein and Michael Pettis argue in "Trade Wars Are Class Wars" (2020) that China's surplus reflects policies that hold down household income

and consumption; others stress China's genuine productivity gains and dense supplier networks. Judgment: both are true at once. For the rest of the world, the result is intense price competition in exactly the goods (electric vehicles, batteries, solar panels, robots) that many developing countries hope to make.

Tariffs: who pays, and the shocks of 2025-2026

A tariff is a tax on imports. It is paid at the border by the importing firm, which then passes some or most of it to its customers in higher prices. The foreign exporter may absorb part of it by cutting its price. Most studies of the 2018-2019 US tariffs on China, such as Amiti, Redding and Weinstein (2019) and Fajgelbaum and coauthors (2020), found that the burden fell mostly on US importers and consumers rather than on Chinese exporters.

In 2025 the United States raised tariffs to levels not seen since the 1930s. Using an emergency law, the International Emergency Economic Powers Act (IEEPA), the administration announced broad "reciprocal" tariffs on most trading partners in April 2025. The Budget Lab at Yale University estimated that, before the courts intervened, the average effective tariff rate faced by US consumers had reached the mid-teens in percent (its estimates ranged from about 14 to 17 percent as methods changed), the highest since the 1930s (Yale Budget Lab, February and March 2026). On February 20, 2026, the US Supreme Court ruled 6-3 in Learning Resources, Inc. v. Trump that IEEPA did not give the president power to impose tariffs. The administration replaced the struck-down tariffs within days with a temporary 10 percent global surcharge under Section 122 of the Trade Act of 1974, in force from February 24, 2026; by law such a surcharge can last only 150 days unless Congress extends it (Congressional Research Service, February and October 2026). Section 122 was itself challenged: on May 7, 2026 the Court of International Trade held that the surcharge was not authorized by that law, although an appeals court paused that ruling while the appeal went ahead (Gibson Dunn client alert, May 13, 2026). When the surcharge expired on July 24, 2026, the administration replaced it with new Section 301 tariffs of 10 or 12.5 percent on imports from 60 economies, based on a finding that those economies failed to ban imports made with forced labor, with no built-in end date (Penn Wharton Budget Model, October 14, 2026; Congressional Research Service, October 2026). Meanwhile, about 166 billion dollars of IEEPA tariffs already collected became subject to refund, and customs had certified about 126 billion dollars of refunds by October 2, 2026 (Penn Wharton Budget Model, October 14, 2026). Notice the lesson about institutions, which Month 2 develops: a court enforcing the limits of a statute changed the price of almost every import into the world's largest consumer market. By August 24, 2026 the Budget Lab put the average statutory tariff rate at 11.0 percent, set to reach 11.8 percent by the end of 2026 under scheduled increases, while the Penn Wharton Budget Model, which measures duties actually collected, put the effective rate at 7.1 percent, with Chinese goods facing about 24 percent (Penn Wharton Budget Model, October 14, 2026). The gap between the two measures is itself a lesson: exemptions, trade-agreement rules and firms changing suppliers mean that headline tariff rates overstate what is actually paid.

For a student of value chains, three effects matter more than the headline numbers. First, uncertainty: firms delay investment when they cannot predict the rules. Second, rerouting: Chinese exports fell to the US but rose to Southeast Asia, Africa, Latin America and Europe, which is why China's global surplus grew. Third, "China plus one": some final assembly moved to Vietnam, Mexico, India and elsewhere, but much of the component supply stayed in China. Month 7 studies these shifts in detail.

Case study: China's export machine and the "China shock"

When China joined the WTO in December 2001, it gained secure access to world markets. Over the next decade its exports exploded. Economists David Autor, David Dorn and Gordon Hanson studied what this did to the United States in a famous 2013 paper, "The China Syndrome" (American Economic Review). They compared US local labor markets that were more or less exposed to Chinese import competition, because of the industries located there. Regions making furniture, toys, textiles and simple electronics lost many manufacturing jobs, and the losses did not quickly disappear: workers did not move easily to other places or industries, and many fell out of the labor force or onto disability benefits. Later work by the same authors estimated that Chinese import competition could explain a sizable share, though not most, of the decline in US manufacturing employment in the 2000s.

The China shock shows three things that the simple Ricardo model hides. First, gains from trade are real but spread thinly over all consumers (cheaper goods), while losses are concentrated on particular towns and workers. Second, adjustment is slow and painful when workers cannot easily move or retrain. Third, politics responds to the concentrated losses, not to the spread-out gains, which helps explain the backlash against trade in rich countries and the tariff wars of 2018-2026.

From China's side, this is the most successful export-led industrialization in history: from simple assembly of imported parts in the 1990s to a dense domestic supplier base, and by 2025 leadership in electric vehicles, batteries, solar panels, shipbuilding and industrial robots. Its 2025 trade surplus of 1.189 trillion dollars was only a little smaller than Saudi Arabia's entire 2025 GDP of about 1.27 trillion dollars, or Indonesia's of about 1.45 trillion dollars (Reuters and South China Morning Post, January 2026; IMF World Economic Outlook database, October 2026). In the first nine months of 2026, China's high-tech manufacturing output grew 14.3 percent year on year, with rapid growth in industrial robots and lithium-ion batteries, even while its property sector kept shrinking (Visual Capitalist summary of Chinese official data, October 2026).

There is a second China shock now under way, aimed less at rich countries than at middle-income ones. Countries such as Brazil, Indonesia, Mexico, South Africa, Thailand and Turkiye that hoped to build car, steel, battery or electronics industries now face Chinese competitors with enormous scale and low prices. Several of them raised tariffs or launched anti-dumping cases against Chinese goods in 2024-2026. The question for the driving theme of this course is sharp: can Africa or Southeast Asia build robot supply chains when China already produces robots, motors, magnets and batteries at a scale and cost no one else matches? We do not answer that yet. But you should already see that the answer will depend less on having ore and more on finding a place in the chain that Chinese scale does not crush, or on partnering with Chinese firms rather than competing head-on.

Key numbers

- World merchandise exports 2025: 26.26 trillion US dollars, up 7 percent (WTO, March 2026).
- World commercial services exports 2025: 9.56 trillion dollars, up 8 percent (WTO, March 2026).
- World goods and services trade 2025, balance-of-payments basis: 34.65 trillion dollars (WTO, March 2026).
- World goods trade volume growth: 4.6 percent in 2025; baseline forecast 1.9 percent for 2026 (WTO, March 2026).
- China's goods exports 2025: 3.77 trillion dollars, about 17.5 percent of world exports excluding intra-EU trade, 14.4 percent including it (WTO, March 2026).

- China's goods trade surplus 2025: 1.189 trillion dollars, a world record (Reuters, January 14, 2026).
- US average statutory tariff rate, August 24, 2026: 11.0 percent, projected 11.8 percent by end-2026 (Yale Budget Lab, August 24, 2026).
- IEEPA tariffs subject to refund after the Supreme Court ruling: about 166 billion dollars; about 126 billion dollars certified for refund by October 2, 2026 (Penn Wharton Budget Model, October 14, 2026).
- US effective tariff rate actually collected, August 2026: 7.1 percent; on Chinese imports: 24.3 percent (Penn Wharton Budget Model, October 14, 2026).

Think like a founder

Comparative advantage applies to firms and people, not just countries. Your startup should do the few things it does relatively best and buy, rent or partner for everything else, even things you could do yourself. For a founder in a developing country, the 2025-2026 trade turmoil creates two kinds of opening. The first is "tariff arbitrage": when tariffs differ by country of origin, value can be captured by whoever can legally and credibly shift assembly or processing to a country with better market access. This is real but fragile, because rules of origin keep tightening and governments punish simple transshipment. The second, more durable opening is building capabilities that make you a trusted supplier regardless of tariffs: quality certification, traceability (proof of where materials came from, increasingly required for batteries and minerals), and close relationships with buyers. Tariffs come and go; a reputation for quality and traceability keeps paying.

Week 3: Global value chains and the smile curve

Core lesson

From making whole products to doing one task

For most of history, a country that exported cars had to make most of the parts at home. That changed after about 1990. Cheaper shipping (the standard container), cheaper communication (the internet), lower tariffs, and the entry of China, India and the former Soviet bloc into the world economy made it possible to split production into stages and place each stage wherever it was cheapest or best. The economist Richard Baldwin calls this the "second unbundling": the first, in the nineteenth century, separated production from consumption, because steamships and railways made goods cheap to move; the second separated the stages of production from each other, because cheap communication made it possible to coordinate a factory in Thailand from an office in Tokyo.

A global value chain (GVC) is the full sequence of activities needed to bring a product from idea to final customer (research, design, raw materials, components, assembly, logistics, marketing, sales and after-sales service) when those activities are spread across at least two countries. The World Bank's World Development Report 2020 estimated that GVC trade (trade in which a good or service crosses at least two borders as it is being made) rose from about 37 percent of world trade in 1970 to a peak of about 52 percent in 2008, then flattened at a bit below half (World Bank, WDR 2020). The report's central message was that joining a GVC let poor countries industrialize without having to build entire industries from scratch. Bangladesh did not need to invent the sewing machine or a global fashion brand; it needed to sew well, cheaply and on time.

Gross trade versus trade in value added

GVCs make normal trade statistics misleading. When a phone assembled in China is shipped to the United States, US trade data record an import from China equal to the phone's full factory price, even if most of that price is made up of chips from Taiwan, Korea and the United States, a screen from Korea and Japanese components. China's actual contribution may be a small slice. Researchers at the OECD and WTO built "trade in value added" (TiVA) databases to trace where value is really created. The typical finding is that bilateral trade deficits measured in value added are often quite different from gross deficits, because much of what looks like "Chinese" exports contains other countries' value added. For our question this is essential: you cannot judge who captures value from a chain by looking at who ships the final box.

Two simple measures help:

- Backward participation: the share of foreign value added inside a country's exports. High for assembly hubs such as Vietnam or Mexico, which import many components.
- Forward participation: the share of a country's exports that are used as inputs in other countries' exports. High for raw material exporters (Chile's copper, Australia's iron ore) and for high-tech component makers (Taiwan's chips, Japan's specialty materials).

A mineral exporter and a chip designer can both have high forward participation, but the value captured per unit is wildly different. That difference is the subject of the smile curve.

Who governs the chain

Value chains have bosses. Gary Gereffi, John Humphrey and Timothy Sturgeon (2005) described five ways chains are governed, depending on how complex the information is that must pass between firms, how easy it is to write it down, and how capable the suppliers are:

- Market: simple products, many buyers and sellers, price does the coordinating (for example, standard steel bars or bulk copper cathode).
- Modular: complex products, but specifications can be written down precisely, and capable suppliers make whole modules to order (electronics contract manufacturing, where a firm such as Foxconn builds to a brand owner's design).
- Relational: complex, hard-to-write-down knowledge that requires trust and frequent contact (for example, advanced machine tools or specialty chemicals).
- Captive: small suppliers depend heavily on one powerful buyer who closely controls them (much garment production for global brands).
- Hierarchy: the lead firm does it in-house because the knowledge is too valuable or hard to transfer (for example, much of chip design or the core of a robot's control software).

The "lead firm", usually the brand owner or the holder of key technology, sets standards, prices and delivery terms, and decides who gets to participate. Being a supplier in a buyer-driven or captive chain can bring jobs and some learning, but it also leaves you exposed: the lead firm can switch suppliers, and it pushes cost cuts down the chain.

The smile curve

In the early 1990s Stan Shih, the founder of the Taiwanese computer company Acer, drew a simple chart that became famous. Put the stages of a value chain on the horizontal axis, from research and design on the left, through components and manufacturing and assembly in the middle, to

distribution, marketing, brands and after-sales service on the right. Put value added or profit margin on the vertical axis. The line curves down in the middle and up at both ends, like a smile.

A text version:

```
value / margin
  high |  R&D, design                          brand, marketing
       |  core IP  \                          /  services, data
       |            \  key components        /   distribution
       |             \                      /
   low |              \__ assembly, routine processing __/
       +-------------------------------------------------------
         upstream (ideas)     production     downstream (customers)
```

Why is the middle low? Three mechanisms:

- Contestability. Routine assembly and simple processing can be done by many firms in many countries. When many suppliers can do a job, competition pushes prices down toward cost, and margins disappear. Capital equipment for assembly can be bought by anyone.
- Scarcity at the ends. Good designs, patents, standards, brands, software platforms and customer relationships are hard to copy. Scarce things earn rents, which is economists' word for income above what is needed to keep a resource in use.
- Bargaining power. The lead firm usually sits at one end (a brand owner or a technology holder). It chooses suppliers, so suppliers compete for its orders; it does not compete for theirs.

Namchul Shin, Kenneth Kraemer and Jason Dedrick tested the idea in the global electronics industry and found evidence consistent with a smile: firms in design and branding, and some specialized component firms, earned higher gross margins than assemblers (Industry and Innovation, 2012).

Where the smile does not hold, and why that matters for robots and AI

The smile curve is a tendency, not a law. It breaks when the middle of the chain is itself hard to copy. The most important example today is advanced chip manufacturing. Taiwan Semiconductor Manufacturing Company (TSMC) does manufacturing, the supposedly low-value middle, yet it is one of the most profitable companies in the world, because building leading-edge chips requires tens of billions of dollars of equipment, decades of accumulated process knowledge, and an ecosystem of suppliers that almost no one else has. The Dutch company ASML, which makes the extreme ultraviolet lithography machines that leading chip factories need, is in a similar position. When a production step requires knowledge that is scarce, the smile flattens or even inverts at that step. Month 8 looks at this in depth.

The same logic applies at the far upstream end, mining. Usually raw ore sits low on the curve because many mines compete and the product is a standardized commodity. But when supply is concentrated (for example, most cobalt comes from one country, and most rare earth processing is in one country) or when prices spike, upstream owners can earn very large rents for a while. Month 5 shows that in critical minerals, the real chokepoint is often not the mine but the refinery, which China dominates. Value capture follows scarcity and control, wherever in the chain they happen to sit.

For robots, a first rough smile looks like this, to be tested with real data in Month 9:

Stage of the robot chain	Typical position on the curve	Why
Mining ore	Low, with occasional price spikes	Commodity; many mines; boom-bust prices
Refining, magnets, battery materials	Low to medium; high where concentrated	Capital and energy intensive; China dominant
Precision components (reducers, servo motors, sensors)	Medium to high	Hard know-how; few top suppliers in Japan, Germany, China
AI chips and compute	Very high	Extreme scarcity of design and fabrication capability
Robot assembly	Low to medium	Increasingly contestable as Chinese makers scale
Control software, AI models, data	High	Hard to copy; network effects; IP
Integration, deployment, service	Medium to high	Close to customer; local knowledge; recurring revenue

Upgrading: climbing the curve

Development scholars talk about "upgrading", meaning moving to higher-value activities. John Humphrey and Hubert Schmitz (2002) described four kinds:

- Process upgrading: doing the same task more efficiently (better quality, less waste).
- Product upgrading: making more sophisticated products (from basic to premium garments, from simple to precision castings).
- Functional upgrading: taking on new, higher-value functions in the chain, such as moving from assembly to design or to owning a brand. This is the hardest and most valuable kind.
- Inter-sectoral (chain) upgrading: using skills from one chain to enter another, as Taiwan moved from televisions to computers to chips.

Lead firms often welcome process and product upgrading by their suppliers, because it lowers their costs. They often resist functional upgrading, because a supplier that learns design or builds a brand becomes a competitor. This is why many countries get stuck in the low middle of the curve, a version of what is often called the "middle-income trap". How East Asian countries broke out is the core of Month 3.

Case study: who captured the value of the iPhone 4

In 2011 Kraemer, Linden and Dedrick published a careful study of who captured value from Apple's iPhone 4 and iPad. They took the 2010 iPhone 4 with a price of 549 US dollars and traced each component back to the firm and country that made it, estimating each firm's gross profit.

Their estimates for the iPhone 4 were striking (Kraemer, Linden and Dedrick, "Capturing Value in Global Networks: Apple's iPad and iPhone", 2011):

Who	Share of the 549 dollar price
Apple's profit	58.5 percent
Cost of materials (inputs)	21.9 percent

Who	Share of the 549 dollar price
Profits of Korean firms (mainly display and memory chips)	4.7 percent
Non-China labor	3.5 percent
Profits of non-Apple US firms	2.4 percent
Labor in China (final assembly)	1.8 percent, about 10 dollars
Profits of EU firms	1.1 percent
Profits of Japanese firms	0.5 percent
Profits of Taiwanese firms	0.5 percent
Unidentified profits	5.3 percent

Yet in US trade statistics, every iPhone 4 counted as an import from China worth its full factory price of a little under 200 dollars. China's assembly workers received about 10 dollars per phone. Apple, which did not make the phone at all, kept more than half of the retail price. For the cheapest iPad, Apple kept about 30 percent of the 499 dollar price, and Chinese labor about 2 percent.

The study's conclusion for policymakers was blunt: "there is little value in electronics assembly", so bringing high-volume assembly back to the US would not be the path to good jobs. The conclusion for managers was also blunt: suppliers that depend heavily on one powerful customer can be made or broken by that customer.

Three cautions keep this case honest. First, these are 2010 numbers; a follow-up study by Dedrick and Kraemer for the World Intellectual Property Organization (2017) found that Apple still captured most smartphone industry profits. Second, gross profit is not pure profit: Apple's share pays for its research, design, software and marketing. Third, China did not stay at 10 dollars per phone. Assembly taught Chinese firms and engineers how to make electronics at scale, and they used that learning to build Chinese phone brands and component makers. Low value capture at first was the price of entry into a learning process. That is the most important point for any developing country: the question is not only "how much value do we capture today?" but "does this position let us learn our way to capturing more tomorrow?"

Key numbers

- Share of world trade that is global value chain trade: about 37 percent in 1970, peak of about 52 percent in 2008, a bit below half since (World Bank, World Development Report 2020).
- iPhone 4 (2010) retail price studied: 549 dollars (Kraemer, Linden and Dedrick, 2011).
- Apple's share of iPhone 4 price: 58.5 percent (same source).
- Chinese assembly labor's share of iPhone 4 price: 1.8 percent, about 10 dollars (same source).
- Korean firms' profit share of iPhone 4 price: 4.7 percent (same source).
- Apple's share of the low-end 2010 iPad price: about 30 percent of 499 dollars (same source).
- World goods trade volume growth in 2025, largely driven by AI-related goods: 4.6 percent (WTO, March 2026).

Think like a founder

The smile curve is a map of where to stand. For a new firm, especially one in a country without big capital, the most promising places are usually near the customer end (services, integration, software, data, distribution) or in a narrow upstream niche where you can own something scarce (a specialized

design, a certified process, a dataset). The middle is where you go if you have scale and cheap capital, or if the middle step itself is hard (like advanced chips). Two practical rules follow. First, before you build a factory, ask what in your plan is scarce and hard to copy. If the honest answer is "nothing", expect thin margins. Second, if you do enter as a supplier in someone else's chain, design for functional upgrading from day one: keep engineers who understand the product, collect data on quality and failure, and build direct relationships with end customers where you can, so that each contract teaches you something you can later sell on your own.

Week 4: How money and capital flow across borders

Core lesson

The balance of payments

Every country keeps a record of its economic transactions with the rest of the world, called the balance of payments. It has two main parts:

- The current account: trade in goods and services, plus income flows (for example interest, dividends and wages paid across borders), plus transfers such as remittances (money migrants send home) and foreign aid.
- The financial account (with a small capital account): purchases and sales of assets across borders, such as factories, shares, bonds, bank loans and deposits, and changes in the central bank's foreign exchange reserves.

Because every transaction has two sides, the two accounts must balance (apart from measurement errors). If a country imports more than it exports and has a current account deficit, it must pay for the difference by selling assets to foreigners or borrowing from them, which shows up as a financial account inflow. If it runs a surplus, it must be acquiring foreign assets. This is the same identity as last week, seen from the money side: a surplus country is a net lender to the world, a deficit country a net borrower.

The foreign exchange market and the dollar

Most cross-border transactions require changing one currency into another. The foreign exchange (FX) market is the largest market in the world. The Bank for International Settlements (BIS), the central bank for central banks, surveys it every three years. In April 2025 FX turnover averaged 9.6 trillion US dollars per day, up 28 percent from 2022. The US dollar was on one side of 89 percent of all trades, the euro of 28.9 percent and the Japanese yen of 16.8 percent. Trading desks in just four places, the United Kingdom, the United States, Singapore and Hong Kong, handled 75 percent of all trading (BIS Triennial Survey, September 2025).

Compare that with world trade: 34.65 trillion dollars of goods and services trade in a whole year in 2025 (WTO). FX trading of 9.6 trillion dollars a day means the FX market turns over the value of a full year of world trade in less than four trading days. Most currency trading is therefore not about paying for trade; it is about investment, hedging (protecting against currency moves) and speculation. This is a central fact of the modern economy: finance is much larger than trade, and capital flows, not trade flows, usually move exchange rates in the short run.

Why the dollar? Because so much trade is invoiced in dollars, so much debt is issued in dollars, and US financial markets are the deepest and most liquid in the world, people hold dollars, which makes

the dollar even more useful, a self-reinforcing network effect. Month 11 studies the dollar system and its challengers in depth.

Types of capital flows

Not all capital flows are alike. They differ in how long they stay and what they bring.

- Foreign direct investment (FDI): buying or building a lasting controlling stake (by convention, at least 10 percent of voting power) in a firm abroad, for example a Japanese carmaker building a factory in Thailand, or a Chinese company building a nickel smelter in Indonesia. FDI is the most stable form, because factories cannot run away in a panic, and it can bring technology, management skills and market access. UNCTAD (United Nations Trade and Development) reported that global FDI rose 6 percent to about 1.6 trillion dollars in 2025, after two years of decline. Inflows to developed economies rose 11 percent to 723 billion dollars; inflows to developing economies rose only 2 percent to 901 billion dollars. Developing Asia received 644 billion dollars, Latin America and the Caribbean 188 billion dollars and Africa about 70 billion dollars. The 20 largest host economies took more than 80 percent of the total, and the least developed countries received just 43 billion dollars, 2.7 percent of global FDI (UNCTAD World Investment Report 2026). The United States was the largest recipient, at about 277 billion dollars.
- Portfolio investment: buying shares and bonds without control. It moves quickly in and out, depending on interest rates and risk appetite.
- Bank lending and other debt: loans from foreign banks, often short-term. Short-term foreign currency debt is the most dangerous kind, as we will see.
- Remittances: money sent home by migrant workers. According to the International Fund for Agricultural Development (IFAD), migrants sent 728.6 billion dollars to low- and middle-income countries in 2025, nearly double the level of a decade earlier, more than four times global official development aid and more than FDI to those countries. India alone received about 150.7 billion dollars (IFAD, Sending Money Home 2026). The average cost of sending money was 6.36 percent of the amount sent, much lower for digital services (about 4.6 percent) than non-digital ones (about 7.3 percent).
- Official flows: aid and loans from governments and international institutions such as the World Bank, the IMF and, increasingly, China's policy banks (Month 11).

One warning: much measured FDI passes through financial centers for tax reasons; UNCTAD reported that the 2025 rise was only 4 percent excluding such "conduit" flows.

Exchange rate regimes and the impossible trinity

A country must choose how to manage its currency. It can let it float (the market sets the price), peg it to another currency (the central bank promises a fixed rate), or do something in between. The "impossible trinity", from work by Robert Mundell and Marcus Fleming in the 1960s, says a country cannot have all three of the following at once:

- A fixed exchange rate.
- Free movement of capital in and out.
- An independent monetary policy (setting its own interest rates to manage its own economy).

It can pick any two. Hong Kong fixes its exchange rate to the dollar and allows free capital flows, so its interest rates must follow US rates. China has long managed its exchange rate and run its

own monetary policy, so it keeps controls on capital flows. The United States, the euro area and Indonesia today let their currencies float, allow capital to move, and set their own interest rates.

The mechanism is simple. If a country with a fixed rate and open capital markets tried to set interest rates below those abroad, investors would sell its currency to buy higher-yielding foreign assets. To defend the peg, the central bank would have to buy its own currency using its foreign exchange reserves. When reserves run low, the peg breaks.

Original sin, currency mismatch and sudden stops

Many developing countries and their firms cannot easily borrow abroad in their own currency, because foreign lenders do not want to bear the risk of it falling. Barry Eichengreen and Ricardo Hausmann called this "original sin". So they borrow in dollars. That creates a currency mismatch: their income is in local currency, their debt in dollars.

Here is a worked example. In mid-1997, one US dollar bought about 2,400 Indonesian rupiah. Imagine an Indonesian property developer who borrowed 100 million dollars. In rupiah, the debt was worth 240 billion rupiah. Now suppose the rupiah falls to 10,000 per dollar. The debt has not changed in dollars, but in rupiah it is now worth 1 trillion rupiah, more than four times as much, while the developer's rents and sales are still earned in rupiah. A company that looked healthy is suddenly bankrupt. When thousands of firms and banks are in that position at once, the currency's fall causes bankruptcies, bankruptcies scare investors, investors sell more of the currency, and the currency falls further. That is a vicious circle.

The economist Guillermo Calvo called the trigger a "sudden stop": foreign capital that was flowing in suddenly stops or reverses. Because the current account must be financed, a sudden stop forces the country to cut imports almost overnight, which means a deep recession.

Terms-of-trade shocks: the 2026 oil shock as a live example

A country's "terms of trade" is the price of its exports relative to the price of its imports. In 2026 the world got a sharp lesson. After the United States and Israel attacked Iran at the end of February 2026 and shipping through the Strait of Hormuz, which normally carries about a fifth of world oil, was badly disrupted, Brent crude rose from about 72 dollars a barrel on February 27 to near 120 dollars by March 9, and peaked at 126.41 dollars on April 30, 2026 (CNBC, April 2026; Reuters via The Business Standard, September 2026). The World Bank called it the largest oil market disruption in history (World Bank Open Data blog, May 2026). Prices fell back in June, rose again with renewed fighting, and were above 110 dollars at the end of October before dropping on hopes of a peace deal (Gulf News, November 2, 2026). For an oil importer, a jump like that widens the current account deficit, weakens the currency and raises inflation all at once. The IMF's October 2026 forecast assumed that oil prices would average about 96 dollars a barrel in 2026, up nearly 41 percent from 2025 (IMF, October 2026, as reported by AP and Asharq Al-Awsat). Here is how that works through an economy, step by step.

Take an invented importing country with GDP of 500 billion dollars that imports, net, 1 million barrels of oil a day, or 365 million barrels a year. At 70 dollars a barrel the bill is about 25.6 billion dollars, 5.1 percent of GDP. At 96 dollars it is about 35.0 billion dollars, 7.0 percent of GDP. Nothing about the country's factories changed, yet it must now send about 9.5 billion dollars more abroad each year, almost 2 percent of GDP, just to buy the same oil. That is a terms-of-trade loss: each unit of exports buys fewer barrels. Four effects follow.

- An income effect. Households and firms pay more for fuel and transport, so they have less to spend on everything else. Real national income falls even if real GDP (the volume of production) hardly moves at first.
- A balance-of-payments effect. The extra 9.5 billion dollars must be financed by exporting more, borrowing, selling assets or running down central bank reserves. If foreign investors are nervous, they demand higher interest rates or leave, which is where a terms-of-trade shock can turn into a sudden stop.
- An exchange-rate and inflation effect. Importers sell local currency to buy dollars for oil, so the currency tends to fall. A weaker currency makes every other import dearer too, so inflation rises by more than the oil price alone would suggest. The IMF's projected rise in world inflation to 4.7 percent in 2026 is largely this mechanism at global scale.
- A policy dilemma. The central bank can raise interest rates to defend the currency and fight inflation, which slows the economy further, or keep rates low and accept a weaker currency and higher inflation. Governments that subsidize fuel face a fiscal version of the same dilemma: the subsidy bill rises exactly when growth weakens.

The same shock is a windfall for exporters, which is why it redistributes value rather than simply destroying it. Every extra dollar paid by the importing country is received by someone: oil-exporting states, oil companies, and especially those who still have a route to market when a chokepoint closes. In 2026 Saudi Arabia's East-West pipeline to the Red Sea let it keep exporting while Hormuz was disrupted, so for a time higher prices more than offset lower volumes (Month 2, Week 3). In the language of Week 3, a disruption makes the bypass route, a storage tank or a spare refinery the scarce thing in the chain, and scarce things capture rents.

Three links to the rest of the course. First, energy is an input into every later stage: smelting aluminum or refining nickel, making chips, and running AI data centers all use large amounts of electricity, so a country's energy terms of trade shape its chances of moving up the chain (Months 4, 6 and 8). Second, metal exporters face the same arithmetic in reverse: when copper or cobalt prices fall, Zambia or the DRC lose income without producing a tonne less (Month 5). Third, a founder in an energy-importing country should treat the oil price as a risk to price, not a background fact: a logistics or manufacturing startup whose costs are in fuel and whose prices are fixed in contracts can be bankrupted by a shock it did nothing to cause, which is why hedging, fuel clauses in contracts and electrification all have real value. Month 4 studies energy in detail.

Case study: the 1997-1998 Asian financial crisis

In the early 1990s Thailand, Malaysia, Indonesia, the Philippines and South Korea were celebrated as "miracle" economies. Foreign money poured in, much of it as short-term bank loans in dollars. Thailand kept the baht pegged to a basket dominated by the dollar, so borrowers felt safe borrowing dollars at lower interest rates than baht. Much of the money went into property and the stock market rather than export industries. Thailand's current account deficit grew large.

In 1996 and early 1997, Thai exports slowed and property prices fell. Investors began to doubt that the peg could hold and sold baht. The Bank of Thailand spent a large share of its reserves defending the currency, much of it through forward contracts. On July 2, 1997, it gave up and let the baht float (IMF, International Capital Markets 1998). The baht fell sharply. Investors then asked: which other countries look like Thailand? Pressure spread to the Philippines, Malaysia and Indonesia, and later to South Korea.

Indonesia was hit hardest. Its macroeconomic indicators looked better than Thailand's: a modest

current account deficit, decent export growth and a government budget in surplus. But Indonesian companies had borrowed heavily in short-term dollars, often without hedging, and the banking system was weak and politically connected (IMF Occasional Paper 178, 1999). After widening its trading band, Bank Indonesia floated the rupiah on August 14, 1997. By early 1998 the rupiah had lost about 81 percent of its value against the dollar compared with July 1997; the baht had lost 56 percent, the Malaysian ringgit 46 percent and the Philippine peso 41 percent (IMF, International Capital Markets 1998). Indonesia's real GDP shrank by about 13 percent in 1998 and inflation reached about 70 percent (IMF Country Report 04/189, 2004). The economic collapse helped end President Suharto's 32-year rule in May 1998, amid riots in which many people died and Chinese-Indonesian communities were targeted.

The IMF organized large rescue packages, about 17 billion dollars for Thailand, 36 billion for Indonesia and about 57 billion for South Korea, with conditions including higher interest rates, budget cuts, bank closures and structural reforms (IMF, Finance and Development, September 1999). Those conditions became one of the most debated episodes in economic policy. Critics, including Joseph Stiglitz, then chief economist of the World Bank, argued that high interest rates and fiscal austerity deepened the recessions and that closing banks without a deposit guarantee in Indonesia triggered bank runs. Defenders argued that the countries needed to restore confidence and that without tight policy the currencies would have fallen even further. Judgment: the IMF's early fiscal tightening was a mistake for countries whose problem was private debt, not government debt, and the IMF later admitted parts of this; but the deeper cause of the crisis was the combination of a fixed exchange rate, open capital accounts, short-term foreign currency borrowing and weak bank supervision, which made the countries vulnerable to a sudden stop no matter what the IMF did afterwards.

The lessons shaped the next 25 years. Asian governments built up huge foreign exchange reserves as self-insurance, much of it invested in US government bonds, which became one pillar of the dollar system (Month 11); most moved to more flexible exchange rates and tighter bank supervision; and firms learned to fear short-term foreign currency debt.

For our driving question, the crisis teaches that capital is the oxygen of industrialization but can also suffocate it. A country trying to build mines, smelters and factories needs large amounts of foreign capital. If that capital comes as stable, long-term equity or FDI, it can build the chain. If it comes as short-term dollar debt, one bad year can wipe out a decade of progress.

Key numbers

- Foreign exchange turnover, April 2025: 9.6 trillion dollars per day, up 28 percent from 2022 (BIS Triennial Survey, September 2025).
- US dollar on one side of FX trades, April 2025: 89 percent (BIS, September 2025).
- Global FDI inflows 2025: about 1.6 trillion dollars, up 6 percent; developing economies 901 billion dollars; Africa about 70 billion dollars (UNCTAD World Investment Report 2026).
- Least developed countries' share of global FDI 2025: 2.7 percent, 43 billion dollars (UNCTAD, 2026).
- Remittances to low- and middle-income countries 2025: 728.6 billion dollars; India 150.7 billion dollars (IFAD, Sending Money Home 2026).
- Average cost of sending a remittance, 2025: 6.36 percent of the amount (IFAD, 2026; secondary summary only, not in IFAD's press release).
- Indonesian real GDP change in 1998: about minus 13 percent (IMF Country Report 04/189, 2004).

- Rupiah depreciation, July 1997 to early 1998: about 81 percent (IMF, International Capital Markets, 1998).
- Brent crude oil: about 72 dollars a barrel on February 27, 2026 to a peak of 126.41 dollars on April 30, 2026 (CNBC, April 2026; Reuters, September 2026).

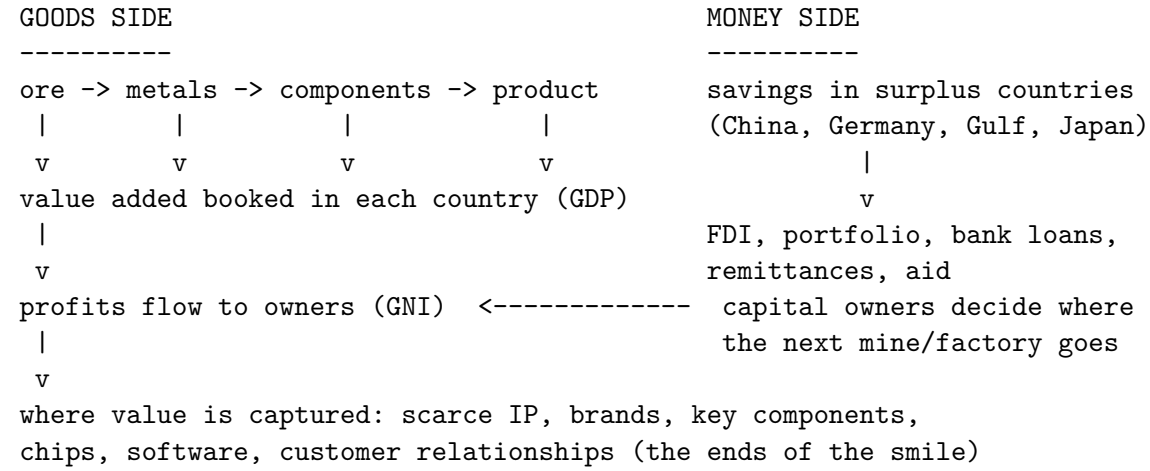
Think like a founder

Every founder in an emerging market is also, whether they like it or not, a currency manager. Three rules come straight out of this Week. First, match the currency of your debts to the currency of your revenues. If you earn rupiah, naira or pesos, borrowing in dollars to save a few points of interest can kill your company in one bad month. Second, prefer patient capital (equity, long-term loans, development finance) for long-lived assets such as factories and processing plants; short-term money belongs only in short-term uses such as working capital. Third, look at the plumbing of money itself as a market. Remittance fees of about 6 percent on more than 700 billion dollars a year mean that tens of billions of dollars are paid every year simply to move money across borders. Cheaper payments, trade finance for small exporters and currency hedging for small firms are large problems where software skills can capture value without owning a mine or factory.

Month synthesis

The four Weeks fit together. Week 1 gave you a ruler: value added. Week 2 explained why production is divided among countries. Week 3 showed how that division now happens task by task, and why profits pile up at the scarce ends of the chain. Week 4 showed the money side, and why the kind of capital a country attracts decides whether it can build lasting capabilities.

Put together, the system looks like this:



And here is a first, rough answer to the course's driving question, in table form. It will be refined every Month.

Link in the chain	What decides value capture	Month where we go deeper
Energy	Cost and reliability of power; who controls fuel routes	4

Link in the chain	What decides value capture	Month where we go deeper
Ore and mining	Ore quality, concentration of supply, price cycles, contracts and taxes	5, 6
Refining and processing	Energy cost, technology, scale; China's dominance	5, 6
Components	Know-how, precision, supplier clusters	7, 9
Chips and compute	Extreme scarcity of design and fabrication	8
Robot assembly	Scale and cost; growing contestability	9
AI models, software, data	Intellectual property, network effects, talent	8, 10
Finance	Cost of capital, currency of debt, dollar system	11
Rules and power	Tariffs, export controls, institutions, geopolitics	2, 3, 12

The main lesson of Month 1 is that location of production and location of value capture are different things, and that the gap between them is shaped by scarcity, bargaining power, ownership and the cost and kind of capital. The question of why some countries manage to own the scarce things and others do not is institutional and historical. That is Month 2.

Common misconceptions

1. "A trade deficit means a country is losing, and a surplus means it is winning." Wrong. A trade balance mostly reflects the gap between a country's saving and its investment. The United States has run deficits for decades while remaining the richest large economy; Germany and China run surpluses partly because their households consume relatively little. A deficit can be a problem when it is financed by short-term foreign currency debt (as in Thailand in 1997), not because deficits are bad in themselves.

2. "If a product says 'Made in China', China captured most of its value." Wrong. Gross trade statistics count the full value of the final product as an export of the assembling country. In the 2010 iPhone 4, Chinese assembly labor received about 1.8 percent of the retail price while Apple kept 58.5 percent (Kraemer, Linden and Dedrick, 2011). Always ask about value added, not shipment value.

3. "Comparative advantage means poor countries should stick to farming and raw materials." Wrong. Comparative advantage describes the best use of today's productivity. It changes as countries invest in skills, infrastructure and firms. South Korea's comparative advantage moved from wigs and plywood to chips and ships in about 40 years. Whether and how governments should speed that change up is debated (Month 3), but nothing in trade theory says advantages are fixed.

4. "Higher GDP growth always means people are better off." Not necessarily. GDP can rise because profits are booked in your country for tax reasons (Ireland 2015), because you are

selling off non-renewable minerals, or because the gains go to a small group. GDP is a measure of production, not of well-being or wealth.

5. "Foreign direct investment always brings technology and good jobs." Sometimes it does; sometimes it brings an enclave that imports its inputs, employs few local workers, sends profits home and teaches little. The effect depends on the sector, the contract terms, local supplier capacity and government policy. A large share of measured FDI is also just money passing through financial centers for tax reasons.

Exercises

1. (Quantitative) A smartphone sells for 400 dollars. A mining company sells metals worth 6 dollars; a refiner turns them into materials sold for 15 dollars; component makers sell parts for 180 dollars total; an assembler sells the finished phone to the brand owner for 210 dollars; the brand owner sells it to retailers for 340 dollars; retailers sell it to consumers for 400 dollars. Compute value added at each stage, check that they add to 400, and compute each stage's share. Which stages look like the "ends of the smile"?

2. (Quantitative) Country X needs 6 hours to make a ton of refined copper and 3 hours to make a solar panel. Country Y needs 30 hours for the copper and 10 hours for the panel. (a) Who has the absolute advantage in each? (b) Compute opportunity costs and identify comparative advantage. (c) Pick a trade price at which both gain and show the hours each saves.

3. (Quantitative) Using the rule of 70, how many years does it take an economy to double at 2 percent, 5 percent and 8 percent growth? If Country A has GDP per person of 5,000 dollars and grows 6 percent a year, while Country B has 40,000 dollars and grows 1.5 percent a year, roughly how many years until A catches B? (Hint: use logarithms, or a spreadsheet.)

4. (Quantitative) A firm earns all its revenue in local currency and has a 50 million dollar loan. The exchange rate moves from 15,000 to 18,000 local units per dollar. By what percentage does the local-currency value of its debt rise? If its annual operating profit is 900 billion local units and the interest rate on the dollar loan is 7 percent, how many times does its profit cover the interest bill before and after the move?

5. (Essay, 600-800 words) "Ireland's GDP is a fiction." Using GDP, GNI and GNI*, explain what is true and what is misleading in this statement, and what it teaches about the location of value capture in modern value chains.

6. (Essay, 600-800 words) Choose one product you use daily (a phone, a motorbike, a pair of shoes). Draw its value chain from raw materials to you, mark where value is captured, and explain why using contestability, scarcity and bargaining power. Separate evidence from guesses.

7. (Essay, 800-1,000 words) Was the IMF's response to the 1997-1998 Asian financial crisis right? Present the strongest case on each side and end with your own reasoned judgment.

8. (Founder exercise) Remittances to low- and middle-income countries were 728.6 billion dollars in 2025 at an average cost of 6.36 percent. Estimate the total annual fees. Then sketch a business that could capture part of this value while lowering costs for migrants. Who are the incumbents, what is scarce, and what regulation stands in your way?

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- Paul Krugman, Maurice Obstfeld and Marc Melitz, "International Economics: Theory and Policy" (many editions). The standard university textbook on trade and international finance; use it for the models behind Weeks 2 and 4.
- Richard Baldwin, "The Great Convergence: Information Technology and the New Globalization" (2016). The clearest account of how global value chains arose and why they let some poor countries industrialize fast.
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- Diane Coyle, "GDP: A Brief but Affectionate History" (2014). A short, honest book on what GDP measures and misses.
- Matthew Klein and Michael Pettis, "Trade Wars Are Class Wars" (2020). A provocative argument that trade imbalances come from inequality and policy choices inside countries; read critically.
- Dani Rodrik, "The Globalization Paradox" (2011). Why full economic globalization, national democracy and national sovereignty are hard to have all at once; useful background for the tariff wars.
- Paul Blustein, "The Chastening: Inside the Crisis That Rocked the Global Financial System and Humbled the IMF" (2001). A journalist's inside story of the Asian financial crisis.

Glossary terms introduced this month

- Gross domestic product (GDP): the market value of final goods and services produced within a country in a period.
- Value added: the value of a firm's output minus the value of inputs bought from other firms; the sum of all value added equals GDP.
- Nominal GDP / real GDP: GDP at current prices / GDP at constant base-year prices, which removes the effect of inflation.
- Purchasing power parity (PPP): an exchange rate that equalizes the cost of a common basket of goods across countries; used to compare living standards.
- Gross national income (GNI): GDP plus income residents earn abroad minus income foreigners earn in the country.
- Modified gross national income (GNI*): Ireland's adjusted GNI that removes the effects of re-domiciled companies and foreign-owned intellectual property and aircraft depreciation.
- Rule of 70: doubling time is about 70 divided by the growth rate.
- Comparative advantage: being able to produce a good at a lower opportunity cost than another producer.
- Heckscher-Ohlin model: theory that countries export goods using intensively the factors (labor, capital, land) they have in abundance.
- New trade theory: explanation of trade based on economies of scale and product variety.

- Dynamic comparative advantage: comparative advantage that changes as a country builds skills, capital and capabilities.
- Effective tariff rate: tariff revenue (or estimated duties) divided by import value.
- Global value chain (GVC): a production process for one product whose stages are spread across two or more countries.
- Trade in value added (TiVA): trade measured by where value is actually added, not by the gross value of shipments.
- Backward / forward participation: foreign value added in a country's exports / a country's value added used in other countries' exports.
- Lead firm: the firm that governs a value chain.
- Smile curve: the pattern in which value added and margins are high at the research and brand ends of a chain and low in the middle.
- Rent (economic): income above the minimum needed to keep a resource in its current use, usually earned from scarcity or control.
- Upgrading (process, product, functional, chain): moving to higher-value activities in or across value chains.
- Foreign direct investment (FDI): cross-border investment that gives lasting control (at least 10 percent of voting power).
- Impossible trinity: a country cannot have a fixed exchange rate, free capital flows and independent monetary policy all at once.
- Original sin: the inability of many countries to borrow abroad in their own currency.
- Currency mismatch: debts in one currency, income in another.
- Sudden stop: an abrupt halt or reversal of foreign capital inflows.
- Terms of trade: export prices relative to import prices.
- Foreign exchange reserves: foreign assets held by a central bank, used to defend the currency or pay for imports in a crisis.

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Month 2: Institutions and Why Nations Get Rich

Month overview

Month 1 gave you the tools to see where value is created and captured in the global economy: value added, comparative advantage, global value chains, the smile curve and the flows of money that connect them. But those tools leave a big question open. Why do some countries end up owning the scarce, high-value parts of the chain (the designs, the patents, the precision components, the brands, the financial power), while others stay stuck exporting raw ore or doing low-margin assembly for decades? Why did South Korea climb the smile curve while countries that were richer than Korea in 1960 did not? Why does diamond-rich Botswana look so different from diamond-rich Sierra Leone, or oil-rich Norway from oil-rich Venezuela?

This Month's answer begins with institutions: the rules, formal and informal, that shape how people can work, invest, own, trade and hold power to account. Week 1 covers the theory of "inclusive" versus "extractive" institutions, for which Daron Acemoglu, Simon Johnson and James Robinson won the 2024 Nobel Memorial Prize in Economic Sciences, and its critics. Week 2 studies the "resource curse" and how a few countries escaped it. Week 3 looks at rentier states that live on resource income rather than taxes, with the Gulf monarchies as the main case, including how the 2026 war involving Iran is testing their plans to diversify into AI and logistics. Week 4 goes back into history: how war built strong states in Europe, why Africa's path differed, and how colonial rule shaped institutions that still affect who captures value from minerals, told through the Congo from King Leopold's rubber to today's cobalt.

The thread that ties this Month to the course question is simple to state and hard to act on: value capture along the chain from ore to robot to AI depends less on what is in the ground than on who makes the rules, who enforces them, and whether those rules let ordinary people and firms invest, learn and keep the rewards. Month 3 then asks the practical follow-up: how have countries actually built the institutions and industries to climb the chain?

Learning goals

By the end of this Month you should be able to:

- Define institutions, distinguish inclusive from extractive economic and political institutions, and explain the mechanisms through which they affect investment, innovation and growth.
- Evaluate the evidence for the institutions view, including the "colonial origins" and "reversal of fortune" studies, and state the strongest criticisms (geography, human capital, culture, and

the China puzzle).
- Explain the resource curse and its main mechanisms (Dutch disease, price volatility, rent-seeking and conflict, and the weakening of the tax link between rulers and citizens), with a worked numeric example.
- Explain why Botswana and Norway largely avoided the resource curse, and what policies and institutions made the difference.
- Describe how rentier states work, using Saudi Arabia and the other Gulf monarchies as cases, and assess their diversification strategies in 2025-2026.
- Explain the "war made the state" argument for Europe, why it applies differently to Africa, and the main channels through which colonialism shaped present-day institutions.
- Connect all of this to the driving question: what institutional conditions would a country in Africa or Southeast Asia need in order to capture value from ore to robot by the 2040s.

Week 1: Inclusive and extractive institutions

Core lesson

What economists mean by institutions

The economic historian Douglass North, who won the Nobel prize in 1993, defined institutions as "the rules of the game in a society", the humanly devised constraints that shape human interaction. They include formal rules (constitutions, laws, property rights, contracts, courts, tax systems, central banks) and informal ones (customs, norms of trust, expectations about whether officials take bribes). Organizations, such as firms, political parties and ministries, are the players; institutions are the rules they play by.

Why should rules matter so much for growth? Because almost every productive act involves a bet on the future. A farmer who plants a tree, a family that pays for a child's schooling, a founder who spends three years building software, a mining company that spends 2 billion dollars on a mine that will produce for 30 years: each gives up something today in hope of a return later. If the return can be taken away by a thief, an official, a powerful rival or a sudden change in the law, people invest less, or they invest only in things that are quick, hidden and easy to move. Institutions decide how safe those bets are.

Inclusive versus extractive

Acemoglu, Johnson and Robinson (often shortened to AJR) built a framework around two types of institutions, set out most fully in Acemoglu and Robinson's book "Why Nations Fail" (2012).

Inclusive economic institutions:

- secure property rights for a broad cross-section of society, not only the elite;
- an unbiased legal system and courts that enforce contracts;
- public services (roads, education, public health) that give people a level playing field;
- freedom to enter new businesses and choose careers.

Extractive economic institutions are designed to take income and wealth from one part of society for the benefit of another: forced labor, monopolies granted to allies, confiscation, rules that block competitors, taxes and fees that fall on the weak and spare the powerful.

Economic institutions do not appear by accident. They are chosen through politics. So AJR also distinguish:

- Inclusive political institutions, which are sufficiently centralized (there is a state able to keep order, collect taxes and provide services) and pluralistic (power is broadly distributed and constrained, so that no small group can rewrite the rules for itself).
- Extractive political institutions, which concentrate power in a narrow elite with few constraints.

The core argument is that extractive political institutions tend to support extractive economic institutions, because the elite uses power to enrich itself, and the wealth then funds its hold on power. This creates a vicious circle. Inclusive political institutions tend to support inclusive economic institutions, which spread wealth and power more widely, which in turn makes it harder for any group to seize control: a virtuous circle.

The mechanisms: why inclusive institutions produce growth

The theory names several concrete mechanisms:

- Investment incentives. Secure property rights make people willing to invest in land, machines, skills and ideas.
- Allocation of talent. When anyone can enter any trade, talent flows to where it is most productive. When jobs and licenses are rationed by family or connections, talent is wasted.
- Creative destruction. Joseph Schumpeter's term for how new firms and technologies replace old ones. Growth requires it, but it threatens incumbents, including political elites whose wealth sits in old industries. Extractive elites therefore often block new technologies. Acemoglu and Robinson give historical examples, such as rulers who opposed railways or printing because they feared the social changes these would bring.
- Credible commitment. A ruler with unconstrained power cannot credibly promise not to expropriate investors tomorrow, because nothing stops them. Paradoxically, constraints on power make a state more trustworthy, and therefore able to borrow and attract investment more cheaply.

The Nobel committee summarized the last point clearly: elites in extractive systems may promise reforms, but "as long as the political system guarantees they will remain in control, no one will trust their promises of future economic reforms" (Royal Swedish Academy of Sciences, press release, October 14, 2024).

The evidence: colonial origins and the reversal of fortune

The hardest problem in this area is cause and effect. Rich countries have better institutions, but maybe being rich produces good institutions, rather than the other way round. AJR's most famous papers try to solve this with a "natural experiment" from colonial history.

In "The Colonial Origins of Comparative Development" (American Economic Review, 2001), AJR argued that Europeans adopted different colonization strategies depending on whether they could settle safely. Where disease killed settlers at high rates (much of tropical Africa, parts of the Caribbean), Europeans set up extractive institutions designed to move resources out with a small European presence. Where settlers survived (the future United States, Canada, Australia, New Zealand), they settled in large numbers and demanded institutions that protected their own property and rights. Those early institutions persisted. AJR used historical settler mortality as a source of

variation in institutions that was not caused by today's income, and found that institutions explain a large share of today's income differences among former colonies.

In "Reversal of Fortune" (Quarterly Journal of Economics, 2002), they showed that among former colonies, the places that were relatively rich and densely populated around 1500 (such as the Aztec and Inca empires and the Mughal lands) tend to be poorer today than the places that were poor and sparsely populated then (such as North America and Australia). Geography did not change, so geography alone cannot explain the reversal. AJR's explanation is that Europeans found it profitable to set up extractive systems where there were many people and existing wealth to exploit, and to set up settler institutions where there were few. The Nobel committee highlighted this "reversal of fortune" as a key finding.

The critics, and why the debate matters

The institutions view is now mainstream, but it has serious critics. You should know their best arguments.

- Geography and disease. Jeffrey Sachs and others argue that tropical disease, poor soils and distance from the sea directly hold back growth, independent of institutions. Even AJR accept that geography matters through its effect on institutions; the dispute is about direct effects.
- The settler mortality data. David Albouy (American Economic Review, 2012) questioned the quality of AJR's historical mortality data, arguing that many numbers were estimated or borrowed from other places and that the results weaken when this is corrected. AJR replied and defended their findings. Judgment: the data are noisier than the original paper suggested, but the broad pattern, that colonial strategy shaped institutions which shaped long-run outcomes, is supported by many other studies with different data.
- Human capital first. Edward Glaeser and coauthors argued in 2004 that settlers brought skills and education, not just institutions, and that education may drive both institutions and growth. Under this view, poor countries often grow first under dictators who invest in schooling, and better institutions come later.
- The China puzzle. China has grown faster for longer than any large economy in history while keeping a one-party political system. Does this disprove the theory? AJR answer that China's growth came after it introduced more inclusive economic institutions (allowing farmers to sell surplus crops, private firms to form, foreign investment to enter) and that growth under extractive political institutions can be fast while a country catches up, but will be harder to sustain once it must innovate at the frontier. Critics answer that China's state has been unusually capable and growth-focused, that local officials were rewarded for growth, and that "extractive" is too blunt a label. Judgment: China shows that a strong, development-focused state can produce decades of catch-up growth without pluralist politics; whether it can sustain frontier innovation in AI, chips and robots under tightening political control is the open question of the 2030s, and the evidence is still coming in. Treat both confident predictions ("China will stall" and "China proves institutions do not matter") with suspicion.

One more point matters for founders and policymakers: institutions are not just laws on paper. Many countries have copied excellent constitutions and mining codes and enforced none of them. What matters is the distribution of power that decides whether rules are enforced, which is why change usually requires a shift in who holds power, not just better legal drafting.

Case study: two Koreas, one people

The Korean peninsula is the closest thing history offers to a laboratory experiment in institutions. Before 1945 Korea was one country with one language, one culture and, broadly, similar geography north and south (the north actually held more of the heavy industry and mineral resources built under Japanese colonial rule). In 1945 it was divided at the 38th parallel between Soviet and American occupation zones. After the Korean War (1950-1953), the North built a communist command economy under the Kim family; the South, after decades of military-led authoritarian rule that nevertheless protected private property and pushed exports, became a democracy in the late 1980s.

The results are stark. The Bank of Korea, South Korea's central bank, estimates that North Korea's nominal gross national income in 2025 was 48.5 trillion won, just 1.8 percent of South Korea's 2,717.1 trillion won, and that North Korea's income per person was about 1.87 million won, about 3.6 percent of South Korea's 52.57 million won, or roughly one twenty-eighth (Bank of Korea, July 31, 2026, as reported by Yonhap). The Bank itself warns that its estimates use South Korean prices and should not be compared directly with other countries' accounts, because North Korea does not publish national accounts. Even allowing for large errors, the gap is enormous: roughly 28 to 1 in income per person. Life expectancy in the South is about ten years longer (Korea Economic Institute of America, 2026). North Korea did grow about 3.5 percent in 2025, its third straight year above 3 percent, but mainly because of expanded economic cooperation with Russia, including arms sales, and trade with China (Bank of Korea, 2026; Korea Economic Institute of America, September 2026).

The Korean case is powerful because it removes most alternative explanations. Geography, culture and history before 1945 were shared. What differed were the rules: whether farmers and firms could own and sell what they produced, whether they could start businesses, trade with the world and keep profits. But the case also complicates any simple story. South Korea was not inclusive in AJR's political sense during its fastest growth: Park Chung-hee, who ruled from 1961 to 1979, was a military dictator. What South Korea had was an elite that tied its survival to export success and allowed broad-based gains, including an early land reform, mass education, and competition among firms in export markets. Month 3 studies how that combination worked and how it later turned into a democracy. The lesson for this Month: the gap between the two Koreas shows that institutions can matter more than resources, culture or geography; the South's own path shows that the route to inclusive institutions is rarely straight.

Key numbers

- 2024 Nobel Memorial Prize in Economic Sciences: Daron Acemoglu, Simon Johnson and James Robinson, "for studies of how institutions are formed and affect prosperity" (Royal Swedish Academy of Sciences, October 14, 2024).
- North Korea nominal GNI 2025: 48.5 trillion won, 1.8 percent of South Korea's (Bank of Korea, July 2026).
- North Korea GNI per person 2025: about 1.87 million won, about 3.6 percent (one twenty-eighth) of South Korea's 52.57 million won (Bank of Korea, July 31, 2026).
- North Korea real GDP growth 2025: 3.5 percent; 2024: 3.7 percent (Bank of Korea, 2025 and 2026).
- North and South Korean populations: about 25.9 million and 51.7 million (Bank of Korea data, as reported in 2026).

Think like a founder

Institutions are not only a topic for governments; they are a business environment you must price. Before you invest time or money in a country, ask four concrete questions. Can I enforce a contract in a reasonable time and at a reasonable cost? Can my property, including intellectual property and data, be taken by an official or a connected competitor? Can a new firm enter the market, or are licenses rationed to insiders? And when rules change, is there a predictable process with a chance to be heard? Where the answers are bad, value capture shifts toward firms that are politically connected, which is a dangerous game for a young founder. Two strategies work better. First, build businesses that strengthen inclusive institutions and profit from them: transparent payments, digital land and business registries, credit scoring that lets small firms borrow, traceability that rewards honest suppliers. Second, structure your company so that the most valuable assets (code, IP, customer relationships, key contracts) are protected by strong institutions, for example through holding structures in jurisdictions with reliable courts, while still doing the real work and creating real jobs locally. Be open about this; hiding it creates its own political risk, as the Ireland case in Month 1 showed.

Week 2: The resource curse

Core lesson

A puzzle

Common sense says that finding oil, copper or diamonds should make a country richer. Often it does in the short run. But since the 1990s, economists have documented a troubling pattern: on average, countries that depend heavily on natural resource exports have grown more slowly over the long run than comparable countries without them, and they are more likely to suffer authoritarian rule, corruption and civil conflict. Jeffrey Sachs and Andrew Warner's studies in the 1990s found that countries with high natural resource exports relative to GDP in 1970 grew more slowly over the following decades. This became known as the "resource curse".

The problem is widespread. UNCTAD reports that two-thirds of developing countries, 95 of 143, were "commodity dependent" in 2021-2023, meaning that more than 60 percent of their merchandise export earnings came from commodities (fuels, minerals and agricultural products). In Africa, 85 percent of countries are commodity dependent (UNCTAD, The State of Commodity Dependence 2025; UNCTAD Trade and Development Board document, 2025).

The extreme modern example is Venezuela, which has the world's largest proven oil reserves and where oil long provided more than 90 percent of export earnings. The IMF estimates that Venezuela's real GDP contracted by more than 75 percent between 2013 and 2021, "the single largest economic collapse for a non-conflict country in almost half a century", and that about 7 million Venezuelans, around 23 percent of the population, had left the country by August 2022 (IMF, Regional Spillovers from the Venezuelan Crisis, 2022). The collapse began with the fall in oil prices in 2014 but was made far worse by years of mismanagement: price controls, money printing, borrowing against future oil, starving the state oil company of investment, and later sanctions.

But the curse is not destiny. Norway, Botswana, Chile, Malaysia, Australia and Canada all have large resource sectors and are among the more successful economies in their regions. The useful question is therefore not "do resources hurt?" but "through which mechanisms can they hurt, and what blocks those mechanisms?"

Mechanism 1: Dutch disease

In the 1960s and 1970s the Netherlands found large natural gas reserves. Gas exports surged, the Dutch currency strengthened, and Dutch manufacturing struggled. The Economist magazine coined the term "Dutch disease" in 1977. The mechanism works like this.

A resource boom brings a flood of foreign currency into the country. Some of it is spent at home, by the government and by people whose incomes rise. Much of that extra spending goes on "non-tradables", goods and services that cannot be imported easily, such as housing, construction, restaurants and local transport. Their prices rise. Wages rise to attract workers into those sectors. At the same time, the flood of foreign currency pushes up the value of the local currency. Both effects squeeze "tradables" other than the resource: manufacturing and farming that compete with imports or sell abroad at world prices they cannot change.

Here is a worked example with invented numbers. A garment exporter sells a batch of shirts abroad for 10 dollars. Its local costs (wages, rent, electricity) are 120,000 local currency units. At an exchange rate of 15,000 units per dollar, its costs equal 8 dollars, so it earns a profit of 2 dollars per batch.

Now a mining boom arrives. The local currency strengthens to 12,000 units per dollar, and local wages and rents rise by 10 percent, so costs rise to 132,000 units. The shirts still sell for 10 dollars, because the world price did not change. But now the costs are 132,000 divided by 12,000, which is 11 dollars. The firm loses 1 dollar per batch. Nothing about the firm got worse; the boom elsewhere made it uncompetitive. Multiply that across thousands of firms and the manufacturing base shrinks.

Why is this a problem rather than simply a shift to a more profitable sector? Because manufacturing is often where learning by doing, technology upgrading and productivity growth happen (the climb up the smile curve from Month 1). Mining typically employs few people per dollar of output and its prices swing. When the resource runs out or prices crash, the country finds that the industries it would need to fall back on have withered.

Mechanism 2: volatility

Commodity prices are far more volatile than prices of manufactured goods or services. Governments that depend on resource revenue tend to spend more in booms (often on projects that are hard to stop, such as public salaries and subsidies) and are then forced into painful cuts or heavy borrowing in busts. Volatility also discourages private investment outside the resource sector, because firms cannot predict the exchange rate or government demand. Botswana's experience in 2024-2026, discussed in the case study, shows that even a well-governed resource economy is exposed: when diamond prices crashed, its economy contracted.

Mechanism 3: rent-seeking, corruption and conflict

Resource wealth, especially "point-source" resources such as oil fields and mines concentrated in a few places, generates large rents (income above the cost of extraction) that are easy for whoever controls the state to capture. That makes control of the state extremely valuable, which can increase the incentive to fight for it, whether by coups, rigged elections or civil war. Paul Collier and Anke Hoeffler's research found that dependence on primary commodity exports was associated with a higher risk of civil war, and "lootable" resources such as alluvial diamonds, which can be dug by small groups with simple tools, have financed rebel movements, as in Sierra Leone and Liberia in

the 1990s. Rents also reward people for spending their energy getting a share of existing wealth (licenses, contracts, jobs in the state oil company) rather than creating new wealth.

Mechanism 4: the broken tax link

This mechanism links directly to Week 3. When a government gets most of its revenue from taxing its citizens, it must bargain with them; citizens who pay taxes demand services and a say in how the money is used. Historically, the slogan "no taxation without representation" captured this bargain. When a government gets most of its revenue from resource rents, it needs its citizens less. It can buy loyalty with jobs and subsidies and fund security forces without asking for consent. The political scientist Michael Ross, in "The Oil Curse" (2012), found that oil wealth is associated with more durable authoritarian rule, less gender equality in work and politics, and more civil conflict in some regions, while noting large differences between countries.

What breaks the curse

The successful cases share several features:

- Good institutions before the boom. Halvor Mehlum, Karl Moene and Ragnar Torvik showed in 2006 that resources tend to hurt growth in countries with weak, "grabber-friendly" institutions but not in those with strong, "producer-friendly" ones. Norway and Botswana both had relatively accountable governments and competent civil services when their resources were found.
- Saving and smoothing. Countries that save a large share of resource revenue in funds invested abroad avoid flooding the domestic economy with currency (limiting Dutch disease), smooth spending through price cycles, and share the wealth with future generations. Norway's Government Pension Fund Global is the best-known example, worth about 22 trillion Norwegian kroner, roughly 2.3 trillion dollars, at the end of September 2026 (Norges Bank Investment Management, October 2026). Under Norway's fiscal rule the government spends, over time, only about the expected real return of the fund, estimated at 3 percent a year; the 2027 budget proposed spending equal to about 2.7 percent of the fund's value (Norwegian government, October 2026).
- Transparent contracts and revenue. Publishing contracts, payments and production data, as promoted by the Extractive Industries Transparency Initiative (EITI), makes it harder to steal.
- Investing in people and broad infrastructure rather than prestige projects.
- A political settlement in which the main groups expect to share the gains over time, so no one has a strong reason to grab everything now.

Judgment: "resource curse" is a slightly misleading name. Resources do not curse; they amplify. In a country with accountable institutions, resource wealth is a large bonus. In a country with extractive institutions, it makes extraction more profitable and power more worth fighting for. That is why this Week sits inside a Month about institutions.

Case study: Botswana, diamonds and the hard part of success

When Botswana became independent from Britain in 1966, it was one of the poorest countries in the world, with only a few kilometers of paved road and very few university graduates. The next year, geologists found diamonds at Orapa, and later at Jwaneng, one of the richest diamond mines ever discovered. Over the following three decades Botswana was among the fastest-growing economies in the world. By the 2020s it had become an upper-middle-income country, with GDP projected by the

IMF at about 22 billion dollars in 2026, which works out to roughly 9,000 dollars per person for a population of about 2.5 million (IMF October 2026 data via Visual Capitalist; author's calculation).

Why did Botswana escape the curse that hit diamond producers such as Sierra Leone, where diamonds financed a brutal civil war from 1991 to 2002? Acemoglu, Johnson and Robinson studied this question in "An African Success Story: Botswana" (2003). Their answer combines history, leadership and institutions:

- Pre-colonial Tswana institutions included forms of consultation (the kgotla, a public assembly where chiefs heard their people) that placed some limits on chiefs' power.
- British colonial rule in Botswana (then Bechuanaland) was light, so these institutions were not destroyed and no large extractive colonial apparatus was built.
- The first president, Seretse Khama, and his allies were cattle owners with an interest in secure property rights and open trade, and they built a competent, merit-based civil service.
- Crucially, the mineral rights were vested in the national state rather than in any one ethnic group's chiefs, which reduced the incentive to fight over them.
- The government negotiated hard with the diamond company De Beers. In 1975 it raised its share of the joint mining company (later renamed Debswana) to 50 percent, and over time it also took a 15 percent stake in De Beers itself.

Botswana's democracy has been stable since independence. Its governing party ruled for 58 years, and in October 2024 it lost an election and handed over power peacefully to President Duma Boko's coalition: a strong test of institutions passed.

But Botswana's story is not finished, and 2024-2026 shows the hard part of success. Diamonds remained roughly three-quarters of merchandise export earnings (72.7 percent in the third quarter of 2025, according to central bank data reported by bne IntelliNews; other estimates run up to 80 percent) and about a third of government revenue. When natural diamond prices collapsed, under pressure from cheap laboratory-grown diamonds (which by 2025 held about 20 percent of the market and sold for 30 to 40 percent less, according to the government's Daily News) and weak demand in China and the United States, Botswana's economy shrank 2.8 percent in 2024 and 0.7 percent in 2025, as real value added in mining fell 14 percent (Statistics Botswana, via bne IntelliNews, November 2026). Debswana cut production and paused some mines, and the country built up a stockpile of about 12 million carats, nearly double its target (Reuters, January 2026). In the first nine months of 2026, De Beers' average realized price fell 37 percent to 97 dollars per carat, though part of that reflected a change in the mix of stones sold (bne IntelliNews, November 2026).

Botswana's government responded by renegotiating its relationship with De Beers. A new agreement signed on February 25, 2025 extended Debswana's mining licenses from August 2029 to July 2054 and set a 10-year sales agreement that can be extended by five more years. The share of Debswana output sold by the state-owned Okavango Diamond Company (ODC) rises in steps: from 25 percent before the deal to 30 percent in 2025-2030 and 40 percent in 2030-2035, reaching 50 percent only in the optional extension period of 2035-2040. The deal also created a Diamonds for Development Fund to finance diversification, with De Beers committing an upfront 1 billion pula plus further contributions linked to Debswana's performance (De Beers Group, May 2025; Reuters, February 25, 2025; Botswana government Daily News, November 2025). In May 2026 the former African Development Bank president Akinwumi Adesina was appointed to chair the fund (African Business, May 29, 2026). Notice what Botswana bargained for: not a bigger share of mining profit, which it already had through the 50:50 joint venture, but control over selling, which is one step closer to the customer end of the smile curve. With Anglo American selling its 85 percent stake in De Beers,

Botswana has been weighing whether to raise its own 15 percent stake to about 25 percent, while the IMF cautioned against taking on more diamond risk given the fiscal situation (IMF Article IV report on Botswana, 2025; bne IntelliNews, November 2026).

The lessons are layered. Good institutions let Botswana turn diamonds into roads, schools, clinics and savings, and avoid conflict. But good institutions did not by themselves produce diversification: after almost 60 years, the economy still rises and falls with one commodity whose value depends on consumers' feelings about natural versus lab-grown stones. Botswana's efforts to move into diamond cutting and polishing created some jobs but captured limited value, because the high-value parts of the chain (brands, jewelry design, retail) sit far away in New York, Antwerp, Mumbai and Hong Kong. Avoiding the curse is necessary; climbing the value chain is a separate, harder task, which is exactly what Months 3 and 6 examine.

Key numbers

- Developing economies that are commodity dependent (over 60 percent of merchandise exports from commodities), 2021-2023: 95 of 143, about two-thirds (UNCTAD, State of Commodity Dependence 2025).
- African countries that are commodity dependent: 85 percent (UNCTAD Trade and Development Board, 2025).
- Venezuela's real GDP fall, 2013-2021: more than 75 percent (IMF, 2022).
- Norway's Government Pension Fund Global: about 22,014 billion kroner, about 2.3 trillion dollars, on September 30, 2026 (NBIM, October 2026).
- Norway's planned fund spending in the 2027 budget: 608.4 billion kroner, about 2.7 percent of fund value, under a fiscal rule based on a 3 percent expected real return (Norwegian Ministry of Finance, October 7, 2026).
- Botswana real GDP: minus 2.8 percent in 2024, minus 0.7 percent in 2025 (Statistics Botswana, via bne IntelliNews, November 2026).
- Diamonds' share of Botswana's merchandise exports and government revenue: roughly three-quarters (72.7 percent of goods exports in the third quarter of 2025; estimates run up to 80 percent) and about one-third (bne IntelliNews, 2026, citing the Bank of Botswana; the same figure is used in Month 6).
- De Beers average realized rough diamond price, first nine months of 2026: 97 dollars per carat, down 37 percent from 155 dollars (De Beers Q3 2026 production report, October 20, 2026).

Think like a founder

The resource curse is, at bottom, a story about people chasing existing rents instead of building new value. For a founder, look for businesses that turn resource wealth into capabilities: environmental monitoring, mine-site automation and maintenance, AI-based exploration and geological data, traceability for responsibly sourced minerals, local supplier development and training. These sit on the higher-margin edges of the chain and build skills that survive when prices fall. Avoid models whose profits depend on a license a change of minister can cancel; if your margin comes from political access rather than from doing something better, you are in the rent game, and the house usually wins. And watch the currency: Dutch disease means your local costs rise in dollar terms during a boom, so build export businesses with enough margin to survive the boom as well as the bust.

Week 3: Rentier states and the Gulf

Core lesson

What a rentier state is

A "rentier" is someone who lives on rent: income from owning something rather than from working or producing. The Iranian economist Hossein Mahdavy used the term "rentier state" in 1970 to describe Iran under the Shah, and Hazem Beblawi and Giacomo Luciani developed the idea in their book "The Rentier State" (1987). A rentier state, in their sense, has three features:

- A large share of government revenue comes from external rents, mainly payments by foreigners for oil or gas, rather than from taxing domestic economic activity.
- Only a small part of the population is involved in generating that rent; most people are involved in distributing or consuming it.
- The government is the main receiver of the rent and decides how to distribute it.

In such a state, the usual relationship between rulers and citizens is turned around. Instead of the state taking money from citizens through taxes and having to justify itself, the state gives money to citizens through public jobs, subsidized fuel and electricity, free healthcare and education, cheap land and housing loans, and sometimes direct transfers. Political scientists summarize the bargain as "no taxation, no representation": citizens accept limited political voice in return for a share of the rent.

How the rentier bargain shapes the economy

The rentier bargain has predictable economic effects:

- Public employment. In the Gulf monarchies, a large share of employed citizens has traditionally worked for the government, often at salaries and conditions far better than private-sector jobs. This makes private-sector work, especially in manufacturing and services, unattractive to citizens.
- A two-tier labor market. Private-sector jobs have been filled largely by foreign workers. Non-citizens are a large majority of the population in the UAE and Qatar. In Saudi Arabia the 2022 census counted 32.2 million people, of whom 13.4 million, or 41.6 percent, were non-Saudis; among non-Saudis, about three in four were men, reflecting a male migrant workforce (General Authority for Statistics, census results announced May 31, 2023, via Saudi Press Agency and the Gulf Labour Markets and Migration programme). Many migrants worked under the "kafala" sponsorship system, which tied a worker's legal status to an employer; human rights groups documented abuses such as passport confiscation and unpaid wages. Several Gulf states have reformed parts of the system since the late 2010s, but enforcement varies.
- Weak non-resource taxes. Gulf states had no income tax and, until the late 2010s, no value-added tax (VAT). Saudi Arabia introduced a 5 percent VAT in 2018 and tripled it to 15 percent in 2020. Building a tax system is not just about money: it requires the state to know its economy and its citizens in detail, which is a form of state capacity.
- Fiscal dependence on oil prices. Budgets swing with oil prices. Analysts talk about a "fiscal breakeven" oil price, the price at which the budget balances. When prices fall below it, the state must cut spending, draw down savings or borrow.

The diversification problem

Every Gulf government knows that oil and gas are finite, that their prices are volatile, and that the global energy transition may reduce long-run demand. So all have "vision" plans to diversify. The difficulty is that the rentier bargain itself works against diversification: citizens prefer public jobs, private firms prefer cheap foreign labor, and many businesses depend on government contracts rather than on competing in export markets. In terms of Month 1, the Gulf economies capture enormous value at the upstream end of the energy chain, but they have struggled to build competitive positions further along any chain other than energy and petrochemicals.

The Gulf states have tried several strategies:

- Downstream energy: refining, petrochemicals, aluminum smelting and fertilizers using cheap gas and power; real successes, but still tied to hydrocarbons.
- Logistics and trade hubs: Dubai built one of the world's busiest ports at Jebel Ali, Emirates airline and a global aviation hub, and free zones with simplified rules for foreign firms. This was the most successful early diversification in the region.
- Finance and sovereign wealth funds: saving oil money in funds that invest worldwide, such as the Abu Dhabi Investment Authority, the Qatar Investment Authority, the Kuwait Investment Authority and Saudi Arabia's Public Investment Fund (PIF), with assets of about 3.4 trillion riyals (about 905 billion dollars) at the end of 2025 according to its annual report (Month 11 studies these funds).
- Tourism, entertainment and sports, including Saudi Arabia's hosting of Expo 2030 and the 2034 FIFA World Cup.
- AI and data centers: the newest bet, using cheap energy, land, capital and close ties to US technology firms to become hubs of computing power.

Gulf AI ambitions and the 2026 war

The AI strategy connects the Gulf directly to the driving question of this course. In May 2025 the UAE and the United States announced a 5-gigawatt "UAE-US AI Campus" in Abu Dhabi, the largest planned outside the United States. Its first part, Stargate UAE, is a 1-gigawatt cluster built by Khazna (part of the Abu Dhabi AI company G42) with partners including OpenAI, Oracle, Nvidia, Cisco and SoftBank; its first 200 megawatts were due in the fourth quarter of 2026 (Middle East Institute, May and September 2026; Enterprise AM, October 2026). Saudi Arabia's PIF created Humain, an AI company that aims for about 1.9 gigawatts of data center capacity by 2030 and over 6 gigawatts by 2034; Saudi official data showed the kingdom's data center capacity rising from 68 megawatts in 2021 to 467 megawatts in the first quarter of 2026 (Fortune, September 2026).

The logic is clear: AI data centers need huge amounts of electricity, and the Gulf has cheap energy, deep pockets and land. But the war that began at the end of February 2026 exposed a new risk. Iranian drones targeted Amazon data centers in the UAE and Bahrain soon after the fighting began (The Economist, September 2026). Iran's Revolutionary Guard publicly declared regional technology companies "legitimate targets" (Middle East Institute, September 2026). Reuters reported in September 2026 that the UAE was revising its 5-gigawatt plan from a single campus to a network of sites spread across the country, with underground construction and air defenses being considered, although work on Stargate UAE continued (Reuters via The Jerusalem Post, September 11, 2026). For anyone thinking about where AI's physical infrastructure will be built in the 2030s and 2040s, the lesson is that cheap energy is necessary but not sufficient: physical security and political stability are now part of the cost of compute.

Norway: the counter-example

Norway shows that a petroleum economy need not become a rentier state. When it found North Sea oil in 1969, it was already a democracy with a high tax-to-GDP ratio, a competent civil service and strong manufacturing, shipping and fishing industries. Oil revenue was added on top of a tax-funded state rather than replacing it. Norway built a state oil company (now Equinor) that learned to operate at the technological frontier, an oil-service industry that exports worldwide, and the savings fund described in Week 2. The difference from the Gulf is not geology. It is that Norway's institutions came first.

Case study: Saudi Arabia's Vision 2030 under fire

Saudi Arabia holds some of the world's largest oil reserves and is the leading member of the Organization of the Petroleum Exporting Countries (OPEC). In April 2016 Crown Prince Mohammed bin Salman launched Vision 2030, a plan to reduce the kingdom's dependence on oil by growing private business, tourism, entertainment, mining, manufacturing and technology; raising the share of Saudi citizens, especially women, in private-sector jobs; and building giga-projects such as NEOM, a planned new region on the Red Sea coast.

Ten years later, the record is mixed but real. The IMF reported that the Saudi economy grew 4.6 percent in 2025, with non-oil GDP growing 4.2 percent, inflation around 2 percent, and an unemployment rate among Saudi nationals of about 7 percent; it credited Vision 2030 with "strengthening the non-oil economy, expanding the role of the private sector, advancing diversification, and delivering labor market gains, including higher female labor force participation" (IMF Article IV press release, July 29, 2026). Official Saudi data put non-oil activities at 57.3 percent of real GDP in the first half of 2026 (Ministry of Finance, Mid-Year Economic and Fiscal report, September 2026).

But the state still depends on oil. The IMF estimates that the oil sector accounts for about 22 percent of GDP, more than half of fiscal revenues and about 60 percent of exports (IMF Country Report 26/210, 2026). Diversification of output has run ahead of diversification of the state's own finances. In 2025, with oil prices lower, the budget deficit widened to 5.8 percent of GDP, far above the 2.1 percent target, and public debt rose to 31.8 percent of GDP (IMF, 2026). The non-oil primary deficit, the IMF's preferred measure of how much the state spends beyond its non-oil income, was 23.3 percent of non-oil GDP in 2025. In plain words: take away oil revenue and interest payments, and the Saudi state spends almost a quarter of the non-oil economy more than it raises from it.

Then came the 2026 war. With the Strait of Hormuz disrupted, Saudi Arabia rerouted oil exports through its East-West pipeline to Red Sea ports, which limited the fall in deliveries; higher oil prices more than offset lower volumes for a time, producing a revenue windfall in the first half of 2026 (IMF, July 2026). In July the IMF cut its 2026 growth projection to 1.7 percent; by October, after the war hit volumes, it projected a contraction of 3.3 percent (IMF World Economic Outlook, October 2026; see Month 12). By the third quarter, however, oil revenue was down 23 percent from a year earlier, while non-oil revenue rose 20 percent, and the quarter's deficit was about 84 billion riyals, around 22 billion dollars (Saudi Ministry of Finance data, reported by Asharq Al-Awsat and Zawya, November 2026). The Ministry of Finance raised its estimate of the 2026 deficit to 4.9 percent of GDP from a budgeted 3.3 percent (Fortune, October 2026).

Under this pressure, the kingdom has made its plans more selective. In April 2026 the PIF's board approved a 2026-2030 strategy focused on six domestic "ecosystems", and the fund's governor confirmed that The Line, the planned 170-kilometer linear city that was the symbol of NEOM, was "no longer a priority" (Reuters, April 15, 2026). Work on The Line was reported to be delayed

until after 2030 (Semafor, May 2026), while industrial, energy and digital work at Oxagon, NEOM's port-industrial zone, continued (Gulf Construction, November 2026). The IMF itself advised Saudi Arabia to "recalibrate the PIF strategy toward more selective capital allocations and a greater private sector role" (IMF, July 2026).

What does this case teach? First, a rentier state can grow its non-oil economy with enough money and determination; progress on tourism, entertainment and women's employment is real. Second, while the state's own revenue depends on oil, the whole strategy depends on oil prices and geopolitics, as 2026 showed. Third, top-down giga-projects risk building things no market wants; the shift toward logistics, mining, AI and data centers reflects a search for returns. Fourth, for the driving question: the Gulf has the capital and energy to become a major node in the AI chain, and possibly in mineral processing, but lacks a large domestic engineering workforce and a dense supplier ecosystem, and sits next to one of the world's most dangerous chokepoints. Whether it captures value in robots and AI, rather than just hosting other people's computers, depends on turning rents into capabilities, the same question Botswana faces with diamonds.

Key numbers

- Saudi real GDP growth: 4.6 percent in 2025; IMF projection for 2026: 1.7 percent (IMF Article IV, July 2026), revised to -3.3 percent in the October 2026 World Economic Outlook.
- Saudi oil sector: about 22 percent of GDP, more than half of fiscal revenue, about 60 percent of exports of goods and services (IMF Country Report 26/210, 2026); oil and oil products are 69 percent of goods exports in the IMF's own summary table (IMF press release 26/267, July 2026).
- Saudi fiscal deficit 2025: 5.8 percent of GDP; public debt 31.8 percent of GDP (IMF, 2026).
- Saudi non-oil primary deficit 2025: 23.3 percent of non-oil GDP (IMF, 2026).
- Saudi non-oil activities' share of real GDP, first half of 2026: 57.3 percent (Saudi Ministry of Finance, September 2026).
- Saudi third-quarter 2026 oil revenue: down 23 percent year on year to 116.5 billion riyals; non-oil revenue up 20 percent to 142.6 billion riyals (Saudi Ministry of Finance, via Asharq Al-Awsat, November 2026).
- PIF size: assets under management of 3.396 trillion riyals, about 905 billion dollars, at the end of 2025 (PIF Annual Report 2025, as used in Month 11); Reuters rounded it to a "925 billion dollar fund" in April 2026.
- Saudi data center capacity: 68 megawatts in 2021 to 467 megawatts in the first quarter of 2026 (Saudi official data via Fortune, September 2026).
- UAE-US AI Campus: 5 gigawatts planned; Stargate UAE first phase 1 gigawatt, first 200 megawatts due in the fourth quarter of 2026 (Middle East Institute, 2026; Enterprise AM, October 2026).

Think like a founder

The Gulf is one of the richest pools of capital in the world looking to diversify, which makes it a natural partner for founders in energy, minerals, logistics and AI, including founders from Africa and Southeast Asia. But Gulf sovereign investors increasingly want capability at home: jobs for citizens, technology transfer, regional headquarters. Two kinds of business fit especially well: those that turn cheap energy into exportable value with better security or a smaller carbon footprint (green metals, resilient data centers, water technology), and those that help rentier economies build capabilities they lack (technical training, software for government services, platforms for small private firms). A

warning: a business whose only customer is a government flush with oil money is exposed to the next oil price crash and change of priorities, as contractors on The Line discovered in 2026.

Week 4: War, European state-building and colonial legacies

Core lesson

"War made the state, and the state made war"

Why did strong, tax-collecting states emerge first in Europe? The sociologist Charles Tilly gave a famous answer: "War made the state, and the state made war" (1975). He developed it in "Coercion, Capital, and European States, AD 990-1992" (1990). The mechanism runs like this.

From about 1500, the "military revolution" (gunpowder weapons, new fortifications, large standing armies and navies) made war far more expensive. Rulers who could not pay for modern armies were conquered by those who could. To pay, rulers needed money on a scale that personal estates and occasional levies could not provide. So they built tax systems, bureaucracies to run them, and ways to borrow. Doing this required bargaining with the people who held wealth: landowners, merchants and towns. In some places this bargaining created representative assemblies with power over taxes. Over centuries, intense competition eliminated most small political units. Europe went from hundreds of independent or semi-independent polities around 1500 to a few dozen states by the twentieth century. The survivors were states with real fiscal and administrative capacity.

Tilly also showed different paths: where coercion dominated (Russia, Prussia), states became bureaucratic and authoritarian; where capital dominated (Venice, the Dutch Republic), rulers depended on merchants; the most successful, such as Britain and France, combined large populations with commercial wealth.

Britain's fiscal-military state and credible commitment

Britain is the classic case. The historian John Brewer, in "The Sinews of Power" (1989), showed that eighteenth-century Britain, often imagined as a state with a small government, actually raised taxes several-fold and built an efficient tax bureaucracy (especially for excise taxes on goods) to fight a series of wars with France. It could also borrow more cheaply than its rivals.

Why? Douglass North and Barry Weingast argued in 1989 that the Glorious Revolution of 1688 gave Parliament, which represented property owners, control over taxation and limited the monarch's ability to seize property or default. The Bank of England was founded in 1694 to manage government borrowing. Because lenders trusted the government to repay, Britain could borrow huge sums cheaply and outspend France in war after war: constraints on power made the state stronger, the "credible commitment" mechanism from Week 1. Historians dispute how sudden the change in 1688 was, but the link between representative control of finance and cheap state borrowing is widely accepted.

Timothy Besley and Torsten Persson, in "Pillars of Prosperity" (2011), formalized why "state capacity" matters for growth. A state with fiscal capacity (the ability to tax broadly) and legal capacity (the ability to enforce contracts and property rights) can provide the public goods (roads, schools, courts, security) that markets need. They also show why such capacity is underbuilt: investing in it pays off only if the group in power expects to benefit, which happens when there are common interests (such as external war) and when institutions make it hard for rulers to grab the gains for themselves.

Today the differences in fiscal capacity are large. The average tax-to-GDP ratio across OECD countries was 34.1 percent in 2024 (OECD, Revenue Statistics 2025). The average across the 38 African countries in the OECD's Africa dataset was 16.1 percent in 2023, with 20 of the 38 below 15 percent, while the averages for Asia and the Pacific and for Latin America and the Caribbean were 19.6 and 21.3 percent (OECD, Revenue Statistics in Africa 2025). A state that collects 10 to 15 percent of GDP in taxes has much less to spend on the engineers, power grids, ports and courts that an ore-to-robot supply chain would need.

Why Africa's states formed differently

The political scientist Jeffrey Herbst, in "States and Power in Africa" (2000), argued that Tilly's European mechanism did not operate in most of Africa, for three reasons:

- Population density. For most of history, much of Africa had abundant land and few people. Rulers competed to control people rather than territory, and people could move away from oppressive rulers. Projecting power over large, thinly populated areas was expensive, so pre-colonial states often had strong centers and weak peripheries.
- Colonial borders. At the Berlin Conference of 1884-1885 and afterwards, European powers drew borders that often cut across ethnic, economic and ecological zones, mainly to settle rivalries among themselves. Colonial states built administration mostly around capitals, ports and mines, and spent little on governing the hinterland.
- The post-1945 norm against changing borders. After independence, the international community and the Organisation of African Unity (founded 1963) treated colonial borders as fixed. That prevented many interstate wars, which is a great good, but it also meant African states rarely faced the existential external competition that forced European states to build tax systems and bureaucracies. They could survive with weak internal control because outsiders recognized them anyway, a condition Robert Jackson and Carl Rosberg called "juridical statehood" (1982).

This is not an argument that wars are good. Tilly's European process killed millions and destroyed many societies, and nobody should want to repeat it. The point is to understand why state capacity is weak in many places, so that it can be built by other, peaceful means: credible tax bargains between governments and citizens, competent civil services, and constitutional limits that make taxpayers trust how their money is spent.

How colonialism shaped institutions: the main channels

Economists and historians have identified several channels through which colonial rule still affects development:

- Settlement versus extraction (AJR, Week 1). Where Europeans settled, they built institutions to protect their own rights; where they did not, they built institutions to extract.
- Forced labor systems. Melissa Dell studied the mita, a forced labor system that the Spanish imposed from 1573 to 1812 to supply workers to the silver mines of Potosi (in today's Bolivia) and the mercury mines of Huancavelica in Peru. Comparing villages just inside and just outside the old mita boundary, she found that areas subject to the mita today have household consumption about a quarter lower and more stunted children, partly because large landed estates, which might have invested in roads and protected peasants, were prevented from forming in mita areas (Dell, Econometrica, 2010).
- The slave trades. Between the sixteenth and nineteenth centuries, about 12.5 million Africans were forced onto ships in the transatlantic slave trade alone, besides the trans-Saharan, Red

Sea and Indian Ocean trades. Nathan Nunn found that the African countries from which the most people were taken are the poorest today (Quarterly Journal of Economics, 2008), and Nunn and Leonard Wantchekon found that ethnic groups more exposed to slave raiding show lower trust today, in neighbors, relatives and local government (American Economic Review, 2011). Low trust makes cooperation, contracts and collective action harder, a direct cost to institutions.

- Indirect rule and chiefs. In much of British Africa, colonial governments ruled through chiefs whom they appointed or empowered and who answered upward to the colonial state rather than downward to their people. Mahmood Mamdani, in "Citizen and Subject" (1996), argued that this created "decentralized despotism" whose effects survived independence.
- Infrastructure built for export. Railways and ports were built to move minerals and crops from the interior to the coast, not to connect African regions to each other. This shaped trade patterns, city locations and the logistics costs that any modern supply chain must still deal with.

The evidence is not all one-way, and an honest course must say so. Melissa Dell and Benjamin Olken studied the Dutch "Cultivation System" in Java (Indonesia), under which, from the 1830s to the 1870s, the colonial state forced villagers to grow sugar for colonial sugar factories. The system was brutally extractive, and its revenues at their peak provided over a third of the Dutch government's revenue. Yet Dell and Olken found that areas close to where the Dutch built sugar factories are today more industrialized, have better infrastructure and are more educated and richer than comparable nearby areas, because the factories and their supporting roads and railways created lasting clusters of economic activity (Dell and Olken, Review of Economic Studies, 2020). Other research on the same system found that places with heavy forced-labor demands had higher mortality in the nineteenth century. Both things are true at once: the system inflicted real suffering on the people who lived under it, and some of the physical and economic structures it created later became foundations for growth. Judgment: colonialism's effects worked through specific channels (institutions, labor coercion, trust, infrastructure, education) whose effects differ by place; blanket claims that colonialism "explains everything" about today's poverty, or that it "did not matter" or "brought development", are both wrong. What matters for a policymaker or founder today is identifying which specific legacies (weak tax capacity, low trust, export-only infrastructure, extractive contracts) still bind, and working on those.

Case study: the Congo, from rubber to cobalt

No country shows the long reach of extractive institutions in the mineral economy more clearly than the Democratic Republic of the Congo (DRC), and no country matters more to the battery and robot supply chain at its upstream end.

At the Berlin Conference of 1884-1885, European powers recognized King Leopold II of Belgium's claim to a vast territory in central Africa. From 1885 to 1908 the "Congo Free State" was in effect Leopold's private property, not a Belgian colony. When world demand for rubber soared with the spread of bicycles and cars in the 1890s, Leopold's agents and concession companies forced Congolese villages to deliver rubber quotas. Those who fell short faced hostage-taking, flogging, burning of villages, killings and mutilation. Reports by the British consul Roger Casement (1904) and campaigns by journalists and missionaries exposed the system. The number of deaths is unknown because there was no census; estimates of population loss run into the millions, and the journalist Adam Hochschild's widely read "King Leopold's Ghost" (1998) discusses an estimate of around 10 million, though historians debate how much of the decline came from killing, famine, flight and

disease. International outrage forced Belgium to annex the territory as a colony in 1908.

Belgian colonial rule was less murderous but still built around extraction. The copper and cobalt belt of Katanga in the south-east was developed by Union Miniere du Haut-Katanga, a company founded in 1906, and became one of the world's great mining regions. The colonial state invested in mines, railways and primary education, but very little in higher education or in preparing Congolese people to govern. When the Congo became independent on June 30, 1960, it had very few university-educated Congolese and almost no experience of self-government. Within days, the army mutinied and Katanga, backed by Belgian mining interests, declared secession. The elected prime minister, Patrice Lumumba, was overthrown and killed in January 1961; a Belgian parliamentary inquiry in 2001 concluded that some Belgian officials bore "moral responsibility" for the circumstances of his death. In 1965 Joseph-Desire Mobutu seized power with Western support and ruled until 1997. He nationalized Union Miniere into the state company Gecamines and then ran the state as a personal patronage machine; the term "kleptocracy" (rule by thieves) is often illustrated with his regime. Copper and cobalt production collapsed as Gecamines was looted and starved of investment. Two wars followed (1996-1997 and 1998-2003), drawing in neighboring countries' armies and causing enormous loss of life, and conflict has continued in the east, where armed groups, including the M23 rebels who took the city of Goma in early 2025, have fought over territory and mineral trade.

Today the DRC sits at the start of the battery and robot chain. It produced about 73 percent of the world's mined cobalt in 2025, followed by Indonesia with 14 percent (US Geological Survey, Mineral Commodity Summaries 2026; Cobalt Institute, 2026). Cobalt is used in many lithium-ion battery cathodes, in superalloys for jet engines and in some magnets. Yet most of the value is captured elsewhere. The DRC mostly exports cobalt as an intermediate product, cobalt hydroxide, and as of 2026 had no commercial plant making cobalt sulfate, battery cathode precursors or cells (Mines.cd, September 2026). China, which mines little cobalt itself, accounted for about 78 percent of global refined cobalt output in 2024 (International Energy Agency figure, via Reuters, February 2026), and Chinese companies own many of the DRC's largest cobalt mines. Alongside the industrial mines, a large artisanal sector employs many thousands of miners digging by hand, where Amnesty International and others documented child labor and deadly accidents in the 2010s. The DRC's GDP was projected at about 122 billion dollars in 2026 (IMF, October 2026), which for a population of over 100 million means roughly 1,100 to 1,200 dollars per person.

In 2025 the DRC government tried to change who captures value. Prices had collapsed because of oversupply, so in February 2025 the DRC suspended all cobalt exports. In October 2025 it replaced the ban with export quotas: about 96,600 tonnes a year for 2026 and 2027, including a base quota of 87,000 tonnes and a strategic quota of 9,600 tonnes, administered by a new regulator, ARECOMS (International Energy Agency policy database, 2026). Cobalt prices more than doubled during 2025 (Lobito Corridor industry site, 2026; Benchmark Mineral Intelligence via Cobalt Institute, 2026). The government then went further. An order signed on June 29, 2026 by the ministers of mines, foreign trade and the economy, made public on August 6, 2026, prohibited exports of copper and cobalt concentrates, while allowing the mines minister to grant one-year waivers in strategic, technical or economic cases (Reuters, August 6, 2026; S&P Global, August 6, 2026). Two details matter. First, the ban covers concentrate (crushed ore with the waste partly removed), not cobalt hydroxide or copper cathode, which are already processed further; companies such as CMOC that export cathode and hydroxide are largely outside its direct scope (Skillings Mining Review, August 9, 2026). Second, it formalizes a de facto ban on unprocessed concentrate exports that had existed since the mid-2010s, under which companies got case-by-case exemptions (Ivanhoe Mines, August 6, 2026; BMI via Mining Weekly, August 12, 2026). It is a ratchet, not a sudden shock.

This is resource nationalism in action, and it tests everything in this Month. Restricting supply can raise prices and government revenue in the short run because the DRC controls so much of the world's mined cobalt. Pushing processing at home could, in principle, move the DRC up the chain. But processing needs reliable electricity, skilled workers, chemicals, capital, security and confidence that rules will not change again; and buyers can respond to high prices by switching to cobalt-free batteries such as lithium iron phosphate, which already dominate many Chinese electric vehicles and storage systems. Whether the DRC's policy succeeds will depend less on how much cobalt lies under Katanga than on whether its institutions can offer investors and its own citizens credible, predictable rules: the lesson of Botswana, Norway and South Korea. Month 6 returns to this as a full case of downstreaming, alongside Indonesia's nickel export ban.

Worked numbers: what the quotas do, and where a tonne of cobalt ends up

The DRC's policy can be read with the tools of Month 1. Start with volumes. The DRC mined about 230,000 tonnes of cobalt in 2025, out of a world total of about 310,000 tonnes (USGS, 2026). The export quota for 2026 is 96,600 tonnes, about 42 percent of the 2025 mine output. In plain words, the country has chosen to let out less than half of what its mines can produce, and to stockpile or process the rest at home.

Now the price. The London Metal Exchange cash price of cobalt averaged about 11.84 dollars a pound in 2024 and about 15 dollars a pound in 2025 (USGS, 2026); an indicative price in early November 2026 was about 20.5 dollars a pound, or about 45 dollars a kilogram (MetalCharts, via Skillings Mining Review, November 2026). Compare two rough values of contained cobalt, at metal prices:

Scenario	Tonnes of cobalt	Price (dollars per pound)	Value of contained cobalt
2024 output at 2024 price	226,000	11.84	about 5.9 billion dollars
2026 quota at November 2026 price	96,600	20.5	about 4.4 billion dollars

The price rose by about 73 percent, but exportable volume fell by about 57 percent, so the value of what can legally leave in a year is lower than in 2024. The policy pays off for the DRC only if one or more of three things happens: the stockpiled cobalt is sold later at good prices; more of the cobalt is processed at home, so that each tonne earns more; or the higher price lifts the value of the DRC's huge reserves (about 6 million tonnes, half the world's, according to USGS) by more than it costs in lost sales now. This is the classic trade-off of a dominant producer, the same one OPEC faces with oil (Month 4): restrict volume and you raise price, but you also invite rivals and substitutes.

Two more caveats keep the numbers honest. First, the DRC does not sell cobalt metal; it mostly sells cobalt hydroxide, an intermediate product priced at a "payable", a percentage of the metal price that rises when hydroxide is scarce and falls when it is plentiful. So the DRC's actual revenue is below the values in the table, and part of the 2025-2026 price gain went to whoever held stocks outside the country. Second, the value that the state captures is not the value of the cobalt but its fiscal share of it (royalties, taxes, dividends of state companies such as Gecamines and the state cobalt trader EGC), plus local wages and purchases: the 29 cents per dollar logic from Month 1, Week 1.

Now follow a tonne down the chain. Cobalt hydroxide from Katanga is trucked to ports such as Dar es Salaam or Durban, shipped mostly to China, refined into cobalt sulfate, combined with nickel and manganese into a cathode precursor, made into cathode material, then into battery cells, then into a battery pack, and finally into a car, a power tool, a phone or a robot. At each stage, value added rises and so does the know-how needed. China's share of refined cobalt output, about 78 percent in 2024 (IEA, via Reuters), means that the steps where know-how and scale are scarce sit mostly outside the DRC.

The final step shows why ore owners have less leverage over the final product than they might hope. A high-nickel battery cathode contains, very roughly, on the order of 0.1 kilograms of cobalt per kilowatt-hour of capacity (author's rough estimate for illustration; Month 5 gives sourced figures by chemistry). A humanoid or mobile robot with a 2 kilowatt-hour battery would then contain about 0.2 kilograms of cobalt. At 45 dollars a kilogram, that is about 9 dollars of cobalt in a machine that might sell for tens of thousands of dollars. Even if the cobalt price doubled, the robot's price would barely move.

That small cost share has two opposite effects, and seeing both is the key insight of this case.

- In the short run it helps the DRC. Because cobalt is a tiny part of the final cost, buyers keep buying even at much higher prices (economists say demand is "inelastic"), so a quota can raise revenue per tonne quickly.
- In the long run it hurts the DRC. Battery makers can redesign products to use less cobalt or none at all, as lithium iron phosphate (LFP) chemistries already do in many Chinese cars and storage systems (USGS, 2026). Every time the DRC pushes the price up, it raises the reward for engineers in Shanghai or Seoul to design cobalt out.

Judgment: the quota is a rational short-run move by a dominant producer in an oversupplied market, and the concentrate ban is a modest step on a path the DRC has followed for a decade. Neither by itself moves the DRC up the smile curve. That would require domestic refining to sulfate and beyond, which needs cheap and reliable electricity, reagents, skilled chemical engineers and rules stable enough that investors will commit for 20 years. The deeper constraint is institutional, not geological, which is the theme of this Month.

Key numbers

- OECD average tax-to-GDP ratio: 34.1 percent in 2024 (OECD, Revenue Statistics 2025).
- Africa average tax-to-GDP ratio (38 countries): 16.1 percent in 2023; 20 of 38 countries below 15 percent (OECD, Revenue Statistics in Africa 2025).
- Asia-Pacific and Latin America-Caribbean average tax-to-GDP ratios: 19.6 and 21.3 percent in 2023 (OECD, 2025).
- DRC share of world mined cobalt: about 73 percent in 2025; Indonesia about 14 percent (USGS, Mineral Commodity Summaries 2026).
- China's share of global refined cobalt output: about 78 percent in 2024 (IEA, Global Critical Minerals Outlook 2025) and 79 percent of 240,000 tonnes in 2025 (Cobalt Institute, Cobalt Market Report 2025, May 2026).
- DRC cobalt export quota for 2026 and 2027: 96,600 tonnes a year, including 87,000 tonnes base and 9,600 tonnes strategic (IEA policy database, 2026; USGS, 2026), compared with DRC mine output of about 230,000 tonnes in 2025 (USGS, 2026).
- DRC ban on copper and cobalt concentrate exports: order signed June 29, 2026, made public August 6, 2026, with one-year waivers possible (Reuters, August 6, 2026).

- Botswana-De Beers deal (February 25, 2025): Debswana licenses extended to 2054; ODC share of sales 30 percent (2025-2030), 40 percent (2030-2035), 50 percent in an optional 2035-2040 extension (De Beers Group, 2025).
- Africans forced onto ships in the transatlantic slave trade, 1501-1866: about 12.5 million embarked, about 10.7 million survived to disembark (Trans-Atlantic Slave Trade Database estimates, as reported by the US National Endowment for the Humanities).
- Dutch Cultivation System revenues at their peak: over one-third of Dutch government revenue (van Zanden 2010, cited in Dell and Olken).

Think like a founder

If you want to build anything in the mineral chain in places like the DRC, Zambia, Guinea or eastern Indonesia, history is part of your risk model and your opportunity. First, legitimacy is a business asset: communities that have lived through generations of extraction judge a new mine or smelter by who benefits, and a "social license to operate" built on local hiring, fair compensation, transparent payments and real grievance processes is cheaper than the conflict that follows without it. Second, traceability is becoming a market requirement: buyers of batteries, electronics and robots increasingly need verifiable proof that minerals are free of child labor and conflict financing, and firms that provide it capture value that used to go to opaque traders. Third, rules will keep shifting as governments try to capture more value, as the DRC's quotas show; founders who design for that (processing partnerships with local firms, clear joint ventures with state companies) align with where politics is going. And remember the ethical floor: institutions are built by legitimate local actors, and outsiders can be partners in that work, never substitutes for it.

Month synthesis

Month 1 showed that value in the global economy is captured at the scarce, hard-to-copy points of a chain. Month 2 shows why some countries own those points and others do not. The pieces fit together like this:

```
HISTORY & GEOGRAPHY
(colonial strategy, war and state-building, slave trades, borders)
        |
        v
POLITICAL INSTITUTIONS  <------  RESOURCE RENTS (oil, diamonds,
(who holds power, and            cobalt) strengthen whoever
 what limits it)                 already holds power
        |
        v
ECONOMIC INSTITUTIONS
(property rights, contracts, entry, taxes, public goods)
        |
        v
INVESTMENT, LEARNING, UPGRADING (climbing the smile curve)
        |
        v
VALUE CAPTURE: owning scarce links of the chain,
or exporting raw rock and importing robots
```

Read the diagram from top to bottom. History and geography shape the starting distribution of power; political institutions decide whether that power is constrained; economic institutions follow from politics. Resource rents enter at the political level: where power is constrained, rents become savings, schools and infrastructure (Norway, Botswana); where it is not, they become a prize to be fought over (Sierra Leone in the 1990s, Mobutu's Zaire, Venezuela). Inclusive economic institutions let firms and workers invest, learn and climb the value chain; extractive ones keep a country exporting raw materials.

A summary of the Month's cases:

Case	Resource or starting point	Institutions	Outcome for value capture
South vs North Korea	Same people and history; North had more industry in 1945	South: export-oriented, private property, later democracy; North: command economy	South owns chips, ships, cars; North's income per person is about 1/28 of the South's
Botswana	Diamonds, very poor at independence	Accountable, competent state; hard bargaining with De Beers	Escaped the curse; still stuck upstream in one commodity
Venezuela	World's largest oil reserves	Increasingly extractive and mismanaged	GDP fell more than 75 percent, 2013-2021
Norway	North Sea oil	Strong democracy and tax state before oil	World's largest sovereign fund; frontier oil-service industry
Saudi Arabia	Vast oil	Rentier monarchy, top-down diversification	Non-oil economy growing, state finances still oil-driven
DRC	Copper and most of the world's cobalt	Extractive from Leopold to Mobutu; weak state today	Exports intermediate products; value captured in China and beyond

For the driving question, the Month's conclusion is this: Africa and Southeast Asia have the ore, and some have cheap energy and young workforces, but building a robot supply chain from ore to finished robot by the 2040s will require institutions that can (1) make credible long-term commitments to investors, (2) tax broadly enough to fund power, transport and education, (3) keep resource rents from becoming a prize for whoever holds power, and (4) allow new firms to enter and grow. Those are not the only requirements (Month 3 adds industrial policy, Month 4 energy, Month 7 supplier ecosystems), but without them, the other ingredients tend to leak away.

Common misconceptions

1. "Countries with lots of natural resources are destined to be rich," or the opposite, "resources are a curse that dooms countries." Both are wrong. Resources amplify the institutions a country already has. Norway and Botswana turned resources into prosperity; Venezuela and

Mobutu's Zaire turned them into disaster. The quality of institutions, especially before the boom, decides which.

2. "Good institutions just mean good laws, so a country can copy them." Laws on paper matter little if the people with power do not want them enforced. Many countries have excellent mining codes and constitutions that are routinely ignored. Institutions are rules plus the distribution of power that enforces them, which is why changing them usually requires political change, not just legal drafting.

3. "Democracy is necessary for fast growth," or "authoritarian states grow faster." Neither is a reliable rule. South Korea, Taiwan and China grew fast under authoritarian rule; many dictatorships were economic disasters. What matters most is whether economic institutions protect broad investment and allow competition, and whether power is constrained enough that rulers cannot simply grab the gains; over the long run, inclusive political institutions make those protections more durable.

4. "Colonialism explains all of today's poverty," or "colonialism is ancient history and no longer matters." Both are too simple. Specific colonial practices (forced labor, slave raiding, extractive concessions, export-only infrastructure, indirect rule) left measurable effects that persist, as the mita, slave trade and Congo cases show. But the effects differ by place, some colonial structures later supported growth (Java's sugar factory areas), and post-independence choices also matter greatly, as the contrast between Botswana and the DRC shows.

5. "Nationalizing or banning raw exports automatically captures more value for the country." Taking control of a resource changes who receives the rent, but capturing more value along the chain requires the capability to process and manufacture competitively: power, skills, capital and credible rules. Mobutu's nationalization of the Congo's copper led to collapse, not to more value. Whether today's export bans succeed depends on what institutions and capabilities stand behind them.

Exercises

1. (Quantitative, Dutch disease) An exporter of furniture sells a set abroad for 400 dollars. Its local costs are 5,400,000 local units, and the exchange rate is 15,000 local units per dollar. (a) Compute its dollar profit. (b) A mining boom causes the currency to appreciate to 13,000 per dollar and local costs to rise 8 percent. Compute the new profit. (c) What percentage cost cut would the firm need to break even?

2. (Quantitative, fiscal rule) A country has a sovereign fund worth 500 billion dollars and an expected real return of 3 percent. Its government spends exactly the expected return each year. (a) How much does it spend? (b) If the fund falls 20 percent in a market crash and the government sticks to 3 percent of the new value, by how much must spending fall? (c) Why might a government choose to spend 2.7 percent rather than 3 percent, as Norway proposed for 2027?

3. (Quantitative, rentier finances) A government has non-oil revenue equal to 15 percent of non-oil GDP and spends 38 percent of non-oil GDP (excluding interest). Compute its non-oil primary deficit. If oil revenue equals 20 percent of non-oil GDP in a good year and 12 percent in a bad year, what is its overall primary balance in each case (as a share of non-oil GDP)?

4. (Quantitative, comparison) Using the Bank of Korea figures, North Korea's GNI per person

was 1.87 million won and South Korea's 52.57 million won in 2025. Compute the ratio. If North Korea grew 3.5 percent a year in per-person terms and South Korea 1.5 percent, how many years would it take for the ratio to fall by half, from about 28 to about 14? What does your answer suggest about the limits of growth without institutional change?

5. (Essay, 800-1,000 words) "China's rise disproves the theory that inclusive institutions are necessary for prosperity." Present the strongest arguments for and against, and give your own reasoned judgment.

6. (Essay, 800-1,000 words) Compare Botswana and the DRC. Both are African mineral exporters with colonial histories. Using the mechanisms from this Month, explain why their outcomes differed so much, and what each could learn from the other in 2026.

7. (Essay, 600-800 words) Could the Gulf states become major producers in the AI value chain, not just hosts of data centers? What institutional and economic barriers stand in the way, and how did the 2026 war change the calculation?

8. (Founder exercise) Pick one mineral-producing country in Africa or Southeast Asia. Using public sources, assess its institutions on four questions: contract enforcement, property security, ease of entry for new firms, and predictability of rule changes. Then propose one business that would profit from improving one of these institutions, and explain who would pay for it.

Reading list

- Daron Acemoglu and James A. Robinson, "Why Nations Fail: The Origins of Power, Prosperity, and Poverty" (2012). The most readable statement of the inclusive-versus-extractive framework, full of historical cases.
- Daron Acemoglu and James A. Robinson, "The Narrow Corridor: States, Societies, and the Fate of Liberty" (2019). Extends the argument to how strong states and strong societies must balance each other.
- Douglass C. North, "Institutions, Institutional Change and Economic Performance" (1990). The foundational short book on what institutions are and why they matter.
- Michael L. Ross, "The Oil Curse: How Petroleum Wealth Shapes the Development of Nations" (2012). The best evidence-based book on how oil affects politics, conflict and gender.
- Paul Collier, "The Bottom Billion" (2007). A clear, opinionated account of the traps, including natural resources and conflict, that keep the poorest countries poor.
- Charles Tilly, "Coercion, Capital, and European States, AD 990-1992" (1990). The classic argument on how war built European states.
- Jeffrey Herbst, "States and Power in Africa: Comparative Lessons in Authority and Control" (2000). Why African state-building followed a different path from Europe's.
- Steffen Hertog, "Princes, Brokers, and Bureaucrats: Oil and the State in Saudi Arabia" (2010). How the Saudi rentier state actually works inside.
- Adam Hochschild, "King Leopold's Ghost" (1998). A gripping history of the Congo Free State and the first international human rights campaign.
- Daron Acemoglu, Simon Johnson and James A. Robinson, "The Colonial Origins of Comparative Development: An Empirical Investigation", American Economic Review (2001). The key empirical paper; read it alongside David Albouy's 2012 critique in the same journal.

Glossary terms introduced this month

- Institutions: the formal and informal rules that structure economic, political and social interaction.
- Inclusive economic institutions: rules that secure property for broad groups, enforce contracts fairly, provide public services and allow free entry.
- Extractive economic institutions: rules designed to transfer income and wealth from many to a few.
- Inclusive political institutions: political systems that are both sufficiently centralized and pluralistic, spreading and constraining power.
- Extractive political institutions: political systems that concentrate power in a narrow elite with few constraints.
- Creative destruction: the process by which new firms and technologies replace old ones.
- Credible commitment: a promise others can believe because the promiser is constrained from breaking it.
- Reversal of fortune: the finding that former colonies that were relatively rich in 1500 tend to be relatively poor today, and vice versa.
- Resource curse: the tendency of resource-dependent countries to experience slower growth, weaker institutions or more conflict.
- Commodity dependence: over 60 percent of merchandise exports from commodities (UNCTAD).
- Dutch disease: the decline of other tradable sectors caused by currency appreciation and rising local costs during a resource boom.
- Sovereign wealth fund: a state-owned investment fund, often financed by resource revenues.
- Extractive Industries Transparency Initiative (EITI): a global standard for publishing payments and contracts in oil, gas and mining.
- Rentier state: a state that lives mainly on external rents, especially from oil or gas, rather than on domestic taxation.
- Non-oil primary deficit: the gap between non-oil revenue and spending excluding interest, a measure of oil dependence.
- Kafala system: a sponsorship system in Gulf states tying migrant workers' legal status to their employer.
- Fiscal-military state: a state organized around raising taxes and borrowing to fight wars, as in eighteenth-century Britain.
- State capacity: a state's ability to tax, enforce laws and provide public goods; fiscal and legal capacity are its main parts.
- Tax-to-GDP ratio: total tax revenue as a share of GDP.
- Mita: the Spanish colonial forced labor system in the Andes (1573-1812).
- Indirect rule: colonial rule through chiefs answerable to colonizers.
- Cultivation System (Cultuurstelsel): the Dutch colonial system in Java (1830s-1870s) forcing villagers to grow export crops.
- Resource nationalism: government action to increase national control over, or the national share of value from, natural resources.
- Social license to operate: ongoing local acceptance of a project.

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Month 3: The Industrialization Playbooks

Month overview

In Month 1 you learned how the global economy is wired: gross domestic product (GDP), trade, global value chains and the "smile curve", which shows that the most value sits at the two ends of a product's life (research and design on one end, brands, distribution and services on the other) while plain assembly in the middle earns the least. In Month 2 you learned why some nations build institutions that let ordinary people invest and innovate, and why others stay trapped by extraction, resource rents or weak states.

This month asks a narrower and more practical question: **how did the countries that got rich actually do it?** Not in theory, but in the specific sequence of policies they used. Every country that industrialized between 1750 and 2025 followed some version of a "playbook". Britain went first, protected by a navy and fed by cheap coal. The United States and Germany copied Britain behind high tariff walls. Japan, South Korea and Taiwan built "developmental states" that forced firms to export and rewarded those that succeeded. China ran the largest industrialization in human history, mixing markets, foreign investment and fierce competition between local governments. Meanwhile much of Latin America and Africa tried to industrialize and stalled, sometimes using tools that looked similar on paper.

Why does this matter to you? Because your driving question, whether Africa or Southeast Asia could build a cheap supply chain from ore to finished robot by the 2040s, is really a question about which playbook can still work in a world where China already makes about 30% of all manufactured goods and where the United States, the European Union, Japan and India are all spending public money to pull factories home. This month gives you the history and the evidence. Month 4 then turns to the base layer that every playbook needs and that no country can skip: **energy**. You cannot run a steel mill, a chip fab or a robot factory without cheap, reliable power, and the history in this chapter is also a history of who had cheap energy and who did not.

Learning goals

By the end of this month you should be able to:

- Explain the three-part "East Asian playbook" (land reform, export discipline, financial control) and why each part matters, using real numbers from Japan, South Korea, Taiwan and China.
- Describe how Britain, the United States and Germany industrialized, including the role of cheap energy, tariffs, banks and colonial extraction.

- Compare a successful industrial policy with a failed one, and name the mechanism that separates them (usually discipline: subsidies tied to performance, especially export performance).
- Summarize the best recent evidence on industrial policy, including Nathan Lane's 2025 study of Korea's heavy industry drive.
- Explain what the US CHIPS Act and the Inflation Reduction Act tried to do, what changed in 2025-2026, and what early results show.
- Apply these lessons to a founder's question: where in a modern supply chain (for example, robots) can a late-developing country or a startup capture value?

Week 1: The First Industrializers: Britain, the United States and Germany

Core lesson

What "industrialization" means

Industrialization is the shift of a country's workers and capital from farming and handicrafts into factories that use machines and external energy (first water and coal, later oil and electricity). The key economic idea is **productivity**: how much output one worker produces in an hour. A handloom weaver in 1780 might produce a few metres of cloth a day. A power-loom worker supervising several machines in 1850 produced many times that. When productivity rises, wages can rise, prices fall, and the country gets richer.

Industrialization is not just "having factories". It is a self-reinforcing process: factories need machines, machines need machine-makers, machine-makers need steel and skilled workers, and all of them need energy and transport. Economists call these **linkages**. A "backward linkage" is demand that a factory creates for its inputs (a car plant needs steel). A "forward linkage" is supply that an industry creates for users downstream (a steel mill makes cheap steel available to shipbuilders). Countries that industrialized well built dense webs of these linkages. Keep this idea in mind; it will come back when we look at robots, whose parts (gear reducers, servo motors, magnets, sensors) each sit in their own web.

Britain: cheap coal, expensive labour, and a strong state

Britain was the first country to industrialize, starting roughly between 1760 and 1830. Why Britain and not China, India or the Netherlands? Historians disagree, but the most influential economic explanation today comes from Robert Allen (*The British Industrial Revolution in Global Perspective*, 2009). His argument is about **relative prices**.

In eighteenth-century Britain, wages were high compared with most of the world, while coal was very cheap, especially near the coalfields of the north. When labour is expensive and energy is cheap, it pays to invent and adopt machines that replace labour with energy. In India or China, where wages were much lower, the same machine would not have paid for itself. A worked example makes the logic clear:

- Suppose a spinning machine costs GBP 100 a year to own and run (interest, repairs, coal) and replaces the work of 5 spinners.
- In Britain, if a spinner earns GBP 20 a year, the machine saves 5 x GBP 20 = GBP 100. It just breaks even, and every improvement to the machine makes it profitable.

- In a region where a spinner earns GBP 5 a year, the machine saves only 5 x GBP 5 = GBP 25. It loses GBP 75 a year. Nobody adopts it, so nobody improves it.

This is why the first machines appeared in Britain. Once British engineers kept improving them (the spinning jenny, the water frame, the mule, the steam engine), machines eventually became cheap enough to pay even in low-wage countries. Allen's point is that the *starting* conditions decided where invention happened first.

Cheap energy was the foundation. Britain sat on large, accessible coal deposits. Coal replaced wood and muscle, and the steam engine turned coal into motion. This is the first appearance of the theme of Month 4: energy is the base layer of everything.

But relative prices are only part of the story. Britain also had:

- **A strong, tax-raising state and navy.** After 1688, Parliament could raise taxes and borrow cheaply. This paid for the Royal Navy, which protected trade routes. (Month 2 covered why war pushed European states to build tax systems.)
- **Protection of its own industry.** The textbook story says Britain was a free-trade country. That is true only *after* it had become dominant. Before that, Britain protected its industries. The Calico Acts (1700 and 1721) restricted imports and use of Indian cotton cloth, which was then better and cheaper than British cloth. The Navigation Acts (from 1651) required that trade with British colonies be carried on British ships. Britain moved to free trade mainly in the 1840s (the repeal of the Corn Laws in 1846), when its factories no longer feared competition. Economist Ha-Joon Chang calls this "kicking away the ladder": climbing up with protection, then telling others to climb without it.
- **Colonial extraction and slavery.** Be honest about this. The raw cotton that fed Lancashire's mills came largely from plantations in the American South worked by enslaved people. British rule in India also destroyed much of India's own textile industry, which had been a world leader in 1700. Economists still debate how much colonial profits *caused* British industrialization (many argue domestic factors mattered more), but there is no debate that colonial power shaped who supplied raw materials and who captured the manufacturing value. This is the earliest version of your driving question: the ore-or-cotton producer versus the factory owner.

By 1860 Britain produced about a fifth of all the world's manufactured goods (19.9%), and by 1880 almost a quarter (22.9%), while having only about 2% of the world's population (Paul Bairoch's estimates, 1982). By 1913 its share had fallen to 13.6% as the United States and Germany (14.8%) caught up. That is the scale of advantage that early industrialization created.

The United States: tariffs, a continental market and land

The United States industrialized mostly between 1820 and 1913, and it did so behind some of the highest tariff walls in the world. Alexander Hamilton, the first Secretary of the Treasury, argued in his *Report on Manufactures* (1791) that a young country needed to protect "infant industries" until they could compete with Britain. This is the **infant-industry argument**: a new industry may be unable to compete at first because it lacks experience, skilled workers and suppliers, but it can become competitive after a period of learning. Protection buys time to learn.

After the Civil War (1861-1865), US average tariffs on dutiable imports stayed roughly between 40% and 50% for decades (Chang, *Kicking Away the Ladder*, 2002, drawing on US historical statistics). At the same time, the United States had advantages no tariff can create:

- **A huge internal market** with free trade between states, connected by railways (the first transcontinental line was completed in 1869).
- **Abundant land and resources**, including coal, iron ore, copper and later oil. By the early 1900s the US was the world's largest producer of many minerals.
- **Mass immigration** that supplied workers and skills.
- **Public investment in knowledge**, such as the land-grant colleges created by the Morrill Act of 1862, which trained engineers and farmers.

By 1913, the United States had become the largest manufacturing economy in the world, with about a third (32.0%) of global manufacturing output, up from 14.7% in 1880 (Bairoch, 1982). It achieved this as a protectionist country. It became the champion of free trade only after 1945, when, like Britain a century earlier, it no longer feared competition.

Note what the tariff did and did not do. The tariff protected the market. But the US market was so large and so competitive internally that firms still had to fight each other hard. Protection without competition breeds lazy firms. Protection *with* fierce internal competition, in a market big enough for many firms to reach large scale, can work. Remember this distinction; it explains much of the difference between East Asia and Latin America later in this chapter.

Germany: catching up with banks, cartels and engineers

Germany started later. Before 1871 it was a patchwork of states. The Zollverein, a customs union formed in 1834, created a common internal market. Friedrich List, a German economist who had lived in the United States, wrote *The National System of Political Economy* (1841), arguing that Germany needed tariffs to grow its industry, just as the US had.

The economic historian Alexander Gerschenkron (*Economic Backwardness in Historical Perspective*, 1962) explained Germany's path with a powerful idea: the **advantages of backwardness**. A latecomer does not have to invent everything. It can copy the most advanced technology and skip earlier stages. But catching up requires investment on a scale that small private savers cannot provide. So latecomers need *institutions that concentrate capital*. In Britain, rich individuals and small partnerships could finance early mills. In Germany, which needed huge steelworks, chemical plants and railways all at once, the answer was **universal banks** (such as Deutsche Bank, founded 1870) that both lent to and owned shares in industrial firms, sat on their boards, and organized mergers. In Russia, which was even more backward, the state itself had to play this role.

Germany also invested heavily in **technical education** and research. Its universities and technical schools produced chemists and engineers, and companies like BASF, Bayer and Siemens built some of the first industrial research laboratories. By 1913 Germany led the world in chemicals and electrical equipment, which were the high-technology industries of the time.

Gerschenkron's insight is directly useful for your question. The later a country starts, the more capital-intensive the frontier industries are, and the more it needs some institution (banks, the state, a sovereign fund, or foreign investors) to concentrate capital. A robot or battery supply chain in 2035 needs far more capital per worker than a cotton mill in 1800. Month 11 will return to who can supply that capital.

The pattern so far

Three early industrializers, three different stories, but four shared features:

Feature	Britain	United States	Germany
Cheap energy	Coal near industry	Coal, then oil	Ruhr coal
Protection while young	Calico and Navigation Acts	Tariffs of 40-50% after 1865	Tariffs after 1879
Capital concentration	Merchants, partnerships, later London markets	Railroads, banks, trusts	Universal banks, cartels
Knowledge investment	Practical tinkerers, engineers	Land-grant colleges	Technical universities, corporate labs

Each country also benefited from something hard to copy: Britain from empire and being first, the US from a vast continent, Germany from coal and a strong scientific tradition. Late developers in the twentieth century had none of these. They had to invent new tools.

Case study: Lancashire cotton and the logic of the first factory

Lancashire, in northwest England, became the world's first great industrial region. Its story shows all the forces of Week 1 at work.

In 1700 India was the world's leading producer of cotton cloth. Indian calicoes and muslins were cheaper and finer than anything Britain could make, and they were so popular in Britain that wool and silk producers lobbied Parliament to restrict them. The Calico Acts did exactly that. This protected domestic demand for *British-made* cotton goods and gave British producers a home market to grow into.

Then came the machines. James Hargreaves's spinning jenny (around 1764), Richard Arkwright's water frame (patented 1769) and Samuel Crompton's mule (1779) multiplied the amount of yarn one worker could spin. Arkwright's Cromford mill (1771) is often called the first modern factory: a building full of machines, driven by one power source, with workers on fixed shifts. After the 1780s, steam engines built by Boulton and Watt let mills move from riversides to towns near coal. Manchester grew from a market town of perhaps 25,000 people in the 1770s to a city of several hundred thousand by the 1850s.

Raw cotton imports tell the story of scale. Britain imported only a few million pounds of raw cotton a year in the 1760s; by the 1830s it imported hundreds of millions of pounds a year, most of it from the American South (figures from standard economic histories of the period; treat the exact numbers as approximate). That cotton was grown by enslaved people. The value chain ran: enslaved labour in Mississippi grew the fibre (the "ore" of the textile age) -> ships carried it to Liverpool -> Lancashire mills spun and wove it -> British merchants sold the cloth worldwide, including to India, whose own industry was being undercut.

Who captured the value? Not the enslaved workers, who were paid nothing. Not, mostly, the Indian weavers, who lost their market. The value went to the mill owners, the machine builders, the shipping and insurance firms, the merchants and banks of Liverpool, Manchester and London, and to the British state through taxes. The fibre producer was at the bottom of the smile curve; the industrial and commercial nodes were at the top.

Three lessons for today:

1. **Protection of the home market came first, then efficiency.** Lancashire grew inside a protected market and only later became a free-trade champion.
2. **Cheap energy turned an invention into an industry.** Water power started the mills; coal and steam scaled them.
3. **Control of the processing and the trading network mattered more than owning the raw material.** This is the same pattern you will see with nickel, cobalt and lithium in Months 5 and 6.

Key numbers: Week 1

- **19.9% and 22.9%**: Britain's share of world manufacturing output in 1860 and 1880 (13.6% by 1913), with about 2% of world population (Bairoch, *Journal of European Economic History*, 1982, Table 10, as reproduced by Crafts and Wolf, CAGE, 2015).
- **32.0%**: US share of world manufacturing output in 1913, up from 14.7% in 1880 (Bairoch, 1982, via Crafts and Wolf, 2015).
- **40-50%**: Approximate range of average US tariffs on dutiable imports for much of 1865-1913 (Chang, *Kicking Away the Ladder*, 2002).
- **1834**: Formation of the German Zollverein customs union.
- **1846**: Britain repeals the Corn Laws, symbolic start of its free-trade era, roughly 80 years after its industrial revolution began.
- **1791**: Hamilton's *Report on Manufactures*, the first full statement of the infant-industry argument.
- **1870**: Founding of Deutsche Bank, a model of the universal bank that financed German heavy industry.

(Historical shares are estimates by economic historians, not official statistics. Treat them as approximate.)

Think like a founder

The first industrial revolution rewarded whoever controlled the **conversion step** (turning cheap fibre and cheap coal into cloth at scale) and the **network** (shipping, finance, distribution). The cotton grower, even when free, captured little.

For a founder in 2026, the equivalent question is: in the chain from ore to robot to AI, where is the conversion step that is about to become cheap because energy or a key input is getting cheaper? Allen's logic still applies. When a factor becomes cheap (solar electricity in Month 4, falling battery costs, cheap AI inference in Month 8), products that use that factor intensively suddenly become profitable in new places. A founder who spots a "relative price shift" early, for example cheap daytime solar power making some electricity-hungry process profitable in a sunny, low-wage country, is doing exactly what Arkwright did. The opportunity is not to own the mine. It is to own the machine that converts the newly cheap input into something people pay for.

Week 2: The Developmental State: Japan and South Korea

Core lesson

A new kind of state

After 1945, a group of East Asian economies grew faster for longer than any countries before them. Japan grew at roughly 9-10% a year in real terms from the mid-1950s to the early 1970s. South Korea and Taiwan grew at roughly 8-10% a year for three decades after the early 1960s. The World Bank's famous 1993 report called this "the East Asian Miracle".

Political scientist Chalmers Johnson (*MITI and the Japanese Miracle*, 1982) gave the model a name: the **developmental state**. In a developmental state, a capable, elite bureaucracy sets national economic goals, chooses industries to promote, and uses its power over credit, imports and licences to push private firms toward those goals. It is not socialism (firms are privately owned and compete), and it is not free-market capitalism (the state actively steers).

Journalist and economist Joe Studwell (*How Asia Works*, 2013) boils the successful East Asian model down to three steps, in order:

1. **Land reform**: break up big estates and give land to the families who farm it.
2. **Manufacturing with export discipline**: protect and subsidize manufacturing firms, but force them to compete in export markets and cut off support to those that fail.
3. **Financial control**: keep the banking system under state direction so savings flow into manufacturing rather than property speculation or consumption.

This week covers Japan and South Korea. Week 3 covers Taiwan and China, and Week 4 looks at countries that tried and failed. As you read, test each step: why does it matter, and what happens if you skip it?

Step 1: Land reform

Why would land reform matter for building factories? There are three mechanisms.

- **Yields rise.** A family farming its own small plot works it intensively, like a garden, and keeps the gains. Studwell shows that small household farms in Asia produce more output per hectare than big estates using hired labour or tenants, especially in rice. Higher food output means cheap food for future factory workers and fewer food imports.
- **Rural demand and savings grow.** Millions of farmers with higher incomes buy fertilizer, bicycles, radios and clothes, creating the first market for local manufacturers. They also save, and those savings flow through banks into industry.
- **Political power shifts.** Big landlords are usually a conservative elite who prefer to invest in land and imports rather than factories. Breaking their power removes a lobby that would block industrialization.

The East Asian land reforms were among the most thorough in history. In Japan, the US occupation authorities pushed a reform (1946-1950) that transferred land from landlords to tenants; tenancy fell from covering roughly half of farmland to roughly a tenth. South Korea carried out land reform in 1949-1950s, partly under the threat of communist North Korea, which had already seized and redistributed land. Taiwan's reform (rent reduction to 37.5% of the harvest in 1949, then the "Land to the Tiller" programme in 1953) is covered next week. (These figures are standard in the development literature, including Studwell; treat exact percentages as approximate.)

Contrast this with the Philippines and much of Latin America, where land reform was blocked or watered down, landlord elites kept power, and farm productivity and rural demand stayed low. Indonesia sits in between; its 1960 Basic Agrarian Law was only partly implemented.

Step 2: Export discipline

This is the heart of the playbook, and the step most often misunderstood.

Every developing country that tries to industrialize protects and subsidizes its new industries. Most fail. The difference in East Asia was **discipline**: support was conditional on performance, and the main measure of performance was **exports**.

Why exports? Because export markets give an honest, hard-to-fake signal. A firm can sell bad products at high prices in a protected home market. It cannot do that in the United States or Europe, where it competes with the best firms in the world. If a Korean firm could sell steel or televisions abroad, it was genuinely becoming competitive. If it could not, the bureaucrats knew it was failing, no matter what its managers claimed.

South Korea under President Park Chung-hee (who seized power in a 1961 military coup and ruled until he was assassinated in 1979) turned this into a system. Note the political reality: Park was a dictator, labour unions were repressed, and dissent was punished. The playbook worked economically, but it was not built democratically. Month 2's lessons about institutions matter here. Later, from 1987, South Korea became a democracy and kept growing, which suggests that authoritarian rule was not *necessary* for the economics, even if it was present at the start.

How export discipline worked in Korea:

- **Export targets.** The government set annual export targets by industry and by firm. Park personally chaired monthly export promotion meetings where officials and business leaders reviewed progress.
- **Rewards for winners.** Firms that met targets got cheap loans, tax breaks, import licences and permission to enter new industries.
- **Punishment for losers.** Firms that failed lost access to credit. Some were forced to merge into stronger firms or went bankrupt.
- **Competition between firms.** The government usually supported two or three firms in each industry, not one, so they competed with each other at home and abroad.

The result was dramatic. South Korean exports grew from $33 million in 1960 and $41 million in 1961 to $119 million in 1964 (the first year above $100 million) and **$10.05 billion in 1977**, a target reached four years ahead of plan (Bank of Korea data in IMF *Finance & Development*, 1971; National Archives of Korea). Exports went from cheap textiles and wigs in the 1960s, to steel, ships and chemicals in the 1970s, to cars and electronics in the 1980s, and to memory chips and smartphones later.

Step 3: Financial control

Industrialization needs huge, patient investment. Market-driven banks in a poor country tend to lend to the safest borrowers: property developers, traders, government. Factories that may take ten years to become profitable get little.

East Asian governments solved this by controlling finance. In 1961 Park's government nationalized Korea's commercial banks. Interest rates were held down, and credit was directed to priority

industries. Capital controls stopped rich Koreans from moving money abroad. In effect, the state took household savings and channelled them into heavy industry at below-market interest rates. Japan used a similar but softer system: the Ministry of Finance and the Bank of Japan guided bank lending, and a huge postal savings system funnelled household deposits into industrial and infrastructure loans.

This was powerful but dangerous. Directed credit with weak monitoring breeds bad loans and politically connected borrowers. Korea's heavily indebted conglomerates (the *chaebol*, such as Samsung, Hyundai, LG and Daewoo) borrowed enormously. When the Asian financial crisis hit in 1997, several chaebol collapsed (Daewoo failed in 1999), and Korea needed an International Monetary Fund (IMF) rescue package of about $57 billion, then the largest in the IMF's history: $21 billion from the IMF, $10 billion from the World Bank, $4 billion from the Asian Development Bank and a bilateral "second line of defence" from the US, Japan and others (ADB; *New York Times*, December 4, 1997). Financial control works best as a temporary tool for catching up. Kept too long, it becomes a source of crises.

Japan: MITI, keiretsu and catch-up

Japan started earlier than Korea. After the Meiji Restoration of 1868, Japan's new government sent officials abroad to study Western technology and institutions, built state factories (and later sold many to private groups that became the *zaibatsu* conglomerates such as Mitsui and Mitsubishi), created a modern army and navy, and built railways. By 1914 Japan had a significant textile and shipbuilding industry. Like Britain, Japan also built an empire, colonizing Taiwan (1895) and Korea (1910), and its colonial rule there was harsh, even as it built some infrastructure and industry in both colonies.

After defeat in 1945, the Ministry of International Trade and Industry (MITI), created in 1949, became the most famous industrial policy agency in the world. MITI's main tools were control over foreign exchange and import licences (in the 1950s, firms needed MITI's approval to buy foreign technology or machines), guidance on investment, help organizing licensing deals with foreign firms, and encouragement of exports. Japan protected its home market for cars and electronics while pushing firms to export.

Japan's industrial groups (*keiretsu*, the postwar successors to the zaibatsu, organized around a main bank) gave firms patient capital and stable suppliers. Toyota developed its famous production system (just-in-time inventory, continuous improvement) in this environment.

A caution: MITI did not always pick winners. It famously tried to discourage Honda from making cars in the early 1960s and tried to consolidate the car industry into fewer firms; the firms resisted, and Honda became a global success. Japan's success came partly from MITI's guidance and partly from fierce competition among private firms that sometimes ignored MITI. The lesson: industrial policy works best when it sets direction and disciplines failure, not when bureaucrats micromanage.

Korea's Heavy and Chemical Industry drive: the best modern evidence

In 1973, Park launched the Heavy and Chemical Industry (HCI) drive, targeting six industries: steel, non-ferrous metals, machinery, shipbuilding, electronics and petrochemicals. Part of the motivation was security: in 1969-1971 the United States announced troop reductions in Korea, and Park wanted a domestic arms industry. Critics at the time, including many economists and the World Bank, thought Korea was far too poor to succeed in capital-intensive heavy industry. Its comparative advantage was in cheap labour, not steel or ships.

For decades, economists argued about whether HCI worked. Korea did grow fast afterwards, but maybe it would have grown fast anyway. The drive also coincided with inflation, debt and a crisis in 1979-1980. Proving cause and effect was hard.

In 2025, economist Nathan Lane published the most careful study to date in the *Quarterly Journal of Economics* (QJE), one of the top economics journals. He collected new data on Korean industries and compared targeted industries with non-targeted ones before, during and after the policy. His findings:

- Targeted heavy and chemical industries **expanded output by more than 100%** relative to non-targeted manufacturing, and labour productivity was **more than 15% higher**.
- The policy shifted Korea's **comparative advantage**: targeted industries became better exporters.
- **Downstream industries benefited too.** Industries that bought inputs from targeted sectors (for example, users of cheaper Korean steel and chemicals) grew faster, through the input-output network. This is the "forward linkage" effect from Week 1.
- **The benefits persisted after the policy ended** in 1979, and some took years to appear.

A separate study by Jaedo Choi and Andrei Levchenko (*Journal of Monetary Economics*, 2025), using firm-level subsidy data, found that subsidized firms kept growing faster than unsubsidized firms for about 30 years after the subsidies ended, and estimated a net welfare gain of roughly 3-4% for Korea.

These studies matter because they move the debate from "industrial policy is always wasteful" to "some industrial policies, well designed, can permanently change what a country is good at." That is not the same as "all industrial policy works." Korea had export discipline, strong state capacity and a security motive that kept leaders focused. Many countries have tried similar policies without those conditions and failed.

The sceptics

Not everyone agrees the developmental state was the key. In the 1990s, economists Alwyn Young and Paul Krugman argued that East Asian growth came mainly from **accumulation** (more workers, more schooling, huge investment in machines) rather than from rising efficiency (what economists call total factor productivity). If true, the "miracle" was mostly hard work and saving, not clever policy. Others pointed out that East Asia also had high literacy, Cold War aid and access to the open US market, which later developers might not have. The World Bank's 1993 report took a middle position: the fundamentals (education, savings, macroeconomic stability) mattered most, and some targeted interventions helped in specific conditions.

Judgment: The weight of evidence now favours a combined view. Fundamentals were necessary but not sufficient; plenty of countries with good schooling and savings did not industrialize as fast. The distinctive East Asian ingredient was the *discipline* attached to support, especially export targets that gave governments an honest signal of which firms were improving. Lane's evidence shows targeted policy can create lasting capabilities. But the policy was run by unusually capable and focused states, and that condition is the hardest to copy.

Case study: POSCO, the steel mill nobody would fund

In the late 1960s South Korea had almost no iron ore, no coking coal and no experience in modern steelmaking. Park wanted an integrated steel mill anyway, because steel was the input for ships,

cars, machinery and weapons.

Foreign lenders said no. An international consortium that had been considering financing the project, and the World Bank, judged that an integrated steel mill in a poor country without raw materials would not be economic. Korea's answer was creative. In 1965 Korea and Japan had normalized relations, and Japan agreed to provide $300 million in grants and $200 million in low-interest government loans over ten years, plus about $300 million in commercial credits, roughly $800 million in all, as a settlement related to its colonial rule (1965 Agreement on the Settlement of Problems concerning Property and Claims; Korea Economic Institute of America). Park's government redirected part of these funds to the steel project, and Japanese steelmakers provided technical assistance. Pohang Iron and Steel Company (POSCO) was founded in 1968 under Park Tae-joon, a former army officer known for extreme discipline. The first blast furnace began operating in 1973.

POSCO had every reason to fail: it imported all its iron ore (from Australia and elsewhere) and coal. Yet it became one of the world's lowest-cost and most efficient steel producers within about a decade. How?

- **Location on the coast.** Importing ore and coal by sea was cheap; the mill was built at a deep-water port, so it did not need domestic raw materials. Owning the mine turned out not to matter. Controlling efficient processing did.
- **Scale and the latest technology.** POSCO built large, modern furnaces from the start, using the "advantage of backwardness" from Week 1.
- **Performance pressure.** POSCO was state-owned, but it was run with export and cost targets and strong management. It sold steel at low prices to Korean shipbuilders and carmakers, creating exactly the downstream linkages Lane's study measures.

POSCO is one of the most important cases for your driving question. It shows that a country **without** the raw material can still capture processing value if it has cheap logistics, scale, technology and discipline. The opposite is also true: a country **with** the ore, such as many African mineral producers, does not automatically capture processing value. Month 6 will look at this directly through nickel in Indonesia and copper and cobalt in Zambia and the Democratic Republic of the Congo (DRC).

Key numbers: Week 2

- **About 9-10% a year**: Japan's approximate real GDP growth from the mid-1950s to the early 1970s (World Bank, *The East Asian Miracle*, 1993, and standard national accounts).
- **$33 million to $10.05 billion**: South Korean exports in 1960 and 1977, an average growth rate of about 39% a year (Bank of Korea via IMF *Finance & Development*, 1971; National Archives of Korea).
- **More than 100%**: Output expansion of Korean heavy and chemical industries targeted in 1973-1979, relative to non-targeted manufacturing (Lane, QJE, 2025).
- **More than 15%**: Higher labour productivity in targeted Korean industries (Lane, 2025).
- **About 30 years**: How long subsidized Korean firms kept outgrowing non-subsidized ones after HCI subsidies ended; welfare gain of roughly 3-4% (Choi and Levchenko, *Journal of Monetary Economics*, 2025).
- **About $57 billion**: Size of the 1997 IMF-led rescue package for South Korea, of which $21 billion from the IMF itself (ADB; *New York Times*, December 1997; some reports at the time said $55 billion).

- **$300m + $200m + about $300m**: Japan's 1965 settlement with Korea (grants, government loans, commercial credits; about $800 million in all), part of which helped finance POSCO (1965 claims agreement, Article I; Korea Economic Institute of America).
- **$36,855**: South Korea's gross national income (GNI) per person in 2025; GDP per person was about $35,800 (Bank of Korea provisional figures, March 10, 2026), compared with roughly $100-160 in the early 1960s in current dollars (World Bank historical data, approximate).

Think like a founder

Korea's chaebol show a pattern founders should study: **use a protected or subsidized position to build capabilities, then compete globally**. Samsung started as a trading company selling dried fish and noodles in 1938. It moved into textiles and sugar, then electronics, then memory chips in the 1980s, when experts said Korea could not compete with Japan and the US. Today Samsung is one of the world's largest makers of memory chips.

The founder lesson is not "get government subsidies". It is that **value capture moved up the smile curve step by step**: from trading, to assembly, to components (memory chips), to brands (Galaxy phones). Each step used profits and skills from the last. For a robotics founder in Southeast Asia or Africa, a realistic path might start with something like system integration (installing and maintaining robots made elsewhere, close to customers), move into making one key component that is heavy, costly to ship or needs local service, and only later attempt full robots.

Week 3: Taiwan and China: Small Firms, Giant Scale

Core lesson

Taiwan: land reform, small firms and a state that built a chip industry

Taiwan followed the same three steps as Korea but with a very different industrial structure. Where Korea built giant conglomerates, Taiwan's economy was built on thousands of small and medium-sized enterprises (SMEs) plus a set of large state-owned firms in upstream industries such as steel, petrochemicals and power.

Land reform first. After the Nationalist government (the Kuomintang, KMT) retreated to Taiwan in 1949, it carried out land reform in three stages: rents were capped at 37.5% of the main crop (1949); public land taken from the Japanese colonial period was sold to tenants (from 1951); and the "Land to the Tiller" Act (1953) forced landlords to sell land above a small limit to the government, which resold it to tenants. Landlords were compensated partly with shares in state-owned companies, which turned some of the old landed elite into industrial shareholders. The KMT could do this more easily because its leaders came from mainland China and had no land in Taiwan, so they were not expropriating themselves. Farm output rose strongly in the 1950s and rural households became savers and consumers.

Export discipline through markets and export zones. In the late 1950s Taiwan shifted from import substitution to export promotion: it unified and devalued its exchange rate, offered tax rebates on imported inputs used for exports, and in 1966 opened one of the world's first export processing zones in Kaohsiung. Firms in the zone could import components duty-free as long as they exported the finished goods. Thousands of small Taiwanese firms learned to make electronics, shoes, toys and bicycles to order for American and Japanese buyers. This is a softer form of export discipline: the buyers' quality demands did the disciplining.

Financial control. Taiwan's banks were mostly state-owned for decades and lent cautiously. Taiwan kept inflation low and avoided the heavy corporate debt that Korea built up, which is one reason it came through the 1997 Asian financial crisis with relatively little damage.

Building technology on purpose. Taiwan's most important industrial policy decision was to create the Industrial Technology Research Institute (ITRI) in 1973. ITRI was a public research organization that acquired foreign technology, trained engineers and then spun off companies. In 1976 ITRI signed a technology transfer agreement with the American firm RCA to learn integrated circuit (chip) manufacturing. Engineers trained in that programme later founded or led United Microelectronics Corporation (UMC, spun off in 1980) and, most famously, Taiwan Semiconductor Manufacturing Company (TSMC, founded 1987). The Hsinchu Science Park (opened 1980) clustered these firms, their suppliers and returning Taiwanese engineers from Silicon Valley in one place.

The payoff came decades later. In 2025, driven by demand for chips used in artificial intelligence (AI), Taiwan's economy grew by an estimated **8.63%** in the first official estimate, revised up to **8.68%** in February 2026, its fastest in 15 years, and its GDP per person reached about **$39,500**, overtaking both South Korea and Japan (Taiwan's statistics agency, the Directorate-General of Budget, Accounting and Statistics, DGBAS, January and February 2026). Part of the overtaking was exchange rates: the yen and won fell more against the dollar than the Taiwan dollar did. A policy choice made in 1973 was still shaping national income in 2025.

China: the largest industrialization in history

China's industrialization since 1978 is the biggest and fastest transformation of an economy ever recorded. China lifted hundreds of millions of people out of extreme poverty and went from a poor, mostly rural country to the world's largest manufacturer. By 2023 China accounted for about **28.8%** of global manufacturing value added (Reuters Breakingviews, November 2025, using World Bank and UN data), and by 2024 its manufacturing output exceeded **$4.6 trillion**, nearly 60% more than the United States (New Zealand Ministry of Foreign Affairs and Trade, June 2026). For a sense of scale: in 2024, China installed **295,000** industrial robots, **54%** of all robots installed worldwide that year (International Federation of Robotics, IFR, *World Robotics 2025*).

How did China do it? Its path both matches and breaks the East Asian playbook.

Land reform, Chinese style. China had already carried out a violent land redistribution after 1949, followed by disastrous collectivization (the Great Leap Forward famine of 1959-1961 killed tens of millions of people). After 1978, the **household responsibility system** gave families control over farming their plots again, while the land stayed formally collective. Farmers could sell surplus output at market prices after meeting state quotas. Grain output and rural incomes rose rapidly in the early 1980s. This was, in effect, a return to Studwell's step 1: small household farming with strong incentives.

Rural industry and local government competition. In the 1980s and early 1990s, "township and village enterprises" (TVEs), owned by local governments, grew explosively, making simple consumer goods. Local officials were evaluated and promoted largely on economic growth in their area, so they competed to attract investment, build roads and industrial parks, and support local firms. This competition between thousands of local governments is one of the distinctive features of China's model. It produced enormous dynamism, and also enormous waste, duplication and debt.

Export discipline through foreign investment. China opened Special Economic Zones (SEZs) from 1980, starting with Shenzhen, then a small town next to Hong Kong. Foreign firms, first

from Hong Kong and Taiwan, then from Japan, Korea, the US and Europe, moved labour-intensive assembly into these zones. China joined the World Trade Organization (WTO) in 2001, which locked in access to world markets, and exports surged. The discipline came from the world market: foreign buyers and investors demanded quality, and Chinese suppliers learned by doing. Many sectors also required foreign firms to form joint ventures with Chinese partners, which helped transfer technology (sometimes against the foreign firm's wishes, a long-running source of conflict with the US and Europe).

Financial control. China's big state banks lent to state-owned enterprises and to local governments' investment projects at controlled interest rates, while capital controls kept savings at home. Household deposit rates were held low, a system sometimes called "financial repression", which effectively taxed savers to fund investment. Investment reached around 40% or more of GDP for many years, one of the highest rates ever recorded in a large economy.

Moving up the value chain: Made in China 2025. In 2015, China launched "Made in China 2025" (MIC2025), a ten-year plan to upgrade ten strategic sectors, including advanced information technology and chips, robotics and machine tools, aerospace, new energy vehicles, power equipment, rail, shipbuilding, new materials and biomedicine. It set targets for domestic market share and self-sufficiency.

How did it turn out? The US-China Economic and Security Review Commission, a body of the US Congress, published an evaluation in November 2025. It concluded that China had "met or exceeded many of the very ambitious" targets, especially in electric vehicles (EVs), electrical equipment, high-speed rail and shipbuilding, where long-term government support, vertically integrated supply chains and economies of scale came together. China fell short in foundational semiconductors and commercial aircraft, sectors dominated by a few foreign firms with very deep technology, but still made significant gains there. The European Union Chamber of Commerce in China, in a more critical April 2025 assessment, said China had achieved technological dominance in only some sectors (shipbuilding, high-speed rail, EVs) and pointed to overcapacity and "involution" (destructive price wars among Chinese firms).

Both assessments are right in different ways. China clearly moved up the value chain: in 2024, Chinese robot makers for the first time sold more industrial robots in China than foreign makers, with a **57%** domestic market share (IFR, 2025). But it also built far more capacity than the world needed in some sectors (solar panels, steel, batteries), driving global prices down and triggering trade defences elsewhere.

Where China breaks the playbook

China differs from Korea and Taiwan in three important ways.

1. **Size.** China's domestic market is so large that firms can reach world scale at home. This makes domestic competition, not just exports, a powerful discipline. Hundreds of EV makers fought each other in China; most will die, and the survivors (such as BYD) are formidable.
2. **Foreign investment.** Korea and Japan mostly kept foreign firms out and licensed technology instead. China, like Taiwan's export zones but on a vastly bigger scale, invited foreign firms in and learned from them.
3. **Local government as entrepreneur.** China's local officials acted almost like venture investors, offering land, cheap power, tax breaks and sometimes equity to firms. This created extraordinary speed (a battery plant can go from groundbreaking to production in well under two years) but also debt and overcapacity.

For your driving question, China's model creates both a ceiling and an opportunity. The ceiling: a latecomer in Africa or Southeast Asia cannot easily match China's scale, supplier density and speed in making motors, reducers, batteries or robots. The opportunity: China's overcapacity means cheap components and machines are available to everyone, and Chinese firms facing tariffs in the US and Europe are looking for factories elsewhere (Month 7 covers "China plus one").

Case study: TSMC and the invention of the foundry

In 1985, Taiwan's government recruited Morris Chang, a Chinese-born American engineer who had spent decades at Texas Instruments, to lead ITRI. Chang proposed a business model that most experts thought was strange: a company that would **only manufacture chips designed by other companies** and never sell chips under its own name. This is the **foundry** model.

At the time, almost all chip companies were "integrated device manufacturers" that designed and made their own chips, like Intel. Chang's insight was that as chip factories (fabs) became more expensive, many companies would want to design chips without owning a fab. A neutral manufacturer that never competed with its customers could serve them all and spread the huge cost of fabs over many customers.

TSMC was founded in 1987. The Taiwanese government, through its development fund, provided roughly half of the initial capital, and the Dutch electronics firm Philips took a large minority stake and contributed technology licences; private Taiwanese investors provided the rest (Chris Miller, *Chip War*, 2022, gives the history). It started with technology a couple of generations behind the leaders.

The model worked because of a feedback loop:

- More customers -> more chips made -> more learning about manufacturing and higher yields (the share of good chips per wafer).
- Higher yields and volume -> lower cost per chip and more profit -> more investment in the next generation of fabs.
- Each new generation costs more (a leading-edge fab now costs tens of billions of dollars), so fewer competitors can keep up, and customers concentrate on the leader.

The foundry model also *created* a whole industry of "fabless" chip designers, such as Nvidia (founded 1993), Qualcomm, AMD (after it sold its fabs) and Apple's chip team, who could design cutting-edge chips without owning factories. By the 2020s TSMC made the large majority of the world's most advanced chips, including the processors that train and run AI models. Month 8 covers this in depth.

What does TSMC teach about value capture? On the smile curve, manufacturing is supposed to be the low-value middle. TSMC turned manufacturing into a high-value position by making it so technically difficult, capital-intensive and scale-dependent that it became a near-monopoly bottleneck. Its gross margins have been far above those of typical manufacturers. The lesson: **the middle of the smile curve is low-value only when manufacturing is easy to copy.** When manufacturing requires rare skills, huge scale and constant learning, the manufacturer can capture as much value as the designer.

Key numbers: Week 3

- **37.5%**: Cap on farm rents in Taiwan's 1949 land reform, followed by the "Land to the Tiller" Act of 1953 (standard histories; Studwell, 2013).

- **1973**: Founding of Taiwan's Industrial Technology Research Institute (ITRI); **1987**: founding of TSMC.
- **8.63%, revised to 8.68%**: Taiwan's real GDP growth in 2025, its fastest in 15 years; GDP per person about **$39,500** (DGBAS advance estimate, January 2026, and preliminary estimate, February 13, 2026).
- **28.8%**: China's share of global manufacturing value in 2023, up from 25.9% a few years earlier (Reuters Breakingviews, November 2025). Estimates differ by source and year: World Bank data give 27.4% for 2025 and UNIDO 32.0% for 2024 (Month 7).
- **More than $4.6 trillion**: China's manufacturing output in 2024, nearly 60% above the US (New Zealand MFAT, June 2026).
- **295,000 and 54%**: Industrial robots installed in China in 2024 and China's share of global installations (IFR, *World Robotics 2025*).
- **57%**: Domestic Chinese robot makers' share of China's own robot market in 2024, up from about 28% a decade earlier (IFR, 2025).
- **2001**: China joins the World Trade Organization.

Think like a founder

Taiwan's small firms and TSMC suggest two founder strategies that matter for robotics and AI.

Strategy 1: be the neutral supplier. TSMC won by never competing with its customers. In robotics, there may be room for a "foundry" equivalent: a firm that makes a key module (for example, an actuator that combines motor, gearbox, sensor and driver) for many robot brands, rather than building its own robot brand and competing with them. Neutral suppliers can aggregate demand from many customers and learn faster.

Strategy 2: plug into the cluster, then move up. Taiwan's SMEs started by making components to order for foreign buyers in export zones. They learned quality, costs and delivery, then some built their own products and brands (Acer, Asus, Giant bicycles). A founder in Semarang or Nairobi can follow a similar path: start by supplying a narrow part or service to a large manufacturer or a foreign brand, learn, and climb.

A warning from China: overcapacity can destroy margins. Before entering any manufacturing segment, check whether Chinese firms already have far more capacity than the world needs. If so, competing on manufacturing cost alone is a losing game. Compete on proximity to customers, service, customization, or a technology edge instead.

Week 4: Why Playbooks Fail, and the Return of Industrial Policy

Core lesson

Latin America: import substitution without discipline

From the 1930s to the 1980s, most of Latin America followed **import substitution industrialization** (ISI). The idea, developed by economists such as Raul Prebisch at the United Nations Economic Commission for Latin America, was that countries exporting raw materials faced falling prices relative to the manufactured goods they imported. To escape, they should build domestic industries that replace imports, protected by high tariffs and quotas.

ISI was not a total failure. Brazil, Mexico and Argentina built car, steel and appliance industries. Brazil grew very fast in its "miracle" years (roughly 1968-1973, with growth near 10% a year).

Brazil's aircraft maker Embraer, created by the state in 1969, became a world leader in regional jets, and its agricultural research agency Embrapa transformed farming in the Cerrado region.

But overall, ISI in Latin America stalled, and the reasons line up neatly with Studwell's three steps:

- **No land reform (in most countries).** Large estates remained, rural inequality stayed extreme, farm productivity grew slowly, and landed elites kept political power. Mexico (after its 1910 revolution) was a partial exception.
- **No export discipline.** Firms were protected but not required to export. Many served only small domestic markets, could not reach efficient scale, and had little reason to improve. Foreign car companies set up plants to jump the tariff wall, producing expensive cars only for the local market.
- **Weak financial control, then debt.** Governments financed industry and deficits by borrowing abroad, especially in the 1970s when oil exporters' surplus dollars were recycled through Western banks. When US interest rates soared in 1979-1981 (a story for Month 11), debt costs exploded. In August 1982 Mexico announced it could not pay, triggering the Latin American debt crisis. The 1980s became the "lost decade": average incomes in the region stagnated or fell, and high inflation (hyperinflation in Argentina, Brazil, Peru and Bolivia at various points) destroyed savings.

The response in the late 1980s and 1990s was the "Washington Consensus": privatization, trade liberalization, fiscal discipline and open capital markets. This ended hyperinflation in most countries but did not restore East Asian-style growth. Latin America today is mostly "middle-income": richer than Africa and South Asia, but far behind Korea and Taiwan, despite starting the 1960s richer than both.

Africa: early hopes, structural adjustment, and new attempts

At independence in the 1950s and 1960s, many African governments also tried to industrialize through ISI and state-owned enterprises. Ghana under Kwame Nkrumah built the Akosombo Dam (completed 1965) to power an aluminium smelter, a classic attempt to process local bauxite. Zambia, Tanzania, Nigeria and others built state factories.

Most of these efforts faltered for reasons that combine Month 2's institutional problems with this month's playbook:

- **Small, fragmented markets.** Colonial borders left many small countries with poor transport links to each other. A factory in a country of 10 million poor people cannot reach world scale without exporting.
- **No export discipline, and overvalued currencies.** Governments often kept their exchange rates artificially high, which made imports cheap (pleasing urban consumers) but made exports, including manufactured exports, uncompetitive.
- **Commodity dependence and debt.** When commodity prices collapsed in the late 1970s and 1980s and interest rates rose, many African governments fell into debt crises.
- **Structural adjustment.** In the 1980s and 1990s, the IMF and World Bank lent on condition of reforms: cutting subsidies, privatizing state firms, opening trade. Many inefficient factories closed. Supporters say the reforms ended unsustainable policies; critics say they were too fast and destroyed industrial capabilities before anything replaced them. Africa's share of manufacturing in GDP fell in many countries, a pattern economists call "premature deindustrialization" (Dani Rodrik's term): countries de-industrializing at much lower income levels than earlier industrializers did.

- **Conflict and political instability**, which scare away the long-term investment that factories need.

There are newer attempts worth knowing. Ethiopia, under a self-declared developmental state model, built state-financed industrial parks after 2014, most famously Hawassa Industrial Park (opened 2016), aimed at garment exporters supplying brands such as PVH (owner of Calvin Klein and Tommy Hilfiger). At its peak Hawassa employed more than 35,000 workers, mostly young women entering formal employment for the first time (*African Business*, July 2026). But the strategy depended on duty-free access to the US market under the African Growth and Opportunity Act (AGOA). When civil war broke out in Tigray (2020-2022), the US suspended Ethiopia from AGOA from January 2022, and many export orders disappeared. Wages were also extremely low, which raised concerns about worker welfare and high turnover.

AGOA itself shows how fragile external market access can be. The programme expired on September 30, 2025, and for just over four months eligible African goods paid full US tariffs. On February 3, 2026, a law restored it retroactively, but only through December 31, 2026 (USTR, February 2026). On September 2, 2026, a two-year extension to December 31, 2028 was signed into law inside a stopgap government funding bill, the Continuing Appropriations and Extensions Act (Brookings, September 2026). Two-year extensions protect existing orders but are too short to justify building new factories, which take a decade to pay off. Ethiopia was not designated eligible for 2026 and the extension did not restore it (CRS, August 2026). Morocco, by contrast, built a car industry (Renault's Tangier plant, opened 2012, and Stellantis in Kenitra) with export discipline toward Europe, cheap logistics through the Tanger Med port, and a free-trade agreement with the European Union. Morocco is now one of Africa's largest car exporters. Month 7 returns to it.

Two African experiments compared: Morocco's cars and Ethiopia's garments

Morocco and Ethiopia both tried, in the 2010s, to do what Korea and Taiwan did in the 1960s: attract foreign factories, export, and learn. One worked far better than the other. Comparing them shows the mechanisms of this chapter at work on African soil.

Morocco: a cluster tied to a big, close market. Renault opened its Tangier plant in 2012, next to the Tanger Med port on the Strait of Gibraltar, a few hours by ship from Spain. Stellantis followed in Kenitra in 2019. By 2025 the numbers were striking:

- Automotive exports of about **154-159 billion dirhams (roughly $16-17 billion)** in 2025, about a third of all Moroccan goods exports and the country's largest export for the third year running, ahead of phosphates (Morocco's foreign exchange office data, reported by le360 and Lemaroc360, 2026). A decade earlier, in 2015, car exports were about 55 billion dirhams.
- Renault alone produced **394,474 vehicles** in Morocco in 2025 and exported 82% of them to 63 countries; the Tangier plant exported 95% of its output (Renault Group Morocco, January 2026).
- A **local integration rate of about 69%** (the share of a car's value sourced inside Morocco), with more than 270 suppliers, and a target of 75% by 2030 attached to Stellantis's 1.2 billion euro expansion at Kenitra (Reuters, July 2025).

Notice how closely this follows the playbook. *Export discipline* is built in: the plants exist to sell into Europe, where Dacia and Peugeot buyers will not accept poor quality. *Logistics* are world-class: Tanger Med is one of the largest container ports in the Mediterranean. *Market access* is long-term and legally secure: Morocco's free-trade agreement with the European Union does not expire every few years. And the *linkage* strategy is explicit: the government judges success by the integration

rate, which pushes suppliers of seats, wiring harnesses, stamped metal and now engines to set up nearby. Wiring-harness exports alone were worth about 40 billion dirhams in 2026 up to late summer (Assahifa, September 2026). That is backward linkage, the idea from Week 1, measured in dirhams.

Ethiopia: cheap labour without the rest of the system. Ethiopia built Hawassa Industrial Park in 2016 with Chinese contractors, offered tax holidays and cheap land and power, and marketed the lowest wages in the garment world. A 2019 study by New York University's Stern Center for Business and Human Rights found entry-level pay of **750 birr, or about $26 a month**, compared with about $95 in Bangladesh, $207 in Kenya and $340 in China (Barrett and Baumann-Pauly, *Made in Ethiopia*, May 2019). At that time Hawassa employed about 25,000 people.

Why did very cheap labour not deliver a Korean-style take-off? Work through the mechanisms:

1. **Labour that is too cheap is not cheap.** At $26 a month, workers could not cover rent, food and transport in Hawassa. Many quit within months. High turnover means factories constantly train new workers who never reach full speed. If a trained sewer produces 100 garments a day and a new one 40, a factory where a third of the staff is always new loses about 20% of its output (one third x 60% of a full worker's output lost = 20%). The wage saving disappears into lost productivity.
2. **No domestic inputs.** Fabric, zips and thread were mostly imported, so the parks added only the sewing step, the bottom of the smile curve. Morocco's integration rate is close to 70%; a garment park that imports almost all its inputs captures a much thinner slice of each product's value.
3. **Market access that could vanish.** The parks were built for duty-free entry to the United States under AGOA. When the US removed Ethiopia from AGOA from January 2022 over the Tigray war, the main reason for many buyers to be there disappeared. Then AGOA itself lapsed for all countries in late 2025. A factory that pays off over ten years cannot rest on a trade preference that may last two.
4. **Conflict and foreign exchange.** The civil war, and a chronic shortage of foreign currency to import inputs, raised risk for every investor.

Ethiopia's parks did not die. Export earnings from the state's industrial parks more than doubled to a record of about $267 million in the 2025/26 fiscal year, as firms diversified to European and other markets (Birr Metrics, July 2026, citing the Industrial Parks Development Corporation). But compare scale: Morocco's car sector exports roughly sixty times as much every year.

What decides the difference? Put the two side by side:

Factor	Morocco (cars)	Ethiopia (garments)
Main attraction offered	Location next to Europe, port, stable trade deal	Lowest wages in the world
Export discipline	Strong: plants built to export to Europe	Present, but fragile when US access was lost
Market access	EU free-trade agreement, no expiry	AGOA preference, suspended in 2022
Local linkages	About 69% integration, 270+ suppliers	Mostly imported inputs
Worker pay and retention	Higher pay, more stable workforce	Very low pay, very high turnover

Judgment: Low wages are the weakest form of comparative advantage, because many countries can offer them and they collapse when workers cannot live on them. Morocco's advantage was built from things that are harder to copy: geography, a port, a permanent trade agreement and a deliberate push for local suppliers. For your driving question, the lesson is direct. An African or Southeast Asian robot supply chain will not be won by offering the cheapest labour; robots are, after all, machines for replacing labour. It will be won by combining a reliable export market, good logistics, cheap and reliable power (Month 4), and a policy that measures success by how much of the product's value is made locally, as Morocco does with its integration rate.

The pattern: what separates success from failure?

Ingredient	East Asian successes	Latin American and many African failures
Land reform	Thorough (Japan, Korea, Taiwan, China after 1978)	Mostly absent or partial
Protection	Yes, temporary	Yes, often permanent
Export discipline	Strong: support tied to exports	Weak: protected firms served small home markets
Exchange rate	Kept competitive	Often overvalued
Finance	Directed to manufacturing; low inflation (Taiwan)	Foreign borrowing; inflation; debt crises
State capacity	Capable, merit-based bureaucracy	Often weaker, more capture by interest groups
External market	Open US market during Cold War	Smaller or less reliable access

The single most important column is **export discipline**. The same tools (tariffs, cheap credit, state firms) produced opposite results depending on whether support was tied to measurable performance in world markets.

The return of industrial policy, 2020-2026

For about 30 years after 1990, "industrial policy" was almost a dirty phrase in Washington and Brussels. That changed after 2020. The COVID-19 pandemic exposed fragile supply chains, China's rise in high technology alarmed the US and Europe, and climate change pushed governments to subsidize clean energy. Industrial policy came back with force.

The numbers are striking. The New Industrial Policy Observatory (NIPO), a joint project of the IMF and the Global Trade Alert team, recorded **more than 2,500** new industrial policy interventions worldwide in 2023 alone, more than two-thirds of which likely discriminated against foreign commercial interests; China, the European Union and the United States accounted for almost half (IMF blog, April 2024). An IMF Staff Discussion Note (March 2025) found that the annual count of industrial policies stayed below 200 for a decade after the 2008 financial crisis, tripled between 2017 and 2018, and reached almost 1,400 by 2022. By 2025, Global Trade Alert reported that industrial policy was shifting toward **trade-restrictive and security-motivated tools** (tariffs, export controls) rather than domestic subsidies.

The US CHIPS Act

The CHIPS and Science Act of 2022 aimed to bring advanced semiconductor manufacturing back to the United States, whose share of global chip manufacturing capacity had fallen from about 37% in 1990 to about 12% in 2020, with almost none of the most advanced chips (Semiconductor Industry Association and Boston Consulting Group, 2020). Its core was **$39 billion** in manufacturing incentives (of which $6 billion could back up to $75 billion in loans), about $11 billion for research and development, and a 25% investment tax credit for building fabs (later raised to 35% by the 2025 tax law).

What happened? According to the US Government Accountability Office (GAO, report GAO-26-109121, August 2026), by July 15, 2026, the Commerce Department had funded **49 projects across 24 companies**. By April 2026 it had disbursed **$13.1 billion**, about **42%** of the roughly $31.2-31.5 billion in direct funding awarded (the report uses $31.2 billion for awards as of April and $31.5 billion for the later total). Companies announced private investments many times larger than the grants (TSMC in Arizona, Samsung and Texas Instruments in Texas, Micron in New York and Idaho, Intel in Ohio and Arizona). But the GAO also found that Commerce had cancelled awards worth $7.8 billion of the $11 billion for research and development, including the award for the National Semiconductor Technology Center, to align with the new administration's priorities.

The most unusual twist came in August 2025, when the Trump administration converted grants to Intel into an **ownership stake**. The US government agreed to buy 433.3 million Intel shares at $20.47 each, about **$8.9 billion**, equal to **9.9%** of the company, funded by $5.7 billion in CHIPS grants that had been awarded but not yet paid plus $3.2 billion from the Secure Enclave defence programme (Intel press release and Reuters, August 22, 2025). Project milestones that Intel had not yet met were removed: the GAO found that Intel's six projects were redefined as complete once Intel had spent $7.865 billion on eligible costs, and Commerce judged that Intel had already done so. This made the US government Intel's largest shareholder, an act more associated with state capitalism than with the American tradition. Supporters argued taxpayers should share in the upside; critics worried about political interference in a private company. Intel's share price later rose strongly: the GAO reported that Commerce's Intel equity was worth about **$44.6 billion** ($102.99 per share) at the market close on July 15, 2026, a paper gain of about five times the $8.9 billion committed (GAO-26-109121, August 2026).

The Inflation Reduction Act and its partial reversal

The Inflation Reduction Act (IRA) of 2022 was the largest climate and clean energy law in US history. Instead of grants, it mainly used **uncapped tax credits**: for clean electricity (production and investment credits), for manufacturing clean energy components in the US (the "45X" advanced manufacturing credit, which paid a fixed amount per battery cell, solar module or wind component made in America), and for consumers buying EVs (up to $7,500 per new vehicle). Because the credits had no cap, the cost depended on how many companies used them. The Congressional Budget Office originally estimated about **$391 billion** over ten years; Goldman Sachs estimated about **$1.2 trillion**, and the Penn Wharton Budget Model about **$1.05 trillion** (Penn Wharton, April 2023).

The IRA did trigger a manufacturing boom. Quarterly investment in US clean technology manufacturing rose from **$2.5 billion** in the third quarter of 2022 to **$14.0 billion** in the first quarter of 2025, mostly in the EV and battery supply chain, and companies announced **380** clean technology factories (Rhodium Group and MIT Clean Investment Monitor, April 2025).

Then politics changed. The "One Big Beautiful Bill Act" (OBBBA), signed on July 4, 2025, cut

much of the IRA. It ended the new EV credit for vehicles acquired after **September 30, 2025**. It required wind and solar projects to start construction by **July 4, 2026**, or be placed in service by the end of **2027**, to claim the clean electricity credits. It added strict "foreign entity of concern" rules to block Chinese-linked supply chains from benefiting, and it phased out the manufacturing credit for critical minerals after 2030 (Congressional Research Service, December 2025; IRS guidance, July 2025). Credits for nuclear, geothermal and storage survived longer.

The lesson for industrial policy is about **credibility**. Factories take years to build and decades to pay off. When a policy can be reversed after three years, investors discount it. East Asian developmental states could promise support for 10-20 years. Democracies with alternating parties find this harder, unless the policy builds a coalition broad enough to survive changes of government.

Industrial policy everywhere else

- **European Union:** the European Chips Act (in force from September 2023, according to the European Commission) aimed to mobilize about EUR 43 billion in public and private investment to double Europe's share of global chip production to 20% by 2030, a target many analysts consider out of reach.
- **Japan:** has committed large subsidies to TSMC's factories in Kumamoto and to Rapidus, a new company trying to make 2-nanometre chips by 2027.
- **India:** the Production Linked Incentive (PLI) schemes, launched in 2020, pay firms a percentage of *incremental sales* in targeted sectors (phones, electronics, batteries, solar modules). Note the design: payment depends on output, a form of performance discipline. India's smartphone exports grew rapidly as Apple's suppliers expanded assembly there.
- **Indonesia:** banned exports of raw nickel ore from 2020 to force processing at home, attracting large Chinese investment in smelters. It is one of the most important experiments for your driving question and gets a full treatment in Month 6.
- **China:** continues to support semiconductors, AI, robotics and other sectors, with large state "big fund" investments in chips.

What the new economics of industrial policy says

Economists Reka Juhasz, Nathan Lane and Dani Rodrik, in "The New Economics of Industrial Policy" (*Annual Review of Economics*, 2024), summarize the modern evidence. Older studies, which simply compared protected and unprotected industries, usually found industrial policy failed. Newer studies, using better data and methods (like Lane's work on Korea), find that some episodes delivered large and lasting gains, while others wasted money. The difference depends on design and on state capacity.

From the evidence and history in this chapter, we can list the design features that matter most:

1. **Discipline and exit.** Support is tied to measurable performance (exports, output, cost reduction), and failing firms lose support.
2. **Competition.** Several firms are supported in each sector, and they compete.
3. **Temporary support with a credible horizon.** Long enough to justify investment, clearly ending eventually.
4. **Linkages.** Target industries with strong downstream users (like steel or chips), so benefits spread.
5. **State capacity and information.** Officials who understand the industry, talk constantly to firms, and are insulated from capture by any single firm. Rodrik calls this "embedded

autonomy": close to business but not owned by it.

6. **Openness to learning from abroad**, through licensing, foreign investment or returning diaspora engineers.

Judgment: Industrial policy is neither a magic formula nor a guaranteed waste. For a late developer, it is probably necessary, because market forces alone tend to keep poor countries specialized in raw materials and simple assembly, exactly the bottom of the smile curve. But it only works with export discipline and a capable state. For Africa and Southeast Asia aiming at robot supply chains in the 2040s, the playbook implies: start with a narrow set of components where you have some cost or location advantage, attract foreign firms to transfer know-how, insist on export performance, and do not try to build everything from ore to finished robot at once.

Case study: The CHIPS Act and the Intel stake, a natural experiment in 2020s industrial policy

The CHIPS Act lets us watch modern industrial policy in real time and compare it with the East Asian playbook.

The problem. In 1990 the US had roughly 37% of the world's chip manufacturing capacity. By 2020 its share had fallen to about 12% (SIA and BCG, 2020), and it made almost none of the most advanced logic chips, which were concentrated at TSMC in Taiwan. Building a leading-edge fab in the US costs more than in Taiwan, partly because of higher construction and labour costs and partly because the supplier ecosystem is thinner.

The tools. Grants (direct funding), loans, a 25% (later 35%) investment tax credit, and research funding. Awards were negotiated project by project with milestones: build the fab, start production, hire and train workers.

Compare with Korea. Korean HCI support was tied to export performance and firms that failed lost support. CHIPS awards were tied to *construction milestones*, not to whether the fabs would be globally competitive. The Intel amendment in 2025 went further, *removing* unachieved milestones in exchange for equity. Under the East Asian logic, this is the opposite of discipline: support continued while performance conditions were dropped. Under a different logic, a government shareholder now has a direct financial interest in Intel's success and a voice (although the stake was described as passive) in its future.

Early results. Disbursement has been slow (42% of awarded direct funding by April 2026), partly because money is paid as projects hit milestones. Several fabs have opened or are ramping, including TSMC's first Arizona fab. Intel's Ohio project was delayed. The research side, which was meant to build long-term capability, was largely cancelled or paused (GAO, August 2026).

The arithmetic of a fab subsidy. Why did the United States have to pay companies to build fabs at all? The SIA and BCG study of 2020 estimated that the ten-year total cost of owning and running a new fab in the United States was roughly **25-50% higher** than in Taiwan, Korea, Singapore or China, and that government incentives abroad explained about **40-70%** of that gap (SIA summary of the BCG report, October 2020). The rest came from higher construction and labour costs and a thinner local supplier base. A worked example, using round illustrative numbers rather than any real project:

- A leading-edge fab costs about $20 billion to build and, over ten years, perhaps another $20 billion to run (people, power, chemicals, maintenance). Total ten-year cost: about $40 billion in Taiwan.

- A 30% US cost premium means the same fab costs about $52 billion over ten years in Arizona: **$12 billion more**.
- A CHIPS grant of, say, 10% of the construction cost ($2 billion) plus the 25% investment tax credit on equipment and buildings (worth perhaps $4-5 billion on a $20 billion build, and more after the 2025 tax law raised the credit to 35%) covers about half of that premium.
- The rest must come from somewhere: customers paying a little more for "made in America" security, the company accepting lower returns, or productivity gains as the US cluster grows.

This explains two things in the GAO record. First, why awards were negotiated project by project: each company's cost gap was different. Second, why disbursement was slow: Commerce paid as milestones were met, so money flowed only when concrete was poured and tools were installed. By April 2026 only about 42% of awarded direct funding had been paid (GAO-26-109121, August 2026).

Did it move the needle? The share of world chip-making capacity located in the United States kept falling after the 2020 study, to about **10% in 2022**. But because of the projects that CHIPS incentives helped trigger, SIA and BCG projected in 2024 that US capacity would roughly triple by 2032 and its share would rise to about **14%**, the first increase in decades; without the policy, they estimated it would have fallen to about 8% (SIA and BCG, May 2024). In other words, the policy is expected to roughly double the US share relative to the no-policy path, but not to bring it anywhere near the 37% of 1990. Projections are not results, so treat the 14% as a forecast to check against reality in the early 2030s.

Did the Intel stake pay? In narrow financial terms, so far, yes on paper. The government committed $8.9 billion for 433.3 million shares at $20.47. At the July 15, 2026 close of $102.99 per share, the GAO valued the stake at about $44.6 billion, a paper gain of about $35.7 billion, or about five times the money committed (GAO-26-109121). The government also holds a five-year warrant to buy another 5% at $20 per share, but only if Intel's ownership of its foundry business falls below 51% (Intel, August 22, 2025). That warrant is a clever piece of design: it discourages Intel from selling control of its US factories, which is the national-security asset the policy cares about.

But a financial return is not the same as an industrial-policy success. The GAO found that, in exchange for the equity, Commerce removed Intel's unmet project milestones and declared all six projects complete once Intel had spent $7.865 billion on eligible costs, which Commerce judged it already had. Commerce no longer requires Intel to report progress on the original milestones (GAO-26-109121). So taxpayers gained an asset whose value depends on Intel's share price, but lost the performance conditions that tied money to specific factories being built on schedule. Under the Korean logic of Week 2, the government swapped *discipline* for *ownership*.

Which is better? Here is the strongest case on each side.

- **For the swap:** milestones measured construction, not competitiveness; a shareholder shares in the upside if Intel succeeds and has a lasting interest in its health; and the warrant protects the strategic asset. The paper gain shows the deal was bought at a very low price.
- **Against the swap:** the share price rose for many reasons, including the AI boom and expectations of government favour, so the gain is not proof that the policy built capability; equity gives the state a reason to steer customers toward Intel, which can weaken the competition that disciplined Korean and Taiwanese firms; and removing milestones sets a precedent that other subsidy recipients may seek.

Judgment: The stake was a good *financial* trade and a weak *disciplinary* one. The test that matters for industrial policy is not Intel's share price in 2026 but whether US fabs reach world-class yields

and costs by the early 2030s and win outside customers such as Apple or Nvidia for leading-edge chips. If they do, the policy will look like Korea's HCI drive. If they do not, it will look like an expensive rescue that happened to make money in a bull market.

Link to value capture. Why do governments care so much about this particular factory step? Because chips are the narrow waist between hardware and AI. Every AI model, every data centre and, increasingly, every robot depends on leading-edge logic and memory chips made by a handful of firms. Whoever controls that step can tax, or deny, everyone above it in the chain. For a country like Indonesia, Vietnam or Kenya, the lesson is not to build a leading-edge fab (the cost gap and the know-how gap are far too large) but to find the less glamorous links where the same logic applies on a smaller scale: chip packaging and testing, power electronics, sensors, or the motor-and-gearbox modules that robots need.

What to watch. Can US fabs reach the yields and costs of Taiwanese fabs? Will customers like Apple and Nvidia move advanced production to the US? Will the government's Intel stake lead to pressure on other companies to use Intel's foundry? The answers will tell us whether rich democracies can run industrial policy with the discipline that East Asian developmental states had.

Link to your driving question. The CHIPS experience shows that even the richest country in the world struggles to rebuild a manufacturing ecosystem once it has moved abroad. Supplier density, skilled workers and process know-how are slow to build. For a poorer country hoping to build a robot supply chain, this is sobering: you cannot buy an ecosystem with one large subsidy. You build it component by component, firm by firm, over decades.

Key numbers: Week 4

- **August 1982**: Mexico's debt moratorium triggers the Latin American debt crisis and the "lost decade" of the 1980s.
- **More than 2,500**: New industrial policy interventions worldwide in 2023, of which about 71% (some 1,800) likely trade-distorting; China, the EU and the US made almost half (IMF Working Paper WP/24/1 and IMF blog, 2024, New Industrial Policy Observatory with Global Trade Alert).
- **$39 billion**: CHIPS Act manufacturing incentives (2022); **49 projects, 24 companies** funded by July 2026; **$13.1 billion** disbursed, about **42%** of awarded direct funding, by April 2026 (GAO-26-109121, August 2026).
- **$8.9 billion for 9.9%**: US government stake in Intel, August 2025, at $20.47 per share (Intel and Reuters, August 22, 2025).
- **$391 billion vs $1.2 trillion**: Original official (Congressional Budget Office) vs Goldman Sachs estimates of the IRA's ten-year climate and energy cost.
- **$2.5 billion to $14.0 billion**: Quarterly US clean technology manufacturing investment, third quarter 2022 to first quarter 2025 (Rhodium/MIT Clean Investment Monitor, 2025).
- **September 30, 2025, and July 4, 2026**: End date for the US new EV tax credit, and start-of-construction deadline for wind and solar to keep IRA credits under the 2025 tax law (CRS, December 2025).
- **December 31, 2028**: New expiry date of AGOA after the September 2026 two-year extension (Brookings, September 2026).

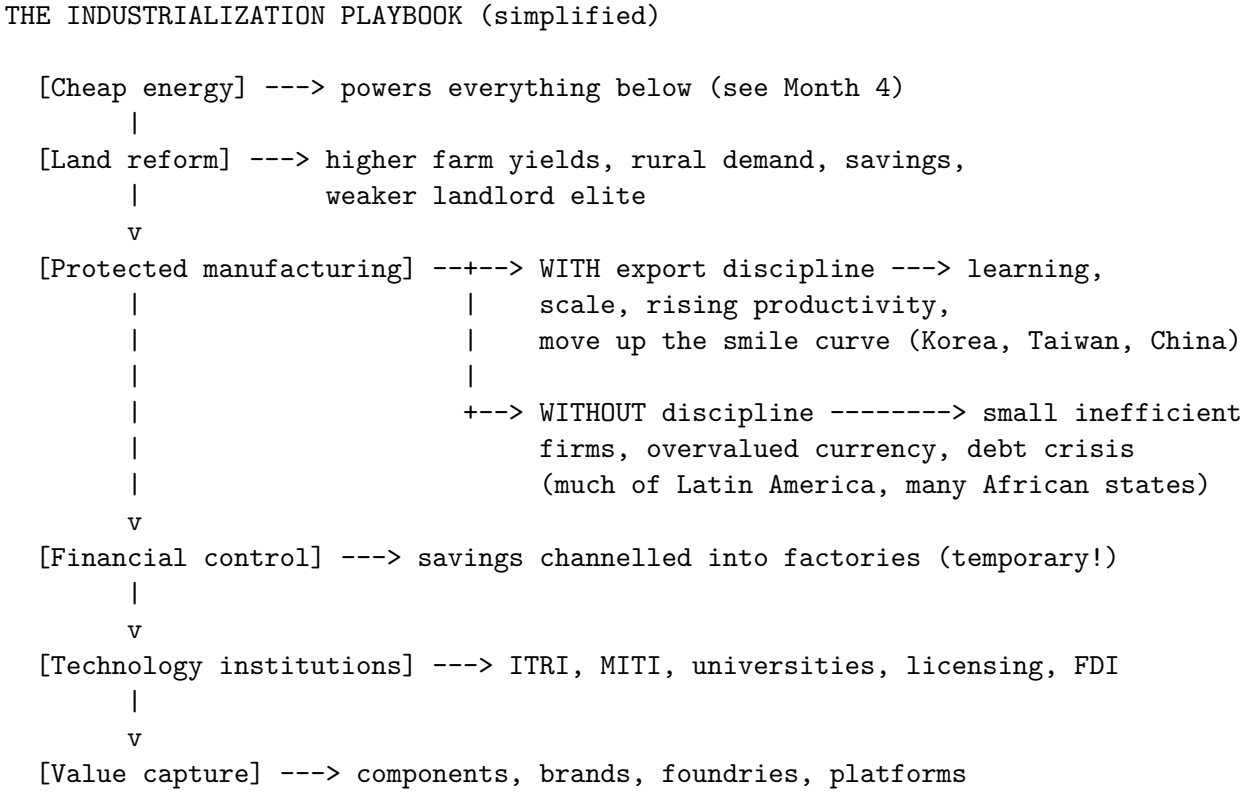
Think like a founder

Industrial policy creates opportunities, but founders who build businesses that only survive on subsidies are taking political risk. The IRA's partial reversal in 2025 stranded some planned factories. A useful rule: **build a business that would still work if the subsidy ended, and treat the subsidy as an accelerator, not the engine.**

Second, notice the opportunity in **compliance and supply-chain tracing**. Rules like the US "foreign entity of concern" restrictions, the EU's battery passport, and local-content rules in India and Indonesia create demand for software and services that prove where materials came from. A founder with AI and data skills could build tools that trace a battery's nickel from mine to cell, or check whether a product qualifies for a subsidy. This sits at the high-value, service end of the smile curve, and it grows every time governments add new industrial policy rules.

Month synthesis

The history of industrialization is a history of countries trying to move up the smile curve: from supplying raw materials and cheap labour to controlling processing, components, technology and brands. Each successful country did it with a combination of cheap energy, a protected period of learning, a way to concentrate capital, and, crucially, discipline that forced firms to become globally competitive.



The table below summarizes the playbooks.

Country	Period	Main tools	Where value was captured
Britain	1760-1850	Cheap coal, protection, navy, empire	Textile machinery, shipping, finance
United States	1820-1913	High tariffs, continental market, land-grant colleges	Steel, oil, mass production
Germany	1850-1913	Tariffs, universal banks, technical education	Chemicals, electrical equipment
Japan	1868-1990	MITI, keiretsu, export push	Cars, electronics, machine tools
South Korea	1961-1997	Land reform, export targets, nationalized banks, HCI	Steel, ships, memory chips, phones
Taiwan	1950-today	Land reform, export zones, ITRI, SMEs	Foundry chips, electronics components
China	1978-today	Household farming, SEZs, FDI, local government competition, MIC2025	EVs, batteries, solar, robots, shipbuilding

Connection to the ore-to-robot-to-AI question. A robot is built from refined metals (copper, rare-earth magnets, steel, aluminium, lithium), components (reducers, motors, sensors, batteries, chips) and software (control systems, increasingly AI). In 2026, China dominates refining and many components; Japan and Germany lead precision reducers and high-end motion control; Taiwan and Korea lead chips; the US leads AI models. A late developer that wants to capture value must find a narrow segment where it can be disciplined by export markets, attract know-how, and grow linkages. Trying to build the whole chain at once, behind high walls without export discipline, repeats the Latin American mistake. And every one of these steps depends on cheap, reliable energy, which is next month's topic.

Common misconceptions

1. **"Rich countries got rich through free trade."** Britain, the United States, Germany, Japan and Korea all protected their industries while they were catching up. They embraced free trade mainly after they became competitive. Free trade can be good policy for mature economies; the historical record shows it was rarely the path to industrialization.
2. **"Industrial policy always fails because governments cannot pick winners."** The best recent evidence (Lane, 2025; Choi and Levchenko, 2025) shows that Korea's heavy industry drive created lasting gains. Industrial policy fails often, but not always. The key is not picking winners perfectly; it is **dropping losers** quickly through discipline.
3. **"East Asia succeeded because of cheap labour."** Many countries have cheap labour and never industrialized. Cheap labour helps at the start, but East Asia's distinctive features were land reform, export discipline, high investment in education and technology, and capable states.

Korea's wages rose rapidly, yet it kept moving into more advanced industries.

4. **"Having the raw material gives you an advantage in processing it."** POSCO built a world-class steel industry with no iron ore or coal. Japan built huge industries with almost no natural resources. Meanwhile many resource-rich countries never developed processing. Logistics, energy, scale, skills and discipline matter more than owning the mine.

5. **"Authoritarian rule is necessary for fast development."** Korea and Taiwan started under authoritarian governments, but both became democracies in the late 1980s and kept growing. Many authoritarian governments (in Africa, Latin America and elsewhere) produced stagnation. What mattered was state *capacity* and *focus on development*, not the absence of elections. And harsh rule had real human costs, which should never be ignored.

Exercises

1. **(Quantitative: relative prices)** Using Allen's logic: a machine costs $5,000 a year to own and run and replaces 4 workers. (a) At what annual wage per worker does the machine break even? (b) In Country A, workers earn $1,500 a year; in Country B, $800. In which country is the machine adopted? (c) Electricity prices fall and the machine's running cost drops to $3,000. Does your answer to (b) change? What does this tell you about how cheap solar power (Month 4) might shift where automation happens?

2. **(Quantitative: export growth)** South Korea's exports were $33 million in 1960 and $41 million in 1961, and $10.05 billion in 1977. Round this to a stylised path of $35 million growing to $10 billion over 15 years, and calculate the average annual compound growth rate. (Hint: solve 35 x (1 + g)^15 = 10,000.) Compare this with a country whose exports grow at 8% a year for 15 years.

3. **(Quantitative: subsidy efficiency)** The CHIPS Act awarded about $31.5 billion in direct funding by mid-2026 and disbursed $13.1 billion. Suppose the funded projects eventually create 25,000 direct jobs. (a) What is the awarded cost per job? (b) Why is "cost per job" an incomplete way to judge a semiconductor subsidy? List at least three other benefits or costs you would count.

4. **(Essay, 600-800 words)** Studwell argues that land reform must come before industrialization. Choose one country that did not carry out thorough land reform (for example, the Philippines, Brazil or Indonesia) and explain how this affected its industrial path. Use at least two numbers with sources.

5. **(Essay, 600-800 words)** Was the US government's 2025 equity stake in Intel a good use of public money? Present the strongest argument for and against, then give your own judgment, referring to the idea of "export discipline" from this chapter.

6. **(Design exercise)** You are an adviser to the government of a Southeast Asian or African country that wants a robot component industry by 2040. Design a five-point industrial policy using the six design features from Week 4. For each point, say what performance measure you would use to discipline firms.

7. **(Founder exercise)** Pick one component of an industrial robot (gear reducer, servo motor, motor driver, encoder, battery pack, or controller). Research who makes most of them today

and in which countries. Where on the smile curve does this component sit? Could a new firm in your country enter this market, and if so, which customers would it start with?

Reading list

- **Joe Studwell, *How Asia Works: Success and Failure in the World's Most Dynamic Region* (2013).** The clearest statement of the three-step playbook (land reform, export discipline, financial control), with vivid comparisons of successes and failures, including Indonesia and the Philippines.
- **Alice Amsden, *Asia's Next Giant: South Korea and Late Industrialization* (1989).** The classic study of how Korea "got prices wrong" on purpose and disciplined firms through performance standards.
- **Robert Wade, *Governing the Market: Economic Theory and the Role of Government in East Asian Industrialization* (1990).** Explains Taiwan's industrial policy in detail and why "governed markets" outperformed pure free markets.
- **Chalmers Johnson, *MITI and the Japanese Miracle* (1982).** The book that defined the "developmental state", tracing Japan's industrial bureaucracy from the 1920s to the 1970s.
- **Ha-Joon Chang, *Kicking Away the Ladder: Development Strategy in Historical Perspective* (2002).** Shows how Britain, the US and Germany used protection while catching up; essential for understanding today's debates about trade rules.
- **Robert C. Allen, *The British Industrial Revolution in Global Perspective* (2009).** The high-wage, cheap-energy explanation for why industrialization started in Britain; directly relevant to Month 4.
- **Alexander Gerschenkron, *Economic Backwardness in Historical Perspective* (1962).** The "advantages of backwardness" essay, explaining why late developers need institutions that concentrate capital.
- **Reka Juhasz, Nathan Lane and Dani Rodrik, "The New Economics of Industrial Policy," *Annual Review of Economics* (2024).** A readable summary of the latest evidence on when industrial policy works.
- **Nathan Lane, "Manufacturing Revolutions: Industrial Policy and Industrialization in South Korea," *Quarterly Journal of Economics* (2025).** The strongest modern evidence that a targeted industrial policy created lasting comparative advantage.
- **Chris Miller, *Chip War: The Fight for the World's Most Critical Technology* (2022).** Tells the story of TSMC, Samsung and the global chip industry; a bridge to Month 8.

Glossary terms introduced this month

- **Advantages of backwardness**: Gerschenkron's idea that late developers can copy the latest technology but need institutions (banks, the state) to concentrate the large capital required.
- **Chaebol**: Large family-controlled Korean business groups such as Samsung, Hyundai and LG.
- **Developmental state**: A state with a capable bureaucracy that actively steers private firms toward national industrial goals using credit, licences and trade policy.
- **Directed credit / financial repression**: Government control of bank lending and interest rates to channel savings into chosen sectors, often by paying savers low interest.
- **Export discipline**: Tying government support (credit, subsidies, protection) to firms' success in export markets, and cutting support to firms that fail.
- **Export processing zone (EPZ) / Special Economic Zone (SEZ)**: An area where firms

can import inputs duty-free and face simpler rules, usually on condition of exporting.
- **Foundry**: A company that manufactures chips designed by other firms (for example, TSMC).
- **Household responsibility system**: China's post-1978 reform giving farm families control over production on their plots.
- **Import substitution industrialization (ISI)**: A strategy of building domestic industries to replace imports, protected by tariffs and quotas.
- **Industrial policy**: Government action aimed at changing the structure of the economy toward particular sectors, technologies or activities.
- **Infant-industry argument**: The case for temporarily protecting a new industry until it gains experience and can compete.
- **Keiretsu / zaibatsu**: Japanese business groups (prewar zaibatsu, postwar keiretsu) linked by cross-shareholdings and often a main bank.
- **Land reform**: Redistribution of farmland from large landlords to the families who farm it.
- **Linkages (backward and forward)**: The demand an industry creates for inputs (backward) and the supply it creates for downstream users (forward).
- **Premature deindustrialization**: Manufacturing's share of the economy peaking and falling at much lower income levels than in earlier industrializers.
- **Production Linked Incentive (PLI)**: India's scheme paying firms a share of their incremental sales in targeted sectors.
- **Universal bank**: A bank that both lends to and owns shares in industrial firms, typical of German industrialization.

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Month 4: Energy, the Base Layer

Month overview

Every chapter of this course rests on one physical fact: **turning matter into useful things takes energy.** Digging ore out of the ground, crushing it, smelting it into metal, rolling the metal into sheets, machining it into gears, running the factory that assembles a robot, and running the data centre that trains the AI model that controls the robot: every step burns energy. Month 3 showed that Britain industrialized first partly because it had cheap coal next to its factories. This month explains why that lesson still holds in 2026, and why it may matter even more in a world of electric vehicles, AI data centres and robots.

We begin with the link between energy and gross domestic product (GDP): how much energy an economy uses, how efficiently, and why the poorest billion people still lack the energy needed to industrialize. Then we study the oil and gas market, the most politically powerful commodity market in the world. 2026 is a dramatic year to study it: a war between the United States and Israel on one side and Iran on the other, which began on February 28, 2026, nearly closed the Strait of Hormuz, through which about a fifth of the world's oil and liquefied natural gas (LNG) normally flowed. Oil prices rose above $120 a barrel and diesel above $200 a barrel. This is a live case study of how energy shocks move through the global economy. Week 3 covers coal, electricity grids and petrostates (countries whose governments depend on oil and gas income), and Week 4 covers the extraordinary fall in the cost of solar power and batteries, which is changing which countries can have cheap energy.

The connection to your driving question is direct. Months 5 and 6 will show that refining nickel, lithium, copper and rare earths is extremely energy-hungry; a smelter is basically a machine for turning electricity into metal. Month 8 will show that AI data centres are among the fastest-growing users of electricity on Earth. Any country that wants to build a supply chain from ore to robot to AI must first answer: **where will our cheap, reliable electricity come from, and who will finance it?**

Learning goals

By the end of this month you should be able to:

- Explain the relationship between energy use and GDP, using units such as exajoules (EJ), terawatt-hours (TWh) and barrels per day, and calculate simple measures like energy per person and energy intensity.
- Describe how oil and natural gas prices are formed, including the role of spare capacity, inventories, OPEC+ (the Organization of the Petroleum Exporting Countries plus allies such

as Russia) and price benchmarks.
- Analyse the 2026 Iran war oil and gas shock with real data: what happened, why prices moved so much, and who won and lost.
- Explain how electricity grids work, why coal remains so large, and what makes a "petrostate" fragile.
- Use learning curves and the levelized cost of electricity (LCOE) to explain why solar and batteries became so cheap, and show with a worked example why the **cost of capital** decides whether cheap solar reaches poor countries.
- Connect energy costs to value capture in the ore-to-robot-to-AI chain, and identify founder opportunities in energy.

Week 1: Energy and the Economy

Core lesson

Units first

Energy statistics use many units, which confuses everyone at first. Here are the ones you need:

- **Joule (J)**: the basic scientific unit of energy. An **exajoule (EJ)** is 10^18 joules (a billion billion). World energy is usually counted in EJ.
- **Watt (W)**: a *rate* of energy use, one joule per second. A **kilowatt-hour (kWh)** is the energy used by running 1,000 watts for one hour (3.6 million joules). Your electricity bill counts kWh. A **terawatt-hour (TWh)** is a billion kWh.
- **Gigawatt (GW)**: a billion watts, used for the *capacity* of power plants. A 1 GW plant running all year at full power would produce 8,760 GWh, or 8.76 TWh.
- **Barrel (bbl)**: 159 litres of crude oil. Oil markets count in **million barrels per day (mb/d)**.
- **Million British thermal units (MMBtu)**: the unit for pricing natural gas.

The difference between **capacity** (GW, how big the machine is) and **energy** (TWh, how much it actually produces) matters enormously. A solar farm produces only when the sun shines, so 1 GW of solar might produce around 1.5-2 TWh a year, while 1 GW of nuclear might produce 7-8 TWh. The ratio of actual output to maximum possible output is called the **capacity factor**.

How much energy does the world use?

According to the Energy Institute's *Statistical Review of World Energy 2026* (published June 2026), global **total energy supply (TES) exceeded 600 EJ in 2025**, up 1.7% from 2024. Every major energy source, including coal, oil, gas, nuclear, hydro and renewables, reached an all-time high for the second year in a row. Fossil fuels supplied about **86%** of the total, a record low share but still the overwhelming majority. Oil supplied about 201 EJ, coal about 166 EJ and natural gas about 151 EJ.

Let us turn this into something human. A worked example:

- World energy supply ~ 600 EJ = 600 x 10^18 J.
- World population ~ 8.2 billion.
- Energy per person ~ 600 x 10^18 / 8.2 x 10^9 ~ 73 billion joules (73 gigajoules, GJ) per year.
- A year has about 31.5 million seconds, so this equals about 73 x 10^9 / 31.5 x 10^6 ~ **2,300 watts**, running continuously.

So the average human uses energy at the rate of about 23 old-fashioned 100-watt light bulbs left on all day and night. But the average hides enormous inequality. A typical American uses several times the world average, while a typical person in much of sub-Saharan Africa uses a small fraction of it. According to the International Energy Agency (IEA), about **730 million people** worldwide still had no electricity at all in 2024, and about 600 million of them lived in sub-Saharan Africa, about 47% of the region's population (IEA, *Financing Electricity Access in Africa*, October 2025). A June 2026 update from the international tracking report put the number at 655 million under a different methodology, with 563 million in sub-Saharan Africa (IEA and partners, *Tracking SDG7 2026*). Either way, roughly four out of five people without power live in sub-Saharan Africa, and the number has barely fallen since 2020 because population growth keeps pace with new connections.

Why energy and GDP move together

Historically, almost every country that got rich multiplied its energy use. The reason is physical: productivity comes from giving each worker more machines, and machines run on energy. A farmer with a tractor, a seamstress with an electric sewing machine, a factory worker supervising robots: each is more productive because external energy multiplies human muscle and attention. Energy historian Vaclav Smil (*Energy and Civilization*, 2017) argues that the whole of modern civilization is built on this multiplication.

But the link is not fixed. As economies grow, two things happen:

1. **Energy intensity falls.** Energy intensity is energy used per dollar of GDP. Rich economies shift toward services (software, finance, health care) that need less energy per dollar of output, and machines become more efficient. The Energy Institute reports that global energy efficiency gains relative to GDP were about **2% in 2025**, far short of the 4% a year targeted at the COP28 climate summit in 2023 (EI, June 2026).
2. **Some energy-intensive production moves abroad.** When rich countries import steel, aluminium and electronics from China, the energy used to make those goods is counted in China's statistics, not theirs.

A simple worked example on energy intensity: if the world uses about 600 EJ and world GDP is roughly $110 trillion (IMF estimates for 2024-2025, at market exchange rates; approximate), then the world uses about 600 x 10^18 J / 110 x 10^12 dollars ~ **5.5 megajoules per dollar** of output. That is about the energy in 0.13 litres of petrol for every dollar of GDP. A country with half the world average intensity is either very efficient or specialized in low-energy activities; a country with double the average is either inefficient or specialized in heavy industry.

The "primary energy" trap: why electricity changes the arithmetic

When you burn coal, oil or gas, most of the energy is lost as heat. A coal power plant converts roughly a third to two-fifths of the coal's energy into electricity. A petrol car engine turns only around a quarter to a third of the fuel's energy into motion at the wheels; the rest heats the air. An electric motor, by contrast, converts around 90% of its electricity into motion.

This means the 86% fossil share in *primary* energy overstates how much useful work fossil fuels do. Carbon Brief, analysing the 2026 Statistical Review, notes that roughly two-thirds of the energy in fossil fuels is lost in combustion. When a country switches from petrol cars to electric cars powered by solar or wind, its primary energy use can fall even as it gets the same useful transport. A worked example:

- A petrol car needs about 60 kWh of fuel energy to travel 100 km (roughly 6.5 litres of petrol; approximate).
- An electric car needs about 15-18 kWh of electricity to travel the same distance.
- If that electricity comes from solar panels, the primary energy needed is about the same 15-18 kWh (statisticians count solar electricity directly). The electric car uses roughly a quarter to a third of the primary energy.

This is why the IEA talks about a coming "Age of Electricity". Global electricity demand grew **3% in 2025**, following 4.4% in 2024, and is projected to grow at least 2.5 times as fast as overall energy demand through 2030, rising from about **28,200 TWh in 2025 to 33,600 TWh in 2030** (IEA, *Electricity 2026*). Drivers include industry, electric vehicles (EVs), air conditioning in hot countries, and data centres. The Energy Institute found that in 2025, for the first time, **all** of the growth in world electricity demand was met by low-carbon sources (renewables and nuclear), and fossil generation fell slightly.

Energy and the ore-to-robot chain

Why does this matter for robots and AI? Because almost every step in the chain is electricity-hungry:

- **Aluminium smelting** uses roughly 13-15 MWh of electricity per tonne of metal (a standard industry figure; approximate). At an electricity price of 3 US cents per kWh, the power cost is about $420 per tonne; at 8 cents, it is about $1,120 per tonne. With aluminium selling for a few thousand dollars per tonne, that difference decides which smelters survive. This is why smelters cluster where power is cheap: Iceland, Norway, the Gulf, China's coal-rich provinces and hydropower regions.
- **Nickel processing** for batteries, especially the Indonesian route of rotary kiln electric furnaces (Month 6), uses huge amounts of electricity, much of it from captive coal plants.
- **Chip fabrication** needs enormous, ultra-reliable power; a single leading-edge fab can use as much electricity as a mid-sized city.
- **AI data centres** used about 415 TWh in 2024, about 1.5% of world electricity, and the IEA estimates this grew about 17% to around 485 TWh in 2025 and will roughly double to about 950 TWh by 2030 (IEA, *Energy and AI*, April 2025, and *Key Questions on Energy and AI*, April 2026). (The Energy Institute's 2026 Review reported a much higher figure, 788 TWh for 2025, up from 658 TWh in 2024, with about 40% in the United States. Its numbers come from S&P Global and include cryptocurrency mining, which the IEA counts separately. When two respected sources disagree by this much, always check their definitions before quoting either.)
- **Robot factories** themselves are less energy-intensive than smelters, but the materials inside a robot (steel, aluminium, copper, magnets, batteries) carry large amounts of "embodied" energy from their production.

The conclusion: a country with expensive or unreliable electricity cannot compete in the energy-intensive middle of the supply chain (refining, smelting, cell manufacturing, chips). It can still do mining at the start and software or services at the end. Cheap power is the ticket to the middle.

Worked example: the energy bill of one robot, from ore to AI

The conclusion above is easy to say and easy to misunderstand. It does *not* mean that a robot is expensive because of the electricity inside it. It means that energy decides *where* some links of the chain can be profitable, while other things (skill, scale, software, brands) decide *who keeps the profit*.

A small worked example shows the difference. The masses below are illustrative assumptions for a mid-sized industrial robot arm, not data for any real model; the energy and cost figures are sourced.

Step 1: the materials. Assume a 200 kg robot arm that contains roughly 120 kg of steel and cast iron, 50 kg of aluminium, 15 kg of copper (motor windings and cables), 2 kg of rare-earth magnets, and a few kilograms of electronics. Primary aluminium needs about 14 MWh of electricity per tonne, so about 14 kWh per kilogram (CRU, 2021; International Aluminium Institute data). Steel made in a blast furnace needs roughly 20 gigajoules of energy per tonne, about 5-6 kWh per kilogram, most of it from coal rather than electricity (a standard industry range; approximate).

- Aluminium: 50 kg x 14 kWh/kg = about **700 kWh** of electricity.
- Steel: 120 kg x 5.5 kWh/kg = about **660 kWh** of (mostly coal) energy.
- Copper, magnets and electronics add more, but less than these two, because their masses are small.

Call the total roughly **1,500-2,000 kWh** of energy embodied in the metal of one arm.

Step 2: what that energy costs. At 3 US cents per kWh, 1,700 kWh costs about $51. At 8 cents it costs about $136. A mid-sized industrial arm sells for tens of thousands of dollars (prices vary widely; treat this as an order of magnitude). So the embodied energy is **well under 1%** of the robot's price, whichever country made the metal. A robot company does not choose its factory location because of the electricity bill.

Step 3: but look at the smelter. For the aluminium smelter that made those 50 kg, the picture is completely different. Electricity is about 30% of all site costs for a primary smelter (CRU, 2021), and European producers put it at 37-40% of production costs (European Aluminium, 2021; ING, 2023). The US Aluminum Association says a smelter needs a 10-20 year power contract at about $40 per MWh (4 cents per kWh) to be globally competitive. Do the arithmetic for a mid-sized smelter producing 300,000 tonnes a year:

- Electricity used: 300,000 t x 14 MWh/t = 4.2 million MWh (4.2 TWh), a large share of the power system of a small country.
- At 3 cents per kWh: 4.2 billion kWh x $0.03 = **$126 million a year**.
- At 8 cents per kWh: 4.2 billion kWh x $0.08 = **$336 million a year**.
- The gap, $210 million a year, is about $700 per tonne, which is roughly a quarter of the price of aluminium (about $2,500-3,000 per tonne in 2025-2026; approximate). No amount of good management closes a gap that large. The expensive-power smelter closes.

Step 4: the battery cell. A robot that moves around (a mobile robot or a humanoid) needs a battery. Making lithium-ion cells takes roughly **20-40 kWh of energy for every 1 kWh of storage capacity** with today's factories (Degen and others, *Nature Energy*, 2023; Fraunhofer FFB). For a 2 kWh robot battery that is about 40-80 kWh, or $1-6 of energy at 3-8 cents. Against a cell price of roughly $40-80 per kWh of capacity in 2025 (BloombergNEF, December 2025), energy is a few percent of the cost of a cell: it matters for a gigafactory's margin, but it is not decisive the way it is for a smelter.

Step 5: the AI. Finally, suppose the robot's intelligence runs partly in the cloud. A 100 MW AI data centre running at 90% average load uses 100 MW x 8,760 hours x 0.9 = about 790,000 MWh (0.79 TWh) a year. At 4 cents per kWh that costs about $32 million; at 10 cents about $79 million. The difference, about $47 million a year for one site, is why AI developers chase cheap, reliable power in places like the US Gulf Coast, the Nordic countries and the Gulf states, and why the IEA now tracks data centres as a major source of electricity demand.

The pattern. Put the steps side by side as "energy needed per dollar of product":

Link in the chain	Energy per unit (approx.)	Price per unit (approx.)	Energy per $ of output
Primary aluminium	14,000 kWh per tonne	$2,500-3,000 per tonne	about 5 kWh
Battery cells	20-40 kWh per kWh	$40-80 per kWh	about 0.3-1 kWh
Finished robot arm	about 1,700 kWh per arm	tens of thousands of $	well under 0.1 kWh

Energy intensity falls by a factor of roughly a hundred as you move from metal to machine. Value per kilogram rises in the opposite direction. This is the physical version of the smile curve from Month 1. The energy-hungry middle (smelting, refining, cell making) is a low-margin business where cheap power is the ticket to survival. The ends (machine design, control software, AI models, brands, service) are where most of the profit sits, and there the main inputs are skills and capital, not kilowatt-hours.

For your driving question, this cuts both ways. A country with cheap clean power, for example from hydro in parts of Africa or solar plus storage near the equator, can win the energy-hungry middle, which brings jobs, exports and industrial learning but thin margins. To capture the larger share of value, it must climb from that base into components, software and brands, the way Korea used cheap POSCO steel to grow its shipbuilders and carmakers (Month 3). Cheap energy gets a country into the chain; it does not, by itself, let the country keep the value.

Case study: China's power machine

No country shows the link between energy and industrialization more clearly than China. In 2025, China's electricity consumption grew by **more than 5%**, the fastest of any major economy, adding demand equal to the entire consumption of Germany in a single year (Energy Institute, June 2026). China already uses more than a third of the world's electricity (its National Energy Administration reported about 9.9 trillion kWh of consumption in 2024, against a world total of roughly 27,000-28,000 TWh; approximate).

How does China supply it? With everything at once.

- **Coal**: China consumed about **4.95 billion tonnes** of coal in 2025, roughly **56%** of the world total (IEA, *Coal 2025*, December 2025). Coal still generated about 58% of China's electricity in 2024 (EI, 2025). China keeps building coal plants partly for **energy security**: after power shortages in 2021 forced factories to shut, Beijing ordered more domestic coal mining and more backup plants.
- **Solar and wind**: China installed somewhere between about **315 and 415 GW** of solar in 2025, depending on the counting method (direct current or alternating current, and which agency), roughly **58-60%** of the world total, plus about **119 GW** of wind, 72% of the world total (Ember, *Global Electricity Review 2026*; IEA PVPS, *Trends in PV Applications 2026*). For scale, China's solar additions in one year exceeded the total installed solar capacity of the United States.
- **The result**: for the first time in about half a century, coal-fired power generation **declined** in both China and India in 2025, in China because of the huge expansion of renewables, in India because of an unusually strong monsoon that boosted hydropower (IEA, *Coal Mid-Year Update 2026*).

China's industrial strategy depends on this power system. Its provinces offer heavy industry long-term electricity contracts at low prices. Its vast transmission network, including ultra-high-voltage lines that carry power thousands of kilometres from coal and renewable bases in the west and north to factories on the coast, lets industry draw on the cheapest sources. China's solar, battery, aluminium, polysilicon and steel industries all grew on cheap, reliable power.

There are costs. China is the world's largest emitter of carbon dioxide, and coal mining and burning have caused severe air pollution and health damage. But from the narrow perspective of industrial competitiveness, China treated electricity as infrastructure for manufacturing first and as a consumer good second. That choice is one of the hidden reasons China dominates the middle of so many supply chains.

Lesson for latecomers: a country that wants processing and manufacturing must plan electricity supply years ahead of the factories. In many African and some Southeast Asian countries, factories run diesel generators because the grid is unreliable, which can cost several times grid prices. No industrial policy from Month 3 can work on top of a broken power system.

Key numbers: Week 1

- **More than 600 EJ**: Global total energy supply in 2025, up 1.7%; fossil fuels about 86% (Energy Institute, *Statistical Review of World Energy 2026*).
- **About 2,300 watts**: Average energy use per person, running continuously (calculated from EI 2026 data and a population of 8.2 billion).
- **730 million**: People without electricity in 2024, about 600 million in sub-Saharan Africa (IEA, October 2025).
- **3% and 4.4%**: Global electricity demand growth in 2025 and 2024 (IEA, *Electricity 2026*).
- **28,200 TWh -> 33,600 TWh**: Global electricity consumption, 2025 to 2030 projected (IEA, *Electricity 2026*).
- **About 485 TWh**: Data centre electricity use in 2025, up 17%, heading to about 950 TWh in 2030 (IEA, *Key Questions on Energy and AI*, April 2026).
- **2%**: Global energy efficiency improvement relative to GDP in 2025, half the 4% COP28 goal (EI, 2026).
- **More than 5%**: China's electricity demand growth in 2025, equal to adding Germany's entire consumption (EI, 2026).

Think like a founder

Energy is a cost for almost every business, but for a few it is the main cost. Founders who understand energy can find two kinds of opportunity.

Opportunity 1: arbitrage cheap energy into products. Where electricity is unusually cheap, energy-intensive products become profitable. Today the cheapest electricity in many sunny countries is midday solar. Businesses that can shift their electricity use to the middle of the day (water pumping and desalination, cold storage, some kinds of materials processing, even some AI computing tasks that are not time-sensitive) can buy power when it is almost free. This is Robert Allen's logic from Month 3, updated: find a newly cheap input and build the machine that turns it into something valuable.

Opportunity 2: sell reliability to people with unreliable power. Hundreds of millions of businesses in Africa and South Asia run on diesel generators or suffer blackouts. Products that

combine solar panels, batteries and smart software to give a factory, hospital or shop reliable power at lower cost than diesel are a large market. The value sits not in the panels (which are a cheap commodity from China) but in the financing, installation, maintenance and software that keep the system running and get paid. Keep this in mind for Week 4.

Week 2: Oil and Gas Markets, OPEC+ and the 2026 War

Core lesson

Why oil is special

Oil is the world's largest single energy source and its most traded commodity. It is special for three reasons.

1. **Transport depends on it.** Almost all ships, planes, trucks and most cars run on oil products (petrol, diesel, jet fuel, fuel oil). There are few short-term substitutes.
2. **It is easy to store and ship**, so it trades globally and has one world price (with regional differences for quality and transport).
3. **Reserves are geographically concentrated**, especially in the Persian Gulf, which makes oil a tool and target of geopolitics.

In 2025 the world used roughly 104-105 million barrels of oil a day (estimates differ slightly between the IEA, the US Energy Information Administration (EIA) and OPEC). That is about 16.5 billion litres a day, or two litres per person per day.

How the oil price is set

Oil prices are set by the balance of supply and demand, but with two features that make them unusually volatile.

Feature 1: Demand and supply are very inelastic in the short run. "Inelastic" means quantity barely responds to price. If oil prices double tomorrow, most people still drive to work and trucks still deliver food; demand falls only slightly. Producers cannot quickly pump more either, since new wells take months or years. Economists estimate short-run demand elasticity at roughly -0.05 to -0.1 (that is, a 10% price rise reduces demand by only 0.5-1%).

Here is why that matters. Suppose 5% of world oil supply suddenly disappears and there are no spare barrels. To make demand fall by 5% to match the lower supply:

- If elasticity is -0.1, the price must rise by about 5% / 0.1 = **50%**.
- If elasticity is -0.05, the price must rise by about 5% / 0.05 = **100%**, that is, double.

A small physical shortage can cause a huge price spike. This simple calculation explains most oil crises.

Feature 2: Buffers decide how big the spike gets. Two buffers absorb shocks:

- **Spare capacity**: production capacity that is not being used but can be switched on within weeks (the IEA defines it as capacity that can be reached within 90 days). Most of the world's spare capacity sits in Saudi Arabia, the United Arab Emirates (UAE), Kuwait and Iraq. Notice the problem: if a shock hits the Gulf itself, the spare capacity may be trapped behind the same chokepoint.

- **Inventories**: stored oil, including commercial stocks and government Strategic Petroleum Reserves. Members of the IEA are required to hold emergency stocks and can release them together.

When buffers are large, a supply loss is absorbed by drawing them down, and prices rise only modestly. When buffers are exhausted, prices must rise until demand is destroyed.

Benchmarks and futures. Oil is priced against benchmark crudes: **Brent** (North Sea, the main global benchmark), **West Texas Intermediate** (WTI, US), and **Dubai/Oman** and **Murban** (Gulf crudes priced for Asian buyers). Most trading happens through **futures contracts**, agreements to buy or sell oil at a set price on a future date. When near-term prices are higher than later prices, the market is in **backwardation**, a sign of tightness (buyers will pay extra to get oil now). Physical cargo prices (such as "Dated Brent") can diverge sharply from futures during a crisis, as we will see.

OPEC and OPEC+

OPEC was founded in 1960 by Iran, Iraq, Kuwait, Saudi Arabia and Venezuela to coordinate production and capture more of the value of their oil, which had been controlled by Western oil companies (the "Seven Sisters"). This is a resource-nationalism story you will meet again in Month 6. In 1973, Arab OPEC members cut production and embargoed the United States and the Netherlands over their support for Israel in the October War; oil prices roughly quadrupled. The 1979 Iranian Revolution caused a second shock. These shocks transferred huge wealth to oil exporters and helped cause the inflation and debt crises of the 1970s and 1980s (Month 3 described how this fed the Latin American debt crisis).

In 2016, after a price collapse caused partly by the boom in US shale oil, OPEC formed a broader alliance with ten non-OPEC producers, led by Russia, known as **OPEC+**. OPEC+ manages supply by setting production targets for members. In 2023, eight members (Saudi Arabia, Russia, Iraq, the UAE, Kuwait, Kazakhstan, Algeria and Oman) made extra "voluntary" cuts. From April 2025, with prices weak, they began unwinding those cuts to regain market share. By September 2026 they had completed restoring 1.65 mb/d of the voluntary cuts made in 2023, with about 2 mb/d of group-wide cuts agreed in 2022 still in place until the end of 2026. The seven core members (the UAE left OPEC and OPEC+ with effect from May 1, 2026, to escape its quota; Reuters, April 28, 2026) then held their combined target at about **31.0 mb/d** for October, November and December (Saudi Arabia 10.478 mb/d, Russia 9.949 mb/d, Iraq 4.431 mb/d), but most were pumping well below target because of the war: in August they produced about 25 mb/d, roughly 5 mb/d below pre-war levels (OPEC statement, November 1, 2026, via *The National*; OPEC data via CNBC, October 4, 2026).

Why does OPEC+ sometimes fail to hold prices up? Because of a classic **cartel problem**: each member gains by quietly producing a bit more than its quota, and high prices encourage competitors outside the cartel (such as US shale, Brazil and Guyana) to grow. In 2025 the Americas produced about 20% more oil than the Middle East, a reversal from two decades earlier, when the Middle East produced 20% more (Energy Institute, June 2026). This shift turned out to matter enormously in 2026.

Natural gas and LNG

Natural gas is harder to move than oil. Over land it travels through pipelines; across oceans it must be chilled to about -162 degrees Celsius to become **liquefied natural gas (LNG)**, shipped in

special tankers, and turned back into gas at the destination. This makes gas markets more regional, with three main price benchmarks: **Henry Hub** in the US, **TTF** (Title Transfer Facility) in Europe, and **JKM** (Japan Korea Marker) for spot LNG in Asia.

Gas matters for industry (fertilizer, chemicals, steel) and for electricity: gas plants can ramp up and down quickly, which makes them the main source of flexible power in many grids. A worked example on gas and power prices: a modern combined-cycle gas plant needs roughly 6.5 MMBtu of gas to produce 1 megawatt-hour (MWh) of electricity (a typical efficiency; approximate).

- At a gas price of $10/MMBtu, the fuel cost is 6.5 x $10 = $65/MWh, or 6.5 US cents per kWh.
- At $25/MMBtu, roughly the Asian spot price in parts of 2026, the fuel cost is 6.5 x $25 = $162.50/MWh, or about **16 cents per kWh**, before paying for the plant itself.

Countries that rely on imported LNG for power (such as Japan, Korea, Pakistan, Bangladesh, and increasingly parts of Southeast Asia) are therefore exposed to global gas shocks. Compare this with the solar costs in Week 4.

Qatar is the giant of LNG. It operates the world's largest LNG complex at Ras Laffan, with roughly 77 million tonnes a year of capacity, and supplies around a fifth of global LNG. Its exports must pass through the Strait of Hormuz.

The Strait of Hormuz

The Strait of Hormuz is a narrow waterway between Iran and Oman connecting the Persian Gulf to the Arabian Sea. Before 2026, ships carried oil equal to roughly **one-fifth of global consumption** through it every day, along with about 20% of world LNG (Reuters, March 2026). Saudi Arabia and the UAE have pipelines that bypass part of the strait (Saudi Arabia's East-West pipeline to the Red Sea port of Yanbu, and the UAE's pipeline to Fujairah), but their capacity is far smaller than normal Gulf exports. Kuwait, Qatar, Bahrain and most of Iraq's exports have no bypass at all. For decades, analysts called a closure of Hormuz the nightmare scenario for the oil market. In 2026 it happened.

Case study: The 2026 Iran war and the largest oil supply shock in history

What happened. On February 28, 2026, the United States and Israel launched joint strikes on Iran. The strikes killed several senior Iranian leaders, including Supreme Leader Ayatollah Ali Khamenei. Iran retaliated against shipping and energy infrastructure across the Gulf and declared it would attack ships transiting the Strait of Hormuz. Tanker traffic collapsed: crude tanker transits fell to four on March 1, against an average of 24 a day since January (Vortexa data reported by Reuters, March 3, 2026). Qatar halted LNG production and declared force majeure (a legal clause excusing a supplier from delivery when events beyond its control prevent it) on March 4. On March 18-19, after Israel struck Iran's South Pars gas field, Iran hit Ras Laffan, destroying two LNG production units ("trains") with 12.8 million tonnes a year of capacity, about 17% of Qatar's LNG exports, for an estimated three to five years, at a cost QatarEnergy put at about $20 billion a year in lost revenue (Reuters, March 19, 2026). (The source summary in this paragraph draws on CNBC's April 2026 timeline and Reuters reports listed in Sources.)

Prices. Brent crude rose from about **$72** a barrel on February 27 to near $120 on March 9, jumped **51% in March** (one of the largest monthly rises on record), and peaked at **$126.41 on April 30** (CNBC, April 2026; Reuters via *The Business Standard*, September 2026). Note the order of events: a two-week ceasefire from about April 7 and talks in Islamabad did *not* end the shock, because

the strait stayed functionally closed and the truce broke down; the peak came at the end of April, after the ceasefire lapsed. A US-Iran arrangement on Hormuz shipping in June then knocked out much of the war premium, and Brent traded in roughly the $75-85 range for parts of June and July (Gulf press reports; approximate), with spikes in between: Brent was above $100 on July 24. Fighting escalated again from late August: attacks by Iran-backed Houthi forces on Saudi energy facilities, the bombing of Saudi Arabia's East-West pipeline on September 11, and a wave of attacks on tankers in October. Brent passed $100 again on September 9 and rose 13.6% in October: on October 30 the expiring December contract settled at **$117.64**, its highest since late April, while the more actively traded January contract settled at **$113.43** (Reuters, October 30, 2026). The gap is a reminder to check *which* contract a headline price refers to. In early November, as US-Iran and Iran-Qatar talks progressed, Brent fell back, settling at **$104.77** on November 2 (Reuters, November 2, 2026; Gulf News, November 2, 2026; CNBC, November 4, 2026). As of this writing (November 2026) there is no final settlement.

Physical flows. Before the war, roughly 20 mb/d of oil passed through Hormuz. According to the IEA, flows in August 2026 were only about **7 mb/d** (its first estimate was 7.6 mb/d; its October report, which says September flows rose by 3.7 mb/d to **10.9 mb/d**, implies about 7.2 mb/d after revision). The September recovery came mainly from ship-to-ship transfers in the Gulf of Oman, which let tankers load outside the most dangerous waters. Separately, the swift repair of Saudi Arabia's East-West pipeline lifted exports that *bypass* the strait by 0.97 mb/d to **5.6 mb/d**. Total Gulf crude exports reached **13.9 mb/d** in September, just over 75% of pre-war levels (IEA, *Oil Market Report*, October 2026). How much output was shut in depends on the month and the source: the US EIA estimated about **5.5 mb/d** of Middle East output shut in during July (Reuters, August 2026), while the IEA said more than **10 mb/d** of Gulf output remained shut in during August, when Saudi output fell to 5.97 mb/d (IEA, *Oil Market Report*, September 2026).

Why diesel went even higher than crude. The Gulf is not only a crude exporter but also a major exporter of refined products such as diesel and LPG (liquefied petroleum gas, used for cooking). Product shipments recovered far more slowly than crude: in September 2026 Middle East refined product exports were 1.9 mb/d against 4.7 mb/d before the war. Combined with Ukrainian drone damage to Russian refineries, global diesel and gasoil availability was 1.5 mb/d lower year on year in the second quarter and 1.7 mb/d lower in the third. Wholesale diesel exceeded **$200 a barrel** in September, reaching all-time highs of around $220 in Europe and the US, nearly double crude prices. Physical "Dated Brent" cargoes traded around **$134** in mid-October, far above futures prices of about $107, showing how desperate refiners were for prompt oil (IEA, October 2026). Diesel matters because it moves trucks, ships, tractors and generators: a diesel shock raises the cost of almost everything.

Buffers. On March 11, 2026, IEA members agreed a collective emergency stock release of **400 million barrels** (Reuters, March 12, 2026). Even so, global observed oil inventories by the end of September were **509 million barrels** below pre-war levels, and on-land stocks were at their lowest since at least 2016 (IEA, October 2026). The buffers were being used up.

Demand destruction. Because supply could not recover fast enough, high prices had to cut demand. The IEA forecast world oil demand would *fall* by **2.6 mb/d in 2026** to about 102.4 mb/d, with diesel a quarter of the drop, and that 2027 demand would still be 0.6% below 2025 (IEA, October 2026). Remember the elasticity calculation: a supply loss of around 5% had to be matched by demand falling roughly 2.5% plus inventory draws, and the price had to rise by tens of percent to achieve that.

Gas. Asian spot LNG prices rose about **143%** in the first weeks, to above $25/MMBtu (Reuters, March 26, 2026), and were still about $26 in September. Europe's TTF gas benchmark rose about 80% in the first days of the war (FT, March 2026). European gas storage reached its lowest level for the time of year since records began in 2011 (Gas Infrastructure Europe data via *bne IntelliNews*, September 2026). Poorer LNG importers such as Pakistan and Bangladesh had cargoes cancelled and were priced out of the spot market.

Winners and losers.

Group	Effect	Why
Gulf producers trapped behind Hormuz (Kuwait, Qatar, Iraq)	Large losses	Volumes collapsed; infrastructure damaged
Saudi Arabia, UAE	Mixed	Some exports via bypass pipelines at high prices, but output and facilities hit
Producers in the Americas (US, Brazil, Guyana, Canada)	Gains	Sold at high prices with no physical disruption; US refiners earned record margins on discounted crude
Russia	Mixed	Higher prices, but refinery damage from Ukraine
Oil-importing Asia (India, Pakistan, Bangladesh, Japan, Korea, much of Southeast Asia)	Large losses	Most Hormuz oil and LNG goes to Asia; import bills soared
Europe	Losses	Diesel and gas price shock; low gas storage
EV and solar owners everywhere	Protected	Electricity from domestic renewables was not affected by Hormuz

Lessons.

1. **Geography is destiny for energy security.** A fifth of world oil depended on one strait. Diversification of routes and suppliers is worth paying for.
2. **Buffers matter more than averages.** In normal years, spare capacity and stocks look like wasted money. In a crisis they are worth hundreds of billions of dollars.
3. **The shift of oil production to the Americas cushioned the shock.** Without 20 years of growth in US shale and South American offshore oil, prices would likely have gone much higher.
4. **Domestic renewables are a security asset, not just a climate policy.** Analysts at Ember noted that this was the second major fossil fuel shock in four years (after Russia's 2022 invasion of Ukraine). Every country with its own solar, wind, hydro or nuclear power was partly insulated. For an energy-importing country like Pakistan, the cheap solar panels of Week 4 are a direct hedge against Hormuz.

Key numbers: Week 2

- **About one-fifth**: Share of world oil consumption, and about 20% of global LNG, that passed through the Strait of Hormuz before the war (Reuters, March 2026; EIA puts 1H 2025 Hormuz

oil flows at 20.9 mb/d, see Month 7).

- **$72 -> $126.41**: Brent crude, February 27, 2026 (pre-war) to its peak on April 30, 2026 (CNBC; Reuters).
- **About 7 mb/d to 10.9 mb/d**: Hormuz oil flows in August and September 2026, against roughly 20 mb/d before the war; bypass exports 5.6 mb/d in September (IEA, *Oil Market Report*, September and October 2026).
- **400 million barrels**: IEA collective emergency stock release agreed unanimously by its 32 members on March 11, 2026, the largest in its history and more than double the 2022 release (IEA press release, March 11, 2026).
- **509 million barrels**: Fall in global observed oil inventories from end-February to end-September 2026 (IEA, October 2026).
- **-2.6 mb/d**: IEA forecast for the fall in world oil demand in 2026, to about 102.4 mb/d (IEA, October 2026).
- **More than $200 a barrel**: Wholesale diesel prices in September 2026 (IEA, October 2026).
- **12.8 million tonnes a year**: Qatari LNG capacity destroyed at Ras Laffan in March 2026, about 17% of exports, for three to five years (Reuters, March 2026).
- **About 20% more**: Oil production in the Americas compared with the Middle East in 2025 (Energy Institute, 2026).

Think like a founder

Energy shocks create businesses as well as destroy them.

Efficiency becomes valuable overnight. When diesel doubles, a logistics company that uses software to cut empty truck journeys by 10%, or a factory that switches its forklifts and generators to batteries charged by solar, saves real money. Founders selling efficiency and electrification find customers much faster during shocks. A good business, though, is one that pays off even at normal prices; shocks only speed up adoption.

Information is valuable in chaos. In 2026, physical oil cargoes traded at prices far from futures, tanker rates exploded, and buyers struggled to know where cargoes were. Companies that track ships by satellite and AI (such as Vortexa and Kpler, whose data appear throughout this chapter) sell exactly this information. This is an example of value capture at the service and data end of the smile curve: these firms own no oil, no tankers and no refineries, yet the whole market pays for their data.

Do not build a business whose survival depends on cheap imported fuel. If your business plan only works with diesel at $80 a barrel, 2026 would have killed it. Stress-test plans against energy shocks.

Week 3: Coal, Electricity Grids and Petrostates

Core lesson

Coal: the fuel that refuses to die

Coal is the dirtiest major fuel, emitting the most carbon dioxide per unit of energy, and many rich countries have nearly stopped using it. Yet world coal use has never been higher. Global coal demand reached a record **8.85 billion tonnes in 2025**, up 0.5% (the IEA's December 2025 estimate; its July 2026 update revised this slightly to 8.84 billion tonnes, up 0.3%), and the IEA forecasts another

record of about 8.94 billion tonnes in 2026, partly because the Hormuz crisis made LNG scarce and pushed some Asian power plants back to coal (IEA, *Coal 2025*, December 2025, and *Coal Mid-Year Update 2026*). China and India together account for about **71%** of world consumption. Global coal production stayed at a record of about 9.1 billion tonnes in 2025.

Why does coal persist?

- **It is cheap and domestic.** China, India and Indonesia all have large coal reserves. Burning your own coal means you do not depend on imported oil or LNG passing through chokepoints like Hormuz. After the 2021 power shortages, both China and India made domestic coal output a pillar of energy security, and both expanded production by hundreds of millions of tonnes.
- **Power plants last decades.** A coal plant built in 2015 may operate until 2055. Asia's coal fleet is young, unlike the old plants being retired in Europe and the US.
- **Industry needs it.** Steelmaking with blast furnaces requires coking coal, and steel production is the second-largest use of coal after power. Much of Indonesia's nickel smelting runs on captive coal plants built inside industrial parks.

But the trend is turning in the biggest markets. For the first time in about half a century, coal-fired power generation fell in both China and India in 2025 (IEA, 2026). The IEA expects global coal demand to plateau and decline slightly by 2030, as renewables, nuclear and a coming wave of LNG supply compete with it. Coal is not disappearing; it is being squeezed at the margin.

The United States moved the other way in 2025: higher gas prices and policies that slowed coal plant retirements lifted US coal power by about 13%, which drove a 3.2% rise in US energy-related emissions, four times China's increase in absolute terms (Energy Institute, June 2026). Energy policy can reverse trends quickly.

How electricity grids work

Electricity is unusual among products: it is very hard to store in large amounts (until recently), so **supply must match demand every second**. If demand exceeds supply, the frequency of the alternating current falls; if the imbalance is large, protective systems cut off parts of the grid to prevent damage, which is a blackout. Grid operators constantly adjust generation up and down to keep the frequency stable (50 hertz in Indonesia, most of Asia, Africa and Europe; 60 hertz in the Americas and parts of Japan).

A grid has three parts:

1. **Generation**: power plants (coal, gas, nuclear, hydro, solar, wind) and increasingly batteries.
2. **Transmission**: high-voltage lines that carry power long distances from plants to regions.
3. **Distribution**: lower-voltage networks that deliver power to homes and businesses.

The merit order. In a competitive power market, plants are dispatched in order of their *marginal cost*, the cost of producing one more MWh. Solar and wind have almost zero marginal cost (sunshine and wind are free), so they run first. Nuclear and coal come next, and gas plants, which have high fuel costs, run last. The price in each hour is often set by the most expensive plant needed to meet demand. A worked example:

- Demand in one hour: 100 GW.
- Available: 40 GW of solar at \$0/MWh, 30 GW of nuclear and hydro at \$10, 20 GW of coal at \$40, 30 GW of gas at \$90.

- Dispatch order: solar (40) + nuclear/hydro (30) + coal (20) = 90 GW. We still need 10 GW, so 10 GW of gas runs.
- The market price for that hour is set by gas: **$90/MWh**, and every generator running receives it.
- At noon on a sunny day, if solar rises to 70 GW, demand is met by solar plus nuclear/hydro, and the price might fall to $10 or lower, even below zero.

This explains two facts that confuse many people. First, expensive gas often sets power prices even in grids with lots of cheap renewables, which is why the 2026 gas shock raised electricity prices in Europe. Second, as solar grows, midday prices collapse, which reduces the revenue of solar farms themselves (this is called **price cannibalization**) and creates a strong incentive for batteries that buy cheap midday power and sell it in the evening.

Capacity versus energy. A grid needs not just energy over the year but enough reliable capacity at the moment of peak demand, which in hot countries is often an evening after sunset when air conditioners are running. This is why grids with lots of solar need storage, flexible plants or demand that can shift.

The grid bottleneck. Building generation is now often easier than building the wires. The IEA estimates global investment in electricity grids at about **$400 billion a year**, less than half the amount spent on generation (IEA, *World Energy Investment 2025*, as reported by Energy Digital and Arab News). In many countries, solar and wind projects wait years for a grid connection. In poorer countries, distribution companies are often financially weak: they lose power to theft and technical losses, charge tariffs below cost for political reasons, and cannot pay generators on time. This creates a cycle of debt and underinvestment that keeps power unreliable.

Who invests in energy, and where

Global energy investment was expected to reach about **$3.3 trillion in 2025**, of which about **$2.2 trillion** went to clean energy (renewables, nuclear, grids, storage, efficiency and electrification), twice the **$1.1 trillion** going to oil, gas and coal (IEA, *World Energy Investment 2025*). But the distribution is very unequal. Africa accounted for only about **2%** of global clean energy investment despite having about **20%** of the world's population (IEA, 2025). The main reason, as Week 4 will show, is the cost of capital.

A few more numbers from the same report show where the money is moving, and why:

- **Electricity has overtaken oil and gas.** Ten years ago, investment in fossil fuel supply was 30% higher than investment in power generation, grids and storage. In 2025 the positions were reversed: the electricity sector was set to receive about **$1.5 trillion**, some 50% more than all spending on bringing oil, gas and coal to market. Solar alone, at about **$450 billion**, was the single largest item in the world's energy investment.
- **China is the largest energy investor by far**, with almost one-third of global clean energy investment, up from a quarter ten years earlier.
- **Africa is going backwards.** Energy investment in Africa in 2025 was about one-third lower than in 2015. Debt service costs across the continent were equivalent to more than **85%** of total energy investment, so money that could build power plants is paying interest instead. International public finance (development banks and aid) provides only about **$32 billion** a year, around 7% of clean energy investment in emerging and developing economies outside China.

(All figures: IEA, *World Energy Investment 2025.*)

The mechanism connects directly to Week 4. A solar farm in Kenya uses the same panels, bought at the same world prices, as one in Spain, but if lenders demand twice the interest rate, the electricity costs much more and fewer projects pencil out. Debt-burdened governments cannot easily offer guarantees, so private investors stay away, so grids stay weak, so the energy-hungry middle of the ore-to-robot chain stays in China. Breaking this loop is less about technology than about finance and institutions, which is why development banks, guarantees and credible contracts appear so often in the "Think like a founder" boxes of this course.

Petrostates: rich in oil, poor in resilience

Month 2 introduced the **resource curse** and **rentier states**: governments that live on income from natural resources ("rents") rather than taxing their citizens. Now we can look at the economics more closely.

A petrostate typically has these features:

- **Oil and gas dominate exports and government revenue.** When prices fall, the budget collapses.
- **The fiscal breakeven price**, the oil price at which the government budget balances, is often high, because governments use oil income to pay large public-sector wage bills and subsidies.
- **Dutch disease**: oil exports push up the value of the currency and wages, which makes manufacturing and agriculture uncompetitive. The name comes from the Netherlands after it discovered North Sea gas in the 1960s and saw its manufacturing weaken.
- **Volatility**: booms encourage spending and borrowing; busts force painful cuts.

The contrast between well-managed and badly managed petrostates is stark:

- **Norway** saves most of its oil revenue in a sovereign wealth fund (the Government Pension Fund Global), invested abroad, and limits how much the government can spend from it each year. The fund is worth roughly $2 trillion (Norges Bank Investment Management; approximate, 2025-2026). By investing abroad, Norway avoids pushing up its own currency (limiting Dutch disease) and saves wealth for future generations. Month 11 covers sovereign funds in detail.
- **Venezuela**, which has the world's largest proven oil reserves, saw output collapse after years of mismanagement, politicization of the state oil company PDVSA, and sanctions. It produced only about **1.16 mb/d in September 2026** (IEA, October 2026), compared with more than 3 mb/d in the late 1990s.
- **Nigeria**, Africa's largest oil producer, has long depended on oil for most of its export earnings and a large share of government revenue, yet suffers from oil theft, refinery failure (it imported most of its fuel for decades despite exporting crude) and fuel subsidies that cost huge sums. In September 2026 its crude output was about **1.42 mb/d** (IEA, October 2026).

The key lesson: **oil wealth helps only if institutions convert it into diversified capital**: infrastructure, education, a savings fund, and new industries. Without those institutions, oil wealth often makes a country poorer and more unstable over time.

Case study: Saudi Arabia, Vision 2030 and the war windfall

Saudi Arabia is the most important petrostate in the world. It holds the largest share of OPEC+ spare capacity, and its state oil company, Saudi Aramco, is one of the most profitable companies

on Earth. Since 2016, Crown Prince Mohammed bin Salman has pursued **Vision 2030**, a plan to diversify the economy away from oil through tourism, entertainment, mining, manufacturing, logistics and enormous "giga-projects" such as NEOM, financed partly by the Public Investment Fund (PIF), the sovereign wealth fund.

The fiscal problem before the war. Diversification is expensive, and it is paid for with oil money. The International Monetary Fund (IMF) estimated Saudi Arabia's fiscal breakeven oil price at about **$92 a barrel in 2025** and about **$87 in 2026**; including the PIF's domestic spending, Bloomberg Economics put it at around $111-114 (Bloomberg, February 2026; AGBI, June 2026). Estimates differ because they count different spending: Fitch used about $94 for 2026, and by October 2026 the bank EFG Hermes reckoned wartime spending had pushed the true figure towards $115. Always ask *which budget* a breakeven covers before quoting it. But Brent averaged only about **$69** in 2025. The result was a 2025 budget deficit of about **276.6 billion riyals** (roughly $74 billion, around 5.5% of GDP), the largest since 2020, funded by borrowing (Saudi Ministry of Finance data reported by Bloomberg, February 2026). OPEC+ had started unwinding production cuts in 2025 partly to regain market share, accepting lower prices.

Then came the war. Higher oil prices should have been a windfall. But Saudi Arabia was also physically in the conflict. Its exports through Hormuz were disrupted; it relied on the East-West pipeline to Yanbu, which was itself bombed on September 11, 2026; and Houthi attacks set Saudi energy installations on fire. Saudi crude output fell to about **5.97 mb/d in August 2026**, against an OPEC+ target of about 10.5 mb/d, before recovering to about **8.3 mb/d in September** (IEA, October 2026). Defence spending rose 26% year on year in the first quarter, and the first-quarter budget deficit was about **125.7 billion riyals ($33.5 billion)**, more than double a year earlier (Ministry of Finance data reported by AGBI, June 2026, and *The GCC Edge*, May 2026). The IMF, in June 2026, expected higher prices to offset volume losses later in the year if shipping through Hormuz normalized, and said Saudi Arabia had fiscal space to respond if the shock lasted longer.

The price-times-volume lesson. An oil exporter's revenue is price x volume. A war that doubles the price but halves the volume leaves revenue roughly unchanged while costs (defence, repairs) rise. Exporters with bypass routes or outside the war zone (the US, Brazil, Norway, Guyana) captured the windfall; exporters inside it captured mostly risk.

What this means for Vision 2030. Analysts increasingly describe the giga-projects as being "rescoped" (scaled back or delayed) as fiscal space tightens. The deeper point for your driving question: Saudi Arabia is trying to use oil rents to buy its way up the value chain into mining, metals processing, manufacturing and even AI data centres (it has abundant cheap land, sun and gas). Whether it succeeds depends on the same factors as in Month 3: export discipline, capable institutions and competitive firms, not just money. Norway shows that saving the rents wisely builds wealth; Saudi Arabia is attempting the harder task of building new industries with them.

Key numbers: Week 3

- **8.84 billion tonnes**: Record global coal demand in 2025, up 0.3% (revised from the December 2025 estimate of 8.85); forecast to rise 1.2% to a new record of 8.94 billion tonnes in 2026 as high LNG prices push power plants back to coal (IEA, *Coal Mid-Year Update 2026*).
- **71%**: Share of world coal consumption in China and India (IEA, *Coal 2025*).
- **About 56%**: China's share of global coal use, about 4.95 billion tonnes in 2025 (IEA, *Coal 2025*).

- **About $400 billion a year**: Global grid investment, less than half of generation investment (IEA, *World Energy Investment 2025*).
- **$3.3 trillion, of which $2.2 trillion clean**: Global energy investment in 2025 (IEA, *World Energy Investment 2025*).
- **2% vs 20%**: Africa's share of global clean energy investment vs its share of world population (IEA, 2025).
- **About $92 and $87**: IMF estimates of Saudi Arabia's fiscal breakeven oil price for 2025 and 2026 (reported by AGBI and *The GCC Edge*, 2026).
- **276.6 billion riyals**: Saudi Arabia's 2025 budget deficit, about 5.5% of GDP (Ministry of Finance via Bloomberg, February 2026).
- **5.97 -> 8.3 mb/d**: Saudi crude output in August and September 2026, against a target of about 10.5 mb/d (IEA, October 2026).

Think like a founder

The grid is where many of the best energy businesses of the next 20 years will be built, because it is the bottleneck.

- **Grid software and flexibility.** As solar grows, the value shifts from producing energy to *moving it in time and space*. Software that controls thousands of batteries, EV chargers, water heaters or factory loads to soak up cheap midday power and reduce evening peaks (a "virtual power plant") can earn money in many markets. This is a natural fit for someone with AI and mathematics skills: it is forecasting and optimization.
- **Fixing distribution in poor countries.** Utilities in many developing countries lose a large share of their power to theft and technical losses. Smart meters, prepaid billing and loss detection with data analytics can turn a money-losing distribution network into a viable one. Companies that do this often need to work closely with governments, which is slow, but the impact is large.
- **Captive power for industry.** Where grids are weak, factories and industrial parks build their own power plants ("captive" power). Indonesia's nickel parks did this with coal. A founder could offer solar-plus-battery captive power to industrial parks, financed on long-term contracts, as a cleaner and increasingly cheaper alternative.

For the petrostates, note the founder opportunity in their diversification plans: Gulf sovereign funds are actively looking for technology partners in mining, processing, manufacturing and AI. A credible startup can sometimes raise large sums from them, but must remember that funding tied to a government's strategy can shift when oil prices fall.

Week 4: Renewables, Storage and the New Cost Curves

Core lesson

The most important price chart of the century

In 2010, a lithium-ion battery pack cost about **$1,474 per kWh** (in today's dollars). By 2025 the average had fallen to **$108 per kWh**, a drop of **93%** (BloombergNEF, *Lithium-Ion Battery Price Survey*, December 2025). Battery packs for stationary storage (batteries attached to the grid) fell even faster, to about **$70 per kWh** in 2025, down 45% in a single year, becoming the cheapest segment for the first time. In China the average pack price was about $84 per kWh, and the lowest observed cell and pack prices were about $36 and $50 per kWh. Lithium iron phosphate (LFP)

packs averaged about $81 per kWh, compared with $128 for nickel manganese cobalt (NMC) packs. These declines came even though battery metal prices rose in 2025; overcapacity, fierce competition and the shift to cheaper LFP chemistry pushed prices down anyway.

Solar has followed a similar path. According to the International Renewable Energy Agency (IRENA), the global average cost of electricity from new utility-scale solar fell by about **90%** between 2010 and 2024, to about **$0.043 per kWh** in 2024; new onshore wind cost about **$0.034 per kWh**. In 2024, **91%** of new utility-scale renewable projects produced electricity more cheaply than the cheapest new fossil fuel alternative (IRENA, *Renewable Power Generation Costs in 2024*, July 2025). Solar modules have been selling at around 8-10 US cents per watt in open markets, and BloombergNEF expects prices to stay around that level through 2026 because of overcapacity (reported December 2025; Payne Institute, August 2026, reports panels below $0.10 per watt).

Why costs fell: learning curves

The explanation is a pattern called the **learning curve** or **Wright's law** (after aeronautical engineer Theodore Wright, who noticed it in aircraft production in 1936): **each time cumulative production doubles, the cost per unit falls by a roughly constant percentage.** For solar modules, the historical learning rate has been around 20% or more per doubling (a widely cited estimate from the academic literature; treat as approximate).

A worked example:

- Suppose a solar module costs $1.00 per watt when the world has produced 100 GW in total, and the learning rate is 20%.
- At 200 GW cumulative: $0.80. At 400 GW: $0.64. At 800 GW: $0.51. At 1,600 GW: $0.41. At 3,200 GW: $0.33.
- Five doublings cut the cost by two-thirds.

Why does this happen? Because producing more teaches manufacturers how to waste less material, make thinner silicon wafers, automate factories, build bigger plants with economies of scale, and improve efficiency. It is the same learning-by-doing that made Korean steel and Taiwanese chips competitive in Month 3. In fact, China's dominance of solar and batteries is a direct product of industrial policy that pushed huge volumes down the learning curve: China installed about 58-60% of the world's new solar in 2025 and makes the large majority of the world's panels and battery cells.

Fossil fuels do not follow learning curves in the same way, because most of their cost is the fuel itself, whose price is set by geology and markets, not by manufacturing learning. A coal plant built today burns coal much as one did 50 years ago.

The scale of deployment

Deployment is now enormous. Counts differ by agency because of methodology (direct-current versus alternating-current capacity, and estimates for China), but all agree 2025 was a record:

- The IEA reports solar additions rose about 12% to **more than 600 GW**, and total renewable additions reached about **800 GW** (IEA, *Global Energy Review 2026*).
- Ember counts **647 GW** of solar (in direct-current terms, up 11% from 582 GW in 2024) plus 167 GW of wind (Ember, March 2026). The IEA's Photovoltaic Power Systems Programme first estimated about **698 GW** (Snapshot, April 2026), then revised this to about **690 GW** in its fuller *Trends* report, bringing cumulative world solar capacity to **2.96 terawatts** (IEA PVPS, September-October 2026). The IEA itself puts cumulative solar at about 2,800 GW.

- IRENA, counting alternating-current capacity, reports 511 GW of solar and 692 GW of all renewables, bringing total renewable capacity to about 5,150 GW, roughly half of all power capacity on Earth (IRENA, March and July 2026).

Solar generation grew about **30%** in 2025 and reached **8.7%** of world electricity, overtaking wind for the first time. Installed battery storage capacity grew **66%** to about **302 GW** (Energy Institute, June 2026).

The intermittency problem, and how storage solves part of it

Solar produces only during the day and less in cloudy seasons; wind varies with weather. A grid that depends heavily on them needs ways to cover the gaps: storage, flexible generation (hydro, gas), transmission that connects regions with different weather, and demand that can shift.

Cheap batteries are solving the **daily** problem (moving midday solar to the evening). Let us estimate the cost with a worked example:

- Installed cost of a grid battery system: IRENA reports total installed costs for battery storage fell 93% from 2010 to about **$192 per kWh** in 2024 (IRENA, 2025). Let us use $200 per kWh.
- Suppose the battery cycles once a day for about 11 years: about 4,000 cycles.
- Ignoring financing for a moment, the capital cost per kWh delivered is $200 / 4,000 = $0.05, plus about 10% losses. So storing a kWh of solar for the evening costs very roughly **5-6 cents**.
- Solar at about 4 cents plus storage at about 5-6 cents gives "evening solar" at roughly **9-10 cents per kWh** before financing costs and margins.

Compare this with the gas example in Week 2: at $25/MMBtu, gas fuel alone costs about 16 cents per kWh. At 2026 LNG prices, solar plus batteries was cheaper than the fuel cost of imported-gas power in many places. Even at $10/MMBtu gas (6.5 cents fuel cost), solar plus storage is close to competitive. And compared with diesel generators, which often cost 25 cents per kWh or more in many African and Asian settings (fuel plus maintenance; a commonly cited range that depends heavily on local diesel prices), it is far cheaper. This is why solar is spreading fastest where people currently pay the most for power.

Batteries do *not* yet solve the **seasonal** problem (storing summer sun for winter in high-latitude countries) at low cost. For that, grids still need hydro, nuclear, long-distance transmission, gas plants used rarely, or new technologies. Tropical countries like Indonesia have a big advantage here: sunshine varies much less across the seasons near the equator than in Germany or northern China.

The cost of capital: why cheap solar is not cheap everywhere

Here is the single most important idea of Week 4 for your driving question. **Solar and wind are capital-intensive**: almost all the cost is paid up front (panels, inverters, land, installation), and the "fuel" is free afterwards. A gas plant is the opposite: cheaper to build, expensive to run. This means the interest rate on the money borrowed to build a solar farm largely decides the cost of its electricity.

A worked example, using the **capital recovery factor (CRF)**, which turns an up-front cost into an equal annual payment over the project's life at a given interest rate:

CRF = r x (1 + r)^n / ((1 + r)^n - 1), where r is the annual cost of capital and n is the number of years.

- A solar farm costs $700 per kW to build (roughly typical for utility-scale solar outside the most expensive markets; approximate) and lasts 25 years.
- In a sunny location with a 20% capacity factor, 1 kW produces 0.20 x 8,760 = 1,752 kWh a year.
- **At a 6% cost of capital** (typical in advanced economies or China): CRF = 0.06 x 1.06^25 / (1.06^25 - 1) = 0.06 x 4.292 / 3.292 ~ 0.0782. Annual capital cost = 0.0782 x $700 ~ $54.8. Per kWh: $54.8 / 1,752 ~ **3.1 cents**. Add about 0.5 cents for operations and maintenance: about **3.6 cents per kWh**.
- **At a 12% cost of capital** (typical in many emerging economies): CRF = 0.12 x 1.12^25 / (1.12^25 - 1) = 0.12 x 17.00 / 16.00 ~ 0.1275. Annual capital cost ~ $89.3. Per kWh: $89.3 / 1,752 ~ 5.1 cents, plus 0.5: about **5.6 cents per kWh**.

The same panels, under the same sun, produce electricity that is about **55% more expensive** simply because money costs more. The IEA's Cost of Capital Observatory finds that the cost of capital for a typical solar plant in emerging and developing economies is **two to three times** higher than in advanced economies and China, and that financing costs made up around **half** of the cost of solar electricity in emerging economies in 2021, compared with 25-30% in advanced economies and China (IEA, Cost of Capital Observatory). In 2024 the IEA's survey put the cost of capital for a 50-100 MW solar project at roughly **8.5-11% in Indonesia** (a median of 9.4%), 10-11.5% in India, 11-13.5% in South Africa and 11.5-12.5% in Brazil, against 5.0-6.5% in advanced economies (IEA, Cost of Capital Observatory dashboard and commentary, September-October 2025). Vietnam's median was 9.0% and the Philippines' 8.0%.

Why is capital more expensive? Investors fear currency depreciation (most panels are bought in dollars but power is sold in local currency), regulatory changes, utilities that pay late, political risk and weak legal enforcement. The IEA's surveys list regulatory risk as the top concern in most markets. Notice the link to Month 2: **institutions show up directly in the price of electricity.** A country that makes contracts credible and utilities solvent can lower its cost of capital, and therefore its electricity prices, without any new technology.

Nuclear, hydro and the "firm" power question

Not all clean power is variable. **Hydropower** is the largest low-carbon source globally and is flexible, but good sites are limited and big dams have serious social and environmental costs. **Nuclear** supplied about 9% of world electricity in 2025 (EI, 2026) and runs around the clock, but new large plants have often been slow and expensive to build outside China and South Korea. AI data centres, which need constant power, have revived interest from technology companies in nuclear power and in proposed small modular reactors, though few of these have yet been built at scale. **Geothermal**, important in Indonesia, Kenya and the Philippines, also provides steady power where geology allows.

In 2025-2026, US policy diverged from the global trend: the 2025 tax law kept credits for nuclear, geothermal and storage longer than for wind and solar (Month 3). Globally, though, solar plus storage is becoming the default choice for new power in most sunny countries because it is the cheapest and fastest to build.

Case study: Pakistan's solar rush

Pakistan, a country of about 240 million people, offers a dramatic example of what happens when cheap solar meets expensive grid power.

The setup. By the early 2020s Pakistan's grid electricity had become very expensive for many users. The government had signed long-term contracts with independent power producers (including Chinese-built coal plants under the China-Pakistan Economic Corridor) that required "capacity payments" whether or not the power was used. Much of its power came from imported LNG and fuel oil, exposing it to global prices, and the rupee lost a large share of its value. Electricity prices roughly doubled in three years (FT, July 2025). High-consumption households and businesses paid **10-18 US cents per kWh**, while the cost of rooftop solar was around **5 cents per kWh** (Payne Institute, August 2026).

The rush. Meanwhile, Chinese solar panels became extremely cheap, below $0.10 per watt, partly because tariffs blocked them from the US and overcapacity in China pushed exporters to find new markets. Pakistan imported about **16.6 GW** of Chinese solar panels in 2024, five times its 2022 imports (Ember data, reported by Reuters, June 2025; the IEA cites a reported 19 GW, about half the country's grid-connected capacity, so counts differ), and about 17.9 GW in its 2024-25 fiscal year; cumulative imports passed roughly **50 GW** by mid-to-late 2025, more than the entire installed capacity of the national grid (Renewables First, December 2025). In the 12 months to June 2025, Pakistan imported more solar panels than all of Africa combined, with one-sixth of Africa's population (Ember data via *Corporate Knights*, October 2025). Factories, farms, mosques and middle-class homes covered their roofs. Solar's share of Pakistan's electricity generation rose from under 2% in 2020 to about 10% in 2024 and about 24% in the first five months of 2025, becoming the largest single source for the first time (Ember, reported by AFP and *Dawn*, July 2025).

The grid problem. This was a citizen-led energy transition, largely unplanned by the government. As customers generated their own power, demand for grid electricity fell: down about 3% in 2024 even though the economy grew about 2.4% (Energy for Growth Hub, December 2025). But the government still owed fixed capacity payments to power producers. With fewer units sold, the fixed costs had to be spread over less demand, raising tariffs for those still fully on the grid (mostly poorer households who could not afford solar), which pushed more wealthy users to solar. Analysts call this a **utility death spiral**. A government report in March 2025 said the solar boom created a "disproportionate financial burden on grid consumers" (AFP, July 2025). In 2025 the government imposed a 10% sales tax on imported panels and moved to revise net-metering rules.

Lessons.

1. **Cheap solar can spread from the bottom up, fast**, when grid power is expensive and panels are cheap. No industrial policy or subsidy was needed.
2. **The value went to consumers and Chinese manufacturers, not to Pakistani industry.** Pakistan imported the panels and the batteries; local value was in trading, installation and financing. This is the smile-curve problem again: the country captured the *use* value but little of the *manufacturing* value.
3. **Grids must be redesigned, not just extended.** Old contracts and tariff structures built for central power plants break when customers become producers. Countries in Southeast Asia and Africa with weak grids should plan for this rather than be surprised by it.
4. **The energy-security payoff was real.** When the 2026 Hormuz crisis cut LNG cargoes to Pakistan, every rooftop panel reduced the country's exposure to the global gas shock.

Pakistan's story is a preview of what may happen across Africa (Nigeria imported about 1.5 GW of Chinese panels in the 12 months to June 2025, a record for the continent, according to the same Ember data) and in parts of Southeast Asia.

Key numbers: Week 4

- **$1,474 -> $108 per kWh**: Average lithium-ion battery pack price, 2010 to 2025, a 93% fall; stationary storage packs about $70/kWh (BloombergNEF, December 2025).
- **$0.043 and $0.034 per kWh**: Global weighted-average cost of electricity from new utility-scale solar and onshore wind in 2024, 41% and 53% cheaper than the cheapest fossil alternative (IRENA, July 2025).
- **91%**: Share of new utility-scale renewable projects in 2024 cheaper than the cheapest new fossil alternative (IRENA, 2025).
- **More than 600 GW**: Solar capacity added worldwide in 2025 (IEA, *Global Energy Review 2026*); 647 GW by Ember's count and about 690 GW by IEA PVPS (revised from 698 GW), but 511 GW in IRENA's alternating-current count (Ember, March 2026; IEA PVPS, September 2026; IRENA, April 2026).
- **8.7%**: Solar's share of world electricity generation in 2025, overtaking wind (EI, 2026).
- **302 GW (+66%)**: Global installed battery storage capacity at end-2025 (EI and KPMG, 2026).
- **2-3 times**: How much higher the cost of capital for solar is in emerging and developing economies than in advanced economies and China (IEA Cost of Capital Observatory).
- **8.5-11%**: Cost of capital for a 50-100 MW solar project in Indonesia in 2024 (IEA Cost of Capital Observatory dashboard).
- **16.6 GW**: Pakistan's imports of Chinese solar panels in 2024 (Ember via Reuters, June 2025).

Think like a founder

Cheap solar panels and batteries are commodities made mostly in China. Margins in panel manufacturing are thin and overcapacity is severe. A founder in Southeast Asia or Africa is unlikely to win by making panels. So where is the value?

- **Financing is the product.** Our worked example showed that the cost of capital can raise solar electricity costs by more than half. A company that can lower financing costs (by bundling many small projects, using mobile payments to cut collection risk, partnering with development banks, or offering pay-as-you-go plans) captures real value. Companies selling solar home systems on mobile-money payment plans in East Africa built large businesses this way.
- **Installation, maintenance and software.** Pakistan's solar rush created thousands of local installers and traders. The more sophisticated layer is software: monitoring systems, predicting failures, optimizing when batteries charge and discharge, and aggregating many small systems into a "virtual power plant" that can sell services to the grid.
- **Productive use.** Cheap daytime solar can power irrigation pumps, cold storage for farmers and fishers, grain mills, and small workshops. A business that sells "cold storage as a service" or "irrigation as a service" uses cheap solar as its input, capturing value from energy rather than selling energy itself. This is also a strong fit for a social-impact business model, because it raises poor farmers' incomes.
- **Energy for the next layer of the chain.** In the long run, the countries that combine cheap solar, good institutions (low cost of capital) and strong grids will be able to host energy-intensive processing (Month 6), battery manufacturing, and data centres. A founder who builds the energy infrastructure for an industrial park is positioning at the foundation of the whole ore-to-robot-to-AI chain.

Month synthesis

Energy is the base layer on which every other layer of the material economy stands. The picture in November 2026 is of two energy systems living side by side: an old system based on fossil fuels that still supplies 86% of primary energy and remains exposed to geopolitical chokepoints like Hormuz, and a new system based on electricity from solar, wind, batteries and other low-carbon sources that is growing much faster, getting cheaper every year, and is mostly manufactured in China.

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ENERGY AS THE BASE LAYER OF THE ORE-TO-ROBOT-TO-AI CHAIN

  AI models & data centres   <-- huge, constant electricity demand (Month 8)
          ^
  Chips & electronics        <-- ultra-reliable power for fabs (Month 8)
          ^
  Robots & machines          <-- motors, magnets, batteries, steel (Month 9)
          ^
  Components & cells         <-- battery cells, magnets, motors (Month 7)
          ^
  Refined metals             <-- smelting & refining: MWh per tonne (Months 5-6)
          ^
  Ore & mining               <-- diesel trucks, crushing, grinding (Month 5)
          ^
  ================== ENERGY ==================
  Old system: oil, gas, coal  -> geopolitical risk (Hormuz), price shocks
  New system: solar, wind, batteries, hydro, nuclear
              -> cheap if capital is cheap; needs grids and storage
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Where does each region stand?

Region	Energy position in 2026	Implication for the ore-to-robot chain
China	Huge coal base plus world-leading solar, wind, grids; makes most panels and batteries	Cheap reliable power underpins dominance in refining, batteries, manufacturing
United States	Energy-rich (shale oil and gas), growing solar, data-centre demand surge	Can host energy-intensive industry; policy swings create uncertainty
Gulf states	Cheap oil and gas, strong sun, but exposed to Hormuz and war	Trying to buy into metals, manufacturing and AI with energy and capital
Europe	Import-dependent for gas; high power prices	Energy-intensive industry under pressure
India	Coal-heavy but fast solar growth; strong grid expansion	Rising potential in processing and manufacturing
Southeast Asia	Mixed: coal (Indonesia, Vietnam), gas imports, strong equatorial sun	Indonesia's nickel processing runs on coal; solar plus lower cost of capital could change this

Region	Energy position in 2026	Implication for the ore-to-robot chain
Sub-Saharan Africa	About 600 million without electricity; high cost of capital; strong sun and hydro potential	Cannot host processing at scale until power and finance improve; biggest upside if they do

The key insight for your driving question: the energy advantage of the 2040s will not belong to whoever has the most oil or coal. It will belong to whoever combines **abundant sun or other clean resources, cheap capital (good institutions), strong grids and a manufacturing base that can use the power.** Sub-Saharan Africa and much of Southeast Asia have the first ingredient in abundance. The second and third are institutional and financial problems, which Months 2, 6 and 11 address. Without solving them, the middle of the ore-to-robot chain (refining, cell making, component manufacturing) will stay where power is cheap and reliable today, mostly China.

Common misconceptions

1. **"Renewables are expensive and need subsidies to compete."** This was true in 2010. In 2024, 91% of new utility-scale renewable projects were cheaper than the cheapest new fossil alternative (IRENA, 2025), and solar plus batteries now beats imported gas or diesel in many places. The real cost barriers today are the cost of capital, grid connections and storage for the hardest hours, not the panels.
2. **"Because fossil fuels supply 86% of primary energy, the energy transition is not happening."** The primary energy share overstates fossil fuels' useful contribution because about two-thirds of their energy is lost as heat. Electricity, which is growing much faster than total energy, met all of its 2025 demand growth from low-carbon sources (EI, 2026). Both statements are true at once: fossil use is still at record levels *and* the new system is growing faster. Watch the growth rates, not just the shares.
3. **"High oil prices are good for oil-exporting countries."** Only if volumes hold up. In 2026, Gulf exporters trapped behind Hormuz saw volumes collapse while prices rose. Revenue is price x volume. And even in normal booms, high prices often trigger overspending and Dutch disease that leave petrostates weaker when prices fall.
4. **"Having oil, gas or coal guarantees cheap energy for industry."** Nigeria exports crude but imported most of its fuel for decades, and many of its factories run on diesel generators. Energy resources only become cheap industrial energy when there are working refineries, power plants, grids and solvent utilities. Institutions turn resources into usable energy.
5. **"Countries with lots of sunshine will automatically have the cheapest solar power."** Sunshine matters, but so does the cost of capital. The same solar farm can produce electricity about 55% more expensively at a 12% cost of capital than at 6% (Week 4 worked example). A cloudy country with cheap finance can beat a sunny country with expensive finance.

Exercises

1. **(Quantitative: energy per person)** Using the Energy Institute figure of about 600 EJ for 2025 and a world population of 8.2 billion, verify the average of about 2,300 watts per person. Then suppose sub-Saharan Africa's 1.2 billion people (approximately) use energy at one-eighth of the world average. How many EJ would be needed to bring them to half the world average? Express your answer as a percentage of current world supply.
2. **(Quantitative: oil shock)** Short-run oil demand elasticity is -0.07. A war removes 4 mb/d of supply from a market of 104 mb/d, and emergency stock releases replace 1.5 mb/d. (a) What is the net supply loss in percent? (b) Approximately how much must the price rise to balance the market? (c) Why did actual 2026 prices rise less than a naive calculation might suggest in some months and more in others? Mention spare capacity, inventories and expectations.
3. **(Quantitative: LCOE and the cost of capital)** A 100 MW solar farm costs $65 million, lasts 25 years and has a 19% capacity factor. Operations cost $1 million a year. Calculate the cost per kWh at a cost of capital of (a) 5%, (b) 9%, (c) 13%. Then explain which policy changes in a country like Indonesia, Kenya or Nigeria could lower the cost of capital from 13% to 9%.
4. **(Quantitative: smelter economics)** An aluminium smelter uses 14 MWh per tonne. Aluminium sells for $2,600 per tonne, and non-energy costs are $1,500 per tonne. (a) What is the maximum electricity price (in cents per kWh) at which the smelter breaks even? (b) Compare this with the solar-plus-storage cost in Week 4 and the gas-power cost at $25/MMBtu. Where in the world could this smelter be profitable?
5. **(Essay, 600-800 words)** "The 2026 Iran war proved that renewable energy is a national security policy, not just a climate policy." Evaluate this statement using evidence from this chapter, including at least one counter-argument (for example, the dependence on China for panels and batteries). Give your judgment.
6. **(Essay, 600-800 words)** Compare Norway and Nigeria as petrostates. Explain, using Month 2's ideas about institutions and this month's ideas about fiscal breakevens and Dutch disease, why oil made one country richer and more resilient and the other more fragile.
7. **(Founder exercise)** Design a business that uses cheap daytime solar power in a Southeast Asian or African town to create a product or service (not just to sell electricity). Estimate the energy it needs per day, the solar and battery capacity required, the up-front cost, and how you would finance it. Who are your customers, and how do they pay?
8. **(Data exercise)** The IEA estimated data-centre electricity use at about 485 TWh in 2025, while the Energy Institute reported 788 TWh. Find out how each defines "data centre" consumption (hint: check whether they include cryptocurrency mining, telecom networks or on-site corporate servers). Write one paragraph on why definitions matter when founders and governments plan for AI demand.

Reading list

- **Vaclav Smil, *Energy and Civilization: A History* (2017).** A deep, numbers-rich history of how energy use shaped every stage of human development; the best single book for understanding energy as the base layer.

- **Daniel Yergin, *The Prize: The Epic Quest for Oil, Money, and Power* (1991).** The classic history of the oil industry, OPEC and oil geopolitics up to 1990; essential background to 1973, 1979 and today's crises.
- **Daniel Yergin, *The New Map: Energy, Climate, and the Clash of Nations* (2020).** Updates the story with shale, LNG, China and the energy transition.
- **Energy Institute, *Statistical Review of World Energy 2026* (June 2026).** The standard free dataset on global energy production, consumption and trade; learn to use its data tables.
- **IEA, *Oil Market Report* (monthly) and *World Energy Outlook 2025* (November 2025).** The IEA's monthly report is the best real-time account of the 2026 oil shock; the *Outlook* gives scenarios to 2050.
- **IEA, *Reducing the Cost of Capital* (2024) and the Cost of Capital Observatory.** Explains why clean energy is more expensive in emerging economies and what can be done; directly relevant to Africa and Southeast Asia.
- **IRENA, *Renewable Power Generation Costs in 2024* (2025).** The best source on the falling costs of solar, wind and storage, with country data.
- **Ember, *Global Electricity Review 2026* (April 2026).** Clear, data-driven analysis of how the world's electricity mix is changing, country by country.
- **IEA, *Energy and AI* (April 2025) and *Key Questions on Energy and AI* (April 2026).** The leading analysis of data-centre electricity demand; a bridge to Month 8.

Glossary terms introduced this month

- **Backwardation**: A market state where near-term prices are higher than later prices, signalling current tightness.
- **Barrel (bbl) / mb/d**: 159 litres of oil; million barrels per day.
- **Brent, WTI, Dubai, Murban**: Benchmark crude oils used to price oil worldwide.
- **Capacity factor**: Actual output of a power plant divided by its maximum possible output over a period.
- **Capital recovery factor (CRF)**: A formula that converts an up-front investment into an equal annual payment over a project's life at a given interest rate.
- **Cost of capital**: The return investors require to finance a project; for capital-intensive projects like solar it largely determines the cost of electricity.
- **Dutch disease**: The weakening of manufacturing and other exports when a resource boom raises a country's currency value and wages.
- **Energy intensity**: Energy used per unit of GDP.
- **Exajoule (EJ)**: 10^18 joules; the unit for national and global energy totals.
- **Fiscal breakeven oil price**: The oil price at which a petrostate's government budget balances.
- **Force majeure**: A contract clause excusing a party from obligations when extraordinary events beyond its control prevent performance.
- **Learning curve / Wright's law**: The pattern that unit costs fall by a roughly constant percentage each time cumulative production doubles.
- **Levelized cost of electricity (LCOE)**: The average cost per kWh of building and operating a power plant over its lifetime.
- **Liquefied natural gas (LNG)**: Natural gas cooled to about -162 degrees Celsius so it can be shipped by tanker.
- **Merit order**: The ranking of power plants by marginal cost, used to decide which run first.

- **OPEC / OPEC+**: The Organization of the Petroleum Exporting Countries, and its wider alliance with Russia and other producers since 2016.
- **Petrostate**: A state whose government and economy depend heavily on oil and gas revenues.
- **Price cannibalization**: The fall in the market value of solar (or wind) power as more of it produces at the same time.
- **Primary energy / total energy supply (TES)**: Energy in its original form (coal, oil, sunlight converted to electricity) before conversion losses.
- **Spare capacity**: Unused oil production capacity that can be brought online within about 90 days.
- **Strategic Petroleum Reserve / emergency stocks**: Government-held oil stocks that can be released in a supply crisis.
- **Utility death spiral**: A cycle in which customers leaving the grid raise costs for those who remain, pushing more to leave.
- **Virtual power plant**: Software that coordinates many small energy devices (batteries, chargers, loads) to act like one power plant.

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Month 5: Critical Minerals from Geology to Market

Month overview

Last month you followed energy, the base layer of every economy. This month we go one layer up the stack, into the rocks. Every machine you care about, from an electric car to a data center to a humanoid robot, is built from a small set of metals that are dug out of the ground in a few places, purified in even fewer places, and priced in markets that most people never see. Copper carries electricity. Lithium, nickel, cobalt, manganese and graphite store it in batteries. Rare earth elements such as neodymium and dysprosium make the powerful permanent magnets inside electric motors, wind turbines and robot joints. Platinum-group metals clean exhaust gases and catalyze chemical reactions, including in hydrogen fuel cells and electrolyzers.

The big idea of this month is a three-way distinction that most news coverage blurs: **reserves** (what is in the ground and economic to mine), **mine production** (what is actually dug up each year), and **refining** (turning ore into a pure chemical or metal that a factory can use). These three layers sit in different countries. The DRC (Democratic Republic of the Congo) mines about 73% of the world's cobalt, but China refines most of it. Chile holds the largest lithium reserves, but Australia mines the most and China turns most of it into battery chemicals. Indonesia now mines about two-thirds of the world's nickel and, unusually, refines most of it at home as well. Whoever controls the narrowest layer, the chokepoint, has power over the whole chain.

The second idea is **price formation**. Minerals are priced on exchanges such as the London Metal Exchange (LME), through negotiated benchmarks, and through private contracts. They swing through violent boom-and-bust cycles because supply takes a decade to respond to demand. Understanding those cycles is the difference between a founder who is crushed by a price crash and one who buys assets cheaply at the bottom.

Month 6 asks the follow-up question: if refining is where much of the leverage lies, can a mining country move into processing (often called downstreaming), and when does that work? Month 9 will take these same metals into the bill of materials of a robot. Keep the driving question in front of you all month: from ore to robot to artificial intelligence (AI), who captures the value, and why?

Learning goals

By the end of this month you should be able to:

- Explain the difference between resources, reserves, mine production and refined production,

and why a country can be rich in one and weak in another.
- Describe where the eight key minerals (copper, lithium, nickel, cobalt, rare earths, graphite, manganese and platinum-group metals) are mined and refined, using 2025 data from the USGS (United States Geological Survey) and the IEA (International Energy Agency).
- Explain the chemistry-to-economics link: why a shift in battery chemistry (from nickel-rich cells to lithium iron phosphate) moves billions of dollars between countries.
- Explain why China dominates mineral refining, using the concepts of scale, learning curves, cost of capital, by-product recovery and state strategy.
- Describe how mineral prices are formed (exchanges, benchmarks, contracts, treatment charges) and why they move in long cycles.
- Do simple calculations: ore grade to tonnage, lithium carbonate equivalent, cost-curve price setting, and the share of a mineral in a final product's cost.
- Judge where in the chain a founder or a developing country could realistically capture value.

Week 1: From rocks to reserves: the geology and economics of supply

Core lesson

Abundance is not the problem; concentration is. Most "critical" elements are not rare in the Earth's crust. Copper is about as common as zinc. Rare earth elements, despite their name, are more abundant in the crust than gold or silver; the USGS notes they are "relatively abundant in the Earth's crust, but minable concentrations are less common than for most other mineral commodities" (USGS MCS 2026, rare earths). The economic problem is that a mineral is only useful when nature has concentrated it into an **ore deposit**, a body of rock rich enough that mining it earns a profit. Geology creates these concentrations through specific processes: hot fluids from magma (porphyry copper deposits in the Andes), weathering in tropical climates that leaves nickel behind in red soils (laterite nickel in Indonesia and the Philippines), evaporation of salty lakes in deserts (lithium brines in the Atacama), or ancient sediments (the sediment-hosted copper-cobalt belt of the DRC and Zambia). Because these processes happened in particular places, deposits cluster. That clustering is the root of geopolitics in minerals.

Grade is the single most important number in mining. The grade is the share of the useful metal in the ore. A typical large copper mine today processes ore of well under 1% copper, often around half a percent. Work through what this means:

- Suppose ore grade is 0.6% copper and the processing plant recovers 88% of it.
- One tonne of ore contains 6 kg of copper, of which 5.28 kg is recovered.
- To produce 1 tonne of copper you must mine and grind about 1,000 / 5.28 = 189 tonnes of ore, plus the waste rock you strip away to reach it, often two or three times as much again.

So a mine producing 300,000 tonnes of copper a year moves well over 100 million tonnes of rock. That is why mining is so energy-hungry and capital-heavy, and why grade decline matters so much: if grade falls from 0.6% to 0.5%, you must move 20% more rock for the same copper. The USGS reports that in 2025 US copper output was hit by "concentrator shutdowns and lower ore grades at multiple mines" (USGS MCS 2026, copper). Falling grades in old districts such as Chile are one reason why the industry's future supply is shifting toward newer, higher-grade deposits such as Kamoa-Kakula in the DRC.

Resources versus reserves. Geologists use two words that newspapers mix up:

- A **resource** is a concentration of a mineral that could plausibly be mined someday. The USGS estimates identified copper resources at about 1.5 billion tonnes and undiscovered copper at about 3.5 billion tonnes (USGS MCS 2026, citing a 2015-based assessment).
- A **reserve** is the part of a resource that has been measured well enough and is economic to extract today, with today's prices, technology and permits. World copper reserves are about 980 million tonnes (USGS MCS 2026).

The key mechanism: **reserves are an economic concept, not a geological one.** If prices rise or technology improves, resources become reserves. If prices fall, reserves shrink. This is why "we will run out of copper in 40 years" arguments are misleading: 980 million tonnes of reserves divided by about 23 million tonnes of annual mine output gives roughly 43 years, but this "reserve life" has stayed in a similar range for decades because exploration and higher prices keep converting resources into reserves. The real constraints are time, cost, permits, water, energy and social licence, not the absolute quantity in the crust.

Production versus refining. Mining produces a concentrate (for copper, typically 25-30% copper), an intermediate (such as cobalt hydroxide or nickel mixed hydroxide precipitate), or a mineral (spodumene for lithium). That product must then be smelted or chemically refined into a pure metal or chemical, such as copper cathode (99.99% copper), lithium hydroxide, nickel sulphate, or separated neodymium oxide. Refining requires large plants, cheap energy and acid, skilled chemical engineers, and buyers nearby. The world map of refining is therefore very different from the map of mines.

The big map, 2025. The table below uses the USGS Mineral Commodity Summaries 2026 (published February 2026, reporting estimated 2025 data). Shares are my calculations from the USGS tables.

Mineral	World mine output 2025	Top miners (share)	Largest reserves	Who refines most
Copper	23 Mt	Chile 23%, DRC 14%, Peru 12%	Chile 180 Mt of 980 Mt	China (14 of 29 Mt refined, 48%)
Lithium	290 kt (excl. US)	Australia 32%, China 21%, Chile 19%, Zimbabwe 10%	Chile 9.2 Mt of 37 Mt	China (most chemicals)
Nickel	3.9 Mt	Indonesia 67%	Indonesia 62 Mt of >140 Mt	Indonesia and China
Cobalt	310 kt	DRC 73%, Indonesia 14%	DRC 6.0 Mt of 12 Mt	China
Rare earths (oxide)	390 kt	China 69%, US 13%, Australia 7%	China 44 Mt of >85 Mt	China (separation and magnets)
Natural graphite	1.8 Mt	China 82%	China 100 Mt of 310 Mt	China (battery anodes)
Manganese (content)	20 Mt	South Africa 38%, Gabon 25%	Australia, South Africa	China (high-purity manganese)

Mineral	World mine output 2025	Top miners (share)	Largest reserves	Who refines most
Platinum	170 t	South Africa 71%, Russia 12%	South Africa (PGM reserves 63,000 t)	South Africa, Russia, others
Palladium	190 t	Russia 44%, South Africa 37%	as above	as above

(Mt = million tonnes; kt = thousand tonnes; t = tonnes. Sources: USGS MCS 2026 chapters for each mineral.)

Notice three patterns. First, **mining is somewhat spread out, refining is not**. The IEA's Global Critical Minerals Outlook 2026 (July 2026) found that, excluding rare earths, the average market share of the top refining country across key energy minerals rose to 72% in 2025, up from 70% in 2023, and that "the top refining countries, Indonesia for nickel and China for other key energy minerals, accounted for over three quarters of total growth in refined supply" over the past two years. Second, **a few countries matter for one mineral each**: the DRC for cobalt, Indonesia for nickel, South Africa for platinum and manganese, Chile for copper and lithium. Third, **China appears everywhere at the refining stage**, even where it mines little (cobalt, where it mined about 2,000 tonnes in 2025 versus the DRC's 230,000).

Why "critical"? A mineral is called critical when two things are true at once: it is economically important (hard to substitute, used in important industries), and its supply is at risk (concentrated, politically exposed, or a by-product). The US government's 2025 critical minerals list, finalized in November 2025, added copper, silver, silicon, potash, lead, rhenium, and, after public comment, metallurgical coal, phosphate rock, boron and uranium (USGS MCS 2026, copper and phosphate chapters). The IEA's 2026 risk assessment ranks gallium, magnet rare earths, yttrium, graphite, tungsten, tellurium, cobalt and germanium as most exposed. "Critical" is therefore a political and economic label, and it changes as technology and geopolitics change.

By-products: the hidden trap. Many small but vital metals are not mined for themselves. Cobalt mostly comes as a by-product of copper (in the DRC) and nickel (in Indonesia). Gallium comes from aluminium refining, germanium from zinc processing, tellurium from copper refining. This creates a strange economics: the supply of a by-product barely responds to its own price, because no one opens a copper mine to get cobalt. The IEA calls modern copper and zinc smelters "strategic processing hubs" for exactly this reason: whoever runs the smelters gets the by-products. We will see in Week 3 why this helps explain China's dominance in gallium and germanium.

Time is the hidden variable. From discovery to production, a new mine commonly takes more than a decade, and often 15 years or more, because of exploration, feasibility studies, financing, environmental permits, community agreements and construction. A refinery is faster to build, typically a few years, but needs feedstock contracts and skilled operators. So when demand jumps, supply cannot follow quickly, prices spike, and then the delayed supply arrives all at once, often after demand has cooled. This lag drives the cycles in Week 4.

Case study: Copper, from Chile's old giants to the DRC's new frontier

Copper is the metal of electrification. Every electric motor, transformer, cable and data center depends on it. The IEA calls copper the market where "the adequacy of supply remains a major

concern" and projects a supply deficit of about 25% by 2035 based on the current project pipeline, improved from about 30% in its 2025 Outlook "as new projects advance, particularly in the DRC and Zambia" (IEA GCMO 2026).

Chile remains the largest producer: 5.3 million tonnes in 2025, down from 5.5 million in 2024, with reserves of 180 million tonnes (USGS MCS 2026). Chile's copper industry grew from the 1990s onward through large open-pit porphyry mines run by both private multinationals and the state company Codelco, which was created after copper nationalization in 1971. But Chile's giant mines are aging. Ore grades have fallen over decades, so each tonne of copper requires more rock, more water (in the world's driest desert, which has pushed miners toward desalinated seawater pumped uphill), and more electricity. Chile also refines a declining share of its own copper: refinery output was about 1.7 million tonnes in 2025 against 5.3 million tonnes of mine output. Most Chilean copper leaves as concentrate, much of it for Chinese smelters.

The DRC is the rising force. Its mine output rose to about 3.2 million tonnes in 2025, from 2.99 million in 2024, overtaking Peru as the second-largest producer (USGS MCS 2026). The Central African Copperbelt, running through the DRC's Katanga region and Zambia, holds sediment-hosted deposits that are often much higher grade than Chilean porphyries. The flagship new mine, Kamoa-Kakula (operated by Ivanhoe Mines with Zijin Mining and the DRC government as partners), is widely described as one of the highest-grade large copper mines in the world. Unusually, the DRC also refines a large share of its own copper into cathode at the mine site using solvent extraction and electrowinning (a chemical method that uses electricity to plate pure copper): USGS shows DRC refinery output of 2.8 million tonnes in 2025, close to its mine output. In 2026 refined copper anode exports from Kamoa-Kakula began moving via the Lobito Corridor rail route to the Atlantic (Euromining, June 2026), which Month 6 examines.

What does this case teach?

1. **Geology sets the starting hand, but institutions decide whether it is played.** The DRC's Katanga deposits were known for a century, through Belgian colonial extraction, the state miner Gecamines' collapse in the 1990s, and war. Production only took off after a 2002 mining code, new foreign investment (much of it Chinese after 2008) and higher prices. Month 2's lessons on institutions apply directly.
2. **High grade beats everything.** A mine with 3% copper ore needs about one-fifth of the rock of a 0.6% mine for the same copper. Lower cost means it survives price crashes, which in turn makes it easier to finance.
3. **Infrastructure is the binding constraint.** The DRC's copper is about 1,300-2,000 km from the nearest ports (Durban, Dar es Salaam, Lobito). Trucks and old railways add hundreds of dollars per tonne. Power shortages limit refining. These are exactly the bottlenecks Month 6 studies.

Key numbers

- World copper mine production: 23 million tonnes in 2025 (USGS MCS 2026).
- World refined copper production: 29 million tonnes in 2025, of which China 14 million tonnes, about 48% (USGS MCS 2026). (Refined output exceeds mine output because it includes refined scrap.)
- World copper reserves: 980 million tonnes; Chile 180 Mt, Australia 100 Mt, Peru 85 Mt, DRC 80 Mt, Russia 80 Mt (USGS MCS 2026).

- DRC copper mine output: 3.2 million tonnes in 2025, up from 2.99 Mt in 2024 (USGS MCS 2026).
- Average top-refiner market share across key energy minerals (excluding rare earths): 72% in 2025, up from 70% in 2023 (IEA GCMO 2026).
- IEA projected copper supply gap in 2035: about 25% (IEA GCMO 2026).
- COMEX (New York) copper price average 2025: a record $4.80 per pound, 14% above 2024 (USGS MCS 2026).

Think like a founder

The mine itself is mostly a game for giant companies and states: it needs billions of dollars and a decade. But mining creates many smaller, high-margin niches around it. Three examples:

- **Exploration and data.** The IEA reports that venture capital investment in minerals recovered in 2025, "increasingly targeting artificial intelligence (AI) technologies that could improve the efficiency of mineral exploration and resource extraction" (IEA GCMO 2026). Finding deposits faster, using machine learning on geophysical and satellite data, compresses the most expensive part of the timeline. A company that cuts exploration cost captures value without owning a mine.
- **Recovery and processing technology.** A mine processing 100 million tonnes a year of 0.6% ore holds 600,000 tonnes of copper in its feed. If you lift recovery from 85% to 90%, you add 5% of 600,000 = 30,000 tonnes of copper a year, worth roughly $300-400 million at 2025-2026 prices of $10,000-13,000 per tonne. Even if a technology captures only a tenth of that as fees, it is a large business. Mining companies pay for technology that does this.
- **Services in frontier regions.** Logistics, power, maintenance, environmental monitoring and training in places like the Copperbelt are scarce. Scarcity is margin.

Value-capture rule: in mining, profit goes to whoever owns the lowest-cost deposit or whoever solves the bottleneck (water, power, transport, permitting) that everyone else is stuck on.

Week 2: The battery metals: lithium, nickel, cobalt, graphite and manganese

Core lesson

How a lithium-ion battery works, in one paragraph. A battery cell has two electrodes. The **cathode** (positive side) is a metal oxide or phosphate that holds lithium ions; the **anode** (negative side) is usually graphite. When you charge the battery, lithium ions move from the cathode through a liquid electrolyte into the graphite anode; when you use the battery, they move back and release energy. The cathode chemistry decides most of the battery's cost, energy density and which metals it needs. Two families dominate:

- **NMC** (nickel manganese cobalt oxide) and its relative **NCA** (nickel cobalt aluminium): high energy density, so longer driving range per kilogram, but they need nickel and cobalt, which are expensive and politically exposed.
- **LFP** (lithium iron phosphate): lower energy density but cheaper, safer, longer-lasting, and made from iron and phosphate, which are abundant. It needs no nickel or cobalt.

Chemistry is economics. Over the last few years, Chinese battery makers such as CATL and BYD improved LFP so much (for example, through cell-to-pack designs that remove packaging

weight) that it took a large share of the electric vehicle (EV) market and dominates stationary energy storage. The consequences rippled across countries. Demand growth for nickel and cobalt slowed relative to forecasts, contributing to the nickel price collapse and the cobalt glut of 2023-2024. Demand for phosphate and lithium rose. Morocco, with about 68% of the world's phosphate rock reserves (50 of 73 billion tonnes, USGS MCS 2026), suddenly became relevant to batteries. This is the most important lesson of this week: **a decision made in a battery laboratory in China can move billions of dollars of value between Indonesia, the DRC and Morocco.**

Global battery demand grew by over 35% in 2025, surpassing 1.5 terawatt-hours (TWh), with battery storage emerging as a major driver; demand for key energy minerals has grown at close to 10% per year in recent years, against about 1% for base metals generally (IEA GCMO 2026, market overview). Lithium demand grew about 25% per year over the past two years.

Lithium: two kinds of deposit, two kinds of economics. Lithium comes from:

1. **Hard rock (spodumene)**, mined mainly in Western Australia, and increasingly in Zimbabwe, Brazil, Mali and China. The mine produces a concentrate (called SC6, about 6% lithium oxide), which is shipped, mostly to China, and converted into lithium hydroxide or carbonate in chemical plants. Mining is relatively fast to start but the conversion step is energy- and acid-intensive.
2. **Brines** in the "lithium triangle" of Chile, Argentina and Bolivia, and in China. Salty groundwater is pumped into huge evaporation ponds, concentrated by the sun over 12-18 months, then processed into lithium carbonate. It is cheap to operate but slow, uses a lot of water in arid areas, and can affect local and Indigenous communities. Newer **direct lithium extraction** (DLE) technologies try to pull lithium out of brine chemically without ponds.

Production in 2025 was about 290,000 tonnes of lithium content outside the US, up 31% from 222,000 tonnes in 2024. Australia led with 92,000 tonnes, then China 62,000, Chile 56,000, Zimbabwe 28,000, Argentina 23,000, Brazil 12,000 and Mali 9,400. Reserves total 37 million tonnes, led by Chile (9.2 Mt) and Australia (8.4 Mt) (USGS MCS 2026). Measured and indicated resources are much larger, about 150 million tonnes, with Argentina (28 Mt) and Bolivia (23 Mt) on top. Bolivia is the classic example of a huge resource that is barely produced: it appears in the resource list but not in the production table.

A unit you must learn: LCE. Lithium markets often talk in tonnes of **lithium carbonate equivalent** (LCE). Lithium carbonate (Li2CO3) is about 18.8% lithium by mass, so 1 tonne of lithium metal content equals about 5.32 tonnes of LCE. The USGS figure of 290,000 tonnes of lithium therefore equals roughly 1.5 million tonnes of LCE. Worked example for spodumene: SC6 contains 6% lithium oxide (Li2O), which is about 46% lithium, so one tonne of SC6 holds about 28 kg of lithium, or about 148 kg of LCE. At 100% recovery you would need about 6.7 tonnes of SC6 per tonne of LCE; with realistic conversion losses, roughly 7.5-8 tonnes. At the USGS-reported Australian spodumene price of about $970 per tonne in November 2025, 7.5 tonnes of concentrate cost about $7,300, against a Chinese lithium carbonate spot price of about $10,300 per tonne in the same month (USGS MCS 2026, lithium). The gap, about $3,000 per tonne, has to pay for conversion chemicals, energy, labour, capital and profit. When the gap is thin, converters in China squeeze miners, or stop buying. This tiny worked example is the whole economics of "who captures value" in lithium.

Nickel: two products from the same metal. About two-thirds of nickel goes into stainless steel; a growing share goes into batteries. These need different products:

- **Class 2 nickel**: lower purity, such as nickel pig iron (NPI) and ferronickel, used for stainless steel.
- **Class 1 nickel**: at least 99.8% purity metal, or battery-grade nickel sulphate, needed for battery cathodes and the only grade deliverable against the LME contract.

Indonesia's laterite ores were long seen as suited mainly to Class 2. Chinese firms, led by Tsingshan, industrialized two routes in Indonesia: **RKEF** (rotary kiln electric furnace) smelting to make NPI cheaply for stainless steel, and **HPAL** (high-pressure acid leaching), which dissolves low-grade limonite ore in sulphuric acid at high temperature and pressure to produce **MHP** (mixed hydroxide precipitate), a battery-grade intermediate containing nickel and cobalt. Nickel matte, made by converting NPI, provided another bridge to battery grade. The result: Indonesia's mine output rose to an estimated 2.6 million tonnes in 2025, about 67% of the world's 3.9 million tonnes, with 62 million tonnes of reserves (USGS MCS 2026). Australia's production fell 54% in 2025 "after multiple companies placed mines into care-and-maintenance status owing to low prices" (USGS MCS 2026). The International Nickel Study Group, cited by the USGS, estimated the primary nickel market in surplus every year since 2022, with a surplus of 182,000 tonnes in 2024 and 189,000 tonnes in the first nine months of 2025. Month 6 tells this story in full.

Cobalt: a by-product with a monopoly producer. Cobalt stabilizes NMC cathodes and is essential in superalloys for jet engines. The DRC mined about 230,000 tonnes in 2025, 73% of the world total, and Indonesia (as a by-product of HPAL nickel) about 44,000 tonnes, 14% (USGS MCS 2026). Most DRC cobalt is a by-product of copper mining, often in the form of cobalt hydroxide shipped to China for refining. There is also a significant artisanal (small-scale, hand-dug) sector in the DRC, with well-documented problems of unsafe work and child labour, which led to supply-chain due-diligence rules and campaigns by buyers. Because cobalt supply depends on copper and nickel decisions, it ran into a glut in 2024 as Indonesian and DRC output both rose. In February 2025 the DRC banned cobalt exports to support prices, and from October 2025 replaced the ban with quotas: 18,125 tonnes for the rest of 2025, and up to 96,600 tonnes per year in 2026 and 2027, including 9,600 tonnes for a strategic reserve (USGS MCS 2026). The IEA reports cobalt prices rose by about 130% between January 2025 and April 2026, "largely due to export restrictions" by the DRC (IEA GCMO 2026). This is a rare example of a producer country successfully using market power; Month 6 asks whether it can last.

Graphite: the anode material nobody talks about. The anode in almost every lithium-ion battery is graphite, either natural (mined) or synthetic (made from petroleum coke or needle coke by heating it to about 3,000 degrees Celsius, an extremely energy-intensive step). China produced about 1.4 million tonnes of the world's 1.8 million tonnes of natural graphite in 2025, about 82%, and dominates the conversion of graphite into **spherical purified graphite** and finished anode material (USGS MCS 2026). New mines are rising in Africa: Tanzania more than doubled output to 75,000 tonnes in 2025, Mozambique produced about 60,000 tonnes and Madagascar about 80,000 tonnes. The USGS notes that the increase in Chinese spherical graphite exports to Indonesia was "likely owing to a Chinese company's new SPG facility that began production in Central Java, Indonesia," showing how Chinese firms are relocating mid-stream steps to Southeast Asia, partly to avoid US tariffs. In 2025 the US Department of Commerce set preliminary anti-dumping duties of 93.5% on Chinese anode material, plus countervailing duties ranging from 11.58% to 721.03% (USGS MCS 2026, graphite).

Manganese: cheap and plentiful, but the high-purity step is concentrated. About 90% or more of manganese goes into steel, so its demand "closely follows the steel industry" (USGS MCS 2026). World production was about 20 million tonnes of manganese content in 2025, led by

South Africa (7.6 Mt), Gabon (5.0 Mt), Ghana (2.0 Mt) and Australia (1.6 Mt). Batteries need **high-purity manganese sulphate**, produced overwhelmingly in China, so even an abundant, Africa-mined metal passes through a Chinese chemical bottleneck before it reaches a cell. The IEA notes manganese as one of the markets where "virtually all supply growth came from the dominant supplier" between 2023 and 2025.

Where is the money in a battery? The IEA gives two numbers worth memorizing: critical minerals account for around one-quarter of battery cell costs but only about 3% of the price of an average EV, and a tripling of battery material prices would increase the final price of EVs and storage systems by around 5% (IEA GCMO 2026). In other words, the mine and the refinery capture a modest slice of the final product's value. The cell maker, the car company and the software and brand on top capture more. But the minerals are a **chokepoint**: if you cannot get graphite anode material, the other 97% of the car cannot be built. The IEA estimates that if battery-grade graphite trade were fully disrupted, over $300 billion per year of downstream production outside China would be at risk. Small value, huge leverage: that is the paradox of critical minerals.

Worked example: the minerals in one kilowatt-hour. The IEA's ratios become much more useful when you can rebuild them yourself from chemistry and prices. Here is how, step by step, for the two main cathode families. The chemistry is standard; the "practical" adjustments are rough engineering assumptions, so treat the results as estimates good to perhaps plus or minus 20%.

Step 1: how much cathode material stores 1 kWh? Energy is charge times voltage. A gram of LFP cathode material delivers roughly 155-160 milliamp-hours of charge in a real cell at an average of about 3.2 volts, so it stores about 0.16 amp-hours x 3.2 volts = about 0.5 watt-hours. To store 1,000 watt-hours (1 kWh) you need about 2 kg of LFP. A gram of NMC 811 (8 parts nickel, 1 manganese, 1 cobalt) delivers about 200 milliamp-hours at about 3.7 volts, or about 0.74 watt-hours, so you need about 1.35 kg per kWh. NMC stores more energy per kilogram, which is exactly why it gives longer driving range.

Step 2: how much of each metal is in that cathode? Use the molar masses from the periodic table.

- LFP is LiFePO4, molar mass about 157.8. Lithium (6.94) is 4.4% of the mass and iron (55.85) is 35%. So 2 kg of LFP holds about 0.09 kg of lithium and 0.7 kg of iron.
- NMC 811 is LiNi0.8Mn0.1Co0.1O2, molar mass about 97.3. Lithium is 7.1% of the mass, nickel 48%, cobalt 6% and manganese 5.6%. So 1.35 kg holds about 0.1 kg of lithium, 0.65 kg of nickel, 0.08 kg of cobalt and 0.08 kg of manganese.

Real cells need somewhat more (process losses, excess lithium, inactive material), so round up: about 0.1 kg of lithium per kWh for either chemistry (about 0.55 kg of lithium carbonate equivalent, using the 5.32 conversion factor), plus about 0.7 kg of nickel and 0.09 kg of cobalt for NMC 811. Both chemistries also need roughly 1 kg of graphite in the anode per kWh (an approximate industry figure), plus copper and aluminium foils.

Step 3: what is it worth? Use rough late-2026 prices: LME nickel about $15.50 per kg (Week 4), the DRC's cobalt export reference price about $38.60 per kg in late October 2026 (Month 6), and Chinese lithium carbonate at roughly $16-17 per kg (Shanghai Metals Market, SMM, quoted battery-grade lithium carbonate at CNY 118,800-126,800 per tonne at the end of September 2026). Lithium moves fast: SMM's spot price hit a 2026 high of CNY 200,500 on May 13, averaged CNY 147,500 in the third quarter, and had fallen to about CNY 90,000 (about $12.60 per kg) by October 30, 2026. At the late-October price the lithium line in the table below falls from about $9 to about $7 per kWh.

Per kWh of cells	LFP	NMC 811
Lithium (about 0.55 kg LCE at about $16/kg)	about $9	about $9
Nickel (0.7 kg at $15.50/kg)	0	about $11
Cobalt (0.09 kg at $38.60/kg)	0	about $3.50
Iron, phosphate, manganese	about $1	under $1
Total cathode metal value	about $10	about $24
Average pack price, 2025 (BloombergNEF)	$81	$128
Cathode metals as share of pack price	about 12%	about 19%

Check this against the IEA's figure that critical minerals are around one-quarter of cell costs. Our numbers are a little lower because they count only the metal content at metal prices, not the higher price of battery-grade chemicals (lithium hydroxide, nickel sulphate) or the graphite, and because a pack costs more than the cells inside it (BloombergNEF's average 2025 cell price was $74 per kWh against $108 for the average pack). The two methods agree on the big picture: minerals are a minority of the cost, and the battery maker's factory, energy, labour, research and margin are the majority.

Step 4: the chemistry shift as a transfer between countries. Now compare the two columns. Switching 1 kWh from NMC 811 to LFP removes about $11 of nickel demand (mostly mined and refined in Indonesia) and about $3.50 of cobalt demand (mostly mined in the DRC and refined in China), while lithium demand barely changes and iron and phosphate demand rises by a dollar's worth. Scale it up. The IEA reports global battery demand above 1.5 TWh in 2025, which is 1.5 billion kWh. If 10% of that demand shifts from NMC to LFP, that is 150 million kWh x about $14 (the nickel and cobalt lost, minus the small gain in iron and phosphate) = roughly $2.1 billion a year of mineral demand removed from nickel and cobalt producers. Every further 10 percentage points moves roughly another $2 billion. That is the arithmetic behind the claim at the start of this week that a laboratory decision in China moves billions of dollars between Indonesia, the DRC and Morocco.

Step 5: who captures the battery's value? Of an $81 LFP pack, roughly $10 is cathode metal. Of that $10, the lithium share is split again between miner and converter: the November 2025 example earlier in this week showed spodumene feedstock worth about $7,300 of the $10,300 price of a tonne of lithium carbonate, so the miner got about 70% and the converter about 30% at that moment, and the split swings with the cycle. The remaining $70 or so of the pack goes to cathode and anode makers, the cell maker, the pack assembler and their suppliers of equipment and energy. In 2025 the average pack in China cost $84 per kWh, while packs in North America and Europe cost 44% and 56% more (BloombergNEF), which tells you that the factory and its ecosystem, not the minerals, decide who is competitive.

Link to robots and AI. A humanoid robot carries a battery of roughly 2 kWh (Month 6 gives the evidence). On the numbers above, that is about $20 to $50 of cathode metals inside a pack worth roughly $160 to $260, in a robot that may sell for tens of thousands of dollars. A data center's battery backup and a grid storage system follow the same arithmetic at a much larger scale; BloombergNEF found stationary storage packs were the cheapest segment in 2025, at $70 per kWh. The lesson for the driving question is the same at every scale: the minerals are a small, volatile slice of the value, the factory captures a larger and more stable slice, and the brand, software and integration on top capture the most.

Case study: The lithium boom and bust, 2021-2026

Lithium is the best recent example of how a commodity cycle works in real life.

The boom (2021-2022). EV sales in China and Europe surged after 2020. Lithium supply was small, and most new projects were years away. Chinese lithium carbonate prices rose roughly tenfold from early 2021. At the peak in November 2022, Fastmarkets assessed Chinese battery-grade lithium carbonate spot at a record CNY 597,500 per tonne, which Reuters converted at about $83,000-88,000 per tonne depending on the date's exchange rate (Reuters, citing Fastmarkets, December 2022 and January 2023). On an annual-average basis the peak was lower: about $68,100 per tonne in 2022 for the series the USGS uses (as compiled from USGS data by a price-history site; I did not open the USGS 2024 chapter itself). The difference between the two numbers is itself a lesson: a spot peak lasts weeks, while contracts and annual averages smooth it. Mining companies, investors and governments rushed in. Australian spodumene producers earned extraordinary margins. Zimbabwe, Mali, Brazil and Nigeria attracted Chinese buyers who paid upfront for deposits. Junior explorers raised money on almost any lithium claim.

The supply response (2023-2025). High prices did exactly what economics predicts: they called forth supply from everywhere at once. Chinese lepidolite mines in Jiangxi, African spodumene (Zimbabwe went from almost nothing to 28,000 tonnes of lithium in 2025, Mali to 9,400 tonnes), Argentine brines (23,000 tonnes) and expanded Australian capacity all came online. Meanwhile EV demand growth slowed in some markets. Prices collapsed. By 2025 the annual average US contract price for lithium carbonate was $9,000 per tonne, down 31% from 2024, and Chinese spot carbonate traded around $9,300-10,300 per tonne through most of the year (USGS MCS 2026). High-cost mines in Australia and elsewhere were closed or delayed. The IEA reports that lithium companies cut investment by around 40% in 2025, and lithium exploration spending fell by about 40-45% (IEA GCMO 2026).

The rebound (late 2025-2026). Then demand surprised on the upside, especially from **battery energy storage systems** (giant batteries attached to solar farms and grids, and increasingly to data centers). The USGS notes that "considerable EV sales growth in China and Europe and increased demand for battery energy storage systems contributed to lithium price increases during the second half of 2025." The IEA reports lithium prices "more than doubled" between January 2025 and April 2026 "amid strong demand from energy storage applications and constrained supply." Supply was constrained partly because projects had been cancelled during the bust, and partly because of policy shocks: Zimbabwe suspended concentrate exports in February 2026 and set a ban on concentrate exports from January 2027.

Lessons.

1. **The cure for high prices is high prices, and vice versa.** High prices create supply; the supply crushes prices; low prices kill investment; the missing investment creates the next shortage.
2. **The low-cost producer survives.** Brine producers in Chile and the best Australian mines kept going through the bust. High-cost lepidolite and marginal spodumene mines did not.
3. **Integration protects margins.** Chinese firms that owned mines, converters and cathode plants could move profit to whichever stage was strong. Pure miners could not.
4. **Demand forecasts are fragile.** Analysts in 2022 projected shortages; by 2024 they projected gluts; by 2026 storage demand had changed the picture again. A founder should plan for a range of prices, not a forecast.

Key numbers

- World lithium mine production (excluding the US): about 290,000 t in 2025, up 31% from 222,000 t in 2024 (USGS MCS 2026).
- World lithium reserves: 37 Mt; resources about 150 Mt (USGS MCS 2026).
- US annual average lithium carbonate contract price: $9,000/t in 2025, down 31% from 2024 (USGS MCS 2026).
- Global battery demand: over 1.5 TWh in 2025, up over 35% (IEA GCMO 2026).
- Critical minerals as share of EV price: about 3%; as share of battery cell cost: about one-quarter (IEA GCMO 2026).
- Indonesia share of world nickel mine output: about 67% (2.6 of 3.9 Mt) in 2025 (USGS MCS 2026).
- DRC share of world cobalt mine output: about 73% in 2025 (USGS MCS 2026).
- China share of world natural graphite production: about 82% in 2025 (USGS MCS 2026).
- Cobalt price change, January 2025 to April 2026: about +130% (IEA GCMO 2026).

Think like a founder

Battery chemistry is where a small team with deep science can change the economics of a whole industry. LFP's rise came from engineering improvements, not new geology. Three places a founder could fit:

- **Chemistry-agnostic businesses.** Build where you win whichever chemistry wins: battery management software, battery recycling (the IEA expects average recycling rates across key energy minerals to rise from about 10% today to close to 20% by 2040), testing, and second-life storage.
- **Mid-stream materials outside China.** Governments in the US, Europe, Japan and Korea are paying a "security premium" for non-Chinese anode, cathode and lithium chemicals (the IEA estimates operating costs are on average around 50% higher outside the dominant supplier). Where there is a subsidy and a guaranteed buyer, a new entrant can survive the first years.
- **Storage integration in emerging markets.** Cheap LFP cells plus solar make reliable power possible where grids are weak. A company that packages, finances and services storage for factories, farms or telecom towers in Africa or Southeast Asia captures value that the cell makers do not.

Watch out: do not build a business whose survival depends on one mineral price staying high. That is a bet, not a business.

Week 3: Magnets, catalysts and the midstream: rare earths, platinum-group metals and China's refining dominance

Core lesson

Rare earths are a family of 17 elements: the 15 lanthanides plus scandium and yttrium. For our purposes, four matter most, because they make the strongest permanent magnets: **neodymium (Nd) and praseodymium (Pr)**, the "light" rare earths that form the bulk of a neodymium-iron-boron (NdFeB) magnet, and **dysprosium (Dy) and terbium (Tb)**, the "heavy" rare earths added in small amounts so the magnet keeps its strength at high temperatures, such as inside a hot motor. NdFeB magnets are in EV traction motors, wind turbine generators, hard disk drives, phone

speakers and vibration motors, missile guidance fins, and the servo motors in industrial robots and humanoid robot joints. When Month 9 opens up a robot actuator, the magnet inside will very likely be NdFeB, and very likely made in China.

The rare earth chain has five steps, and the hard part is in the middle.

1. **Mining** ore that contains a mix of rare earths (for example, bastnaesite at Mountain Pass in California or Bayan Obo in Inner Mongolia; ionic clays in southern China and Myanmar, which are rich in the heavy elements).
2. **Concentration** into a mixed rare earth concentrate.
3. **Separation** into individual oxides. This is the chokepoint. Rare earths are chemically almost identical, so separating them requires hundreds of stages of solvent extraction, one stage after another, each shifting the mix a tiny amount. It requires precise chemical engineering know-how, large volumes of acids and solvents, and careful handling of radioactive thorium and uranium found in some ores.
4. **Metal making**: reducing oxides into metals and alloys.
5. **Magnet manufacturing**: making the alloy into powder, pressing it in a magnetic field, sintering, and treating it (for example, with "grain boundary diffusion," which places scarce dysprosium exactly where it is needed).

China mined about 270,000 tonnes of rare earth oxide in 2025, 69% of the world's 390,000 tonnes; the US mined about 51,000 tonnes; Australia 29,000; Myanmar (Burma) about 22,000 (USGS MCS 2026). But China's share of separation and magnet making is much higher than its share of mining. The IEA reports that China's share of rare earth refining fell from over 90% in 2023 to 85% in 2025, as new US projects and higher Malaysian output came online, and projects that it could fall to about 70% by 2035 if planned projects outside China are built on schedule (IEA press release on the GCMO 2026, July 16, 2026). Note that this is an average across all rare earths: for the heavy rare earths (dysprosium, terbium) China's share remains much higher. China also makes the large majority of the world's NdFeB magnets (industry estimates commonly put it above 90%; treat this as approximate). Myanmar's heavy-rare-earth ore, much of it mined in conflict-affected Kachin State, is mainly shipped to China for separation.

How did China win the midstream? This is one of the most important stories in modern industrial policy. Several mechanisms worked together:

- **State strategy and patience.** From the 1980s China treated rare earths as strategic. A widely quoted line attributed to Deng Xiaoping in 1992 compares China's rare earths to Middle Eastern oil. Whatever the exact wording, policy followed: state research institutes, consolidation of producers into a few large state groups, and production quotas.
- **Low costs, including low environmental costs.** For decades Chinese separation plants operated with weak environmental enforcement, producing toxic and radioactive waste around Baotou and in the southern clay districts. Prices were so low that Western producers could not compete, and many closed in the 1990s and 2000s, including, for a period, Mountain Pass in the US. This is an uncomfortable fact: part of China's cost advantage was paid for by its own people's health and land.
- **Learning curves and clustering.** Once nearly all separation was in China, engineers, equipment makers, chemical suppliers and magnet producers clustered there, so costs kept falling and know-how deepened. A newcomer elsewhere faces not just one missing plant but a missing ecosystem. The IEA notes that in magnet making, grain boundary diffusion is "highly patented, with only one equipment supplier outside China and equipment costs that

are reportedly more than ten times higher."

- **Cost of capital.** Chinese state banks and policy funds lend at lower rates and accept lower returns, which matters hugely for capital-heavy plants that may lose money for years.
- **Downstream demand at home.** China's own EV, wind, electronics and motor industries became the world's largest buyers, so a Chinese refiner always had a customer down the road.

From dominance to leverage: the export-control era. In 2010, during a dispute with Japan, Chinese exports of rare earths to Japan were reported to have been disrupted, and prices spiked; Japan responded by diversifying, notably by financing Lynas in Australia (which then built separation capacity in Malaysia). For the next decade China's dominance was mostly a quiet commercial fact. That changed in 2023-2025. China placed export controls on gallium and germanium (2023), graphite (late 2023), and then rare earths. The USGS records the sequence: in April 2025 China added export controls on alloys, compounds, metals and oxides of samarium, gadolinium, terbium, dysprosium, lutetium, scandium and yttrium; in October 2025 it expanded controls to europium, holmium, erbium, thulium and ytterbium; in November 2025 it suspended the October controls for one year, while the April controls remained in effect (USGS MCS 2026).

The IEA calls 2025 "the year when the economic risks of highly concentrated supply chains materialised at scale." The April 2025 controls forced some automakers outside China to cut production or temporarily halt lines. The October 2025 package went further: it would have reached products made outside China containing Chinese-origin rare earths above a 0.1% value threshold, or made with Chinese technology. The IEA estimates that full implementation could put about $6.5 trillion per year of downstream production outside China at risk, across cars, high-tech, defence and energy. China also announced controls on battery cathode materials, precursors, graphite anode materials and battery manufacturing equipment in October 2025. The IEA counts that the number of mineral tariff codes under Chinese export controls has tripled since 2023.

As of early November 2026, the situation was this: Presidents Trump and Xi agreed a one-year truce at Busan, South Korea, on October 30, 2025. On September 23, 2026, as Xi arrived for a state visit to Washington, US Treasury Secretary Scott Bessent announced that the truce, due to expire on November 10, 2026, would be extended by two months to January 10, 2027 (CNBC, September 23-24, 2026); China's commerce ministry confirmed the same date on September 28 without naming rare earths. Trade lawyers noted that as of late October neither government had yet published the legal texts carrying the suspensions past November 10 (E+E Leader, October 30, 2026). The April 2025 controls on seven medium and heavy rare earths were never suspended and remain in force. Licensing continues, but slowly and selectively.

Price splits: one metal, two prices. Export controls create a two-tier market. The IEA reports that in Europe, prices for gallium and the heavy rare earths dysprosium and terbium were around five times higher than Chinese domestic prices, and germanium almost three times higher (IEA GCMO 2026). Trade press reported dysprosium oxide in Rotterdam at about $1,250-1,675 per kg in late August 2026, against about $270-330 per kg free on board China (Fastmarkets, as reported by CSS Eurasia, October 2026). A Chinese magnet maker therefore buys its inputs for a fraction of what a European rival pays. This is a powerful mechanism: **export controls do not just restrict supply; they hand a permanent cost advantage to the controlling country's downstream industry.** That is why they are a tool of industrial policy, not only of diplomacy.

The response: governments become investors. The most striking example is the US deal with MP Materials (owner of Mountain Pass) in July 2025. The US Department of Defense bought $400 million of convertible preferred equity (potentially about 15% of the company), lent $150 million for

heavy rare earth separation, guaranteed a **price floor of $110 per kg for NdPr** for ten years, and committed that 100% of the magnets from a new "10X" magnet plant would be bought by defense and commercial customers (MP Materials SEC filing, July 2025). MP's own results show a realized NdPr price of $52 per kg in the first quarter of 2025 and $57 per kg in the second quarter (MP Materials quarterly results; a Reuters report at the time attributed the $52 figure to the second quarter), so the floor was roughly twice the prevailing Chinese-set market level. The floor works like a **contract for difference**: if the market price is $70, the government pays MP $40 per kg; if prices later exceed $110 after the plant is at full capacity, the government gets 30% of the upside. The IEA recommends exactly these tools: contracts for difference, price floors, offtake guarantees and strategic stockpiles. It estimates that diversifying magnet rare earth supply chains would need about $60 billion of investment over the next decade, and that stockpiling 11 high-risk materials would cost countries outside the dominant supplier less than $900 million a year net.

And yet, small numbers again: the IEA estimates that rare earths are about 40% of a permanent magnet's cost but less than 1% of a vehicle's value, and that a tripling of rare earth prices would raise the cost of a car by just 0.1%. **A modest "security premium" is affordable for the final buyer.** The barrier is not affordability but the willingness of someone to sign a long contract at the higher price.

Platinum-group metals (PGMs). The six PGMs (platinum, palladium, rhodium, ruthenium, iridium, osmium) are superb catalysts. Their biggest use is in catalytic converters that clean car exhaust; they are also used in chemical and petroleum refining, glass making, electronics, hard disks, and in hydrogen technology: platinum in fuel cells and in proton exchange membrane (PEM) electrolyzers, and iridium in electrolyzer anodes. Supply is extremely concentrated in two countries: South Africa produced about 120 tonnes of the world's 170 tonnes of platinum in 2025, and Russia about 84 of 190 tonnes of palladium (USGS MCS 2026). South African output fell about 9% in 2025 "owing to declining palladium prices, higher costs associated with deep-level mining, and ongoing disruptions to the supply of electricity" (USGS MCS 2026). That last phrase connects Month 4 directly to this month: an electricity crisis (South Africa's load-shedding) becomes a metals supply shock. PGM mines in South Africa's Bushveld Complex are among the deepest in the world, with high labour and energy costs. Prices rose in 2025: ruthenium up 53%, platinum 25%, rhodium 24%, palladium 11%, while iridium fell 9% partly because of "the waning of investor enthusiasm in the hydrogen power market" (USGS MCS 2026). PGMs face a long-term demand question: battery EVs have no catalytic converter, so palladium and rhodium demand will shrink as EVs spread, while hydrogen demand is uncertain.

The strategic minor minerals and the AI connection. The IEA's 2026 Outlook has a special focus on "strategic minor minerals" such as gallium, germanium, indium, antimony, tungsten and tellurium: tiny markets with outsized importance for semiconductors, robotics, AI hardware, aerospace and defence. Gallium is used in power electronics (gallium nitride chargers and power supplies) and radio-frequency chips; germanium in fibre optics, infrared optics and some semiconductors. Prices for these minerals "more than doubled" between 2024 and early 2026, and tungsten prices rose sixfold (IEA GCMO 2026). The link to Month 8 is direct: AI data centers need power electronics, optical interconnects and cooling, all of which touch these minor metals. **The ore-to-AI chain is real, and its weakest links are often the smallest markets.**

Why the midstream is the leverage point. Put the pieces together. Mining is spread over many countries because geology is spread out. Final manufacturing is spread over many countries because markets are spread out. But refining and separation have high fixed costs, strong economies of scale, deep tacit know-how, large chemical and environmental footprints, and the need for cheap

energy and acid. Those features favour concentration in one place that is willing to tolerate low returns and environmental costs for strategic reasons. The IEA's 2026 analysis of project pipelines confirms the imbalance even now: outside the dominant supplier, planned refining and downstream capacity lags behind planned mining; for rare earths, planned magnet production is only about one-third of expected mined supply by 2035, and planned cathode capacity is only about one-third of projected lithium mining capacity. **The world outside China is building mines faster than it is building the factories that turn their output into usable materials.** For a country like Indonesia, Brazil, Malaysia or Morocco, this gap is the opportunity Month 6 explores.

Case study: China's 2025 rare earth controls and the automotive scramble

The best way to see the midstream's leverage is to follow a single shock.

April 2025. In the middle of a tariff war with the United States, China's Ministry of Commerce added seven medium and heavy rare earths (samarium, gadolinium, terbium, dysprosium, lutetium, scandium and yttrium) and their products, including magnets containing them, to its export licensing system (USGS MCS 2026). Exporters now needed a licence for each shipment, with end-user information. The rules did not ban exports; they slowed and filtered them.

May-June 2025. Licences came slowly. Carmakers and suppliers outside China, who had held only weeks of magnet inventory because just-in-time supply had always worked, began to run short. The IEA records that the controls had "significant impacts across downstream industries, forcing some automakers to reduce utilisation rates or temporarily halt operations" (IEA GCMO 2026). The reason is mechanical: a modern car contains dozens of small electric motors (seats, windows, pumps, steering, braking) and, in EVs, a traction motor, most using NdFeB magnets, many with dysprosium or terbium for heat resistance. A part worth a few dollars stopped a product worth tens of thousands.

October 2025. China announced a much broader package: five more rare earths, rare earth production equipment and technology, controls on battery materials and equipment, and an extraterritorial rule reaching foreign products with 0.1% or more of Chinese rare earth content by value. This was modelled on how the US controls semiconductors, which reach foreign-made chips using US technology. It was a clear signal that China could use the same legal weapon in reverse.

November 2025. After the Busan meeting between the US and Chinese leaders on October 30, 2025, China suspended the October package for one year, until November 10, 2026, while the US suspended some of its own measures (USGS MCS 2026; trade-law analyses, 2026). The April controls stayed. In September 2026 the two governments announced an extension of the truce to January 10, 2027, though trade lawyers noted in late October 2026 that the formal legal texts extending the suspensions had not yet been published (see sources).

Effects in 2026. Chinese rare earth export volumes fell about 11% in January-August 2026 compared with a year earlier, while their value rose about 54%, and China's rare earth magnet exports to the US averaged about 504 tonnes a month in 2026, against 621 tonnes in 2024 (CSS Eurasia, citing Chinese customs data, October 2026; I could not open the original customs data). Prices for heavy rare earths outside China stayed several times higher than inside. Meanwhile non-Chinese capacity grew: the IEA reports that rare earth refining was the one mineral where concentration fell modestly between 2023 and 2025, thanks to new US projects and production increases in Malaysia (Lynas), "highlighting the role of targeted policy and investment support in enabling diversification."

Lessons for the driving question.

1. **Value is not the same as leverage.** The rare earth market is tiny compared with copper (a few billion dollars of oxide a year against hundreds of billions for copper), yet it gave China leverage over trillions of dollars of downstream output.
2. **Leverage is strongest when the substitute takes years.** A new separation plant plus magnet plant takes three to seven years and needs know-how that is scarce outside China.
3. **Using leverage triggers diversification.** Every use of an export control raises the price that rivals will pay to escape it. China's challenge is to use the tool enough to win concessions without destroying its long-run dominance. This is a repeated game, and it will shape Month 12's scenarios.
4. **For a robotics founder, the magnet is a strategic input.** If you build actuators outside China, your bill of materials depends on a political decision in Beijing. Designing motors that use less dysprosium, or none (for example, some ferrite magnet or induction motor designs), is not just engineering; it is risk management.

Key numbers

- World rare earth mine production: 390,000 t of rare earth oxide in 2025; China 270,000 t (69%), US 51,000 t, Australia 29,000 t, Myanmar 22,000 t (USGS MCS 2026).
- World rare earth reserves: more than 85 Mt; China 44 Mt, Brazil 21 Mt, Australia 6.3 Mt (USGS MCS 2026).
- Downstream production outside China at risk from full rare earth export controls: about $6.5 trillion per year (IEA GCMO 2026).
- European prices of dysprosium and terbium versus Chinese domestic prices: around five times higher (IEA GCMO 2026).
- Rare earth share of magnet cost: about 40%; of a car's value: less than 1% (IEA GCMO 2026).
- US price floor for NdPr from MP Materials: $110 per kg for 10 years from Q4 2025 (MP Materials SEC filing, July 2025).
- Estimated investment to diversify magnet rare earth supply chains: about $60 billion over the next decade (IEA GCMO 2026).
- World platinum mine output: 170 t in 2025, South Africa 120 t; palladium 190 t, Russia 84 t (USGS MCS 2026).
- Tungsten price increase, 2024 to early 2026: about sixfold (IEA GCMO 2026).

Think like a founder

The midstream is where governments are now willing to pay for diversification, and where engineering talent is scarce. That combination creates openings:

- **Process technology licensing.** Better separation chemistry (fewer stages, less waste), magnet recycling (recovering NdFeB from old motors and hard drives), and dysprosium-reduction techniques can be sold to many plants worldwide. The IEA expects rare earth magnet recycling to benefit from growing end-of-life volumes of early EVs and wind turbines.
- **Design-around businesses.** Motor and actuator designs that cut heavy rare earth use, or substitute ferrite, sell to every robot and EV maker who fears the next export control.
- **The "trusted midstream" in a neutral country.** Malaysia's role in Lynas's separation, and the IEA's note that Malaysia increased rare earth production, show that a Southeast Asian country with chemical industry, legal stability and acceptable environmental regulation

can host a non-Chinese midstream. The country captures processing value; the founder who builds the logistics, services or engineering around it captures a slice too.

The key strategic question for any such business: who is your guaranteed buyer at a price above the Chinese price? Without an offtake contract, a price floor, or a defense customer, a non-Chinese midstream plant is usually undercut. The MP Materials deal is a template: equity, debt, price floor and offtake together.

Week 4: How mineral prices are made: exchanges, benchmarks, cost curves and cycles

Core lesson

Three ways a mineral gets a price.

1. **Exchange-traded metals.** Copper, aluminium, nickel, zinc, lead and tin trade on the **London Metal Exchange (LME)**, founded in 1877 and owned since 2012 by Hong Kong Exchanges and Clearing. They also trade on the **Shanghai Futures Exchange (SHFE)** and, for copper, on **COMEX** in New York (part of CME Group). These exchanges publish transparent prices that physical contracts all over the world refer to. For example, a nickel ore contract in Indonesia, a copper concentrate sale from Peru, and a cable maker's purchase in Vietnam may all be priced as "LME price plus or minus a premium."
2. **Benchmarks and price reporting agencies.** Lithium, cobalt, rare earths, graphite and many minor metals have small, opaque markets with no deep futures contract (although futures on lithium and cobalt now exist on some exchanges). Prices are discovered by **price reporting agencies** such as Fastmarkets, Benchmark Mineral Intelligence, S&P Global Platts, Argus, Asian Metal and Shanghai Metals Market, which survey buyers and sellers and publish assessments. The USGS cites Benchmark Mineral Intelligence for its lithium price data.
3. **Negotiated contracts.** Long-term offtake contracts between a mine and a refiner, or a refiner and a battery maker, often set price by formula (for example, a percentage of a benchmark), with a volume commitment. Governments increasingly set prices too: Indonesia publishes a reference price (HMA) for nickel ore that sets royalties and a minimum domestic ore price; the DRC publishes a weekly cobalt export reference price.

How an exchange works, simply. The LME is a market for **futures**: contracts to buy or sell a fixed quantity (for example, 6 tonnes of nickel) of a specified quality for delivery on a future date, at a price fixed today. Most futures are never delivered; they are closed out before the date. Who uses them?

- **Hedgers** reduce risk. A copper mine sells futures to lock in its price for next year's output; a cable maker buys futures to lock in its input cost. Each gives up possible gains to avoid possible losses.
- **Speculators** (funds, traders) take the other side, betting on price direction. They provide liquidity, so hedgers can always find a counterparty.
- **Physical delivery** keeps the paper price honest. If the futures price is far above the physical price, someone will buy metal, put it in an LME-approved warehouse, and deliver it against the contract. LME warehouse stocks are therefore watched closely: rising stocks signal surplus, falling stocks signal tightness.

Clearing and margin. A **clearing house** (LME Clear) stands between buyer and seller, guaran-

teeing trades. To protect itself, it demands **margin**: cash collateral that rises when prices move against a trader. In a fast price spike, a trader who is short (has sold futures) may face enormous margin calls, which can force them to buy back futures at any price, pushing prices even higher. That feedback loop is called a **short squeeze**, and it is the subject of this week's case study.

Contango and backwardation. If future prices are higher than today's spot price, the market is in **contango**, which usually signals plentiful supply (you are paid to store metal). If spot is higher than future prices, the market is in **backwardation**, which signals immediate scarcity (people pay extra to get metal now). Traders read the shape of this "forward curve" like a doctor reads a pulse.

Treatment and refining charges (TC/RCs): the price of processing. For copper, a mine usually sells concentrate to a smelter. The price is the LME copper price for the contained copper, minus a **treatment charge** (dollars per tonne of concentrate) and a **refining charge** (cents per pound of copper) paid to the smelter for its service. When concentrate is plentiful, smelters can demand high charges; when concentrate is scarce, smelters compete for it and charges fall. Here is the remarkable 2026 situation, from the IEA: "Benchmark copper smelter fees were settled at USD 0 per tonne in 2026, the lowest level ever agreed in annual negotiations, while spot charges have remained negative since 2024." A negative charge means the smelter pays the miner for the privilege of processing its concentrate. Why would anyone do that? Because China expanded smelting capacity much faster than mine supply grew. The IEA reports that since 2005 China has accounted for over 90% of growth in global copper smelting, raising its share of global capacity from about 15% to 50% by 2025, with utilization around 85% in China but below 70% outside China. Smelters survive on by-products (sulphuric acid, gold, silver, and minor metals) and on state support. **This is how value capture shifts along the chain: when the midstream is overbuilt, the miner captures more of the copper price; when mines are overbuilt, the smelter does.** It also shows why "just build a smelter" is not automatically a path to riches: a smelter that is a price-taker in a crowded market may earn nothing.

The cost curve: how the marginal producer sets the price. Arrange every mine in the world from lowest cost to highest cost per tonne, and plot cumulative production on the horizontal axis. You get an upward staircase called the **cost curve** or industry supply curve. In the long run, the price tends toward the cost of the **marginal producer**, the most expensive mine still needed to meet demand.

Worked example. Suppose world nickel demand is 3.5 million tonnes, and the cost curve is:

Producer group	Volume (Mt)	Cash cost ($/t)	Cumulative (Mt)
Indonesian NPI and HPAL	2.4	10,000	2.4
Other low-cost (Russia, some Canada)	0.5	13,000	2.9
Mid-cost (Philippines ore, New Caledonia, Brazil)	0.5	16,000	3.4
High-cost (Australia, older sulphide mines)	0.4	20,000	3.8

(These cost numbers are illustrative round numbers chosen to show the mechanism, not measured data.)

To meet 3.5 Mt of demand, you need all of the first three groups (3.4 Mt) plus 0.1 Mt from the high-cost group. So the price must be around $20,000 to keep those last mines open. Now suppose Indonesia adds 0.4 Mt of low-cost supply. Cumulative supply from the first three groups becomes 3.8 Mt, enough to meet demand without the high-cost group. The price can fall toward $16,000, and the Australian mines close. This is almost exactly what happened between 2022 and 2025: the USGS reports that Australia's nickel output fell 54% in 2025 as mines went into care and maintenance, and that the LME nickel cash price fell 11% in 2025 after falling from its 2022 highs. The low-cost newcomer did not just win market share; it **lowered the price for everyone**, which transferred value from producers everywhere to buyers (mainly Chinese stainless steel and battery makers). Indonesia's own response in 2026, cutting its ore mining quota (the RKAB) from 379 million wet tonnes approved for 2025 to 260-270 million wet tonnes (later revised to about 290-300 million), was an attempt to move back up the cost curve and raise the price. As of early November 2026, LME nickel traded around $15,000-16,000 per tonne, with LME stocks near 278,000 tonnes, and Indonesia's energy minister was publicly urging producers to cut output (S&P Global, Antara, Yieh, November 2026). Being the dominant low-cost producer gives you market share; turning that into higher prices requires discipline that is hard to enforce.

Why mineral prices move in long cycles. Combine three facts: demand is driven by investment booms (construction, electrification, China's growth), which come in waves; supply takes 10-15 years to respond for new mines; and mines, once built, keep producing as long as price covers cash cost, even at a loss on total cost. The result is the classic **commodity cycle**:

```
High prices -> exploration and new projects approved
     ^                                      |
     |                                      v   (5-15 years)
Shortage <- projects cancelled <- Low prices <- new supply arrives
```

Some analysts argue that electrification and AI data centers are driving a new copper **supercycle** (a price wave lasting decades, like the 2000s boom driven by Chinese urbanization). Evidence for: copper prices set records, with the LME three-month contract reaching an intraday high of about $14,500 per tonne on January 29, 2026, and averaging about $14,000-14,500 per tonne in August-September 2026 (Benchmark Mineral Intelligence via market press; monthly averages from a commodity data service); and the IEA reports that base metal prices rose by about one-third between January 2025 and April 2026, with copper investment rising 8% in 2025 while other mineral investment fell. Evidence against: high prices encourage substitution (aluminium for copper in cables and radiators, as the USGS notes), thrift (using less copper per unit), and recycling; and Chinese property construction, a huge copper user, is weak.

Judgment: copper has the strongest structural demand case of any metal in this chapter, because electrification, grids and data centers cannot avoid it and its supply growth is constrained by grades, permits and water. But even copper will cycle. A founder or investor should assume prices that are high on average in the late 2020s and 2030s but with crashes of 30-40% along the way.

Policy is now a price driver. Traditionally, mineral prices were explained by supply and demand. In 2025-2026, policy decisions moved prices as much as markets did: the DRC's cobalt ban and quotas; China's export controls; Indonesia's ore quotas, reference prices and proposed royalty and export tax changes; Zimbabwe's lithium export suspension; US tariffs on copper and aluminium (aluminium tariffs rose to 50% for most countries by June 2025, and the USGS attributes the 2025 COMEX copper record mainly to "uncertainty regarding the implementation of tariffs on U.S. imports of copper materials"). Even war matters: the IEA notes that the 2026 conflict in the Middle

East and closure of the Strait of Hormuz disrupted aluminium (the region produces about 8% of world aluminium) and sulphur (about one-quarter of global supply, with half of seaborne sulphur trade passing through Hormuz). Sulphur makes sulphuric acid, which is used to leach copper, nickel, cobalt, lithium and rare earths; China curbed sulphuric acid exports in May 2026, and in some cases acid costs overtook energy costs as the largest cost item in mineral processing. **Month 4's energy shock became Month 5's mineral cost shock through a chemical you probably never thought about.**

Volatility hurts the countries that most need stability. For a mining-dependent economy, a large price fall can slash government revenue and trigger a debt or currency crisis. Month 2's resource curse and Month 11's sovereign wealth funds give the long-run answers: save in booms, spend steadily, diversify.

Case study: The LME nickel short squeeze of March 2022

Background. By early 2022 nickel was already tight: LME stocks were low and prices were rising with EV hopes. Tsingshan Holding Group, the Chinese stainless steel and nickel giant building much of Indonesia's nickel industry, had reportedly built a very large short position (bets that the price would fall), partly as a hedge on its expected Indonesian output. Reports put the position at roughly 100,000-200,000 tonnes (S&P Global Market Intelligence, April 2022). Its logic was reasonable: Indonesia's production was about to flood the market. But much of Tsingshan's output was nickel pig iron and matte, not the Class 1 nickel that can be delivered on the LME. A short position is safe only if you can deliver or can afford the margin.

The squeeze. Russia's invasion of Ukraine on February 24, 2022 raised fears of losing Russian nickel (Russia's Nornickel is a major Class 1 producer). Prices rose. As prices rose, short holders received margin calls; to reduce risk, they bought back futures, pushing prices even higher, which caused more margin calls. The LME's independent review later described "reinforcing cycles of buying" in which "rising prices led to market participants facing rapidly growing margin calls, which prompted further buying to reduce risk." Over three trading days the price rose by over 270%. On Monday March 7 it rose about 69%; in the early hours of Tuesday March 8, in thin Asian trading, it more than doubled again to over $100,000 per tonne, with three-month nickel touching $101,365 (Reuters; FCA, 2025).

The intervention. At 8:15 a.m. London time on March 8, the LME suspended nickel trading and then cancelled all nickel trades executed on or after midnight that day, with an aggregate notional value reported at about $12-13 billion (Reuters reported $12 billion; a Clifford Chance briefing on the FCA notice cites about $13.37 billion). The market stayed closed for about eight days. An analysis by the US Office of Financial Research found the price surge threatened to put 12 of LME Clear's 45 clearing members into default and exhaust its default fund. Some hedge funds that had been on the winning side sued over the cancellation; the LME won in the English courts. In March 2025 the UK Financial Conduct Authority (FCA) fined the LME GBP 9.2 million, its first fine against a recognised investment exchange, for failing to have adequate systems and controls for severe market stress (FCA, March 2025).

Consequences. Trust in the LME nickel price was damaged. Trading volume fell; some buyers shifted to other price references, including Indonesian and Chinese benchmarks. The LME's nickel market did not reopen in Asian trading hours until March 2023 (FCA). Meanwhile Indonesia's supply wave did arrive, and prices fell back to around $15,000 per tonne by mid-2025, less than a sixth of the spike.

Lessons.

1. **Paper markets and physical markets can disconnect.** The exchange contract covers only Class 1 nickel, while most new supply was Class 2. Tsingshan had plenty of nickel, but not the kind the contract accepted.
2. **Leverage plus thin liquidity equals fragility.** Futures let traders control large positions with small margin; in a squeeze, small margin becomes huge cash calls.
3. **Who sets the price matters for value capture.** If Indonesian and Chinese producers increasingly price off their own benchmarks (Indonesia's HMA, Chinese MHP and NPI prices), price-setting power moves east, along with the information advantage that comes with it.
4. **Rules can change in the middle of the game.** Cancelling trades protected the system but broke an expectation of finality. In minerals, political and institutional risk is never zero, even in London.

Key numbers

- LME three-month nickel peak: $101,365/t on March 8, 2022, more than double the previous day's close, most of the rise in little over an hour (Reuters, 2022; FCA, March 2025).
- LME fine by the FCA: GBP 9.2 million, March 2025 (FCA).
- Copper benchmark treatment charge for 2026: $0 per tonne, the lowest ever annual settlement; spot charges negative since 2024 (IEA GCMO 2026).
- China's share of global copper smelting capacity: from about 15% in 2005 to 50% in 2025 (IEA GCMO 2026).
- Smelter utilization 2025: about 85% in China versus below 70% elsewhere (IEA GCMO 2026).
- LME three-month copper intraday record: $14,527.50/t on January 29, 2026, an 11% one-day jump, closing near $13,700 (Benchmark Mineral Intelligence; Reuters, January 29, 2026).
- Indonesia nickel ore quota: 379 million wet tonnes approved for 2025 (320.37 million actually mined); 260-270 million initially for 2026, revised on October 29, 2026 to 290-300 million (Ministry of Energy and Mineral Resources, via Indonesia Business Post; The Oregon Group).
- Base metal prices (aluminium, copper, tin): up about one-third between January 2025 and April 2026 (IEA GCMO 2026).
- Critical mineral investment: down 9% in 2025; lithium companies down about 40%; copper-focused companies up 8% (IEA GCMO 2026).

Think like a founder

Price risk is the silent killer of mineral businesses. Founders who understand price formation can build on it:

- **Hedging and risk tools for small players.** Large miners hedge on the LME; small processors, recyclers and manufacturers in emerging markets often cannot. A business that offers simple hedging, inventory financing or price-indexed contracts to small firms in Southeast Asia or Africa serves a real need.
- **Price data and transparency.** Price reporting agencies are profitable businesses because opaque markets need trusted prices. New, smaller markets (recycled battery materials, regional scrap, minor metals) still lack them. Data is a classic way to capture value without owning a mine.
- **Counter-cyclical investing.** The best time to buy mineral assets, equipment or companies is in the bust, when the IEA reports that exploration and investment collapse. Patient capital

captures the next boom. As a future founder-chief executive officer, remember this when you raise money: raise in the boom, buy in the bust.

- **Structure contracts to survive the cycle.** If your robotics or battery business depends on a metal, use long-term contracts, dual sourcing and designs that can switch materials. The value you protect is your margin.

Month synthesis

The month's four weeks fit into one picture: a chain of layers, each with different economics, sitting in different countries.

LAYER	WHAT HAPPENS	ECONOMICS	WHO DOMINATES (2025)
Geology	Nature concentrates metals	Luck; fixed location	Chile, DRC, Australi Indonesia, China, S.
Mining	Dig, crush, concentrate	Capital-heavy, 10-15 yr lead time, grade is king	Spread out, but one per mineral (DRC Co,
Refining / separation	Smelt, leach, separate, purify to battery or magnet grade	Scale, know-how, cheap energy and acid, low cost of capital, by-products	China (most minerals Indonesia (nickel)
Components	Cathode, anode, magnet, cable, catalyst	Learning curves, clusters, close to customers	China, then Korea, J
Products	EV, battery, robot, turbine, data center	Brand, software, design, distribution	China, US, EU, Japar

Three rules summarize the value-capture logic:

Rule	Mechanism	Example
The chokepoint captures leverage, not always profit	The narrowest layer can stop the whole chain	Rare earths are less than 1% of a car's value but controls put $6.5 trillion of output at risk (IEA)
The low-cost producer sets the terms	Marginal cost sets price; new low-cost supply pushes out high-cost rivals	Indonesian nickel closed Australian mines (USGS)
Overbuilt stages earn little	Competition for scarce inputs transfers margin to the input owner	Copper smelter fees at $0 in 2026 (IEA)

For the driving question, the implication is uncomfortable but clear. A country or company that only digs captures a modest and volatile slice. A country that refines captures more, but only if it is low-cost and has a buyer; otherwise the refining margin can vanish. Durable value is captured by whoever combines a scarce position (low-cost ore, a chokepoint process, or a protected market) with the ability to move downstream into components and products. China built exactly this combination over 30 years. Month 6 asks whether Indonesia, the DRC, Zimbabwe, Chile or Morocco can build some version of it, and Month 9 asks what it means for the cost of a robot.

Common misconceptions

1. **"Rare earths are rare."** They are not especially rare in the crust; minable concentrations are less common, and separation is technically hard. The scarcity is in processing capacity and know-how, mostly in China, not in geology.
2. **"Reserves tell us when we will run out."** Reserves are an economic category that grows with price and technology. Copper's reserve life has stayed in a similar range for decades. The real constraints are time, cost, permits, water, energy and social acceptance.
3. **"The country with the most reserves controls the market."** Chile has the largest lithium reserves but Australia mines more, and China refines most of the world's lithium chemicals. Bolivia has one of the largest lithium resources and produces almost none. Control comes from production, refining and contracts, not from what sits in the ground.
4. **"Higher mineral prices will make EVs and robots unaffordable."** The IEA estimates critical minerals are about 3% of an EV's price and rare earths less than 1% of a car's value. Price spikes hurt intermediate makers (cell and magnet factories) much more than final buyers. The bigger risk is physical unavailability, not price.
5. **"Exchanges like the LME always give the true price."** Exchange prices reflect only the grades deliverable against the contract and can disconnect from the physical market, as the 2022 nickel squeeze showed. In small markets, prices come from surveys by price reporting agencies and can be thin and opaque.

Exercises

1. **Grade arithmetic (quantitative).** A copper mine processes ore at 0.45% copper with 87% recovery. (a) How many tonnes of ore are needed per tonne of copper? (b) If the strip ratio (waste rock to ore) is 2.5 to 1, how many tonnes of total rock are moved per tonne of copper? (c) If grade falls to 0.40% with the same recovery, by what percentage does ore throughput have to rise to keep copper output constant?

2. **Lithium conversion margin (quantitative).** Using the USGS November 2025 figures (spodumene SC6 at about \$970/t, Chinese lithium carbonate spot about \$10,300/t) and assuming 7.7 t of SC6 per tonne of LCE, calculate the gross margin per tonne of LCE available for conversion costs and profit. If conversion costs are \$2,500/t, what is the converter's profit per tonne? What happens to that profit if spodumene rises 30% while carbonate rises only 10%?

3. **Cost curve (quantitative).** Using the illustrative nickel cost curve in Week 4, find the market-clearing price if demand rises to 3.7 Mt. Then add 0.3 Mt of new Indonesian supply at \$10,000/t and recompute. Who gains and who loses from the new supply?

4. **Share of value (quantitative).** A humanoid robot sells for \$30,000. Its actuators contain NdFeB magnets worth \$400, of which rare earths are 40% of cost (IEA ratio). If rare earth prices triple, by how much does the robot's cost rise, in dollars and as a percentage? What does this imply about how a robot maker should think about magnet supply risk versus magnet price risk?

5. **Essay (600-800 words).** "China's dominance of mineral refining is mainly the result of state subsidies." Evaluate this claim using at least four mechanisms from Week 3 (state strategy, environmental costs, learning curves and clustering, cost of capital, domestic demand). Give a judgment.

6. **Essay (600-800 words).** Explain how the 2026 closure of the Strait of Hormuz affected the cost of producing copper, nickel and lithium in places far from the Middle East. Trace the chain step by step, using the IEA's evidence on sulphur and sulphuric acid.
7. **Policy design.** You advise a government outside China that wants a domestic rare earth magnet plant. Using the MP Materials deal and the IEA's policy tools, design a support package (equity, loans, price floor, offtake, stockpile). Estimate the maximum annual cost to taxpayers if the market price of NdPr stays at \$60/kg and the plant produces 1,000 tonnes of NdPr a year under a \$110/kg floor.
8. **Minerals per kilowatt-hour (quantitative).** Using the per-kWh table in Week 2, recompute the cathode metal value of 1 kWh of NMC 811 if cobalt falls to \$25 per kg and nickel rises to \$20 per kg. By how much does the gap between NMC 811 and LFP change? If 1.5 TWh of batteries are made a year, what is the annual swing in nickel and cobalt revenue between your case and the base case?

Reading list

1. **U.S. Geological Survey, *Mineral Commodity Summaries 2026* (February 2026).** The standard free source for world production and reserves of over 90 minerals; learn to read its two-page chapters.
2. **International Energy Agency, *Global Critical Minerals Outlook 2026* (July 2026).** The best single overview of supply, demand, refining concentration, export controls and policy tools for energy minerals.
3. **Ed Conway, *Material World: The Six Raw Materials That Shape Modern Civilisation* (2023).** A vivid, readable tour of sand, salt, iron, copper, oil and lithium, showing how physical materials underpin the digital economy.
4. **Javier Blas and Jack Farchy, *The World for Sale: Money, Power and the Traders Who Barter the Earth's Resources* (2021).** How commodity traders such as Glencore and Trafigura capture value between mine and market.
5. **Siddharth Kara, *Cobalt Red: How the Blood of the Congo Powers Our Lives* (2023).** A harsh first-hand account of artisanal cobalt mining in the DRC; read it to understand the human cost behind battery supply chains (it is an advocacy book, so read critically).
6. **Henry Sanderson, *Volt Rush: The Winners and Losers in the Race to Go Green* (2022).** A journalist's account of the battery metals boom and China's rise in the supply chain.
7. **Guillaume Pitron, *The Rare Metals War* (English edition 2020).** A provocative account of rare earths and the hidden environmental costs of green technology; useful for the China midstream story.
8. **Ernest Scheyder, *The War Below: Lithium, Copper, and the Global Battle to Power Our Lives* (2024).** A Reuters reporter on the conflicts between mining, communities and environment, mainly in the Americas.
9. **Oliver Wyman for the LME, *Independent Review of Events in the Nickel Market in March 2022* (2023).** The detailed anatomy of the nickel squeeze; essential for understanding how exchanges can fail.

Glossary terms introduced this month

- **Anode / cathode**: the negative and positive electrodes of a battery; graphite is the usual anode, metal oxides or phosphates the cathode.
- **Backwardation / contango**: futures curve shapes where spot is above future prices (scarcity) or below them (surplus).
- **By-product**: a metal recovered as a secondary output of mining or refining another metal (for example, cobalt from copper).
- **Class 1 / Class 2 nickel**: high-purity nickel (at least 99.8%, battery and LME grade) versus lower-purity products (nickel pig iron, ferronickel) used mainly for stainless steel.
- **Clearing house and margin**: the entity guaranteeing exchange trades, and the collateral it demands from traders.
- **Concentrate**: an upgraded product from a mine, richer in metal than ore but not yet refined.
- **Contract for difference / price floor**: a payment that tops up a producer's revenue when the market price is below an agreed level.
- **Cost curve**: producers ranked from lowest to highest cost; the marginal producer's cost tends to set the long-run price.
- **Export control**: a government licensing requirement or ban on exports of a good or technology.
- **Grade**: the share of a useful metal in ore.
- **HPAL (high-pressure acid leaching)**: a process that dissolves laterite nickel ore in hot, pressurized sulphuric acid to make battery-grade intermediates.
- **LCE (lithium carbonate equivalent)**: the standard unit for lithium quantities; 1 tonne of lithium is about 5.32 tonnes of LCE.
- **LFP / NMC**: lithium iron phosphate and nickel manganese cobalt battery cathode chemistries.
- **LME (London Metal Exchange)**: the main global exchange for base metals.
- **MHP (mixed hydroxide precipitate)**: an intermediate product containing nickel and cobalt, made by HPAL.
- **NdFeB magnet**: neodymium-iron-boron permanent magnet, the strongest commercial magnet type.
- **Price reporting agency**: a company that surveys trades and publishes price assessments for opaque markets.
- **Reserve / resource**: the economically mineable part of a deposit today, versus the larger amount that might be mined someday.
- **RKEF (rotary kiln electric furnace)**: a process for smelting laterite nickel ore into nickel pig iron or ferronickel.
- **Short squeeze**: a rapid price rise forced by traders buying back short positions to meet margin calls.
- **Strategic minor minerals**: small-market minerals such as gallium, germanium, indium, antimony and tungsten with critical uses.
- **TC/RC (treatment and refining charges)**: fees paid by miners to smelters for processing concentrate.

Sources

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Month 6: Processing, Downstreaming and Resource Nationalism

Month overview

Month 5 ended with a puzzle. Mining is spread across many countries, but refining is concentrated, mostly in China, and in nickel's case in Indonesia. The refining layer gives leverage over everything downstream: the batteries, the magnets, the robots, the data centers. So why don't the countries that dig the ore simply refine it themselves and keep the value?

Many are now trying. The policy has several names. In Indonesia it is called **hilirisasi**, usually translated as downstreaming. In Africa it is often called **beneficiation** or **local value addition**. Economists call the broader tendency **resource nationalism**: governments asserting more control over natural resources through ownership stakes, higher taxes and royalties, export bans or quotas, local processing requirements, and renegotiated contracts. Indonesia banned nickel ore exports in 2020 and bauxite in 2023. The DRC banned cobalt exports in 2025 and then imposed quotas. Zimbabwe banned raw lithium ore exports in 2022 and set a ban on lithium concentrate from January 2027. Botswana renegotiated its diamond deal with De Beers. Chile brought its state copper company into lithium. Guinea revoked bauxite licences from a company that had not built a promised refinery.

This month asks four questions: what are the economics of a smelter or refinery; who captured the value in Indonesia's nickel ban; how African producers (the DRC, Zambia, Zimbabwe, Botswana, Guinea, Morocco) are using their bargaining power; and when downstreaming works or fails. We end with a scorecard you can apply to any country and mineral.

This connects back to Month 3's industrialization playbooks (export discipline, learning by doing, state capacity) and Month 4's energy economics (because a smelter is mostly an energy-conversion machine). It leads into Month 7, where we follow processed materials into manufacturing supply chains, and Month 9, where they become robots. For your driving question, this month is the hinge: it tests whether Africa or Southeast Asia can climb from ore to finished product, or whether they will mostly host the dirty middle of someone else's chain.

Learning goals

By the end of this month you should be able to:

- Explain the cost structure of smelting and refining (energy, reagents, capital, labour, by-products) and why margins in processing can be thin or even negative.

- Calculate simple processing economics: energy cost per tonne of aluminium, value added per tonne of nickel at each stage, and the effect of treatment charges on a smelter.
- Describe Indonesia's nickel downstreaming policy, its results (output, exports, investment, jobs) and its costs (environment, dependence on Chinese capital, oversupply, lower prices).
- Compare resource nationalism strategies in the DRC, Zambia, Zimbabwe, Botswana, Guinea, Morocco and Chile, and explain why outcomes differ.
- Explain the strategic logic of transport corridors such as the Lobito Corridor and its rival, TAZARA.
- Apply a downstreaming scorecard to judge whether a country's policy is likely to succeed.
- Identify where a founder could capture value in a processing economy without owning a smelter.

Week 1: The economics of smelting and refining

Core lesson

What processing actually is. Ore leaves the mine as rock or concentrate. Turning it into a usable material requires one of two families of technology, and often both:

- **Pyrometallurgy** (fire-based): heating ore in furnaces until chemical reactions separate the metal. Copper smelting, nickel pig iron production in RKEF (rotary kiln electric furnace) plants, and iron and steel making are pyrometallurgical. These processes need large amounts of heat or electricity and produce gases such as sulphur dioxide, which modern smelters capture and turn into sulphuric acid.
- **Hydrometallurgy** (water and chemistry-based): dissolving the metal out of ore with acids or other solutions and then recovering it, for example by precipitation, solvent extraction or electrowinning (using electricity to plate metal onto a cathode). HPAL (high-pressure acid leaching) for nickel, lithium conversion, rare earth separation, and much DRC copper and cobalt processing are hydrometallurgical. These need reagents (acid, lime, solvents), water and careful waste management.

A smelter is an energy-conversion machine. The clearest example is aluminium. Bauxite is refined into alumina (aluminium oxide) in the Bayer process, and then alumina is split into aluminium metal by electrolysis in the Hall-Heroult process. The electrolysis step is so electricity-hungry that aluminium is often called "congealed electricity": the industry commonly cites roughly 13-15 megawatt-hours (MWh) of electricity per tonne of aluminium (a widely used industry range; exact figures vary by plant). Work through the arithmetic:

- Suppose a smelter uses 14 MWh per tonne.
- If electricity costs \$30 per MWh (cheap hydropower or captive coal), power costs 14 x 30 = \$420 per tonne.
- If electricity costs \$90 per MWh (typical industrial grid power in many countries), power costs \$1,260 per tonne.
- At an illustrative aluminium price of \$2,500 per tonne, the difference of \$840 per tonne is about a third of revenue, often larger than the entire profit margin.

So aluminium smelters cluster where power is cheapest: Chinese coal and hydro provinces, the gas-rich Gulf, and hydro-rich Iceland, Norway and Quebec. China produced about 45 million of the world's 74 million tonnes of primary aluminium in 2025 (USGS MCS 2026). The country with bauxite does not necessarily win: **Guinea** mined about 150 million tonnes of bauxite in 2025, about a third of the world total and more than any other country, but produced only about 360,000 tonnes

of alumina, a tiny fraction of China's 93 million tonnes (USGS MCS 2026). Guinea has the ore; China, which buys most of it, has the power, refineries and customers.

The value-added staircase. Each processing step adds value, but not equally, and the way the numbers are usually quoted hides more than it shows. Start with the headline claim, then rebuild it properly.

The headline claim comes from Indonesian commentary: in 2020 the price of nickel ore on world markets was around $83 per tonne of wet ore, while ferronickel sold for more than $13,000 per tonne, so downstreaming "multiplied value 157 times" (NEXT Indonesia Center, June 2025, using official data). The comparison is broken, for two reasons. First, the two numbers measure different things. A wet tonne of laterite ore contains only around 10-12 kg of nickel, while $13,000 is close to the price of a whole tonne of nickel metal in 2020 (the LME average that year was roughly $13,800 per tonne), so it is probably quoted per tonne of contained nickel. You are comparing a tonne of dirt with a tonne of metal. Second, as we will see, most of the higher price pays for energy, coal, capital and profit, not for the nickel.

Rebuilding the ladder per kilogram of nickel. The right unit is value per kilogram of contained nickel at each step. Work through the first rung with real 2026 numbers:

- Indonesia's government reference price (HMA) for 1.6% nickel ore was about $57 per wet tonne in late October 2026, while LME nickel traded around $15,500 per tonne, or $15.50 per kilogram (SMM; Yieh, November 2026).
- Laterite ore is wet. Assume about 35% moisture (a typical figure for Indonesian saprolite; treat it as an assumption). One wet tonne is then about 0.65 dry tonnes, and the 1.6% grade applies to the dry ore: 650 kg x 1.6% = about 10.4 kg of nickel.
- So the ore price is $57 / 10.4 kg = about $5.50 per kilogram of contained nickel.

That is about 35% of the LME value of the same nickel. The true multiplier from ore to refined metal, per unit of nickel, is roughly 3 times, not 157 times. This matters for policy: it means Indonesian mine owners, who sell ore at the reference price, already capture a substantial slice, and that the processing margin above them is a few dollars per kilogram, not hundreds.

The rungs above, and who is paid at each one. Now follow one tonne of contained nickel through an RKEF plant that makes nickel pig iron (NPI) for stainless steel. The ore price and LME price below are real; every other line is an illustrative assumption chosen to be in a plausible range, so that you can see the mechanism. Do not quote these as measured data.

Line (per tonne of nickel in NPI)	Value ($)	Who receives it	Stays in Indonesia?
Revenue: NPI sold at about 88% of LME per unit of nickel (assumption)	13,640	Plant owner	(revenue)
Ore: about 107 wet tonnes at $57 (10.4 kg Ni per tonne, 90% recovery)	6,100	Indonesian mine owners, royalties to state	Yes

Line (per tonne of nickel in NPI)	Value ($)	Who receives it	Stays in Indonesia?
Electricity: about 40 MWh at $50/MWh from captive coal (assumption)	2,000	Captive plant, using mostly Indonesian coal	Mostly
Coke, electrodes, spare parts (assumption)	1,000	Largely imported from China	Mostly no
Wages and local services (assumption)	600	Indonesian and foreign workers, contractors	Partly
Depreciation and interest on capital (assumption)	1,500	Foreign lenders and equipment makers	No
Operating profit (what is left)	2,440	Owner (in the parks, usually a Chinese-controlled firm)	Mostly no, except taxes

Three lessons fall out of this table. First, the biggest single item that stays in Indonesia is the ore itself, which Indonesia would have earned (at a lower world price) even without a smelter. Second, because Indonesia is one of the world's largest coal producers, the energy line also stays mostly at home, which is unusual: in most countries a smelter's power bill is partly a payment to foreign fuel suppliers. Third, the profit line is real, but it flows mostly to foreign owners, minus corporate tax, and many smelter investors received multi-year tax holidays, which shrinks even that tax share in the early years. If we add up the "Yes" and "Mostly" lines plus taxes on profit, perhaps 60-70% of the $13,640 stays in the country in this illustration. That is far better than raw ore exports, but far less than the export statistics imply when they count the whole $13,640 as an Indonesian gain.

The battery branch. The same tonne of nickel can instead go up a different ladder: HPAL to mixed hydroxide precipitate (MHP), then to battery-grade nickel sulphate, then to precursor cathode active material (pCAM), then to cathode active material, then into a cell. Each rung adds a smaller margin per kilogram of nickel but adds other materials, more chemistry and more intellectual property. By the time nickel sits inside an NMC cell, it is a minor part of the value. Week 2 of Month 5 showed that an NMC 811 cell needs roughly 0.7 kg of nickel per kilowatt-hour (kWh); at $15.50 per kilogram that is about $11 of nickel in a pack that BloombergNEF found sold for an average of $128 per kWh in 2025 (BNEF, December 2025). So nickel is under a tenth of the pack price. The rest pays for lithium, cobalt, graphite, copper and aluminium foils, separators, electrolytes, factory capital, energy, labour, research and profit, much of it at the cell maker.

Why this is the key to value capture from ore to robot. The pattern is general: the share of final value that the ore holder can capture shrinks at each rung, while the share that goes to whoever owns the know-how, the factory capital and the customer relationship grows. A robot or an AI server is the top of many such ladders at once. A country that stops at the first rung captures a large share of a small number. A country that climbs captures a smaller share of much larger numbers, but only if its own firms and workers own the rungs. That is why, in the rest of this week, we look at processing as an energy and capital business rather than as a mineral business, and why Weeks 2 to 4 keep asking not only "was the plant built?" but "who owns it, who supplies it and who buys from it?"

Value added is revenue minus the cost of inputs bought from others, and that is what a country's GDP actually gains. If the coal, the acid, the machinery, the engineers and the capital all come from abroad, the domestic value added is much smaller than the export value suggests. Exercise 2 at the end of this month asks you to do this calculation yourself.

The cost structure of a processing plant. Typical components, in rough order of importance:

1. **Capital cost (capex)**: furnaces, autoclaves, power plants, acid plants, tailings facilities. An HPAL plant is one of the most capital-intensive and technically difficult plants in mining; early HPAL projects outside China (for example in Australia and Papua New Guinea) suffered large cost overruns and slow ramp-ups. Capital must be repaid with interest, so the **cost of capital** matters as much as the engineering. A plant financed at 4% (a Chinese policy bank) has a very different cost per tonne from one financed at 12% (a frontier-market project).
2. **Energy**: electricity and heat. Indonesia's nickel parks run largely on **captive coal power**, coal plants built inside the industrial park, which gave them cheap power but high carbon emissions.
3. **Reagents**: sulphuric acid, lime, coal or coke as a reductant, solvents. HPAL uses very large quantities of sulphuric acid, made by burning sulphur. The IEA reports that the 2026 Middle East conflict disrupted sulphur supplies (the region supplies about one-quarter of global sulphur, and half of seaborne sulphur trade passes through the Strait of Hormuz), China curbed sulphuric acid exports in May 2026, and "acid costs overtaking energy costs" became the largest cost item in some critical mineral processing (IEA GCMO 2026). An unverified industry report (August 2026) puts about 80% of Indonesia's sulphur imports from the Middle East.
4. **Labour and skills**: fewer workers than mining per tonne, but more specialized: metallurgists, chemical engineers, instrumentation technicians.
5. **Feedstock**: the ore or concentrate itself, bought from mines (or from the company's own mine).
6. **By-product credits**: revenue from secondary products such as sulphuric acid, gold, silver, cobalt, or minor metals. These can make or break a smelter.

The scale and learning trap. Processing has strong **economies of scale** (a bigger furnace or autoclave costs less per tonne) and strong **learning curves** (the tenth plant built by the same team is far cheaper and faster than the first). That is why the IEA finds capital costs for refining projects are "20% to over 150% higher outside the dominant supplier," and operating costs about 50% higher on average, "driven by feedstock and energy prices," compounded by "technical and skills constraints, infrastructure gaps, and lengthy permitting processes" (IEA GCMO 2026). This is the heart of the downstreaming dilemma: the country that wants to start processing is, almost by definition, the high-cost newcomer facing all these handicaps at once.

Thin margins and the treatment-charge squeeze. Month 5 showed that copper smelter fees (treatment and refining charges) fell to a benchmark of $0 per tonne in 2026, with spot charges negative since 2024, because China built smelting capacity faster than mines produced concentrate (IEA GCMO 2026). Work through what this means for a new smelter:

- A custom smelter buys concentrate containing 1 tonne of copper (with 25% copper concentrate, that is 4 tonnes of concentrate). Traditionally it might earn, say, $80 per tonne of concentrate in treatment charge (4 x $80 = $320) plus a refining charge of 8 cents per pound of copper (about 2,205 pounds x $0.08 = $176), roughly $500 per tonne of copper in total (illustrative numbers in line with historical benchmarks).

- If those fees go to zero, the smelter's processing revenue on the copper itself is zero. It must cover all its operating costs from by-products (acid, precious metals) and any "free metal" (small amounts of metal it recovers beyond what it pays the miner for).
- The IEA warns that "many custom smelters outside China could face growing economic pressure," with utilization outside China below 70% in 2025.

So a country that forces its copper miners to smelt locally, without having lower costs than Chinese smelters, may simply transfer losses to its own smelter. Downstreaming is not automatically value-creating; it depends on whether the local plant is competitive.

Why might a country still want processing? Four good reasons, even when margins are thin:

1. **Linkages.** A smelter creates demand for power, logistics, maintenance, chemicals and engineering services, and supplies inputs to the next stage (stainless steel, battery materials). Economists call these **backward and forward linkages**. They are where long-run industrial capability is built (Month 3's lesson from Korea and Taiwan).
2. **Bargaining power.** A country that refines is harder to bypass than one that only ships ore. Indonesia's position in nickel is far stronger than Guinea's in bauxite.
3. **By-products and strategic minerals.** Smelters recover cobalt, gallium, germanium and other minor metals that the IEA now calls strategically important.
4. **Fiscal capture.** A processing industry in the country can be taxed more easily and transparently than ore exports priced by foreign buyers through opaque transfer prices.

And four bad reasons, which are common:

1. **Prestige** and political slogans ("we will no longer export dirt").
2. **Mistaking export value for value added**, as explained above.
3. **Rent-seeking**: export bans create valuable licences and quotas that politically connected firms can capture.
4. **Ignoring opportunity cost**: the capital, power and skilled workers used in a marginal smelter could earn more elsewhere.

Case study: China's copper smelting build-out and the negative-fee world

Between 2005 and 2025 China accounted for over 90% of the growth in world copper smelting, raising its share of global smelting capacity from about 15% to 50% (IEA GCMO 2026). China mines relatively little copper (about 1.8 million tonnes in 2025), but refined about 14 million tonnes, roughly 48% of the world total (USGS MCS 2026). It achieved this by importing concentrate from Chile, Peru, the DRC and elsewhere and processing it at home.

Why did China build so much? Several mechanisms:

- **Security of supply.** As the largest copper consumer, China wanted control of the refined metal its factories need.
- **Policy finance and local government incentives.** Provincial governments competed for large industrial projects that bring jobs, tax and GDP growth, often with land, power and financing support.
- **By-products and clustering.** Chinese smelters became hubs that also recover gold, silver, sulphuric acid and minor metals, which feed other Chinese industries (fertilizer, chemicals, electronics).
- **Scale and learning.** Repeated construction by Chinese engineering firms drove costs down; Chinese companies also became exporters of smelting technology and construction services.

The consequence for everyone else. When smelting capacity outgrows mine supply, smelters bid against each other for concentrate, and treatment charges collapse. In 2026 the annual benchmark was settled at zero. Smelters in Japan, Korea, Europe, India and elsewhere saw their main revenue stream disappear; some cut output. The IEA notes that zinc and lead smelter fees also turned negative. Meanwhile miners capture more of the copper price. **In this market, the midstream is overbuilt and value has shifted upstream to the mines.**

What this means for downstreaming countries. Chile, Peru and the DRC all face political pressure to smelt more copper at home. In this market environment, a new smelter in Chile or Peru would have to compete for concentrate against Chinese smelters willing to accept zero or negative fees. It would need either much lower costs, strong by-product revenue, guaranteed feedstock from a domestic mine, or a subsidy. Indonesia's approach, discussed next week, used a different tool: since 2023-2024, Indonesia has required copper concentrate to be processed domestically, which forced Freeport Indonesia to build the Manyar smelter in Gresik, East Java: a roughly $4.2 billion plant designed to treat 1.7 million tonnes of concentrate and make about 600,000 tonnes of copper cathode a year, described as the world's largest single-line copper smelter. Its short life shows how fragile a forced midstream can be: a fire in October 2024, a restart in May 2025, then a halt when the September 8, 2025 mud rush at Freeport's Grasberg mine cut concentrate supply. Feeding resumed in August 2026 and about 5,000 tonnes of cathode came out in September 2026, with full operation tied to Grasberg's recovery, targeted for late 2027 (CNBC Indonesia, September 15, 2026; Antara, October 2026). The older PT Smelting plant also halted for about two weeks from August 8, 2026 for furnace repairs (Reuters via Mining.com, August 2026). A captive feedstock solves the competition problem, but the cost is borne by the mining company and, through lower taxes or dividends, by the state.

The deeper lesson. Chinese overcapacity is a recurring pattern: in solar panels, steel, aluminium, batteries, and now copper smelting. When China decides a stage is strategic, it can push that stage into oversupply, collapsing margins for everyone, including Chinese firms, but leaving China with the largest and most experienced industry when the shakeout ends. A newcomer country that targets the same stage must expect to fight that dynamic. It is often wiser to target a stage where China is not overbuilding, or where geography gives a lasting cost advantage (cheap clean power, ore that cannot be exported).

Key numbers

- China share of global copper smelting capacity: about 15% in 2005, 50% in 2025; over 90% of growth since 2005 (IEA GCMO 2026).
- Copper smelter benchmark treatment charge, 2026: $0 per tonne (IEA GCMO 2026).
- Capital cost of refining projects outside the dominant supplier: 20% to over 150% higher; operating costs about 50% higher (IEA GCMO 2026).
- World primary aluminium production 2025: about 74 Mt; China about 45 Mt (USGS MCS 2026).
- Guinea bauxite output 2025: about 150 Mt (34% of world); alumina output about 0.36 Mt (USGS MCS 2026).
- China alumina output 2025: about 93 Mt of the world's 150 Mt (USGS MCS 2026).
- Middle East share of global sulphur supply: about one-quarter; share of seaborne sulphur through Hormuz: about half (IEA GCMO 2026).
- Indonesian nickel ore (1.6% Ni) reference price late October 2026: about $57 per wet tonne; LME nickel about $15,500/t (SMM; Yieh, November 2026).

Think like a founder

Processing plants themselves are for large firms and states, but processing economies are full of founder-sized problems:

- **Energy and acid as a service.** Every smelter cluster needs reliable power, increasingly low-carbon power, and reagents. A company that supplies renewable power purchase agreements, battery storage, or on-site acid regeneration to smelters captures value from the cost items that dominate processing.
- **Process optimization software.** AI and machine learning applied to furnace control, autoclave uptime, reagent dosing and predictive maintenance can raise output and cut costs by a few percent. On a plant with $1 billion of annual revenue, 2% is $20 million a year.
- **Skills and training.** The IEA lists skills shortages as a major barrier outside China. A training company that produces certified metallurgical technicians for Indonesia or the Copperbelt fills a real gap; so does a recruitment and placement business for such workers.
- **By-product recovery.** Smelters increasingly live on by-products; recovering gallium, germanium, cobalt or scandium from residues turns waste into revenue.

The founder's rule: in processing, value goes to whoever reduces the dominant cost (energy, acid, capital, downtime) or creates a new revenue stream from what others throw away.

Week 2: Indonesia's nickel ban: the biggest downstreaming experiment

Core lesson

The policy. Indonesia's 2009 Mining Law (Law No. 4/2009) required domestic processing of minerals, with a five-year transition. In January 2014 Indonesia banned exports of nickel ore. In 2017 it partly relaxed the ban, allowing limited exports of low-grade ore (below 1.7% nickel) for companies building smelters. In 2019 it announced that the full ban would return from January 1, 2020, two years earlier than planned (WTO panel report, DS592, 2022). The logic: Indonesia has the world's largest nickel reserves (62 million of more than 140 million tonnes, USGS MCS 2026). If foreign buyers want Indonesian nickel, they must come and process it in Indonesia.

The investors who came. Chinese companies, not Western miners, came at scale. Tsingshan Holding Group, a private Chinese stainless steel maker, partnered with Indonesia's Bintang Delapan Group to build the **Indonesia Morowali Industrial Park (IMIP)** in Central Sulawesi from 2013 onward. The **Indonesia Weda Bay Industrial Park (IWIP)** in North Maluku and parks in Southeast Sulawesi followed, with firms such as Huayou, Lygend, GEM and CNGR building RKEF and HPAL plants. Research by Transparency International Indonesia and C4ADS, cited by CSIS (Center for Strategic and International Studies), estimates that Chinese firms, often through affiliates in Hong Kong and Singapore, control about 75% of Indonesia's nickel refining capacity (CSIS, September 2025). IMIP's own disclosure lists its shareholders as Shanghai Decent Investment Group (a Tsingshan subsidiary) with 49.69%, PT Bintang Delapan Investama with 25.31%, and PT Sulawesi Mining Investment with 25% (IMIP website, FAQ). Because Sulawesi Mining Investment is itself a Tsingshan-Bintang Delapan joint venture, analysts such as the Business and Human Rights Resource Centre calculate Tsingshan's effective stake at about 66% and Bintang Delapan's at about 34%.

Why Chinese firms? Four reasons: they had the RKEF and HPAL know-how (built in China in the

2000s and 2010s); they had a guaranteed customer (China's stainless steel and battery industries); they had access to cheap long-term financing and Chinese equipment; and they were willing to build in remote islands with captive coal power and imported workers, moving fast with tolerance for risk. Western firms were slower and more cautious about environmental, social and governance (ESG) risks.

The results: what went right. By almost any headline measure, the ban achieved its stated goal:

- **Production.** Indonesia's nickel mine output reached an estimated 2.6 million tonnes in 2025, about 67% of world supply, up 13% in 2025 alone (USGS MCS 2026). CSIS reports Indonesia accounted for 58% of global production in 2024.
- **Smelters.** The number of nickel smelters rose from 2 in 2014 to 54 operating by mid-2025, with 120 more under construction and 16 in planning (NEXT Indonesia Center, June 2025, citing the Ministry of Energy and Mineral Resources). Counts differ by source and definition: the government reported 48 operating in September 2023, and Transparency International Australia counted "over 60" operating or under construction in 2025, so treat any single number as approximate.
- **Exports.** Former President Joko Widodo said the policy raised the export value of nickel from Rp 17 trillion to Rp 510 trillion (cited by CSIS). Ferronickel exports rose from $2.9 billion in 2019 to $4.7 billion in 2020 and $14.1 billion in 2024, and total nickel-product exports reached $22.1 billion in 2024 (NEXT Indonesia Center, June 2025). Figures built directly from Statistics Indonesia (BPS) customs codes differ a little (Kompas). MHP exports, which began in 2021, reached $2.66 billion in 2023, giving Indonesia 77.3% of world MHP exports (ERIA, citing UN Comtrade).
- **Investment and jobs.** Foreign direct investment (FDI) into Indonesia rose 44.2% between 2021 and 2022 to $45.6 billion (CSIS). IMIP alone hosted 53 companies and employed about 86,000 Indonesians, 16,000 Chinese workers and 24,000 contractors in 2025 (Rest of World, July 2025, citing an IMIP spokesperson).
- **Moving into batteries and stainless steel.** Indonesia became a major stainless steel producer; CSIS reports Chinese firms drove Indonesian stainless steel output from 4 million to 7.5 million tonnes between 2020 and 2024. The government and investors launched battery projects, including a CATL-linked integrated battery project of about $6 billion that broke ground in 2025 (National Bureau of Asian Research, January 2026). And the USGS notes a Chinese company's new spherical graphite facility in Central Java, showing battery mid-stream steps moving into Indonesia.

The results: what went wrong, or is still unresolved.

- **Dependence on one country.** China supplies much of the capital, technology, equipment and buyers. CSIS reports that in 2024 China took 82% of Indonesia's nickel exports, supplied about 70% of Indonesia's mining machinery imports and 80-90% of its refining machinery imports, and supplied much of the sulphuric acid for HPAL. The buyer concentration gives Chinese purchasers what economists call **monopsony** power (the power of a dominant buyer). Indonesia moved up the chain, but much of the profit, technology and decision-making stayed with Chinese firms.
- **Mostly low-grade products.** CSIS notes that Indonesia's refining industry mainly produces lower-grade products (nickel pig iron, ferronickel, matte, MHP) that need further processing, much of it in China, to become battery-grade nickel sulphate or cathode material. The highest-value steps, cathode and cell manufacturing, have been slower to come.
- **Oversupply and low prices.** Indonesia's success flooded the market. The global primary

nickel market has been in surplus every year since 2022 (International Nickel Study Group, via USGS MCS 2026). LME nickel fell from over $48,000 per tonne in early 2022 to about $15,000 by mid-2025 (Rest of World, 2025). This hurt competitors (Australia's output fell 54% in 2025), but it also cut Indonesia's own export earnings per tonne and royalties. In 2026 the government cut ore mining quotas (RKAB) from 379 million wet tonnes approved for 2025 to 260-270 million, then revised them to 290-300 million in October. Prices rose in the first half of 2026 (LME nickel briefly reached $20,000 per tonne in May 2026, according to Mining.com), then fell back. In November 2026 the energy minister, Bahlil Lahadalia, publicly blamed Indonesian overproduction for low prices, estimated global oversupply at about 800,000 tonnes, and urged producers to cut output (Antara and S&P Global, November 2, 2026).

- **Environment.** The parks run on captive coal power, which makes Indonesian nickel among the most carbon-intensive in the world and raises concerns for buyers subject to carbon rules. Deforestation, sediment runoff into coastal waters and fisheries, and tailings disposal (including HPAL tailings) are major issues. One researcher's community health survey near IMIP reported high asthma rates among young children and a rise in pulmonary disease near smelters, and recorded at least 21 flooding and mudslide incidents in the Morowali region in 2022 (New America, September 2025; this is one author's fieldwork, not an official survey). In September 2026 IMIP said water shortages had forced some nickel pig iron production cuts (Mining.com, September 2026).
- **Labour and safety.** IMIP and other parks have suffered serious industrial accidents, including a furnace explosion at the PT Indonesia Tsingshan Stainless Steel plant inside IMIP on December 24, 2023, which killed 21 workers, 8 of them Chinese, during furnace repairs (AP, February 2024), as well as labour disputes, including tensions between Indonesian and Chinese workers. Rest of World reported layoffs in 2025 as low prices and the shift to LFP batteries hit demand.
- **Trade law.** The European Union challenged the ban at the WTO (World Trade Organization). In November 2022 a WTO panel ruled that the export ban and domestic processing requirement violated Article XI of GATT (the General Agreement on Tariffs and Trade) and were not justified by any exception (WTO DS592 panel report, November 30, 2022). Indonesia appealed on December 8, 2022, but because the WTO Appellate Body has been non-functional since 2019, the appeal went "into the void," and the ban stayed in place. This shows a structural fact of 2020s trade: **the WTO can rule against export bans but cannot currently enforce its rulings when the losing country appeals.**

Who captured the value? Let us be precise about the driving question.

Actor	What they gained	What they gave up or risk
Indonesian state	More tax base, FDI, exports, jobs, geopolitical relevance	Lower royalties per tonne as prices fell; environmental liabilities; dependence on China
Indonesian workers and regions	Tens of thousands of jobs in Sulawesi and Maluku; local business	Pollution, accidents, rising local costs, layoffs in downturns
Chinese firms (Tsingshan, Huayou and others)	Control of the world's lowest-cost nickel supply; profits; market power	Exposure to Indonesian policy changes; price collapse
Chinese battery and stainless industries	Cheap nickel inputs	Little

Actor	What they gained	What they gave up or risk
Rival producers (Australia, New Caledonia, Philippines)	Little	Mine closures, job losses
Global battery and stainless buyers	Lower prices	Concentration risk; carbon and ESG concerns

Judgment: Indonesia's nickel policy is the most successful downstreaming policy of the 21st century so far, measured by output, investment and exports. It worked because Indonesia had something no buyer could do without (the world's largest, cheapest nickel ore), met an investor (Tsingshan) with the technology and the market, and stayed firm under international pressure. But much of the value was captured by Chinese firms and by global buyers through lower prices, and the environmental costs fall on Indonesian communities. The next test is whether Indonesia can turn a processing enclave into an industrial ecosystem with Indonesian-owned firms, Indonesian engineers, cleaner power and access to markets beyond China.

Case study: The bauxite sequel and the limits of copying yourself

After nickel, Indonesia applied the same tool to bauxite (aluminium ore): exports of bauxite were banned from June 2023. Did it work as well?

The facts. Indonesia's bauxite production "declined sharply following Indonesia's 2023 ban on bauxite exports" (USGS MCS 2026). Indonesia produced about 10 million tonnes of bauxite in 2025, far below Guinea's 150 million or Australia's 97 million. Alumina refining capacity grew slowly: in April 2025 a 1-million-tonne-per-year alumina refinery in Mempawah, West Kalimantan, shipped its first alumina to an aluminium smelter in North Sumatra, and a 500,000-tonne-per-year primary aluminium smelter in North Kalimantan was expected to begin operating by the end of 2025 (USGS MCS 2026). By mid-2026, bauxite investment was rising sharply, overtaking nickel investment in one quarter according to an industry report.

Why was bauxite harder? Compare the conditions:

1. **Market power.** In nickel, Indonesia had about a fifth of world mine supply before the ban and the largest reserves. In bauxite, Indonesia was a smaller player (about 5% of global production in 2023, according to an LPEM and NRGI policy paper), and buyers could switch to Guinea or Australia. A ban by a small supplier mostly hurts the supplier.
2. **Energy intensity.** Aluminium smelting needs enormous, cheap and reliable power. Indonesia's best low-cost options are coal (carbon-intensive) and hydropower in Kalimantan and Sumatra, which takes years to build.
3. **The investor match.** Nickel downstreaming matched Indonesia with Chinese stainless and battery firms that urgently wanted the ore. In aluminium, China already had huge alumina and smelting capacity and plenty of Guinean bauxite, so the pull was weaker.

Lesson. An export ban is only powerful if the rest of the world cannot easily replace you. Downstreaming worked in nickel because Indonesia was indispensable; it was much slower in bauxite because it was not. When you evaluate any country's ban, the first question is: **what share of world supply does it control, and how fast can buyers find substitutes?**

Key numbers

- Indonesia nickel mine output: about 2.6 Mt in 2025, about 67% of world (USGS MCS 2026).
- Indonesia nickel reserves: 62 Mt of more than 140 Mt worldwide (USGS MCS 2026).
- Nickel smelters in Indonesia: 2 in 2014, about 54 operating by 2025 (NEXT Indonesia Center, 2025).
- Ferronickel exports: $2.9 billion in 2019, $14.1 billion in 2024 (NEXT Indonesia Center, 2025).
- Indonesia share of world MHP exports: 77.3% in 2023 (ERIA, citing UN Comtrade).
- Chinese-controlled share of Indonesian nickel refining capacity: about 75% (CSIS, 2025, citing C4ADS).
- Share of Indonesia's nickel exports going to China: 82% in 2024 (CSIS, 2025).
- IMIP workforce: about 86,000 Indonesians, 16,000 Chinese workers, 24,000 contractors (Rest of World, 2025).
- Nickel ore mining quota (RKAB): 379 million wet tonnes approved for 2025 (320.37 million actually mined); 260-270 million initially for 2026, revised on October 29, 2026 to 290-300 million (Ministry of Energy and Mineral Resources, via Indonesia Business Post).
- WTO panel ruling against the nickel export ban: November 30, 2022; Indonesia appealed December 8, 2022 (WTO DS592).

Think like a founder

Indonesia's $20-billion-plus nickel export sector, built in under a decade, has gaps a founder with local knowledge could fill:

- **Decarbonizing the parks.** Buyers in Europe, the US, Japan and Korea increasingly want low-carbon nickel. Solar, storage, and grid connections for nickel parks, plus carbon measurement and certification, are large markets. Whoever qualifies Indonesian nickel for premium buyers captures part of a "green premium."
- **Local supplier substitution.** CSIS shows that 70-90% of machinery comes from China. Maintenance, spare parts, refurbishment, and eventually local manufacture of simpler equipment are classic import-substitution opportunities.
- **Traceability and ESG data.** Battery passport rules in the EU require supply-chain data. A software company that tracks ore from mine to smelter to cathode, with satellite and sensor data, serves both miners and buyers.
- **Community and environmental services.** Water treatment, tailings monitoring, clinics and training in Sulawesi and Maluku, increasingly demanded by regulators and buyers.

Caution: the nickel price cycle and policy shifts (quotas, royalties, reference prices) can hit any business tied to the sector. Build businesses that serve the industry in both booms and busts, such as maintenance, water, and compliance, rather than ones that only work at high nickel prices.

Week 3: Africa's bargaining power: the DRC, Zambia, Zimbabwe, Botswana, Guinea and Morocco

Core lesson

A short history of the bargain. Month 2 told the colonial part of this story. After independence, many nationalized mines collapsed under debt and mismanagement (Zambia's copper in the 1980s and 1990s, the DRC's Gecamines in the 1990s); in the 1990s and 2000s, under creditor pressure,

countries privatized and wrote investor-friendly codes with low royalties. Production recovered, but governments felt they received too small a share. Since about 2010, and very strongly since 2020, the pendulum has swung back toward the state: higher royalties, state equity stakes, local processing requirements, and export controls.

Economists describe this swing with the **obsolescing bargain**. Before a mine is built, the investor has the power: it can choose among many countries, and the host government needs its capital. After the mine is built, the capital is sunk in the ground and cannot leave, so the government gains power and can renegotiate. Governments that renegotiate too aggressively scare off the next investor; governments that never renegotiate leave money on the table. The art is to use bargaining power at the right moment, predictably, and within the law.

The DRC: using a near-monopoly. The DRC mines about 73% of the world's cobalt and is the second-largest copper miner (3.2 million tonnes in 2025) (USGS MCS 2026). Its 2018 mining code raised royalties and classified cobalt as a "strategic substance" with a higher royalty. In 2024-2025, a glut from DRC copper-cobalt mines (especially CMOC's Tenke Fungurume and Kisanfu) and from Indonesian HPAL had pushed cobalt prices to multi-year lows. In February 2025 the DRC's regulator for strategic minerals, ARECOMS, suspended cobalt exports; in October 2025 the suspension was replaced with quotas: 18,125 tonnes of contained cobalt for the rest of 2025, and 96,600 tonnes per year for 2026 and 2027 (a base quota of 87,000 tonnes, or 7,250 tonnes per month, plus a 9,600-tonne strategic quota at ARECOMS's discretion) (USGS MCS 2026; IEA policy database). For comparison, the DRC mined about 230,000 tonnes in 2025, so the quota is well under half of recent output.

The results show both the power and the limits of a monopoly producer:

- **Prices rose sharply.** The IEA reports cobalt prices rose about 130% between January 2025 and April 2026, "largely due to export restrictions" by the DRC. Prices reached nearly $58,000 per tonne in the first quarter of 2026 (Ecofin, citing IEA data).
- **Miners adapted.** Glencore's own-sourced cobalt output fell 52% in the first nine months of 2026 to 13,700 tonnes, with cobalt "increasingly being retained in solution" (that is, left unrecovered or stored) because of the quotas and a focus on copper (Mining.com, October 2026). The IEA says a projected cobalt supply gap has emerged for the next decade because of the quota.
- **Then prices fell again.** Exports bunched up as quota deadlines approached (DRC exports of contained cobalt were about 41,140 tonnes in the first half of 2026), Chinese buyers drew on inventories, and battery demand for cobalt-rich cells was weak. Cobalt hydroxide prices fell about 31% in the third quarter of 2026, and the DRC's own weekly export reference price fell from about $55,600 per tonne in August to about $38,600 in late October 2026 (Bankable/Benchmark Mineral Intelligence; The Oregon Group, citing the DRC Ministry of Foreign Trade, October 2026). ARECOMS said it might reduce quotas further "if the imbalance between supply and demand were to persist or deteriorate" (Fastmarkets, August 2026).

The DRC's deeper goal is to refine more cobalt and copper at home, and to attract battery precursor plants. Its regulator has said that volumes above the quota "can go to the strategic reserve or they can be processed in the DRC if investors want to come" (S&P Global, May 2026). The obstacles are severe: unreliable electricity, very long transport routes to the coast, insecurity in the east, and a governance environment that investors see as risky. These are exactly the conditions Week 1 identified as raising processing costs.

Zambia: the copper comeback and the debt trap. Zambia, the DRC's neighbour on the Copperbelt, produced about 940,000 tonnes of copper in 2025, up from 823,000 in 2024, and refined about 270,000 tonnes (USGS MCS 2026). After nationalization, collapse, privatization and swings in tax policy, Zambia became in 2020 the first African country to default during the COVID-19 pandemic, followed by a long G20 Common Framework restructuring. The government now targets 3 million tonnes of copper a year by 2031. Note two things about the numbers. First, Zambia's mines ministry reported 2025 output of 890,346 tonnes, up 8% from 825,513 tonnes and short of its own 1-million-tonne target for the year (Reuters, January 27, 2026); the USGS estimate of 940,000 tonnes above is an earlier estimate on a slightly different basis. Second, going from about 0.89 million to 3 million tonnes in six years requires growth of about 22% a year, every year: (3 / 0.89) raised to the power 1/6 is about 1.224. Output in the first half of 2026 grew less than 1%. Treat the target as an aspiration that signals policy direction, not as a forecast. Zambia also borrows to join the Lobito Corridor: the African Development Bank approved a $255 million loan and $10 million grant in August 2026 (AfDB; the parliamentary brief gives about $252 million). Zambia's lesson: **policy volatility itself is a cost.** Each sudden tax change raised the risk premium investors demanded, and so reduced the long-run investment and revenue the policy was meant to increase.

Zimbabwe: from raw ore ban to sulphate plants. Zimbabwe became Africa's largest lithium producer in just a few years, reaching about 28,000 tonnes of lithium content in 2025, about 10% of world output (USGS MCS 2026). Chinese companies bought most of the main deposits during the 2021-2023 boom: Huayou Cobalt (Arcadia), Sinomine (Bikita), Yahua (Kamativi) and others. Zimbabwe's policy moved in steps:

1. **December 2022**: ban on exports of raw lithium-bearing ore without a permit, the first such ban in Africa (IEA policy database).
2. **June 2025**: announcement that lithium concentrate exports would be banned from January 1, 2027, so that only lithium sulphate or higher products could be exported (IEA policy database).
3. **February 2026**: sudden suspension of all raw mineral and lithium concentrate exports, citing "malpractices and leakages" (under-declaration of value); in April 2026 exports resumed under company quotas, with a 10% export tax on concentrate and written commitments to build sulphate plants (Reuters, April 2026). The 10% rate comes from Zimbabwe's Finance Act No. 7 of 2025, in force from January 2026, which zero-rates lithium sulphate to reward processing. You may see "16%" quoted: that is not the tax rate but producers' own calculation of the combined burden of the 10% tax plus a 3% export levy, because neither can be deducted when corporate income tax is assessed (Prospect Lithium Zimbabwe, quoted by Mining Zimbabwe, October 2026).
4. **April 2026**: Huayou's Prospect Lithium Zimbabwe shipped Africa's first consignment of lithium sulphate from its $400 million Arcadia plant, with capacity of about 50,000 tonnes a year (Reuters, April 2026).

By autumn 2026 only one sulphate plant was operating, it could not process other miners' concentrate, Sinomine's planned plant at Bikita had slipped to July 2027, and the government refused to delay the January 2027 ban (Reuters, July 2026; Africa Briefing, October 2026). Zimbabwe illustrates a middle case: it forced a first step of processing (concentrate to sulphate), which adds real value and jobs, but the step is still an intermediate that goes mainly to China for conversion into battery-grade chemicals, the plants are Chinese-owned, and the abrupt policy changes created uncertainty and stranded concentrate. The lithium price bust and rebound (Month 5) made the timing even harder.

Botswana: the patient bargainer. Botswana, Month 2's success story, runs its diamond mines through **Debswana**, a 50-50 joint venture with De Beers, and owns 15% of De Beers itself. Diamonds

were 72.7% of goods exports in the third quarter of 2025 (bne IntelliNews, citing the central bank), about a third of government revenue and roughly a quarter of GDP (estimates vary by source and year). Botswana has also pursued beneficiation: in 2013 De Beers moved its global rough diamond sorting and sales operation from London to Gaborone, and the government's own Okavango Diamond Company (ODC) sells a share of Debswana output independently.

In February 2025, after tough negotiations, Botswana and De Beers signed a new ten-year sales agreement and extended Debswana's mining licences by 25 years to 2054. ODC's share of Debswana output rises from 30% (up from 25% before) for the first five years, to 40% in years six to ten, and to 50% in any five-year extension; De Beers committed BWP 1 billion (about $75 million) upfront to a Diamonds for Development Fund, with potential contributions up to BWP 10 billion over ten years (De Beers 2025 sustainability report; bne IntelliNews, September 2026; JCK, June 2025). Botswana is now weighing a larger stake in De Beers as Anglo American sells its 85% holding (Reuters, November 2025; bne IntelliNews, November 2026).

But Botswana's bargaining power is being undercut by technology: **lab-grown diamonds** have collapsed natural diamond demand and prices. Debswana's production fell to about 17.9 million carats in 2024 and was planned at about 15 million carats in 2025, and foreign-exchange reserves fell sharply (Reuters, June 2025; industry analysis, November 2026). Lesson: **a strong bargain over a product whose demand is collapsing is still a weak position.** Bargaining power comes from being indispensable, and substitution can remove it.

Guinea: when the state calls the investor's bluff. Guinea holds the world's largest bauxite reserves (7.4 billion of 29 billion tonnes) and mines about a third of world output, but refines almost none (USGS MCS 2026). For years governments required bauxite miners to build alumina refineries, and companies promised them but rarely delivered. In August 2025, "following a dispute over the construction of an alumina refinery," Guinea's government revoked the bauxite mining licences of a subsidiary of Emirates Global Aluminium (EGA), a United Arab Emirates-based producer, and reallocated the concessions to a state-backed mining company (USGS MCS 2026). This is resource nationalism at its sharpest: the state used licence revocation to enforce a processing promise. Whether it brings a refinery or simply scares future investors depends on what happens next, and Guinea's military-led government (in power since a 2021 coup) adds political risk that investors price in.

What explains the differences? Line up the cases:

Country, mineral	Share of world mine supply (2025)	Main tool	Outcome so far
DRC, cobalt	73%	Export ban, then quotas	Prices up sharply, then partly reversed; little new domestic refining yet
Zambia, copper	about 4%	Tax changes, then stability; corridor investment	Output rising after a decade of volatility
Zimbabwe, lithium	about 10%	Ore ban, concentrate ban, export tax, quotas	First sulphate plant; others delayed; Chinese-owned

Country, mineral	Share of world mine supply (2025)	Main tool	Outcome so far
Botswana, diamonds	major producer (Debswana)	Joint venture and negotiated sales share	Strong bargain, but demand collapse from lab-grown diamonds
Guinea, bauxite	34%	Licence revocation to enforce refinery promises	Uncertain
Indonesia, nickel (Week 2)	67%	Ore export ban	Massive processing build-out, mostly Chinese-owned

(Shares from USGS MCS 2026, my calculations; Zambia's copper share is 0.94 of 23 Mt.)

Three variables explain most of the variation: **market share** (can buyers go elsewhere?), **processing cost** (does the country have cheap power, acid, logistics and skills, or a partner who brings them?), and **policy credibility** (is the rule stable enough that investors will sink capital?). We formalize this in Week 4.

Case study: Morocco's OCP, a state champion that climbed the chain

If Guinea shows a country with ore but no processing, Morocco shows the opposite: a state company that turned a mineral into an industrial empire.

The resource. Phosphate rock is the source of phosphorus, one of the three main plant nutrients (with nitrogen and potassium). Without phosphate fertilizer, global food production would fall sharply; it cannot be substituted. Morocco (including the disputed territory of Western Sahara, a political issue investors and buyers must take into account) holds about 50 billion tonnes of the world's 73 billion tonnes of phosphate rock reserves, about 68% (USGS MCS 2026). China produces more phosphate rock (about 110 million tonnes in 2025 versus Morocco's 36 million) but uses most of it at home and has restricted fertilizer exports at times.

The company. OCP (originally Office Cherifien des Phosphates), founded in 1920 under the French protectorate, is wholly state-controlled. For decades it mainly exported phosphate rock. From the 1960s and especially in the 2000s and 2010s under chief executive Mostafa Terrab, it moved downstream: it built large phosphoric acid and fertilizer complexes (notably at Jorf Lasfar on the Atlantic coast), a slurry pipeline replacing rail transport of rock, and a global sales network, especially to Africa, India and Brazil. It now sells mostly processed products (phosphoric acid and fertilizers such as DAP, MAP and TSP) rather than raw rock.

The numbers. In 2025 OCP reported revenue of MAD 113.9 billion (about $12.27 billion), up from MAD 97.0 billion in 2024, and EBITDA (earnings before interest, taxes, depreciation and amortization) of MAD 43.2 billion (about $4.62 billion), a 38% margin (OCP full-year 2025 results). Fertilizer revenue grew 15%, helped especially by exports to India, and volumes of triple superphosphate (TSP), a fertilizer that does not require ammonia, rose 48% and reached 30% of fertilizer export sales, up from 22% in 2024, with OCP citing strong TSP demand from East Asia and Brazil (OCP full-year 2025 results, April 1, 2026). Net debt was MAD 119 billion, 2.76 times EBITDA. Fitch Ratings expects OCP's fertilizer capacity to rise from 15.4 million tonnes to 20 million tonnes by 2027 (Fitch, November 2025).

The next step: green inputs and batteries. OCP's main imported input is ammonia, made from natural gas, which exposes it to gas price shocks (Month 4). Its response is a roughly $13 billion green investment programme for 2023-2027: 100% renewable electricity by 2027, desalinated and treated water (it says it has used 100% non-conventional water since early 2025), and a target of up to 1 million tonnes of green ammonia, made from renewable hydrogen; a Tarfaya green ammonia project was estimated at about $7 billion (OCP press releases; 7news Morocco, August 2026). OCP also aims to supply phosphate materials for LFP batteries, since the "P" in LFP is phosphate. Month 5 showed that the shift to LFP made phosphate a battery mineral.

Why did OCP succeed where many state miners failed?

1. **Dominant, low-cost reserves.** Like Indonesia in nickel, Morocco is hard to bypass in the long run.
2. **Long-term, commercially run strategy.** OCP is state-owned but managed with market discipline, issues international bonds, publishes audited results and holds an investment-grade-adjacent credit rating (Fitch: BB+). It reinvests heavily.
3. **Market building.** OCP did not wait for buyers; it built customized fertilizer products for African soils and invested in distribution in African markets, creating demand for its processed products.
4. **Energy strategy.** It is attacking its biggest cost exposure (ammonia and energy) with domestic renewables, turning Morocco's sun and wind into a cost advantage.

Weaknesses and critiques. OCP's operations in Western Sahara raise legal and political issues, and some buyers and investors avoid phosphate from the territory. Fertilizer prices are cyclical; net profit fell about 13% in 2025 despite record revenue, mainly due to higher taxes. Green ammonia projects are expensive and unproven at scale.

Judgment: OCP is Africa's strongest example that a state-owned mineral company can capture processing value, but its success rests on preconditions (a dominant reserve, a disciplined management culture, access to capital, and investment in demand) that few other countries share. It is a model for Morocco-like situations, not a template that can be copied everywhere.

Key numbers

- DRC share of world cobalt mine output: about 73% (230,000 t of 310,000 t) in 2025 (USGS MCS 2026).
- DRC cobalt export quota: 96,600 t per year in 2026 and 2027 (base 87,000 t plus 9,600 t strategic) (USGS MCS 2026; IEA).
- Glencore own-sourced cobalt output, January-September 2026: 13,700 t, down 52% from 28,500 t (Glencore Q3 report, via Mining.com, October 29, 2026).
- Zambia copper mine output: 940,000 t in 2025, up from 823,000 t (USGS MCS 2026 estimate); 890,346 t according to Zambia's mines ministry (Reuters, January 2026); government target 3 Mt by 2031.
- Zimbabwe lithium output: about 28,000 t in 2025 (USGS MCS 2026); 10% export tax on concentrate from January 2026, zero on sulphate; concentrate export ban from January 1, 2027 (IEA; Reuters; Mining Zimbabwe).
- Huayou Arcadia lithium sulphate plant: $400 million, about 50,000 t per year capacity (Reuters, April 2026).
- Botswana ODC share of Debswana output: 30% for five years, then 40%, up to 50% in any extension; licences extended to 2054 (De Beers; bne IntelliNews).

- Guinea bauxite: about 150 Mt mined in 2025 (34% of world); 7.4 Bt reserves (USGS MCS 2026).
- Morocco phosphate rock reserves: 50 Bt of 73 Bt worldwide (68%) (USGS MCS 2026).
- OCP 2025 revenue: MAD 113.9 billion (about $12.27 billion); EBITDA margin 38% (OCP, 2026).

Think like a founder

African governments now insist on local processing, content and ownership but lack the firms to supply them, a founder's opening, especially for Africans and partners who respect local ownership:

- **Local content suppliers.** Mining codes in the DRC, Zambia, Zimbabwe and elsewhere increasingly require local procurement. Companies that provide engineering services, catering, transport, security, equipment maintenance or laboratory testing to mines and smelters can grow fast if they meet international standards.
- **Power for processing.** The single biggest obstacle to processing in the DRC and Zambia is reliable power. Mini-grids, solar-plus-storage for mine sites, and power trading across the Southern African Power Pool are large markets with development-bank support.
- **Agritech built on processed minerals.** OCP shows how a mineral can be turned into a product for millions of farmers. Soil testing, customized fertilizer blending and farmer financing in Africa are businesses where software and local distribution capture value downstream of the mine.
- **Compliance and traceability.** Buyers need proof that cobalt is not from child labour, that lithium is declared at its true value, that diamonds are conflict-free. Traceability platforms, auditing and certification are businesses that grow with every new rule.

A founder working in these countries must take political risk seriously: sudden bans, quota systems and licence revocations can change a business overnight. Diversify across countries, keep contracts clear and lawful, and build relationships with legitimate local partners and institutions rather than with individual politicians.

Week 4: Lithium in Chile, the Lobito Corridor, and when downstreaming works

Core lesson

Chile's lithium model: the state as partner, not owner-operator. Chile has the world's largest lithium reserves (9.2 million of 37 million tonnes) and produced about 56,000 tonnes of lithium in 2025, about 19% of world output (USGS MCS 2026). In Chile lithium has been legally reserved to the state since 1979 and is "not concessible," meaning private firms cannot simply acquire mining concessions for it; they must lease rights from the state development agency, Corfo, or sign special contracts (La Tercera, October 2026). For decades two private companies, SQM (Sociedad Quimica y Minera de Chile) and Albemarle, produced lithium in the Salar de Atacama under Corfo leases that were renegotiated in the late 2010s to give the state much higher royalties.

In 2023 President Gabriel Boric announced a **National Lithium Strategy**: the state would take a majority role in strategic salt flats through public-private partnerships, while other salt flats would be opened to private companies under special contracts. The flagship deal is the Codelco-SQM partnership, signed in May 2024 to develop the Salar de Atacama from 2025 to 2060. The joint

company, Novandino, was created in December 2025: Codelco holds 50% plus one share, SQM runs operations until 2030 and Codelco from 2031, and SQM gains access to the salar beyond its old 2030 lease expiry (Econosur, July 2026). The Chilean government says the state will capture about 70% of the operating margin of new production in 2025-2030 and about 85% from 2031, through payments to Corfo, taxes and Codelco's profits (Government of Chile, December 2025). In the first half of 2026, Corfo received $791 million from lithium agreements, up from $150 million a year earlier, of which Novandino paid $670 million, helped by the lithium price rebound (Nueva Mineria, August 2026). In July 2026 Novandino filed for environmental approval of a roughly $3 billion "Salar Futuro" project, built around direct lithium extraction technology and reinjection of treated brine, aiming to reduce freshwater use (Mining.com, July 2026). Codelco has also partnered with Rio Tinto at the Maricunga salt flat (Codelco 50.01%, with up to $900 million of Rio Tinto investment).

Notice what Chile did and did not do. It did **not** ban exports or force a new processing stage; Chile already processes its brine into lithium carbonate and hydroxide at home (the IEA notes that Latin America refines only about one-fifth of its mined output of key energy minerals, "with the exception of lithium where integrated chemical processing has developed"). Instead, Chile used its ownership of the resource to **raise its share of the rent** (the surplus profit from a scarce resource) while keeping an experienced private operator in charge, and to push for more water-efficient technology. A new government took office in March 2026 and has been reviewing the strategy's private contracts (La Tercera, October 2026), so policy continuity is the next test.

What Chile has not achieved is moving further downstream into cathodes and cells. Earlier plans to attract cathode makers with preferential lithium prices produced little. Chile's market is far from the big battery factories in China, Korea and the US, and cathode plants want to sit near cell plants. **Geography and market access limit downstreaming even when the resource is dominant.** The IEA estimates that if Latin America refined its lithium, nickel, cobalt, graphite and rare earths locally and processed two-thirds of its copper in the region, the economic value generated could rise by nearly 50% to about $220 billion by 2035, but only by addressing "high financing costs, infrastructure gaps in power and water, technology and skills constraints" (IEA GCMO 2026).

Corridors: the infrastructure of value capture. For landlocked mineral regions, transport to the sea can cost hundreds of dollars a tonne and take weeks, so great-power competition in Africa is partly a competition over railways: the US- and EU-backed Lobito Corridor to the Atlantic versus the Chinese-rehabilitated TAZARA to the Indian Ocean (this week's case).

When does downstreaming work? A scorecard. Drawing together Months 3, 4, 5 and 6, here are eight conditions. Score each from 0 (absent) to 2 (strong).

1. **Indispensability**: the country's share of world mine supply and reserves, and how hard it is for buyers to substitute (another country's ore, another chemistry, recycling). Indonesia in nickel and the DRC in cobalt score high; Indonesia in bauxite scores low.
2. **Cost competitiveness of the processing step**: cheap and reliable energy, reagents, water, and logistics. Gulf aluminium (cheap gas) and Moroccan fertilizer (adjacent reserves, coastal plants) score high; DRC refining (weak grid, long routes) scores low today.
3. **A capable partner**: an investor with the technology, the capital and the market. Tsingshan in Indonesia, Huayou in Zimbabwe, De Beers in Botswana, SQM in Chile.
4. **Market access**: the processed product must have a buyer, ideally more than one. Intermediates that only one country's factories can use leave the producer exposed to monopsony.
5. **Policy credibility**: stable, legal, predictable rules, enforced fairly. Sudden bans and revocations raise investors' required returns.

6. **Fiscal and ownership capture**: does the state (and do domestic firms) capture a meaningful share of the rent through taxes, royalties, equity and domestic ownership? Chile and Botswana score high; Indonesia's nickel parks score lower because of foreign ownership.
7. **Linkages and skills**: does the processing stage train local engineers, create local suppliers and lead to the next stage (stainless steel, cathodes, cells)?
8. **Environmental and social licence**: can the activity be done without destroying land, water, health and community trust, which would eventually stop it?

A policy with high scores on 1, 2 and 3 tends to produce processing capacity quickly (Indonesia nickel). Whether that capacity becomes national development depends on 6, 7 and 8. A policy that scores low on 1 or 2 usually fails or produces expensive, subsidized plants (Indonesia bauxite in its early years; many past African refinery promises).

Judgment: downstreaming works best as a **bargain**, not as a ban. The export ban is a tool to start negotiations when you are indispensable; it should be paired with a clear path for investors (stable taxes, infrastructure, power), requirements that raise domestic capture over time (local equity, local suppliers, skills, research), and environmental standards that buyers will accept. Countries that are not indispensable should focus instead on lowering processing costs (power, logistics, skills) and on regional cooperation, such as pooling ore from several countries into a shared processing hub, rather than on bans.

What about building robots from ore in Africa or Southeast Asia by the 2040s? Apply the scorecard to the inputs of a robot (Month 9 will go deeper):

Robot input	Key minerals	Best-placed developing region (2026 evidence)	Main gap
Structure and gears	Steel (iron ore, nickel, manganese), aluminium	Indonesia (nickel, stainless steel, growing aluminium); South Africa and Gabon (manganese)	Precision machining and reducers
Motors and actuators	NdFeB magnets (rare earths), copper	Malaysia (rare earth separation), Copperbelt and Indonesia (copper)	Magnet making know-how, heavy rare earths
Batteries	Lithium, nickel, cobalt, graphite, manganese, phosphate	Indonesia (nickel, cobalt, graphite anode in Central Java), Zimbabwe (lithium), Morocco (phosphate), Tanzania and Mozambique (graphite)	Cathode and cell plants, chemistry choice
Electronics and AI compute	Copper, gallium, germanium, silicon, many minor metals	Malaysia (chip packaging and testing), Vietnam	Advanced chips (Month 8)

Southeast Asia, led by Indonesia and Malaysia, already hosts several mid-stream links (nickel, cobalt, graphite, rare earth separation, chip packaging) next to the East Asian manufacturing cluster. Africa

holds more of the ore but fewer mid-stream links and much weaker power and transport. Judgment: by the 2040s, Southeast Asia has the better chance of hosting a fairly complete ore-to-robot chain, mainly because of proximity to the East Asian manufacturing ecosystem; Africa's best route is regional processing hubs (Copperbelt copper and cobalt, Moroccan phosphate and batteries, southern African manganese and platinum) linked by corridors, with robots and electronics assembled later as markets and skills grow. Neither region is likely to do it alone; the realistic model is partnership with whichever industrial power offers technology, capital and markets on terms that build local capability.

Worked example: following one humanoid robot back to the mine. The table above says where the inputs could come from. It does not say how much money sits at each stage, and that is what decides whether hosting a stage is worth fighting for. Let us follow one humanoid robot back down the chain. Real figures are cited; the rest are labelled assumptions, and Month 9 will refine them.

Step 1: the magnets. The Foundation for American Innovation (FAI), surveying published estimates, finds that a humanoid robot uses about 2-4.5 kg of NdFeB magnet across its roughly 30-40 joint motors, containing about 0.57-1.3 kg of neodymium and praseodymium (NdPr) and about 35-79 grams of dysprosium (FAI, September 2026). An average electric car uses about 1.4 kg. The lower figure fits robots with high-ratio gearboxes (more common in Western designs); the higher figure fits "quasi-direct-drive" joints with bigger motors and little gearing (more common in Chinese designs, where magnets are cheap). So a design choice about gears is also a choice about how exposed you are to Chinese export controls.

What is that magnet content worth? MP Materials realized about $57 per kg for NdPr oxide in the second quarter of 2025, and the US government's floor is $110 per kg (Month 5). So the NdPr in one robot is worth roughly $30 to $140, depending on design and on which price you pay. The IEA says rare earths are about 40% of a magnet's cost (IEA GCMO 2026), so the finished magnets are worth roughly 2.5 times their rare earth content: about $80 to $360 per robot.

Step 2: the battery. Tesla's 2022 Optimus prototype carried a 2.3 kWh, 52-volt pack (Tesla AI Day, as reported by IEEE Spectrum, October 2022). Take 2 kWh as a round assumption. At BloombergNEF's 2025 average pack prices of $81 per kWh for LFP and $128 for NMC, the pack costs about $160 to $260. The mined minerals inside it (Month 5, Week 2) are worth roughly $20 to $50.

Step 3: the structure and wiring. A humanoid weighs around 60 kg (Adamas Intelligence, as reported by InvestorNews, 2025). Assume about 35 kg of aluminium and steel and 3 kg of copper (assumptions). At rough 2026 metal prices of $2.5-3 per kg for aluminium, under $1 per kg for steel, and about $13 per kg for copper, this metal is worth roughly $100 to $150.

Step 4: the "brain". A robot needs an onboard AI computer. Nvidia priced its Jetson AGX Thor developer kit, aimed at humanoids, at $3,499 when it launched in August 2025, and the production module at $2,999 each in orders of 1,000 (CNBC; TechSpot, August 2025). The chip inside is made by TSMC in Taiwan from silicon, copper, gallium and other minerals worth a few dollars.

Now add it up for a robot that sells for $30,000 (an assumption in the middle of the range of announced prices, which run from under $10,000 for small Chinese models to far more for Western industrial humanoids).

Layer	Rough value per robot ($)	Share of a $30,000 robot	Where it is mostly captured (2026)
All mined and refined metal content (magnet rare earths, battery metals, aluminium, steel, copper)	about 150-350	about 0.5-1%	Mines in many countries; refining mainly China
Finished magnets	about 80-360	about 0.3-1%	China (most magnet making)
Battery pack	about 160-260	under 1%	China, Korea
Actuators (motors, precision reducers, encoders, drivers), including the magnets	assumption: 8,000-12,000	about 25-40%	China, Japan, Germany
AI computer and sensors (cameras, force sensors)	assumption: 3,000-6,000	about 10-20%	US chip designers, Taiwan fabs, Japan and China for sensors
Assembly, software, AI models, brand, sales and margin	the remainder, about 12,000-19,000	about 40-60%	Whoever owns the robot company and its models

The rows overlap on purpose: the metal content in the first row is physically inside the magnets, battery and actuators below it, so the shares do not add to 100%. The actuator and computer rows are rough assumptions, consistent with industry commentary that actuators are the largest single hardware cost block in a humanoid; Month 9 will test them properly.

Three conclusions follow, and they are the core of the driving question.

1. **The ore is the smallest number in the robot.** All the mined content together is about 1% of the price, close to the IEA's finding that critical minerals are about 3% of an EV's price and rare earths less than 1% of a car's value. A country that only mines, even a country that mines everything a robot needs, captures around 1% of the robot.
2. **The leverage sits in the middle; the money sits at the top.** Magnets are less than 1% of the robot's value, yet without them the actuators, which are a quarter or more of the value, cannot be built. That is why export controls on magnets are powerful. But the largest pools of value are the actuator, the computer and above all the software, models and brand. In 2026 those are captured in China, Japan, Germany, Korea, Taiwan and the US.
3. **The bridge from ore to robot is the actuator.** For Indonesia, Malaysia or a Copperbelt country, the realistic climb is not "mine to robot" in one jump. It is mine, then refined metal, then a component that sits inside the actuator: copper wire and motor windings, NdFeB magnets (if Malaysia's separation and Indonesia's or Vietnam's light rare earth supply can be joined to magnet-making know-how), or battery packs. A country that hosts magnet and motor making captures perhaps 5-10 times more value per robot than one that only mines, and it begins to build the skills (precision machining, motor design, power electronics) that the top of the chain needs.

Here is the arithmetic of a 2040 scenario. Suppose the world builds 10 million humanoids a year by the late 2030s (Bank of America's projection for 2035, cited by FAI) at an average price of $20,000.

That is a $200 billion a year market. A Southeast Asian bloc that supplied all of the mined content of 20% of those robots would earn about 1% of $40 billion, or $0.4 billion a year. If it also made the magnets, motors and battery packs for them, it would capture perhaps 10-15% of that $40 billion, or $4-6 billion a year, plus the engineering skills that lead to the next step. That tenfold gap, not the mines themselves, is the prize for which downstreaming policy should aim.

Judgment: for a founder, the lesson is to aim at the actuator layer and the services around it (motor design, magnet recycling, testing, maintenance), not at the mine, and to design for substitution (ferrite magnets or higher gear ratios) so that a single country's export decision cannot stop your production line.

Case study: The Lobito Corridor and the race for the Copperbelt's exit

The geography. The Copperbelt of the southern DRC and northern Zambia is one of the world's richest mineral regions and one of the most landlocked. Ore and metal travel for weeks by truck or rail, through congested borders, to Durban, Dar es Salaam, Walvis Bay, Beira or Lobito. The Benguela Railway, built under Portuguese colonial rule from Lobito into the interior in the early 20th century, was destroyed during Angola's civil war (1975-2002) and rebuilt with Chinese financing in the 2000s, but carried relatively little mineral traffic.

The Western-backed bet. In 2022-2023 the US and the European Union named the **Lobito Corridor** a flagship of their infrastructure initiatives (the US Partnership for Global Infrastructure and Investment and the EU's Global Gateway), as a way to give Western buyers a route for Copperbelt copper and cobalt that does not depend on Chinese-controlled logistics. The project has two parts:

1. **The Angolan line.** A 30-year concession for the existing roughly 1,300 km Benguela line from Lobito to the DRC border at Luau, plus the Lobito mineral terminal, was awarded to **Lobito Atlantic Railway (LAR)**, a joint venture of the commodity trader Trafigura, the Portuguese construction group Mota-Engil and the rail operator Vecturis. On December 17, 2025, LAR signed a $753 million financing package in Washington: a $553 million 15-year senior loan from the US International Development Finance Corporation (DFC) and up to $200 million from the Development Bank of Southern Africa (DBSA) (DFC press release, December 17, 2025). The Africa Finance Corporation, which advised the borrower, announced that the financing reached full financial close in early July 2026 (AFC, July 3, 2026; GTR, July 2026); a later DBSA statement gave a figure of $786 million, so expect small differences between sources as fees and facilities are counted differently. The DFC says the investment is expected to raise Lobito's transport capacity tenfold, to 4.6 million tonnes a year, and to cut the cost of moving critical minerals by up to 30% (DFC press release, December 17, 2025). LAR drew about $300 million in June 2026 and expected to double cargo volumes in 2027 (Bloomberg, September 2026). First copper-cobalt shipments to US customers via Lobito were reported in the first quarter of 2026, and refined copper anode exports from Kamoa-Kakula began moving via the corridor by June 2026 (Afrimintel; Euromining, 2026).
2. **The Zambian extension.** A new railway of roughly 830 km (about 550 km in Zambia and 280 km in Angola) would connect Zambia's Copperbelt directly to the Benguela line, creating a shorter route. The Africa Finance Corporation (AFC) leads development; in August 2026 the African Development Bank approved a $255 million loan and $10 million grant for Zambia's part of the programme (AfDB, August 2026), and in October 2026 the AFC named preferred bidders for the construction contracts: a Mota-Engil and Odebrecht joint venture and a consortium including Webuild and Ictas (International Railway Journal; Semafor, October

2026). Construction contracts and full financing were not yet complete as of late October 2026; AFC has said it targets completion around 2030.

The Chinese-backed rival. TAZARA, the 1,860 km Tanzania-Zambia Railway, was built with Chinese support in the 1970s to give Zambia an outlet that avoided white-minority-ruled Rhodesia and South Africa. By the 2020s it had decayed, carrying under 500,000 tonnes of freight a year, a fraction of its design capacity. In September 2025 the TAZARA Authority signed a 30-year concession with China Civil Engineering Construction Corporation (CCECC) for a roughly $1.2 billion rehabilitation (TM Africa; IntelliNews, October 2026). In late October 2026 Angola and Tanzania announced they would pursue a link between Lobito and TAZARA, potentially creating a rail route from the Atlantic to the Indian Ocean, with no financing or timetable yet (Zitamar, November 2026; Kilimo Kwanza, October 2026).

Who captures the value? Follow the money:

- **Trafigura** gets a rail concession plus trading flows: it can buy Copperbelt metal and move it on its own railway, capturing logistics and trading margins.
- **The DRC and Zambia** get lower transport costs, which raise the value of their ore at the mine (a 30% cut in transport costs on a route costing several hundred dollars per tonne is significant for lower-grade deposits), and a second option against monopoly routes. Zambia, however, takes on more debt.
- **Angola** gets transit fees, port activity and a chance to diversify beyond oil (Month 4's petrostate problem).
- **Western buyers and governments** get traceable, non-Chinese logistics for critical minerals.
- **China** keeps TAZARA, its large mining stakes in the DRC (CMOC, Zijin and others) and its smelters, so much of the metal moving on the Lobito line may still end up in Chinese hands. Competition between corridors mostly benefits the African countries, which gain bargaining power and lower costs.

Risks. African railways have often failed through underinvestment, poor maintenance, cross-border friction and political interference; this one also needs three governments to cooperate, mines to keep running, and power and processing along the line. A corridor that only exports concentrate faster is a better colonial railway; one that also carries processed metal, farm goods and manufactures is a development corridor.

Judgment: the Lobito Corridor is one of the most promising pieces of infrastructure for the driving question in Africa, because it lowers costs, creates competition between global powers, and could anchor processing hubs along its route. But it will only change who captures value if the countries along it use the lower costs to build processing and local industry, rather than only exporting raw material more efficiently.

Key numbers

- Chile lithium reserves: 9.2 Mt of 37 Mt; production about 56,000 t in 2025 (USGS MCS 2026).
- Chilean state capture of operating margin from Novandino: about 70% in 2025-2030 (Government of Chile), about 85% from 2031 (Government of Chile; SQM chief executive, May 2024, via Econosur).
- Corfo lithium receipts, first half of 2026: $791 million, of which Novandino $670 million (Nueva Mineria, August 2026).
- Latin America refines about one-fifth of its mined output of key energy minerals; local refining could raise value about 50% to about $220 billion by 2035 (IEA GCMO 2026).

- Lobito Atlantic Railway financing: $753 million ($553 million DFC, $200 million DBSA), signed December 2025, financial close July 2026 (GTR; DBSA).
- Benguela line length: about 1,300 km; target capacity about 4.6 Mt per year, tenfold today's; transport cost cut of up to 30% (DFC press release, December 2025).
- Zambia Lobito Railway: about 830 km new line; AfDB $255 million loan approved August 2026 (AfDB).
- TAZARA: 1,860 km; freight below 500,000 t per year before rehabilitation; about $1.2 billion Chinese-led concession signed September 2025 (TM Africa; IntelliNews).
- Public finance commitments for critical minerals in advanced economies: about $65 billion in 2025, over four times the 2023 level (IEA GCMO 2026).

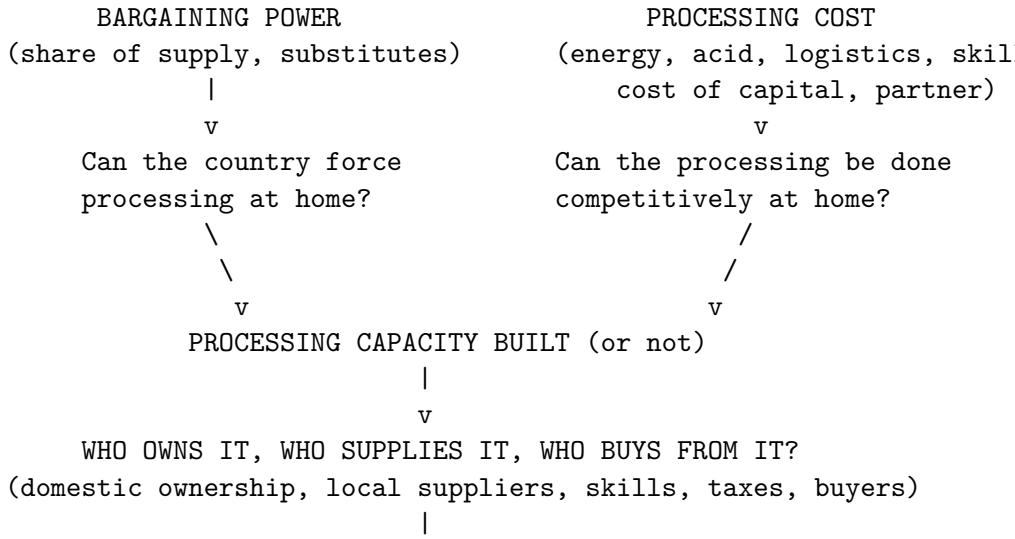
Think like a founder

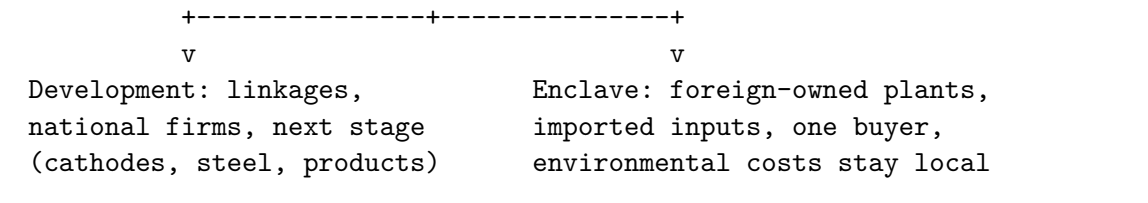
Corridors and state partnerships create clusters of opportunity:

- **Corridor businesses.** Along a new railway, the profits are often in the dry ports, warehouses, cold chains, truck feeders, customs brokerage and digital freight platforms around it. A founder who builds the "last 200 km" of logistics around Lobito, or a digital platform that books and tracks cargo across three countries' borders, captures value from the corridor's traffic.
- **Water and direct lithium extraction technology.** Chile's Salar Futuro is built around DLE and brine reinjection; water-saving extraction is a global market.
- **Partnership structuring.** As countries copy Chile's public-private model, states and companies need advisers and governance tools (escrow accounts, transparent royalty calculation, independent audits) that make these partnerships trustworthy.
- **Regional processing hubs.** The ideal Africa or Southeast Asia downstreaming project often combines ore from several countries. A company that aggregates feedstock (for example, recycled batteries and concentrate from multiple sources) to feed one efficient plant solves the scale problem that individual countries cannot.

Month synthesis

Month 6's pieces fit into one logic: a mining country's share of the value depends on its bargaining power, its processing costs, and how much of the processing it actually owns.





Case	Bargaining power	Processing cost	Ownership and linkages	Overall verdict (2026)
Indonesia, nickel	Very high	Low (captive coal, Chinese know-how)	Mostly Chinese; growing local supply chain	Capacity success; development partly achieved
Indonesia, bauxite	Low	Medium to high	Mixed	Slow; costly early years
DRC, cobalt	Very high	High (power, transport)	Mostly foreign (Chinese and Swiss firms)	Price power used; little new processing yet
Zimbabwe, lithium	Moderate	Medium	Chinese-owned plants	First step achieved; risky timing
Botswana, diamonds	High but eroding	Low (sorting, sales)	Strong state share	Good bargain; demand collapse threatens it
Guinea, bauxite	High in ore	High	Weak	Enforcement by revocation; uncertain
Morocco, phosphate	Very high	Low	State champion (OCP)	Clear success
Chile, lithium	High	Low (brine processing already at home)	State-private partnership	Rent capture success; further downstream limited

The connection to the driving question: the share of value that stays in an ore-producing country rises at each step only if the country owns or taxes that step and builds the next one. Processing is necessary but not sufficient. Months 7 and 9 will show that the biggest margins sit even further downstream, in components, machines and brands, where the processing countries must eventually compete.

Common misconceptions

1. **"Export value equals value added."** A jump from $1 billion of ore exports to $20 billion of processed exports does not mean the country gained $19 billion. Imported coal, acid, machinery, foreign capital costs and profits paid to foreign owners must be subtracted. Domestic value added is often a fraction of the headline.
2. **"An export ban always works."** Bans work only if buyers cannot easily go elsewhere and if processing at home is reasonably competitive. Indonesia's nickel ban worked because Indonesia was indispensable; its bauxite ban was much slower because it was not.

3. **"Processing is always more profitable than mining."** Margins depend on competition. In 2026 copper smelter fees hit zero because smelting is overbuilt; miners captured more of the copper price than smelters did.
4. **"Resource nationalism scares away all investment."** It depends on credibility. Botswana and Chile captured more rent while keeping investors; the problem is unpredictable, retroactive or arbitrary changes, which raise the risk premium for everyone.
5. **"Chinese investment means a country loses control."** Chinese firms brought the technology, capital and markets that made Indonesian and Zimbabwean processing possible when others would not. The question is whether the host government negotiates local ownership, skills, environmental standards and tax capture. Control is negotiated, not given or taken by default.

Exercises

1. **Aluminium power economics (quantitative).** A smelter uses 14 MWh per tonne. Compare annual electricity costs for a 500,000 t/yr smelter at \$25, \$50 and \$90 per MWh. If the aluminium price is \$2,500/t and non-power costs are \$1,400/t, at which electricity prices is the smelter profitable? Which countries in Month 4 have power that cheap, and why?
2. **Value added versus export value (quantitative).** A nickel pig iron plant exports \$1 billion of product a year. It buys \$250 million of imported coal and coke, \$150 million of imported electrodes and spare parts, and \$300 million of nickel ore from local mines; it pays \$50 million in local wages and \$100 million in interest to foreign banks; its profit of \$150 million goes to a foreign owner. Calculate (a) the plant's value added, (b) the part of value added that stays in the country, and (c) the total domestic gain including the ore purchases. Compare with the \$1 billion export figure.
3. **Cobalt quota arithmetic (quantitative).** Suppose the DRC exported 200,000 t of cobalt at \$25,000/t before the ban, and under quotas exports 96,600 t. What cobalt price is needed for quota-era export revenue to match pre-ban revenue? Using the 2026 price path in Week 3 (about \$55,600/t in August, about \$38,600/t in October), when was the DRC better off, and when worse? What other costs and benefits (stockpiles, royalties, relations with buyers) should be included?
4. **Corridor economics (quantitative).** A Copperbelt mine ships 300,000 t of copper cathode a year. Trucking to Durban costs \$350/t; the upgraded Lobito route is 30% cheaper. How much does the mine save each year? If the mine's ore grade falls so that its cost rises by \$500/t, how much of the increase does the corridor offset?
5. **Scorecard essay (800-1,000 words).** Choose one country and mineral not covered in depth this month (for example, the Philippines and nickel, Malaysia and rare earths, Tanzania and graphite, Argentina and lithium, Mongolia and copper). Apply the eight-condition scorecard from Week 4, with evidence for each score. Recommend a policy, and state your judgment.
6. **Essay (600-800 words).** "Indonesia's nickel downstreaming mainly benefited China." Evaluate this claim using the value-capture table in Week 2. Consider the state, workers, regions, Chinese firms, global buyers and rival producers.
7. **Policy memo (one page).** You advise Zambia's government. Write a memo recommending how to use the Lobito Corridor to capture more value from copper and cobalt, covering power, processing, local suppliers, debt, and relations with both the US-backed and Chinese-backed corridors.
8. **Founder exercise.** Identify one business that could be built along the Lobito Corridor or

in an Indonesian nickel park without owning a mine or smelter. Describe the customer, the problem, why now, how you would make money, and the biggest political or price risk.

Reading list

1. **International Energy Agency, *Global Critical Minerals Outlook 2026* (July 2026), especially the policy and Latin America chapters.** The best current evidence on refining costs outside China, policy tools and the value of local processing.
2. **CSIS (Richard Gray and Duc Minh Nguyet Nguyen), "Indonesian Industrialization: Downstreaming Up the Value Chain" (September 2025).** A short, data-rich analysis of what Indonesia's nickel policy achieved and its dependence on China.
3. **SOAS Department of Development Studies case study, "Processing Power: How Indonesia Used Industrial Policy to Move up the Global Nickel Value Chain" (2026).** A development-economics view of the nickel ban as industrial policy (I found this through search and did not read it in full).
4. **WTO, *Indonesia: Measures Relating to Raw Materials* (DS592), Panel Report (November 2022).** The legal case against export bans, and a window into how trade law treats downstreaming.
5. **Raymond Vernon, *Sovereignty at Bay* (1971).** The classic source of the "obsolescing bargain" between multinational investors and host governments.
6. **Paul Collier, *The Plundered Planet* (2010).** How poor countries can manage natural resources for development: discovery, taxation, saving and investing.
7. **Tom Burgis, *The Looting Machine* (2015).** A journalist's account of corruption and extraction in African resource economies, including the DRC and Guinea; read it to understand the governance risks.
8. **Natural Resource Governance Institute, *Natural Resource Charter* (2nd edition, 2014).** Twelve principles for governing resources well, used by many governments and civil society groups.
9. **OECD, *Due Diligence Guidance for Responsible Supply Chains of Minerals from Conflict-Affected and High-Risk Areas* (3rd edition, 2016).** The standard framework behind cobalt, tin and gold supply-chain checks; essential for any founder in minerals.

Glossary terms introduced this month

- **Backward and forward linkages**: demand a plant creates for local inputs (backward) and supplies it provides to later industries (forward).
- **Beneficiation / downstreaming (hilirisasi)**: processing raw minerals domestically before export.
- **Captive power**: a power plant built for and owned by an industrial site, often coal-fired in Indonesian nickel parks.
- **Corridor (development or trade corridor)**: a coordinated transport route (rail, road, port) linking a resource or industrial region to markets.
- **Domestic processing requirement**: a rule that minerals must be processed in the country before export.
- **Electrowinning**: recovering metal from a solution by passing electricity through it, plating the metal onto a cathode.
- **Enclave**: an industrial zone with few links to the rest of the domestic economy.

- **Export quota**: a limit on the quantity of a good that may be exported in a period.
- **Hydrometallurgy / pyrometallurgy**: water-and-chemistry-based versus heat-based metal processing.
- **Lithium sulphate**: an intermediate lithium chemical, between concentrate and battery-grade carbonate or hydroxide.
- **Monopsony**: a market with one dominant buyer, which can push prices down.
- **Obsolescing bargain**: the shift of bargaining power from investor to host government once capital is sunk.
- **Rent (resource rent)**: the surplus profit from a scarce natural resource above normal returns on capital and labour.
- **Resource nationalism**: government assertion of greater control and revenue from natural resources.
- **Value added**: revenue minus the cost of inputs bought from other firms; the contribution to GDP.

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Month 7: Manufacturing and Global Supply Chains

Month overview

In Month 5 you followed minerals out of the ground, and in Month 6 you watched countries fight over who gets to smelt and refine them. This month we follow those refined materials into factories, where they become motors, circuit boards, phones, cars and, eventually, robots. Manufacturing is the middle of the chain from ore to robot to artificial intelligence (AI). It is where most physical value is added, where most industrial jobs sit, and where geopolitics now bites hardest.

Three facts frame the month. First, one country, China, now makes roughly 27 to 32 percent of the world's manufacturing value added, depending on how you measure it, and ran a record goods trade surplus of about 1.19 trillion US dollars in 2025 (Reuters, January 2026). Second, the rest of the world is trying, with mixed success, to move some production elsewhere: to Vietnam, Thailand, Malaysia, India and Indonesia ("China plus one"), and to Mexico and North Africa ("nearshoring"). Third, the arteries that connect factories, the shipping lanes and chokepoints, have been unusually unstable: Houthi attacks closed most of the Red Sea route to container ships from late 2023 to mid-2026, and the war that began on 28 February 2026 has left the Strait of Hormuz close to shut for commercial shipping for most of this year (IEA, 2026).

On top of this sits the trade war. Between 2018 and 2026 the United States raised tariffs to their highest average level since the 1930s, then had a large part of them struck down by its own Supreme Court in February 2026, then rebuilt many of them under other laws.

Why does this matter to a future founder? Because the question "could Africa or Southeast Asia build a cheap robot supply chain by the 2040s?" is mostly a manufacturing question. Robots are made of motors, gears, magnets, batteries, sensors and chips. Whoever has the dense supplier network to make those parts cheaply and quickly will capture a large share of the value. Month 8 then zooms into the most concentrated part of that chain, semiconductors and AI compute, and Month 9 opens up the robot itself.

Learning goals

By the end of this month you should be able to:

- Explain why manufacturing clusters form and why they are so hard to copy, using the ideas of agglomeration, supplier density and learning by doing.

- Read the main numbers on China's manufacturing scale and say what they do and do not prove.
- Compare Vietnam, Thailand, Malaysia, India, Indonesia, Mexico and Morocco as alternative production bases, including what each one actually captures in value added.
- Describe how goods physically move (containers, ports, chokepoints) and estimate how a shipping shock changes the cost of a product.
- Tell the story of the US-China trade war from 2018 to November 2026, including the February 2026 Supreme Court ruling and the tariffs that replaced the struck-down ones.
- Use the "smile curve" to locate where value is captured in a supply chain, and apply it to a robot.
- Judge which manufacturing niches a small company from Southeast Asia or Africa could realistically enter.

Week 1: China's manufacturing machine and the economics of clusters

Core lesson

What "manufacturing value added" means

When economists say a country "makes" a share of the world's goods, they usually mean manufacturing value added (MVA). Value added is the value of what a factory sells minus the value of the inputs it bought from other firms. If a Vietnamese factory buys 80 dollars of parts and sells a finished laptop for 100 dollars, its value added is 20 dollars, even though the export statistics record 100 dollars of exports. This difference between gross trade (what crosses the border) and value added (what was actually created inside the country) is the single most important idea for this month. Keep it in mind every time you read a headline about exports.

Using value added, here is how the world's factories were shared out recently:

Measure	China's share	Year	Source
World Bank data, current US dollars	27.4% (China 4.82 trillion of a world 17.60 trillion)	2025	World Bank WDI, via chinainfigures.com
World Bank data, current US dollars	about 28% (peak 31% in 2021)	2024	Hinrich Foundation, World Bank
UNIDO estimate	32.0%	2024	UNIDO SDG 9 report 2025
UNIDO forecast	45%	2030	UNIDO 2024, cited by Chatham House 2026

Why do the numbers differ? Because the share depends on exchange rates and price adjustments. China's currency weakened against the dollar in 2022-2025 and Chinese factory prices fell (deflation), so in current dollars China's share dipped from its 2021 peak even while its real output kept growing. UNIDO's measure adjusts for prices differently and shows a higher share. The honest summary is: China makes somewhere between a quarter and a third of the world's manufactured value, more than the United States (about 17 percent), Japan, and Germany combined (Hinrich Foundation, October 2025).

For comparison, in 2000 China made only about 6 percent of world MVA (UNIDO). Few economic shifts in history have been this large and this fast.

China is also the world's largest exporter. In 2025 its goods exports were 3.77 trillion dollars, about 17.5 percent of world merchandise trade excluding trade inside the European Union (WTO Annual Report 2026). Its imports were roughly flat at about 2.58 trillion dollars, giving a surplus of 1.189 trillion dollars, the first time any country passed 1 trillion (Reuters; BBC, January 2026). Exports to the United States fell about 20 percent in 2025 (Caixin, January 2026), but exports to Southeast Asia, Africa and Latin America rose faster than the overall rate (NBC News, citing China's customs administration).

Why factories cluster: the three forces

In 1890 the British economist Alfred Marshall noticed that industries clump together in "industrial districts". He named three forces, and they still explain most of what you see in Shenzhen, Penang or Tangier.

1. **Labor pooling.** When many firms in one industry sit together, workers with specialist skills (mould makers, surface-mount technicians, quality engineers) gather there. A firm can hire quickly and a worker can switch jobs without moving house. Both sides take less risk.
2. **Input sharing (supplier density).** A cluster supports specialist suppliers that no single factory could keep busy: a company that only makes injection moulds, one that only anodises aluminium, one that only makes tiny screws. The bigger the cluster, the more specialised and cheaper these suppliers become.
3. **Knowledge spillovers.** Ideas leak. Engineers meet at lunch, change employers, copy good ideas from the factory next door. Learning travels faster across a street than across an ocean.

Economists call the combined effect "economies of agglomeration": the cost of producing in a place falls as more producers locate there. This creates a self-reinforcing loop. More firms make the cluster more attractive, which attracts more firms. That is why a cluster, once established, is very hard to dislodge, and why a newcomer country cannot simply copy the wages and expect the factories to come.

A worked example: the cost of distance from suppliers

Imagine you design a small robotic arm and need a custom aluminium bracket changed three times during development.

- In Shenzhen, the supplier is 30 minutes away. You send a drawing in the morning, get a machined sample the next day, and test it. Three revisions take about one week.
- In a city with no machining cluster, you order from abroad. Each revision needs about 10 days including shipping and customs. Three revisions take about a month, plus courier costs.

If your engineering team costs 20,000 dollars a month, the slow location burns roughly 15,000 dollars extra in salaries for one bracket, before counting the cost of reaching the market three weeks later. Multiply that by the hundreds of custom parts in a robot and you see why hardware start-ups have flocked to Shenzhen for two decades. The numbers here are illustrative, but the mechanism, iteration speed, is real and is the main reason hardware founders say clusters matter more than wages.

Learning by doing and scale

A second force is learning by doing. Each time a factory doubles its cumulative output, its unit cost tends to fall by a roughly constant percentage. This is often called the experience curve or Wright's Law (after Theodore Wright, who observed it in aircraft production in 1936). You met cost curves for solar panels and batteries in Month 4. The same logic applies to almost every manufactured product.

Worked example: suppose a robot gearbox costs 200 dollars at a cumulative output of 100,000 units, with a 20 percent learning rate (each doubling cuts cost by 20 percent).

- At 200,000 units: 200 x 0.8 = 160 dollars
- At 400,000 units: 160 x 0.8 = 128 dollars
- At 800,000 units: 128 x 0.8 = 102 dollars
- At 1.6 million units: about 82 dollars

A producer with eight times the cumulative volume has a cost advantage of almost 50 percent. China's home market and export volumes give its firms this head start in product after product: solar panels, batteries, electric vehicles (EVs), drones, and increasingly robots. A new entrant in Africa or Southeast Asia starts at the expensive end of the curve, which is why it usually needs either a protected niche, a cost advantage elsewhere (cheap energy, proximity to a market), or partnership with an experienced producer.

The role of the Chinese state

Clusters do not appear only by market forces. China's government at central, provincial and city level has provided cheap land, cheap credit from state banks, subsidised electricity, infrastructure built ahead of demand, and targeted funds for strategic industries such as semiconductors. Estimates of the cost of this support vary widely; one academic study of European Union-China relations cites official data suggesting China spends close to 2 percent of GDP on industrial policies (Jean et al., 2025), and other estimates are higher. Measuring this is hard, because much support comes through below-market loans and local government deals that are not published.

Two consequences follow. First, Chinese producers can price aggressively, sometimes below what competitors consider sustainable. Critics call this "overcapacity" and a "China shock 2.0", a sequel to the first China shock of the 2000s, when Chinese imports cost the United States hundreds of thousands of manufacturing jobs (Autor, Dorn and Hanson, 2013). Second, intense domestic competition (Chinese commentators call it "involution", meaning brutal price wars) forces Chinese firms to cut costs and export.

Judgment: "overcapacity" is partly real and partly a label for "a competitor is better than us". China's surplus in 2025 was not mainly caused by subsidies; it was caused by weak household consumption at home (imports stayed flat) combined with genuinely world-class productivity in some sectors. Both things are true at once. For a founder, the practical point is that you should never assume a Chinese competitor's price will rise. Plan for it to fall.

What China's dominance means for value capture

Being the world's factory does not automatically mean capturing most of the value. The next section on the iPhone shows why. But China has been moving up. In 2000 it mostly assembled foreign designs. By 2025 Chinese firms (BYD in EVs, CATL in batteries, DJI in drones, Unitree in robots,

Huawei in telecoms equipment) owned brands, designs and patents. This is the move up the "smile curve", which you met in Month 1: value is high at the two ends of the chain (research, design and brand on the left; marketing, service and software on the right) and low in the middle (assembly). The most important change of the last 15 years is that China now sits at both ends in several industries, not just in the middle.

Case study: The iPhone, Apple and who captures the value

The iPhone is the most studied product in supply chain economics, and it teaches three lessons.

Lesson 1: The assembler captures little. A well-known study by Kenneth Kraemer, Greg Linden and Jason Dedrick of the University of California estimated how the value of an iPhone 4 sold in 2010 was divided. Apple's profit took about 58.5 percent of the sale price. Labor costs in China, where the phone was assembled, were about 1.8 percent. Component makers in Korea, Japan, Taiwan and elsewhere took most of the rest (Kraemer, Linden and Dedrick, 2011). The trade statistics, however, recorded the full factory price as a Chinese export to the United States. So the "US trade deficit with China" in iPhones mostly reflected value created in California, Seoul and Tokyo, not Shenzhen. This is the gross trade versus value added lesson in one product.

Lesson 2: Assembly at scale still builds capability. Apple did not simply buy cheap labor. As Patrick McGee documents in "Apple in China" (2025), Apple sent large numbers of its own engineers to Chinese suppliers for years, trained workers, and paid for advanced machinery inside supplier factories. The knowledge stayed in China. Chinese firms that learned on Apple's production lines (such as Luxshare, which now assembles iPhones, and many component makers) later supplied Chinese phone brands like Huawei and Xiaomi. The value of the assembly was not the 1.8 percent margin. It was the skills.

Lesson 3: Moving is slow and partial. From 2017 Apple began assembling iPhones in India, and the US tariffs of 2025 sped up the move. Apple's suppliers (Foxconn, Tata Electronics and Pegatron) assembled about 55 million iPhones in India in 2025, up from 36 million in 2024, roughly a quarter of global output (Bloomberg via Mobile World Live; BusinessToday, April 2026). Smartphones became India's largest export category for the first time in 2025, at 30.13 billion dollars, with iPhones making up about 76 percent (Financial Express, February 2026). For a while in 2025, nearly all of Foxconn's Indian iPhone exports went to the United States (Reuters, June 2025).

But note what moved: final assembly. Many of the components inside an India-assembled iPhone still come from China, Taiwan, Korea and Japan. India's value added per phone is still modest, though rising as component makers follow. This is the typical pattern of "China plus one": assembly moves first, the deeper supplier base moves slowly, if at all.

What this means for the driving question. If a Southeast Asian or African country assembles robots in the 2040s using imported motors, reducers and chips, it may capture only a few percent of the value, like China in 2010. The prize is to use assembly as a training ground and then move into components and design, as China did. That takes decades and deliberate policy, and it is not guaranteed.

Key numbers: Week 1

- China's share of world manufacturing value added: 27.4% (2025, World Bank WDI via chinainfigures.com); 32.0% by UNIDO's estimate (2024, UNIDO SDG 9 report 2025).
- China's share of world MVA in 2000: about 6% (UNIDO, 2025).

- United States' share of world MVA: about 17% (2024, Hinrich Foundation using World Bank data).
- China's goods exports: 3.77 trillion US dollars, 17.5% of world trade excluding intra-EU trade (2025, WTO Annual Report 2026).
- China's goods trade surplus: 1.189 trillion US dollars, a world record (2025, China customs via Reuters).
- China's exports to the US: down about 20% (2025, Caixin citing China customs).
- iPhones assembled in India: about 55 million, roughly 25% of global output (2025, Bloomberg via Mobile World Live).
- India's smartphone exports: 30.13 billion US dollars, about 76% from Apple (2025, Financial Express).
- Chinese labor's share of an iPhone 4's value: about 1.8%; Apple's profit share about 58.5% (2010, Kraemer, Linden and Dedrick 2011).

Think like a founder: Week 1

- **Do not fight the cluster; plug into it.** A young hardware company from Semarang, Nairobi or Lagos will almost certainly prototype faster and cheaper by using Shenzhen suppliers in the early years. Your own value can sit at the ends of the smile curve: design, software, local distribution and after-sales service.
- **The service end is defensible.** Chinese firms can out-manufacture almost anyone, but they find it harder to provide local installation, maintenance, financing and customer support in thousands of small markets. A company that installs, maintains and finances robots or solar systems for local factories and farms captures value that the manufacturer cannot easily reach.
- **Watch the learning curve.** If you build a product whose cost falls 20 percent with each doubling of volume, the first company to reach scale wins. Ask: who already has the volume? If the answer is a Chinese giant, pick a different product or partner with them.
- **Think in value added, not revenue.** A business that imports 90 percent of its parts has thin margins and little bargaining power. A business that owns the design, the software or the customer relationship does not.

Week 2: China plus one in Asia: Vietnam, Thailand, Malaysia, India and Indonesia

Core lesson

Why firms diversify away from China

"China plus one" means keeping production in China while adding at least one other country, so that a single shock (a tariff, a lockdown, a political crisis) cannot stop the whole business. Four forces pushed firms in this direction after about 2015:

1. **Rising Chinese wages.** In 2019 Japanese companies in China paid a typical experienced factory worker a monthly base salary of about 493 dollars, already above Thailand (446) and far above Vietnam (236) (JETRO, 2020). Labor-intensive work such as garments, shoes and furniture began moving to cheaper places first.
2. **Tariffs.** The 2018-2019 US tariffs on Chinese goods, and the much larger 2025 tariffs, made "Made in China" more expensive for the US market specifically.

3. **Pandemic lockdowns.** China's zero-COVID policy (2020-2022) shut ports and factories at short notice. Companies learned that concentration is a risk.
4. **Geopolitics and customers.** Western governments and big customers started asking suppliers for production outside China, especially for anything linked to defense, telecoms or chips.

The result shows clearly in US data. China's share of US goods imports fell from about 22 percent in 2017 to 9 percent by the end of 2025, and real US imports from China dropped 28 percent in 2025 alone (Bown, PIIE, March 2026). Most of that share was picked up by Taiwan (+4.1 percentage points since 2017), Vietnam (+3.7) and Mexico (+2.3) (PIIE, 2026).

Gross trade moved; value added moved less

Here is the catch. A Stanford study found that from 2017 to 2022, China's direct share of US imports fell from 21 percent to 16 percent, but China's share of the value added inside US imports barely moved, from 15.7 percent to 15.3 percent (Stanford Center on China's Economy and Institutions, 2025-2026). How? Because many of the Vietnamese, Mexican and Thai goods now shipped to the United States contain Chinese parts, Chinese materials and Chinese machinery. The final assembly moved; the supply chain behind it often did not.

A simple example. Suppose a Vietnamese factory exports a 100-dollar electric fan to the United States. Inside are a motor, a circuit board and plastic resin worth 55 dollars, imported from China. Vietnamese labor, local parts, factory overhead and profit add 45 dollars. The US records 100 dollars of imports from Vietnam and zero from China. In value-added terms, 55 dollars still came from China. When people say "China plus one", they often mean "China plus one assembly location".

Vietnam's own trade data confirms this. In 2025 Vietnam ran a record surplus of about 134 billion dollars with the United States and a deficit of about 116 billion dollars with China (VietnamPlus, citing the National Statistics Office, January 2026). The two numbers are connected: Vietnam imports Chinese inputs and exports finished goods to America.

Comparing the main "plus one" countries

Country	Main manufacturing strengths	Typical factory wage (Japanese firms, mean monthly base, Aug 2025)	Key weakness
Vietnam	Phones, laptops, electronics assembly, textiles, furniture, shoes	302 USD	Depends on imported inputs; foreign firms dominate exports; power shortages in 2023
Thailand	Cars and parts, hard disk drives, petrochemicals, food	490 USD	Ageing workforce; Japanese auto base disrupted by Chinese EVs

Country	Main manufacturing strengths	Typical factory wage (Japanese firms, mean monthly base, Aug 2025)	Key weakness
Malaysia	Semiconductor assembly and testing, electronics	about 2,160 ringgit, roughly 490 USD (Aug 2024 survey; above Thailand's 437 that year)	Small labor pool; needs to move into chip design
India	Phones (iPhone), pharmaceuticals, cars, software	329 USD	Infrastructure gaps, complex regulations, high tariffs on inputs
Indonesia	Nickel processing, cars and motorbikes, textiles, food	385 USD	Logistics costs, rigid labor rules, little electronics base
China (for comparison)	Almost everything; deepest supplier base	654 USD (Aug 2024 survey)	Rising wages, tariffs, geopolitical risk

Wages for Thailand, Indonesia, India and Vietnam are from JETRO's FY2025 survey of Japanese-affiliated companies (August 2025 data, converted to dollars at August 2025 exchange rates), as compiled by ASIA DESK (October 2026); the medians are lower (for example 276 USD in Vietnam and 285 USD in India). The Malaysia and China figures are from the previous (FY2024) survey, August 2024 data (JETRO Kuala Lumpur, January 2025; JETRO Biz News, December 2024), so treat the comparison as approximate. These are averages for regular factory operators with about three years of experience, not minimum wages. The striking gap is China: its operators earned roughly twice Vietnam's pay, yet China kept most of its factories, which tells you how much supplier density is worth.

Notice that wages are not the whole story. Indonesia's average wage is below Thailand's, but Thailand has a far deeper auto supplier base. India's wage is close to Vietnam's, but India has historically been slower to build export manufacturing because of infrastructure, regulation and tariffs on imported components. The JETRO survey also shows Indian wages rising fastest, with expected base pay increases above 9 percent a year (JETRO 2025 survey via ASIA DESK).

Thailand: when a new technology resets the cluster

Thailand spent 50 years building Southeast Asia's largest auto cluster around Japanese brands (Toyota, Honda, Isuzu, Mitsubishi). Japanese brands held about 92 percent of the Thai market in 2010 (Straits Times, 2025). Then EVs arrived. Because EVs depend on batteries, electronics and software rather than engines and gearboxes, Chinese firms had the advantage. Japanese brands' share of the Thai market fell from about 88 percent in 2020 to 69.3 percent in 2025, while Chinese brands rose from about 3.2 percent to 21.2 percent, and Chinese brands took 90.5 percent of battery-EV sales (MarkLines data cited in China Daily, August 2026). BYD opened a plant in Rayong with capacity of 150,000 vehicles a year, costing about 490 million dollars (China Daily, 2026).

The lesson: a technology shift can make a decades-old cluster partly obsolete, because the valuable parts change. Thai engine and transmission suppliers lose orders; battery and electronics suppliers, many of them Chinese, arrive. Nikkei Asia reported in 2025 that the number of Chinese auto parts

makers in Thailand had tripled. For the robot question this is a warning and an opportunity: when robots become mass products, the clusters that matter will be those for motors, reducers, sensors and batteries, not the old auto clusters as such.

Malaysia: the back end of the chip chain

Malaysia handles about 13 percent of the world's semiconductor assembly, testing and packaging (ATP), most of it in Penang, and is roughly the sixth-largest semiconductor exporter with about 7 percent of global semiconductor exports (Bank Negara Malaysia, 2025; Malaysia Semiconductor Industry Association, 2026). Electrical and electronics products made up about 40 percent of Malaysia's exports in 2024, worth about 601 billion ringgit (InvestPenang, citing Department of Statistics Malaysia). Integrated circuit exports rose more than 24 percent in 2025 to about 389 billion ringgit on the back of AI demand (Straits Times, May 2026).

But assembly and testing capture only about 6 percent of the semiconductor industry's global value added, according to Bank Negara Malaysia's own analysis. Design and front-end manufacturing capture far more. Malaysia's government talks openly about moving from "Made in Malaysia" to "Made by Malaysia", meaning owning designs rather than only packaging other firms' chips. Month 8 returns to this.

India: the big bet

India is the only "plus one" candidate with a population and home market comparable to China's. Its Production Linked Incentive (PLI) scheme, launched in 2020, pays manufacturers a percentage of their extra sales in chosen sectors such as phones, electronics, pharmaceuticals and solar modules. The phone scheme is the clearest success: iPhone exports in the year to March 2025 were about 1.5 trillion rupees, roughly 17.5 billion dollars, double Apple's PLI target (Business Standard, April 2025).

Judgment: India will be the most important new manufacturing location of the next 20 years, but it will not "replace" China. It is more likely to become a second large hub for its own market plus selected exports, much as Mexico is for North America. Its weakness is the depth of its supplier base; its strength is a huge engineering workforce and a domestic market large enough to support learning curves.

Indonesia: commodity strength, manufacturing gap

Indonesia (one case among many, but relevant to the reader) shows the opposite pattern to Vietnam. As Month 6 explained, its nickel export ban pulled in massive Chinese investment in smelting, so Indonesia captures more value from its ore than before. But in labor-intensive and electronics manufacturing, Indonesia has lagged Vietnam badly, mainly because of higher logistics costs across an archipelago, more rigid labor regulations, and less integration into electronics supply chains. In the US Section 301 tariffs of July 2026, Indonesia was placed in the lower 10 percent group (USTR, July 2026), which helps a little. The open question is whether the nickel-to-battery chain can be extended to battery packs, motors and eventually EVs and robots. That would be the first true "ore to machine" chain in Southeast Asia.

Case study: Vietnam, the most successful "plus one"

The numbers. In 2025 Vietnam's goods trade reached 930 billion dollars, up 18.2 percent, with exports of 475 billion dollars and a surplus of about 20 billion dollars (Vietnam Customs; National Statistics Office, January 2026). The United States was its largest export market at about 153 billion dollars (Vietnam News, January 2026). Vietnam's economy grew about 8 percent in 2025 (Reuters, January 2026). For a country of about 100 million people, exports worth close to its entire GDP are extraordinary.

How it happened. Vietnam's rise rests on four pillars:

1. **Samsung's bet.** Samsung had assembled TVs in southern Vietnam since the 1990s, but the decisive step came in March 2008, when Bac Ninh province licensed its first phone factory (initial registered investment 670 million dollars). Production began in April 2009, and a second phone complex in Thai Nguyen started in 2014 (Vietnam Government portal; VietNamNet, 2014). By June 2025 the two plants had made 2 billion phones, and by mid-2026 their cumulative exports reached 500 billion dollars on cumulative investment of about 24 billion dollars (Vietnam Government portal, August 2025 and August 2026). Suppliers followed Samsung, creating a cluster.
2. **Trade agreements.** Vietnam signed an unusually wide set of free trade agreements, including with the European Union and the trans-Pacific CPTPP, giving factories tariff-free access to many markets.
3. **Openness to foreign investment.** Vietnam welcomed foreign firms, offered tax breaks and industrial parks near ports, and kept wages competitive. Disbursed foreign direct investment (FDI) reached an estimated 27.6 billion dollars in 2025, the highest in five years (VCCI, citing the National Statistics Office).
4. **Geography.** Northern Vietnam borders southern China. Parts can be trucked from Guangdong in a day. Vietnam is, in effect, an extension of the Chinese supply chain with lower tariffs into the US.

The weakness: who captures the value? Foreign-invested firms produced 367 billion dollars, or 77.3 percent, of Vietnam's 2025 exports; domestic firms' exports actually fell 6.1 percent to about 108 billion dollars (VnEconomy, January 2026). The domestic sector ran a trade deficit of about 29 billion dollars, while the foreign-invested sector ran a surplus of about 49 billion dollars (VietnamNet, January 2026). Vietnam's own trade officials acknowledge that domestic firms "had yet to engage deeply in higher-value stages and remained dependent on imported inputs" and that value retained in Vietnam "remained low, mainly concentrated in assembly" (VCCI, 2026).

The political risk. The US surplus makes Vietnam a target. In 2025 Washington accused Vietnam of allowing Chinese goods to be "transshipped" (relabelled to avoid tariffs) and set higher tariffs on transshipped goods. In July 2026 Vietnam was placed in the higher 12.5 percent group of the new Section 301 tariffs (Kelley Drye, July 2026).

What to learn. Vietnam proves that a poor country can plug into global supply chains fast if it is open, stable, close to the cluster, and well connected by trade deals. It also shows the limits: plugging in is not the same as owning. Vietnam's next challenge, like China's in 2010, is to grow domestic firms that supply components and own brands. Whether it can do that will determine whether it becomes the next Korea or stays an assembly platform.

Key numbers: Week 2

- China's share of US goods imports: about 22% (2017) to 9% (end of 2025) (PIIE, March 2026, citing US Census Bureau).
- Real US imports from China: down 28% in 2025 (PIIE, March 2026).
- China's share of value added in US imports: 15.7% (2017) to 15.3% (2022), versus direct share 21% to 16% (Stanford SCCEI China Brief).
- Vietnam goods exports: 475 billion USD, up 17%; trade 930 billion USD (2025, Vietnam Customs and NSO).
- Vietnam surplus with the US: about 134 billion USD; deficit with China about 116 billion USD (2025, NSO via VietnamPlus).
- Foreign-invested firms' share of Vietnam's exports: 77.3% (2025, NSO via VnEconomy).
- Monthly base wage of experienced factory operators at Japanese firms: Thailand 490, Indonesia 385, India 329, Vietnam 302 USD (August 2025, JETRO FY2025 survey).
- Malaysia's share of global chip assembly, testing and packaging: about 13%; share of global semiconductor exports 7.3% (2024, Bank Negara Malaysia *Economic and Monetary Review 2024*, Box 1; MSIA).
- Japanese brands' share of Thailand's car market: about 88% (2020) to 69.3% (2025); Chinese brands 3.2% to 21.2% (MarkLines via China Daily, 2026). Toyota Thailand's tally, omitting some Chinese brands, gives 18.2% (JETRO).

Think like a founder: Week 2

- **Follow the inputs, not the headlines.** If you want to sell into a new factory cluster (in Vietnam, Thailand or Indonesia), find out which inputs are still imported from China. Every imported input is a potential local business: packaging, moulds, cables, fixtures, testing services, industrial maintenance.
- **Technology shifts reopen closed doors.** Thailand's EV shake-up shows that when a product changes (engine to battery, manual factory to robot factory), the old supplier hierarchy breaks. Founders should watch for these moments; incumbents are slow to adapt.
- **Use the "plus one" demand.** Multinationals moving into Southeast Asia need local partners for logistics, compliance, HR, automation and quality control. Selling picks and shovels to the companies relocating is often safer than competing with them.
- **Domestic value added is the policy prize.** Governments in Vietnam, Indonesia and Malaysia actively want local firms in higher-value stages. A founder who can credibly claim "local design, local software, local components" may get incentives, procurement contracts and patient capital.

Week 3: Nearshoring: Mexico, North Africa and the question of Africa

Core lesson

What nearshoring is and why it happens

Nearshoring means moving production closer to the final customer, usually to a neighbouring country with lower wages: Mexico for the United States, Morocco, Turkiye and Eastern Europe for the European Union. It differs from "China plus one" because the main goal is not only diversification but also speed and lower transport risk.

The economics are simple. A container from Shanghai to Los Angeles takes two to three weeks at sea plus port and rail time; a truck from Monterrey to Texas takes a day. When shipping is cheap and reliable, distance matters little and the cheapest factory wins. When shipping is expensive or unreliable, as it was in 2021 and again in 2024-2026, distance becomes a cost. Add tariff preferences (Mexico's free trade with the US and Canada; Morocco's free trade with the EU) and political trust, and nearshoring can beat lower Asian wages.

A worked example of total landed cost. Suppose a US company buys an industrial motor:

Cost element	From China	From Mexico
Factory price	100 USD	115 USD
Ocean or truck freight per unit	4 USD (normal times)	2 USD
US tariff	China: PIIE puts the average US tariff on Chinese goods at 35.2% in August 2026 (use 35%, so 35 USD)	0% if it meets USMCA rules of origin
Inventory and delay cost (extra weeks of stock)	3 USD	1 USD
Total	142 USD	118 USD

With tariffs, Mexico wins even though its factory price is 15 percent higher. Without the tariff, China wins (107 versus 118). This is why US tariff policy drives Mexican investment so strongly, and why uncertainty about that policy is so damaging. (The 35 percent is an average across all goods; the rate on a particular motor depends on its tariff code and on which Section 301, 232 and other duties apply, and it changed several times in 2025-2026, as Week 4 explains.)

Mexico: the biggest nearshoring winner, with a question mark

Mexico became the largest US trading partner in goods in 2023 and kept the position in 2024 and 2025. In 2025 US-Mexico goods trade reached 872.8 billion dollars, and total goods and services trade 976.1 billion dollars (Congressional Research Service, 2026). Mexico supplied about 534 billion dollars, or 15.6 percent, of all US goods imports (CargoFax, from US Census data). The United States accounts for about 60 percent of Mexico's total trade, worth 752 billion dollars (CRS, 2026).

Mexico recorded record foreign direct investment of 40.87 billion dollars in 2025, up 10.8 percent, though most was reinvested earnings of existing firms; brand-new investment was about 7.38 billion dollars (Mexico's Secretaria de Economia, via Rio Times). A striking detail: in 2025 US imports of computer equipment from Mexico exceeded passenger vehicles for the first time (Wilson Center, August 2026), driven by AI servers assembled for US data centers. Foxconn announced in 2024 that it was building the world's largest AI server assembly plant for Nvidia in Mexico (PIIE, 2026). As PIIE notes, Mexico's focus on assembly means it captures much less of the value of an AI server than Taiwan, which makes the chips.

Mexico's question mark is policy. At the first six-year review of the United States-Mexico-Canada Agreement (USMCA) on 1 July 2026, the United States declined to renew the agreement in its current form (USTR, July 2026). That does not end it; the agreement stays in force, with annual reviews, until it expires in 2036 unless all three countries agree to extend it (CRS, September 2026). But it means that the rules for investing in Mexico are now uncertain for a decade. In addition, from 24 July 2026 the US imposed a 10 percent Section 301 tariff on Mexican goods that do not meet

USMCA rules; about 88 percent of US imports from Mexico in early 2026 were USMCA-compliant and therefore exempt (CRS, August 2026). Some textile and industrial parts makers are reportedly evaluating moves to Vietnam and other Asian locations because of this uncertainty and Mexico's judicial reforms (Mexico Business News, September 2026).

Judgment: Mexico's position is structurally strong, because geography cannot be tariffed away and the integrated North American auto and electronics chains are very costly to unwind. But the 2026 review shows that nearshoring based on a trade agreement is only as reliable as the politics behind the agreement. Investors should expect Mexico to keep growing as a manufacturing base, with periodic political shocks.

North Africa and Eastern Europe: Europe's nearshore

Europe's nearshore has two halves. Central and Eastern Europe (Poland, Czechia, Slovakia, Hungary, Romania) became Germany's extended factory after 2004, especially for cars. North Africa and Turkiye are the lower-cost southern ring. Morocco is the standout, examined in the case study below. Egypt and Tunisia have smaller clusters in auto wiring harnesses, textiles and electrical goods.

The logic is the same as Mexico's: free trade with the EU, a truck or short ferry ride to European customers, and wages well below European levels. The European Union's push to "de-risk" from China and to meet carbon rules adds a new reason, since shorter supply chains emit less transport carbon and can use North African solar and wind power.

Africa's manufacturing gap

Africa as a whole accounted for about 3.2 percent of world GDP but only about 2.0 percent of world manufacturing value added in 2024 (UNIDO, December 2025). For a continent with roughly one in six of the world's people (UN population estimates), this is the central development problem. Most African economies export raw materials (oil, minerals, cocoa, coffee) and import manufactured goods.

Why has manufacturing not taken off? The main reasons, discussed across Months 2, 3 and 6, are:

- **Expensive and unreliable electricity.** Factories need steady power. Many African firms run diesel generators, which multiplies energy costs.
- **High transport and logistics costs.** Many countries are landlocked; roads, railways and ports are weak; border crossings are slow. Intra-African trade is low.
- **Small, fragmented markets.** Each national market is too small to support learning curves. The African Continental Free Trade Area (AfCFTA) aims to fix this but is still early.
- **Competition from Asia.** Since the 2000s, African producers of shoes, clothes and simple goods have faced Chinese and later Bangladeshi and Vietnamese competition.
- **Policy instability and conflict.** Investors fear sudden changes in rules, currency crises and violence.

Ethiopia shows both the promise and the fragility. Its government built industrial parks, most famously the Hawassa Industrial Park (opened in July 2016, built by a Chinese contractor and anchored by the US clothing group PVH), to attract garment exporters serving the US under the African Growth and Opportunity Act (AGOA), which gives eligible African countries duty-free access to the US market. The plan was ambitious: the World Bank recorded a target of 60,000 jobs at Hawassa and a government projection of about 1 billion dollars of exports a year, but by early 2019 the park employed about 25,000 people (World Bank, Hawassa Industrial Park Community Impact

Evaluation), and at its peak more than 35,000 (International Growth Centre, 2025). A World Bank review of Ethiopia's parks put their net exports at about 141 million dollars in 2020/21, roughly 11 percent of Ethiopia's manufacturing exports (World Bank, 2022). Then on 1 January 2022 the US suspended Ethiopia from AGOA over human rights violations in the Tigray war. A National Bank of Ethiopia study found that 18 foreign companies left the parks and about 11,500 jobs were lost, 4,321 of them at Hawassa (NBE, 2025, as reported by the IGC and The Reporter). Ethiopia's total exports to the US fell 31.7 percent in 2023 (IGC, using US Census data). An IGC study that followed 1,410 laid-off Hawassa workers found that, a year later, their labour income had halved and extreme poverty had nearly doubled. AGOA itself expired on 30 September 2025, was retroactively renewed through December 2026 in February 2026 (CRS), and Congress cleared a further extension to 2028 in September 2026 (CSIS, September 2026), but Ethiopia remains suspended. Some firms survived by shifting sales to Europe and Asia, but the parks never came close to the original targets.

The lesson for founders and governments: manufacturing that depends on a single foreign trade preference is fragile. Building on regional markets, reliable energy and multiple export destinations is slower but sturdier.

Can Africa or Southeast Asia build a robot supply chain? A first pass

We can now make a first, manufacturing-based assessment. Think of a robot's supply chain in four layers:

Layer	Example parts	Who leads today	Hard to enter?
Materials	Rare-earth magnets, copper wire, battery chemicals, steel	China (magnets over 90% of global production, IEA 2024)	Very hard, energy-intensive
Components	Servo motors, precision reducers (gearboxes), sensors, batteries, chips	China, Japan, Germany, Korea, Taiwan	Hard, needs precision clusters
Assembly	Putting the robot together, testing	China, Japan, Germany, Korea; increasingly anywhere	Moderate
Integration and service	Installing robots in factories, programming, maintenance	Local firms everywhere	Easiest, highest local value

Southeast Asia has a head start in components and assembly: Thailand's auto parts base, Malaysia's electronics and chip back end and its rare earth refinery, Vietnam's electronics assembly, Indonesia's nickel and battery chain. Africa's strengths are in raw materials (cobalt, copper, manganese, graphite, platinum, rare earths) and in a few clusters such as Morocco's auto industry and South Africa's mining and auto base.

Following one kilogram of rare earth oxide into a robot joint

Tables of "layers" can hide the mechanism, so let us follow one input through the chain and watch where value is added and why. The numbers below are a teaching model: the price floor and cost

shares come from the sources named, but the rest are rounded illustrations, not industry data.

Step 1: oxide. A mine and separation plant produce neodymium-praseodymium (NdPr) oxide, the main rare earth input to magnets. The US government guaranteed MP Materials a floor of 110 dollars per kilogram for ten years from 2025, about twice the price MP had recently received in a market set by Chinese supply (MP Materials, July 2025; Reuters, as discussed in Month 5). So call the oxide worth 110 dollars per kilogram outside China, and less inside China.

Step 2: metal and magnet. The oxide is reduced to metal and alloyed with iron and boron, then pressed, sintered (baked) and machined into magnets. A sintered NdFeB magnet is roughly 30 percent rare earth by weight, so one kilogram of magnet needs very roughly 0.35 kilograms of NdPr oxide, about 39 dollars' worth at the floor price. The IEA estimates that rare earths are about 40 percent of a permanent magnet's cost (cited in Month 5). That implies a magnet cost of very roughly 100 dollars per kilogram (39 divided by 0.4). In other words, the magnet maker roughly doubles to triples the value of the oxide it buys, through energy-intensive processing, tight quality control and know-how about grain structure and coatings that took decades to build. This is where China holds about 94 percent of sintered magnet production (IEA data, Month 9).

Step 3: motor. A frameless brushless motor for a robot joint might contain 20 to 30 grams of magnet, worth only 2 to 3 dollars, yet the motor sells for far more because of copper windings, precision machining, balancing and testing. McKinsey's breakdown of a typical integrated robot actuator (Month 9) puts the motor at only 10 to 20 percent of actuator cost.

Step 4: actuator. The motor is combined with a precision reducer (30 to 50 percent of actuator cost), a driver board (15 to 20 percent), sensors (5 to 10 percent) and housings and bearings (10 to 20 percent). Suppose an actuator costs 400 dollars. Then the motor is about 60 dollars, the reducer about 160, the driver about 70, the sensors about 30 and the mechanics about 60. The magnet inside, worth perhaps 2 to 3 dollars, is under 1 percent of the actuator's cost.

Step 5: robot. A humanoid or advanced arm has dozens of actuators (Month 9 counts 25 to 80 or more motors in a humanoid), plus batteries, computers, cameras, software and a brand. With 40 actuators, a robot holds roughly 0.8 to 1.2 kilograms of magnet, made from oxide worth about 30 to 45 dollars. If the robot sells for 30,000 dollars, the oxide is about 0.1 percent of the final price.

Stage	Rough value (teaching model)	Share of final robot price	What creates the value	Who leads today
NdPr oxide in one robot	about 30-45 USD	about 0.1%	Geology, separation chemistry	China; Australia, US, Malaysia (refining)
Magnets in one robot	about 80-120 USD	about 0.3-0.4%	Processing know-how, energy	China (about 94%)
Actuators (say 40 x 400 USD)	about 16,000 USD	about half	Precision machining, reducers, control electronics	Japan, Germany, China, Korea
Robot hardware, software and brand	about 30,000 USD	100%	System design, AI, integration, distribution	US, China, Japan, Korea, Germany

Two lessons follow, and they pull in opposite directions.

Lesson 1: value grows by orders of magnitude as you move downstream. A country that only mines and exports oxide captures a fraction of one percent of the robot's value. Even a magnet plant captures less than one percent. The big money is in actuators, especially reducers, and in the system, software and brand. This is the smile curve of Month 1 drawn in kilograms and dollars.

Lesson 2: the cheapest input can still hold the most power. Because the magnet is under 1 percent of the actuator's cost, a robot maker will pay almost any price to get magnets rather than stop production. That is why China's 2025 export licences could shut car factories, and why the US was willing to guarantee MP Materials about twice the market price. Low value share plus no substitute equals high bargaining power. In practice, though, that power is used as a geopolitical weapon or a bargaining chip rather than turned into higher prices in normal times, because Chinese magnet makers compete fiercely with one another.

What this means for Africa and Southeast Asia. A rare earth or nickel producer that wants a larger share should not dream of jumping straight to finished robots. The realistic ladder is: (1) separate and refine at home, using cheap clean power and long-term offtake contracts with buyers outside China who want non-Chinese supply (Malaysia's Lynas plant is the regional example); (2) add magnet or battery-material production with a technology partner, accepting that the margin is thin but the strategic value is high; (3) move into motors and battery packs, where local assembly for regional markets is plausible; (4) only then reach for reducers and actuators, the precision steps where Japan, Germany and China have decades of learning-curve advantage (Week 1). Each step takes roughly five to ten years, which is why the 2040s, not the 2020s, is the honest horizon.

Judgment: from this month's evidence, Southeast Asia is far better placed for components by the 2040s, because it already has precision clusters (Thai auto parts, Malaysian electronics, Vietnamese assembly) and a refining foothold. Africa's realistic path is processed materials, plus assembly and service for its own fast-growing market, with Morocco the most plausible African component hub because it already makes wiring, seats and electrical parts for European carmakers. Month 9 tests this judgment against a full robot bill of materials.

Case study: Morocco's automotive cluster

From nothing to Africa's largest car exporter. In the early 2000s Morocco produced few cars. In 2012 Renault opened a large plant at Melloussa near Tangier; Stellantis (then PSA) followed with a plant at Kenitra in 2019. In 2025 Morocco's automotive exports were 159.2 billion dirhams, about 17 billion US dollars at the 2025 average rate of roughly 9.3 dirhams per dollar, its largest export category for the third year running and about a third of all goods exports of 470 billion dirhams (Office des Changes, annual foreign trade report, October 2026; exchange rate from Le360, May 2026). That was slightly below the revised 2024 figure of 159.6 billion dirhams. (Earlier provisional monthly data had shown 154.5 billion; annual reports revise monthly data, so always check which version a number comes from.) Inside the total, vehicle assembly ("construction") fell 13.5 percent, while wiring harnesses rose 9 percent and seats and interiors 11.5 percent, a reminder that much of the value is components, not finished cars. Morocco accounted for about 52 percent of Africa's vehicle exports by volume in 2025; together with South Africa it shipped about 869,000 vehicles abroad (Le360, May 2026).

The port. The cluster grew around Tanger Med, a port complex 14 kilometres across the Strait of Gibraltar from Spain. In 2025 Tanger Med handled 11.1 million containers (measured in TEU, twenty-foot equivalent units), up 8.4 percent, about 161 million tonnes of cargo, and 526,862 vehicles, of which about 327,569 were Renault exports (Tanger Med Port Authority, February 2026). Renault

describes it as Africa's and the Mediterranean's leading port and the 17th largest in the world (Renault Group, June 2026).

Why it worked. Several conditions came together:

1. **Free trade with the EU** (association agreement in force since 2000) and with the US (free trade agreement since 2006).
2. **A purpose-built port and free zones** with fast customs (the truck platform reports average export transit of about 3 hours, Tanger Med 2026 brochure).
3. **Proximity.** Factories near Tangier can reach Spanish and French plants by truck and ferry in a day or two, fitting just-in-time production.
4. **Supplier development.** The government pushed suppliers (wiring harnesses, seats, interiors, stamping) to locate nearby. A large share of the export value is components, not only finished cars (Without Borders Systems, 2026).
5. **Political stability and a consistent long-term industrial strategy** across several governments.

The limits. Vehicle exports through Tanger Med fell 12 percent in 2025, which the port called a cyclical slowdown (Tanger Med, 2026). Morocco's cars are mostly European-designed small petrol and diesel models; the shift to EVs threatens this base unless Morocco attracts battery and EV production. It is trying: Chinese battery and materials firms have announced plants in Morocco, partly because Morocco has phosphates (useful for lithium iron phosphate batteries, as Month 6 discussed with OCP) and free trade access to both the EU and the US. Local design and brand ownership remain minimal; the value captured is mostly wages, supplier margins and port fees.

What to learn. Morocco shows that an African country can build a world-class manufacturing cluster in about 20 years when it combines a great port, trade agreements, stability and focus. It also shows that the cluster is controlled by foreign brands, so the next step, local engineering and component design, is still ahead. For the robot question, Morocco is the African country best placed to make robot components (motors, wiring, batteries) for the European market by the 2040s.

Key numbers: Week 3

- US-Mexico goods trade: 872.8 billion USD; goods and services 976.1 billion USD (2025, CRS).
- Mexico's share of US goods imports: 15.6%, about 534 billion USD (2025, US Census via CargoFax).
- Mexico's FDI: record 40.87 billion USD, up 10.8%; new investment 7.38 billion USD (2025, Secretaria de Economia via Rio Times).
- USMCA: not renewed at 1 July 2026 review; remains in force with annual reviews until 2036 unless extended (USTR; CRS, 2026).
- Share of US imports from Mexico that were USMCA-compliant: 88% (January-May 2026, CRS).
- Africa's share of world MVA: about 2.0%; share of world GDP 3.2% (2024, UNIDO).
- Morocco automotive exports: 159.2 billion dirhams, about 17 billion USD (2025, Office des Changes annual report, October 2026).
- Tanger Med: 11.1 million TEU, about 161 million tonnes, 526,862 vehicles (2025, Tanger Med Port Authority).
- Ethiopia industrial park job losses after AGOA suspension: about 11,500 jobs, 18 foreign firms exited (2022-2023, National Bank of Ethiopia study via IGC, 2025).

Think like a founder: Week 3

- **Location is a product feature.** If your customers are in Europe, a Moroccan or Egyptian base can beat Asia on speed. If they are in the US, Mexico. If they are in Southeast Asia, a base inside ASEAN avoids tariffs and shipping delays. Decide where your customer is before deciding where to build.
- **Policy risk is a real cost.** Ethiopia's AGOA loss and Mexico's USMCA uncertainty show that a business built on one trade preference can collapse overnight. Diversify markets early, and favour products that are competitive even without preferences.
- **Service and integration travel with factories.** Every new auto or electronics cluster needs automation integrators, maintenance firms, training providers, testing labs and software. These are capital-light businesses a young founder can start, and they become the seedbed for local engineering capability.
- **Africa's domestic market is the long game.** Africa's population is projected to grow strongly through 2050. Products designed for African customers (cheap, robust, repairable, solar-powered) are an area where African founders have an insight advantage over Chinese or European firms.

Week 4: Ships, chokepoints and the trade wars of 2018-2026

Core lesson

The container: the machine that made global supply chains possible

Before the 1960s, cargo was loaded piece by piece by dock workers, which was slow, costly and prone to theft. The standard steel shipping container, popularised by the American trucking entrepreneur Malcolm McLean from 1956, changed this. Goods are packed once at the factory, and the sealed box moves by truck, crane, ship and train without being opened. Marc Levinson's book "The Box" (2006) argues that the container cut the cost of loading and moving goods so much that it made today's global supply chains possible.

Today about 80 percent of the volume of world goods trade travels by sea (UNCTAD). Seaborne trade reached about 12,720 million tonnes in 2024 (UNCTAD Review of Maritime Transport 2025). Container capacity is measured in TEU (twenty-foot equivalent units); the largest ships now carry more than 24,000 TEU (for example COSCO's OOCL Portugal, 24,188 TEU, gCaptain, September 2026).

How cheap is shipping? In normal times, very. Before the pandemic, Drewry's World Container Index (WCI), a benchmark of spot rates on eight major routes, averaged about 1,420 dollars per 40-foot container in 2019. A 40-foot container can hold several thousand pairs of shoes or a few hundred microwave ovens. Worked example: if a container holds 5,000 pairs of shoes and costs 1,420 dollars to ship, freight adds about 28 cents per pair. Even at the September 2021 record of 10,377 dollars (Drewry), freight added about 2 dollars per pair, noticeable for cheap shoes but small for a phone. On 29 October 2026 the WCI stood at 4,429 dollars, with Shanghai to Los Angeles at 7,514 dollars and Shanghai to Rotterdam at 3,686 dollars (Drewry, October 2026).

The lesson: freight rarely decides where expensive goods are made, but it matters greatly for cheap, bulky goods (furniture, building materials, simple consumer goods) and for reliability. A factory that cannot predict when its parts will arrive must hold extra inventory, which ties up cash.

Chokepoints: where the world's arteries narrow

Most sea trade passes through a few narrow passages. If one closes, ships must take long detours or stop.

Chokepoint	What passes	Oil flow before the 2026 war	Recent disruption
Strait of Malacca (between Malaysia, Singapore and Indonesia)	Trade between the Indian Ocean and East Asia	23.2 million barrels a day, 29% of maritime oil (1H 2025)	None major; piracy risk
Strait of Hormuz (between Iran and Oman)	Gulf oil and LNG exports	20.9 million barrels a day (1H 2025)	Near-closed since 28 Feb 2026
Bab el-Mandeb and Red Sea to Suez Canal	Asia-Europe container trade	About 4 to 5 million barrels a day	Houthi attacks from late 2023; partial return of container lines from mid-2026
Panama Canal	US East Coast to Asia	About 2 million barrels a day	Drought restrictions 2023-2024
Cape of Good Hope	Detour route around Africa	9.3 million barrels a day (2024)	Busy because of Red Sea diversions

Oil flow data from the US Energy Information Administration (EIA), "World Oil Transit Chokepoints", updated March 2026.

The Red Sea, 2023-2026. From late 2023 Yemen's Houthi movement attacked merchant ships in the Red Sea. Container lines diverted around the Cape of Good Hope, adding about 10 to 15 days to Asia-Europe voyages (IFS, October 2026). From early February 2024 to mid-2026, container ship capacity transiting the Suez Canal averaged only about 141,000 TEU a week, roughly 11 percent of the 2023 level (BIMCO citing Clarksons Research, October 2026). Average voyage distances rose from 4,831 miles in 2018 to 5,245 miles in 2024 (UNCTAD, 2025). Longer voyages need more ships, so the diversion soaked up shipping capacity and raised rates. In the first two weeks of October 2026, container capacity through Suez recovered to about 753,000 TEU a week, just over half the 2023 level, as major lines returned (BIMCO, October 2026), despite continuing Houthi gains on land in Yemen.

Hormuz, 2026. On 28 February 2026 the United States and Israel launched strikes on Iran, beginning what the IEA calls "the largest supply disruption in the history of the global oil market" (IEA Oil Market Report, March 2026). Flows through Hormuz fell from about 20 million barrels a day to a trickle in March, and Gulf producers cut output by at least 10 million barrels a day (IEA, March 2026). Two different "Brent" prices tell the story. Front-month Brent futures, the price usually quoted in the news, rose from about 72 dollars on 27 February to a peak of 126.41 dollars on 30 April (Reuters, April 2026), the figure used in Month 4. Dated Brent, the price of physical cargoes loading in the next 10 days to a month, went much higher: a record 144.42 dollars on 7 April, when futures were trading near 109 dollars (Platts data via CNBC and Bloomberg, April 2026). A gap of more than 25 dollars between the two (EIA, April 2026) means refiners were paying a huge premium for oil they could get now; the physical market was short of barrels, not just nervous. By September

2026, ship-to-ship transfers and pipelines bypassing the strait had restored Gulf crude exports to over 75 percent of pre-war levels, but refined product exports remained below half (IEA, October 2026). Wholesale diesel prices exceeded 200 dollars a barrel in September (IEA, October 2026). In early November 2026 only about 9 AIS-visible commercial ships a day were crossing Hormuz, about 10 percent of the pre-crisis 84.8 a day, and container lines were avoiding it (hormuz.now, 4 November 2026; TradeWinds, November 2026). Yet oil was still moving: crude and product flows through the strait averaged about 10.3 million barrels a day in October, against about 19 million before the war (Vortexa data via Argus, November 2026). Count the ships and the strait looks closed; count the oil and it is about half open, because a few very large tankers carry enormous volumes, many sail in US-escorted convoys with their transponders switched off, and pipelines bypass the strait altogether. Brent futures settled at 104.77 dollars on 2 November 2026 (Reuters), well above the pre-war level.

Why does this matter for manufacturing? Diesel moves trucks and ships; oil and gas feed plastics, fertilisers and chemicals; Gulf LNG powers factories in Asia. The IEA expects world oil demand to fall by 2.6 million barrels a day in 2026 because high prices are slowing economies (IEA, October 2026). For Asian manufacturers that depend on Gulf energy, 2026 has been a year of higher costs and rationing. It also strengthens the case for electrification and local renewable energy (Month 4), because a factory running on local solar power is insulated from chokepoint shocks.

A worked example: how a shipping shock reaches a product price

A Vietnamese electronics firm exports a 300-dollar robot vacuum to Europe, 1,000 units per 40-foot container.

- Normal freight (about 1,500 dollars per container): 1.50 dollars per unit, 0.5 percent of the price.
- Crisis freight (say 6,000 dollars): 6 dollars per unit, 2 percent of the price.
- Extra 12 days in transit because of a detour, at a 10 percent annual cost of capital on 300,000 dollars of goods: about 1,000 dollars per container, 1 dollar per unit.
- Diesel and plastic costs rise, adding perhaps 2 to 4 dollars per unit in materials.

Total shock: perhaps 8 to 10 dollars per unit, about 3 percent. Painful for a firm with a 5 percent margin, survivable for one with 20 percent. This is why high-margin products ride out shipping shocks and low-margin commodity manufacturing gets squeezed.

The trade war, 2018-2026: a timeline

Round one (2018-2019). The first Trump administration imposed tariffs of up to 25 percent on about 370 billion dollars of Chinese goods under Section 301 of the Trade Act of 1974 (a law allowing tariffs against "unfair" foreign practices), plus tariffs on steel and aluminium under Section 232 (national security). China retaliated. The Biden administration (2021-2025) kept most tariffs, added export controls on chips (Month 8) and raised tariffs on Chinese EVs to 100 percent in 2024.

Round two (2025). After returning to office in January 2025, President Trump used the International Emergency Economic Powers Act (IEEPA), a 1977 law for national emergencies, to impose tariffs: first on Canada, Mexico and China over fentanyl, then on 2 April 2025 ("Liberation Day") "reciprocal" tariffs on almost every country. Tariffs on China rose by 145 percentage points by April (PIIE, 2026), before a partial truce in May. Yale's Budget Lab estimated the average effective US tariff rate after the April 2025 announcement at 22.5 percent, the highest since 1909 (Yale Budget Lab, April 2025). Sector tariffs under Section 232 on cars (25 percent), steel, aluminium, copper and later pharmaceuticals were added.

China strikes back with export controls (2025). On 4 April 2025 China restricted exports of rare-earth permanent magnets and several rare-earth elements. On 9 October 2025 it announced much wider controls, including licences for foreign-made products containing even small amounts of Chinese rare earths. It also briefly blocked exports from the chipmaker Nexperia's Chinese operations in October 2025, causing chip shortages at carmakers (PIIE, 2026). See the case study.

The Busan truce (30 October 2025). At a meeting in Busan, South Korea, Presidents Trump and Xi agreed a one-year truce. The US cut its fentanyl-related tariff on China from 20 to 10 percent, which the White House described as lowering the combined tariff on Chinese goods from about 57 to 47 percent (trade-weighted averages calculated by the Congressional Research Service put the average lower, at about 39 percent, because many goods were exempt), kept the higher reciprocal tariff suspended, and paused its "Affiliates Rule" extending export controls to subsidiaries of blacklisted firms. China suspended its October 2025 rare-earth controls for a year, issued general licences for rare earths, gallium, germanium, antimony and graphite, suspended retaliatory tariffs and agreed to buy US soybeans (White House fact sheet, November 2025; Reuters, November 2025). During Xi's state visit to Washington on 23-25 September 2026 the truce, due to expire on 10 November 2026, was extended to 10 January 2027 (China Ministry of Foreign Affairs, September 2026; AP, 23 September 2026).

The Supreme Court ruling (20 February 2026). In Learning Resources, Inc. v. Trump, the Supreme Court held 6-3 that IEEPA does not authorise the President to impose tariffs, because taxing imports is a power of Congress (CRS, February 2026). The IEEPA tariffs, both the fentanyl and the "reciprocal" ones, were void. Importers began claiming refunds of the roughly 166 billion dollars collected under IEEPA; US Customs had certified about 126 billion dollars of refunds by 2 October 2026 (Penn Wharton Budget Model, October 2026, citing court filings). Section 232 and Section 301 tariffs were not affected.

The replacements (2026). The same day, the President imposed a 10 percent global tariff under Section 122 of the Trade Act of 1974, a never-before-used provision allowing tariffs of up to 15 percent for 150 days to address balance-of-payments problems (Covington, February 2026). On 7 May 2026 the Court of International Trade ruled the Section 122 tariff unlawful, but the ruling was stayed on appeal (Gibson Dunn, May 2026). When Section 122 expired on 24 July 2026, it was replaced by Section 301 tariffs of 10 or 12.5 percent on 60 economies found to have failed to ban imports made with forced labor (USTR, July 2026). The 10 percent group includes Canada, Mexico, India, Indonesia, Malaysia, Cambodia and Bangladesh; the 12.5 percent group includes China, Vietnam, Thailand, the Philippines, Singapore, Brazil, South Africa and Morocco; the EU, Japan, Korea, Taiwan and Switzerland have caps that include existing duties (USTR; Kelley Drye, July 2026). Yale's Budget Lab put the average statutory tariff rate at about 11.1 percent after this switch, set to rise to about 11.8 percent by the end of 2026 under current law (Yale Budget Lab, July 2026).

How high is a tariff, really? Three measures and one question of who pays

Headlines quote tariff numbers that seem to disagree: 145 percent, 47 percent, 35 percent, 24 percent, 7 percent. They are usually all correct, because they measure different things. A founder who imports parts needs to know which number matters for which decision.

1. **The headline (statutory) rate on a product.** This is the rate written in the law for one product from one country, adding up every layer. In mid-2026 a Chinese electric motor might face the old Section 301 duty from 2018-2019 (often 25 percent), plus the new 12.5 percent

Section 301 "forced labor" duty, plus any Section 232 duty if it is classified as a car part or contains steel. The layers stack, so the rate on one product can be far above the average.

2. **The trade-weighted average statutory rate.** Take the statutory rate on every product and weight it by how much of that product was imported in a base year. PIIE's tracker does this: the average US tariff on Chinese goods was 3.1 percent in January 2018, about 19 percent after the first trade war, 126.1 percent at the peak in early May 2025, and 35.2 percent on 22 August 2026, covering 89 percent of goods (PIIE, August 2026). The CRS, using a slightly different method, gives about 36.5 percent for July 2026. This is the best measure of policy intent.
3. **The effective (collected) rate.** Divide the duties actually collected by the value of goods actually imported. The Penn Wharton Budget Model calculates that the US collected an average of 7.1 percent on all imports in August 2026, and 24.3 percent on imports from China (Penn Wharton Budget Model, October 2026). This is lower than the statutory rate for two reasons: exemptions (81.4 percent of imports from Canada and Mexico claimed duty-free USMCA treatment in August 2026), and substitution, because importers stop buying the most heavily taxed goods. A tax that is high enough to stop the trade collects nothing at all.

The gap between measures 2 and 3 is itself a lesson. When China's statutory average jumped in April 2025, Chinese goods simply stopped arriving for a few weeks, so the effective rate stayed far below the headline. The tariff "worked" not by raising money but by diverting trade, mostly to Vietnam, Mexico and Taiwan.

Who pays? The importer writes the cheque to US Customs, but the economic burden can be shifted. A New York Federal Reserve study explains the logic with a simple example (Amiti, Flanagan, Heise and Weinstein, February 2026). A foreign exporter sells a good for 100 dollars and the US imposes a 25 percent tariff:

- If the exporter keeps its price at 100, the importer pays 125. The whole burden falls on the US buyer ("100 percent pass-through").
- If the exporter cuts its price to 80, the importer still pays 100 (80 plus 20 of duty). The whole burden falls on the foreign seller.
- If the exporter cuts to 96, the duty is 24 and the importer pays 120. The US side bears 20 of the 24 dollars, about 83 percent.

Using detailed import data, the study found that US importers bore about 94 percent of the burden of the 2025 tariffs in January to August 2025, falling to about 86 percent by November as some foreign exporters began cutting prices. Applied to the average December 2025 tariff of 13 percent, that implies US import prices for tariffed goods roughly 11 percent higher than for untaxed goods (13 x 0.86). The earlier 2018-2019 tariffs showed almost complete pass-through as well.

Why do foreign exporters absorb so little? The answer is market power and alternatives. A Chinese exporter of a commodity product with many other buyers worldwide has little reason to cut its price for the US; it can sell elsewhere, which is exactly what happened as Chinese exports to Southeast Asia, Africa and Latin America rose in 2025. A US buyer who has qualified one supplier for a precision part cannot switch quickly, so it pays. The burden falls on whichever side has fewer options, which is the same logic as the value-capture rule of this month: whoever is hardest to replace keeps the money.

A worked example for a founder. Suppose a small robot company in Texas imports 1,000 servo motors a year from Shenzhen at 40 dollars each and sells finished robots for 2,000 dollars, with motors, magnets and gearboxes making up 20 percent of its cost.

- Import bill: 40,000 dollars. A 35 percent average tariff adds 14,000 dollars, if the Chinese supplier passes all of it on.
- If the supplier absorbs 14 percent of the burden (the November 2025 estimate), the company still pays about 12,000 dollars more.
- Option A: switch to a Vietnamese supplier at 46 dollars, facing a 12.5 percent tariff (about 52 dollars landed versus 54 from China). Saving: about 2 dollars a motor, before counting the cost of qualifying a new supplier and the risk that the Vietnamese motor contains Chinese magnets that customs may treat as transshipment.
- Option B: redesign to use fewer motors, or motors with less rare earth content. This takes a year but protects against the next tariff change and the next export control.

The lesson is that a tariff mostly works as a tax on the importer in the short run and as a signal to redesign supply chains in the long run. For an exporter in Southeast Asia or Africa, the opportunity is not that tariffs make you cheaper than China (often they do not), but that they create a premium for a supplier whose content is verifiably non-Chinese, which requires traceability, local value added and documentation.

What did it all achieve? The US goods deficit was about 1.23 trillion dollars in 2025, slightly larger than in 2024 (CargoFax, from Census data). Imports from China fell sharply, but imports from the rest of the world rose, especially AI computing equipment from Taiwan and Mexico (PIIE, 2026). China's global surplus hit a record. Tariff revenue rose by an estimated 214.7 billion dollars above the 2022-2024 average by February 2026 (Yale Budget Lab, April 2026), and imported core goods prices rose about 1.5 percent in 2025 (Yale Budget Lab). Two sides of the debate:

- **Supporters** argue tariffs reduce dependence on China, raise revenue, push firms to invest in the US, and give leverage in negotiations.
- **Critics** argue tariffs are taxes paid mostly by US importers and consumers (the New York Fed study above found US importers bore about 86 to 94 percent of the 2025 burden), that they did not shrink the overall deficit because the deficit reflects US saving and investment, not trade rules (see Month 1 and Month 11), and that they push Chinese goods into third markets, worsening trade tensions elsewhere.

Judgment: the tariffs succeeded in one narrow goal, reducing direct US dependence on Chinese final goods, and largely failed at their broad stated goal, shrinking the US trade deficit. They also showed that tariffs alone cannot fix dependence on Chinese inputs, as the magnet crisis proved. Their largest lasting effect may be uncertainty: when tariff rates change by law, court ruling and executive order several times a year, companies delay investment or build duplicate supply chains, which raises costs for everyone.

Case study: The rare-earth magnet squeeze of 2025

The product. Neodymium-iron-boron magnets are the strongest permanent magnets. They sit in EV motors, wind turbines, phone speakers, hard drives, car seat motors, windshield wipers, missiles, and in almost every servo motor in a robot. China accounted for over 90 percent of global production of these magnets in 2024 and over 90 percent of mining and processing of the underlying rare earths (IEA, cited by PIIE 2026). Month 5 explained why: decades of investment, tolerance of pollution, and price pressure that drove Western producers out of business.

The squeeze. On 4 April 2025, in response to US tariffs, China required export licences for seven rare-earth elements and magnets. Monthly magnet exports to the United States fell to nearly zero in May (PIIE, 2026). Ford's CEO told Bloomberg in June: "We have had to shut down factories.

It's hand-to-mouth right now" (quoted in PIIE, 2026). Because the controls were global, carmakers in Europe, Japan and India also shut lines; Suzuki halted production of its Swift model in Japan (Nikkei, cited in PIIE). After a US-China deal, exports recovered by July. In October 2025 China announced broader controls and, separately, blocked exports from Nexperia's Chinese plants after a dispute over the Dutch government's control of the company, which hit Honda, Stellantis and others (PIIE, 2026). The Busan truce suspended the October controls for a year, but China's April 2025 controls on seven rare earths stayed in place (TariffWarfare, October 2026).

Why trade data missed it. As Chad Bown of PIIE points out, standard trade statistics did not reveal that US carmakers depended on Nexperia, because chips are not recorded by end use or origin of the wafer. A 2024 US government survey found 44 percent of companies did not know whether their chips came from China (PIIE, 2026). The magnet dependence was visible, but policy ignored it.

What it means.

1. **Small, cheap parts can stop large, expensive products.** A magnet worth a few dollars can halt a 40,000-dollar car. In a robot, magnets, reducers and chips are the equivalent.
2. **Supply chain power comes from concentration, not size.** China did not need to stop selling cars or phones. It needed only to control one chokepoint input. This is the same logic as the US export controls on advanced chips and chipmaking tools, which Month 8 covers. Both great powers now treat chokepoint inputs as weapons.
3. **The response is diversification, which is slow.** The US, Japan, Australia, the EU and others announced magnet and rare-earth processing projects (for example, in July 2025 the US Department of Defense bought 400 million dollars of convertible preferred stock and warrants in MP Materials, about 15 percent of the company on an as-converted basis, guaranteed a price floor of 110 dollars per kilogram for its NdPr oxide for ten years, and promised that all magnets from a planned "10X" plant would be bought for ten years; Month 5 analyses the deal in detail. The US also hosted critical minerals ministerials in early 2026, PIIE). New mines and magnet plants take five to ten years.

For the driving question. Any country hoping to build robots from ore to finished machine must either secure magnets from China on stable terms, develop its own processing, or design motors that use fewer rare earths. Southeast Asian and African countries that hold rare-earth deposits (Malaysia hosts the Lynas processing plant; several African countries have deposits) have a potential opening, but only if they also master the refining and magnet-making steps that Month 6 showed are energy-intensive and polluting.

Key numbers: Week 4

- Share of world goods trade volume carried by sea: about 80% (UNCTAD, 2025).
- Seaborne trade: 12,720 million tonnes (2024, UNCTAD RMT 2025).
- Drewry WCI: 1,420 USD per 40ft (2019 average); record 10,377 USD (23 Sep 2021); 4,429 USD (29 Oct 2026) (Drewry).
- Strait of Malacca oil flow: 23.2 million barrels/day, 29% of maritime oil; Hormuz 20.9 million barrels/day (1H 2025, EIA).
- Suez container capacity: about 11% of 2023 level (Feb 2024 to mid-2026); about 52% in early Oct 2026 (BIMCO/Clarksons).
- Brent futures: about 72 USD (27 Feb 2026) to peak 126.41 USD (30 Apr 2026) (Reuters); Dated Brent record 144.42 USD (7 Apr 2026, Platts via CNBC); futures settled at 104.77 USD

(2 Nov 2026, Reuters).

- Average US tariff rate: 22.5% estimate after 2 April 2025 announcement (Yale Budget Lab); about 11.1% statutory after 24 July 2026, 11.8% scheduled by end-2026 (Yale Budget Lab); 7.1% effective, duties collected divided by imports (Aug 2026, Penn Wharton Budget Model).
- IEEPA tariffs collected: about 166 billion USD; refunds certified about 126 billion USD (by 2 Oct 2026, Penn Wharton Budget Model citing CBP court filings).
- Average US tariff on Chinese goods: 35.2% (trade-weighted statutory, 22 Aug 2026, PIIE); effective rate actually collected on Chinese imports 24.3% (Aug 2026, Penn Wharton Budget Model).
- China's share of global neodymium magnet production: over 90% (2024, IEA via PIIE).

Think like a founder: Week 4

- **Resilience is a service you can sell.** After the Red Sea, Hormuz and the magnet squeeze, manufacturers will pay for supply chain mapping, alternative sourcing, inventory analytics and risk monitoring. AI tools that read customs data, shipping data and supplier records to find hidden dependencies (like Nexperia) are a real business opportunity.
- **Design out chokepoints.** Products that use fewer contested inputs (motors with less or no rare earths, batteries without cobalt, designs that accept chips from several suppliers) are easier to make outside China and less exposed to export controls.
- **Local energy is a competitive advantage.** A factory or data center powered by local solar and storage is protected from oil and LNG shocks. In 2026 that protection has been worth a great deal.
- **Never build a business plan on one tariff rate.** US tariffs changed legal basis four times in 18 months. Build in a margin of safety, sell to several markets, and treat any trade preference as a bonus, not a foundation.

Month synthesis

This month showed that manufacturing is where the physical chain from ore to robot becomes a product, and that the value captured depends on position, not just participation.

A simple map of the manufacturing layer

```
ORE & ENERGY (Months 4-6)
       |
       v
PROCESSED MATERIALS  -- China dominant (magnets >90%, refining)
       |
       v
COMPONENTS  -- clusters: China (all), Japan/Germany (precision),
       |       Korea/Taiwan (chips, batteries), Malaysia (chip back end)
       v
ASSEMBLY  -- moving: China -> Vietnam, India, Mexico, Thailand, Morocco
       |       (gross exports move fast; value added moves slowly)
       v
SHIPPING & CHOKEPOINTS  -- Malacca, Hormuz, Red Sea/Suez, Panama
       |       (2024-2026: Red Sea diversion, Hormuz near-closure)
```

```
       v
MARKETS  -- shaped by tariffs, trade deals, export controls
            (US tariffs 2018-2026; China's mineral controls 2025)
```

Where value is captured: the smile curve applied to a robot

Stage	Share of value (rough, illustrative)	Who captures it today	Opportunity for Southeast Asia / Africa
Research, design, software, brand	High (perhaps 30-40%)	US, China, Japan, Germany, Korea firms	Software and local-market design
Key components (motors, reducers, sensors, chips, batteries)	High (perhaps 30-40%)	China, Japan, Germany, Taiwan, Korea	Malaysia, Thailand (some); Morocco (some)
Final assembly	Low (perhaps 5-10%)	China, increasingly others	Vietnam, India, Indonesia, Morocco
Integration, installation, maintenance, financing	Medium to high, recurring (perhaps 15-25%)	Local firms everywhere	Most accessible for local founders

These shares are illustrative estimates for teaching, based on the smile-curve logic and the iPhone evidence; actual shares vary by product. Month 9 builds a real robot bill of materials.

How the four weeks connect

- **Week 1** explained why clusters form and why China's dominance is self-reinforcing (agglomeration plus learning curves plus state support).
- **Week 2** showed that "China plus one" moves assembly and gross exports faster than value added; Vietnam is the success story with a value-capture problem.
- **Week 3** showed nearshoring's strength (geography) and weakness (policy risk), with Morocco as Africa's model and Ethiopia as a cautionary tale.
- **Week 4** showed that the arteries (shipping, chokepoints) and the rules (tariffs, export controls) can change suddenly, and that power comes from controlling chokepoint inputs.

The thread connecting all four: **value goes to whoever controls the hardest-to-replace step.** In 2025 that was rare-earth magnets (China). In Month 8 you will see the other great chokepoint: advanced chips and chipmaking tools, controlled mainly by Taiwan, the Netherlands, Japan, Korea and the United States.

Common misconceptions

1. **"Factories go wherever wages are lowest."** Wrong. If that were true, the poorest countries would have the most factories. Firms weigh wages against productivity, supplier density, infrastructure, energy reliability, logistics, tariffs and political risk. Bangladesh has lower wages than Vietnam but far less electronics manufacturing; Thailand has higher wages than Indonesia but a deeper auto cluster.

2. **"A country's exports show what it makes."** Only partly. Gross exports include imported parts. Vietnam's 475 billion dollars of exports contain large amounts of Chinese, Korean and Japanese value. Always ask about value added.
3. **"Tariffs make the trade deficit shrink."** Not reliably. The US goods deficit was about 1.23 trillion dollars in 2025, slightly larger than in 2024, despite the highest tariffs in decades. The overall deficit reflects national saving and investment (Month 1). Tariffs mostly shift which countries the deficit is with.
4. **"China plus one means China is losing its manufacturing."** No. China's global exports and surplus hit records in 2025. Its share of US imports fell, but its share of value added in US imports barely moved, because Chinese inputs flow through third countries. China is becoming the supplier to the suppliers.
5. **"Shipping costs decide where things are made."** Rarely for high-value goods. Freight is usually under 1 to 2 percent of an electronics product's price. What matters more is reliability, speed and tariffs. Freight matters most for cheap, heavy or bulky goods.

Exercises

1. **Value added calculation (quantitative).** A Mexican factory imports 70 dollars of Chinese parts and 10 dollars of US parts, and exports a finished server component to the US for 120 dollars. (a) What is Mexico's value added? (b) How much US import value is recorded as coming from Mexico? (c) How much Chinese value is inside the US import? (d) Explain why the US-China bilateral trade deficit understates US dependence on China.
2. **Learning curve (quantitative).** A Chinese firm has produced 4 million robot servo motors at a cost of 40 dollars each. An Indonesian start-up has produced 50,000 at 120 dollars. Assume both follow a 15 percent learning rate. (a) How many doublings does the start-up need to reach 40 dollars, ignoring the Chinese firm's further progress? (b) What cumulative volume is that? (c) What does this imply for the start-up's strategy?
3. **Landed cost (quantitative).** Rebuild the Week 3 China-versus-Mexico motor table with a US tariff on China of 10 percent instead of 35 percent, and freight from China at 12 dollars per unit (a crisis level). Which location wins? At what tariff rate on China are the two locations equal?
4. **Shipping shock (quantitative).** Using the robot vacuum example, recalculate the shock for a 40-dollar product (a cheap fan) with 2,000 units per container. What percentage of the price is the shock? Which kinds of manufacturers are most hurt by chokepoint closures?
5. **Essay (800-1,000 words).** "Vietnam is the next South Korea." Evaluate this claim using evidence on value added, domestic firms, FDI dependence and policy. What would Vietnam need to do differently from today to make it true?
6. **Essay (800-1,000 words).** Compare Morocco and Ethiopia as African manufacturing strategies. Why did one build a durable cluster and the other suffer a collapse after losing a trade preference? What would you advise a country such as Kenya, Nigeria or Tanzania?
7. **Essay with a judgment (800 words).** Was the 2025-2026 US tariff policy a success? Define success first, then use at least five numbers from this chapter, and give a clearly marked judgment.
8. **Founder exercise.** Choose one robot component (servo motor, reducer, battery pack, sensor, controller board). Map who makes it today, which inputs are chokepoints, and where in Southeast Asia or Africa a new producer could plausibly start. Write a one-page business concept with your target customer and your source of advantage.

Reading list

- **Marc Levinson, "The Box: How the Shipping Container Made the World Smaller and the World Economy Bigger" (2006, 2nd ed. 2016).** The best history of containerisation; explains why transport costs collapsed and supply chains globalised.
- **Richard Baldwin, "The Great Convergence: Information Technology and the New Globalization" (2016).** Explains how cheaper communication, not just cheaper transport, let rich-country firms split production across borders; the core theory behind global value chains.
- **Patrick McGee, "Apple in China: The Capture of the World's Greatest Company" (2025).** A detailed account of how Apple built and depended on China's manufacturing base, and what that taught Chinese firms.
- **Dan Wang, "Breakneck: China's Quest to Engineer the Future" (2025).** An insider's argument that China is an "engineering state" that builds fast, with costs and benefits; useful for understanding China's manufacturing culture.
- **Joe Studwell, "How Asia Works" (2013).** Covered in Month 3, but essential here for why export discipline and manufacturing created East Asian success.
- **World Bank, "World Development Report 2020: Trading for Development in the Age of Global Value Chains" (2020).** The standard policy overview of global value chains, with evidence on how countries join and upgrade.
- **Kenneth Kraemer, Greg Linden and Jason Dedrick, "Capturing Value in Global Networks: Apple's iPad and iPhone" (2011 working paper, University of California, Irvine).** The classic value-capture study used in Week 1.
- **UNCTAD, "Review of Maritime Transport 2025: Staying the Course in Turbulent Waters" (2025).** Authoritative annual data on shipping, ports, chokepoints and freight costs.
- **Chad P. Bown, "The Trump-China trade wars: Five takeaways from US imports in 2025" (PIIE, March 2026).** A clear, data-rich summary of how tariffs reshaped US imports and how China weaponised magnets and chips.
- **Soumaya Keynes and Chad P. Bown, "How to Win a Trade War: An Optimistic Guide to an Anxious Global Economy" (Simon & Schuster, May 2026).** A witty, accessible guide by an FT columnist and a PIIE trade economist to tariffs, rare-earth controls, chip curbs and how governments and firms should respond.

Glossary terms introduced this month

- **Agglomeration economies:** the cost advantages firms gain from locating near many similar firms (shared workers, suppliers and knowledge).
- **AGOA (African Growth and Opportunity Act):** a US law giving eligible sub-Saharan African countries duty-free access to the US market for many products.
- **Assembly, testing and packaging (ATP), also OSAT:** the "back end" of chipmaking, where finished silicon is cut, packaged and tested; OSAT means outsourced semiconductor assembly and test.
- **Chokepoint:** a narrow passage (a strait or canal) or a single critical input whose blockage disrupts a whole system.
- **China plus one:** a strategy of keeping production in China while adding at least one other country to reduce risk.
- **Effective tariff rate:** the total customs duties collected (or implied) divided by the value of imports.

- **Export controls:** government rules restricting the sale of goods or technology abroad, usually for security reasons.
- **FDI (foreign direct investment):** investment by a firm in one country that gives it lasting control of a business in another, such as building a factory.
- **Gross trade vs value added:** gross trade counts the full value of goods crossing a border; value added counts only the value created in that country.
- **IEEPA (International Emergency Economic Powers Act):** a 1977 US law used in 2025 to impose tariffs; the Supreme Court ruled in February 2026 that it does not allow tariffs.
- **Landed cost:** the total cost of getting a product to the buyer, including factory price, freight, duties, insurance and inventory cost.
- **Learning curve (Wright's Law):** the pattern that unit costs fall by a roughly constant percentage each time cumulative output doubles.
- **MVA (manufacturing value added):** the value created by manufacturing, equal to output minus purchased inputs.
- **Nearshoring:** moving production to a country close to the final market, such as Mexico for the US.
- **PLI (Production Linked Incentive):** India's scheme paying manufacturers a share of incremental sales in selected sectors.
- **Rules of origin:** the rules that decide which country a product "comes from" for tariff purposes, often based on how much value was added locally.
- **Section 122, Section 232, Section 301:** sections of US trade law allowing temporary balance-of-payments tariffs (122), national-security tariffs (232) and tariffs against unfair foreign practices (301).
- **Smile curve:** a chart showing that value added is high at the start (research, design) and end (brand, service) of a value chain and low in the middle (assembly).
- **TEU (twenty-foot equivalent unit):** the standard unit of container capacity; a 40-foot container is 2 TEU.
- **Transshipment:** routing goods through a third country, sometimes with minimal processing, to change their apparent origin and avoid tariffs.
- **USMCA:** the United States-Mexico-Canada Agreement, the 2020 successor to NAFTA.

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Month 8: Semiconductors, Compute and the AI Supply Chain

Month overview

Month 7 showed how factories cluster and how value is captured in ordinary manufacturing. This month we look at the most extreme case of all: the semiconductor, or chip. A modern AI chip starts as quartz sand and ends as a device with tens of billions of transistors, each smaller than a virus. The chain that makes it crosses dozens of countries, and at several steps only one or two companies on Earth can do the job.

Chips are now the most important manufactured input to the economy you will build your career in. Global chip sales were about 796 billion US dollars in 2025 (World Semiconductor Trade Statistics, WSTS, March 2026) and are forecast to reach about 1.5 trillion dollars in 2026, almost doubling in one year, mainly because of AI (WSTS, June 2026). One company, Nvidia, sold 215.9 billion dollars of products in its fiscal year to January 2026 (Nvidia 10-K). Four US technology companies plan to spend roughly 720 to 740 billion dollars on capital investment in 2026, most of it on AI data centers (company guidance, late October 2026). And those data centers used about 485 terawatt-hours of electricity in 2025, about as much as a large country, a figure the International Energy Agency (IEA) expects to roughly double by 2030 (IEA, April 2026).

This month answers four questions. Week 1: how is a chip designed and made, and why is the leading edge so concentrated in Taiwan? Week 2: why are the machines that make chips, especially ASML's lithography tools, the ultimate chokepoint, and how have the United States and China used export controls since 2022? Week 3: why have memory and packaging, once boring parts of the industry, become the bottlenecks of the AI boom in 2025-2026? Week 4: what do data centers need (electricity, water, land, capital), and how do AI model companies actually make money?

The thread is the same as in every month: who captures the value? In AI the answer so far is clear, and it is not the countries that supply the sand, the copper, the electricity or the land. Month 9 then takes the chips and the models and puts them into robots.

Learning goals

By the end of this month you should be able to:

- Describe the semiconductor value chain (design, design software, equipment, materials, fabrication, packaging and testing) and say which countries and firms lead each step.

- Explain, with numbers, why leading-edge chipmaking is so concentrated (fab costs, learning curves, EUV lithography).
- Tell the story of US export controls on chips and China's response from 2022 to November 2026, and judge what they achieved.
- Explain why high-bandwidth memory (HBM) and advanced packaging became bottlenecks, and what the 2026 memory price surge means for buyers.
- Estimate the electricity, capital and water needs of an AI data center, and describe what a host country such as Malaysia gains and risks.
- Explain the economics of AI models: training versus inference, falling cost per task, and why revenue and spending are both exploding.
- Locate the points in the AI supply chain where a founder from Southeast Asia or Africa could realistically capture value.

Week 1: How chips are made, and why the leading edge lives in Taiwan

Core lesson

What a chip is

A semiconductor is a material, usually silicon, whose ability to conduct electricity can be switched on and off. A chip (or integrated circuit) is a small slab of silicon carrying billions of tiny switches called transistors, connected by microscopic copper wires. Transistors are combined to store information (memory chips) and to perform calculations (logic chips, such as the central processing unit, CPU, in your laptop, or the graphics processing unit, GPU, that trains AI models).

The industry groups chips roughly into three families:

- **Logic:** processors that compute: CPUs, GPUs, AI accelerators, and the "systems on a chip" in phones. Logic was the largest category, about 300 billion dollars of sales in 2025 (WSTS).
- **Memory:** chips that store data: DRAM (dynamic random-access memory, the fast working memory) and NAND flash (long-term storage, as in a phone's storage). Memory sales were about 230 billion dollars in 2025 and are forecast to reach about 800 billion dollars in 2026 (WSTS, June 2026), a story we tell in Week 3.
- **Analog, discrete and other:** chips that handle power, radio signals and sensors, made on older, cheaper processes. These are less glamorous but essential for cars, factories and robots.

A useful fact from Deloitte: in 2025 the world bought about 1.05 trillion chips at an average price of about 74 cents each. But AI chips, which account for roughly half of industry revenue in 2026, are fewer than 20 million units, about 0.2 percent of chips sold (Deloitte, 2026 Semiconductor Industry Outlook). The money is concentrated in a tiny number of extremely valuable chips.

The value chain, step by step

Making a chip takes six main steps. Each is dominated by a few firms in a few countries.

1. **Design software and intellectual property (IP).** Chip designers use electronic design automation (EDA) software, dominated by two US firms, Synopsys and Cadence, and license basic building blocks (for example, processor designs from the UK-based, Japanese-owned Arm).

2. **Chip design.** Companies such as Nvidia, AMD, Apple, Qualcomm, Broadcom, and increasingly Google, Amazon and Huawei design chips. Many are "fabless": they design but do not manufacture.
3. **Equipment.** Factories need machines for lithography (printing patterns), deposition (adding thin layers), etching (removing material), cleaning and inspection. ASML (Netherlands), Applied Materials, Lam Research and KLA (United States) and Tokyo Electron (Japan) dominate.
4. **Materials.** Ultra-pure silicon wafers, specialty gases and chemicals, mostly from Japan, Korea, Taiwan, Germany and China.
5. **Fabrication ("front end").** The wafer is processed in a fabrication plant (fab) through hundreds of steps over two to three months. TSMC (Taiwan) leads, followed by Samsung (Korea) and Intel (United States); China's SMIC is the most advanced Chinese foundry.
6. **Assembly, testing and packaging ("back end").** The finished wafer is cut into individual chips (dies), which are packaged and tested. Firms such as ASE (Taiwan) and Amkor (United States), and large clusters in China, Taiwan and Malaysia, do this work. For AI chips, a new kind of "advanced packaging" has become critical (Week 3).

How is value shared across these steps? A widely cited estimate by Boston Consulting Group (BCG) and the US Semiconductor Industry Association (SIA), reproduced by the World Trade Organization, divides the industry's value added in 2019 as follows:

Step	Share of industry value added (2019)
Design of logic chips (mostly fabless)	30%
Design of memory chips	17%
Design of analog, discrete and other chips	9%
Wafer fabrication	19%
Manufacturing equipment	12%
Assembly, packaging and test	6%
Materials	5%
Design software and core IP	3%

Source: WTO, Global Value Chain Development Report 2023, Figure 4.2, citing BCG and SIA (2021). Note that memory design and manufacturing are usually done by the same firm, so the memory share overlaps with fabrication in practice.

This is the smile curve again. Design captures more than half of the value. Assembly and testing, the step that Malaysia, the Philippines and Vietnam do most of, captures about 6 percent. Bank Negara Malaysia makes the same point about its own country (Month 7).

Why the leading edge is so concentrated

Chips are classified by their "process node", a name like 7 nanometers (nm), 3 nm or 2 nm. The names no longer measure any physical feature exactly; they are marketing labels for generations, each packing more transistors into the same area with better speed and lower power use. "Leading edge" means the newest generations, below about 10 nm.

In 2019 about 92 percent of the world's leading-edge logic capacity (below 10 nm) was in Taiwan, with the rest in South Korea (BCG, 2021). By 2022 BCG and SIA estimated Taiwan's share at about 69 percent of sub-10 nm logic capacity, and projected it would fall to about 47 percent by 2032

as the United States, Japan and Europe build fabs (SIA and BCG, May 2024). Even so, Taiwan will remain the center for years.

Three forces explain the concentration:

1. **Enormous fixed costs.** A leading-edge fab module with capacity of about 20,000 wafers a month costs roughly 25 to 35 billion dollars to build and equip (industry estimate cited by Tom's Hardware, July 2026). Only firms with huge, predictable demand can fill such a factory.
2. **Learning curves and yield.** "Yield" is the share of chips on a wafer that work. A new process may start with low yield; every month of production teaches engineers how to fix defects. The firm that has produced the most wafers at a node learns fastest. This is the same experience-curve logic from Month 7, but with stakes of tens of billions of dollars.
3. **The foundry model and trust.** TSMC, founded in 1987 by Morris Chang with Taiwanese government support, pioneered the "pure-play foundry": it makes chips for others and never competes with its customers by selling its own. This made it the natural partner for every fabless designer, giving it more volume, which made it better, which gave it more customers. Samsung and Intel, which also design their own chips, have found it harder to win that trust.

A worked example: the economics of a wafer

The numbers below are illustrative, not TSMC's actual prices, which are confidential.

Suppose a 2 nm wafer (300 millimeters across) costs the designer about 30,000 dollars. A large AI accelerator die might measure about 800 square millimeters. A 300 mm wafer has an area of about 70,000 square millimeters, but square dies do not fit perfectly into a circle, so you might get about 60 candidate dies.

- If yield is 50 percent, 30 good dies: 30,000 / 30 = 1,000 dollars per good die.
- If yield improves to 80 percent, 48 good dies: 30,000 / 48 = 625 dollars per good die.

Now compare that with the selling price. An AI accelerator built from one or two such dies, plus memory and packaging, sells for tens of thousands of dollars. The silicon die is a small share of the price. Most of the value is in the design, the software ecosystem (Nvidia's CUDA), the memory, the packaging, and the scarcity. This explains why Nvidia earns far higher margins than TSMC, even though TSMC does the hardest manufacturing.

Also notice how yield works: the bigger the die, the more likely it contains a defect. That is one reason the industry is moving to "chiplets": several smaller dies packaged together, which yield better than one giant die.

Moore's Law and its costs

In 1965 Gordon Moore, later a co-founder of Intel, observed that the number of transistors on a chip doubled about every year, later revised to every two years. This "Moore's Law" held for decades and is why a phone today outperforms a 1990s supercomputer. It has slowed: each new node now costs much more to develop, and the performance gain per node is smaller. The industry keeps progress going through new transistor shapes (TSMC's 2 nm uses "nanosheet" transistors), chiplets, advanced packaging, and specialised chips designed for one job, such as AI. For a founder, the practical lesson is that compute will keep getting cheaper per unit of work, but the cost of the most advanced factories rises, so the industry keeps concentrating.

Case study: TSMC, the world's most important factory

The numbers. In 2025 TSMC's revenue rose 35.9 percent in US dollar terms to 122.42 billion dollars, with a gross margin of 59.9 percent (TSMC 4Q25 Management Report, January 2026). Advanced technologies (7 nm and below) made up 74 percent of its wafer revenue, up from 69 percent in 2024; 3 nm alone was 24 percent (TSMC, 2025 Annual Report). Its 2 nm process entered high-volume production in the fourth quarter of 2025 with "good yield", and enhanced versions (N2P and A16) were scheduled for volume production in the second half of 2026, with A14 planned for 2028 (TSMC 2025 Annual Report). TSMC estimates it held 40 percent of what it calls the "Foundry 2.0" industry in 2025 (which includes packaging and testing), up from 34 percent in 2024 (TSMC 2025 Annual Report, chapter 2). Customers based in North America produced 74 percent of its revenue in the fourth quarter of 2025 (TSMC 4Q25 Management Report).

How it invests. TSMC planned capital expenditure of 52 to 56 billion dollars for 2026, up from about 41 billion in 2025 (Futurum; The Diligence Stack, January 2026), and raised the 2026 budget to 60 to 64 billion dollars in July 2026 (Aterio and Tom's Hardware, July 2026). For comparison, China's leading foundry SMIC spent a record 8.1 billion dollars in 2025 (Nikkei Asia, February 2026). TSMC's spending in one year is roughly five to eight times SMIC's.

Going abroad. Under US pressure and with US subsidies (up to 6.6 billion dollars in CHIPS Act grants, NIST), TSMC built fabs in Arizona. Its announced US commitment rose from 12 billion dollars in 2020 to 65 billion, then to 165 billion in March 2025 (TSMC press release), and to about 265 billion dollars with another 100 billion announced in July 2026 for at least four more 2 nm-class fabs (Tom's Hardware; Aterio, July 2026). It also has fabs in Japan (Kumamoto) and a joint venture in Germany (Dresden). However, its most advanced processes still start in Taiwan, and the Taiwanese government has an interest in keeping it that way.

The geopolitical risk. Because Taiwan makes most of the world's leading-edge logic chips, any conflict or blockade around Taiwan would be an economic catastrophe far larger than the Hormuz closure of 2026 (Month 7). This is sometimes called Taiwan's "silicon shield": the idea that the world's dependence on TSMC deters an attack. Judgment: the shield is real but weakening, because the US and others are deliberately building alternatives, and an aggressor might calculate that dependence cuts both ways. Founders building hardware businesses should treat a Taiwan disruption as a low-probability, very-high-impact risk and avoid single-sourcing critical chips where possible.

What to learn. TSMC shows that value can be captured in manufacturing, not only in design, if the manufacturing is hard enough. Its gross margin of about 60 percent is closer to a software company than to an assembler. The lesson for the driving question: assembly (Month 7) captures little, but extreme-precision manufacturing with decades of accumulated know-how captures a lot. The path from one to the other took Taiwan 40 years of focused policy, education and investment.

Key numbers: Week 1

- Global semiconductor sales: 795.6 billion USD, up 26.2% (2025, WSTS, March 2026).
- Forecast global semiconductor sales: 1.51 trillion USD, up about 90% (2026, WSTS Spring forecast, June 2026).
- Logic chip sales: about 300 billion USD (2025, WSTS).
- Chips sold: about 1.05 trillion units at about 0.74 USD average; AI chips about 0.2% of units (2025-2026, Deloitte).
- Taiwan's share of leading-edge (sub-10 nm) logic capacity: about 92% (2019, BCG); about 69% (2022, SIA/BCG); projected 47% (2032, SIA/BCG).

- Share of semiconductor value added in assembly, packaging and test: about 6%; in logic design about 30% (2019, BCG/SIA via WTO 2023).
- TSMC revenue: 122.42 billion USD, gross margin 59.9% (2025, TSMC 4Q25 Management Report).
- Cost of a 2 nm-class fab module (about 20,000 wafers/month): about 25-35 billion USD (2026 industry estimate via Tom's Hardware).
- SMIC capital expenditure: record 8.1 billion USD (2025, Nikkei Asia).

Think like a founder: Week 1

- **Do not try to build a fab.** The capital needed is beyond almost any start-up or developing country. But design is far cheaper: a team of skilled engineers with EDA licenses and access to a foundry can design a specialised chip. Small chip-design firms (for sensors, power management, edge AI) have emerged in India, Vietnam and Malaysia.
- **Older chips are an opportunity.** Robots, cars, solar inverters and appliances mostly use mature-node chips (28 nm and above) for power, sensing and control. These are cheaper to design and make, less affected by export controls, and in demand in Southeast Asia's growing electronics clusters.
- **Your scarcest resource is talent.** Taiwan's success rested on thousands of engineers trained at home and in the US. For a founder in Indonesia or Kenya, hiring and training hardware and chip-design engineers may be harder than raising money.
- **Know your exposure.** If your product needs leading-edge chips, your supply depends on Taiwan, on US export rules and on the AI boom's appetite for capacity. Design for flexibility: support more than one chip supplier where you can.

Week 2: The machines behind the machines, and the chip war

Core lesson

Lithography: printing with light

Lithography is the step that "prints" circuit patterns onto silicon. A machine shines light through a mask (a stencil of the circuit) and through lenses onto a wafer coated with a light-sensitive chemical. The smaller the wavelength of light, the smaller the features that can be printed. For decades the industry used deep ultraviolet (DUV) light with a wavelength of 193 nanometers, using tricks such as printing the same layer several times ("multi-patterning") to make smaller features.

Extreme ultraviolet (EUV) lithography uses light with a wavelength of 13.5 nanometers. Making that light is extraordinarily hard: a laser hits tiny droplets of molten tin tens of thousands of times a second, creating a plasma that emits EUV light, which is then reflected by mirrors of almost unimaginable smoothness (Zeiss and ASML often say that if one were scaled up to the size of a country like Germany, its largest bump would be well under a millimeter high). EUV light is absorbed by air, so the whole process happens in a vacuum. Developing EUV took ASML and its partners (including the German optics firm Zeiss and the US laser company Cymer, which ASML bought) roughly two to three decades and billions of euros, with research funding from Intel, Samsung and TSMC.

Only ASML makes EUV machines. A standard ("Low-NA") EUV machine costs roughly 170 million euros or more, and the newest "High-NA" machine more than 350 million euros, about 410 million dollars (Tom's Hardware, July 2026; Bloomberg, April 2026). NA means numerical aperture, a

measure of how much light the lens collects; higher NA prints finer features. Every leading-edge chip in the world, whether made by TSMC, Samsung or Intel, depends on ASML's EUV tools.

ASML in numbers

In 2025 ASML's total net sales were 32.7 billion euros, with a gross margin of 52.8 percent and net income of 9.6 billion euros (ASML Q4 2025 results, January 2026). It recognized revenue on 48 EUV systems worth 11.6 billion euros, up 39 percent, and DUV systems worth 12.0 billion euros, down 6 percent (ASML investor call, January 2026). Its order backlog at the end of 2025 was 38.8 billion euros. It expected 2026 sales of 34 to 39 billion euros (ASML, January 2026).

China is an important customer for DUV machines, which can still be sold there with licenses. China's share of ASML's sales fell from 41 percent in 2024 to 33 percent in 2025, and ASML's finance chief expected about 20 percent in 2026 (South China Morning Post, January 2026). Chinese firms bought heavily in 2023-2024, partly to stockpile tools before controls tightened.

A telling detail: in April 2026 TSMC said it would not use High-NA EUV in production through 2029 because the machines are too expensive, preferring to extract more from existing tools (Bloomberg, April 2026). Even the world's leading chipmaker weighs cost against capability. Intel, trying to catch up, adopted High-NA earlier.

Why equipment is a chokepoint

Equipment sits upstream of every fab. If you cannot buy the machines, you cannot build a leading-edge fab, no matter how much money you have. The equipment market is itself concentrated: the US, Japan and the Netherlands together lead in almost every tool category (SIA and BCG, 2024). This is why the US, which no longer makes most leading-edge chips, still holds enormous power over the industry: many critical tools contain US technology, and US export rules can reach foreign products made with US technology.

The chip war: US export controls 2022-2026

Since 2022 the United States has used export controls to slow China's progress in advanced chips and AI. The main steps:

Date	Action
October 2022	US Commerce Department's Bureau of Industry and Security (BIS) restricts exports of advanced AI chips (such as Nvidia's A100) and chipmaking equipment to China, and restricts US citizens from helping Chinese advanced fabs.
2023	Netherlands and Japan add their own controls on advanced lithography and other tools. October 2023 rules close loopholes; Nvidia's China-specific A800 and H800 chips are banned, and Nvidia designs the weaker H20.
December 2024	Controls extended to high-bandwidth memory (HBM) and more equipment.

Date	Action
January 2025	Biden administration issues the "AI Diffusion Rule", a global licensing system for AI chips by country tier.
April-May 2025	Trump administration requires licenses for Nvidia's H20; in May announces it will rescind the AI Diffusion Rule.
August 2025	US licenses H20 exports to China in a deal reported to give the US government a share of revenue (Reuters, August 2025).
September 2025	BIS "Affiliates Rule" extends Entity List restrictions to firms at least 50% owned by listed companies; China's regulator says Nvidia violated anti-monopoly law.
November 2025	Under the Busan truce, the US suspends the Affiliates Rule for one year.
December 2025 - January 2026	President Trump announces Nvidia may sell its H200 chip to China with the US government taking 25%; a BIS rule effective 15 January 2026 moves H200-class exports to case-by-case review with conditions, including US-based testing (Federal Register, January 2026).
January-March 2026	Chinese customs reportedly block H200 imports and tell firms not to buy; in March Beijing reportedly approves purchases (CSIS; Reuters, March 2026).
May 2026	US clears about 10 Chinese firms, including Alibaba, Tencent and ByteDance, to buy up to 75,000 H200s each, but no deliveries had yet been made (Reuters, May 2026). BIS closes a loophole allowing advanced chip sales to Chinese-headquartered firms' subsidiaries abroad (Reuters, May 2026).
August 2026	Nvidia says H200 shipments to China were less than 1% of its data center revenue and it assumes no China data center compute revenue in its outlook (Nvidia Q2 FY2027 results; The Register, September 2026).

The logic of the controls changed over time. The Biden administration's goal was to keep China several generations behind in AI hardware and chipmaking tools. The Trump administration has treated chip access partly as a bargaining chip and a revenue source, allowing second-best chips (H20, then H200) to be sold for a fee while keeping the best (Blackwell, Rubin) barred. A Congressional Research Service analysis questioned the legal basis for charging export fees, since US law does not allow direct export taxes; the arrangement routes chips through US territory for testing so that a 25 percent import tariff applies instead (CRS, March 2026; Reuters, May 2026).

China's response: build everything at home

China has responded in three ways.

1. **Domestic chips.** Huawei designs Ascend AI accelerators. Huawei does not publish output, and estimates differ widely, which is itself a lesson in how to read numbers. Epoch AI estimates that Huawei produced about 800,000 Ascend units in 2025, against about 4.5 million AI chips sold by Nvidia, and gives a median estimate of 1.5 million Ascend units for 2026, against 5.9 million for Nvidia (Epoch AI, September 2026). Bloomberg reported a plan for about 600,000 Ascend 910C chips in 2026 and up to 1.6 million dies across the line, and Reuters a target of about 750,000 of the new Ascend 950PR, which entered volume production in early 2026 (as reported by TrendForce and Tom's Hardware, 2026). At the low end, the US Commerce Department estimated Huawei's 2025 capacity at under 200,000 chips (as reported in industry summaries, 2026). Two facts make the higher numbers less impressive than they look. First, much of the 2024-2025 output used logic dies made by TSMC before controls bit (about 2.9 million dies, according to SemiAnalysis) and stockpiled Samsung and SK hynix memory; Epoch expects that stockpile to be used up in 2026, after which high-bandwidth memory from China's CXMT becomes the binding limit. Second, each Ascend chip is weaker: Epoch puts the Ascend 950 at roughly half the performance of Nvidia's H100, a chip that began shipping in 2022, so Huawei's 2026 output equals less than 4 percent of Nvidia's in computing power. In September 2026 a Huawei executive claimed Ascend sales in China had overtaken Nvidia's (The Register, September 2026), which is plausible precisely because Nvidia was largely shut out of China; Huawei expected about 12 billion dollars of AI chip revenue in 2026 (Tom's Hardware, May 2026).
2. **Domestic fabs.** SMIC produces 7 nm-class chips using older DUV tools and multi-patterning, which is slower and yields worse than EUV. Chinese firms reportedly aim to raise 7 nm and 5 nm-class output from under 20,000 wafers a month to 100,000 within one to two years (Nikkei via TrendForce, February 2026). Goldman Sachs estimated China's self-supply of advanced chips at about 8 percent of demand in 2025, potentially rising to about 66 percent by 2035 (Goldman Sachs via Chosun Biz, August 2026). These are projections, not facts.
3. **Domestic tools and memory.** China funds domestic equipment makers (Naura, AMEC, SMEE) and memory makers (CXMT for DRAM, YMTC for flash), and has used its own export controls on gallium, germanium and rare earths (Month 7) as counter-pressure.

How large is the gap? The Council on Foreign Relations argued in 2026 that the best US AI chips were about five times more powerful than the best Chinese chips by one standard measure, and that the gap could widen to about 17 times by late 2027 (CFR, 2026). Huawei's own executives frame the issue differently: "Even though our chips may be less advanced, at least their supply is assured" (Eric Xu, quoted in The Register, September 2026).

Judgment: export controls slowed China's access to the very best AI hardware and forced Chinese firms to use less efficient tools, which raises their costs. They did not stop Chinese AI progress, as models such as DeepSeek showed, and they gave Beijing a powerful reason to build a fully domestic chip industry, which over a decade may reduce the leverage the controls depend on. The 2025-2026 policy of selling second-best chips for a fee has been confused in execution: China itself discouraged purchases. Both superpowers now treat chips the way earlier generations treated oil.

What this means for third countries

Countries in Southeast Asia, the Middle East, Africa and Latin America are caught between. US rules restrict exports of advanced AI chips to some destinations and require buyers to prevent diversion to China. Malaysia and Singapore have prosecuted cases of suspected chip smuggling. At the same time, Chinese firms offer cheaper alternatives (Huawei chips, Chinese cloud services) without US conditions. Judgment: for most developing countries the sensible strategy is to stay usable by both blocs where legally possible, comply carefully with export rules to keep access to US chips, and invest in skills and data centers that add local value regardless of whose chips are inside.

Case study: ASML and the single point of failure

Origins. ASML began in 1984 as a joint venture between the Dutch electronics company Philips and the equipment maker ASM International. In its early years it was a small player behind Japan's Nikon and Canon. Its bet on EUV in the late 1990s and 2000s looked risky; Nikon and Canon eventually abandoned EUV. ASML succeeded by acting as a systems integrator: it designs and assembles the machine, but relies on thousands of specialised suppliers, most importantly Zeiss in Germany for the mirrors and Cymer in the US for the light source. A single EUV machine has hundreds of thousands of parts and is shipped in dozens of containers or several cargo planes.

Why nobody can copy it quickly. Chris Miller's book "Chip War" (2022) describes the EUV machine as perhaps the most complex machine humans have built in volume. To replicate it, a rival would need not only ASML's designs but its supplier network, each of which holds its own decades of know-how. This is agglomeration (Month 7) at its most extreme: the value lies not in one firm but in a web of firms that learned together.

The political lever. Because ASML is Dutch, the US needed the Netherlands to agree to restrict sales to China. Since 2019 ASML has not been able to ship EUV machines to China, and since 2023 the Dutch government has restricted some advanced DUV machines too. This shows that controlling a chokepoint requires allies: the US alone could not have blocked China's access to lithography.

The economics. ASML's 2025 gross margin was 52.8 percent and it plans to raise EUV prices to capture more of the value its tools create (Tom's Hardware, July 2026). Its customers are few (TSMC, Samsung, Intel, SK hynix, Micron and Chinese firms for DUV), and each is dependent on it. This is near-monopoly pricing power, limited mainly by the risk that customers delay purchases, as TSMC did with High-NA.

What to learn. The deepest chokepoints are not raw materials, which can be found elsewhere, but accumulated knowledge embedded in a network of firms. For a country hoping to move up the value chain, the lesson is sobering: the frontier of the semiconductor industry is not something you buy; it is something you grow over decades. But it also shows that a small country (the Netherlands has under 20 million people) can hold one of the most valuable positions in the world economy if it specialises deeply.

Key numbers: Week 2

- ASML total net sales: 32.7 billion EUR; gross margin 52.8%; net income 9.6 billion EUR (2025, ASML results).
- ASML EUV systems recognized: 48, worth 11.6 billion EUR, up 39% (2025, ASML).
- China's share of ASML sales: 41% (2024) to 33% (2025), expected about 20% (2026) (SCMP citing ASML CFO).

- Price of a High-NA EUV machine: over 350 million EUR, about 410 million USD (2026, Bloomberg).
- Price of a standard Low-NA EUV machine: from about 170 million EUR (2026, Tom's Hardware).
- H200 export arrangement: US government share 25%; up to 75,000 chips per approved Chinese buyer (Dec 2025 to May 2026, Reuters; CRS).
- H200 shipments to China: under 1% of Nvidia data center revenue (quarter to July 2026, Nvidia via The Register).
- China's advanced-chip self-supply: about 8% of demand (2025), projected about 66% (2035) (Goldman Sachs via Chosun Biz, August 2026).
- Huawei Ascend output: about 800,000 units (2025) and a median estimate of 1.5 million (2026), versus Nvidia about 4.5 million (2025) and 5.9 million (2026); Huawei 2026 compute under 4% of Nvidia's (Epoch AI, September 2026).

Think like a founder: Week 2

- **Compliance is a competitive advantage.** If you build AI products in Southeast Asia, your access to US chips and cloud services depends on following export rules. A start-up with clean, documented compliance can partner with US firms that a sloppier competitor cannot.
- **Do not bet everything on one bloc.** Build software that can run on more than one kind of hardware (Nvidia, AMD, Google TPUs, Huawei Ascend, or cheaper inference chips). This reduces risk if rules change.
- **Specialist suppliers can be hugely valuable.** ASML's power rests on thousands of niche suppliers. In robotics, the same pattern will emerge: a firm that makes one critical component extremely well can capture outsized value.
- **Watch the second-best market.** When the best chips are restricted or scarce, there is demand for good-enough alternatives and for software that makes older or cheaper chips more efficient. That is where many non-US, non-Chinese founders can compete.

Week 3: Memory and packaging, the hidden bottlenecks

Core lesson

Why AI is hungry for memory

An AI accelerator is a calculating engine, but it can only calculate as fast as it can be fed data. Large AI models have hundreds of billions of numbers ("parameters") that must be moved from memory to the processor again and again. If memory is slow, the expensive processor sits idle. Engineers call this the "memory wall".

The solution is high-bandwidth memory (HBM): several DRAM chips stacked vertically, like floors of a building, connected by thousands of tiny vertical wires called through-silicon vias (TSVs), and placed right next to the processor on the same package. HBM moves data far faster than ordinary memory, but it is much harder to make: each stack combines many chips, and a defect in any layer can ruin the whole stack. Each new generation (HBM3E, then HBM4 from 2026) adds layers and speed.

Three companies make almost all the world's DRAM and HBM: Samsung and SK hynix of South Korea, and Micron of the United States. China's CXMT is growing in ordinary DRAM. SK hynix bet early on HBM and became Nvidia's main supplier, holding about 64 percent of HBM revenue in

the second quarter of 2025. By the second quarter of 2026 its share had fallen to about 50 percent as Samsung recovered to about 33 percent, with Micron at about 18 percent (Counterpoint Research via Aju Press and Seoul Economic Daily, October 2026).

The memory supercycle of 2025-2026

Memory has always been a boom-and-bust industry. When prices are high, the three makers build factories; a few years later supply floods the market and prices crash. AI broke the usual rhythm. From the second half of 2025, data center buyers wanted so much HBM and server DRAM that the memory makers shifted their production lines to those products, which left too little ordinary DRAM for phones and PCs. Prices exploded:

- Conventional DRAM contract prices rose about 45 to 50 percent in the fourth quarter of 2025 alone (TrendForce, February 2026).
- In the first quarter of 2026 they rose another 93 to 98 percent, and total DRAM industry revenue jumped 81 percent in one quarter to about 97 billion dollars (TrendForce, June 2026).
- In the second quarter of 2026 DRAM industry revenue rose another 59.5 percent to about 154.7 billion dollars; Samsung led with a 39.4 percent share (TrendForce, September 2026).
- TrendForce expects server DRAM contract prices to rise about 270 percent over 2026 as a whole, after a 64 percent rise in the second half of 2025 (TrendForce, August 2026).

The WSTS now forecasts the memory market to grow about 250 percent in 2026, to more than 800 billion dollars, making memory larger than logic for the first time in years (WSTS, June 2026). TrendForce estimates that DRAM and NAND flash together will make up 47 percent of large cloud companies' capital spending in 2026 and 68 percent in 2027 (TrendForce, August 2026). The IEA notes that the HBM shortage is expected to last at least until the end of 2027 (IEA, April 2026).

A worked example: who pays when memory prices triple?

Suppose before the boom a mid-range laptop cost 800 dollars to make, of which 16 GB of DRAM cost 40 dollars and storage 40 dollars.

- If DRAM prices triple, memory now costs 120 dollars: +80 dollars.
- If storage prices rise 50 percent: +20 dollars.
- Total extra cost: about 100 dollars, or 12.5 percent of the original cost.

A laptop maker with a 5 percent margin cannot absorb that; it must raise prices, cut memory (8 GB instead of 16 GB), or accept losses. This is why the Straits Times reported in May 2026 that laptops could cost much more because of the "AI-fuelled RAMpocalypse" (Straits Times, May 2026), and why TrendForce warned of weak consumer demand as notebook prices reflected higher component costs (TrendForce, September 2026). The AI boom is, in effect, outbidding ordinary consumers for the same factory capacity. In a country like Indonesia, where many buyers are price-sensitive, this can slow the spread of computers and phones, a real cost of the AI boom that rarely appears in headlines.

Advanced packaging: the new bottleneck

Packaging used to mean putting a chip into a protective case with pins. For AI chips it means something far more complex. TSMC's CoWoS (Chip on Wafer on Substrate) places one or more processor dies and several HBM stacks side by side on a thin silicon "interposer" with dense wiring,

then mounts the whole assembly on a substrate. This lets processor and memory talk at enormous speeds. Nvidia's and other AI accelerators depend on it.

CoWoS capacity became the binding constraint on AI chip supply. TSMC does not publish its CoWoS capacity, so all figures are industry estimates:

Period	Estimated TSMC CoWoS capacity (wafers per month)	Source
End of 2024	about 35,000	industry estimates via newtype, INDmoney (2026)
End of 2025	about 72,500-75,000 (range of estimates 65,000-80,000)	SemiVision via Silicon Analysts; TrendForce, December 2025
End of 2026	about 120,000-135,000 (KGI 120,000; Bernstein 135,000; "over 130,000" per Commercial Times)	KGI, September 2026; Bernstein, July 2026; TrendForce, June and October 2026
End of 2027	about 185,000-200,000 (target)	KGI, September 2026; Commercial Times via TrendForce, October 2026

Even with capacity nearly quadrupling in two years, analysts estimate that CoWoS demand still exceeded supply by roughly 10 to 20 percent through 2026, and lines were reported fully booked (TrendForce, June 2026; KGI, September 2026). Note how much the estimates differ: a 15,000-wafer gap between two respected analysts is about 12 percent of the total, because TSMC publishes no figure and analysts count capacity differently (for example, whether to include steps outsourced to partners). TSMC began outsourcing more steps to packaging firms such as ASE and Amkor (TrendForce, August 2026) and was reported to be preparing CoWoS price increases of up to 20 percent (TrendForce, October 2026). Advanced packaging was slightly over 10 percent of TSMC's revenue in 2025 (TSMC 4Q25 call, via Silicon Analysts). Epoch AI data reported by industry analysts suggests that the top four AI chip designers consumed about 90 percent of global CoWoS capacity and HBM supply in 2025 while accounting for only about 12 percent of advanced logic die production (vlsi.kr, October 2026, citing Epoch AI; not independently verified).

Why packaging matters for Southeast Asia

Remember that assembly, packaging and test captured only about 6 percent of semiconductor value added in 2019 (Week 1). Advanced packaging changes this. It is high-precision, capital-intensive and scarce, and it commands rising prices. Malaysia (Penang), Singapore, Taiwan and increasingly Vietnam and the Philippines host packaging and testing plants. Judgment: advanced packaging is the most realistic way for Southeast Asia to move up the semiconductor value chain in the next 10 to 15 years, because it builds on existing back-end clusters and does not require a leading-edge fab. Malaysia's National Semiconductor Strategy explicitly targets this move; whether firms there can win advanced-packaging work from TSMC, ASE and Amkor is the key question for Penang's next decade.

Case study: The 2026 memory squeeze

Setting. By mid-2026, memory had become the single biggest driver of chip industry growth. Gartner forecast memory would account for 54 percent of semiconductor revenue in 2026, up from 27 percent in 2025 (Gartner, August 2026). Samsung, which had lost DRAM leadership to SK hynix in 2025 by being slow on HBM, recovered by moving quickly into HBM4 and by having the most capacity for ordinary DRAM just as its price soared (TrendForce, September 2026; Counterpoint, August 2026).

Contracts and power. In a shortage, the seller has power. Memory makers signed long-term agreements (LTAs) with cloud companies covering years of supply; Samsung was reported to have locked up about 70 percent of some capacity through 2031, and spot prices for HBM3E were reported at four to five times contract levels (TrendForce news, September 2026). TrendForce forecast that HBM average prices would rise about 121 percent in 2027 as HBM4 takes a larger share (TrendForce, September 2026).

Who wins and who loses.

- **Winners:** the three memory makers, whose profits surged; South Korea's export earnings; equipment makers selling to memory fabs.
- **Squeezed:** PC and phone makers, consumers, smaller AI start-ups and data center operators without long-term contracts, and carmakers and industrial firms that need ordinary memory.
- **Mixed:** cloud companies, which pay more for memory but secure supply that smaller rivals cannot.

The cycle question. Will this end in a crash, as past memory booms did? Memory makers are building new fabs, but a DRAM fab takes two to three years, and HBM yields improve slowly. Most analysts in late 2026 expected tight supply through 2027 (IEA; TrendForce). Judgment: some correction is likely by 2028-2029 when new capacity arrives, because the memory industry has never escaped its cycle for long. But AI demand has raised the floor: the market is unlikely to return to 2023 price levels.

What to learn. In the AI supply chain, the bottleneck moves: first GPUs, then CoWoS packaging, then HBM and ordinary memory, then electricity (Week 4). Each time, whoever controls the current bottleneck captures extra profit. For a founder, the skill is to anticipate the next bottleneck rather than chase the current one. For the robot question, it is a warning: in the 2030s and 2040s, if robots need large amounts of on-board memory and compute, they will compete with data centers for the same chips, and the robot builders without long-term contracts will pay the highest prices.

Key numbers: Week 3

- Memory market forecast: more than 800 billion USD, up about 250% (2026, WSTS, June 2026).
- Conventional DRAM contract prices: up about 93-98% quarter on quarter (1Q 2026, TrendForce).
- DRAM industry revenue: about 97 billion USD (1Q 2026) and 154.7 billion USD (2Q 2026, up 59.5% quarter on quarter) (TrendForce); conventional DRAM contract price growth expected to slow to 13-18% in 3Q 2026.
- HBM revenue shares: SK hynix 50% (down from 64% a year earlier), Samsung 33% (from 15%), Micron 18% (2Q 2026, Counterpoint Research). Overall DRAM revenue shares in 2Q 2026: Samsung 39.4%, SK hynix 24.9%, Micron 23.3% (TrendForce; Counterpoint gives 38/25/24,

with China's CXMT at 10%).
- Memory share of large cloud companies' capital spending: 47% (2026) and 68% (2027 forecast) (TrendForce, August 2026).
- HBM shortage expected to last until at least end-2027 (IEA, April 2026).
- TSMC CoWoS capacity estimates: about 35,000 (end-2024), about 75,000 (end-2025), about 120,000-135,000 (end-2026), about 185,000-200,000 (end-2027 target) wafers/month (industry estimates; TSMC publishes none).
- Advanced packaging share of TSMC revenue: slightly over 10% (2025, TSMC via Silicon Analysts).
- Memory share of semiconductor revenue: 27% (2025) to 54% (2026 forecast) (Gartner, August 2026).

Think like a founder: Week 3

- **Design for scarcity.** If you build AI products, make them efficient in memory use. Smaller models, compression ("quantization") and smart caching reduce your costs when memory is expensive. Efficiency is not just engineering pride; in 2026 it is a margin.
- **Lock in supply early.** Large firms win shortages through long-term contracts. Small firms can partner with cloud providers or regional data center operators that already hold supply, rather than buying hardware on the spot market.
- **Watch for the bust.** Memory booms end. A founder who plans for memory prices to fall again in 2028-2030 could design products (cheap edge AI devices, robots) that become affordable exactly when the cycle turns.
- **Packaging and testing services are a real opportunity.** Southeast Asian engineers in Penang, Batam or Ho Chi Minh City can build businesses in test equipment, inspection software, and specialised packaging for sensors and power chips, areas where incumbents are stretched.

Week 4: Data centers, electricity and the economics of AI models

Core lesson

The scale of the build-out

An AI data center is a building full of servers, each holding several accelerators, linked by fast networks, cooled by air or liquid, and fed by enormous amounts of electricity. The money spent on them in 2025-2026 is without precedent in the private sector.

- The capital expenditure of five large technology companies exceeded 400 billion dollars in 2025 and was expected to rise another 75 percent in 2026; the IEA notes this is now larger than global investment in oil and gas production (IEA, Key Questions on Energy and AI, April 2026).
- By October 2026, Amazon guided to about 220 billion dollars of capital spending for 2026, Alphabet to 195-205 billion, Microsoft to about 170 billion (cut from 175 billion on 28 October because some third-party data center capacity was delayed) and Meta to 135-145 billion, roughly 720-740 billion dollars together (company guidance reported by CNBC, 28-29 October 2026). Adding other large spenders, Business Insider estimated 2026 spending of 803-828 billion dollars (Business Insider, October 2026).
- Nvidia, the main beneficiary, reported revenue of 215.9 billion dollars in its fiscal year ended

January 2026, of which data center revenue was 193.7 billion (Nvidia 10-K and press release, February 2026). In the quarter ended 26 July 2026 it reported revenue of 96.2 billion dollars, data center revenue of 89.0 billion (up 117 percent from a year earlier), and a gross margin of 75 percent, and guided to 108 billion dollars for the next quarter (Nvidia, August 2026).

Where does the money go? Roughly, an AI data center's capital cost splits among the chips and servers (the largest share), networking, memory, the building, power equipment (transformers, switchgear, backup generators, batteries) and cooling. In 2026 memory's share rose sharply (Week 3). Nvidia also announced partnerships with large investment firms to mobilise over 500 billion dollars of outside capital for AI infrastructure (Nvidia, August 2026), a sign that the build-out now relies on capital markets, not just company profits.

Electricity: how much, and where?

The IEA estimates data centers used about 415 terawatt-hours (TWh) of electricity in 2024, about 1.5 percent of world consumption, of which the United States accounted for 45 percent, China 25 percent and Europe 15 percent (IEA, Energy and AI, April 2025). Consumption grew 17 percent in 2025 to about 485 TWh, with AI-focused data centers growing 50 percent (IEA, April 2026). The IEA's central projection has data center consumption roughly doubling to about 950 TWh by 2030, around 3 percent of global electricity demand, with AI-focused data centers tripling (IEA, April 2026).

Three things make this harder than the global percentage suggests:

1. **Concentration.** Data centers cluster in a few places (northern Virginia, Dublin, Johor), where they can take a large share of local electricity and strain the grid.
2. **Density.** Between 2020 and 2025 the power density of AI servers rose about 11 times; by 2027 a single rack the size of a refrigerator could draw as much peak power as 65 households (IEA, April 2026).
3. **Speed mismatch.** A data center can be built in 18 to 24 months; a high-voltage transmission line or power plant can take five to ten years. In the United States, developers are building gas turbines on site to avoid waiting for grid connections; the IEA estimates 15 to 27 gigawatts (GW) of on-site gas could power data centers by 2030, and notes gas turbine orders rose about 70 percent in 2025 (IEA, April 2026).

The good news is efficiency. The IEA reports that the energy used per AI task has dropped by at least an order of magnitude per year in recent years, and that a simple text query now uses less electricity than running a television for the same time. The bad news is that new uses, such as video generation, reasoning models and AI agents, can use hundreds or thousands of times more energy per query (IEA, April 2026). This is a classic case of the Jevons paradox: when something becomes cheaper to use, people use much more of it, so total consumption can rise even as efficiency improves.

A worked example: powering and paying for a 100-megawatt AI data center

Assume a data center with 100 megawatts (MW) of IT load, running at 80 percent average utilisation, with cooling and other overhead adding 25 percent (a power usage effectiveness, PUE, of 1.25).

- Average power drawn: 100 MW x 0.8 x 1.25 = 100 MW.
- Annual electricity: 100 MW x 8,760 hours = 876,000 MWh = 0.876 TWh.
- At 80 dollars per MWh, the electricity bill is about 70 million dollars a year.

How does that compare with the hardware? If the accelerators, servers and networking cost on the order of 3 to 4 billion dollars and wear out (become obsolete) over about five years, the hardware cost is roughly 600 to 800 million dollars a year. Electricity is perhaps a tenth of that. These figures are illustrative, but the conclusion is robust: for AI data centers, chips dominate cost, not power. That is why companies will pay high prices for electricity if it lets them start running expensive chips sooner. Speed to power, not the price of power, is the main location factor, which favours places with spare grid capacity and fast permitting.

Water is the other resource. AMRO, the regional economic monitor, estimates that a 100 MW data center can consume about 4.2 million litres of water a day for cooling (AMRO via TNGlobal, July 2026), though newer liquid-cooling designs and recycled water can reduce this sharply.

The economics of AI models: training versus inference

AI models cost money in two ways:

- **Training** is the one-off cost of building a model: running thousands of accelerators for weeks or months on huge amounts of data. Frontier training runs now cost hundreds of millions to billions of dollars. Epoch AI has estimated that the cost of training frontier models has grown by more than two times a year (Epoch AI data, as compiled in industry summaries).
- **Inference** is the cost of running the model each time someone uses it. Each request is cheap, but billions of requests add up. As usage explodes, inference is becoming the larger share of compute demand.

Two trends collide:

1. **The price of a given capability is collapsing.** Epoch AI found that the cost of achieving a given level of AI performance on its benchmarks fell about 47 percent per quarter since 2023, roughly 13 times per year, faster than any other transformative technology it compared, including electricity, batteries, computing and DNA sequencing (Epoch AI, "The plunging price of thought", September 2026). In one example, the cost of reaching a particular score on a PhD-level science test fell about 725-fold in under 18 months (Epoch AI).
2. **Demand and revenue are exploding anyway.** Anthropic told investors its annualised revenue run-rate passed 65 billion dollars at the end of July 2026, up from about 47 billion in May, with preliminary second-quarter revenue of 11.5 billion dollars, 14 times a year earlier (CNBC and Bloomberg, August 2026). OpenAI told investors its run-rate was approaching 50 billion dollars at the end of September 2026 (Axios and Financial Times, October 2026). These two numbers are not directly comparable, as the next section explains, but together the two firms were earning at a rate of well over 100 billion dollars a year, roughly triple a year earlier (Epoch AI, August 2026).

How can both be true? Because cheaper capability unlocks new uses (coding agents, customer service, research assistance), and because customers keep paying for the newest, best models, whose prices fall more slowly. Epoch AI found that costs fall fastest right after a capability becomes state of the art (about 66 percent per quarter) and more slowly two years later (about 32 percent per quarter), which suggests that "supra-normal profits in LLM service provision may be fleeting for any given model" (Epoch AI, September 2026). In other words, a model company must keep building new frontier models to keep charging premium prices, which means it must keep spending on training compute. This creates a treadmill: revenue grows fast, but so does the cost of staying at the front.

A worked example: the unit economics of a token

A token is a small piece of text, roughly three-quarters of an English word. Suppose an AI company sells access to a mid-sized model at 2 dollars per million input tokens and 10 dollars per million output tokens, prices typical of mid-tier models in late 2026 (published price lists compiled by CoinCentral, 2026).

- A customer-service conversation uses 3,000 input tokens and 500 output tokens.
- Revenue: 3,000/1,000,000 x 2 + 500/1,000,000 x 10 = 0.006 + 0.005 = 0.011 dollars, about one US cent.
- If the company's compute cost to serve that conversation is 0.004 dollars, its gross margin on inference is about 64 percent.

Now suppose a competitor releases a model that is just as good at half the price. The company must cut prices or lose customers. If it cuts to half, revenue per conversation falls to about 0.0055 dollars and its margin drops to about 27 percent unless its own costs also fall. This is why AI companies race to make inference cheaper (better chips, smaller models, efficient software), and why some analysts describe a "margin collapse" in the middle of the market even as frontier models stay profitable. Reported estimates of inference gross margins at leading labs vary widely, from about 40 to 85 percent depending on the firm, period and method (for example, Longbridge's analysis of reported Anthropic figures, October 2026); none of these are audited public figures, so treat them as rough.

How AI companies count revenue, and where one dollar of AI spending ends up

Two numbers in the previous section look precise but hide a trap. In September 2026 several outlets reported that OpenAI's run-rate was about 70 billion dollars; in October, OpenAI's own figure to investors turned out to be about 50 billion (Axios, October 2026). Nothing was hidden; the difference was accounting. Learning to read this kind of number is part of thinking like a founder or investor.

Run-rate is not revenue. An annualised run-rate takes revenue from a recent month (or quarter) and multiplies it to a year: 5.4 billion dollars in one month becomes a run-rate of about 65 billion. For a company growing fast, the run-rate at the end of a year is far above the revenue it actually booked during that year. Anthropic's preliminary revenue for the whole second quarter of 2026 was 11.5 billion dollars (CNBC, August 2026), equal to a run-rate of 46 billion for the quarter as a whole, while its run-rate at the end of July was over 65 billion. Both numbers are honest; they measure different moments. Public-market investors prefer actual recognised revenue; private start-ups prefer run-rates because they flatter growth.

Gross versus net. Suppose a business customer pays 100 dollars for an AI model through a cloud provider's marketplace, and the cloud provider keeps a share. Axios explains that Anthropic's accounting records the full 100 dollars as revenue and lists the cloud provider's cut as a cost, while OpenAI records only its own share of such partner sales (Axios, October 2026). Both methods follow US accounting rules (GAAP); the choice depends on who is judged to control the customer relationship. The 70-billion figure was an attempt to "gross up" OpenAI's revenue to Anthropic's method for comparison. The lesson: before you compare two companies' revenue, ask what each one counts.

Following one dollar down the stack. Now we can trace where money paid for AI goes. The split below is a teaching model built from the margin figures in this chapter, not a measured flow;

real splits differ by firm and year.

1. A company pays an AI lab 100 dollars for model access.
2. Suppose the lab spends about 50 dollars on computing to serve that demand (inference), and much more on training future models. Most of that compute is rented from a cloud company.
3. The cloud company uses its revenue to pay for data centers. In 2026 the four largest US cloud and platform companies guided to roughly 720 to 740 billion dollars of capital spending, most of it on AI (company guidance, October 2026). Of each dollar of AI data center capital spending, the largest single share goes to accelerators and servers.
4. Nvidia sells those accelerators with a gross margin of about 75 percent (quarter to July 2026). So of each 100 dollars of Nvidia revenue, about 25 dollars pays for everything Nvidia buys to make the product: silicon wafers and CoWoS packaging from TSMC, high-bandwidth memory from SK hynix, Samsung and Micron, substrates, boards and assembly.
5. TSMC keeps a gross margin of about 60 percent on what it is paid (2025), and spends part of the rest on tools from ASML (gross margin about 53 percent in 2025), Applied Materials, Lam, KLA and Tokyo Electron.
6. Further down are the electricity, land, water, copper and silicon. In the 100 MW worked example above, electricity was about a tenth of the annual hardware cost.

The pattern is the same as the robot joint in Month 7: the further you move from the scarce, hard-to-copy step, the thinner the margin. A host country that supplies power, land and water to a data center earns its share as a utility earns: steady, regulated and small relative to the value of what runs inside. The scarce steps (accelerator design and software ecosystem, leading-edge fabrication, HBM, EUV tools) earn margins of 50 to 75 percent.

The revenue gap. One more calculation shows why investors argue about an "AI bubble". If around 720 billion dollars of capital spending in 2026 were all depreciated over five years (servers often over about five to six years, buildings much longer), it would create roughly 140 billion dollars a year of depreciation cost, before electricity, staff and a return on capital. A 10 percent return on 720 billion is another 72 billion. So the AI services built on this spending need to generate gross profit on the order of 200 billion dollars a year, and growing, as later years' spending piles up. In late 2026 the two leading labs together had revenue run-rates of a bit over 100 billion dollars, and cloud companies also earn AI revenue directly and use AI inside their own products such as search and advertising, which is hard to measure. Optimists point to revenue roughly tripling each year; pessimists point out that capital spending is also rising fast and that much of it is financed by debt and outside capital. Judgment: the numbers do not prove a bubble, but they show that AI revenue must keep growing several-fold for today's spending to earn normal returns, so a founder should not assume that cheap, subsidised compute will last forever.

Judgment: the economics of AI models look like a mix of the airline and pharmaceutical industries. Like pharmaceuticals, frontier development is extremely expensive, and the newest product earns high margins for a short time. Like airlines, the mature product becomes a commodity with intense price competition. The largest, most durable profits so far have gone not to model makers but to the suppliers of scarce hardware: Nvidia (75 percent gross margin), TSMC (about 60 percent), ASML (about 53 percent) and, in 2026, the memory makers.

Data centers in Southeast Asia: what does a host country capture?

Southeast Asia has become a major data center region. As of 2026, Malaysia had about 1.3 GW of operational capacity with a development pipeline of over 5 GW, Singapore about 1.1 GW, and

Indonesia about 470 MW operational with nearly 2 GW in development (Cushman & Wakefield via The Star, October 2026). Different sources give somewhat different totals. The case study below looks at Johor, the center of this boom.

What does a host country gain? Construction jobs and spending; land sales and rents; electricity sales; some skilled technical jobs; tax revenue; and better digital infrastructure for local firms. What does it not gain automatically? The chips are designed in the US and made in Taiwan and Korea; the AI models are owned by US or Chinese firms; the profits flow to the cloud companies. A data center is, in value-capture terms, closer to a mine or a port than to a design center: it uses local land, energy and water to serve a global value chain controlled elsewhere. Judgment: data centers are worth hosting if the host country prices its power, water and land properly, insists on renewable energy and local skills development, and uses the infrastructure to build local AI businesses. They are a poor deal if they are subsidised with cheap power that households then pay for.

Case study: Johor, the data center boom next door to Singapore

How it started. Singapore, short of land and power, imposed a moratorium on new data centers from 2019 to 2022. Operators moved across the causeway to Johor, Malaysia, which offered land, power, fibre connections and proximity to Singapore's business networks (The Star; AMRO, 2026). Johor drew an estimated 35 billion dollars in investment (Reuters, July 2026). In early 2021 Johor had about 10 MW of data center capacity (Straits Times, August 2025). By 2026 estimates of its operational capacity ranged from about 825 MW to about 1 GW, with around 2 GW under construction and several more gigawatts planned (BMI via The Edge, October 2026; Roland Berger via TNB news clip, October 2026). Johor accounts for around 80 percent of Malaysia's operational data center capacity (AMRO, July 2026).

The strain. Wood Mackenzie estimates data centers could account for about 40 percent of Johor's electricity demand by 2035 (Wood Mackenzie, June 2026). BMI estimates Johor's data centers will need about 30 TWh a year by 2031, around 18 percent of Peninsular Malaysia's electricity supply, up from 6 percent in 2026 (BMI via The Edge, October 2026). Malaysia's energy ministry projects peak data center demand in Peninsular Malaysia could reach 8.39 GW by 2030 (Malay Mail, October 2026). In late 2025 Johor banned new approvals for the most water-intensive basic data center types, which can use as much as 50 million litres a day (Straits Times, November 2025). From February 2026 Malaysia began restricting new non-AI data center investments because of their power and water use (AMRO, July 2026), and 2026 saw the country's first community protests against data centers (Reuters, July 2026).

What Malaysia is trying to capture. Malaysia is attempting to turn the boom into more than land and power sales: requiring renewable energy (one operator signed a 500 MW renewable power deal with Tenaga Nasional in 2025, Business Times), building a special economic zone with Singapore (the Johor-Singapore Special Economic Zone), and linking data centers to its semiconductor strategy in Penang. Whether local firms can capture higher-value work (AI services, chip design, engineering) is the open question.

What to learn. Johor shows both sides of the AI infrastructure boom for a middle-income country. It brings investment fast, but it competes with households and factories for electricity and water, and most of the value of what runs inside the buildings is captured elsewhere. For Indonesia, which is also attracting data centers (Batam, near Singapore, and the Jakarta area), the Johor experience is a direct lesson: plan the grid and water first, price resources properly, and bargain for local capability.

Key numbers: Week 4

- Capital spending of five large tech companies: over 400 billion USD (2025), expected +75% (2026) (IEA, April 2026).
- 2026 capital spending guidance: Amazon about 220, Alphabet 195-205, Microsoft about 170, Meta 135-145 billion USD, total about 720-740 (late October 2026, company guidance via CNBC).
- Nvidia revenue: 215.9 billion USD, data center 193.7 billion USD (fiscal year to January 2026, Nvidia 10-K).
- Nvidia quarterly revenue: 96.2 billion USD; data center 89.0 billion USD; gross margin 75.0% (quarter to 26 July 2026, Nvidia).
- Data center electricity: about 415 TWh (2024), about 485 TWh (2025), about 950 TWh projected (2030) (IEA, 2025 and 2026).
- Cost of a given AI performance level: falling about 47% per quarter, about 13x per year (2023-2026, Epoch AI, September 2026).
- Annualised revenue run-rates: Anthropic over 65 billion USD (end of July 2026, CNBC/Bloomberg); OpenAI about 50 billion USD (end of September 2026, Axios/FT); the two use different accounting for sales through cloud partners.
- Malaysia data center capacity: 925.6 MW live, 2.0 GW under construction, 3.1 GW planned (October 2026, BMI via Malay Mail).
- Johor data centers' projected share of Peninsular Malaysia electricity: about 6% (2026) to 18% (2031) (BMI via The Edge, October 2026).

Think like a founder: Week 4

- **Build on falling prices.** If the cost of a given AI capability falls about 13 times a year, a product that is too expensive to run today may be cheap in 12 to 18 months. Design for where costs will be, not where they are, but make sure you can survive until then.
- **Do not compete with frontier labs on models.** Their training budgets are in the billions. Compete on data, distribution, local language, local regulation, trust and integration into real workflows, where a local founder has an edge.
- **Energy is a business.** Data centers need power fast. Founders who can develop solar, storage, grid services or efficient cooling in Southeast Asia or Africa sell to one of the fastest-growing customer groups in the world.
- **Think "sovereign" but practical.** Governments want local AI capacity. A founder who offers local hosting, local-language models and compliance with local data laws can win public contracts, but must avoid building expensive infrastructure that sits idle.

Month synthesis

The AI supply chain as a stack

```
ENERGY, LAND, WATER  (local: US, China, Malaysia, Gulf, Nordics...)
       |
DATA CENTERS & CLOUD  (Amazon, Microsoft, Google, Meta, Oracle; CoreWeave;
       |               Chinese clouds)  capex ~720-830 bn USD in 2026
       |
AI MODELS  (OpenAI, Anthropic, Google, Meta, DeepSeek, Alibaba...)
       |    cost per capability falling ~13x/yr; revenue >100 bn USD/yr run-rate
```

```
       |
ACCELERATORS & SYSTEMS  (Nvidia ~75% gross margin; AMD; Google TPU; Huawei)
       |
MEMORY (HBM, DRAM)  (SK hynix, Samsung, Micron)  2026 supercycle
       |
ADVANCED PACKAGING  (TSMC CoWoS; ASE; Amkor)  2025-26 bottleneck
       |
FABRICATION  (TSMC ~60% gross margin; Samsung; Intel; SMIC)
       |
EQUIPMENT  (ASML EUV monopoly; AMAT, Lam, KLA, TEL)
       |
MATERIALS & MINERALS  (silicon, gallium, germanium, rare earths, copper)
       (Months 5-6)
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Who captures value in 2025-2026?

Layer	Example firm	Margin signal (2025-2026)	Concentration
Equipment	ASML	Gross margin 52.8% (2025)	Sole EUV supplier
Fabrication	TSMC	Gross margin 59.9% (2025)	Most leading-edge logic
Memory	Samsung, SK hynix, Micron	Revenue up 60-80% per quarter in 1H 2026	Three firms
Accelerators	Nvidia	Gross margin 75.0% (quarter to July 2026)	Dominant, with rising rivals
Cloud and data centers	Hyperscalers	Spending 720-830 bn USD (2026); returns still uncertain	Five US firms plus Chinese clouds
AI models	Frontier labs	Fast revenue growth; margins debated, prices falling	A handful of labs
Host countries	Malaysia, Indonesia	Land, power, construction, some jobs	Many competing locations

The pattern: value concentrates where substitutes are fewest (EUV tools, leading-edge fabs, HBM, Nvidia's ecosystem). It thins out where many suppliers compete (data center land, electricity in some places, older chips, and increasingly AI model access itself).

Connecting to the driving question

Month 7 showed that robot assembly is likely to spread, but components and materials are concentrated. This month shows that the "brain" of a robot, its chips and the AI models it runs, is even more concentrated. A country in Southeast Asia or Africa in the 2040s that assembles robots will probably buy its AI chips from a handful of firms in the US, Taiwan, Korea or China and license its AI models from a few labs. Judgment: the most realistic routes to capturing value are (1) advanced

packaging and testing in Southeast Asia, (2) mature-node and specialised chip design for robots and sensors, (3) data center hosting with local energy, bargained for hard, and (4) applications and services built on cheap AI, where local knowledge is the advantage. Leading-edge fabs and frontier models are out of reach for all but a few nations.

Common misconceptions

1. **"Chips are a commodity, so they do not matter much."** Most chips are cheap (about 74 cents on average), but the most advanced ones are among the most valuable and concentrated products ever made. A shortage of a few critical chips can stop entire industries, as the auto chip shortage of 2021 and the Nexperia episode of 2025 showed.
2. **"With enough money, any country can build a TSMC."** Money is necessary but not sufficient. Leading-edge chipmaking requires decades of learning, a dense supplier network, thousands of experienced engineers and access to ASML's tools. Even Intel and Samsung, with huge budgets, have struggled to match TSMC.
3. **"Export controls have stopped China's AI."** They slowed China's access to the best chips and raised its costs, but Chinese firms continue to produce competitive models and are building domestic chips. Controls are a delay, not a wall.
4. **"AI will soon use most of the world's electricity."** The IEA projects data centers at around 3 percent of global electricity in 2030. The real problem is local: concentration in a few grids and the speed mismatch between data centers and power infrastructure.
5. **"Cheaper AI means less spending on AI."** The opposite has happened. The cost per unit of capability falls about 13 times a year, yet spending and usage are rising rapidly, because cheaper AI unlocks new uses (the Jevons paradox).

Exercises

1. **Wafer economics (quantitative).** A 300 mm wafer costs 25,000 dollars and fits 120 candidate dies of 400 square millimeters. (a) At 70 percent yield, what is the cost per good die? (b) If the die were split into two 200 square millimeter chiplets, 250 candidate chiplets fit and yield rises to 85 percent. What is the silicon cost for one pair of good chiplets? (c) Why might chiplets still cost more overall?
2. **Data center power (quantitative).** A planned 300 MW (IT load) data center in Indonesia runs at 75 percent utilisation with a PUE of 1.3. (a) Calculate its annual electricity use in TWh. (b) If Indonesia's electricity consumption is about 300 TWh a year, what share would this one site use? (c) What would you require of the developer if you were the energy minister?
3. **Memory shock (quantitative).** A phone costs 200 dollars to make, of which memory is 25 dollars. Memory prices rise 150 percent. (a) What is the new cost? (b) What percentage increase is that? (c) If the phone sells for 215 dollars (a margin of about 7 percent over the 200-dollar cost), what happens to its profit if it cannot raise prices?
4. **Token economics (quantitative).** An AI tutoring start-up in Semarang charges students 5 dollars a month. Each student uses 2 million input tokens and 400,000 output tokens a month. Model prices are 0.10 dollars per million input and 0.50 dollars per million output tokens. (a) What is the compute cost per student? (b) What is the gross margin? (c) What happens if usage triples? (d) What if model prices fall 13-fold next year?
5. **Essay (800-1,000 words).** "Taiwan's silicon shield makes war less likely." Evaluate this argument, using evidence on TSMC's role, overseas fab construction, and the strategic interests

of the US and China.

6. **Essay (800-1,000 words).** Were US export controls on advanced chips to China from 2022 to 2026 a success? Define success, use at least five facts from this chapter, and give a clearly marked judgment.
7. **Essay with a judgment (800 words).** Should Indonesia or Malaysia welcome more AI data centers? Use the Johor case, the power and water numbers, and the value-capture table. Propose three policy conditions.
8. **Founder exercise.** Pick one bottleneck in the AI supply chain (memory, packaging, power, cooling, skilled engineers, local-language data). Write a one-page business plan for a company from Southeast Asia or Africa that would profit from that bottleneck in the next five years, and explain what happens to your business if the bottleneck disappears.

Reading list

- **Chris Miller, "Chip War: The Fight for the World's Most Critical Technology" (2022).** The best single history of the semiconductor industry, from Silicon Valley to TSMC and the US-China conflict.
- **Stephen Witt, "The Thinking Machine: Jensen Huang, Nvidia, and the World's Most Coveted Microchip" (2025).** A readable history of how Nvidia became the central company of the AI boom.
- **SIA and BCG, "Emerging Resilience in the Semiconductor Supply Chain" (May 2024).** Data-rich report on where each step of chipmaking happens and how it may shift by 2032.
- **CSET (Saif M. Khan, Alexander Mann, Dahlia Peterson), "The Semiconductor Supply Chain: Assessing National Competitiveness" (2021).** A clear map of the value chain and each country's position, written for policymakers.
- **IEA, "Energy and AI" (April 2025) and "Key Questions on Energy and AI" (April 2026).** The most authoritative global analysis of data center electricity demand.
- **Epoch AI, "The plunging price of thought" (September 2026), by Luke Emberson and David Roodman.** Careful measurement of how fast AI costs are falling, with honest discussion of limitations.
- **WTO and partners, "Global Value Chain Development Report 2023" (2023), chapter 4.** Places semiconductors in the wider story of global value chains and resilience.
- **TSMC Annual Report 2025 (2026).** Reading a real annual report teaches you how the world's most important factory describes its own business, risks and strategy.

Glossary terms introduced this month

- **Accelerator (AI accelerator):** a chip specialised for AI calculations, such as a GPU or Google's TPU.
- **Advanced packaging:** techniques that place several chips (for example a processor and memory stacks) very close together in one package with dense connections; TSMC's CoWoS is the leading example.
- **CoWoS (Chip on Wafer on Substrate):** TSMC's advanced packaging technology used for most AI accelerators.
- **DRAM (dynamic random-access memory):** fast working memory used in computers, phones and servers.

- **EDA (electronic design automation):** software used to design chips.
- **EUV (extreme ultraviolet) lithography:** chip-printing technology using 13.5-nanometer light; made only by ASML.
- **Export controls:** government restrictions on selling certain goods or technology abroad.
- **Fab (fabrication plant):** a factory that processes silicon wafers into chips.
- **Fabless:** a chip company that designs chips but outsources manufacturing.
- **Foundry:** a company that manufactures chips designed by others (TSMC is the largest).
- **GPU (graphics processing unit):** a processor with many parallel cores, originally for graphics, now the main chip for AI.
- **HBM (high-bandwidth memory):** stacked DRAM placed next to a processor to provide very fast data transfer.
- **Hyperscaler:** a very large cloud computing company (Amazon, Microsoft, Google, Meta, Oracle).
- **Inference:** running a trained AI model to produce outputs.
- **Jevons paradox:** when greater efficiency lowers the cost of using a resource, total use can rise.
- **Lithography:** printing circuit patterns onto silicon using light.
- **Moore's Law:** the observation that the number of transistors on a chip doubles about every two years.
- **Node (process node):** a generation of chipmaking technology, labelled in nanometers (for example 3 nm, 2 nm).
- **PUE (power usage effectiveness):** total data center power divided by the power used by IT equipment; 1.0 is ideal.
- **Token:** a small chunk of text (about three-quarters of an English word) used to measure AI model input and output.
- **Training:** the process of building an AI model by adjusting its parameters on large amounts of data.
- **Transistor:** a tiny electronic switch, the basic building block of a chip.
- **TWh (terawatt-hour):** one billion kilowatt-hours, a unit for measuring large amounts of electricity.
- **Wafer:** a thin disc of silicon, usually 300 mm across, on which many chips are made at once.
- **Yield:** the share of chips on a wafer that work correctly.

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Month 9: Robotics, Anatomy and Industrial Economics

Month overview

Last month you followed the chain from sand to silicon: chip design, fabrication plants, lithography machines, memory, packaging, data centres and the electricity that feeds them. That chain produces intelligence that lives inside computers. This month the intelligence gets a body. A robot is the place where everything in this course meets: the copper and rare earths of Month 5, the refining and processing politics of Month 6, the factory clusters and supply chains of Month 7, and the chips and AI models of Month 8. If you want to know who captures value "from ore to robot to AI", the robot is the object you must take apart.

We will do exactly that. Week 1 separates the robots that already exist at scale (about five million industrial robot arms working in factories) from the humanoid robots that fill the news. Week 2 opens the body: motors, gearboxes called reducers, screws, magnets, sensors, batteries and computers, and traces each part back to its minerals and its suppliers. Week 3 is about money: the bill of materials (the list of parts and their costs), how costs fall as production grows, and why a robot built with a Chinese supply chain costs roughly a third of one built without it. Week 4 maps the players by country and asks the question you care about most: could Africa or Southeast Asia build a cheap robot supply chain, from ore to finished robot, by the 2040s?

The main lesson of the month is simple to state and hard to act on. In robotics, value does not sit mainly in the final assembly. It sits in a few precision components (reducers, screws, force sensors, magnets), in the software and AI that make the body useful, and in the services that keep robots running. The country or company that controls one of those chokepoints captures far more than the one that bolts the parts together. Next month (Month 10) turns from the machines to the people: what happens to jobs, wages and inequality when these machines spread, and who owns them.

Learning goals

By the end of this month you should be able to:

- Explain the difference between industrial robots, collaborative robots, mobile robots and humanoids, and use the International Federation of Robotics (IFR) statistics correctly, including their limits.
- Name the main parts of a robot joint (motor, reducer, screw, encoder, driver) and explain what each does and why it costs what it does.

- Trace a humanoid robot's main parts back to their minerals (neodymium, dysprosium, copper, lithium, nickel, steel) and to the countries that dominate each stage.
- Read a bill of materials and use a learning curve (Wright's law) to estimate how costs could fall with volume.
- Compare the robot industries of China, Japan, Germany, South Korea and the United States, and explain why China's share is rising.
- Judge, with numbers, which layers of the robot value chain Southeast Asia or Africa could realistically capture by the 2040s, and which they almost certainly cannot.

Week 1: What a robot is, and where robots actually work

Core lesson

Four kinds of robot

The word "robot" covers very different machines. Economists and investors who mix them up make bad forecasts, so start with clear categories.

1. **Industrial robots.** These are the robot arms you see in car and electronics factories. The International Federation of Robotics (IFR), the industry body that has collected robot statistics since the 1990s, follows the international standard ISO 8373: an industrial robot is an "automatically controlled, reprogrammable, multipurpose manipulator, programmable in three or more axes" used in industrial automation. Most are fixed to the floor, work inside safety fences and repeat precise motions: welding, painting, picking parts, assembling circuit boards. They are mature technology. The first one, Unimate, went to work at a General Motors plant in 1961.
2. **Collaborative robots ("cobots").** These are smaller arms built to work next to people without a fence. They move more slowly, sense contact and stop. They are easier to program and cheaper to install, which matters for small firms.
3. **Mobile robots and service robots.** These include warehouse robots that carry shelves, autonomous mobile robots (AMRs) that move parts around factories, cleaning robots, delivery robots, agricultural robots and four-legged ("quadruped") inspection robots. The IFR counts professional service robots separately from industrial robots.
4. **Humanoids.** These are robots built roughly in human shape: two legs (or a wheeled base), two arms, hands and a head full of sensors. The bet behind them is that a human-shaped machine can use spaces, tools and workflows designed for humans, so one general-purpose robot could replace many special-purpose ones. That bet has not yet paid off at scale.

Why does the distinction matter for economics? Because the first category is a real, large, measured industry, while the fourth is mostly a forecast. In 2025 factories installed 603,307 industrial robots worldwide, a record (IFR, World Robotics 2026). Epoch AI, a research group, estimates that only about 16,000 humanoids were produced in the whole world in 2025 (Epoch AI, April 2026). The industrial arm market is about forty times larger in units, and much larger again in value, than the humanoid market. When you read that "robots will transform the economy", always ask: which robots?

The global numbers

The IFR's World Robotics 2026 report, released on 24 September 2026, gives the latest full-year picture (data for 2025):

- **Installations:** 603,307 new industrial robots in 2025, up 11% from 2024. After four years at around 550,000 a year, demand broke out to a new record. The previous record was 552,946 units in 2022.
- **Operational stock:** 5,079,078 industrial robots working worldwide at the end of 2025, up 9%. The IFR expects the stock to pass six million in 2028.
- **Market value:** US$18.6 billion for the robots themselves in 2025 (this excludes software, peripherals and system integration, which are usually larger).
- **Geography:** Asia installed 457,315 units (76% of the world total). China alone installed 354,226 units, 59% of the world. Europe installed 73,054 (down 14%). The Americas installed 57,044 (up 14%), of which the United States took 38,428.
- **Customers:** the electrical and electronics industry bought 142,235 robots, the automotive industry 138,429, and metal and machinery 108,571.
- **Forecast:** 655,000 installations in 2026 and 806,000 in 2029, about 7% a year (an IFR projection, not a fact).

Two other figures from the IFR are worth remembering because they show how unequal the map is. In 2025, the whole of Africa installed 519 industrial robots and Australia 569 (IFR, "Exciting Times for Robot Suppliers!", 28 September 2026). China installed about 680 times as many as all of Africa. And the IFR calls Japan "the world's largest exporter of robots", even though Japan's own domestic installations fell 19% to 36,219 units in 2025.

A quick worked example shows what the market value figure means. US$18.6 billion divided by 603,307 robots is about US$30,800 per robot on average. That is the price of the arm and its controller. A finished robot cell (the arm plus grippers, sensors, safety fencing, conveyors, programming and installation) usually costs several times more (an industry rule of thumb rather than an IFR statistic; treat it as rough). This is the first sign of a pattern that runs through the whole month: the physical machine is only part of the value. Integration, software and service often earn as much or more.

Robot density, and a lesson about statistics

The most quoted robot statistic is **robot density**: the number of operational industrial robots per 10,000 employees in manufacturing. It lets you compare a giant country with a small one. On the IFR's April 2026 release (World Robotics 2025 data for 2024):

Economy	Robots per 10,000 manufacturing workers (2024)
South Korea	1,220
Singapore	818
Germany	449
Japan	446
Sweden	377
United States	307
Chinese Taipei (Taiwan)	302
China	166
Mexico	62

Source: IFR press release "Robot Density Surges in Europe, Asia, and Americas", 8 April 2026.

A density of 1,220 means Korea has one robot for roughly every eight manufacturing workers. Korea's

two biggest robot-buying industries, electronics and cars, are exactly the industries where Korea is world class.

Now look closely at China. This figure looks like a typo, but it is real, and it is one of the most instructive statistical revisions of the decade. On 22 October 2025 an IFR blog post summarising World Robotics 2025 listed China's 2024 density as 567, third in the world behind Korea and Singapore and ahead of Germany (449) and Japan (446), with a global average of 177. (A year earlier, the IFR had put China's 2023 density at 470.) On 8 April 2026 the IFR's official density press release gave China's 2024 figure as 166, sixth in Asia and 22nd in the world, "based on updated labor market data issued by China's National Bureau of Statistics", and still a 17% rise on the year before on the new basis. The global average fell at the same time from 177 to 132. The robots did not change: Korea, Singapore, Germany and Japan have exactly the same numbers in both documents. Only the denominator, the number of Chinese manufacturing workers, was revised up sharply. Some analysts explain the gap by the old basis counting only workers in urban non-private firms (roughly 37 million people) while China's economic census records over 100 million manufacturing workers; treat that explanation as secondary analysis, not an IFR statement. A quick check: if the robot count stayed the same and density fell from 567 to 166, the implied workforce grew by a factor of 567 / 166, about 3.4.

This is a valuable lesson for a future founder or investor. Every ratio has a numerator and a denominator, and either can be wrong. China's huge manufacturing workforce means that even with about two million robots in operation (around 4.5 times Japan's stock, per the IFR), its robots per worker are below those of Korea, Germany, Japan or the United States. Both statements are true at once: China has by far the most robots, and China is not (yet) the most automated manufacturing economy. Which one matters depends on your question. If you sell robots, the stock and the flow of installations matter. If you study productivity per worker, density matters.

Why firms buy robots: the basic economics

A firm buys a robot when the robot's cost per hour of useful work is lower than the cost of the labour it replaces, after adjusting for quality and flexibility. Five forces drive that comparison:

1. **Wages.** The higher and faster rising the wage, the sooner a robot pays off. This is why Korea, Japan and Germany automated early.
2. **Demography.** Aging and shrinking workforces raise wages and make hiring harder. The IFR repeatedly links robot demand to "demographic change".
3. **Precision and quality.** Robots weld and place components with consistency humans cannot match at high speed. In electronics, some tasks are impossible by hand.
4. **Capital costs.** Robots are capital goods. When interest rates are high, the yearly cost of owning one rises. When governments subsidise them (through tax credits or cheap loans), adoption speeds up.
5. **Policy and relocation.** The IFR's 2026 outlook says reshoring, tariffs and "the need for more resilient supply chains" are moving factories to high-wage countries, and high-wage factories need more automation.

Here is a worked payback example. All inputs are illustrative assumptions, chosen to be roughly realistic, not sourced data:

- A welding robot cell costs US$150,000 installed and lasts 8 years. Maintenance is US$5,000 a year.
- It replaces two workers per shift on two shifts: four workers.

- Case A, a high-wage country: each worker costs US$50,000 a year including benefits. Labour saved: US$200,000 a year. Payback: about 0.8 years.
- Case B, a middle-income country: each worker costs US$6,000 a year. Labour saved: US$24,000 a year, minus US$5,000 maintenance, gives US$19,000. Payback: about 7.9 years, nearly the life of the robot.

The same robot is an obvious buy in Germany and a marginal one in a lower-wage country, unless quality, speed or a buyer's requirements force automation. This explains a puzzle: the countries with the most to gain from cheap labour (and the most workers needing jobs) are the slowest to adopt robots, while rich aging countries adopt fastest. It also tells you why the humanoid industry is obsessed with cost. A humanoid that costs US$100,000 is a toy for rich labs. One that costs US$15,000 and works reliably for five years would change the payback arithmetic even in middle-income economies.

Case study: How China became the world's robot market, then a robot maker

In 2013 China became the world's largest market for industrial robots, but most of the robots it bought were foreign: Japanese (Fanuc, Yaskawa, Kawasaki), European (ABB, KUKA) and some Korean. Chinese factories were the customers; foreign firms captured the margins.

Three things changed that.

First, policy. China's industrial strategy from about 2015 (known as "Made in China 2025") named robotics a priority sector, and later plans set targets for domestic content. Local governments subsidised robot purchases and robot makers.

Second, the purchase of a crown jewel. In 2016 the Chinese home-appliance maker Midea bought the German robot maker KUKA, one of the "big four" industrial robot firms. (McKinsey lists Midea's acquisition of KUKA among the deals showing that "supply chain access will significantly shape the physical-AI landscape".)

Third, the domestic supplier base grew around the world's largest customer. Chinese robot brands such as Estun, Inovance and Efort, and component makers for reducers, servo motors and controllers, learned by supplying a huge home market. The IFR's numbers show the result. The share of domestic suppliers in Chinese installations was around 30% from 2019 to 2021, then "began to accelerate steeply in 2022". Since 2024 more than half of robots installed in China have been Chinese brands: 57% in 2024 and 55% in 2025, when Chinese makers installed about 195,000 units at home (IFR, September 2026).

The next step is exports. Xinhua reported, citing customs data, that China became a net exporter of industrial robots for the first time in 2025, and that its industrial robot exports reached 6.29 billion yuan in the first half of 2026, up 18.6% year on year, shipped to 141 countries and regions. The IFR's supplier committee chair (an executive at Yaskawa, a Japanese company) wrote in September 2026 that domestic Chinese brands "now leverage this by exporting their products overseas putting pressure on established suppliers from Japan, Korea and Europe".

What should you learn from this? The pattern is the same one you saw for solar panels and electric vehicle batteries in Months 6 and 7. China first became the biggest customer, then used that market (plus policy, finance and fierce domestic competition) to grow domestic suppliers, then those suppliers became exporters with lower prices. The customer became the competitor. For any country hoping to build a robot industry, the key question is: where is your home market? A supplier needs customers to learn on.

Key numbers

Figure	Value	Year	Source
Industrial robots installed worldwide	603,307	2025	IFR World Robotics 2026
Operational stock of industrial robots	5,079,078	end-2025	IFR World Robotics 2026
China's installations (share of world)	354,226 (59%)	2025	IFR World Robotics 2026
Chinese brands' share of China's installations	55%	2025	IFR press release, Sept 2026
Industrial robots installed in all of Africa	519	2025	IFR, Klos blog, Sept 2026
Robot density, South Korea	1,220 per 10,000 workers	2024	IFR, April 2026
Robot density, China (revised)	166 per 10,000 workers	2024	IFR, April 2026
Market value of robots installed	US$18.6 billion	2025	IFR World Robotics 2026
Humanoids produced worldwide (estimate)	about 16,000	2025	Epoch AI, April 2026
IFR installation forecast	806,000	2029	IFR (projection)

Think like a founder

The robot arm itself is a crowded, mature product with established giants and aggressive Chinese challengers. Few founders should try to build a new robot arm company. But look at where adoption is blocked. The IFR's supplier committee says many small and medium enterprises "still find robot deployment too complex", and that "function packages, engineering tools, application wizards and pre-trained skill sets are key factors". In other words, the bottleneck in many markets is integration and software, not hardware.

That suggests opportunities: a system integrator that specialises in one industry (for example food processing, furniture or garment finishing) in a region where robot density is low; software that lets a factory worker teach a robot a task by demonstration; robot-as-a-service leasing, where a small factory pays per hour of robot work instead of buying a US$150,000 cell. The social-impact angle is real too: robots taking the most dangerous jobs (grinding, painting with toxic fumes, heavy lifting) while workers move into supervision and maintenance. The value capture lesson: in a market where hardware is becoming a commodity, own the customer relationship and the know-how of deployment.

Week 2: Anatomy of a robot, from joint to mineral

Core lesson

The robot as a stack

Engineers sometimes describe a humanoid as "mind, body and soul" (Bank of America Institute, March 2026). The "mind" is the AI that perceives the world and plans tasks. The "body" is the hardware: actuators, hands, sensors, battery and structure. The "soul" is the motion-control system that keeps the robot balanced and turns plans into precise movements thousands of times per second. An economist would rather call it a stack of layers, each with its own suppliers, margins and chokepoints:

1. **Actuators** (the muscles): motor, reducer or screw, driver electronics, encoder and bearings, packaged as a joint module.
2. **Hands** (dexterous manipulators): dozens of tiny actuators plus touch sensors.
3. **Sensors** (the senses): cameras, depth sensors, inertial measurement units (IMUs, which sense tilt and acceleration), force and torque sensors, tactile skins.
4. **Compute and control**: an AI computer for perception and planning, plus real-time controllers in each joint.
5. **Energy**: battery pack, power electronics, thermal management.
6. **Structure**: frame, housings, covers, wiring.
7. **Software and AI models**: from low-level motor control up to "vision-language-action" models that turn a spoken instruction into movement.

McKinsey (April 2026) estimates the hardware cost split of a typical humanoid as follows: actuators 40 to 60% of the bill of materials, sensing and perception 10 to 20%, compute and control 10 to 15%, structure 5 to 10%, and batteries 5 to 10%. The single most important fact in this table is the first line. **A humanoid is mostly a collection of expensive joints.** The "brain" chip is a small part of the hardware cost.

Inside a joint

Every joint of a robot has the same problem to solve. Electric motors are good at spinning fast with little force (torque). A knee or a shoulder needs the opposite: slow, strong, precise movement. So between the motor and the limb sits a transmission that trades speed for torque. McKinsey breaks a typical integrated actuator into five elements:

- **Gearbox: 30 to 50% of actuator cost.** This is the reducer, the heart of the joint.
- **Driver: 15 to 20%.** Electronics that control the current to the motor, increasingly safety-rated.
- **Motor: 10 to 20%.** Usually a frameless brushless DC motor with permanent magnets.
- **Mechanical parts: 10 to 20%.** Housings, shafts, precision bearings.
- **Sensors: 5 to 10%.** Encoders that measure the joint's exact angle, torque sensors, temperature sensors.

A worked example of what "reduction" means. Suppose a motor spins at 3,000 revolutions per minute and produces 1 newton-metre of torque. A reducer with a ratio of 100 to 1 makes the output turn at 30 revolutions per minute, but with up to about 100 newton-metres of torque (minus losses to friction). That is the difference between a motor that could barely turn a door handle and a joint that can lift a box. The precision of the reducer decides how accurately the arm ends up where it

should. If the gears have even a little "backlash" (play between the teeth), a robot arm one metre long can miss its target by millimetres, which is fatal in electronics assembly.

The three reducers and the screw

There are three main kinds of high-precision reducer, plus a linear alternative:

- **Strain wave gears** (often called harmonic drives, after the Japanese company Harmonic Drive Systems whose brand became the generic name). A flexible steel ring with teeth is pushed into an oval shape by a rotating cam and meshes with a rigid outer ring. They are compact, light and have almost no backlash, so they are used in wrists, fingers and smaller joints. Their weakness: the flexible ring can fail when the robot is hit or falls.
- **Cycloidal or RV reducers.** These use lobed discs rolling inside a ring of pins. They are stiff and strong, used in the base, shoulders and elbows of heavy industrial arms, and in high-torque humanoid joints.
- **Planetary gearboxes.** Small "planet" gears circle a central "sun" gear. They are cheaper and common in lower-cost humanoids and in load-bearing joints such as thighs.
- **Planetary roller screws.** These turn rotation into straight-line motion, like a precise screw jack. Some humanoids use them as linear actuators in the legs; analyst teardowns of Tesla's Optimus describe roller screws in its leg joints (secondary reports, treat as indicative). They are a narrow high-precision niche with few suppliers.

Epoch AI's April 2026 study compiled global production for each. Roughly: planetary gearboxes about 3 million a year, cycloidal/RV reducers about 2 million a year, strain wave gears about 12 million a year. Its reference humanoid uses 6 planetary and 4 cycloidal reducers. A worked example of a supply ceiling: 3,000,000 planetary reducers divided by 6 per robot is 500,000 humanoids a year, if every planetary reducer on Earth went into humanoids (which it would not; they are needed for industrial robots and machine tools too). Epoch concludes that reducers are "the closest to a binding constraint", capping humanoid production at about 500,000 a year with today's supply, roughly 30 times current output. By contrast, cameras (7 billion a year), sensors, bearings and battery cells are produced at such scale that robots would barely dent supply.

Epoch also notes the direction of travel. Precision reducers "are not semiconductors": tolerances are in micrometres, not nanometres, and they do not need clean rooms. Chinese manufacturers are expanding fast at lower margins, and Epoch treats a doubling of reducer output by 2028 and quadrupling by 2030 as reasonable floors. So the chokepoint is real but breakable, which is exactly what makes it a contested source of value.

Hands: the hardest part

A human hand has more than 20 degrees of freedom (independent directions of movement) and very sensitive touch. McKinsey reports that Tesla's Optimus has 28 joint actuators in its body plus 50 more in its Gen 3 hands. Hands are where three immature supply chains overlap: tiny actuators, force sensing and tactile sensing, for which "no dominant design has yet emerged". In Morgan Stanley's teardown-style estimate of an Optimus Gen 2-class robot built with Chinese suppliers in 2025, dexterous hands cost about US$15,000 out of a total of about US$45,500 (reported by J.P. Morgan Private Bank, August 2026). One third of the robot's material cost is in its hands.

Sensors and compute

Cameras and depth sensors benefit from the smartphone and car industries, so they are cheap and plentiful. The hard sensors are **six-axis force/torque sensors**, which measure the push and twist at a wrist or ankle. McKinsey notes they are "calibration intensive", made by a limited number of firms such as ATI Industrial Automation and OnRobot, and benefit little from car or consumer electronics spillover.

For compute, most humanoid makers use off-the-shelf AI modules, most prominently NVIDIA's Jetson series, plus automotive-grade microcontrollers. Tesla is the exception, adapting its car self-driving computer for Optimus. Chipmakers are now designing robot-specific processors: Qualcomm launched a humanoid processor in January 2026 (McKinsey). In Morgan Stanley's China-built 2025 estimate, "chip plus software" is about US$3,000 of US$45,500, under 7% of material cost. The brain is cheap to build per unit. As you learned in Month 8, the expensive part of AI is training the model in data centres, which is paid once and spread over every robot that uses it. That is a very different cost structure from the joints, which must be paid for again in every unit.

Batteries and energy

A humanoid's battery is tiny compared with a car's, roughly a few kilowatt-hours instead of 50 to 100, so it is a small cost share. The challenge is not cost but performance: bursts of high power for dynamic movement, heat inside a small body, and safety when the robot falls. Battery makers are building robot-specific products. McKinsey reports that LG Energy Solution supplies the cells for Boston Dynamics' Atlas, and that Samsung SDI showed solid-state batteries designed for humanoids in March 2026, targeting mass production in the second half of 2027. Short battery life (often a few hours) is one reason humanoids are not yet productive across full factory shifts.

From joint to mineral

Now connect the robot to the earlier months of this course. Each component sits at the end of a mineral chain:

Robot part	Key materials	Who dominates the processing stage
Permanent magnets in motors	Neodymium, praseodymium; dysprosium and terbium for heat resistance; iron, boron	China: about 91% of magnet rare earth refining and 94% of sintered magnet production (IEA, 2024 data)
Motor windings, cables	Copper	Refining spread more widely, but China is the largest refiner (see Month 5)
Reducers, screws, bearings	High-grade alloy and bearing steel, precision machining	Japan, China, Germany; China holds about 40% of precision bearings (McKinsey)
Batteries	Lithium; nickel, cobalt, manganese or iron phosphate; graphite	China: 80% or more of many midstream battery stages, about 95% for some cathode materials (IEA, Oct 2025)

Robot part	Key materials	Who dominates the processing stage
Structure	Aluminium, magnesium, plastics, carbon fibre	Widely available; China is largest producer of aluminium and magnesium
Chips	Silicon, plus the whole Month 8 supply chain	Taiwan, South Korea, US, Netherlands, Japan at different stages

The pattern from Months 5 and 6 repeats. Mining is spread over several countries. Refining and the next step (magnets, cathodes) are concentrated in China. A robot maker anywhere in the world depends on Chinese processing for at least its magnets and probably its battery materials.

Case study: The magnet shock of 2025 and the reducer race

Two stories show how chokepoints work in real life, one in materials and one in precision parts.

The magnet shock. On 4 April 2025, in response to US tariffs, China's Ministry of Commerce introduced export licensing for seven medium and heavy rare earth elements (samarium, gadolinium, terbium, dysprosium, lutetium, scandium and yttrium) and their compounds, metals and magnets. Licences were slow. The IEA reports that "as export volumes fell sharply in April and May, many carmakers in the United States, Europe, and elsewhere struggled to obtain permanent magnets, with some forced to cut utilisation rates or even temporarily shut down factories". Even after volumes recovered, rare earth prices in importing countries stayed high, "with European prices reaching up to six times those in China". On 9 October 2025 China announced a much wider set of controls, including on foreign-made products containing Chinese rare earth materials or made with Chinese technology, and on refining equipment. After a US-China meeting at the end of October 2025, China suspended the October measures for a year, but the April licensing system stayed in force (Reuters, 30 October 2025). The legal instrument, MOFCOM and Customs Announcement No. 70 of 2025, suspends the October package "until November 10, 2026". On 23 September 2026 the US Treasury Secretary said the wider US-China truce would be extended to 10 January 2027, and on 28 September 2026 China's Ministry of Commerce confirmed that the Kuala Lumpur arrangement, under which both sides had suspended some measures until 10 November 2026, would run to 10 January 2027. But that statement did not mention rare earths or export controls, and as of early November 2026 no announcement amending No. 70 had been published (China Prospect, checked against MOFCOM texts on 1 November 2026). So the extension to January 2027 is political, not yet legal: watch MOFCOM's announcement list around 10 November 2026. Two further details matter. The April 2025 licensing of the seven heavy rare earths, which is what bites for dysprosium- and terbium-containing magnets, was never suspended. And a separate suspension of China's ban on gallium, germanium and antimony exports to the United States runs to 27 November 2026 (Announcement No. 72 of 2025).

Why does this matter for robots? A humanoid has 25 to 80 or more motors, nearly all using neodymium magnets. McKinsey notes that Elon Musk "has publicly noted that magnet supply constraints have affected Optimus production". In 2024 China exported 58,000 tonnes of rare earth magnets (IEA). Building alternatives is slow: the IEA says new rare earth mines take on average around eight years from discovery to production, and planned magnet capacity outside China is even thinner than mining or refining. New magnet plants started in 2025 in the United States (MP

Materials) and Estonia (Neo Performance Materials), with projects in Korea, Vietnam and Germany. The lesson: the country that controls a processing chokepoint can use it as a weapon of trade policy, and the target cannot quickly escape.

The reducer race. For three decades two Japanese firms dominated precision reducers. Nabtesco says it has "approximately 60%" of the global market for precision reduction gears in the joints of medium-size to large industrial robots (its own estimate, on its website). Harmonic Drive Systems (HDS), which invented the commercial strain wave gear, has long led that segment, but estimates of its share vary with what is measured: one analysis puts it at about 85% of global harmonic reducers in 2023, another at about 58% of global revenue (and 40 to 50% of units) in 2024, and a Chinese financial report cites "roughly 40%" for 2024. Treat these as a range, not a fact: the differences come from counting units versus revenue and from whether Chinese domestic output is fully included. HDS's whole company had net sales of about 59.6 billion yen (roughly US$400 million) in the year to March 2026, with reduction gears about 78% of that (company results as reported by trade press). These firms built their positions through decades of metallurgy, heat treatment and precision grinding: knowledge that sits in machines and people, not in blueprints.

Chinese challengers are now closing in, and the numbers show exactly how a precision chokepoint erodes. Leaderdrive (Suzhou, listed on Shanghai's STAR Market in 2020) grew revenue 47.3% in 2025 to about 571 million yuan, with net profit attributable to shareholders up 121% to about 124 million yuan. QY Research puts its 2025 share of the global harmonic reducer market at about 12% (second in the world) and its share of Chinese shipments at 27.5% (first). Its humanoid customers reportedly include Unitree, UBTech and AgiBot. Zhejiang Laifual, which listed in Hong Kong in June 2026, says in its listing document that it ranked second in China by 2025 shipments with 21.4% of units, but only 12.9% of revenue, and that its reducers reached a rated service life above 10,000 hours. Industry reports often cite 15,000 to 20,000 hours for Japanese benchmarks (not independently verified).

Deepening: how a chokepoint gets competed away

The interesting part is not that Chinese firms caught up, but what happened to the money when they did. Follow three numbers.

1. **Price.** According to Bernstein, as reported in Chinese financial press, Leaderdrive's average harmonic reducer price fell from about 1,900 yuan in 2017 to about 1,100 yuan in 2024, a fall of about 42% in seven years. Laifual's average selling price fell from about 802 yuan in 2023 to about 573 yuan in 2025, a fall of about 29% in two years.
2. **Margin.** Leaderdrive's consolidated gross margin fell from 52.2% in 2021 to 36.9% in 2025 and 32.3% in the first half of 2026, even as revenue grew fast. In that half-year its gross margin was about 30% on Chinese sales but about 45% on overseas sales.
3. **Losses at the number two.** Laifual more than doubled revenue in 2025, to about 261 million yuan, yet reported a loss of about 171 million yuan for the year.

(These Leaderdrive and Laifual figures come from company filings as summarised by financial news services; the original filings were not read for this book, so treat them as well-reported but secondary.)

What is the mechanism? A chokepoint earns high margins only while customers have few qualified alternatives. Once a second and third supplier pass qualification, the buyer can play them against each other, and the "scarcity rent" (the extra profit that exists only because supply is scarce) is competed away. In China this happened unusually fast for three reasons that link straight back to Week 1. First, a huge home market of robot makers was willing to try domestic parts, which

gave Chinese reducer makers millions of hours of field feedback, the learning-by-doing that Japan's firms once had to themselves. Second, policy and supply-chain security pushed robot makers to "dual source" with at least one domestic supplier. Third, capital was cheap: STAR Market and Hong Kong listings let challengers fund capacity before they were profitable, which is exactly what Laifual's loss shows.

So where did the value go? Mostly downstream, to robot makers and their customers. A small worked example makes this concrete. Suppose a mid-size humanoid uses 14 strain wave gears (a hypothetical number for illustration; real designs vary). At 1,900 yuan each, the reducers cost 26,600 yuan; at 1,100 yuan they cost 15,400 yuan; at 573 yuan they cost about 8,000 yuan. Against a robot that sells for about 85,000 yuan (roughly Unitree's G1 list price in China after its 2025 cut, per its filing), the reducer bill fell from about 31% of the selling price to about 9%. That 18,600-yuan difference did not disappear: it became lower robot prices (more buyers), higher robot-maker margins, or both. This is why Unitree, which makes its own reducers, and its customers benefited from a price war that hurt pure-play reducer suppliers.

There is a second lesson in the overseas margin gap. Leaderdrive still earns about 15 percentage points more on exports than at home. Western and Japanese robot makers pay more because qualification is slow and switching is risky: a reducer failure on a car line can stop a factory. The chokepoint has not vanished; it has split into a commodity tier (cheap, good-enough parts, mainly for Chinese humanoids and light arms) and a premium tier (long-life, certified parts for high-duty industrial use and medical or semiconductor equipment). Value now sits less in "being able to make a reducer" and more in **proven reliability data and qualification with demanding customers**.

Finally, the dependency moves down a layer. One industry analysis claims China still imports roughly 90% of the precision machine tools used to make reducers, mostly from Japan (secondary source; treat as indicative). If true, the next chokepoint in the reducer chain is the grinding and gear-cutting machine, not the gear. This is a pattern you will see again and again in this book: when a component is commoditised, ask who makes the machines that make it.

Judgment: Precision reducers are a "breakable" chokepoint. For a founder or a country, entering the commodity tier now means a price war against well-funded Chinese firms; the attractive entry points are the premium tier (with a long, patient qualification path), the machine tools and metrology behind reducers, or integrating reducers into complete joint modules where software and testing add value.

Put the two stories together. Japan's chokepoint is skill and quality in a precision mechanical part. China's chokepoint is the refining of materials and the scale of manufacturing. Japan's chokepoint is eroding because China has the customers to learn on. China's chokepoint is protected by geology, environmental cost (refining rare earths produces radioactive waste, as the IEA notes) and an eight-year lead time for alternatives. That asymmetry is a key to the geopolitics of robotics.

Key numbers

Figure	Value	Year	Source
Actuators' share of humanoid bill of materials	40-60%	2026 estimate	McKinsey, April 2026
Gearbox share of one actuator's cost	30-50%	2026 estimate	McKinsey, April 2026

Figure	Value	Year	Source
Reducers per humanoid	20-40	2026	Epoch AI
Humanoid ceiling from current planetary/RV reducer supply	about 500,000 a year	2026	Epoch AI
Actuators in Tesla Optimus	28 in body, about 50 in Gen 3 hands	2026	McKinsey
China's share of magnet rare earth refining	about 91%	2024	IEA, Oct 2025
China's share of sintered permanent magnet production	94%	2024	IEA, Oct 2025
Rare earth magnets exported by China	58,000 tonnes	2024	IEA, Oct 2025
Nabtesco share, reducers for medium/large industrial robot joints	about 60%	company estimate	Nabtesco website
Average lead time for new rare earth mines	about 8 years	2025	IEA

Think like a founder

Value in a robot concentrates where three things overlap: high cost share, few qualified suppliers, and slow capacity growth. Today that means reducers and roller screws, force and tactile sensors, dexterous hands and rare earth magnets. McKinsey's advice to suppliers is blunt: design wins at the prototype stage "convert into production incumbency once architectures stabilize". The German supplier it describes went from no humanoid presence to a key actuator supplier in under two years by signing co-development deals early.

For a founder outside China, two strategies are realistic. One is a **component niche** where precision matters more than scale and where you can be qualified by several robot makers: a better tactile sensor, a more robust strain wave gear, a calibration service for force sensors, or a magnet-free motor design that escapes the rare earth chokepoint (carmakers such as BMW and Renault already make rare-earth-free motors, per Reuters). The other is **recycling and recovery**: magnets and batteries from retired robots, motors and electronics are a future source of rare earths and cobalt outside Chinese control. Both require deep engineering, not just software, which is a reason to study materials science and control engineering alongside AI.

Week 3: Bills of materials, cost curves and the economics of a humanoid

Core lesson

Reading a bill of materials

A **bill of materials (BOM)** is the list of every part in a product with its cost. It is not the price the customer pays. Between the BOM and the price sit assembly labour, factory overhead, research and development, warranty, sales costs and profit. Analysts like BOMs because they show where money flows to suppliers, and therefore which supplier industries will grow if the product succeeds.

The most widely quoted humanoid BOM comes from Morgan Stanley's teardown-style estimate of a Tesla Optimus Gen 2-class robot (data as of December 2025, published in J.P. Morgan Private Bank's August 2026 summary). It compares three cases, in thousands of US dollars:

Part	China supply chain, 2025	China supply chain, 2034 (about 1 million sold a year)	Non-China supply chain, 2025
Actuators	22.0	7.0	58.0
Dexterous hands	15.0	6.0	47.0
Feet	4.0	1.0	16.0
Chip plus software	3.0	1.0	7.0
Vision	0.4	0.2	0.5
Battery	0.1	0.1	0.3
Others	1.0	1.0	3.0
Total	**45.5**	**16.3**	**131.8**

Source: Morgan Stanley Research, as reported by J.P. Morgan Private Bank, 14 August 2026.

Four lessons jump out.

1. **Actuators plus hands are 81% of the China-built BOM in 2025** (37.0 of 45.5). If you want to know where hardware money goes, it goes into joints.
2. **The non-China robot costs 2.9 times more.** This is not mainly about cheap labour, since the table counts parts, not assembly. It reflects supplier depth, competition, scale and clustering. McKinsey stresses that China's advantages "are not primarily labor-driven" but come from "scale economies, component clustering, supplier competition, and reuse of mature EV-grade ecosystems".
3. **The brain is cheap per unit.** Chip plus software is 3.0 of 45.5, about 7%.
4. **Costs are expected to fall by about two thirds by 2034** in the China case, from 45.5 to 16.3, as volumes reach about one million a year.

Other estimates differ in detail but agree in shape. Bank of America Global Research estimates that a China-built humanoid's BOM was about US$35,000 in 2025, falling below US$17,000 by 2030, while pilot-stage robots elsewhere cost US$90,000 to US$100,000 (Bank of America Institute, 12 March 2026). It projects that in 2030 linear actuators will be 27% of the BOM, rotary actuators 24%, dexterous hands 19%, the control system 10%, vision and sensors 4% and energy 4%. McKinsey gives a range of US$30,000 to US$150,000 for a typical humanoid BOM today and calls "under US$20,000" the widely cited long-term target.

Prices on the market

Prices tell a similar story. McKinsey notes that Unitree publicly lists its G1 humanoid starting at about US$13,500, and that Tesla expects Optimus costs to fall to roughly US$20,000 to US$30,000 at scale. J.P. Morgan Private Bank describes low-cost humanoids at US$5,000 to US$10,000 that "can look impressive in controlled demonstrations" but whose real dexterity, battery life and autonomy are "meaningfully constrained". So the market already has humanoids that cost less than a used car. What it does not yet have is a cheap humanoid that does reliable, useful work for a full shift. Price and capability are separate problems.

Learning curves: why costs fall

In 1936 an American aircraft engineer, Theodore Wright, observed that each time cumulative production of an aircraft doubled, the labour needed per plane fell by a roughly constant percentage. This regularity, now called **Wright's law** or the learning curve, has held for many manufactured goods: each doubling of cumulative output cuts unit cost by a fixed share, the "learning rate". Solar panels and lithium-ion batteries are famous examples, with learning rates often estimated at around 20% (meaning costs fall about a fifth with each doubling).

The formula is simple. If C1 is the cost when cumulative output is Q1, then at output Q2:

C2 = C1 x (Q2 / Q1) ^ b, where b = log2(1 - learning rate).

Worked example. Suppose humanoid cumulative production was about 20,000 units by the end of 2025 (an assumption based on Epoch's estimate of about 16,000 produced in 2025 alone plus a few thousand earlier). Suppose a China-built BOM of US$45,500 then. What learning rate would bring it to Morgan Stanley's US$16,300 when cumulative production reaches, say, 3 million (a plausible total if annual sales climb to about one million by 2034)?

- Number of doublings: 3,000,000 / 20,000 = 150, and log2(150) is about 7.2 doublings.
- Required fall: 16,300 / 45,500 = 0.358.
- So (1 - learning rate) ^ 7.2 = 0.358, which gives 1 - learning rate of about 0.867, a learning rate of about 13%.

A 13% learning rate is modest compared with solar or batteries, so the forecast is not crazy on its own terms. The real uncertainty is the other variable: will cumulative production actually reach millions? Learning curves describe what happens if volume arrives. They do not create the demand. This is the central trap in humanoid forecasting: cost falls only if people buy, and people buy only if cost falls and the robots are useful. Economists call this a chicken-and-egg or coordination problem. McKinsey describes it as a "scaling dilemma": "low volumes prevent suppliers from investing in dedicated production lines, yet without cost reductions, end user demand remains constrained."

Where does the cost reduction come from?

Learning curves are a summary; the mechanisms underneath are concrete:

- **Design simplification.** Fewer actuators, fewer custom parts, cheaper materials where precision is not needed. Chinese makers often use planetary gears where Western prototypes use costlier harmonic drives.
- **Shared supply chains.** McKinsey points out that motors, power electronics, magnets, bearings and batteries overlap with the electric vehicle value chain. Morgan Stanley analysts

wrote in 2025 that humanoids could be "a third wave of growth for auto parts suppliers" (CNBC, June 2025).

- **Automation of production.** Today many humanoid parts are machined one at a time on computer-controlled (CNC) machines, a prototype method. Die-casting and stamping become possible once designs freeze.
- **Supplier competition.** Each new reducer or motor maker that qualifies pushes prices down. Chinese reducers sell at roughly one half to two thirds of import prices, according to industry reports.
- **Standardisation.** When interfaces between parts are standard, suppliers can sell the same module to many robot makers and spread their fixed costs.

The economics of a robot hour

What matters to a buyer is not the BOM but the cost of an hour of useful work. Here is a worked example with explicit assumptions:

- Purchase price US$40,000 (a BOM of about US$25,000 plus maker's costs and margin).
- Life of 5 years, working 2 shifts of 8 hours on 300 days a year: 4,800 hours a year, 24,000 hours in total.
- Capital cost per hour: 40,000 / 24,000 = about US$1.67.
- Maintenance, repairs, software subscription and electricity: assume US$6,000 a year, or US$1.25 per hour.
- Total: about US$2.90 per robot hour.

Now the critical adjustment: productivity. If the robot does only 40% as much useful work per hour as a trained human (because it is slower, drops things, needs human help), the cost per human-equivalent hour is 2.90 / 0.40 = about US$7.30.

Compare this with labour costs. US$7.30 an hour is below the total hourly cost of a factory worker in the United States, Germany or Japan, so even a slow robot could pay off there if it is reliable. It is above the hourly labour cost in most of Southeast Asia, South Asia and Africa. The result shows two thresholds that matter more than any headline price. First, reliability and productivity: raising the robot's useful output from 40% to 80% of a human's halves its effective cost. Second, utilisation: a robot that runs three shifts instead of two cuts capital cost per hour by a third. This is why the first big humanoid markets will be high-wage, multi-shift operations (car plants, warehouses, logistics), which matches Counterpoint Research's estimate (cited by Bank of America) that 72% of humanoid installations in 2027 will be in logistics, manufacturing and automotive.

How big will the market be? Forecasts disagree enormously

J.P. Morgan Private Bank averaged forecasts from Morgan Stanley, Bank of America, Barclays, McKinsey and Bain (as of July 2026). For the whole robotics market (industrial robots, drones, humanoids and autonomous vehicles), the average base case is US$2.5 trillion of annual sales by 2035, from about US$100 billion in 2025, with a bear case of US$0.5 trillion and a bull case of US$8 trillion. For humanoids alone, the average is roughly US$300 billion by 2035, against about US$2 billion in 2025. Bank of America forecasts humanoid shipments rising from about 20,000 in 2025 to 1.2 million in 2030 and 10 million in 2035, and as many as 3 billion humanoids in operation by 2060.

A sixteen-fold gap between the bear and bull cases tells you the honest answer: nobody knows. The IFR, which represents the established robot industry, is cautious. Its World Robotics 2026 summary

says: “Most humanoids remain prototypes or pilot-stage technologies rather than industrial-grade products. At the current stage of technology, industrial robots are economically superior for industrial production tasks.” Its supplier committee chair adds that investors “should carefully monitor how quickly the bet on millions of humanoid robots in industrial applications pay off. Market pull must follow the current technology push.”

Judgment: Production volumes will grow fast through the late 2020s, because Chinese makers are scaling and costs are falling; J.P. Morgan reports Chinese government data showing the three leading Chinese humanoid makers on track for more than 100,000 units in 2026, about five times the previous year. But the IFR is right that today’s demand is mostly research, education, demonstration and marketing, not productive labour. The decisive variable is not price, which will fall, but reliability and task competence, which depend on AI progress (Month 8) and on real-world data. A reasonable expectation is that by the early 2030s humanoids are a real but niche industrial tool, numbering in the low millions worldwide, with household use further away. Treat any forecast of billions of humanoids as a scenario, not a prediction.

Case study: Unitree, the profitable humanoid maker

Unitree Robotics, founded in 2016 in Hangzhou by the engineer Wang Xingxing, began with four-legged robots and became the world’s best-selling humanoid maker. Its listing on the Shanghai Stock Exchange’s STAR Market in August 2026 opened its books to the public, which makes it one of the few humanoid companies whose economics you can actually check.

Growth. According to its prospectus (reported by Xinhua, 10 August 2026), Unitree’s revenue rose from 159.13 million yuan in 2023 to 392.77 million yuan in 2024 and 1.70 billion yuan in 2025, more than a tenfold rise in two years. Net profit was 278.21 million yuan in 2025; profit attributable to shareholders after removing non-recurring items was 590.75 million yuan. Overseas sales were 731.66 million yuan, 43.65% of main-business revenue.

Volumes. The prospectus says Unitree’s humanoid shipments exceeded 5,500 units in 2025 (pure humanoids, excluding wheeled dual-arm robots). Reuters reported that the prospectus claimed this was 32.4% of the global humanoid market. From 2023 to 2025 it sold 33,294 quadruped robots. (Note that different documents count “sold” and “shipped” differently; Xinhua reports 5,632 humanoids “sold” over 2023-2025, so be careful when combining figures.)

Margins. A company statement carried by the Shanghai Stock Exchange’s news service put the gross margin of its core businesses at 60.13% in 2025. Unitree’s reply to the exchange’s pre-listing questions (published 20 March 2026, as analysed by Inside Deep Tech) is more precise: the humanoid gross margin was 62.4% in the first half of 2025 and 63.9% in the third quarter, while quadrupeds earned 56.5% and then 53.4%. That is an extraordinary margin for a hardware firm. Part of the reason is vertical integration: Unitree designs and makes its own motors, reducers, encoders, drivers and structural parts. The filing says bought-in components (battery cells, compute modules, storage, some lidars, cameras and some dexterous hands) were only 14% to 18% of total cost, excluding optional high-specification lidar and hands. Part is pricing power from being first at scale.

Falling prices. The filing gives the clearest public learning-curve data in the humanoid industry. Unitree sold 5 humanoids in 2023 at an average of about 593,400 yuan, 410 in 2024 at about 260,700 yuan, and 3,551 in the first nine months of 2025 at about 167,600 yuan. List prices fell in the same way: the full-size H1 launched in August 2023 at 499,800 yuan, the newer H2 starts at 199,000 yuan, the mid-size G1 came down from 99,000 to 85,000 yuan, and the small R1, launched in July 2025 at a starting price of 39,900 yuan (Reuters, 25 July 2025), had its starting price cut to 29,900 yuan on 24

June 2026, when Unitree also began selling it from stock (Xinhua, 25 June 2026). The 29,900-yuan entry point is the lighter R1 Air version; Unitree's own price list shows R1 Air at 29,900 yuan and the standard R1 at 39,900 yuan. Reuters noted that the shift to the lower-priced G1 "trimmed gross margin" for the company as a whole, because the cheaper product mix lowered average margins, even though humanoid margins stayed above 60%.

Deepening: reading Unitree's learning curve

Use Week 3's learning-curve tool on these numbers. Cumulative humanoid sales went from 5 (end of 2023) to 415 (end of 2024) to about 3,966 (September 2025): roughly three doublings from 415 to 3,966 (415 x 2 x 2 x 2 = 3,320, and a little more). Over that stretch the average price fell from 260,700 to 167,600 yuan, a 36% fall. If all of the price fall were learning (it is not; part is a shift to smaller models), that would be a fall of about 14% per doubling (0.86 x 0.86 x 0.86 is about 0.64), similar to the 13% learning rate implied by the Morgan Stanley forecast earlier this week, and a little below the roughly 20% often estimated for solar panels and batteries. The more striking fact is that **gross margin stayed above 60% while prices fell**. That means unit costs fell at least as fast as prices. When a firm makes its own key parts, every gain in its motor, reducer and assembly lines shows up as its own cost reduction, rather than as a supplier's.

Now ask the value-capture question for a single robot. Take a G1 at about 85,000 yuan and a gross margin of about 63%. Then cost of goods is about 31,500 yuan and gross profit about 53,500 yuan. If bought-in parts are about 16% of cost, Unitree pays outside suppliers only about 5,000 yuan per robot; the remaining 26,500 yuan of cost is its own labour, materials and depreciation, much of it on in-house actuators. Compare a hypothetical assembler that buys complete joint modules from suppliers who each earn 30% gross margins: the same parts would cost it far more, and the supplier margin would leave the assembler's books. That is the whole argument of this month in one calculation: in a young industry, the firm that owns the joint owns the margin.

Who buys? This is the sobering part. For the first nine months of 2025, the filing splits humanoid revenue three ways: research and education about 438 million yuan (73.6%), commercial and consumer about 104 million yuan (17.4%), and industrial application about 54 million yuan (9.0%). Within that industrial slice, Unitree says, enterprise reception and tour-guide use account for roughly 50% to 70% (Reuters, March 2026). Work the arithmetic through: robots doing manufacturing or inspection work earned somewhere between about 2.7% and 4.5% of humanoid revenue (30-50% of 9%). The trend is upward, but slowly: research and education was 100% of humanoid revenue in 2023 and 86.3% in 2024. Epoch AI makes the same point for the whole industry: "most current humanoid purchases are not driven by productive use", but by research and marketing. Note also that market-share figures depend on who counts: the filing quotes one report giving Unitree 32.4% of about 17,000 humanoids shipped worldwide in 2025, and another giving 37.6% of a smaller total, and it says itself that research firms lack a "unified and comparable" data source.

Valuation. Unitree offered 40.45 million new shares at 150.80 yuan, 10% of its enlarged share capital of 404.5 million shares, raising about 6.10 billion yuan gross. That implies a market value at the offer price of about 404.5 million x 150.80 = roughly 61 billion yuan (close to US$9 billion), a price-to-earnings ratio of about 219 on 2025 net profit (Xinhua). Offline subscriptions were about 2,760 times the shares on offer. On its first trading day, 19 August 2026, the shares closed at 845 yuan, up 460% (Xinhua), which values the company at roughly 404.5 million x 845 = about 342 billion yuan, or about 200 times 2025 revenue. A Chinese financial report in early September 2026 put its market value at about 237 billion yuan (secondary source), and on 19 October 2026 the shares closed at 495.92 yuan, a market value of about 200 billion yuan (Sci-Tech Innovation Board

Daily via Longbridge), so the price had fallen a long way from the first-day close. At these prices investors were not paying for today's tour-guide robots. They were paying for an option on the humanoid future, and for a Chinese "national champion". Meanwhile growth was slowing: first-half 2026 revenue rose 48.5% to about 1.15 billion yuan, but net profit after non-recurring items fell 19.3% to 244 million yuan (attributable net profit was 274 million yuan), which the company attributed to higher research and sales spending, a larger base and more competition (The Standard, 18 August 2026).

What it teaches. Unitree shows that a humanoid maker can be profitable now, but mainly by selling to researchers, universities, companies wanting a showpiece, and other robot developers. It shows the power of vertical integration in a young industry where no component supplier ecosystem yet exists. And it shows how China's capital markets (the STAR Market, state-linked investors, local governments) can fund scaling before the product is proven, which McKinsey calls "capital structures that can support scaling before full commercial validation".

Key numbers

Figure	Value	Year	Source
Humanoid BOM, China supply chain (Optimus Gen 2-class)	US$45,500	2025	Morgan Stanley via J.P. Morgan PB
Same robot, non-China supply chain	US$131,800	2025	Morgan Stanley via J.P. Morgan PB
Projected China BOM at about 1 million units a year	US$16,300	2034	Morgan Stanley via J.P. Morgan PB
China-built humanoid BOM (BofA)	US$35,000, falling below US$17,000	2025, 2030	Bank of America Institute, Mar 2026
Unitree G1 list price	from about US$13,500	2026	McKinsey, citing Unitree
Unitree revenue	1.70 billion yuan (from 392.77 million)	2025 (2024)	Unitree prospectus via Xinhua
Unitree humanoid shipments	more than 5,500	2025	Unitree prospectus via Xinhua
Unitree humanoid average selling price	260,700 yuan to 167,600 yuan	2024 to Jan-Sep 2025	Unitree filing via Inside Deep Tech
Unitree humanoid gross margin	62.4% (H1) and 63.9% (Q3)	2025	Unitree filing via Inside Deep Tech
Share of Unitree humanoid revenue from research and education / industry	73.6% / 9.0%	Jan-Sep 2025	Unitree filing via Inside Deep Tech
BofA humanoid shipment forecast	1.2 million (2030), 10 million (2035)	forecast	Bank of America Institute
Average analyst forecast, humanoid sales	about US$300 billion	2035	J.P. Morgan PB average of 5 banks/consultancies

Figure	Value	Year	Source
Robotics venture funding	US$40.7 billion	2025	CB Insights via McKinsey

Think like a founder

Unitree's 60% gross margin did not come from assembling other people's parts. It came from owning the joint. In a young industry with no mature supplier ecosystem, vertical integration is how a company captures value and moves fast. Once the industry matures and standard modules appear (as happened in personal computers and is happening in electric vehicles), assembly margins usually collapse and value moves to the owners of key components, the software platform and the brand.

So a founder must ask: at what stage of maturity will my product enter the market? If you enter in 2030 with a generic humanoid assembled from purchased modules, you will compete with Chinese makers on price and lose. If you enter with a vertical solution, meaning a robot plus software plus service for one job in one industry where you know the customer deeply (for example, robots that sort recyclable waste, or that do night-shift inventory in pharmacies), you can capture value that the big makers ignore. The robot-hour calculation above is your best friend: always know your cost per useful hour and your customer's cost per labour hour.

Week 4: The players, adoption by country, and the Africa and Southeast Asia question

Core lesson

The national robot industries

Each major robot nation has a different model. Knowing these models helps you see where a newcomer could fit.

Japan: the precision incumbent. Japan built the modern industrial robot industry in the 1970s and 1980s. Its firms (Fanuc, Yaskawa, Kawasaki, Nachi, Epson, Denso and others) make arms; Nabtesco and Harmonic Drive Systems make the reducers inside many of the world's robots, Japanese and foreign. The IFR calls Japan "the world's largest exporter of robots" and "the world's predominant robot manufacturing country". Japan's model is deep, slow-accumulated precision skill and long relationships with car and electronics makers. Its weakness is its home market: Japanese installations fell 19% to 36,219 in 2025, and the IFR expects no major recovery soon.

Germany and Europe: engineering, integration and a shrinking market. Germany is the world's fifth-largest robot market and installed fewer than 25,000 units in 2025, down 8%, with its car industry in retreat; Europe as a whole fell 14% (IFR). European strengths are in factory software, automation systems and precision parts. Ownership has shifted: KUKA belongs to China's Midea since 2016, and in October 2025 SoftBank of Japan agreed to buy ABB's robotics division for US$5.4 billion, saying its "next frontier is physical AI" (Reuters, cited by McKinsey). European suppliers are repositioning for humanoids: McKinsey describes a large German automotive supplier signing three humanoid actuator partnerships in five months (with Neura Robotics, the UK's Humanoid and China's Leju), and Bosch partnering with Neura.

South Korea: the world's most automated factories, and a chaebol bet on humanoids.

Korea has the highest robot density in the world (1,220 per 10,000 workers in 2024). Its installations have been flat at about 30,000 a year since 2019, but the IFR expects new car investments to lift demand from 2027. Hyundai owns Boston Dynamics, maker of the Atlas humanoid, which McKinsey reports is scheduled to deploy in Hyundai's US factories from 2028 with batteries from LG Energy Solution. Korea's model is integration inside giant business groups that are both robot makers and robot users.

The United States: AI, capital and customers, but a thin hardware base. The US became the second-largest robot market in 2025, with 38,428 installations (IFR). It leads in AI models, AI chips and capital: McKinsey cites the Stanford AI Index figure of about US$109 billion of private AI investment in the US in 2024 against about US$9.3 billion in China. Humanoid developers include Tesla (Optimus), Figure AI, Agility Robotics and Apptronik, with NVIDIA supplying compute and simulation tools and Google DeepMind and others building robot AI models. The weakness is hardware supply. Epoch AI notes that new factories take "2+ years in most Western countries" against 6 to 9 months at best in China, and Morgan Stanley's estimate that a humanoid built without Chinese suppliers costs nearly three times as much shows the dependence. Agility's RoboFab factory, a flagship American plant, produces around 150 to 500 humanoids a year against a stated target of 10,000 (Epoch AI).

China: the scale manufacturer, now climbing the value chain. China has the largest robot market (59% of 2025 installations), more than half of it supplied by Chinese brands, and is now a net exporter of industrial robots (Xinhua, citing customs data). In humanoids, J.P. Morgan Private Bank estimates China accounts for roughly 75% of the market and produces about 70% of the world's actuators. McKinsey reports China filed about 7,700 humanoid-related patents over five years, against roughly 1,560 in the US and 1,100 in Japan, and that China committed a US$138 billion state venture capital guidance fund to AI, robotics and other high-tech sectors. Leading humanoid makers include Unitree, AgiBot, UBTech, Galbot, Leju and Fourier, and car companies such as XPeng are also building humanoids. China's weaknesses are frontier AI chips (limited by US export controls, Month 8) and, still, the very top end of precision parts.

India and Brazil: fast-growing users. India installed almost 10,500 robots in 2025, up 15%, with installations growing 27% a year on average from 2020 to 2025 (IFR). Brazil installed almost 4,300, up 38%, with its car industry's robot purchases up 212%, which the IFR attributes most likely to investment by Chinese carmakers. Note the mechanism: Chinese manufacturing investment abroad brings robot demand with it.

Who captures value in a robot?

Put the anatomy of Week 2 and the BOM of Week 3 together with this map of players, and you can draw a rough "smile curve" (Month 1) for robotics. At the left end, high value: AI models, chip design, robot system design and key intellectual property. At the bottom of the smile, low margin: generic metal parts, wiring, final assembly. At the right end, high value again: system integration, software subscriptions, maintenance and fleet services, and the brand that owns the customer. Precision components (reducers, screws, force sensors, magnets) are a special case: they are physical manufacturing, normally the low middle of the smile, but they earn high margins because few firms can make them well. These are chokepoint components, and they sit in a raised "bump" in the middle of the smile.

The upstream mineral layer has its own logic, which you studied in Months 5 and 6: mining earns resource rents that swing with commodity prices, refining is a scale-and-energy business dominated

by China, and neither captures much of the value of the final robot. Rare earth materials are a tiny share of a robot's cost, but without them the robot cannot be built, which is exactly why control of them is a weapon and not a profit centre.

The Africa and Southeast Asia question

Now the question at the heart of this course. Could Africa or Southeast Asia build a cheap robot supply chain, from ore to finished robot, by the 2040s? Let us test each layer with evidence.

Layer 1: ores. Both regions are rich. Southeast Asia: Indonesia mined about 2.6 million tonnes of nickel in 2025, roughly 67% of the world's 3.9 million tonnes, and Indonesia was also the second-largest cobalt producer with about 14% (USGS Mineral Commodity Summaries 2026, nickel and cobalt chapters); on the same USGS figures, Indonesia and the Philippines together produced about 74% of the world's mined nickel in 2025 (the OECD's April 2026 note gives 68%, probably on an earlier year's figures); Vietnam holds about 4% of world rare earth reserves; Myanmar has become an important producer of heavy rare earths such as dysprosium and terbium. Africa: the Democratic Republic of the Congo (DRC) mined about 73% of the world's cobalt in 2025 (USGS); the DRC and Zambia are major copper producers; South Africa dominates platinum-group metals and manganese; and the IEA lists new rare earth mining projects in Tanzania among the diversification pipeline. Verdict: both regions can supply ores, though neither has much of the magnet rare earth chain except Myanmar (where mining has serious governance and environmental problems) and future African projects.

Layer 2: refining and materials. Southeast Asia is well ahead. Indonesia's nickel export ban forced processing at home (Month 6), and Malaysia hosts one of the few industrial-scale rare earth refineries outside China; the IEA describes Malaysia as the "distant second" in rare earth separation and refining. The IEA also lists Vietnam among countries with magnet manufacturing projects. Africa mostly exports concentrates and intermediate products; the DRC's cobalt is largely refined in China (USGS notes that most of China's refined cobalt comes from partially refined material imported from the DRC and Indonesia). Refining needs cheap, reliable electricity and strict waste management, two things many African mining regions lack today (Month 4 and Month 6).

Layer 3: precision components. Neither region makes high-end reducers, roller screws or force sensors today. Southeast Asia has relevant adjacent industries: Thailand's automotive parts cluster, and the electronics and semiconductor assembly industries of Malaysia, Vietnam, Thailand and the Philippines (Month 7). These give a base of machining, quality control and supplier management. Africa's base is thinner: the car industries of South Africa and Morocco are the main exceptions.

Layer 4: assembly and the home market. This is decisive. Industrial robot installations in all of Africa were 519 in 2025 (IFR), smaller than Brazil's car industry alone. Southeast Asia installs thousands a year; for example Thailand installed about 3,600 in 2023 according to IFR data compiled by Statbase. Chinese robot exporters already send about 28% of their exports by value to Southeast Asia, according to one industry analysis (not independently verified), because Chinese factories are relocating there. Recall the lesson of the China case: suppliers learn on a home market. Southeast Asia is becoming a robot-using region. Africa is not yet.

Layer 5: AI and software. Both regions will largely use AI models trained elsewhere. The rise of open robot AI models (NVIDIA's GR00T family and others, Month 8) lowers the barrier: a firm in Semarang, Hanoi or Nairobi can fine-tune an open model on local tasks. This layer is the most open to newcomers because it needs talent and data more than factories.

A summary table, with my subjective assessment of what is achievable by the mid-2040s:

Layer	Southeast Asia, mid-2040s	Africa, mid-2040s
Ores (nickel, cobalt, copper, rare earths)	Yes, already	Yes, already
Refining and battery/magnet materials	Likely, with Chinese, Japanese or Korean partners	Possible in a few places with cheap power (partial)
Precision components	Possible for mid-range parts	Unlikely except niches
Robot assembly for regional market	Likely	Possible in one or two hubs (Morocco, South Africa, maybe Egypt or Kenya)
AI, integration, services	Likely	Likely in major cities
Full chain inside the region, cost-competitive with China	Unlikely	Very unlikely

Judgment: A complete, cost-competitive "ore to robot" chain inside either region by the 2040s is unlikely. The economics of agglomeration (Month 7) mean that China's dense clusters will keep a large cost lead for decades, and the precision-component layer takes many years of learning. But that is the wrong goal. The realistic and valuable goal is to own several layers and to connect them to partners for the rest. Southeast Asia's best path is "China plus one" for robots: refine nickel and rare earths, make battery materials and magnets, assemble robots for the region's own fast-growing factories, and build integration and service firms. Africa's best path starts from its mines: mining is a dangerous industry that is already automating, so African engineering firms can become integrators and service providers of mining robots and drones, then build regional assembly in one or two hubs with reliable power and ports. In both regions the layer most open to a young founder is the top one: integration, software and service, where knowledge of local customers matters more than capital. This judgment would change if robot designs became so standardised and modular that assembly from purchased modules became the main source of value, or if China's export controls made the rest of the world build parallel supply chains at any cost.

Case study: Indonesia's nickel and the battery-to-robot question

Indonesia is one case among many, but it is the clearest test of whether controlling an ore can pull the rest of the chain into a country. As Month 6 described, Indonesia banned exports of nickel ore from 2020, forcing miners to build smelters at home, mostly with Chinese investment. The result: Indonesia's share of world nickel mining rose to about two thirds by 2025, and refined nickel products and battery precursors now flow from Indonesian industrial parks.

What does this mean for robots? Nickel goes mainly into stainless steel and into some battery chemistries (nickel-manganese-cobalt). A humanoid's battery is a few kilowatt-hours and a small share of its cost (US$100 to US$300 in Morgan Stanley's estimates). So even if every humanoid battery used Indonesian nickel, nickel would capture a tiny share of robot value. The lesson is sobering and important: **owning a major input does not mean capturing the value of the final product.** The value of a humanoid sits in its joints, hands, sensors and software. Indonesia's downstreaming policy captured more of the value of nickel. It did not, by itself, create an electronics, precision-machinery or robotics industry. For that, a country needs a different set of policies: engineering education, supplier development programmes, a domestic market of robot users, and investment rules that require technology learning, not just processing.

The same lesson applies to the DRC and cobalt, to Chile and lithium, and to any country that hopes minerals will carry it up the value chain. Minerals give bargaining power. They do not give capabilities. Capabilities have to be built separately, deliberately, and usually with foreign partners.

Key numbers

Figure	Value	Year	Source
US robot installations (now 2nd largest market)	38,428	2025	IFR World Robotics 2026
Japan robot installations	36,219 (down 19%)	2025	IFR
India robot installations, average growth 2020-25	about 10,500; 27% a year	2025	IFR
China's share of global humanoid market	about 75%	2026	J.P. Morgan Private Bank
Humanoid-related patents over 5 years: China / US / Japan	7,700 / 1,560 / 1,100	to 2025	SCMP via McKinsey
SoftBank's purchase of ABB Robotics	US$5.4 billion	announced Oct 2025	Reuters via McKinsey
Indonesia's share of world nickel mining	about 67%	2025	USGS MCS 2026
DRC's share of world cobalt mining	about 73%	2025	USGS MCS 2026
Industrial robots installed in Africa	519	2025	IFR
Factory build time, China vs West	6-9 months (best case) vs 2+ years	2026	Epoch AI

Think like a founder

If you want to build in robotics from Southeast Asia, three opportunities stand out, in order of how quickly a small team could start:

1. **Integration and service for regional factories.** As Chinese, Japanese and Korean firms move production into Southeast Asia, they need local engineers to install, program and maintain robots. A firm that does this well for one industry can grow into a regional platform. Low capital, high know-how, and the customer relationship is yours.
2. **Robot AI for local tasks.** Fine-tuning open robot models for tasks common in the region (handling agricultural produce, garment finishing, sorting recyclables) uses your AI skills directly. The data you gather from real deployments becomes a moat.
3. **Materials and recycling.** Recovering rare earths, cobalt and nickel from retired motors and batteries is capital-intensive but strategically valued by governments trying to escape China's chokepoints, which means access to development finance and patient capital (Month 11).

For social impact, the strongest case is robots for dangerous, dirty work where labour is scarce or

workers are harmed: mining, waste sorting, heavy agriculture, chemical handling. There, automation saves lives and raises the wages of the workers who remain as operators, which links directly to next month's question of who gains and who loses.

Month synthesis

This month followed the value chain from rock to robot. The picture fits on one page:

```
ORES              REFINING            COMPONENTS              ROBOT              USE
nickel, cobalt    rare earth oxides   magnets, motors         design + AI        integration
copper, REE       battery materials   reducers, screws        assembly           service, fleets
lithium, steel    alloy steels        sensors, chips, hands   brand              data
   |                  |                   |                     |                  |
 resource rent     scale + energy      CHOKEPOINT MARGINS      platform/brand     recurring revenue
 (volatile)        (China ~90% for     (Japan, China;          (US AI, China      (whoever owns
                   magnets)            few qualified firms)    scale, Korea)      the customer)
```

Layer	Share of robot value (rough)	Who leads now	Barrier to entry
Minerals	very small	DRC, Indonesia, Chile, Australia, China	Geology, capital, politics
Refining and materials	small	China	Energy, waste handling, scale, 8-year lead times
Precision components	large (actuators 40-60% of BOM)	Japan (quality), China (scale)	Decades of process know-how, customers to learn on
Chips and AI models	small per unit, large fixed cost	US, Taiwan; China catching up	Capital, talent, export controls
Final assembly	modest margin unless vertically integrated	China	Supplier cluster
Integration and services	large over the robot's life	Fragmented, local	Customer knowledge

Three conclusions tie the month together. First, the industrial robot arm is a mature, measurable, growing industry dominated by China as market and increasingly as supplier; the humanoid is an emerging industry whose cost will fall but whose usefulness is not yet proven. Second, value concentrates in chokepoint components and in the software, data and service layers, not in minerals or assembly. Third, China's advantage is a system (supplier clusters, a giant home market, state capital, fast factory building and control of rare earth processing), and systems cannot be copied quickly. A newcomer region wins by choosing layers, not by trying to copy the whole system. Next month asks what all these machines do to human work: if robots and AI become cheap, who earns the income they produce?

Common misconceptions

1. **"The AI chip is the most expensive part of a humanoid."** Wrong. In Morgan Stanley's China-built estimate, chips and software are about 7% of the material cost; actuators and hands are about 81%. The AI is expensive to train, not to install per unit.
2. **"China has the most automated factories in the world."** China has by far the most robots (about two million operational) and installs more than half of the world's new ones, but its robot density, after the IFR's April 2026 revision, is 166 per 10,000 workers, far below Korea (1,220), Germany (449) or Japan (446). Quantity and intensity are different measures.
3. **"Humanoids are already replacing factory workers at scale."** Not yet. The IFR says most humanoids remain prototypes or pilots and that industrial robots are "economically superior" for production tasks today. Unitree's own filing shows that in the first nine months of 2025 about 74% of its humanoid revenue came from research and education and only about 9% from industrial applications, of which half to seven tenths was reception and tour-guide work.
4. **"A country rich in robot minerals will capture the robot industry."** Minerals are a tiny share of a robot's value. Indonesia's nickel and the DRC's cobalt give bargaining power but not capabilities in precision engineering, electronics or AI.
5. **"Cheap labour countries will be the last to be affected by robots, so they can ignore them."** Partly wrong. Low wages do slow adoption (see the payback example), but robots change where factories locate: cheap robots in rich countries can bring production home ("reshoring"), which reduces the demand for labour-intensive exports from poorer countries. Month 10 develops this.

Exercises

1. **Robot density (quantitative).** A country has 120,000 operational industrial robots and 9.5 million manufacturing employees. Calculate its robot density. If its statistical office revises employment up to 14 million, what is the new density? Explain in three sentences why the revision does not change the number of robots and what it does change.
2. **Payback (quantitative).** A robot cell costs US$120,000, lasts 7 years and costs US$4,000 a year to maintain. It replaces three workers. Calculate the payback period if each worker costs (a) US$45,000 a year, (b) US$12,000 a year, (c) US$4,000 a year. At what annual labour cost per worker does payback equal the robot's life?
3. **Learning curve (quantitative).** A humanoid BOM is US$40,000 when cumulative production is 25,000 units. With a learning rate of 15%, what is the BOM at 400,000 units? At 6.4 million units? How many doublings are needed to reach US$15,000?
4. **Supply ceiling (quantitative).** A humanoid design uses 8 planetary reducers and 6 strain wave gears. World output is 3 million planetary reducers and 12 million strain wave gears a year, and robots can use at most 30% of each. Which component binds, and what is the maximum annual humanoid output?
5. **Robot hour (quantitative).** Repeat the Week 3 robot-hour calculation for a robot costing US$25,000 that runs three shifts and reaches 70% of human productivity. Compare the result with the hourly labour cost of a factory worker in a country you choose (find a source and cite it).
6. **Essay (600 words).** "Japan's chokepoint in reducers is eroding while China's chokepoint in rare earth magnets is not." Explain the mechanisms behind each, using the ideas of learning-by-doing, home markets and lead times.

7. **Essay (800 words).** Choose one country in Southeast Asia or Africa. Using the five layers of Week 4, write a 15-year strategy for capturing value in the robot chain. State which layers you would not try to enter, and why.
8. **Founder exercise.** Pick one dangerous or dirty job in your region. Estimate the number of workers, their wages and injury risks, and the cost of a robot that could do the task. Would you sell the robot, rent it per hour, or sell a service? Justify with numbers.

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- Richard Baldwin, *The Great Convergence* (2016). The economics of global value chains and the smile curve, which you need to reason about who captures value.

Glossary terms introduced this month

- **Actuator:** the complete joint module (motor, transmission, driver, sensors) that makes a robot move.
- **Autonomous mobile robot (AMR):** a robot that moves around a facility on its own, for example carrying parts.
- **Bill of materials (BOM):** the list of a product's parts and their costs, excluding assembly, overhead and profit.
- **Cobot (collaborative robot):** a robot arm designed to work safely near people without a fence.
- **Cycloidal (RV) reducer:** a stiff, strong gearbox using lobed discs, used in heavy robot joints.
- **Degrees of freedom:** the number of independent ways a robot or limb can move.
- **Dexterous hand:** a robot hand with many independently controlled fingers and touch sensing.
- **Encoder:** a sensor that measures the exact angle or position of a joint.
- **Industrial robot:** under ISO 8373, an automatically controlled, reprogrammable, multipur-

pose manipulator programmable in three or more axes.
- **Learning rate / Wright's law:** the fixed percentage by which unit cost falls each time cumulative production doubles.
- **NdFeB magnet:** neodymium-iron-boron permanent magnet, the strongest common magnet type, used in robot and electric vehicle motors.
- **Planetary gearbox:** a compact, relatively cheap gearbox with gears circling a central gear.
- **Planetary roller screw:** a high-load screw that turns rotation into straight-line motion.
- **Reducer:** a transmission that lowers speed and raises torque between motor and joint.
- **Robot density:** operational industrial robots per 10,000 manufacturing employees.
- **Six-axis force/torque sensor:** a sensor measuring force and twist in all directions at a wrist or ankle.
- **Strain wave gear (harmonic drive):** a compact, zero-backlash reducer using a flexible toothed ring.
- **System integrator:** a firm that designs, installs and programs complete robot systems for a customer.
- **Torque:** rotational force; what a joint needs to lift or push.
- **Vertical integration:** a company making its own key components instead of buying them.

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Month 10: AI, Robots and the Labor Economy

Month overview

Month 9 took a robot apart and followed the money through its joints, magnets and software. This month asks the question that follows from it: when machines that can see, move and reason become cheap, what happens to the people who used to do that work? Who gets the income the machines produce?

This is not a side question for the course's driving theme of value capture from ore to robot to AI. It is the final step of that chain. Every economy divides its income between labour (wages and salaries) and capital (profits, rents and interest paid to the owners of machines, land, firms and intellectual property). Months 5 to 9 traced how value is divided between countries and firms along the supply chain. Month 10 traces how it is divided between workers and owners inside each country, and between rich and poor countries as automation changes where things are made.

Week 1 builds the economics of productivity: what economists mean by growth, why new technologies take decades to show up in the statistics, and why respected economists disagree tenfold about how much AI will add to output. Week 2 introduces the most useful tool in the field, the task framework, which separates automation (machines taking over tasks) from augmentation (machines making people more productive) and explains why the same technology can raise or lower wages. Week 3 looks at the evidence from 2023 to 2026 on jobs, wages and inequality, including the first signs that AI is changing hiring for young workers, and asks who owns the machines. Week 4 uses history (the Luddites, the horse, the bank teller) and growth theory to think about the future, and then examines the policy toolbox: education, social insurance, taxes on automation, basic income and shared ownership of capital.

The month ends with a framework that Month 11 (money, finance and sovereign wealth funds) and Month 12 (scenarios to 2050) will build on. Keep one question in mind throughout, because it matters for you as a future founder: if your company replaces human work with AI or robots, where does the saved money go, and who decides?

Learning goals

By the end of this month you should be able to:

- Explain total factor productivity (TFP), the productivity J-curve and Hulten's theorem, and use them to judge forecasts of AI's effect on growth.

- Use the task framework (displacement, productivity and reinstatement effects) to explain why automation can raise or lower wages and employment.
- Summarise the best 2023-2026 evidence on AI and robots in the labour market, including its limits, and distinguish exposure estimates from measured effects.
- Explain the labour share of income and why ownership of capital decides who benefits from automation.
- Draw lessons from at least three historical episodes of technological change and say where the analogies break down.
- Evaluate the main policy responses (education, social insurance, robot taxes, basic income, universal basic capital, sovereign funds) with evidence and trade-offs.
- Connect labour-market outcomes to the course's question of value capture from ore to robot to AI.

Week 1: Productivity, growth and what AI could add

Core lesson

Growth in one equation

Why are some people rich and others poor, and why are people today richer than their great-grandparents? Economists start with a simple accounting idea. Output (GDP, the value of all goods and services produced) depends on three things: how much labour is used, how much capital (machines, buildings, computers) is used, and how efficiently they are combined. The third part is called **total factor productivity (TFP)**. It captures technology, organisation, skills and anything else that lets the same labour and capital produce more.

The growth model developed by Robert Solow in 1956 made a striking point that still shapes the debate: in the long run, adding more machines per worker runs into diminishing returns. The first tractor on a farm doubles output; the tenth adds little. Sustained growth in income per person therefore comes mostly from rising TFP, which means from better technology and better ways of using it. That is why economists care so much about whether AI raises TFP. If it does, it can lift living standards permanently. If it only substitutes for labour at slightly lower cost, it may change who earns the income without making the pie much bigger.

A small worked example of growth accounting. Suppose in one year an economy's capital stock grows 4%, its labour hours grow 1%, and capital earns 40% of income while labour earns 60%. If output grows 3%, how much came from TFP?

- Contribution of capital: 0.40 x 4% = 1.6 percentage points.
- Contribution of labour: 0.60 x 1% = 0.6 percentage points.
- TFP growth is the remainder: 3% - 1.6% - 0.6% = 0.8%.

TFP is measured as a "residual", what is left after counting inputs. This makes it powerful (it captures everything we cannot name) and fragile (any measurement error in inputs or outputs ends up in it).

General-purpose technologies and the J-curve

Some technologies change one industry. Others, called **general-purpose technologies (GPTs)**, spread across the whole economy and keep improving: the steam engine, electricity, the computer, the internet. The 2023 paper by Tyna Eloundou, Sam Manning, Pamela Mishkin and Daniel Rock

was titled "GPTs are GPTs" to argue that generative pre-trained transformers (the language models behind ChatGPT) have the traits of a general-purpose technology.

General-purpose technologies share an awkward pattern: their effect on measured productivity is slow at first. When American factories first electrified around 1900, they replaced the central steam engine with a central electric motor and kept the old layout. The big gains came only decades later, when factories were redesigned around small motors at each machine. Erik Brynjolfsson, Daniel Rock and Chad Syverson formalised this as the **productivity J-curve**: firms first spend heavily on "intangible" investments (redesigning processes, training people, building software and data) that statistics do not count as output, so measured productivity looks weak; later the payoff arrives and measured productivity may even overstate the improvement. The curve dips, then rises, like the letter J.

This matters for reading 2026 data. A study published by Anders Humlum and Emilie Vestergaard (NBER Working Paper 33777, revised March 2026) linked surveys of AI chatbot adoption in Denmark to official records of earnings and hours. They found "rapid currents" (most employers in exposed occupations had adopted chatbot initiatives, and workers reported productivity benefits), yet "precise null effects on earnings and recorded hours", ruling out effects larger than 2% two years after ChatGPT's launch. What changed was the structure of work: new tasks in content generation, AI oversight and AI integration. Their summary: "Technological change reshapes work well before it surfaces in earnings or hours." That is the bottom of the J-curve, observed in real time.

Micro evidence: big gains in specific tasks

At the level of individual tasks, the gains can be large. The best-known study, by Erik Brynjolfsson, Danielle Li and Lindsey Raymond (published in the Quarterly Journal of Economics in 2025), followed 5,179 customer-support agents at a software company as an AI assistant was introduced in stages. Access to the tool raised productivity, measured as issues resolved per hour, by 14% on average, with a 34% improvement for novice and low-skilled workers and "minimal impact on experienced and highly skilled workers". The authors' interpretation is important: the AI "disseminates the best practices of more able workers and helps newer workers move down the experience curve". In other words, the AI captured the tacit knowledge of the best workers and handed it to beginners. Customer sentiment improved and employee turnover fell.

Notice the double edge. If AI makes a three-month beginner as good as a two-year veteran, the employer needs to pay less for experience, and may need fewer people. The same mechanism that helps novices today could reduce the value of experience tomorrow. Keep this in mind for Week 3.

Macro forecasts: from 0.66% to 7%

How do task-level gains add up to the whole economy? Daron Acemoglu (who shared the 2024 Nobel memorial prize in economics with Simon Johnson and James Robinson) made the most influential sceptical estimate in "The Simple Macroeconomics of AI" (NBER 2024; Economic Policy, 2025). His method uses **Hulten's theorem**: if a technology lowers costs in some tasks, the gain in aggregate productivity is roughly the share of the economy's tasks affected multiplied by the average cost saving in those tasks.

A worked example of the logic, with illustrative inputs (not Acemoglu's exact numbers):

- Share of all work tasks exposed to AI: 20%.
- Share of exposed tasks where using AI is actually profitable within 10 years: 25%.

- So 20% x 25% = 5% of all tasks are affected in a decade.
- Average cost saving on those tasks: 15%.
- Aggregate cost saving, and therefore TFP gain over 10 years: 5% x 15% = 0.75%, less than 0.1% a year.

Acemoglu's own estimate was "no more than a 0.66% increase in total factor productivity (TFP) over 10 years", and possibly below 0.53%, because early evidence comes from "easy-to-learn tasks", while future gains depend on harder tasks with "no objective outcome measures from which to learn successful performance".

Other forecasters are far more optimistic. Goldman Sachs economists Joseph Briggs and Devesh Kodnani estimated in 2023 that generative AI could raise global GDP by 7% over ten years and expose the equivalent of 300 million full-time jobs to automation; by 2026 the bank's baseline was a 15% rise in the level of labour productivity in developed economies once AI is fully adopted (Goldman Sachs, 2023 and 2026). Some researchers go much further, modelling scenarios in which AI automates most cognitive work within a decade (see Week 4).

Why the tenfold disagreement? Three assumptions drive it:

1. **Breadth:** what share of tasks AI can do. Optimists include future capabilities; Acemoglu uses what is profitable with today's evidence.
2. **Depth:** the cost saving per task. A tool that saves 15% of the time is very different from one that does the whole task.
3. **New tasks and new products:** Hulten's theorem only counts cost savings on existing tasks. If AI creates entirely new goods (new medicines, new materials, new software), gains can be much larger. Acemoglu also warns that some new tasks may have "negative social value", such as algorithms designed for online manipulation.

Judgment: For the period up to about 2030, the evidence supports the cautious end: measurable but modest aggregate gains, concentrated in software, customer service, writing and analysis, with large gains for specific firms and workers. Beyond 2030 the honest range is wide, because it depends on whether AI moves from assisting with tasks to completing whole jobs reliably, and whether robots (Month 9) extend this to physical work. A good analyst holds both views: sceptical about near-term macro statistics, open about long-term transformation.

Baumol's twist: where value goes when things get cheap

One more mechanism completes the growth picture. In the 1960s William Baumol observed that a string quartet needs four players and the same time to play Beethoven as it did in 1800, even though factory productivity has multiplied many times. Because musicians must still be paid wages comparable to factory workers, the relative price of live music rises over time. This is **Baumol's cost disease**: sectors where productivity grows slowly (care for the elderly, teaching, haircuts, nursing) become more expensive relative to sectors where it grows fast (electronics, manufactured goods).

For AI and robots, Baumol's idea has a surprising implication. If AI makes software and documents almost free, their share of GDP may shrink, and spending shifts toward whatever remains hard to automate: in-person care, physical services, trusted relationships, scarce land and energy. The bottleneck sectors capture a growing share of value. This links back to the material economy of Months 4 to 6: if intelligence becomes abundant, energy, minerals, land and the physical capacity to act in the world (robots, factories, grids) become relatively more valuable. It also suggests a

counter-intuitive point for a founder: the most valuable businesses in an AI-abundant world may be those that own the scarce physical or relational piece, not the abundant cognitive piece.

Case study: The call centre, from Fresno to Manila

Customer service is one of the clearest test cases for AI and work, and it connects rich and developing countries.

The experiment. The Brynjolfsson, Li and Raymond study described above took place in exactly this kind of work: customer-support agents answering software questions by chat. A 14% average productivity gain and a 34% gain for novices is large by the standards of any management intervention.

The economic choice for the firm. Suppose a firm has 1,000 agents handling 1,000,000 chats a month. With a 14% productivity gain, it can handle the same volume with about 1,000 / 1.14 = 877 agents. It has three options: (a) keep 1,000 agents and serve customers faster or offer more services; (b) let attrition shrink the workforce to about 880 over a year without layoffs; (c) cut more deeply as AI improves, moving toward fully automated chat with humans handling only hard cases. Which option it chooses depends on demand. If cheaper, better service brings more customers (demand is "elastic"), employment may hold up. If demand is fixed, jobs fall. This is the general rule of the whole month: **productivity gains raise employment only if demand grows enough to absorb them.**

The development dimension. Call centres and other business process outsourcing (BPO) are a major source of good jobs in some developing economies. The Philippines is the leading example: industry figures put the Philippine BPO workforce at about 1.68 million in 2025, up 4% from 1.62 million despite fears about AI, with 1.73 million projected for 2026 (CX Association of the Philippines, via Philstar, June 2026). The IMF's 2024 analysis found that about 40% of jobs in emerging markets and 26% in low-income countries are exposed to AI, against about 60% in advanced economies. Exposure is lower in poor countries because more people work in agriculture and manual jobs. But the jobs that are exposed in middle-income countries are often exactly the good, urban, English-speaking service jobs that offered a path into the middle class: call centres, back-office processing, transcription, basic software testing.

So far, the feared collapse has not happened, and some firms report using AI to raise quality rather than cut staff. But the mechanism is clear. Offshoring moved service jobs to places where labour was cheap. AI can make the labour content so small that labour cost stops mattering, and the work can be done anywhere by software. The first rung of the ladder that India and the Philippines climbed in the 1990s and 2000s may be weaker for countries that hoped to climb it in the 2030s.

Key numbers

Figure	Value	Year	Source
Productivity gain from AI assistant, customer support (average / novices)	14% / 34%	study published 2025	Brynjolfsson, Li and Raymond, QJE
Acemoglu upper estimate of AI's TFP gain over 10 years	0.66% (or below 0.53%)	2024	NBER WP 32487

Figure	Value	Year	Source
Goldman Sachs estimate of global GDP boost from generative AI	7% over 10 years	2023	Goldman Sachs
Goldman Sachs baseline labour productivity uplift at full adoption	about 15%	2025-2026	Goldman Sachs
Largest effect on Danish earnings/hours ruled out, 2 years after ChatGPT	2%	2026 revision	Humlum and Vestergaard, NBER WP 33777
Global employment exposed to AI (IMF)	almost 40%	2024	IMF
Exposure in advanced / emerging / low-income economies	about 60% / 40% / 26%	2024	IMF
Philippine BPO workforce	about 1.68 million (1.73 million projected for 2026)	2025	CXAP via Philstar

Think like a founder

The call centre case gives a founder's rule: AI creates the most value where it spreads expert knowledge to many less experienced people. That was the mechanism in the 34% gain for novices. Ask where in your world expert knowledge is scarce and expensive: agricultural advice for smallholder farmers, diagnosis support for rural health workers, legal and tax guidance for small businesses, maths tutoring for students without good teachers. In each case an AI that turns a beginner into a competent practitioner raises incomes rather than replacing workers, because demand for these services is huge and mostly unmet.

The value capture question is then about pricing. If you sell the tool to an employer, the employer may capture the gain by cutting staff. If you sell it to the worker (the farmer, the health worker, the small business owner), the worker captures more of it. Social-impact founders should design business models where the person whose productivity rises keeps a meaningful share of the gain. That is a choice, not an accident.

Week 2: Automation versus augmentation, and the task framework

Core lesson

Jobs are bundles of tasks

The most useful idea in the modern economics of technology is that a job is not one thing but a bundle of tasks. A bank teller counts cash, checks identity, answers questions, sells products and handles complaints. A radiologist reads scans, talks to other doctors, decides on further tests and

explains results. Technology rarely replaces a whole job at once. It takes over some tasks, changes others and creates new ones.

Daron Acemoglu and Pascual Restrepo built this into a model that separates three forces:

1. **The displacement effect.** When machines take over tasks that workers used to do, demand for labour in those tasks falls. This pushes wages and employment down.
2. **The productivity effect.** Automation lowers costs, which lowers prices, raises real incomes and increases demand for goods and services, including for tasks still done by people. This pushes labour demand up, in the same industry or elsewhere.
3. **The reinstatement effect.** Technology creates new tasks in which humans have an advantage: programming the robot, maintaining it, designing new products, managing data, caring for customers in new ways. This pushes labour demand up.

The net effect on workers depends on the balance. If displacement is large and productivity gains are small, workers lose. If productivity gains are large and new tasks appear quickly, workers can gain even as old tasks vanish.

Augmentation and automation

This leads to a crucial distinction. **Automation** technologies replace human labour in a task (a welding robot, a self-checkout machine, an AI that writes routine reports with no human). **Augmentation** technologies make humans more productive in a task (a power tool, a spreadsheet, the AI assistant in the call centre that helps an agent answer better). The same AI model can be used either way; the choice depends on how firms design jobs.

David Autor, Caroline Chin, Anna Salomons and Bryan Seegmiller measured both kinds of innovation for the United States from 1940 to 2018, linking patents to occupations and tracking new job titles over eight decades (NBER Working Paper 30389; Quarterly Journal of Economics, 2024). Their findings are worth remembering:

- **The majority of current US employment is in job specialties that did not exist in 1940.** Most of today's work is "new work".
- New work appears in response to innovations that complement an occupation's output (augmentation) and to rising demand; innovations that automate tasks slow the creation of new work.
- Augmentation innovations raise an occupation's labour demand; automation innovations reduce it.
- Most worrying: "the demand-eroding effects of automation innovations have intensified in the last four decades while the demand-increasing effects of augmentation innovations have not."
- The location of new work has shifted: from middle-paid production and clerical jobs in 1940-1980 to high-paid professional jobs and, second, low-paid services since 1980. This is the "hollowing out" of the middle.

So-so automation

Acemoglu adds a warning about what he calls **"so-so technologies"**: automation that is just good enough to replace workers but not good enough to raise productivity much. A self-checkout machine is the classic example. It displaces cashiers (displacement effect) but barely lowers costs (small productivity effect) and creates few new tasks (small reinstatement effect). From society's point of view it is the worst combination: workers lose and the economy gains little. Firms may still adopt it

if it is slightly cheaper for them, especially if the tax system favours buying machines over hiring people (Week 4).

Worked example. A shop pays a cashier US$30,000 a year. A self-checkout system costs US$27,000 a year to own and run, and handles the same customers, though some customers are slower and theft rises slightly. The firm saves US$3,000 a year, a 10% cost saving on that task. The cashier loses US$30,000 of income and must find other work. The gain to society (lower costs) is US$3,000; the private loss to the worker, at least temporarily, is ten times larger. If the worker finds an equally good job quickly, the loss is small. If not, it is large. "So-so" automation is profitable but not progress.

Measuring exposure: what the numbers do and do not mean

Since 2023 many studies have estimated how "exposed" jobs are to AI. Their headline numbers are often misread as forecasts of job losses. They are not. Exposure means AI could perform or speed up a significant share of a job's tasks. Whether that leads to job loss, higher pay or new work depends on the three effects above.

- **Eloundou, Manning, Mishkin and Rock (2023):** about 80% of the US workforce could have at least 10% of their tasks affected by large language models, and about 19% of workers could see at least half of their tasks affected. With LLMs alone about 15% of tasks could be done significantly faster at the same quality; with software built on top of LLMs, 47 to 56%. Exposure is higher for higher-income jobs, the opposite of earlier automation waves that hit factory and clerical workers hardest.
- **IMF (2024):** almost 40% of global employment exposed; about 60% in advanced economies, of which "roughly half" may benefit from AI integration and half may face lower labour demand.
- **International Labour Organization, ILO (2025):** one in four workers worldwide is in an occupation with some generative AI exposure; 3.3% of global employment is in the highest exposure category. Women are more exposed (4.7% in the highest category against 2.4% for men), mainly because they are over-represented in clerical work, the most exposed occupational group. Exposure is about one in three jobs in high-income countries but only one in ten in low-income countries. The ILO's conclusion is that "transformation, not replacement, is the most likely outcome".
- **Goldman Sachs (2023-2026):** about two-thirds of US occupations exposed to some degree; tasks equal to about 25% of US work hours could be automated; the bank's baseline is that 6 to 7% of US workers will eventually be displaced, with unemployment rising about half a percentage point during the transition.

Three lessons from comparing these numbers. First, they differ mainly in definitions, not in data. Second, they all show exposure concentrated in office, clerical and professional work, which is new: for two centuries automation mostly hit manual and routine work. Third, exposure in poorer countries is lower, but that is partly because their workers are in jobs (farming, construction, informal services) that are not yet exposed to AI, though some may be exposed to robots later.

Robots versus AI

It helps to compare the two technologies of Months 9 and 10:

Feature	Industrial robots (1990s-2020s)	Generative AI (2023-)	Humanoid and mobile robots (2030s?)
Tasks affected	Routine manual: welding, painting, assembly	Cognitive: writing, coding, analysis, customer contact	Non-routine manual: handling, logistics, care support
Workers most exposed	Manufacturing production workers	Clerical, professional, young graduates	Warehouse, logistics, cleaning, some care work
Where	Factories in rich and middle-income countries	Offices everywhere with internet	First high-wage, multi-shift sites
Cost to adopt	High capital cost per robot	Low marginal cost per user	Falling, still high (Month 9)
Speed of spread	Slow (years per factory)	Fast (software)	Slow at first, limited by production (Month 9)

The key difference is speed. A robot requires a physical purchase and installation. An AI model can be rolled out to millions of users in weeks. That is why AI's labour effects, if they come, could arrive faster than robotics ever did.

Case study: Robots in American and German towns

The best natural experiment on robots and jobs compares two rich countries that adopted industrial robots heavily from the 1990s.

The United States. Acemoglu and Restrepo studied US local labour markets from 1990 to 2007 (Journal of Political Economy, 2020). They measured each area's exposure to robots using national robot adoption in each industry and the local mix of industries, and checked that exposed areas did not have different trends before 1990. Their estimate: "one more robot per thousand workers reduces the employment to population ratio by about 0.18-0.34 percentage points and wages by 0.25-0.5 percent." In the NBER summary, the arrival of one new industrial robot in a local labour market coincides with an employment drop of about 5.6 workers. They also showed that the effect of robots was distinct from imports from China and Mexico, offshoring and other technologies.

Germany. Wolfgang Dauth, Sebastian Findeisen, Jens Suedekum and Nicole Woessner applied similar methods to Germany, which has far more robots per worker than the US (449 per 10,000 manufacturing workers in 2024, against 307 for the US, by IFR data from Month 9). Their findings (Journal of the European Economic Association, 2021) were very different. They found "no evidence that robots cause total job losses". Each robot replaced about two manufacturing jobs, accounting for roughly 275,000 lost manufacturing jobs from 1994 to 2014, about 23% of the decline in German manufacturing employment. But those losses were "fully offset by additional jobs in the service sector". Incumbent workers were not pushed out; they were more likely to stay with their employer, often switching to new tasks in the same plant. The adjustment fell on young people, who found fewer new manufacturing jobs and shifted from vocational training toward university. Wages of some groups fell, the labour share declined and productivity rose.

Why the difference? Several explanations fit the evidence, and they are about institutions, not technology:

- **Worker voice and job protection.** German firms have works councils and strong unions; displacing incumbents is costly and negotiated, so firms retrain and redeploy.
- **Apprenticeship and training systems** that make it easier to move workers into new tasks.
- **Export demand.** German manufacturers sold to booming world markets, so productivity gains turned into more output and more jobs (the productivity effect), not just fewer workers.
- **The type of robot use.** In Germany robots were concentrated in the car industry, which was expanding; in parts of the US, robots arrived in declining regions already hit by import competition.

The lesson is powerful: **the same technology produced job losses in one rich country and job reallocation without net losses in another.** Institutions, demand and policy decided the outcome. Technology sets the menu; society chooses from it.

Key numbers

Figure	Value	Year	Source
US workers with at least 10% of tasks exposed to LLMs	about 80%	2023	Eloundou et al.
US workers with at least half of tasks exposed	about 19%	2023	Eloundou et al.
Global workers in occupations with some GenAI exposure	about 1 in 4	2025	ILO Working Paper 140
Share in highest exposure category, women / men	4.7% / 2.4%	2025	ILO
Goldman Sachs estimate of US workers eventually displaced	6-7% (range 3-14%)	2025-2026	Goldman Sachs
Effect of one more robot per 1,000 workers on US employment-to-population ratio	-0.18 to -0.34 points	1990-2007 data	Acemoglu and Restrepo, JPE 2020
Manufacturing jobs replaced per robot in Germany	about 2 (offset by services)	1994-2014 data	Dauth et al.
German manufacturing jobs lost to robots	about 275,000 (23% of decline)	1994-2014	Dauth et al.
Share of current US employment in job types that did not exist in 1940	majority	2018 data	Autor, Chin, Salomons and Seegmiller

Think like a founder

The task framework is also a product design framework. Every AI or robot product sits somewhere between pure automation (remove the human) and pure augmentation (make the human better). Pure automation is easier to sell to buyers who want to cut costs, and it can be very profitable. But it is also where "so-so" products live, where political backlash concentrates, and where your product competes directly on price with human wages, which are low in much of the world.

Augmentation products have a different logic: they raise the value of a person's time, so they can be sold even where wages are low, because they make a worker able to do things they could not do before. A founder in a country with many workers and few experts should usually look for augmentation: tools that let a nurse do what a doctor used to do, a technician do what an engineer used to do, a farmer act on information only an agronomist had. Those products create new tasks (reinstatement) and raise wages, which is both better business in labour-abundant economies and better social impact.

Week 3: Wages, inequality and who owns the machines

Core lesson

The labour share: how the pie is divided

Every year a country's income is split between labour and capital. The **labour share** is the fraction paid to workers as wages, salaries and benefits; the rest goes to the owners of capital as profits, interest and rents. Automation pushes on this split directly. When a firm replaces a worker with a machine, the money that used to be a wage becomes a cost of capital (paid to the robot maker and its shareholders) and, if the machine is cheaper, extra profit (paid to the firm's shareholders).

The ILO reports that the global labour income share fell by 0.6 percentage points from 2019 to 2022 and has stayed flat since, "compounding a long-running downward trend". If the share had stayed at its 2004 level, labour income would have been US$2.4 trillion larger in 2024 alone (ILO, World Employment and Social Outlook, September 2024). The ILO lists technological advances, including automation, as one cause alongside the pandemic, and warns that AI "could deepen inequality" without policies to share gains.

Acemoglu's 2024 paper reaches a similar conclusion for AI specifically. Because AI's effects are spread across more demographic groups than earlier factory automation, he expects it to increase wage inequality less than robots did. But "AI is predicted to widen the gap between capital and labor income". The IMF's 2024 assessment is blunter: "In most scenarios, AI will likely worsen overall inequality."

A worked example shows the mechanism. A logistics company earns US$10 million a year in revenue. Before automation it pays US$6 million in wages to 150 workers, US$2 million for other costs, and earns US$2 million in profit. The labour share of its value added (wages plus profit) is 6 / (6 + 2) = 75%. It installs robots that let it handle the same volume with 75 workers. Robots cost US$1.5 million a year (depreciation, financing and maintenance, paid to the robot maker and lenders).

- New wages: US$3 million. New robot costs: US$1.5 million. Other costs: US$2 million.
- New profit: 10 - 3 - 1.5 - 2 = US$3.5 million.
- The firm's labour share of value added falls from 75% to 3 / (3 + 3.5 + 1.5) = 37.5%, counting robot costs as capital income to someone.

- The 75 displaced workers lose US$3 million. The firm's owners gain US$1.5 million. The robot maker's owners and lenders gain US$1.5 million.

If prices fall because of competition, consumers capture some of the gain too, and if demand rises the firm may hire again (the productivity effect). But the first-round effect is clear: income moves from workers to the owners of the firm and the owners of the robot maker. **Who owns those shares is therefore the central question of the automation economy.**

Who owns the capital?

Ownership of financial capital is very concentrated almost everywhere. In the United States, the country with the deepest stock market, the Federal Reserve's Distributional Financial Accounts show the wealthiest 1% of households held about 50.9% of corporate equities and mutual fund shares in the second quarter of 2026, and the top 10% held about 88%; the bottom half held around 1% (Federal Reserve data via FRED; compiled figures, not all fetched directly). In most developing countries, where few households own any shares at all, ownership of firms is even more concentrated in founding families, the state and foreign investors.

Now connect this with the value chain from Month 9. The profits of robots and AI flow to:

- the owners of AI model companies and chip designers (mostly in the US, with China catching up);
- the owners of chokepoint components (Japanese, Chinese and some European firms);
- the owners of robot makers and integrators (increasingly Chinese);
- the owners of the firms that deploy robots and AI (wherever they operate);
- governments, through taxes on all of the above, if they can collect them.

If a factory in Central Java replaces workers with Chinese robots running an American AI model, the wage income that disappears was Indonesian; the capital income that replaces it flows largely to shareholders in China and the United States, unless the factory owner or the state captures it. This is the labour-market version of the course's driving question. Automation is not just a domestic issue of workers against owners. It is also an international question of which countries own the machines.

Superstar firms and winner-take-most markets

Digital technologies have another feature that concentrates income: very low marginal costs. Once an AI model is trained, serving one more customer costs little. This favours the largest firms, which can spread fixed costs over the most customers and gather the most data. David Autor, David Dorn, Lawrence Katz, Christina Patterson and John Van Reenen showed that the fall in the US labour share from the 1980s was driven largely by a shift of activity toward "superstar firms" with high profits and low labour shares (Quarterly Journal of Economics, 2020). The AI industry, with its enormous training costs and network effects (Month 8), looks even more prone to winner-take-most outcomes than earlier digital industries.

The 2025-2026 evidence on jobs: canaries, not an avalanche

What has actually happened to employment since ChatGPT launched in November 2022? The best evidence as of late 2026 points in one direction for the overall economy and another for young workers.

No economy-wide collapse. The Budget Lab at Yale, tracking US data and updating in October

2026, found that "the occupational mix is not yet changing in ways that clearly align with the introduction of AI", that measures of AI usage "show no connection to changes in employment or unemployment", and that a statistical analysis "does not yet clearly indicate an AI-related labor market footprint". The Danish study by Humlum and Vestergaard found the same for earnings and hours.

A widening gap for young workers in exposed jobs. Erik Brynjolfsson, Bharat Chandar and Ruyu Chen use payroll data from ADP, a large US payroll processor. Their August 2026 update reports six facts:

1. No widespread, economy-wide job displacement associated with AI.
2. Employment among workers aged 22 to 25 in highly AI-exposed occupations is about 19% below where it would be if it had kept pace with similarly aged workers in less-exposed occupations. Experienced workers show no comparable gap.
3. The gap has widened steadily: 15% at the July 2025 data and 19% as of June 2026. In levels, employment of 22 to 25 year olds in the two most exposed groups of occupations fell about 11% from November 2022 to June 2026, while it grew about 10% in the least exposed groups.
4. The adjustment works mainly through reduced hiring of young workers, not through firing.
5. Declines are concentrated where AI is used to automate tasks; where it complements workers, employment is flat or rising, especially for experienced workers.
6. So far the adjustment shows up in employment rather than base pay.

They add a new mechanism: employment fell among young workers in occupations relying on **codified knowledge** (formal knowledge from textbooks and written procedures) and rose among experienced workers in occupations relying on **tacit knowledge** (learned through practice and mentorship). AI is good at reproducing knowledge already written down; it is weaker at experience-based judgement. The authors are careful: these are "descriptive patterns, not causal estimates", the gap shrinks when education is accounted for, and some trends began before generative AI.

Goldman Sachs economists reported in October 2026 that workers displaced from AI-exposed jobs since 2025 took about a month longer to find new work than other displaced workers, and were about 10 percentage points more likely to still be unemployed a year later (Goldman Sachs economist Pierfrancesco Mei, as reported by Fortune, October 30, 2026).

Judgment: The evidence as of late 2026 is consistent with the German robot story of Week 2, not with mass unemployment: incumbents are mostly protected, and the adjustment falls on the young, through fewer entry-level openings. This is less visible than layoffs but has long-term consequences. If junior roles are where people acquire tacit knowledge, cutting them may weaken the pipeline of future experts. It also means that the people most affected are exactly your generation.

Global inequality: will automation pull up the ladder?

Month 3 described how countries such as South Korea, Taiwan and China grew rich through labour-intensive manufacturing exports: textiles, then electronics assembly, then more complex goods. Economists worry that automation could weaken this path. Dani Rodrik documented **premature deindustrialisation**: developing countries reaching their peak manufacturing share of employment at much lower income levels than earlier industrialisers, and then seeing that share fall (Journal of Economic Growth, 2016). Robots add a new pressure: if a robot in Germany or the United States can produce a T-shirt or assemble a phone at a cost close to a worker in a low-wage country, the reason to move production to the low-wage country weakens, and "reshoring" becomes viable.

The IFR's 2026 outlook explicitly links robot demand to relocation of manufacturing to high-wage countries (Month 9).

The IMF warns that AI "could worsen inequality among nations", because richer economies have the digital infrastructure and skills to adopt it. Its AI Preparedness Index ranked Singapore, the United States and Denmark highest among 125 countries.

There are counter-forces. Cheap AI tools can spread expertise to poor countries faster than any previous technology (the augmentation path). Cheap robots, if made in China at US$15,000 or less (Month 9), may become affordable for middle-income manufacturers too. And aging is shrinking the workforce of China, Korea, Japan and Europe, which creates labour shortages that automation fills rather than labour surpluses that it worsens. Africa and South Asia, with young and growing populations, face the opposite problem: they need jobs for hundreds of millions of young people. For them, the main question is not whether robots take existing jobs but whether automation abroad closes the route to new ones.

Case study: The missing first rung

Imagine three graduates in 2026: one in the United States, one in the Philippines, one in Indonesia.

The US graduate studied economics and hoped to start as a junior analyst writing reports and building spreadsheets. The ADP data show that employment of 22 to 25 year olds in highly exposed occupations is about 19% below its counterfactual path. Firms are not firing their senior analysts; they are hiring fewer juniors, because a senior analyst with AI tools can produce what used to need a team. The graduate may find work, but later, in a less exposed role, or at a smaller firm.

The Philippine graduate hoped to join a call centre or back-office firm. So far, the industry has kept growing (about 1.68 million workers in 2025 by industry figures, with 1.73 million projected for 2026). But firms are adopting AI assistants that make each agent more productive. Whether employment keeps growing depends on whether demand grows faster than productivity. If clients use AI directly, demand for outsourced human work may fall.

The Indonesian graduate studied engineering and hoped to join a manufacturer. Robot adoption in Southeast Asia is still low, so manufacturing jobs are not being automated quickly. But new factories, often relocated from China, arrive with more automation than older ones, so each new factory creates fewer jobs per dollar invested than factories did twenty years ago.

What the three cases share: **the first rung of the career ladder is where the pressure falls.** That rung has two roles. It provides income, and it provides training. Junior work is where people learn the tacit knowledge that AI cannot easily copy. If firms stop hiring juniors because AI does the junior tasks, they save money now but may face a shortage of experienced people later, and individuals lose the path that turns education into expertise.

What can a young person do? The evidence suggests three practical rules. First, aim for work that combines codified knowledge with tacit, experience-based skills: building things, dealing with people, judging situations, owning outcomes. Second, use AI as an apprentice uses a master: to learn faster, not to avoid learning. Third, look for organisations, including your own start-up, where you get responsibility early, because responsibility is how tacit knowledge is acquired.

Key numbers

Figure	Value	Year	Source
Fall in global labour income share	0.6 percentage points	2019-2022	ILO WESO, Sept 2024
Extra labour income in 2024 if share had stayed at 2004 level	US$2.4 trillion	2024	ILO
Employment gap, US 22-25 year olds in high-AI-exposure jobs	about 19% below counterfactual	June 2026	Brynjolfsson, Chandar and Chen
Same gap one year earlier	15%	July 2025	Brynjolfsson, Chandar and Chen
Change in employment of 22-25 year olds, most vs least exposed jobs	-11% vs +10%	Nov 2022-June 2026	Brynjolfsson, Chandar and Chen
Share of US corporate equities and fund shares held by top 1% / top 10%	about 50.9% / 88%	Q2 2026	Federal Reserve DFA via FRED
AI Preparedness Index top scorers	Singapore, US, Denmark	2024	IMF
Extra time to find work for workers displaced from AI-exposed jobs	about 1 month (and about 10 points more likely to be unemployed a year later)	2025-2026	Goldman Sachs via Fortune

Think like a founder

If automation shifts income from wages to capital, then **ownership design is a social-impact lever available to every founder.** You decide, at the start, who owns your company. Options with real track records include:

- **Employee share ownership:** giving workers equity, so that when automation raises profits, the people whose jobs change share in them.
- **Customer or user ownership (cooperatives):** common in agriculture and finance; a farmers' cooperative that owns its sorting robots captures the robot dividend for its members instead of paying it to an outside owner.
- **Revenue sharing with the people whose expertise trains your AI.** If your AI learns from the best farmers or nurses, a share of revenue can flow back to them, just as the call centre AI captured the best agents' knowledge.

None of these is free: they complicate fundraising, and investors may demand a discount. But they let a founder build a company where automation increases the incomes of the people involved rather than only the owners. Month 11 will show how founders raise money and how such structures interact with venture capital.

Week 4: History, growth theory and the policy toolbox

Core lesson

Three historical analogies, and where they break

People have feared that machines would destroy work for at least two centuries. History is the best evidence we have, but each analogy has limits.

1. The Luddites and Engels' pause. Between 1811 and 1816, textile workers in the English Midlands and North smashed the new knitting frames and power looms that were destroying their skilled trade; the British government sent soldiers and made frame-breaking punishable by death. The usual lesson is that the Luddites were wrong: industrialisation made Britain far richer, and employment grew. The economic historian Robert Allen gives a more careful answer. He documented what he called **"Engels' pause"** (after Friedrich Engels, whose 1845 book *The Condition of the Working Class in England* described the misery of Manchester workers): between 1780 and 1840 British output per worker rose by about 46% while the average real wage rose by only about 12%, and in the core years 1800 to 1830 real wage growth was zero. The extra income went to profits: the profit rate roughly doubled (Allen, Explorations in Economic History, 2009). Only after about 1840 did wages catch up: between 1840 and 1900 output per worker rose about 90% and real wages about 123%. So in the long run the optimists were right, but a whole generation of workers lived and died in the "short run". Lesson: the transition can last a working lifetime, and who captures the gains during it is a political question.

2. The bank teller. When automated teller machines (ATMs) spread in the United States from the 1970s, many expected bank teller jobs to vanish. The economist James Bessen showed in his book *Learning by Doing* (2015) that teller employment did not fall for decades: ATMs lowered the cost of running a branch, so banks opened more branches, and tellers shifted from counting cash to selling and advising. This is the productivity and reinstatement effects at work. Lesson: partial automation of a job often changes it rather than eliminates it. Caution: the later spread of online and mobile banking did reduce teller numbers. Analogies depend on the period you choose.

3. The horse. The economist Wassily Leontief offered a darker analogy in the early 1980s. Horses were essential workers for centuries; each new technology (better carts, better roads) made them more useful. Then the internal combustion engine arrived, and the horse population in the United States, which peaked in the 1910s, collapsed within a few decades. Horses did not "retrain". If machines become better than humans at nearly every task, could humans go the way of horses? The analogy breaks in one important way: humans, unlike horses, own property, vote and set the rules. Horses could not demand a share of tractor profits. Humans can, through ownership, taxes and politics. Lesson: if a technology really does substitute for almost all human labour, then the distribution of income depends almost entirely on the ownership of capital and on political institutions, which is exactly the course's value capture question.

4. Keynes's grandchildren. In 1930 John Maynard Keynes wrote "Economic Possibilities for our Grandchildren". He named "technological unemployment" as a temporary problem of adjustment and predicted that within a century living standards in rich countries would be four to eight times higher, so people might work perhaps fifteen hours a week. The first part came true: incomes in rich countries rose roughly as he predicted. The second did not: people chose more consumption rather than much more leisure, and working hours fell far less. Lesson: even when technology makes a society rich, how the gains are used (more goods, more leisure, more equality) is a social choice.

Deepening: inside Engels' pause, and what it says about AI

The Luddite story is usually told as a morality tale. Allen's paper is more useful than that because it is a working model with numbers, and its mechanism maps closely onto the AI and robot economy of this course. It is worth understanding properly.

The facts. Allen's growth accounting for Great Britain (his Table 1) splits the industrial revolution into four periods:

Period	Growth of output per worker, per year	Growth of real wage, per year
1760-1800	0.26%	0.39%
1800-1830	0.63%	0.00%
1830-1860	1.12%	0.86%
1860-1900	1.03%	1.61%

Source: Robert C. Allen, "Engels' pause", Explorations in Economic History 46(4), 2009, Table 1.

Over the same years labour's share of national income fell from about 60% in the late eighteenth century to around 50% in the middle of the nineteenth, while capital's share more than doubled, from about 20% to over 40%. The gross return on capital rose from near 10% to about 15% in the early nineteenth century and above 20% by mid-century.

A worked example shows what this meant for a worker. Index output per worker and the real wage at 100 in 1780. By 1840 output per worker is about 146 but the wage is about 112. Had labour's share stayed constant, the wage would also have been 146. So by 1840 the average worker received roughly 23% less (112 / 146 is about 0.77) than he or she would have under "equal sharing" of growth, and that gap persisted through a full working life of about forty years.

The mechanism. Allen's explanation has three links, and each one has a modern counterpart.

1. **The new technology was hungry for capital, and capital could not easily replace labour.** Allen estimates that the elasticity of substitution between capital and labour (how easily one can be swapped for the other) was only about 0.2, far below the value of 1 assumed in simple textbook models. Industrialisation needed factories, machines, canals, railways, housing and city infrastructure in roughly fixed proportion to workers. When invention accelerated after 1800, the economy suddenly needed much more capital than it had. Capital became scarce, so its return rose, and with a low elasticity of substitution a higher return meant a bigger share of national income for capital owners.
2. **Profits financed the catch-up.** Workers' household budgets from the period show they saved almost nothing; savings came from capitalists and landlords. Allen estimates that capitalists saved about 20% of profits (and landlords about 14% of rents). As profits rose, total investment rose too: the investment rate climbed from about 6% of national income in 1760 to about 11% in the 1830s and 1840s. Inequality was, in his words, "intrinsic to the growth process": the rising profit share was what paid for the capital the new technology needed.
3. **Population growth stretched the capital further.** Britain's population was booming, so each new worker also needed new capital. When Allen re-runs his model without the population boom, the pause almost disappears. When capital accumulation finally caught up with technology, around the 1840s, wages began to rise in line with productivity, and the "modern" pattern of shared growth began.

Allen's numbers have critics. Gregory Clark argued that real wages grew faster and output more slowly than the standard estimates, which would make workers the main winners. Allen replies, in an appendix to the same paper, that even Clark's own wage series shows a pause, only a shorter and milder one. The existence of a lag between productivity and wages is widely accepted; its size remains debated.

What carries over to AI and robots. Now apply the three links to today.

- *Capital hunger.* Frontier AI is extraordinarily capital-hungry: chips, data centres, power plants and grids (Month 8), and, for physical AI, robot factories and precision components (Month 9). If this capital is scarce relative to what the new technology can use, its owners earn high returns, just as British factory owners did. This is the economic meaning of the chokepoints in Month 9: they are where scarcity returns pile up during a pause.
- *Who saves and owns.* In 1800 almost all capital income went to a small class. Today, as Week 3 showed, ownership of company shares in the United States is also highly concentrated. If AI raises the capital share, the gains flow mainly to that concentrated group, unless ownership is broadened (pension funds, sovereign funds, employee shares, the Alaska-style dividend in this week's case study).
- *How fast capital catches up.* Modern capital markets are global and deep, so in principle accumulation can catch up much faster than in 1830. But physical bottlenecks (chip fabs, transformers, reducers, rare earth magnets, permitting) slow it down, and every bottleneck extends the period during which owners of scarce capital capture the gains.

There is one crucial difference. In Allen's model, workers eventually won because capital and labour were **complements**: once there was enough capital, more of it made each worker more productive, so wages rose. If advanced AI and robots become close **substitutes** for human labour across most tasks (the high-substitution scenario of Korinek and Suh below), capital accumulation would not end the pause; it would deepen it. Whether AI behaves more like the steam engine (a complement that needed workers) or more like the tractor did for the horse is the single most important unknown in this month.

Judgment: The most likely near-term pattern is a partial Engels' pause: productivity gains from AI arrive faster than wage gains for many workers, particularly young workers in codified-knowledge jobs (the "canaries" of Week 3), while returns concentrate in owners of compute, chips, energy and key robot components. Unlike in 1830, the length and severity of the pause is something policy can shorten, by widening ownership of the new capital and by removing physical bottlenecks to building it, rather than something societies must simply wait out. For a founder in Southeast Asia or Africa, the practical lesson is that during a pause it is better to own a piece of the scarce capital or a hard-to-copy complementary skill than to sell labour that the new capital substitutes for.

What growth theory says about a world with very capable AI

Standard growth models assume that some tasks always need human labour. If that assumption fails, the theory changes. Anton Korinek and Donghyun Suh modelled scenarios for a transition to artificial general intelligence (NBER Working Paper 32255, 2024). Their key variable is whether there is an endless supply of tasks too complex for machines, or whether machines eventually can do everything humans can.

- If the space of tasks is unbounded and humans keep finding new things machines cannot do, wages keep rising with productivity, as in the past.

- If AI and robots can eventually perform all tasks, wages can rise at first (while humans remain the bottleneck in some tasks) and then fall, potentially sharply, once machines can do the remaining tasks more cheaply than the cost of keeping a human. In that world, output can grow very fast while labour's share of income heads toward zero.

You do not need to believe the second scenario is likely to see its value as a stress test. It shows that the long-run outcome for workers is not decided by technology alone, but by three things: whether new human tasks keep appearing, how fast machines can be built (Month 9's factory and reducer bottlenecks matter here), and who owns the machines.

The policy toolbox

Governments have several tools. Each has evidence, costs and limits.

1. Education, training and mobility. The IMF recommends "comprehensive social safety nets" and "retraining programs for vulnerable workers". Retraining works best when it is close to real jobs (apprenticeships, employer-linked courses), as Germany's system showed in the robot case. Generic programmes for displaced middle-aged workers have a mixed record. Education policy also has to change what it teaches: if AI handles codified knowledge, schools must build judgement, practical skills and the ability to learn quickly.

2. Social insurance. Unemployment insurance, wage insurance (topping up the pay of workers who take lower-paid new jobs) and portable benefits reduce the private cost of displacement. They matter most in countries where losing a job means losing health care or falling into poverty, and least developed in countries with large informal sectors, where most workers have no insurance at all.

3. Fixing tax biases toward automation. In many countries labour is taxed more heavily than machines: wages carry income and payroll taxes, while equipment and software receive depreciation allowances and investment credits. Acemoglu, Andrea Manera and Restrepo argued that the US tax code favours automation, encouraging "so-so" technologies that would not be profitable without the tax advantage (Brookings Papers on Economic Activity, 2020). They estimated that moving from the US tax system of the 2010s to an optimal taxation of capital and labour would raise employment by about 4% and the labour share by about 0.8 percentage points. Removing that bias is less dramatic than a "robot tax" but has broad support among economists.

4. Robot taxes. In 2017 South Korea, then and now the most robot-dense economy, reduced the corporate tax credit for investment in productivity-raising automation: from 3% to 1% for large firms and from 5% to 3% for mid-sized firms, keeping 7% for small firms (Seoul Economic Daily and Korea Times, August 2017). The press called it the world's first "robot tax", though it was a smaller subsidy, not a tax. The European Parliament rejected a proposal for a robot tax in 2017, citing risks to innovation. Problems with robot taxes are practical: defining a "robot" (is a dishwasher a robot? is an AI model?), avoiding a brake on productivity, and stopping firms from moving investment abroad. A later study summarised by the Windfall Trust policy atlas (Kang, Lee and Quach, 2024) reports that the Korean reform reduced automation investment and raised employment; treat that as a single study, not settled evidence.

5. Cash transfers and basic income. A universal basic income (UBI) pays every citizen a regular amount with no conditions. Critics worry it reduces work. Evidence from experiments is more reassuring than critics expect, though no experiment has tested a permanent national UBI in a rich country. In Kenya, a long-term experiment run with the charity GiveDirectly found no reduction in work; recipients shifted from wage work toward running their own small businesses

(Banerjee, Faye, Krueger, Niehaus and Suri, 2023; summary not fully verified). Finland's 2017-2018 basic income experiment found small employment effects and better wellbeing (Finnish government, 2020; summary not fully verified). The biggest obstacle is cost: a meaningful UBI in a rich country costs a large share of GDP, which is why it is often discussed together with taxes on the capital income that automation creates.

6. Shared ownership of capital: universal basic capital and sovereign funds. If the problem is that automation shifts income from labour to capital, one direct answer is to spread the ownership of capital. Ideas include "baby bonds" that give every child an investment account, employee ownership, and public or sovereign wealth funds that hold stakes in firms and pay dividends to citizens. Proposals for an "AI sovereign wealth fund" appeared in the United States in 2026 (for example, a bill summary from Senator Bernie Sanders' office; reported, not verified in detail). Month 11 examines sovereign wealth funds (Norway, Saudi Arabia's PIF, Singapore, Indonesia's Danantara) in depth. The case study below shows the simplest working example.

7. Worker voice and co-determination. Germany's works councils and board-level employee representation shaped how robots were adopted (Week 2). Giving workers a formal say in how technology is introduced tends to steer firms toward redeployment and augmentation rather than pure displacement, at some cost in speed.

Judgment: No single tool is enough. The evidence of the last decade suggests a combination: remove tax biases that subsidise so-so automation; invest heavily in training linked to real jobs; strengthen social insurance, especially for young entrants who bear most of the adjustment; and, as AI and robots spread, build mechanisms for broad ownership of the capital that earns the new income. Taxing robots directly is the weakest option, because it is hard to define and slows the productivity gains that make everything else affordable. Spreading ownership is the most powerful and the least tested.

Case study: From oil dividend to robot dividend, Alaska

Resource economics (Months 2, 5 and 6) and robot economics turn out to have the same structure. In both, a valuable asset (an oil field, or a fleet of robots and AI models) produces large income with few workers. The question in both is who receives that income: a few owners, the state, or every citizen.

In 1976 the US state of Alaska, newly rich from oil in Prudhoe Bay, amended its constitution to create the **Alaska Permanent Fund**, which saves a share of oil revenues and invests them in a diversified portfolio of stocks, bonds and other assets. Since 1982 the fund has paid an annual **Permanent Fund Dividend** to almost every resident, adult or child, with no conditions. It is the longest-running universal cash transfer in a rich economy.

Three lessons follow.

It did not reduce work. Damon Jones and Ioana Marinescu studied the dividend's effect on the labour market and found no reduction in overall employment; part-time work rose somewhat, which they attribute to extra local spending creating jobs (American Economic Journal: Economic Policy, 2022). This is the strongest real-world evidence that a universal dividend need not make people stop working.

It turned a resource rent into citizen capital. The money did not go only to oil companies and politicians; part of it became a permanent, diversified fund owned on behalf of all Alaskans.

Norway did the same on a much larger scale with its Government Pension Fund Global (Month 11). Contrast this with the "resource curse" countries of Month 2, where oil rents fed elites and conflict.

It is politically durable but politically contested. Because every resident receives the dividend, it is very popular; but its size is fought over every year in the state budget, especially when oil prices fall. Shared ownership needs strong institutions to survive.

Now apply the lesson to robots. Imagine a country in 2040 where robot fleets and AI systems produce a large share of output. A "robot dividend" could be funded by a public fund that owns stakes in domestic robot and AI firms, by a share of taxes on capital income or by royalties on public data and spectrum, and paid to every citizen, as Alaska does with oil. The mechanism is not science fiction; it already exists for a natural resource. The open questions are the ones from Month 2: will institutions be inclusive enough to share the rent, or extractive enough to capture it?

Key numbers

Figure	Value	Year	Source
Luddite machine-breaking	1811-1816	historical	standard histories
Engels' pause: British output per worker vs real wage	+46% vs +12%	1780-1840	Allen, Explorations in Economic History, 2009
Capital's share of British national income	about 20% to over 40%	late 1700s to mid-1800s	Allen, 2009
Keynes's forecast of living standards in 100 years	4 to 8 times higher	1930	Keynes
South Korea automation tax credit, large firms	cut from 3% to 1%	2017	Seoul Economic Daily; Korea Times
South Korea automation tax credit, mid-sized firms	cut from 5% to 3% (small firms kept 7%)	2017	Seoul Economic Daily
Alaska Permanent Fund Dividend	paid annually since 1982	1982-	Alaska; Jones and Marinescu
Alaska dividend effect on total employment	no reduction	2022 study	Jones and Marinescu, AEJ: Policy
IMF policy priority for emerging and developing economies	digital infrastructure and digitally competent workforce	2024	IMF

Think like a founder

Policy is not only for governments. Founders design small versions of these policies every day: how much to train staff, whether to share equity, whether to automate a job or redesign it. Three practical habits follow from this month.

1. **Run the three-effect test on every automation decision.** For each AI or robot feature you build, estimate the displacement effect (which tasks disappear), the productivity effect (how much costs fall and whether demand will grow) and the reinstatement effect (what new tasks it creates). If displacement is large and the other two are small, you are probably building "so-so" automation: profitable now, fragile later.
2. **Design for the first rung.** If your company uses AI to avoid hiring juniors, you save money but stop creating future experts. Consider structured apprenticeships where juniors learn with AI rather than being replaced by it. This is good for society and often good for the firm, because experienced staff are scarce.
3. **Think in dividends.** If your business earns a rent from a scarce asset (data, a model, a robot fleet, a platform), decide early who shares in it: employees, users, the communities whose data or labour trained your system. Founders who build shared ownership into their structure are building the private-sector version of Alaska's fund.

Month synthesis

This month followed the income produced by machines to the people who receive it. The core logic fits in one diagram:

```
NEW AI / ROBOT CAPABILITY
        |
        v
  Which tasks?  -->  DISPLACEMENT (workers lose tasks)
        |         -->  PRODUCTIVITY (costs fall, demand rises?)
        |         -->  REINSTATEMENT (new human tasks?)
        v
  NET LABOUR DEMAND  (depends on demand elasticity, institutions, speed)
        |
        v
  INCOME SPLIT:  wages  <---->  capital income
                                      |
                        who owns the capital?
         domestic workers / founders / state funds / foreign shareholders
        |
        v
  POLICY & INSTITUTIONS: training, insurance, tax design, ownership, worker voice
```

Two uncertainties decide which world we get. This 2x2 table is a simplified version of the scenario framework that Month 12 develops:

	Ownership concentrated	Ownership broad
Gradual AI and robotics ("diffusion")	Slow productivity gains; rising inequality; young workers squeezed; politics sour	Modest gains widely shared; Nordic-style adjustment; training and insurance do most of the work

	Ownership concentrated	Ownership broad
Rapid, broad automation ("takeoff")	Very high output, falling labour share, income flows to owners of models, chips and robot fleets; high risk of political backlash	Very high output shared through funds, dividends and universal capital; work becomes more optional

How this links to the whole course:

Layer of the chain	Who captures value (Months 5-9)	What it means for workers (Month 10)
Ore and refining	Mining states, refiners (China)	Few jobs; rents shared only if institutions are inclusive
Components and robots	Japan, China, chokepoint firms	Skilled manufacturing jobs in a few clusters
Chips and AI models	US, Taiwan, China	Very few, very well-paid jobs; large capital income
Deployment	Firms everywhere	Displacement and new tasks; outcome set by institutions
Ownership	Shareholders, states, funds	Decides whether automation raises or lowers most people's incomes

The month's main conclusion: technology determines what is possible; institutions and ownership determine who benefits. The same robots produced job losses in American towns and reallocation in German ones. The same oil produced a curse in some countries and a citizen dividend in Alaska and Norway. AI and robots are the next asset class whose rents must be divided. Month 11 turns to the financial system that will decide how: money, capital flows, sovereign funds and venture capital.

Common misconceptions

1. **"Technology always creates as many jobs as it destroys, so there is nothing to worry about."** Over long periods new work has appeared (most US jobs today are in job types that did not exist in 1940), but transitions can last decades (Engels' pause), the new jobs may go to different people and places, and since the 1980s automation's job-destroying effects have grown while augmentation's job-creating effects have not (Autor and co-authors).
2. **"If 40% of jobs are exposed to AI, 40% of jobs will disappear."** Exposure means AI can affect a substantial share of tasks. The IMF itself expects roughly half of exposed jobs in advanced economies to benefit. Measured effects through 2026 are small in aggregate and concentrated among young workers.
3. **"AI will hit low-skilled workers hardest, like earlier automation."** This wave is different: exposure is highest in clerical, professional and higher-paid office work, and among young graduates entering codified-knowledge occupations. Manual work is more exposed to robots, which spread more slowly.

4. **"A robot tax is a simple way to protect jobs."** Robots are hard to define, taxes can slow useful productivity growth, and investment can move abroad. South Korea's "robot tax" was in fact a smaller subsidy. Removing tax biases that favour machines over workers is a more defensible step.
5. **"Basic income makes people stop working."** Evidence from Alaska, Kenya and Finland shows small or no reductions in employment, though none of these tested a full national UBI in a rich country, and cost remains the main obstacle.
6. **"Poor countries are safe because their jobs are less exposed to AI."** Lower exposure partly reflects that their workers are in agriculture and informal work. The bigger risk is indirect: automation in rich countries can make reshoring viable and weaken the export-manufacturing route to development.

Exercises

1. **Growth accounting (quantitative).** An economy's output grows 2.5% in a year. Capital grows 3% and has an income share of 35%; labour hours grow 0.5%. Calculate TFP growth. Then suppose AI adds 0.3 percentage points to TFP growth each year for ten years. By how much is output higher after ten years than without AI?
2. **Hulten's theorem (quantitative).** Using the Week 1 method, estimate the 10-year TFP gain if 30% of tasks are exposed, 40% of exposed tasks are profitably automated, and cost savings are 25%. Compare your result with Acemoglu's 0.66% and Goldman Sachs's 7% of GDP. Which assumption explains most of the difference?
3. **The labour share (quantitative).** Rework the Week 3 logistics example assuming that competition forces the firm to cut prices so that revenue falls to US$9 million. Recalculate profit and the labour share. Who captures the gain from automation now?
4. **Demand elasticity (quantitative).** A call centre gets a 25% productivity boost. If demand for its service rises 10% because of better, cheaper service, how does employment change? If demand rises 40%? At what increase in demand does employment stay constant?
5. **Essay (700 words).** Why did robots reduce employment in US local labour markets but not total employment in Germany? Use at least three mechanisms and say which one you find most convincing, and why.
6. **Essay (800 words).** "The horse analogy is wrong because humans own capital and vote." Evaluate this claim. Under what conditions would it fail?
7. **Policy design (essay, 1,000 words).** Design a "robot dividend" for a middle-income country of your choice by 2040. Where would the money come from (which taxes, which public stakes)? How would you protect it from political capture? Use the Alaska case and Month 2's ideas on institutions.
8. **Founder exercise.** Pick one job in your city that AI or robots could affect within ten years. List its tasks, mark each as likely automated, augmented or unaffected, and design a product that augments the worker rather than replacing them. Estimate who captures the value.

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Economic Policy, 2025). The sceptical benchmark for AI's growth effect; read it to learn Hulten's theorem in action.
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- International Labour Organization, "Generative AI and Jobs: A Refined Global Index of Occupational Exposure", ILO Working Paper 140 (2025). A global, gender-aware view of exposure beyond rich countries.
- John Maynard Keynes, "Economic Possibilities for our Grandchildren" (1930). A short, classic essay; compare his predictions with what actually happened.
- Thomas Piketty, *Capital in the Twenty-First Century* (2014). The long history of capital and inequality; useful background for why ownership matters.

Glossary terms introduced this month

- **Augmentation:** use of technology that makes workers more productive in a task rather than replacing them.
- **Automation:** use of technology to perform a task previously done by a person.
- **Baumol's cost disease:** the tendency for prices of low-productivity-growth services to rise relative to high-productivity-growth goods.
- **Codified knowledge:** formal knowledge written in documents and textbooks; easier for AI to reproduce.
- **Displacement effect:** the fall in labour demand when machines take over tasks.
- **Engels' pause:** the period in early industrial Britain when output per worker rose but real wages stagnated.
- **Exposure (to AI):** the share of a job's tasks that AI could perform or speed up significantly; not a forecast of job loss.
- **General-purpose technology:** a technology that spreads across the economy, keeps improving and spawns complementary innovations.
- **Hulten's theorem:** the aggregate productivity gain from a cost-saving technology roughly equals the share of activity affected times the cost saving.
- **Labour share:** the share of national income paid to workers.
- **Premature deindustrialisation:** developing countries reaching peak manufacturing employment at lower incomes than earlier industrialisers.
- **Productivity effect:** the rise in labour demand from lower costs and higher incomes after automation.

- **Productivity J-curve:** the pattern where measured productivity dips during early investment in a new technology and rises later.
- **Reinstatement effect:** the rise in labour demand from new tasks created by technology.
- **Robot tax:** any tax or reduced subsidy aimed at the use of robots or automation.
- **So-so automation:** automation that displaces workers but barely raises productivity.
- **Superstar firm:** a highly profitable, dominant firm with a low labour share.
- **Tacit knowledge:** knowledge learned through experience and practice; harder for AI to reproduce.
- **Total factor productivity (TFP):** output growth not explained by more labour or capital; captures technology and efficiency.
- **Universal basic capital:** giving every citizen ownership of a share of productive capital.
- **Universal basic income (UBI):** a regular, unconditional cash payment to every citizen.

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Month 11: Money, Finance and Sovereign Capital

Month overview

For ten months we have followed physical things: electrons, ore, molten metal, wafers, motors and the people who work with them. This month we follow the thing that decides which of those physical projects get built at all: money. A nickel smelter, a chip fab, a robot factory and a data center have one thing in common. Each needs a very large amount of money up front, years before it earns anything back. Whoever supplies that money, at what interest rate, in which currency, and with what conditions, captures a large share of the value that the project later creates.

This is why finance sits so close to the centre of our driving question: who captures the value in the chain from ore to robot to AI? A country can own the richest nickel deposit on Earth (Month 5) and still lose most of the value if it must borrow in dollars at 10% while a competitor borrows at 3%. A founder can invent a better robot gripper (Month 9) and still lose control of the company if she raises money on bad terms.

The month has four weeks. Week 1 explains the dollar system: why one country's currency is the money of the whole world, and what that does to everyone else. Week 2 explains interest rates and capital flows: why money rushes into and out of countries like Indonesia, and why that creates crises. Week 3 explains debt: governments, companies, development banks and Chinese lenders, and what happens when borrowers cannot pay. Week 4 explains the two kinds of long-term risk capital that matter most for building new industries: sovereign wealth funds (state investors such as Norway's oil fund, Saudi Arabia's PIF and Indonesia's Danantara) and venture capital (the money that funds startups).

This month connects backward to Month 2 (institutions decide whether savings are invested well or stolen), Month 4 (oil money creates sovereign funds and petrodollars) and Month 6 (downstreaming needs cheap long-term capital). It connects forward to Month 12, where finance becomes a weapon of geopolitics: sanctions, frozen reserves, export controls and rival payment systems.

A note on timing. "Today" in this book is November 2026. The world in which you read this month is unusual: a war between the United States and Iran since February 2026 has pushed oil above $100 a barrel, the US Federal Reserve raised interest rates in September 2026 for the first time since 2023, the 10-year US government bond yield is above 5%, and Indonesia's stock market faces a possible expulsion from the main emerging-market indexes. These are not distractions from the lesson. They are the lesson, happening live.

Learning goals

By the end of this month you should be able to:

- Explain why the US dollar dominates global reserves, trade and finance, using the ideas of network effects, safe assets and the "exorbitant privilege", and judge the real evidence on "de-dollarization".
- Explain how a change in US interest rates travels to Jakarta, Lagos or Sao Paulo through capital flows, exchange rates and borrowing costs, and use the "impossible trinity" to explain what a central bank like Bank Indonesia can and cannot do.
- Calculate how interest rates and growth rates decide whether a government's debt rises or falls, and explain why developing countries pay far more to borrow than rich ones.
- Describe the main types of sovereign wealth fund, compare Norway, Saudi Arabia, Abu Dhabi, Singapore and Indonesia, and explain what makes a state investor create value or destroy it.
- Explain how venture capital works (fund structure, power law, dilution) and lay out the realistic funding paths for a founder in Southeast Asia or Africa.
- Show, with a worked example, how the cost of capital decides who captures value in a capital-heavy industry such as smelting, chipmaking or robot manufacturing.

Week 1: The dollar system

Core lesson

What money does, and why the world uses one country's money

Economists say money has three jobs. It is a **medium of exchange** (you pay with it), a **unit of account** (you measure prices in it) and a **store of value** (you save in it). Inside Indonesia, the rupiah does all three jobs for daily life. But when an Indonesian nickel company sells to a Chinese battery maker, when Pertamina buys crude oil from Saudi Arabia, or when Bank Indonesia holds savings for a rainy day, the money used is usually the US dollar.

The numbers show how strong this dominance is:

- **Foreign exchange (FX) trading.** Each day in April 2025 about $9.6 trillion of currencies were traded worldwide, and the dollar was on one side of 89% of all trades, according to the Bank for International Settlements (BIS) Triennial Survey. The euro was on one side of about 29%, the yen 17%, and China's renminbi 8.5%. (Each trade has two currencies, so shares add up to 200%.)
- **Central bank reserves.** The International Monetary Fund (IMF) reports that official foreign exchange reserves totalled $13.22 trillion in the second quarter of 2026. Of the reserves where the currency is reported, 56.7% were in dollars, 20.6% in euros and only 2.1% in renminbi.
- **Trade invoicing.** Roughly half of world trade is invoiced in dollars, even though the United States makes only about a tenth of world exports. Commodities (oil, copper, nickel on the London Metal Exchange) are almost all priced in dollars.

Why would 190-plus countries use the money of one of them? There are four reasons, and they reinforce each other.

1. Network effects. A currency is like a language or a social network: it is useful because others use it. If every bank in Jakarta, Lagos and Lima can trade rupiah, naira and soles against the dollar, but not directly against each other, then the dollar becomes the bridge. A Nigerian importer buying

Indonesian palm oil may sell naira for dollars and then dollars for rupiah, because the naira-rupiah market is tiny and expensive. This "vehicle currency" role means trading through the dollar is cheaper, which makes more people trade through the dollar, which makes it cheaper still.

2. Deep, safe, liquid markets. A central bank holding reserves needs to put them somewhere safe that it can sell quickly in a crisis without moving the price. The US Treasury market (US government debt) is the largest bond market in the world. Debt held by the public was about $32.4 trillion in October 2026 (US Joint Economic Committee). No other safe asset comes close in size. The euro area has no single government bond; it has separate German, French and Italian bonds with different risks. China's bond market is large but has capital controls, so foreigners cannot be sure they can take their money out.

3. Institutions and law. The dollar is backed by a large, rich economy, an independent central bank (the Federal Reserve, or "Fed"), courts that enforce contracts, and New York law, under which much international debt is written. Month 2 argued that inclusive institutions make nations rich. The same logic applies to currencies: investors trust a currency when they trust the institutions behind it.

4. History and inertia. After the Second World War, the Bretton Woods agreement (1944) tied other currencies to the dollar and the dollar to gold at $35 per ounce. When the United States ended gold convertibility in August 1971, many expected the dollar's role to collapse. It did not, partly because there was no alternative, and partly because of the "petrodollar" arrangements of the 1970s, in which oil was priced in dollars and Gulf exporters recycled their surpluses into US assets. Once a system is built, switching is costly.

The exorbitant privilege and its price

In the 1960s, French finance minister Valery Giscard d'Estaing called the dollar's position America's "exorbitant privilege". What does the United States get?

- **Cheaper borrowing.** Because the world needs safe dollar assets, foreigners buy US government debt even at relatively low interest rates. Research generally finds this lowers US borrowing costs, though estimates vary.
- **Borrowing in its own currency.** The US government and US companies borrow mostly in dollars, so a fall in the dollar does not increase their debt. Most developing countries cannot do this, a problem economists Barry Eichengreen and Ricardo Hausmann named **"original sin"**: the inability to borrow abroad in your own currency.
- **Financial power.** Almost all dollar payments eventually pass through US banks. This gives the US government the ability to cut people, companies or countries out of the dollar system. That is the basis of financial sanctions (Week 1 case study, and Month 12).

There is also a price. To supply the world with dollar assets, the United States must, over time, sell more assets to foreigners than it buys, which tends to mean running trade deficits. The Belgian-American economist Robert Triffin pointed out in 1960 that the country issuing the world's reserve currency faces a dilemma: if it stops supplying dollars, the world economy is starved of liquidity; if it supplies too many, confidence in the currency eventually weakens. This **Triffin dilemma** helps explain why American politicians in 2025-2026 complained both that the strong dollar hurt US factories and that foreigners might stop buying US debt.

Is the dollar declining? The evidence on "de-dollarization"

You will read many claims that the dollar is collapsing. Be careful. Look at what the data show.

The dollar's share of global reserves has fallen slowly, from about 71% in 2000 to about 57% in 2026. The Federal Reserve Bank of New York reported in September 2026 that the share fell from 64% in 2015 to 56% in 2025, but that this aggregate decline mostly reflects the choices of "a handful of large reserve holders", not a broad global move away from the dollar. Meanwhile, the dollar's share of FX trading actually rose slightly, from 88.3% in 2019 to 89.2% in 2025 (BIS data, published by Banco de Mexico).

Some reserve managers have moved into gold. In 2025, gold overtook US Treasuries as a share of total official reserves, but the IMF noted this was "driven almost entirely by gold price valuation effects": gold's price rose, so the gold already held was worth more. It was not mainly new buying.

A subtle point about reserve shares: the IMF measures reserves in dollars. If the euro rises 10% against the dollar, euro reserves are suddenly "worth" more dollars, and the euro's share rises even if no one bought a single euro. So a falling dollar share sometimes means only a falling dollar price. In the second quarter of 2026, however, the IMF said exchange-rate effects were small, and the euro's share rose to 20.6% "due mostly to active purchasing". That is a real, if modest, shift.

The renminbi is the most-discussed challenger. Its reserve share was about 2.1% in mid-2026, below its peak in 2022. China does use renminbi for more of its own trade and lending: Boston University found that in 2024 all of Kenya's new Chinese infrastructure loans were denominated in renminbi, unlike the dollar loans of the 2010s. But a currency cannot become a global reserve while its government controls who may move money in and out.

Judgment: the dollar's dominance is eroding at the edges, mainly in reserves and in trade between countries that fear US sanctions, but its core roles as the vehicle currency of trading and the main safe asset are intact in 2026. The more likely future through the 2030s is a dollar that is still first but less overwhelming, with the euro, gold and regional payment links (Month 12, Week 2 covers BRICS payment plans) gaining at the margins. For a founder, the practical rule is simple: if your inputs or customers are priced in dollars, your business is a dollar business, whatever currency your bank account uses.

A new dollar channel: stablecoins

A **stablecoin** is a digital token on a blockchain designed to stay worth exactly one dollar. The issuer holds safe dollar assets (cash, short-term US Treasury bills) as backing. By late October 2026, the total stablecoin market was about $305 billion, with Tether's USDT at about $183 billion and Circle's USDC at about $72 billion (CoinDesk, citing on-chain data).

On July 18, 2025 the United States enacted the GENIUS Act (Guiding and Establishing National Innovation for U.S. Stablecoins Act), its first federal law for payment stablecoins. It allows only approved ("permitted") issuers to issue payment stablecoins in the United States and requires them to hold reserves one-for-one in safe assets, such as cash, bank deposits and Treasury bills maturing within 93 days. The law itself takes effect on January 18, 2027. On October 21, 2026 the US bank regulator, the Office of the Comptroller of the Currency (OCC), became the first agency to publish its final implementing rule, covering reserves, redemption (generally within two business days), capital, custody and supervision, also effective January 18, 2027. Rules on anti-money-laundering and sanctions are being written separately with the Treasury.

Why does this matter for our story? Two reasons. First, stablecoins spread the dollar to places banks do not reach: a freelancer in Lagos or Semarang can hold "digital dollars" on a phone. This deepens dollarization in emerging markets, which can weaken local central banks' control. Second, stablecoin issuers have become large buyers of US Treasury bills. The San Francisco Fed reported in September 2026 that the two largest issuers' holdings of Treasury securities had grown more than tenfold in five years. Who captures value here? The issuer earns the interest on the reserves (around 3.8% in late 2026 on government money-market funds) while holders usually earn nothing. Stablecoins are a fresh example of the old exorbitant privilege, now partly privatized.

Case study: Russia 2022, when reserves stop being yours

In February 2022, after Russia invaded Ukraine, the United States, the European Union, the United Kingdom, Japan and others froze a large part of the Russian central bank's foreign reserves, about $300 billion according to the US Congressional Research Service and Russia's own finance ministry, roughly half of its gold and foreign-exchange reserves. These were not loans; they were Russia's own savings, held as deposits and bonds in Western financial systems. Overnight, Russia could not use them to defend the rouble or pay for imports.

This event changed how central banks think about reserves. Reserves are meant to be the safest money a country has. Russia learned that "safe" depends on politics: an asset held inside another country's legal system is only as safe as your relationship with that country. Several consequences followed:

- **Gold buying rose.** Central banks, especially in emerging markets, bought more gold, which no foreign government can freeze if it is stored at home. The World Gold Council's 2026 survey found that 74% of central bank respondents expected the dollar's reserve share to be lower in five years.
- **Payment workarounds grew.** Russia and China expanded trade in renminbi and roubles, and China's Cross-Border Interbank Payment System (CIPS) grew. India bought Russian oil and paid partly in other currencies.
- **But the dollar did not collapse.** As shown above, the dollar's share of FX trading rose between 2022 and 2025. Most countries concluded that the risk of being frozen applies mainly to countries in open conflict with the West, and that the dollar's advantages still outweigh the risk.

There is also a cost to the West. Every use of financial sanctions gives others a reason to build alternatives. This is a slow-moving trade-off: sanctions are powerful today partly because the dollar is dominant, and using them repeatedly may reduce that dominance tomorrow.

What this teaches about value capture: the issuer of the reserve currency captures not just money (cheaper borrowing) but power (the ability to exclude). For countries building ore-to-robot supply chains, this creates a design question. A country that sells nickel to China in renminbi and buys machinery from Germany in euros and pays for chips in dollars is exposed to three currency systems and three sets of politics. Diversifying payment channels is a form of insurance, but it has costs in liquidity and fees.

Key numbers

Figure	Value	Year	Source
Daily global FX turnover	$9.6 trillion	April 2025	BIS Triennial Survey
Dollar's share of FX trades (one side)	89%	April 2025	BIS
Total official FX reserves	$13.22 trillion	Q2 2026	IMF COFER
Dollar share of allocated reserves	56.7%	Q2 2026	IMF COFER
Euro share of allocated reserves	20.6%	Q2 2026	IMF COFER
Renminbi share of allocated reserves	2.1%	Q2 2026	IMF COFER
Dollar share of reserves, 2015 vs 2025	64% to 56%	2015-2025	Federal Reserve Bank of New York
Stablecoin market value	about $305 billion	October 2026	CoinDesk
US federal debt held by the public	$32.43 trillion	October 5, 2026	US Joint Economic Committee

Think like a founder

- **Price in the currency of your costs.** If your robot uses Japanese reducers and Chinese motors priced in dollars, and you sell in rupiah or naira, a 15% currency fall can wipe out your margin. Many hardware startups in emerging markets die from currency mismatch, not bad products. Solutions: dollar-linked pricing for business customers, local sourcing of parts, or hedging contracts (where available and affordable).
- **Cross-border payments are still expensive.** Moving money between emerging-market currencies often costs several percent and takes days, because it goes through dollar correspondent banks. This friction is a business opportunity, which is why fintech companies in payments (and now regulated stablecoin rails) attract investment. But it is a heavily regulated area: compliance, licensing and trust are the real moat, not the technology.
- **The value capture lesson:** in the money layer, the biggest value goes to whoever owns the network everyone must pass through: the reserve currency, the payment rail, the stablecoin float. A founder rarely builds such a network, but can build the tools that sit on top of it for underserved users.

Week 2: Interest rates, capital flows and crises

Core lesson

The interest rate is the price of time

An interest rate is the price of using someone else's money for a period of time. If you borrow 100 today and repay 106 in a year, the price of one year was 6%. Every investment decision in this textbook (a mine, a smelter, a fab, a robot factory) compares money spent now with money earned later. The interest rate is the exchange rate between "now" and "later".

The most important interest rate in the world is the **federal funds rate**, the rate at which US banks lend to each other overnight, which the Federal Reserve sets as a target range. On October 28, 2026, the Fed kept that range at 3.75% to 4.00%, by a 10-2 vote. Both dissenters wanted to raise it further, and the statement said "inflation remains elevated". This followed a rate rise in September 2026, the first since July 2023, as the oil shock from the Iran war pushed up prices.

Longer-term rates matter even more for big projects. The yield on the 10-year US Treasury bond touched 5.34% on October 1, 2026, its highest level since 2002, before closing at about 5.24% (Reuters; CNBC). This yield is the "risk-free" benchmark against which almost every long-term investment in the world is priced. When it rises, every project's hurdle gets higher.

How a Fed decision reaches Semarang

Suppose the Fed raises rates. Here is the chain of cause and effect, step by step:

1. **Dollar assets pay more.** A global investor can now earn more by holding US Treasury bills, with almost no risk.
2. **Money leaves riskier places.** Some investors sell Indonesian government bonds (known as SBN) or Indonesian stocks and move the money into dollars. This is a **capital outflow**.
3. **The rupiah weakens.** To sell rupiah assets and buy dollars, investors sell rupiah. More sellers than buyers means the rupiah's price falls. On November 2, 2026, Bank Indonesia's reference rate (JISDOR) was Rp18,161 per dollar, the weak end of its recent range.
4. **Imports and dollar debts get more expensive.** Fuel, wheat, machinery and chips, all priced in dollars, cost more rupiah. Companies with dollar debts must find more rupiah to repay them.
5. **The central bank responds.** Bank Indonesia (BI) can raise its own rate to make rupiah assets more attractive, sell some of its dollar reserves to support the rupiah, or use other tools. BI raised its policy rate (the BI-Rate) by a total of 100 basis points (1 percentage point) in three steps: from 4.75% to 5.25% on May 20, 2026, to 5.50% on June 9, and to 5.75% on June 18. It then held the rate at 5.75% at its July, August, September and October 2026 meetings. At the October 20-21 meeting it raised its deposit facility rate to 5.00%, cut banks' reserve requirements in stages, and made its currency-hedging instruments cheaper for foreign investors. BI says these hedging incentives attracted about $9.9 billion of foreign inflows between June and October 20, 2026, and its reserves stood at $146.3 billion at the end of September (Bank Indonesia, October 21, 2026).
6. **Local borrowing gets costlier.** Higher BI rates and higher bond yields (Indonesia's 10-year government bond was about 7.45% in late October 2026) make loans more expensive for Indonesian companies and households. Growth slows.

So a decision made in Washington changes the interest rate a small business in Semarang pays for a loan. This is not unique to Indonesia. It is the normal condition of most emerging economies in a dollar-centred system.

The impossible trinity

Why can't Bank Indonesia simply ignore the Fed? The answer is one of the most useful ideas in international economics: the **impossible trinity** (also called the trilemma), developed from the work of Robert Mundell and Marcus Fleming in the 1960s. A country can have at most two of these three things at once:

1. **A fixed (or stable) exchange rate.**

2. **Free movement of capital** in and out of the country.
3. **An independent monetary policy** (setting its own interest rate for its own economy).

Examples:

Country (simplified)	Chooses	Gives up
Hong Kong	Fixed rate to US dollar + free capital	Independent interest rate (follows the Fed)
China	Managed exchange rate + own interest rate	Fully free capital movement (capital controls)
United States, euro area	Free capital + own interest rate	Fixed exchange rate (currency floats)
Indonesia	Mostly free capital + own interest rate	A fully fixed rate (rupiah floats, but BI smooths it)

Indonesia in practice tries to get "a bit of all three": it lets the rupiah float but intervenes to smooth sharp moves, keeps capital mostly open, and sets its own rate. The cost is that when pressure is strong, BI must choose. Raise rates and slow the economy, or let the rupiah fall and import inflation. In October 2026 it chose a middle path: about half of the 38 economists surveyed by Bloomberg had expected a rate rise to 6.00% instead.

Hot money, carry trades and sudden stops

Not all capital flows are equal.

- **Foreign direct investment (FDI)** is when a foreign company builds or buys a factory, mine or business. It is hard to pull out quickly. A Chinese-built nickel smelter in Sulawesi cannot be moved to Vietnam in a week.
- **Portfolio flows** are purchases of stocks and bonds. They can leave in hours. Economists call the fast-moving part "hot money".
- **Bank loans** sit in between.

A popular hot-money strategy is the **carry trade**: borrow in a low-interest currency (historically the Japanese yen) and invest in a high-interest currency (such as the rupiah, Brazilian real or Mexican peso). If an investor borrows yen at 1% and buys Indonesian bonds at 7%, she earns 6% a year, as long as the rupiah does not fall more than 6% against the yen. Carry trades work well in calm times and unwind violently in panics: everyone rushes for the exit at once, the high-yield currency crashes, and that triggers further selling.

A **sudden stop** is when foreign capital that had been flowing in abruptly stops or reverses. Countries that rely on foreign money to fund a current account deficit (spending more abroad than they earn) are most exposed.

History's warning: the Asian Financial Crisis of 1997-98

The most important financial event in modern Indonesian history is the Asian Financial Crisis. In the early 1990s, foreign money poured into Thailand, Indonesia, South Korea and Malaysia. Local banks and companies borrowed heavily in dollars, often short-term, because dollar interest rates were lower than local ones, and the local currencies were kept roughly stable against the dollar. This was a currency mismatch on a national scale: dollar debts, local-currency incomes.

In July 1997, Thailand could no longer defend the baht and let it float. It fell sharply. Investors then asked: which other country looks similar? Money fled Indonesia. The rupiah, around 2,400 per dollar in mid-1997, collapsed to the range of 16,000 to 17,000 at its worst points in January and June 1998, a fall of about 85% at the bottom. Companies with dollar debts were suddenly bankrupt, because their debts in rupiah had increased about six-fold. Banks failed. Indonesia's economy shrank by about 13% in 1998 (IMF), the worst decline among the crisis countries, prices of food soared, and the political crisis ended the Suharto government in May 1998.

The lessons Indonesia (and much of Asia) learned shaped policy for a generation:

- **Hold large reserves** as insurance, so you never again depend on emergency loans with harsh conditions.
- **Let the currency float**, so pressure is released gradually rather than in one crash.
- **Limit short-term foreign-currency borrowing** by banks and companies.
- **Build domestic bond markets**, so the government can borrow in rupiah.

These lessons are why the rupiah's slide from roughly 15,000 to over 18,000 per dollar between 2023 and 2026 has been painful but not catastrophic: debts are more in rupiah, reserves are larger, and the currency floats.

Value capture through the cost of capital: a worked example

Here is the single most important calculation in this month. It shows why finance decides who captures value in capital-heavy industries.

Imagine two identical nickel refineries (or robot-reducer factories, or solar-panel lines). Each costs $1 billion to build and will run for 20 years. The only difference is the cost of capital: the interest rate investors demand.

- Plant A is financed at 5% a year (typical for a strong company in a rich, stable country, or one backed by a state bank).
- Plant B is financed at 12% a year (typical for a project in a country with currency risk, political risk and a shallow local capital market).

To repay the $1 billion with interest over 20 years in equal annual payments, we use the annuity formula: annual payment = investment x r / (1 - (1 + r)^-n).

- At 5%: $1,000 million x 0.0802 = about **$80 million per year**.
- At 12%: $1,000 million x 0.1339 = about **$134 million per year**.

Plant B must earn $54 million more every year, just to pay for its money. If both plants produce 50,000 tonnes a year, Plant B's capital cost is about $1,070 higher per tonne. In a commodity market where both sell at the same world price, Plant A can undercut Plant B, or earn far more profit at the same price.

Now you can see why "who captures value" is so often decided by finance rather than geology or engineering. China's dominance in refining (Month 5) and in solar, batteries and robots rests partly on cheap, patient capital from state banks, local governments and policy funds. Rich countries capture value by being cheap places to borrow. Poor and middle-income countries, even when they own the ore, often pay a "risk premium" that transfers much of the value to their lenders. Lowering the cost of capital, through better institutions, stable money, deeper local markets and credible contracts, is one of the most powerful development policies there is.

Case study: Indonesia's double squeeze in 2026

Indonesia in 2026 is a live example of everything in this week's lesson.

The external squeeze. The Iran war that began on February 28, 2026 sent oil prices sharply higher: Brent crude rose from about $72 a barrel before the war to a peak of $126.41 (front-month futures) on April 30, 2026 (Reuters), fell back to the low $70s after a ceasefire and a June memorandum of understanding, then rose above $100 again when fighting resumed in July. In late October 2026 Brent traded above $110 (on October 30 the expiring December contract settled at $117.64 and the new front-month January contract at $113.43), before falling 7.6% to settle at $104.77 on November 2 on hopes of peace talks (Reuters, November 2, 2026). Indonesia is a net oil importer, so higher oil prices widen its trade deficit and raise the cost of fuel subsidies. Meanwhile, rising US rates and a 10-year Treasury yield above 5% pulled money toward the United States. Both pressures pushed the rupiah down.

The domestic squeeze. In January 2026, MSCI, the index company whose emerging-markets index is used as a benchmark for about $10 trillion of investments, warned that Indonesia might be downgraded from "emerging" to "frontier" status. Its concerns were about how the Indonesian stock market works: opaque ownership, a small "free float" (the share of a company's stock actually available to trade), suspected coordinated trading, and weak enforcement. On October 15, 2026 (New York time, the early morning of October 16 in Jakarta), MSCI went further and proposed reclassifying Indonesia directly to "standalone" status, which would remove it from both its emerging and frontier indexes. MSCI said that despite announced reforms, "significant concerns remain around market dysfunction, including continued suspected coordinated trading behavior and lack of evidence of meaningful enforcement", and that a move to frontier status "would not properly recognize the depth of the market problems". It also kept a freeze on adding Indonesian stocks or raising their index weights. The Jakarta Composite Index fell 11.6% on October 16, its biggest one-day fall since the 1998 crisis, and another 5% on October 19, wiping about $65 billion off the value of Indonesian stocks (CNA). UBS estimated that a standalone classification could trigger about $4 billion of forced selling by index-tracking ("passive") funds, plus up to $15 billion from active funds; Goldman Sachs estimated $5.6 billion of passive outflows.

Why can an index company have such power? Because of how modern investing works. Trillions of dollars sit in funds that simply copy an index. If a country leaves the index, those funds must sell, whatever they think of the companies. Active managers are judged against the same index, so many follow. MSCI's classification is effectively a gate to a huge pool of global capital.

The policy response. Indonesian authorities raised the minimum free-float requirement gradually toward 15%, required disclosure of shareholders above 1%, imposed fines of 2.1 trillion rupiah (about $116 million) on entities linked to market manipulation, and asked police to investigate. President Prabowo Subianto said MSCI "intended to intimidate us". MSCI said it follows transparent, rules-based methods. It is taking feedback until April 30, 2027 and will announce its decision on or before June 30, 2027, and it said it expects to go ahead with the downgrade unless the consultation finds "clear evidence of material improvements in market governance and functioning".

What to learn from it.

1. **Capital flows respond to institutions, not just interest rates.** Indonesia's growth was solid and its debt moderate, yet foreign investors pulled back because they doubted the fairness of the market's plumbing. This connects directly to Month 2: extractive or opaque institutions raise the cost of capital for everyone in the country, including honest companies.

2. **The cost of a weak market falls on future founders.** If the stock market is excluded from global indexes, Indonesian companies find it harder and more expensive to raise equity, and venture investors find it harder to exit through local listings (Week 4). The value lost is not only today's $65 billion; it is the higher cost of capital for years.
3. **Defiance and reform are different strategies.** National pride is understandable, but the investors who decide are not persuaded by speeches. The countries that escaped downgrades in the past (or won upgrades, like China's inclusion in MSCI from 2018) did so by changing rules and enforcing them.

Judgment: the MSCI episode is, in the long run, an opportunity as much as a threat. If it forces real improvements in market transparency and enforcement, Indonesia's cost of capital could fall below where it was before. If the response is mainly political, the country risks a structurally higher cost of capital at exactly the moment it is trying to finance downstreaming and digital industries.

Key numbers

Figure	Value	Year	Source
US federal funds target range	3.75%-4.00%	October 28, 2026	Federal Reserve
10-year US Treasury yield	about 5.24% close (5.34% intraday, highest since 2002)	October 1, 2026	Reuters; CNBC
Bank Indonesia policy rate (BI-Rate)	5.75%	October 21, 2026	Bank Indonesia
Rupiah reference rate (JISDOR)	Rp18,161 per dollar	November 2, 2026	Bank Indonesia
Indonesia 10-year government bond yield	about 7.45%	October 21, 2026	The Business Times
Jakarta Composite fall on MSCI proposal	-11.6% in one day	October 16, 2026	CNA / Reuters
Assets benchmarked to MSCI Emerging Markets	about $10 trillion	2026	CNA
Indonesia's weight in MSCI EM index	0.46%	2026	JPMorgan via CNA
Brent crude before the Iran war	about $72 per barrel	February 27, 2026	CNBC
Brent futures peak	$126.41 per barrel	April 30, 2026	Reuters
Brent futures settle after peace-talk hopes	$104.77 per barrel	November 2, 2026	Reuters
Rupiah at worst of Asian crisis	about 16,000-17,000 per dollar	mid-1998	Historical record

Think like a founder

- **Your cost of capital is partly national, partly personal.** You cannot change the Fed, but you can lower your own risk premium: audited accounts from day one, clean ownership,

contracts in enforceable jurisdictions, dollar revenues if your costs are in dollars. Investors pay more for companies they can trust.

- **Build for a world where money is not cheap.** The era of near-zero interest rates (2009-2021) rewarded growth at any cost. With the 10-year Treasury above 5%, investors value profits sooner. A business plan that reaches positive cash flow quickly is worth more in 2026 than one with a beautiful story about 2035.
- **Look for opportunities created by high capital costs.** If capital is expensive in your country, businesses that help others use capital better (equipment leasing, robot-as-a-service, shared factories, inventory finance) can capture value. A small factory that cannot afford a $50,000 robot might pay $1,500 a month to rent one. The company that owns the robots and has cheaper access to capital earns the spread.

Week 3: Debt, development finance and Chinese lending

Core lesson

A world built on debt

Debt is a promise to pay later. It lets a government build a port before it has collected the taxes the port will generate, a company build a factory before it has sold a single product, and a family buy a house before it has saved the full price. Used well, debt pulls future income forward into productive investment. Used badly, it pulls future income forward into consumption, waste or theft, and leaves the future to pay the bill.

The world has a lot of it. The Institute of International Finance (IIF) estimated total global debt (governments, households, non-financial companies and financial companies) at a record $365.5 trillion in the first half of 2026, about 310% of world output. Emerging-market debt rose by $6.5 trillion in those six months to about $110.6 trillion, led by China. The IIF warned that the ratio to output looked stable only because higher inflation was raising nominal output, calling this "an illusion of stability".

The United States is the largest single borrower. Gross federal debt passed $40 trillion in 2026 and stood at $40.25 trillion on October 5, 2026. The federal deficit for fiscal year 2026 (ending September 30) was about $1.99 trillion, and net interest payments topped $1 trillion for the first time: $1,051.75 billion, up from about $971 billion in fiscal 2025 and more than the $917 billion spent by the Department of Defense (US Treasury Monthly Treasury Statement, via the Joint Economic Committee). Gross interest on the public debt, before subtracting interest the Treasury pays to its own trust funds, was more than $1.3 trillion (CNBC). The CBO projects debt held by the public rising from about 101% of output in 2026 to about 120% in 2036. If interest rates stay 1.5 percentage points above its baseline, interest costs rise much faster.

Why does a US problem matter to you? Because the US government competes with every other borrower in the world for savings. When it borrows $2 trillion a year at rising rates, it pulls up the "risk-free" rate that every project from Jakarta to Nairobi is measured against (Week 2).

The arithmetic of debt: when does debt explode?

There is a simple equation that every finance minister and founder should know. Let d be government debt as a share of output (GDP), r the average interest rate on that debt, g the growth rate of

nominal output, and *pb* the primary balance (tax revenue minus spending excluding interest, as a share of output; positive is a surplus). Then, approximately:

change in debt ratio (approximately) = (r - g) / (1 + g) x d - pb

What it means in words: if the interest rate is higher than the growth rate, debt grows faster than the economy unless the government runs a primary surplus. If growth is higher than the interest rate, debt shrinks as a share of the economy even with a small deficit.

Worked example. A country has debt of 60% of output, pays 8% interest, and its nominal economy grows 5% a year. With a balanced primary budget (pb = 0):

(0.08 - 0.05) / 1.05 x 60 = 1.7 percentage points a year.

So debt rises from 60% to about 61.7% in one year, and keeps rising, without the government doing anything new. To keep the ratio stable, it needs a primary surplus of 1.7% of output, which means cutting spending or raising taxes. Now flip it: same debt, but 4% interest and 7% growth: (0.04 - 0.07) / 1.07 x 60 = -1.7 points. Debt falls on its own.

This is why the *same* debt level can be safe for one country and dangerous for another. Japan's government debt is about 199% of output (IMF, 2026), yet for decades it paid very low interest in its own currency, so r stayed below g. A frontier economy with debt at 60% of output, paying 10% in dollars while growing 4%, can be in crisis. Rich countries pay low r; poor countries pay high r. That gap is one of the largest hidden transfers of value in the world economy.

Developing countries: paying more than they receive

The World Bank's International Debt Report (December 2025) gave a stark picture of the 2022-2024 period, when global interest rates rose:

- Developing countries paid out $741 billion more in principal and interest on external debt than they received in new financing between 2022 and 2024, the largest gap in at least 50 years.
- The combined external debt of low- and middle-income countries hit a record $8.9 trillion in 2024, including $1.2 trillion owed by the 78 poorest countries eligible for the World Bank's concessional lending arm, the International Development Association (IDA).
- These countries paid a record $415 billion in interest alone in 2024.
- Bond investors returned in 2024, providing $80 billion more than they took out, but at interest rates "around 10%, about double those before 2020".
- Government-to-government (bilateral) lenders, including China, took in $8.8 billion more than they lent.
- Among the 22 most indebted countries (external debt above 200% of export revenue), an average of 56% of people could not afford a healthy minimum diet.

Notice the direction of money. Textbooks say capital should flow from rich countries (where it is abundant) to poor countries (where it is scarce and returns should be higher). For much of 2022-2024 it flowed the other way. Economists call this the "Lucas paradox", after Robert Lucas, who asked in 1990 why more capital does not flow to poor countries. The answers include risk, weak institutions, currency instability and the high cost of enforcing contracts: the same factors behind the risk premium in Week 2.

Development finance: the public lenders

Between private markets and grants sits **development finance**: lending by public institutions on better terms than markets offer.

- **Multilateral development banks (MDBs)**: the World Bank Group, the Asian Development Bank (ADB), the African Development Bank (AfDB), the Asian Infrastructure Investment Bank (AIIB, led by China), and others. They borrow cheaply on world markets thanks to the backing of member governments and their high credit ratings, then lend to governments at lower rates and longer terms. The World Bank's IDA lends to the poorest countries at very low interest or gives grants. In 2024 the World Bank provided a record $18.3 billion more in new financing to IDA-eligible countries than it received back, plus $7.5 billion in grants (World Bank).
- **Development finance institutions (DFIs)** that lend to private companies: the World Bank's International Finance Corporation (IFC), Britain's British International Investment, the US International Development Finance Corporation, Germany's DEG, and similar.
- **Export credit agencies**, which finance purchases of their own country's goods (China Exim Bank, Japan's JBIC, the US Export-Import Bank).

The key mechanism to understand is **leverage of public money**. A development bank with $10 billion of shareholder capital can borrow and lend several times that amount. And **blended finance** uses a small amount of cheap or risk-absorbing public money to attract a larger amount of private money. For example, a DFI might agree to take the first 10% of losses on a solar project, which makes the project safe enough for a pension fund to lend the other 90%.

Development finance matters for our driving question because the projects that let a country move up the chain (power plants, transmission lines, ports, railways, industrial parks, vocational schools) often cannot get private finance at reasonable cost. The Lobito Corridor (Month 6), a railway from the copper belt of Zambia and the Democratic Republic of the Congo to Angola's Atlantic coast, depends on US, European and African development finance precisely for this reason.

Chinese lending: rise, peak and retreat

From the mid-2000s to the late 2010s, China became the world's largest bilateral lender to developing countries, mostly through two state policy banks, China Development Bank and the Export-Import Bank of China. Much of it was linked to the Belt and Road Initiative launched in 2013.

Boston University's Chinese Loans to Africa database estimates that from 2000 to 2024, 42 Chinese lenders signed 1,319 loan commitments worth $180.87 billion with 49 African governments and seven regional institutions. Annual lending peaked at about $28.8 billion in 2016. Since 2020 it has not exceeded $5 billion a year. In 2024 it was just under $2.1 billion across only six projects, with Angola receiving about $1.45 billion of that for power transmission and roads. Boston University's 2026 bulletin found that net flows have turned negative: African countries now repay China more each year than they receive in new disbursements.

Why did China lend so much, and why did it pull back?

- **Why it lent:** to find uses for China's huge savings and its construction companies' spare capacity; to secure resources (oil in Angola, copper and cobalt in Zambia and the DRC); to build political relationships; and because Western lenders were not financing much infrastructure in Africa at the time. Chinese loans were often fast, without the governance conditions attached by the IMF or World Bank, and tied to Chinese contractors.

- **Why it pulled back:** defaults and restructurings (Zambia, Ghana, Sri Lanka, Ethiopia); China's own slowing economy and property crisis; criticism of "debt traps"; and a shift toward smaller, commercially viable projects, renminbi-denominated loans, lending through local banks, and direct investment rather than loans.

The "debt-trap diplomacy" debate. One side argues China deliberately lent to weak governments to seize strategic assets when they could not pay. The most cited example is Sri Lanka's Hambantota port, leased to a Chinese company for 99 years in 2017. The other side, including researchers at Johns Hopkins and Chatham House, argues that the evidence for a deliberate strategy is weak: Hambantota's lease was negotiated to raise cash, most of Sri Lanka's debt was owed to Western bondholders, and Chinese lenders have generally restructured rather than seized assets.

Judgment: the "debt trap" as a deliberate master plan is not well supported by the evidence. But Chinese lending did have real problems: opacity (confidential contracts, collateral clauses), tied procurement that captured much of the construction value for Chinese firms, and lending to projects with weak economics. The lesson for borrowers is the same regardless of the lender: the danger is not who lends, but borrowing in foreign currency for projects that do not generate foreign-currency income, under contracts the public cannot see.

When borrowers cannot pay: restructuring

When a government cannot pay its debts, it **defaults** and then **restructures**: negotiates with creditors to reduce the amount, lower the interest, or extend the time to repay. This used to be fairly simple when most debt was owed to Western banks or governments in the "Paris Club". Now creditors include bondholders scattered around the world, Chinese policy banks, commercial banks, and commodity traders. Each wants others to take bigger losses.

In 2020, the Group of Twenty (G20) created the **Common Framework** to coordinate restructurings for poor countries, bringing China and traditional creditors to the same table. It has been slow. Only four countries applied (Chad, Zambia, Ethiopia, Ghana). By April 2026, the IMF and World Bank reported that most cases begun in 2021-22 were "largely completed", with only some commercial creditors outstanding. In 2024, developing countries restructured $90 billion of external debt, the most since 2010 (World Bank).

Case study: Zambia, copper, default and a slow recovery

Zambia is Africa's second-largest copper producer and sits at the heart of the Central African copper belt, which you studied in Months 5 and 6. Copper is essential for electrification, data centers and robots. Zambia should, in theory, be well placed to capture value from the energy transition. Its debt story shows why that is not automatic.

The borrowing boom. After debt relief in the mid-2000s, Zambia's debt was low. In the 2010s, with copper prices high and global interest rates low, it borrowed heavily: Eurobonds (dollar bonds sold to international investors) from 2012, plus loans from Chinese lenders for roads, airports, power and telecoms. Some projects were useful; others were overpriced or poorly chosen. Copper prices fell in 2015, the kwacha weakened, and dollar debts grew heavier in local-currency terms.

The default. In November 2020, during the COVID-19 crisis, Zambia became the first African country of the pandemic era to default on its Eurobonds. In January 2021 it applied for debt treatment under the G20 Common Framework.

The slow negotiation. The restructuring took years, partly because creditors disagreed about fair burden-sharing: Chinese lenders wanted bondholders to take similar losses; bondholders and Western governments wanted transparency about Chinese loan terms; and the IMF needed everyone's agreement to approve its programme. By early 2026, the IMF reported that Zambia had reached agreement on 94% of the external debt included in the restructuring, and that public debt had fallen from 133% of GDP in 2023 to 86% by mid-2025, taking the agreements into account. On January 27, 2026 the IMF board completed the sixth and final review of Zambia's 38-month programme (approved in August 2022), bringing total IMF disbursements to about $1.7 billion. The IMF still judged the debt "sustainable but at high risk" of distress. Two details show how slow "done" can be: by August 2026 a Zambian parliamentary brief reported that agreements covered 94% of the debt but only 60% were fully executed, and on October 9, 2026, after new fiscal pressure from the oil shock and election-year spending, IMF staff agreed a new 36-month programme worth about $1.47 billion, still subject to board approval. Zambia was the first country to restructure under the Common Framework, so its experience became the template for the others.

What it teaches.

1. **Commodity booms tempt borrowing in the wrong currency.** When copper is high, dollar debt looks cheap. When copper falls, the debt becomes crushing. This is the resource curse (Month 2) in financial form.
2. **Owning the ore is not enough.** Zambia exports mostly copper concentrate and refined copper. Most of the value from turning copper into wire, motors, batteries and robots is captured elsewhere. And foreign mining companies capture much of the mining profit. Zambia captures taxes and royalties, which are volatile.
3. **Restructuring is costly even when it "works".** Years of uncertainty froze investment. Being "almost complete" five years after default shows how slow the current system is.
4. **There are better models.** Chile's copper stabilization fund saved revenue in boom years to spend in busts. Botswana's diamond revenues were managed through rules and savings (Month 6). Zambia's experience is an argument for counter-cyclical rules: save in booms, borrow in local currency, and match debt to projects that earn the currency they must repay.

Key numbers

Figure	Value	Year	Source
Total global debt	$365.5 trillion (about 310% of world GDP)	H1 2026	IIF Global Debt Monitor
US gross federal debt	$40.25 trillion	October 5, 2026	US Joint Economic Committee
US net interest payments	$1,051.75 billion (gross interest more than $1.3 trillion)	Fiscal year 2026	US Treasury Monthly Treasury Statement via JEC; CNBC
Developing countries' net debt outflow	$741 billion	2022-2024	World Bank International Debt Report 2025
External debt of low- and middle-income countries	$8.9 trillion	2024	World Bank

Figure	Value	Year	Source
Interest paid by developing countries	$415 billion	2024	World Bank
Chinese loan commitments to Africa, cumulative	$180.87 billion	2000-2024	Boston University GDP Center
Chinese loan commitments to Africa	just under $2.1 billion	2024	Boston University GDP Center
Zambia public debt	133% of GDP to 86%	2023 to mid-2025	IMF
Zambia debt covered by restructuring agreements	about 94% (60% fully executed)	2026	IMF; Zambia Parliamentary Budget Office

Think like a founder

- **Debt is a tool for assets that earn; equity is for risk.** If your business owns robots, vehicles or inventory that produce predictable cash, borrowing to buy them can be smart: you keep ownership. If your business is an unproven idea, debt is dangerous because repayments come before revenue. A common founder mistake is using short-term loans to fund long-term growth.
- **Match currencies.** Never borrow in dollars to fund a business that earns only rupiah, naira or pesos, unless you can hedge. Zambia's lesson applies to a company as much as a country.
- **Development finance is underused by founders.** DFIs and MDBs run programmes for small and medium enterprises, climate projects, and digital infrastructure in Southeast Asia and Africa, often through local banks. Blended-finance structures can make a project in a high-risk country financeable. A founder who understands these instruments can access capital competitors do not even know exists. The value capture insight: whoever can access the cheapest long-term capital can afford to build the infrastructure others must then rent.

Week 4: Sovereign wealth funds and venture capital

Core lesson

This week covers two very different kinds of investor that share one feature: both can, in principle, take a long view and fund things that ordinary banks will not. A **sovereign wealth fund (SWF)** is an investment fund owned by a government. **Venture capital (VC)** is private money that buys stakes in young, risky companies hoping a few will grow enormous. One is the state's patient capital; the other is the market's risk capital. Together, they decide a great deal about which new industries a country builds.

Part A: Sovereign wealth funds

Why governments create investment funds A government ends up with a large pool of money to invest for several reasons, and each produces a different kind of fund:

Type	Purpose	Example
Savings fund	Turn a temporary resource (oil, gas) into permanent financial wealth for future generations	Norway's Government Pension Fund Global
Stabilization fund	Smooth government spending when commodity prices swing	Chile's Economic and Social Stabilization Fund
Reserve investment fund	Earn more on surplus foreign reserves than cash would pay	Singapore's GIC, China Investment Corporation
Strategic or development fund	Invest at home to build industries and diversify the economy	Saudi Arabia's PIF, Abu Dhabi's Mubadala
Holding company for state firms	Own and manage state-owned enterprises commercially	Singapore's Temasek, Indonesia's Danantara

In 2008 a group of SWFs agreed the **Santiago Principles**, 24 voluntary guidelines on governance, transparency and investing for financial (not political) reasons. They matter because the biggest risk of a state fund is that it becomes a piggy bank for politicians.

Norway: the benchmark Norway discovered North Sea oil in 1969. Instead of spending the revenue, it created a fund (by law in 1990, first deposit in 1996) to invest oil money abroad. Its value was 22,014 billion Norwegian kroner (about $2.35 trillion) on September 30, 2026, down from 22,683 billion at the end of June, mostly because a stronger krone shrank the kroner value of foreign assets (NBIM). That is roughly five times Norway's mainland economy. It owns on average about 1.5% of every listed company in the world. In the first half of 2026 it returned 9.4%, earning a record 1,753 billion kroner, driven largely by Asian technology stocks; in the third quarter it lost 0.5% as interest rates rose.

Four design choices make Norway the benchmark:

1. **Invest abroad, not at home.** Spending oil money domestically would overheat the economy and push up the krone, harming other industries (the "Dutch disease" of Month 2). Investing abroad avoids this.
2. **A spending rule.** The government may spend, on average over time, only the expected real return of the fund, set at 3% a year. In 2026 the structural non-oil deficit was estimated at 2.7% of the fund. The capital is never touched.
3. **Passive, diversified, transparent.** The fund mostly tracks global indexes (about 72% in stocks at mid-2026), publishes every holding, and is run by the central bank under rules set by parliament.
4. **Political consensus.** The rules have survived changes of government for three decades.

The result: oil wealth was converted into a permanent financial endowment that now supplies about a quarter of Norway's public spending. Its chief executive warned in 2026 that the fund could lose most of its value in an extreme global crash, a reminder that even the best fund is exposed to the same markets as everyone else.

Saudi Arabia's PIF: the state as builder Saudi Arabia's Public Investment Fund (PIF) follows almost the opposite model. Under Vision 2030, launched in 2016, it was turned from a sleepy holding company into the main engine of economic diversification. Its assets under management grew from

about $150 billion in 2015 to about 3.4 trillion riyals (about $905 billion) at the end of 2025. A large part of that growth came from the government transferring 16% of Saudi Aramco's shares to PIF, worth around $320 billion (Arab Gulf States Institute in Washington analysis).

In 2025, PIF's assets fell slightly, by about 1% to 3.396 trillion riyals (about $905 billion), the first decline this decade and short of its 4 trillion riyal target, as Saudi stocks performed poorly. PIF does not headline a single-year return: its annualized total shareholder return since 2017 fell from 7.2% at the end of 2024 to 5.8% at the end of 2025 (PIF; Reuters, August 17, 2026), which implies a 2025 return of roughly -4% (our calculation from those two averages, not a PIF figure). Domestic investments made up 76% of its assets, international investments 20%. PIF owns stakes in more than 200 companies and is building giga-projects such as NEOM, plus new industries: electric vehicles, mining (through Ma'aden and Manara), tourism, and artificial intelligence through its company Humain, which plans to build data center capacity with hundreds of thousands of Nvidia chips. In April 2026 its board approved a 2026-2030 strategy that shifts from "rapid scale-building to sustained value creation", focusing on six domestic "ecosystems" and dropping explicit asset-size targets.

The strengths of the PIF model: speed, scale and the ability to build industries that markets would not start alone. The weaknesses: concentration in one country, mixing commercial and political goals, giga-projects that have been scaled back, and lower returns than a passive global portfolio. PIF's annualized shareholder return since 2017 is 5.8%, by its own report.

Abu Dhabi and Singapore: mixed models **Abu Dhabi** runs several funds with different jobs. The Abu Dhabi Investment Authority (ADIA), one of the world's largest funds, invests globally like Norway but with more private equity, real estate and infrastructure. ADIA does not publish its size; the consultancy Global SWF estimated it at about $1.13 trillion at the end of 2025. Mubadala reported assets under management of 1.4 trillion dirhams (about $385 billion) at the end of 2025, up 17% in a year, with a ten-year annual return of 10.3%. It is more strategic and active, and founded the AI investment firm MGX and the AI company G42, partners in the 1-gigawatt Stargate UAE data center cluster within a planned 5-gigawatt US-UAE AI campus in Abu Dhabi. The Middle East Institute estimated that the UAE directed $148 billion into AI at home and abroad from the start of 2024 to late 2026. Abu Dhabi uses oil-era savings to buy a position in the AI era: a clear attempt to move from the energy layer to the compute layer of our value chain.

Singapore has no oil. Its two funds were built from budget surpluses and high national savings. GIC manages the government's foreign reserves; it does not publish its size, but reports a 20-year annualized real return of 3.4% to March 2026. Temasek owns and manages companies such as Singapore Airlines and DBS Bank and invests globally; its net portfolio value was S$518 billion (about US$401 billion) at March 31, 2026, with a 20-year shareholder return of 6.8%. Temasek's history is important for Indonesia: it is the closest model for a holding company that runs state firms commercially, and its credibility rests on decades of professional management, published accounts and a constitutional safeguard that limits how governments can draw on past reserves.

Indonesia's Danantara: the newest experiment Danantara (full name Daya Anagata Nusantara) was created by Law No. 1 of 2025 and launched in February 2025, reporting directly to President Prabowo Subianto. It was given a mandate to manage state assets worth around $900 billion across roughly 1,000 state-owned companies, including the big state banks (Mandiri, BRI, BNI), the energy giants Pertamina and PLN, and firms in mining, logistics, telecoms and fertilizer. The President said in 2026 that its assets were about $1 trillion. Its idea: take the dividends that

state companies used to pay to the budget, and invest them for higher returns and in national priorities.

Its structure matters, so it is worth being precise. At the top sits the agency itself (BPI Danantara). Below it are two state-owned operating companies. PT Danantara Asset Management (DAM) is the "operational holding": it now holds the government's controlling shares in the state firms (for example, 52% of Bank Mandiri and about 52% of Telkom Indonesia at the end of 2025, according to those companies' own filings) and restructures them. PT Danantara Investment Management (DIM) is the "investment holding": it takes the dividends and invests them for returns. A presidential decree signed in April 2026 and published at the end of May created a third arm, a development investment fund "for national development and public services that have social and economic impacts", which may receive money from the state budget as equity (Reuters via Kontan, June 2026). That third arm is important: it allows the budget and the fund to mix, which is exactly what the Santiago Principles warn about.

How much money is actually invested? Much less than the headline. The "about $1 trillion" figure is the book value of the state companies Danantara owns, not cash. DIM's own financial statements, filed with the Singapore Exchange for a planned $5 billion note programme, showed total assets of about 140 trillion rupiah (roughly $8 billion) at April 30, 2026, of which about 88% was cash and deposits, mostly placed in the big state banks, and only about 16.6 trillion rupiah (about $1 billion) was in investments (IDNFinancials and Katadata, November 2026). A sovereign fund that parks most of its cash in banks it also owns is, for now, closer to a holding company than an investor.

In its first eighteen months, Danantara:

- planned to invest up to $14 billion in 2026, mainly in Indonesia, funded largely by dividends (Reuters, January 2026);
- raised money through "Patriot Bonds" sold to Indonesian business groups at below-market interest rates, and in June 2026 issued its first international dollar bond, $1.5 billion, with orders of about $4.6 billion;
- built a public-markets portfolio of about $3 billion, roughly 65% stocks, 30% bonds and 5% hedge funds, about a quarter in Indonesia (Reuters, October 2026);
- invested abroad, including $250 million in the US oil producer HighPeak Energy in 2026, a food-processing partnership with Brazil's JBS, and about $533 million in a hotel and land in Makkah;
- merged four state-owned asset managers (from Mandiri, BRI, BNI and PNM), with about 170 trillion rupiah under management in June 2026, into one firm, PT Danareksa Investment Management Indonesia, effective October 1, 2026 (DealStreetAsia).

The case study below examines the risks and the potential.

What makes a state investor create value? Looking across these funds, a few patterns stand out:

1. **Clear purpose.** A fund that tries to maximize returns, save for the future, rescue failing firms, fund the president's priorities and support the stock market all at once will usually do none of them well.
2. **Separation from daily politics.** Professional boards, published rules for withdrawals, and protection from being used to fund the budget or bail out connected businesses.
3. **Transparency.** Audited accounts, published holdings, and clear reporting to parliament. Norway and Temasek publish; many funds do not.

4. **Patience with discipline.** Being willing to wait for long-term returns, but still killing projects that fail.

The negative example everyone cites is Malaysia's 1MDB, a state development fund from which billions of dollars were stolen in the 2010s, leading to criminal convictions in Malaysia and the United States.

Part B: Venture capital and how founders raise money

How a venture fund works A **venture capital fund** is a pool of money raised by a VC firm (the **general partners**, or GPs) from investors called **limited partners** (LPs): pension funds, university endowments, sovereign wealth funds, insurance companies, and wealthy families. A typical fund lasts about ten years. The GPs usually charge a yearly management fee of about 2% of the fund and keep about 20% of the profits (called **carried interest**, or "carry"). This "2 and 20" model is a common pattern, not a law.

The core fact about venture returns is the **power law**: a tiny number of investments produce almost all the gains. A fund might invest in 25 startups. Perhaps 15 fail or return little, 8 return one to three times the money, and 1 or 2 return 50 or 100 times. That one big winner pays for the whole fund. This explains VC behaviour that confuses founders: investors want companies that could become enormous, not just good businesses. A company that will reliably earn $5 million a year in profit is wonderful for its founder but useless to a VC fund that needs one investment to return $300 million.

Simple fund math: a $100 million fund aiming to return three times its money must return $300 million. If it owns 10% of its best company at exit, that company must be worth $3 billion for that single investment to return the target.

The 2025-2026 boom, and how concentrated it is Global venture funding was $425 billion in 2025, up 30% from 2024, according to Crunchbase. About 50% went to artificial intelligence companies ($211 billion). Five companies (OpenAI, Scale AI, Anthropic, Project Prometheus and xAI) raised $84 billion between them, 20% of all venture money in the world. US-based companies received 64%. In the first three quarters of 2026, global venture funding reached a record $679 billion; in the third quarter alone, AI startups took 64% of the $159 billion invested.

Compare the regions closest to our driving question:

- **Southeast Asia** startups raised $5.37 billion across 461 equity deals in 2025, about a quarter of the 2021 peak, with Singapore-headquartered companies taking about 78% of the value (DealStreetAsia and Kickstart Ventures). Indonesia raised only about $356 million in 2025, down from roughly $6.9 billion in 2021 (Indonesia Startup Report 2026). In the first half of 2026, Southeast Asia's total jumped to $7.25 billion, but one data-center company's $4.5 billion round made up 62% of it.
- **Africa**: Partech counted $4.1 billion raised by African tech startups in 2025 (up 25%), of which $2.4 billion was equity and a record $1.6 billion was debt. Another tracker, Africa: The Big Deal, counted $3.2 billion. Kenya, Egypt, South Africa and Nigeria took about 82% of the money.

So the world's venture capital is overwhelmingly concentrated in the United States and in AI. Southeast Asia and Africa together receive roughly 2-3% of global venture money, even though they

hold about a quarter of the world's population. For a founder in those regions, this is a fact to plan around, not complain about.

Stages and dilution: a worked example A startup typically raises money in stages:

- **Pre-seed and seed**: from the founders themselves, friends, angels (wealthy individuals), accelerators and seed funds. Often tens of thousands to a few million dollars. Commonly done with a **SAFE** (Simple Agreement for Future Equity), which converts into shares at a later round.
- **Series A, B, C. . .**: larger rounds from VC funds as the company proves its product and grows.

Each time the company sells new shares, existing owners own a smaller percentage. This is **dilution**. Example:

1. Two founders own 100% of a robotics software startup.
2. They create an employee option pool of 10% to attract engineers. Founders now own 90%.
3. Seed round: they raise $1 million at a $4 million "pre-money" valuation. Post-money value is $5 million; the seed investors own $1M / $5M = 20%. Founders now own 90% x 80% = 72%.
4. Series A: they raise $6 million at a $24 million pre-money valuation. Post-money is $30 million; Series A investors own 20%. Founders now own 72% x 80% = 57.6%.

Note that the founders' share fell from 100% to 57.6%, but the value of their stake rose from almost nothing to 57.6% x $30 million = about $17 million on paper. Dilution is fine if each round increases the value of the company more than it reduces your percentage. The danger lies in the **terms**: liquidation preferences (investors get their money back first in a sale), board control, and anti-dilution clauses can matter more than the headline valuation.

Funding paths for founders outside Silicon Valley For a founder in Semarang, Nairobi, Lagos or Hanoi, the realistic menu is wider than VC:

Source	Good for	Watch out for
Customers (revenue, pre-orders)	Proving demand, keeping control	Slower growth
Grants and competitions	Early research, prototypes	Small amounts, reporting burden
Angels and accelerators	First outside money and mentors	Terms on SAFEs; choose people, not just money
Local and regional VC	Growth capital, local networks	Small fund sizes; limited follow-on money
Impact investors	Businesses with measurable social benefit	Must report impact; return expectations vary
Venture debt, revenue-based finance	Companies with steady revenue	Repayment pressure
Development finance (via local banks or funds)	Infrastructure, climate, SMEs	Paperwork, slower decisions
Corporate partners	Hardware and industrial pilots	Dependence on one big customer

An important structural issue: many Southeast Asian and African startups incorporate a holding company in Singapore, Delaware or Mauritius, because many international investors require a familiar legal system. This is why Singapore captures most Southeast Asian venture value in the statistics: much of it is Indonesian, Vietnamese or Malaysian business, held through Singapore companies. Value capture through law, again.

How the two halves connect Sovereign funds are among the largest limited partners in global venture funds. Temasek, GIC, Mubadala, PIF and others put billions into US and Chinese VC and growth funds, and increasingly invest directly in AI companies. This means oil and savings from the Gulf, Norway and Singapore flow, through venture funds, into AI labs in California. The value chain of this textbook runs through finance too: petrodollars become venture dollars become GPUs become AI models. Whether a Southeast Asian or African sovereign fund chooses to invest in its own region's founders, or follows global capital to the United States, is one of the most consequential choices those funds make.

Case study: Danantara, between Temasek and 1MDB

Danantara is the boldest attempt in Indonesia's history to manage state wealth like a professional investor. It is too young to judge, but we can lay out the bull and bear cases.

The bull case: a Temasek for Indonesia. Indonesia's state companies are enormous but often inefficient: overlapping businesses, political appointments, and dividends that disappeared into the budget. Consolidating them under one professional owner could:

- improve governance and returns, as Temasek did for Singapore's state firms from the 1970s;
- free capital to co-invest with foreign partners in downstreaming (nickel processing, battery materials), energy and digital infrastructure, lowering the cost of capital for strategic projects (Week 2);
- give Indonesia a credible partner for foreign sovereign funds and companies; and
- act as a long-term buyer in the domestic capital market (Danantara said in October 2026 it could step in during stock selloffs if valuations were attractive).

Its early international bond, three times oversubscribed, and partnerships with established global managers suggest foreign investors are willing to give it a chance.

The bear case: political control and hidden risks. Critics, including many international investors, worry that:

- reporting directly to the President, with a broad mandate to drive national priorities, risks mixing commercial and political goals (Reuters reported in June 2026 that a rapid expansion of its mandate was testing the fund's capacity);
- "Patriot Bonds" sold to business groups at below-market rates blur the line between investment and political obligation;
- concentrating the banks, energy and mining companies in one entity creates one very large point of failure; and
- transparency is still limited: the state audit board began auditing its 2025 accounts only in September 2026;
- most of the investment arm's cash sat in deposits at state banks that Danantara itself controls, which earns interest but does little to lower anyone's cost of capital; and
- the new development arm can receive budget money as equity, blurring the line between a commercial investor and a government spending agency.

Malaysia's 1MDB is the warning every analyst raises. Semafor noted in September 2026 that "graft and political influence are particular concerns among international investors whose goodwill Indonesia, which runs both a budget deficit and a trade deficit, needs."

How to judge it over the next five years. Watch for: (1) audited, published consolidated accounts; (2) independent board members with real power; (3) clear rules on what Danantara must pay to the budget and what it can keep; (4) returns measured against a public benchmark; (5) whether loss-making projects are stopped; and (6) whether it invests in Indonesian startups and industrial supply chains on commercial terms.

Judgment: Danantara's design gives it the potential to lower Indonesia's cost of capital for strategic industries, which, after this month's lessons, is perhaps the most valuable thing a state fund can do. But its governance is not yet strong enough to give investors confidence that it will be run like Temasek rather than captured by politics. Its credibility will be built by rules and audited numbers, not by size. The same test applies to any sovereign fund that Southeast Asian or African countries create to finance their place in the ore-to-robot chain.

Key numbers

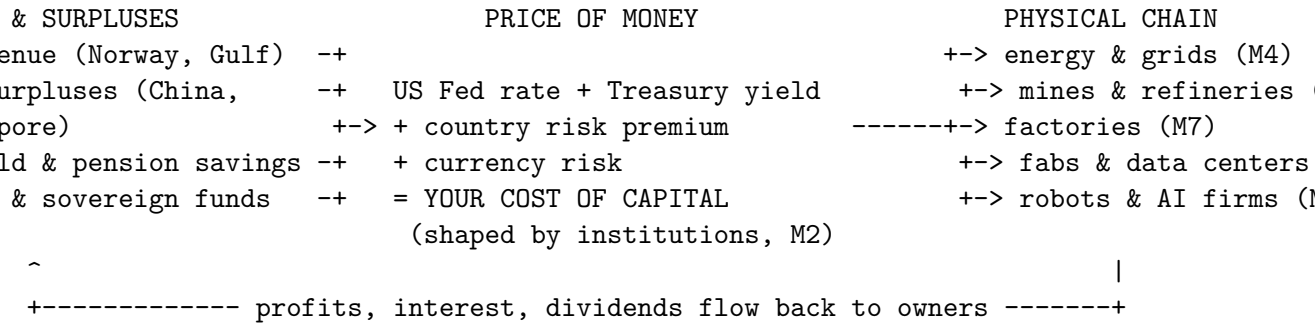
Figure	Value	Year	Source
Norway GPFG market value	22,014 billion kroner (about $2.35 trillion)	September 30, 2026	NBIM
Norway fiscal spending rule	3% of fund (2.7% used in 2026)	2026	Morningstar DBRS
Saudi PIF assets under management	3.396 trillion riyals (about $905 billion)	End-2025	PIF Annual Report 2025
PIF share invested domestically	76%	2025	PIF via Asia Asset Management
Temasek net portfolio value	S$518 billion (about US$401 billion)	March 31, 2026	Temasek Review 2026
Mubadala assets under management	1.4 trillion dirhams (about $385 billion)	End-2025	Mubadala
ADIA assets (estimate; not disclosed)	about $1.13 trillion	End-2025	Global SWF
Danantara Investment Management total assets	about 140 trillion rupiah (about $8 billion), about 88% cash	April 30, 2026	DIM statements via IDNFinancials
Danantara international bond	$1.5 billion (about $4.6 billion orders)	June 2026	Danantara
Global venture funding	$425 billion; AI about 50%	2025	Crunchbase
Global venture funding, first 9 months	$679 billion	2026	Crunchbase
Southeast Asia startup funding	$5.37 billion	2025	DealStreetAsia / Kickstart Ventures
African tech funding (equity + debt)	$4.1 billion	2025	Partech

Think like a founder

- **Know what kind of company you are building, then choose your money.** A robotics company selling to factories in Southeast Asia may never fit the VC power law, but could be a very good, profitable business funded by customers, equipment finance and development lenders. Do not take VC money unless you want, and can credibly aim for, a very large outcome.
- **Sovereign funds are customers and partners, not just investors.** Danantara, PIF and Mubadala own the companies (energy, banks, mines, ports) that buy industrial software, robots and AI services. In many emerging economies, selling to state-linked enterprises is the fastest path to scale, but also brings political risk.
- **The value capture lesson:** in the capital layer, value flows to those who (a) control cheap long-term money and (b) own equity in the companies that win. Founders capture value by owning a meaningful share of a company that becomes essential. Protect your equity by raising only what you need, at stages where you have proven something.

Month synthesis

Finance is the circulatory system of the material economy. Every layer from Months 4 to 10 needs capital up front, and the price and source of that capital shape who captures value.



Layer	Who supplies capital	Typical cost (2026)	Who captures value
Reserve currency	US Treasury market	10-year about 5%	US borrowers, payment networks
Strategic industry in a strong state	State banks, policy funds	Low	Owners of cheap capital (e.g., Chinese refiners)
Project in an emerging market	Bond investors, DFIs	High (often 8-12%+ in dollars)	Lenders take a large share
Startup	Angels, VC, sovereign LPs	Very high (equity)	Founders and early owners of winners

Three ideas tie the month together. First, **the cost of capital is a competitive weapon**: the $1 billion refinery example in Week 2 shows a 7-point difference in interest rates is worth over $50 million a year. Second, **institutions set the risk premium**: Indonesia's MSCI crisis and Zambia's default show that trust, transparency and contract enforcement are financial assets. Third, **the dollar system transmits US decisions everywhere**, so emerging economies must build buffers

(reserves, local-currency debt, floating exchange rates) before they can safely take on large industrial projects. For Africa and Southeast Asia to build ore-to-robot chains by the 2040s, cheap and patient capital, raised and governed credibly, matters as much as ore and labor.

Common misconceptions

1. **"The dollar is about to be replaced by BRICS or the renminbi."** The dollar's reserve share is slowly falling (about 57% in 2026), but it is on one side of 89% of FX trades, and the renminbi holds about 2% of reserves. Erosion, not collapse.
2. **"High government debt always causes a crisis."** What matters is the interest rate compared with growth, the currency of the debt, and who holds it. Japan carries about 199% of GDP with low interest in yen; Zambia defaulted at far lower levels because its debt was in dollars at high rates.
3. **"Chinese loans are a deliberate debt trap."** Evidence for a master plan is weak; most problem debt in crisis countries was owed to many creditors, including Western bondholders. The real problems were opacity, foreign-currency borrowing and weak projects, regardless of lender.
4. **"A sovereign wealth fund automatically makes a country richer."** A fund only moves wealth around. It creates value if it is well governed and invests well; it destroys value if it is captured by politics (1MDB).
5. **"Venture capital is the only way to fund a startup."** Most successful businesses in the world never take VC. VC suits a small subset of companies that can grow enormous; customers, debt, grants and development finance fund the rest.

Exercises

1. **(Quantitative)** A country has government debt of 80% of GDP, pays an average interest rate of 9%, and its nominal GDP grows 6% a year. What primary balance does it need to keep its debt ratio stable? What if interest falls to 5%?
2. **(Quantitative)** Recalculate the Week 2 refinery example for a $2 billion battery-materials plant with a 25-year life, financed at 4% versus 11%. What is the difference in annual capital cost? Express it per tonne if the plant makes 60,000 tonnes a year.
3. **(Quantitative)** A founder owns 100% of a company, creates a 12% option pool, raises $500,000 at a $2 million pre-money valuation, then $4 million at a $16 million pre-money valuation. What percentage does she own after both rounds, and what is her stake worth on paper?
4. **(Analysis)** Using the impossible trinity, explain what options Bank Indonesia had in October 2026, and why it chose to hold its policy rate while raising its deposit facility rate. What would you have done, and why?
5. **(Essay, 800 words)** Is the dollar's dominance good or bad for developing countries? Consider costs (original sin, US rate shocks, sanctions risk) and benefits (cheap trade finance, deep markets). Give a "Judgment".
6. **(Essay, 800 words)** Design the governance rules for a sovereign wealth fund in a mineral-rich African country that wants to finance a battery and motor supply chain. Borrow from Norway, Temasek and Chile; avoid 1MDB.
7. **(Research)** Find the latest MSCI decision on Indonesia (expected by June 2027). What reforms did Indonesia make, and how did foreign flows respond?

Reading list

1. Barry Eichengreen, *Exorbitant Privilege: The Rise and Fall of the Dollar* (2011). The clearest history of why the dollar dominates.
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5. Deborah Brautigam, *The Dragon's Gift: The Real Story of China in Africa* (2009). Evidence-based account of Chinese finance in Africa, a corrective to myths.
6. World Bank, *International Debt Report 2025*. The best data on developing-country debt flows.
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Glossary

- **Basis point**: one-hundredth of a percentage point (0.01%).
- **Blended finance**: using public or philanthropic money to reduce risk so private investors will fund a project.
- **Capital outflow / inflow**: money leaving / entering a country to buy assets.
- **Carry trade**: borrowing in a low-interest currency to invest in a high-interest one.
- **Carried interest ("carry")**: the share of profits (often 20%) kept by venture fund managers.
- **Common Framework**: G20 process (2020) to coordinate poor countries' debt restructurings.
- **Cost of capital**: the return investors require to fund a project; the "price" of money for that project.
- **Dilution**: the fall in an owner's percentage when a company issues new shares.
- **Exorbitant privilege**: the advantages the US gains from issuing the world's main reserve currency.
- **Free float**: the share of a company's stock available for public trading.
- **Impossible trinity (trilemma)**: a country cannot have a fixed exchange rate, free capital movement and independent monetary policy all at once.
- **Limited partner (LP) / general partner (GP)**: investors in / managers of a venture fund.
- **Original sin**: the inability of a country to borrow abroad in its own currency.
- **Power law**: a distribution where a few outcomes account for most of the total.
- **Primary balance**: government revenue minus non-interest spending.
- **SAFE**: Simple Agreement for Future Equity, a common early-stage investment contract.
- **Santiago Principles**: voluntary governance standards for sovereign wealth funds (2008).
- **Sovereign wealth fund (SWF)**: a government-owned investment fund.
- **Stablecoin**: a digital token designed to hold a fixed value, usually one US dollar.
- **Triffin dilemma**: the conflict between supplying the world with reserve currency and keeping confidence in it.

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Month 12: Geopolitics, Scenarios and Strategy, 2026-2050

Month overview

Eleven months ago we started with a simple question: how do energy, ores, processing, manufacturing, chips, robots, AI, finance and geopolitics fit together, and who captures the value? We have now walked the whole chain. This final month asks what happens to that chain when great powers fight over it, and what a young founder should do about it.

In economics textbooks, trade is usually described as a positive-sum game: both sides gain. That is true, and it remains the most important lesson of Month 1. But in 2025 and 2026 the world was reminded of a second truth: interdependence is also a source of power. Whoever controls a bottleneck that others cannot do without can use it as a weapon. China used its near-monopoly on rare earth magnets. The United States used its control of advanced chips, the dollar system and its own huge consumer market. Iran, in the war that began in February 2026, used its position on the Strait of Hormuz.

The month has four weeks. Week 1 studies the US-China rivalry and the toolkit of economic statecraft: tariffs, export controls, sanctions, investment screening, and the chokepoints that make them work. Week 2 studies blocs, chokepoints and the options of the "Global South" (the large group of developing and middle-income countries), using the 2026 Hormuz crisis as a stress test of the whole system. Week 3 builds scenarios to 2050 for Africa, Southeast Asia, the Gulf, Latin America and India, and returns to the course's sharpest question: could Africa or Southeast Asia build a cheap robot supply chain from ore to finished robot by the 2040s? Week 4 is the capstone. It asks what a founder should build, then draws the whole twelve-month system on one page.

This month uses every earlier month; if a concept here feels thin, go back to the month where it was built.

A warning about forecasting. No one knows what 2050 looks like. In 2001, few predicted that China would become the world's largest manufacturer within a decade. The point of scenarios is not to predict, but to prepare: to see which choices look good across many possible futures. That is how good founders, investors and governments think under uncertainty.

Learning goals

By the end of this month you should be able to:

- Explain the main tools of economic statecraft (tariffs, export controls, sanctions, investment screening) and why they work only where there is a chokepoint.
- Summarize the US-China trade conflict from 2018 to late 2026 (told in detail in Month 7) and judge whether the truce is stable.
- Analyse the 2026 Strait of Hormuz crisis as a system-wide shock, tracing its effects through energy, food, chips, data centers and finance.
- Evaluate the realistic options for middle powers and the Global South: alignment, hedging, bloc-building (including BRICS) and specialization.
- Build and use simple scenarios for 2026-2050, and apply them to Africa, Southeast Asia, the Gulf, Latin America and India.
- Give a reasoned judgment on whether Africa or Southeast Asia can build an ore-to-robot supply chain by the 2040s, and which segments are realistic first.
- Use the full twelve-month framework to decide where a founder can create and capture value, and draw the whole system from energy to geopolitics.

Week 1: The US-China rivalry and the tools of economic statecraft

Core lesson

From "engagement" to rivalry

For about thirty years after 1990, the dominant idea in Washington was "engagement": bring China into the world trading system, and it would grow richer and, many hoped, more open. China joined the World Trade Organization (WTO) in 2001. Its exports exploded, multinational companies built their supply chains around it (Month 7), and the two economies became deeply intertwined.

By the late 2010s, both governments had changed their view. American leaders concluded that China had not opened up politically, that its industrial policies (Month 3) gave its firms unfair advantages, and that its rise in advanced technology threatened US leadership. Chinese leaders concluded that dependence on American technology, especially chips, was a strategic weakness that the United States would exploit. Each side began to see interdependence not as insurance but as vulnerability.

The result is a rivalry fought largely with economic tools. It is not a "new Cold War" in the strict sense: the Soviet Union and the United States barely traded, while the US and China still traded hundreds of billions of dollars of goods in 2026. It is better described as **managed rivalry within interdependence**, with periods of escalation and truce.

The toolkit

Governments use five main economic weapons. Each is worth understanding as a mechanism.

1. **Tariffs**: taxes on imports. They raise the price of foreign goods, protect domestic producers, and raise government revenue. Who pays? Mostly the importing country's companies and consumers, unless the exporter cuts its prices to stay competitive. Studies of the 2018-2019 US tariffs found they were almost completely passed through to American importers and consumers (Amiti, Redding and Weinstein; Month 7).
2. **Export controls**: bans or license requirements on selling certain goods or technologies abroad. The US uses these mainly for advanced chips, chipmaking equipment and AI technology. China uses them for critical minerals: gallium, germanium, graphite, antimony and rare earths.

3. **Sanctions**: bans on dealing with named people, companies or countries, often enforced through the dollar banking system (Month 11's Russia case study).
4. **Investment screening**: reviewing or blocking foreign investment in sensitive sectors. The US Committee on Foreign Investment in the United States (CFIUS) screens inbound investment; new US rules restrict American investment into Chinese AI, chips and quantum firms. China introduced new rules on its own companies' outbound investment in 2026.
5. **Entity lists**: lists of specific foreign companies that may not receive certain goods. The US "Entity List" has been used against Huawei and hundreds of other Chinese firms.

The key insight, developed by political scientists Henry Farrell and Abraham Newman, is called **weaponized interdependence**. These tools only work where one side controls a **chokepoint**: a node in the network that others cannot easily avoid. The United States controls the dollar payment system and much of the world's most advanced chip technology (with allies such as the Netherlands, Japan, South Korea and Taiwan). China controls the processing of many critical minerals and the manufacture of rare earth magnets. A tariff on toys is weak, because toys can be made in many places. A ban on rare earth magnets or advanced chips is strong, because alternatives take years to build.

This connects directly to our value-capture question. In Month 1, we saw the "smile curve": value concentrates at the ends of the chain (design and brands on one side, distribution and services on the other), not in the middle (assembly). Week 1 adds a second rule: **power concentrates at chokepoints**, wherever they sit on the curve. Rare earth refining is not glamorous and earns modest margins in normal times, but in a crisis it decides whether the world's car and robot factories run.

Timeline: 2018 to November 2026

Month 7 (Week 4) tells the tariff story in detail. Here we keep only the milestones that matter for strategy:

Date	Event
2018-2019	First Trump administration tariffs on roughly $360-376 billion of Chinese goods (estimates vary); China retaliates
2019-2024	US restricts Huawei; from October 2022, sweeping US export controls on advanced chips and chipmaking tools, tightened in 2023 and 2024
2023-2025	China restricts gallium, germanium, graphite and antimony, then rare earths (April and October 2025)
April-May 2025	"Liberation Day" tariffs; US-China tariffs escalate to triple digits, then a 90-day pause after Geneva talks
October 30, 2025	Busan truce for one year; China suspends its October 2025 rare earth controls
February 20, 2026	Supreme Court rules 6-3 (Learning Resources, Inc. v. Trump) that IEEPA does not allow tariffs; a temporary 10% Section 122 tariff follows on February 24

Date	Event
July 24, 2026	Section 301 "forced labor" tariffs of 10% or 12.5% on 60 economies replace Section 122 (China and Vietnam 12.5%, Indonesia and India 10%)
September 23-25, 2026	Xi state visit to Washington: truce extended two months to January 10, 2027; Boards of Trade and Investment begin work; a US-China AI dialogue is set up
November 18-19, 2026	APEC leaders' meeting in Shenzhen, where Trump and Xi are expected to meet again

The February 2026 ruling is Month 2's lesson about institutions in action: the Court held that IEEPA's power to "regulate ... importation" does not include the power to tax imports, which belongs to Congress. The administration rebuilt tariffs under other laws (Sections 122, 232 and 301); Vietnam's 20% "reciprocal" rate, for example, became a 12.5% Section 301 tariff. Courts constrained the tool but did not end the policy. For businesses, trade rules in 2026 are made and remade through law, politics and negotiation at the same time.

What is negotiable and what is not

The most important analysis of the September 2026 summit, by the law firm Wilson Sonsini, makes a distinction worth remembering. The summit relaxed tariffs on "non-sensitive" goods such as toys, small appliances, farm products and cosmetics. But "many regulatory trade controls related to national security ... remain firmly in place", and "neither side appears to have scaled back its recent expanded use of many such controls." The US Trade Representative described advanced US semiconductors as the "crown jewels" that were outside the talks. China, meanwhile, temporarily banned helium exports in July 2026 and tightened controls on drone-related items destined for the United States in August 2026.

So the rivalry has two layers:

- **The commercial layer** (consumer goods, farm products, ordinary manufacturing): negotiable, with tariffs going up and down as bargaining chips.
- **The strategic layer** (advanced chips, AI, critical minerals, magnets, defense-related technology, data): largely non-negotiable, with controls expanding on both sides.

Judgment: the most likely path through 2030 is a continued pattern of short truces in the commercial layer and steady hardening in the strategic layer. The original Busan truce ran for a year; its first extension, in September 2026, ran for only two months. A grand bargain that removes export controls on chips and minerals is unlikely while both governments see technology leadership as a question of national security. For countries and companies outside the two powers, this means the strategic layer of the value chain (chips, AI compute, magnets, battery materials) will increasingly split into two partly separate systems, while the commercial layer stays more global.

Who gains from the split?

When two giants restrict trade with each other, some third countries benefit as production moves. This is the "China plus one" strategy of Month 7. The clearest winners so far:

- **Vietnam**: total exports rose 17% in 2025 to $475 billion, with exports to the United States of about $153 billion, up 28% (Vietnam's statistics office, via Reuters and The Diplomat). Chinese investment in Vietnam rose sharply after the 2018-2019 tariffs (Federal Reserve), but most of the 2026 export surge came from electronics made by firms such as Samsung, LG, Intel and Apple's suppliers (CSIS).
- **Mexico**: benefited from proximity to the US and the USMCA trade agreement, but on July 1, 2026 the United States declined to renew the USMCA for a fresh 16-year term. It stays in force with annual reviews and expires on July 1, 2036 unless all three countries extend it (USTR; Month 7), which has made long-term investment more cautious.
- **India**: growing electronics assembly (especially smartphones) and fast overall growth, which the IMF projected at 7.0% for 2026 (IMF WEO October 2026; India's figures are on a fiscal-year basis).

But the winners face a catch. The US has introduced higher tariffs for "transshipment", goods that are merely relabeled after passing through a third country, and rules of origin that require real local value. Countries that only assemble Chinese components gain less, and face more political risk, than those that build genuine local supply chains. This is Month 6's downstreaming lesson in a new form: **capturing value from geopolitical shifts requires real capability, not just a different flag on the box**.

Case study: rare earths as a weapon, 2025-2026

Months 5, 7 and 9 told this story from the mineral, trade and robot sides; here we look only at what it teaches about leverage. Rare earth elements are seventeen metals. Four of them, neodymium, praseodymium, dysprosium and terbium, are used to make the powerful permanent magnets inside electric vehicle motors, wind turbines, hard drives, drones, missiles, and the servo motors that move robot joints (Month 9). A humanoid robot may contain dozens of such motors.

The chokepoint. According to the International Energy Agency (IEA), in 2024 China accounted for about 60% of the world's mined production of magnet rare earths, 91% of refining, and 94% of sintered permanent magnet production. Two decades earlier, China made only about half of the world's permanent magnets. Month 5 explains how state support, tolerance of toxic and sometimes radioactive refining waste, and low prices built that dominance.

The weapon used, and the leverage it bought. Month 7 (Week 4) tells the story of the 2025 magnet squeeze. In short: in April 2025 China required export licences for seven heavy rare earths and magnets, and in October 2025 it announced controls reaching even foreign-made parts containing Chinese-sourced rare earths. Some car factories in Europe, the US and elsewhere cut output or paused lines for lack of magnets (IEA). The October controls arrived weeks before the Busan summit, where China suspended them for a year in exchange for US tariff cuts and the suspension of a US rule on subsidiaries of listed Chinese companies. Many analysts concluded that the rare earth threat was China's most effective bargaining tool. The IEA estimated in 2026 that if the controls were implemented in full, up to $6.5 trillion a year of downstream production outside China would be at risk.

The response. Japan, the United States, Australia, the European Union and others accelerated support for mines, refineries and magnet plants outside China. The IEA reported that China's share of rare earth refining fell slightly in 2025, with new US projects and higher output in Malaysia. But the IEA also warned that by 2035, existing and announced projects outside China would cover only about half of mining needs, a quarter of refining needs, and less than a fifth of magnet demand

outside China. It estimated that about $60 billion of investment over a decade would be needed to build diversified supply chains, which is "modest compared with the scale of potential economic losses."

Lessons for our driving question.

1. **Midstream processing is where power hides.** Most people think of mines (upstream) or finished products (downstream). The chokepoint was in refining and magnet-making, the middle of the chain.
2. **Chokepoints decay when used.** Each use of the weapon gives others a reason and political support to build alternatives. China's share of refining is already slipping. But building alternatives takes a decade, which is why the weapon was effective in 2025.
3. **This is an opening for third countries.** Malaysia already hosts the largest rare earth refinery outside China (operated by Australia's Lynas). Vietnam, Brazil and several African countries have rare earth deposits. A country that offers reliable, environmentally responsible processing could capture a strategic role that markets and governments will pay a premium for. But the technology, skills and equipment are concentrated in China, and the environmental costs are real.

Key numbers

Figure	Value	Year	Source
China share of mined magnet rare earths	about 60%	2024	IEA
China share of rare earth refining	91%	2024	IEA
China share of sintered permanent magnets	94%	2024	IEA
Economic activity outside China at risk from full controls	up to $6.5 trillion per year	2026 estimate	IEA
Investment needed to diversify rare earth chains	about $60 billion over a decade	2026 estimate	IEA
US-China truce expiry after two-month extension	January 10, 2027 (was November 10, 2026)	September 2026	Wilson Sonsini / Reuters
Reciprocal tariff cuts agreed at Washington summit	about $30 billion of goods each way	September 2026	Wilson Sonsini
Supreme Court vote striking down IEEPA tariffs	6-3	February 20, 2026	AP / Congressional Research Service
Vietnam total exports	$475 billion (+17%)	2025	Vietnam statistics office via Reuters
Vietnam exports to the US	about $153 billion (+28%)	2025	Vietnam statistics office via The Diplomat

Think like a founder

- **Map your chokepoints before your competitors do.** For any product, list every input that comes from one country or one company. A robot startup that depends on Chinese magnets, Japanese reducers and US chips (Month 9) has three geopolitical risks. Designing products that can switch suppliers, or that use less of the scarcest inputs (for example, motors that need less dysprosium), is a competitive advantage.
- **Compliance is a product.** Export controls, sanctions, rules of origin and new requirements such as the European Union's battery passport create demand for software and services that track where materials come from. Every manufacturer in Vietnam, Mexico, Morocco or Indonesia that wants to sell to the US or EU must prove origin. Traceability is a growing market.
- **The value capture lesson:** in a fragmenting world, the "trusted supplier" premium becomes real. Customers will pay more for a supply chain they believe will not be cut off. Countries and companies that can credibly offer neutrality, transparency and reliability can capture value that pure cost competition would not give them.

Week 2: Blocs, chokepoints and the choices of the Global South

Core lesson

Physical chokepoints: the geography of the material economy

Week 1 dealt with technological chokepoints. The material economy also has geographic ones: narrow places where a large share of the world's trade must pass. Month 7 introduced them. The most important are:

- **The Strait of Hormuz**, between Iran and Oman: in normal times about 20 million barrels a day of crude oil and oil products, roughly a fifth of world oil consumption and a quarter of seaborne oil trade (EIA; IEA), plus large volumes of liquefied natural gas from Qatar.
- **The Strait of Malacca**, between Indonesia, Malaysia and Singapore: the main route for oil and goods between the Indian Ocean and East Asia, including most of China's oil imports.
- **The Suez Canal and Bab el-Mandeb**, linking Europe and Asia through the Red Sea.
- **The Taiwan Strait**: not a trade chokepoint in the usual sense, but next to the island that produces the large majority of the world's most advanced chips (Month 8).
- **The Panama Canal**, linking the Atlantic and Pacific.

The economics of a chokepoint are simple. Ships can take longer routes, but the extra distance costs time, fuel and insurance, and some routes have no practical alternative at all. Pipelines can bypass a strait, but only up to their capacity. When a chokepoint closes, prices jump far more than the share of supply lost, because short-term demand for oil, gas and food is very inelastic: people and factories cannot quickly use less.

Case in motion: the 2026 Iran war

The war that began with US and Israeli strikes on Iran on February 28, 2026 is the largest test of the material economy since Russia's 2022 invasion of Ukraine; the case study below follows it through the system. The IMF's October 2026 World Economic Outlook projected that the Middle East and Central Asia region would shrink by 0.3% in 2026, with Saudi Arabia contracting 3.3% (after 4.6% growth in 2025), the UAE 3.9%, and Kuwait, Qatar and Bahrain facing double-digit

falls, while world growth slowed to 3.1%. The IMF assumed Hormuz would reopen gradually from March 2027 (IMF; The National). Saudi Arabia did better than its neighbours for a physical reason: its East-West pipeline lets it export from the Red Sea.

Blocs: who lines up with whom?

When great powers compete, other countries face pressure to choose. In practice, very few choose fully. The world of 2026 looks less like two blocs and more like a set of overlapping clubs:

Grouping	Members (simplified)	What it does
G7	US, Japan, Germany, UK, France, Italy, Canada (+EU)	Coordinates rich democracies; sanctions; critical minerals partnerships
BRICS	Brazil, Russia, India, China, South Africa, Egypt, Ethiopia, Iran, UAE, Indonesia, and Saudi Arabia (listed, but has not formally accepted)	Forum for large emerging economies; New Development Bank; talks on payments
ASEAN	11 Southeast Asian states	Regional trade and diplomacy; formally neutral
African Union and AfCFTA	55 African states	Continental free trade area, still being implemented
Bilateral and minilateral deals	US-Japan, US-UAE AI partnership, China-ASEAN, etc.	Specific security, technology and resource deals

BRICS deserves careful treatment, because claims about it are often exaggerated. It expanded in 2024 (Egypt, Ethiopia, Iran and the UAE, with Saudi Arabia invited at the same time) and Indonesia joined as a full member in 2025, so official BRICS documents count 11 members. Even membership is fuzzy: Saudi Arabia has never formally accepted its 2023 invitation, and its foreign minister took part in the 2026 New Delhi summit as an invited country, although BRICS chairs including India list it as a member (Indian Express, September 15, 2026). The group also has ten "partner countries", including Malaysia, Thailand, Vietnam and Nigeria. Intra-BRICS exports reached about $1.17 trillion in 2024 (UNCTAD, as cited by the Hindu BusinessLine). But BRICS members disagree on core questions. At the New Delhi summit on September 12-13, 2026, India pushed hard for linking members' payment systems and central bank digital currencies. The final declaration acknowledged the work of a BRICS Payment Task Force but, as the Business Standard and Rediff reported, omitted any firm commitment to payment interoperability. India's government repeated that there is no proposal for a BRICS currency. On the Iran war, the declaration called for restraint but avoided condemning either side, reflecting divisions among members that include both Iran and the Gulf states it had attacked.

Judgment: BRICS is a useful forum for emerging economies to coordinate and to signal that they will not simply follow Washington. It is not a bloc that can replace the dollar system or act together in a crisis, because its members are often rivals (India and China) or on opposite sides of a war (Iran and the UAE or Saudi Arabia).

The four strategies of middle powers

How should a country that is neither the United States nor China navigate this world? There are four broad strategies, and most countries mix them.

1. **Alignment**: choose one side for security and technology. Japan, South Korea and the Philippines rely on US security guarantees and accept US technology controls. The UAE, after a long debate, broke with Chinese technology suppliers to gain access to advanced US AI chips: in July 2026, the US Commerce Department eased export rules for the UAE, and Nvidia chips began arriving for the Stargate UAE data center cluster.
2. **Hedging**: keep good relations with both, avoid hard choices, and extract benefits from each. This is the traditional strategy of most of Southeast Asia, including Indonesia, Vietnam, Malaysia and Thailand, and of India, Brazil, Turkey and much of Africa and the Gulf. Indonesia joined BRICS in 2025 while also courting US investment; its sovereign fund Danantara (Month 11) invested in a US oil producer in 2026.
3. **Bloc-building**: create regional groups that increase bargaining power. ASEAN, the African Continental Free Trade Area (AfCFTA) and Gulf cooperation are examples. Small countries negotiate better together.
4. **Specialization in indispensability**: become so important at one point in the chain that both sides need you. Taiwan in advanced chips, the Netherlands in lithography machines (ASML), Singapore in finance and logistics, and possibly Indonesia in nickel (Month 6) are examples. Being indispensable gives protection, but also makes you a target.

The best strategy depends on geography, size and capabilities. **Hedging works best for countries that have something both sides want**. Indonesia's nickel, India's market and talent, Vietnam's factories, the Gulf's capital and energy, and Brazil's food and minerals give them bargaining power. Countries without such assets have less room.

Value capture in a fragmented world

Fragmentation has costs. An IMF staff study (2023) estimated that trade fragmentation could cost from 0.2% of world output in a mild scenario to up to 7% in a severe one, and that adding technological decoupling could push losses to 8-12% of GDP in some countries. Emerging and low-income economies are most at risk, because they lose access to the cheapest technology and the knowledge that flows with trade. But fragmentation also redistributes value:

- **Countries that host "trusted" capacity** (Vietnam, Mexico, India, Morocco, Malaysia) capture investment that would otherwise have gone to China.
- **Owners of chokepoints** capture rents and bargaining power, until alternatives are built.
- **Duplication creates demand**: building a second rare earth chain, a second chip supply chain and second data centers for "sovereign AI" means more orders for equipment, construction and engineering, often paid for by governments.
- **Consumers and the poorest countries pay**, through higher prices and lost efficiency.

Case study: the Strait of Hormuz, 2026, a stress test of the whole system

The shock, in brief. Month 4 (Week 2) tells the war's oil story in detail. The essentials: after US and Israeli strikes on Iran on February 28, 2026, traffic through Hormuz largely stopped. Brent crude, about $72 a barrel the day before, rose 51% in March alone (CNBC), and front-month Brent futures peaked at $126.41 on April 30 (Reuters). By April the IEA counted global supply losses of

12.8 million barrels a day since February, with output from the affected Gulf countries 14.4 million barrels a day below pre-war levels (IEA Oil Market Report, May 2026). An April ceasefire and a June memorandum briefly brought prices back to pre-war levels, but fighting resumed in July, and in September drones hit Saudi Arabia's East-West pipeline, the main route around Hormuz. Brent settled at $104.77 on November 2 on hopes of peace talks (Reuters).

How the shock travelled through the material economy. This is the point of the case study. Follow the chain:

1. **Energy (Month 4)**: oil and LNG prices jumped worldwide. Energy-importing countries, including India, Japan, Korea, much of Europe and Indonesia, faced higher import bills and weaker currencies.
2. **Food**: the Gulf exports large amounts of fertilizer, especially urea made from natural gas. Disruptions raised fertilizer costs, feeding through to food prices months later.
3. **Minerals and industry (Months 5-7)**: higher energy prices raised the cost of energy-intensive smelting and refining. Shipping and insurance costs rose on routes near the Gulf.
4. **Chips and AI (Month 8)**: Iranian strikes damaged Amazon Web Services data centers in Bahrain and the UAE in March 2026. Iran warned that Stargate UAE could be a target. Yet the Gulf's AI ambitions continued: Saudi Arabia's Humain set out plans in September 2026 for about 6 gigawatts of data center capacity by 2034, and the first Nvidia chips arrived in the UAE.
5. **Finance (Month 11)**: the oil shock pushed up inflation. The Federal Reserve raised rates in September 2026 for the first time since 2023, and the 10-year US Treasury yield rose above 5%. Emerging-market currencies, including the rupiah, weakened.
6. **Geopolitics**: the war strained BRICS unity, deepened Gulf-US technology ties (the Middle East Institute argued the war had acted as a catalyst for closer Gulf-US AI cooperation), and gave both the US and China new reasons to secure energy routes.

What it teaches.

- **Energy is still the base layer.** A shock to one strait moved interest rates in Washington and the price of a loan in Semarang.
- **Diversified routes are insurance.** Saudi Arabia's East-West pipeline and the UAE's pipeline to Fujairah, built decades earlier, limited the damage.
- **Electrification reduces exposure, but creates new dependencies.** A country running on domestic solar, wind and batteries is less exposed to Hormuz, but more exposed to China's dominance of solar panels, batteries and magnets (Week 1). There is no dependency-free energy system, only a choice of which dependencies to manage.
- **The Gulf is trying to move up the chain.** Even under fire, Gulf states kept investing in AI compute. Their bet is that oil money, cheap energy and US partnerships can turn energy exporters into compute exporters (Week 3).

Key numbers

Figure	Value	Year	Source
Oil through Hormuz in normal times	about 20 million barrels/day (about 20% of consumption)	2024-2025	EIA; IEA
Brent before the war	about $72/barrel	February 27, 2026	CNBC / BBC

Figure	Value	Year	Source
Brent rise in March 2026	+51% in one month	March 2026	CNBC
Brent futures peak	$126.41/barrel	April 30, 2026	Reuters
Global oil supply loss since February	12.8 million barrels/day	April 2026	IEA Oil Market Report
Brent after peace-talk hopes	$104.77/barrel (settle)	November 2, 2026	Reuters
World growth forecast	3.1%	2026	IMF WEO October 2026
Middle East and Central Asia growth	-0.3%	2026	IMF WEO October 2026
Saudi Arabia growth	-3.3%	2026	IMF WEO October 2026
BRICS members	11 listed, Saudi Arabia not formally accepted (plus 10 partner countries)	2026	BRICS India 2026; Indian Express
Intra-BRICS exports	about $1.17 trillion	2024	UNCTAD via Hindu BusinessLine

Think like a founder

- **Build for shocks, not averages.** A business plan that assumes oil at $70 and interest rates falling would have been wrong twice in 2026. Stress-test your costs at oil $120, your currency 20% weaker and interest rates 2 points higher. If the business survives, investors will trust it.
- **Energy independence is a product.** Factories, cold-storage operators and data centers in energy-importing countries now value on-site solar, batteries and efficiency more than before. Selling and financing "energy-as-a-service" to small and medium enterprises is a business that both cuts costs and reduces national exposure to chokepoints.
- **The value capture lesson:** in a world of shocks, value flows to those who supply resilience: alternative routes, backup capacity, storage, insurance and flexibility. These businesses are boring in calm years and essential in crises. Founders who build them should price for the crisis, not the calm.

Week 3: Scenarios to 2050

Core lesson

How to think about the future without pretending to predict it

Scenario planning was developed in the 1960s and 1970s, most famously at the oil company Shell, whose planners had considered a scenario of sudden oil supply cuts before the 1973 oil shock, and so reacted faster than rivals. The method is simple:

1. Identify the **driving forces** that will shape the future.
2. Separate **near-certainties** (things that will almost surely happen, such as population aging) from **critical uncertainties** (important things that could go either way).

3. Pick the two most important uncertainties and cross them, producing four distinct, internally consistent futures.
4. For each future, ask: who wins, who loses, what signals would tell us we are heading there?
5. Look for **robust strategies**: choices that do reasonably well in all four futures.

Near-certainties to 2050

Some trends are slow and predictable enough to treat as near-certain:

- **Demography.** The UN's World Population Prospects 2024 projects Africa's population rising from about 1.5 billion in 2024 to about 2.5 billion by 2050, when more than one in four people on Earth will be African. Sub-Saharan Africa is expected to add about 625 million working-age people over the next quarter-century, more than three-quarters of the global increase (World Bank analysis via VoxDev). East Asia's working-age population, including China's, will fall significantly; India will age later; Thailand and Vietnam are aging quickly.
- **Electrification.** Electricity's share of final energy use will keep rising, driven by electric vehicles, heat pumps, industry and data centers (Months 4 and 8). This raises demand for copper, aluminum, nickel, lithium, graphite and rare earths (Month 5).
- **Automation.** Industrial robot installations reached a record 603,307 units in 2025, up 11%, and the global stock of operating industrial robots passed 5 million, according to the International Federation of Robotics (IFR). The IFR forecasts about 806,000 annual installations by 2029. Whatever happens with humanoid robots, the use of machines in factories, warehouses and farms will rise.
- **Aging in rich countries and China**, which creates labor shortages and demand for automation, care services and migration.

Critical uncertainties

Two uncertainties matter most for our driving question:

1. **The US-China relationship**: does it stay a managed rivalry with trade in most goods (as in 2026), or harden into a deep split with two separate technology systems and frequent use of economic weapons?
2. **The speed of AI and robot diffusion**: do AI and robots spread fast and cheaply around the world, making robots affordable to small factories in Nairobi or Surabaya by the 2030s? Or do they stay expensive, concentrated in rich countries and China, limited by energy, chips, skills and regulation?

Crossing them gives four scenarios:

	Fast, cheap diffusion of AI and robots	**Slow, concentrated diffusion**
Managed rivalry	**A. "Open Automation"**: Chinese and other suppliers sell cheap robots and open AI models worldwide; Southeast Asia and parts of Africa adopt quickly; manufacturing spreads but needs fewer workers per factory	**B. "Slow Convergence"**: trade stays open, but robots and frontier AI stay expensive; labor-cost advantages still matter; Africa and South Asia can still follow the classic export-manufacturing path, more slowly
Deep split	**C. "Two Stacks"**: two technology systems (US-led and China-led) compete to sell cheap AI and robots to the Global South; countries that hedge well get both; those forced to choose pay more	**D. "Fortress World"**: blocs, high tariffs, export controls and expensive technology; supply chains shorten and regionalize; poorer countries are squeezed out of both markets and technology

Which is most likely? In late 2026, the evidence points toward a mix of A and C: the US-China relationship is unstable but not broken, and robot prices are falling fast. Chinese vendors made about 97% of global humanoid shipments in the first half of 2026 (Smart Analytics Global), and Unitree's revenue per humanoid fell from about 593,400 yuan in 2023 to about 166,400 yuan in 2025 (its prospectus; Month 9). Industrial-grade machines still cost far more than entry models, but the trend is steep.

Judgment: plan for a world that looks like scenario C, "Two Stacks", with periods that feel like A. That means cheap robots and AI will be available to emerging economies, but technology choices will carry political weight, and supply chains for the strategic layer (chips, magnets, battery materials) will be partly duplicated. The worst scenario for poorer countries, D, is possible but not the base case.

Regional scenarios

Below, each region gets a short assessment: its starting position in 2026, its main assets, its main constraints, and a "plausible good path" and "plausible bad path" to 2050. Growth forecasts are from the IMF's October 2026 World Economic Outlook unless stated.

Africa. Starting point: sub-Saharan Africa is projected to grow 4.5% in both 2026 and 2027, with resource exporters doing better than non-resource economies in the oil shock (IMF). It holds huge mineral wealth: copper and cobalt in the DRC and Zambia, platinum and manganese in South Africa, graphite in Mozambique and Tanzania, bauxite in Guinea, lithium in Zimbabwe, phosphates in Morocco (Months 5-6). It also has the world's fastest-growing workforce. Constraints: weak electricity supply in many countries, high cost of capital (Month 11), small and fragmented national markets, poor transport links, and institutions that range from strong to very weak (Month 2).

- *Good path*: AfCFTA slowly creates larger markets; a few anchor countries (Morocco, Egypt, Kenya, Ethiopia, Rwanda, South Africa, Nigeria, Ghana, Cote d'Ivoire) build industrial corridors with reliable power; mineral countries capture first-stage processing (copper cathode,

cobalt hydroxide refining, battery precursors) through projects like the Lobito Corridor; cheap solar and batteries solve power shortages in industrial zones; Africa becomes the main growth market for cheap robots and AI services.
- *Bad path*: automation in rich countries and China removes the labor-cost advantage before African manufacturing takes off ("premature deindustrialization", a term from economist Dani Rodrik); minerals are exported raw; debt crises recur; young people face unemployment and migration pressure.

Morocco is the best example of the good path so far (Months 6 and 7 tell its story). Its automotive exports reached 159.2 billion dirhams (about $17 billion) in 2025, its largest export (Office des Changes). Gotion High-Tech, in which Volkswagen is a major shareholder, began commissioning its lithium iron phosphate (LFP) battery plant in Kenitra in the week of October 20, 2026; Gotion describes a 20 gigawatt-hour first phase, while the African Development Bank, which lent 100 million euros, describes the financed first phase as 10 gigawatt-hours (Morocco World News; AfDB). Morocco combines phosphate (a raw material for LFP batteries), access to Europe and decades of industrial-zone building.

Southeast Asia. Starting point: the IMF raised its 2026 growth forecast for the five largest ASEAN economies (Indonesia, Malaysia, the Philippines, Singapore, Thailand) to 4.4%, helped by the AI and electronics boom. Malaysia is a major hub for chip assembly and testing. Indonesia dominates world nickel supply and has built a large nickel processing industry (Month 6), mostly with Chinese capital and technology. Thailand is a regional car-making hub now attracting Chinese electric vehicle makers. Constraints: most high-value components, equipment and technology are still foreign-owned; Indonesia's capital markets and governance face investor doubts (Month 11's MSCI case); countries compete with each other for the same investment.

- *Good path*: the region becomes the main "plus one" manufacturing base, deepening local supplier networks; Indonesia moves from nickel matte toward battery cells and motors; Vietnam and Malaysia move from assembly toward components and design; Singapore provides capital, law and engineering services; ASEAN cooperation creates a regional robot and EV supply chain.
- *Bad path*: the region stays an assembly platform for Chinese, Japanese, Korean and US firms; local firms capture little; transshipment rules and US tariffs punish it; environmental damage from nickel processing triggers backlash; aging (in Thailand and Vietnam) arrives before high income.

The Gulf. Starting point: the war hit hard in 2026, with the IMF projecting a 3.3% contraction for Saudi Arabia in 2026 and a strong rebound in 2027. Assets: cheap energy, very large sovereign wealth funds (Month 11), and new technology partnerships with the US. Constraints: physical vulnerability to Iranian attack, small domestic labor forces, and dependence on US chips and security.

- *Good path*: the Gulf becomes a global hub for AI compute, powered by cheap gas and solar, and a financier of mining and processing in Africa and Asia; oil revenue is converted into ownership of the next layers of the value chain.
- *Bad path*: recurring conflict deters investment; giga-projects fail; the oil-to-compute bet is undercut by cheaper compute elsewhere or by energy-efficient AI.

Latin America. Starting point: slow growth, with the IMF projecting 2.3% for the region in 2026, 2.2% for Brazil and 1.5% for Mexico. Assets: lithium (Chile, Argentina, Bolivia), copper (Chile, Peru), food, renewable energy potential, and Mexico's position next to the US market. Constraints: the USMCA's uncertain future, weak productivity growth, and a long history of commodity cycles (Month 3's failures).

- *Good path*: Mexico deepens its integration with the US in electronics, cars and medical devices; Chile and Argentina capture lithium refining with cheap solar power; Brazil uses clean power for green industry.
- *Bad path*: the region repeats the commodity cycle, exporting raw lithium and copper during booms and stagnating during busts.

India. Starting point: the fastest-growing major economy, with the IMF projecting 7.0% growth in 2026 and 2027, and nominal GDP of about $4.1 trillion. Assets: a huge domestic market, engineering and software talent, digital public infrastructure (the UPI payments system), and growing electronics assembly. Industrial robot installations reached 10,500 in 2025, up 15% (IFR). Constraints: manufacturing remains a small share of the economy, logistics and land are difficult, and its workforce is large and mostly low-skilled.

- *Good path*: India becomes the second great manufacturing base after China, specializing in electronics, pharmaceuticals and engineering, and becomes a major supplier of AI services and software for the Global South.
- *Bad path*: India's growth stays service-led and its manufacturing ambitions stall, leaving too few good jobs for its young workforce.

Signposts: how to tell which future is arriving

A scenario is only useful if you can tell, year by year, which one you are drifting toward. Shell's planners called these early warning signs "signposts". For our two critical uncertainties, here are signposts you can check in the news every few months, with the reading in late 2026.

Signpost	Points toward a deep split (C or D)	Points toward managed rivalry (A or B)	Reading in late 2026
Length of US-China truces	Truces lapse, or are renewed for weeks	Multi-year deals with working institutions	Mixed: the one-year Busan truce was extended by only two months, but new Boards of Trade and Investment exist
Export controls on chips and minerals	New controls added every year on both sides	Controls frozen or narrowed	Toward split: Wilson Sonsini found neither side scaled back its controls
Rules of origin and transshipment penalties	Stricter "local content" rules for third countries	Third countries treated as neutral	Toward split: Section 301 tariffs now vary by country (10% or 12.5%)
Payment systems	Large trade settled outside the dollar system	Dollar share of FX trading stable	Toward managed: dollar on one side of 89% of FX trades (Month 11)

Signpost	Points toward fast diffusion (A or C)	Points toward slow diffusion (B or D)	Reading in late 2026
Price of a basic humanoid or cobot	Falls 20-30% a year	Flat or rising	Fast: Unitree's revenue per humanoid fell about 72% from 2023 to 2025
Open-weight AI models	Near-frontier models free to download and fine-tune	Best models closed and export-controlled	Mostly fast, with growing political limits
Robot installations outside China, Japan, Korea, US, Germany	Double-digit growth for years	Stagnant	Slow so far: all of Africa installed about 519 industrial robots in 2025 (Month 9)
Electricity cost and reliability in emerging industrial zones	Cheap solar plus storage in industrial parks	Blackouts and diesel back-up	Mixed: solar and battery costs keep falling (Month 4), grids lag

Two lessons come out of this table. First, the world in 2026 is not in one box. It is moving toward a split in the strategic layer and toward fast diffusion in the price of hardware, which is why the judgment above chose scenario C with periods that feel like A. Second, signposts disagree, and that is information: the cheapest robots in history are arriving with the hardest trade rules in decades, so a founder can sell cheap hardware into markets that need help proving origin, compliance and safety.

Running one strategy through the four futures

The real use of a scenario grid is to stress-test a decision. Take a decision that many governments in Africa and Southeast Asia are weighing: should the country (or a company in it) build a plant that turns local mineral ore into battery cathode material, the most valuable step between the mine and the cell (Months 5 and 6)? Run it through each box.

- **A, "Open Automation".** Demand for batteries and robots grows fast and trade is open. A cathode plant competes head-on with Chinese plants that are bigger, cheaper and highly automated. It survives only if it has a real cost edge (cheap clean power, ore next door) or a long contract with a carmaker. Risk: high. Reward if it works: moderate.
- **B, "Slow Convergence".** Demand grows more slowly and Chinese overcapacity keeps prices low, as in the battery-materials glut of 2023-2025. Margins are thin for years. This is the box in which many downstreaming plants lose money.
- **C, "Two Stacks".** Western carmakers pay a premium for cathode that is not made in China, and rules of origin reward local value added. A plant in Morocco or Indonesia with credible traceability can win long contracts at above-market prices. This is the best box for the plant, but only if the plant's ownership, technology and power supply are acceptable to the buyer's government.
- **D, "Fortress World".** Blocs close. A plant that depends on Chinese technology cannot sell to the US or EU, and one that depends on Western buyers may lose access to Chinese equipment and expertise. The plant becomes hostage to politics.

The plant is not robust: it does well in one box out of four. Now run a second decision through

the same grid: building the skills and companies that install, maintain and finance automation in local factories, mines and farms. It earns money in A (more machines to install), in B (fewer, more expensive machines that need careful use), in C (both stacks need local integrators) and even in D (machines that cannot easily be replaced must be kept running). That is what "robust" means: not the highest return in any one world, but an acceptable return in all of them.

This does not mean nobody should build cathode plants. It means that a plant is a bet on scenario C and should be designed for it: long offtake contracts signed before construction, traceable and audited supply chains, ownership structures acceptable to the main buyers, and cheap patient capital (Month 11), because a high-interest loan turns a bet that might pay off into one that cannot.

What would change the judgment

Three things would move this textbook's base case. A durable US-China agreement that narrows export controls on chips and minerals would push the world from C toward A, rewarding open, low-cost manufacturing and weakening the "trusted supplier" premium. A Taiwan crisis, or a long closure of the Strait of Malacca, would push it toward D in months rather than years, and every strategy that depends on one route or one supplier would be tested at once. And a breakthrough in robot learning that lets cheap hardware do useful work in messy environments without expensive integration would shrink the integration layer that this chapter recommends, and shift value toward whoever owns the best models and the most deployment data. A founder should check these three signposts as often as she checks her sales numbers.

Case study: can Africa or Southeast Asia build an ore-to-robot supply chain by the 2040s?

Step 1: break the robot into layers. From Month 9, a typical robot (industrial arm or humanoid) contains: structural parts (aluminum, steel, plastics); actuators (electric motors with rare earth magnets, precision reducers, bearings); power (batteries and power electronics); sensors (cameras, force sensors, encoders); compute (chips); and software and AI. Behind each sit the earlier layers: energy, mining, refining, and manufacturing equipment.

Step 2: score each region on each layer in 2026. The table below is a judgment, based on the evidence in Months 4-11 (scale: Strong, Emerging, Weak, Absent).

Layer	Southeast Asia (esp. Indonesia, Vietnam, Malaysia, Thailand)	Africa (esp. Morocco, South Africa, DRC/Zambia, Kenya)
Energy (cheap, reliable power)	Emerging (coal-heavy; solar growing)	Weak overall; Strong in a few places (Morocco, Ethiopia hydro, South Africa when grid works)
Mining of key minerals	Strong (nickel, tin, bauxite, some rare earths)	Strong (copper, cobalt, manganese, platinum, graphite, phosphate)
Refining and processing	Strong in nickel (Indonesia); Emerging in rare earths (Malaysia)	Emerging (copper cathode in DRC/Zambia; battery materials in Morocco)
Batteries	Emerging (Indonesia, Thailand cell plants)	Emerging (Morocco's Gotion plant from 2026)

Layer	Southeast Asia (esp. Indonesia, Vietnam, Malaysia, Thailand)	Africa (esp. Morocco, South Africa, DRC/Zambia, Kenya)
Motors and magnets	Weak to Emerging	Weak
Precision reducers, bearings	Weak	Absent to Weak
Electronics and chip packaging	Strong (Malaysia, Vietnam assembly and test)	Weak
Robot assembly	Emerging (some local integrators)	Weak
Software, AI and integration	Emerging (Singapore, Vietnam, Indonesia talent)	Emerging (Kenya, Nigeria, Egypt, South Africa)
Cost and depth of capital	Mixed (Singapore deep; others shallow)	Weak (high cost, Month 11)

Step 3: compare with the leader. China installed 354,000 industrial robots in 2025, 59% of the world total, and Chinese suppliers held 55% of their home market (IFR). It controls about 94% of rare earth magnet production (IEA), has a dense network of reducer, motor, sensor and battery suppliers, and has the scale that drives costs down (Month 7's economies of agglomeration). Any region trying to build a full ore-to-robot chain is competing with that.

Step 4: identify the realistic sequence. History (Month 3) shows that no country builds a whole chain at once. South Korea and Taiwan went from assembly to components to design over 30-40 years. China took about 25 years from export assembly to world-leading robot and EV industries. A realistic sequence for Southeast Asia or Africa:

1. *2026-2032*: robot **users and integrators** (installing, programming and maintaining imported robots for local factories, farms, mines and warehouses); first-stage **mineral processing**; **battery cells** and packs; **software and AI services**.
2. *2032-2040*: **components** with lower technical barriers: motor assembly, simple gearboxes, structural parts, battery management, sensors; robot **assembly** for regional markets using imported high-precision parts.
3. *2040-2050*: selective **high-precision components** (reducers, magnets) where a region has a resource or skill advantage; regional **robot brands** for specific uses (agriculture, mining, logistics) where local knowledge matters.

Step 5: judgment.

- **A complete, cheap, independent ore-to-robot chain, from mine to magnet to reducer to chip to finished robot, inside Africa or Southeast Asia by the 2040s is very unlikely.** No region outside China has it today, including the United States, Europe and Japan, each of which depends on imports for some layers. Chips alone make full independence unrealistic.
- **Southeast Asia has a realistic chance to build most of the chain except leading-edge chips by the 2040s**, because it already has strong mining and processing (Indonesia), electronics (Malaysia, Vietnam), car and battery industries (Thailand, Indonesia), capital (Singapore), and proximity to Chinese, Japanese and Korean suppliers who are relocating. The main risks are governance, the cost of capital, and remaining an assembly platform for foreign firms rather than building local champions.
- **Africa's realistic path is narrower and slower**: become a major processor of battery and magnet raw materials (copper, cobalt, manganese, graphite, phosphate, rare earths), build

battery and vehicle assembly in a few anchor countries (Morocco first), and become a large market and integration hub for cheap imported robots in agriculture, mining and logistics. Electricity and capital are the binding constraints; solve those, and the rest becomes possible.
- **The most valuable layer for both regions is not the hardware, but the integration and application layer**: knowing how to make robots and AI useful in local conditions, languages, crops, mines and factories. That layer has low capital requirements, captures value through services and software, and cannot easily be imported. It is also where a founder can start.

Key numbers

Figure	Value	Year	Source
Africa population	about 1.5 billion to about 2.5 billion	2024 to 2050	UN WPP 2024 via UNECA
Sub-Saharan Africa working-age increase	about 625 million	2025-2050	World Bank via VoxDev
Global industrial robot installations	603,307 (+11%; press release: more than 600,000)	2025	IFR World Robotics 2026
Global operational stock of industrial robots	about 5.08 million (press release: 5 million, +9%)	2025	IFR
China's robot installations	354,000 (59% of world)	2025	IFR
India's robot installations	10,500 (+15%)	2025	IFR
Global humanoid shipments	about 19,100 (97% Chinese vendors)	H1 2026	Smart Analytics Global
Unitree revenue per humanoid sold	about 166,400 yuan (about $23,400)	2025	Unitree prospectus via ICM Intelligence
Unitree R1 starting price	cut from 39,900 to 29,900 yuan	June 2026	Xinhua
Morocco automotive exports	159.2 billion dirhams (about $17 billion)	2025	Office des Changes annual report (Oct 2026)
India GDP growth forecast	7.0%	2026	IMF WEO October 2026

Think like a founder

- **Start where the chain is weakest locally but demand is strongest.** In Southeast Asia and Africa, the scarcest capability is not robot manufacturing; it is robot and AI integration for small and medium firms. That is a services business with low capital needs and a path to proprietary software and data.
- **Ride the scenarios that are robust.** Building integration capability, local-language AI, training and maintenance networks works in all four scenarios. Building a magnet factory works only in some, and needs state-level capital.

- **The value capture lesson:** in the 2030s and 2040s, value in emerging markets will be captured by those who control the customer relationship and the know-how of deployment, even if the hardware comes from China, Japan or Korea. Over time, the integrator with deep customer knowledge is well placed to move backward into components, just as many Asian manufacturers moved from assembly to design.

Week 4: Capstone. What should a founder build?

Core lesson

Five questions for finding value

After twelve months, we can compress the course into five questions a founder should ask about any business idea in the material economy:

1. **Where is the scarce capability?** Value and power gather at chokepoints (Week 1 of this month) and at the ends of the smile curve (Month 1, Week 3). Ask: what in this chain is hard to copy, slow to build, or controlled by few? If your business sits in the easy-to-copy middle, you will compete on price forever.
2. **Who owns the customer?** The company that understands the customer's problem best, and is trusted to solve it, can switch suppliers underneath. Hardware can be bought; relationships and know-how cannot.
3. **What is your cost of capital, and can you lower it?** Month 11 showed that a few percentage points of interest decide who wins in capital-heavy businesses. If you are in a capital-heavy business, you need cheap, patient money (development finance, sovereign partners, customers paying upfront). If you cannot get it, choose a lighter business.
4. **Is it robust across scenarios?** Will it still work if US-China relations worsen, oil doubles, interest rates rise, or robots get ten times cheaper? Robust businesses survive the futures you did not predict.
5. **How does it create measurable social value, and does that value strengthen the business?** For a for-profit social-impact company, the best designs are those where impact and profit come from the same action: a farmer's yield rises *because* she uses your service, a factory's workers earn more *because* your robots make them more productive (Month 10). If impact and profit pull in opposite directions, one will lose.

Seven opportunity areas, 2027-2040

These are not recommendations to start any particular company. They are areas where the analysis of this course suggests value will be created, and where a founder from Southeast Asia or Africa may have an advantage. Each comes with its value-capture logic.

Area	What it is	Why value is captured	Main risk
Robot and AI integration for small and medium firms	Selecting, installing, programming and maintaining robots and AI for factories, warehouses, farms and mines	Owns the customer and deployment know-how; hardware is a commodity input; recurring service revenue	Low barriers at first; must build proprietary tools and data

Area	What it is	Why value is captured	Main risk
Robotics-as-a-service and equipment finance	Renting robots, solar systems or machines monthly instead of selling them	Exploits the cost-of-capital gap (Month 11): whoever finances cheaply earns the spread	Needs cheap capital and good credit assessment
Local-language and domain AI	AI models and agents for local languages, regulations, crops, health systems and small businesses	Data and context that global firms lack; trust	Global AI firms may move in; must stay specialized
Energy-as-a-service for industry	On-site solar, batteries and efficiency for factories, cold chains and data centers	Reduces exposure to oil shocks (Week 2); predictable contracts	Capital-heavy; currency mismatch on imported panels and batteries
Traceability and compliance	Software to prove origin, carbon content and labor standards for minerals, batteries and manufactured goods	Required by US and EU rules; switching costs once embedded	Regulatory changes; needs credibility
Skills and certification	Training technicians, robot operators, AI engineers and quality inspectors	The binding constraint in every robot chain is skilled people (Month 10)	Hard to scale profitably; must be tied to jobs
Cross-border trade finance and payments	Payments, invoicing and working-capital finance for exporters and importers	Friction in non-dollar payments is costly (Month 11); network effects	Heavy regulation; must be licensed

Notice a pattern. Most of these are **light on capital and heavy on knowledge**, sit **close to the customer**, and **use** the hardware layers rather than trying to build them. Hardware matters, but a founder without cheap patient capital should enter where capital needs are lowest and learning is fastest, then move backward into hardware once there is revenue, data and credibility. This is the path many great industrial companies followed, as the case study shows.

Worked example: the unit economics of robots-as-a-service

Here is one opportunity area worked through with numbers. All numbers are hypothetical but realistic in order of magnitude; change them and redo the sums for your own case.

The idea. A start-up in a mid-sized industrial city in Southeast Asia or Africa installs robot cells (a Chinese-made collaborative arm, a gripper, safety equipment, programming and training) in small factories, and charges a monthly fee instead of selling the robot. This is robotics-as-a-service (RaaS), the model suggested in Month 11 for places where capital is expensive.

Step 1: the customer's side. Suppose a factory packs or finishes goods worth $40,000 a month on one line. The task is done by two operators, one per shift, each paid roughly $300 a month. A

robot cell rented at $750 a month costs more than the two wages it might replace. So in a low-wage economy the robot does not win on wages. It wins elsewhere. If the robot's consistency cuts the scrap and rework rate from 4% of output to 1%, the factory saves 3% of $40,000, or $1,200 a month. Add more stable output during labour shortages and festival seasons, and the deal is attractive even if no worker is fired: the two operators can be moved to quality control or machine minding (Month 10's augmentation path). This is the first lesson: **in low-wage markets, sell quality, consistency and safety, not labour replacement.** A pitch built on "replace your workers" will fail where workers are cheap, and it creates political enemies where they are not.

Step 2: the start-up's side. Suppose each installed cell costs the start-up $20,000 and lasts five years. Service, spare parts and insurance cost $2,200 a year per cell. Revenue is $750 x 12 = $9,000 a year if the cell is rented all year. The start-up must also pay for its capital, using the annuity formula from Month 11, Week 2.

Cost of capital	Annual capital cost per cell	Profit per cell per year (full use)	Profit per cell per year (85% use)
8% (development-finance loan in dollars)	about $5,000	about $1,800	about $440
14% (local bank loan in local currency)	about $5,800	about $970	about $380 loss
20% (expensive short-term credit)	about $6,700	about $110	about $1,240 loss

Read the table slowly. The product, the customer and the robot are identical in every row. Only the price of money changes, and it decides whether the business works at all. At 8% the start-up earns a margin of about 20% when every robot is busy. At 20% it barely breaks even, and it loses money as soon as one robot in seven sits idle between customers. This is Month 11's refinery example shrunk to the size of a start-up: **the cost of capital is part of the product.**

Step 3: the risks that hide in the spreadsheet.

1. **Currency mismatch.** The robot is bought in dollars or yuan; the rent is paid in rupiah, naira or dong. If the local currency falls 15%, the capital cost measured in local money rises 15%, which at 8% wipes out almost half the margin. Solutions: borrow in local currency where possible, price contracts with an exchange-rate adjustment clause for larger customers, or use the cheaper hedging that some central banks offer.
2. **Utilisation.** The single most important operating number is the share of robots earning rent. Idle robots still cost money. A RaaS business is really a logistics and sales business: moving cells quickly from one customer to the next.
3. **Customer credit.** Small factories sometimes pay late; a record of who pays is an asset that later supports cheaper borrowing.
4. **Supplier concentration.** If every cell uses the same Chinese arm, a new export control or tariff (Week 1) can stop the business. Qualify a second supplier early, even at a slightly higher cost.

Step 4: where the value capture really sits. After a few hundred installations, the start-up knows things no robot maker knows: which tasks robots handle well in local factories, which grippers fail in humid conditions, how long local operators take to learn, and which customers pay. That

knowledge lets it write better software, negotiate better hardware prices, and borrow more cheaply, because lenders can see a record of steady payments. This is how an integrator climbs the smile curve from the customer end (Month 1) and how, decades later, it might move backward into components, following the BYD path described in the case study below.

The general lesson for any idea in the table. Before you fall in love with a product, write down four numbers: what the customer gains each month, what you earn per unit, what your money costs, and what share of your assets is earning at any time. If the business only works at the cheapest cost of capital and full utilisation, it is not yet a business. Either find cheaper money (development finance, customer prepayment, a strategic partner) or redesign the offer until it survives the middle row of the table.

What a founder can control, and what they cannot

You cannot control: US-China relations, oil prices, the Fed's interest rate, the dollar's dominance, whether your country is in or out of an MSCI index, or whether there is a war in the Gulf.

You can control: where in the chain you enter; whether your costs and revenues are in the same currency; how much you depend on any single supplier or country; how quickly you reach positive cash flow; how clean your accounts and governance are (which lowers your cost of capital); which skills you build; and whether your product makes people more productive rather than simply replacing them.

The best founders spend their energy on the second list while watching the first.

A note on institutions and legitimacy

Month 2's conclusion still holds: in every Week 3 scenario, the difference between the good and bad path was usually institutional. Founders strengthen institutions not as political actors but as builders: companies that pay taxes, publish honest accounts, treat workers fairly and refuse bribes create constituencies for good rules.

Case study: BYD, from batteries to an integrated empire

The start. BYD was founded in Shenzhen in 1995 by Wang Chuanfu, a chemist, to make rechargeable batteries for mobile phones. It undercut Japanese battery makers by redesigning production lines to use more workers and simpler, home-made equipment. It entered the car business in 2003 by buying a small state-owned carmaker.

Moving along the chain. Over two decades, BYD built or bought capabilities across the electric vehicle stack: battery cells (its "Blade" lithium iron phosphate battery), electric motors, power electronics, its own power semiconductors (BYD Semiconductor makes insulated-gate bipolar transistors, or IGBTs, and silicon carbide modules), software, and complete vehicles, plus interests in lithium supply. A teardown of its Seal sedan by UBS in 2023 estimated that about 75% of the car's components were made in-house, against roughly 30% for a typical legacy carmaker, and Rhodium Group estimated in 2026 that BYD makes nearly 80% of its core components itself, more than double Tesla's share, saving about $2,369 per Seal in supplier mark-ups (CNBC, March 2026). During the chip shortage of 2021-2022, when many carmakers stopped production, BYD could supply its own power chips.

The result. In 2025, BYD sold 4,602,436 new energy vehicles (electric and plug-in hybrid), up 7.7%,

and remained the world's best-selling new energy vehicle brand for the fourth year. Its overseas sales passed 1 million for the first time, about 1.05 million, more than double the 2024 level, with factories being built or opened in Brazil, Thailand, Hungary, Turkey and elsewhere.

The limits. Vertical integration is not magic. BYD's net profit fell about 19% in 2025 as a brutal domestic price war squeezed margins (CnEVPost). Its overseas expansion faces tariffs in the US and EU and political suspicion. Integration also means BYD carries all the capital and all the risk.

Lessons for the course.

1. **Start with one component where you have a real cost or capability advantage.** BYD did not start by building cars. It started with batteries, where it could beat incumbents.
2. **Use each layer to fund and learn the next.** Batteries funded cars; cars created demand for in-house chips and motors.
3. **Integration captures value when the chain is immature or unreliable.** When suppliers are few, slow or politically risky, owning the layer is safer. When the chain matures, specialists may become cheaper again.
4. **The national ecosystem mattered.** BYD benefited from China's huge market, supplier networks, cheap capital and industrial policy (Months 3 and 7). But the sequencing lesson travels.

For a founder in Semarang, Nairobi or Ho Chi Minh City, the BYD lesson is not "build a car company." It is: find the component, service or software where you can be the best in your market, become indispensable to your customers, and let that position carry you along the chain.

Key numbers

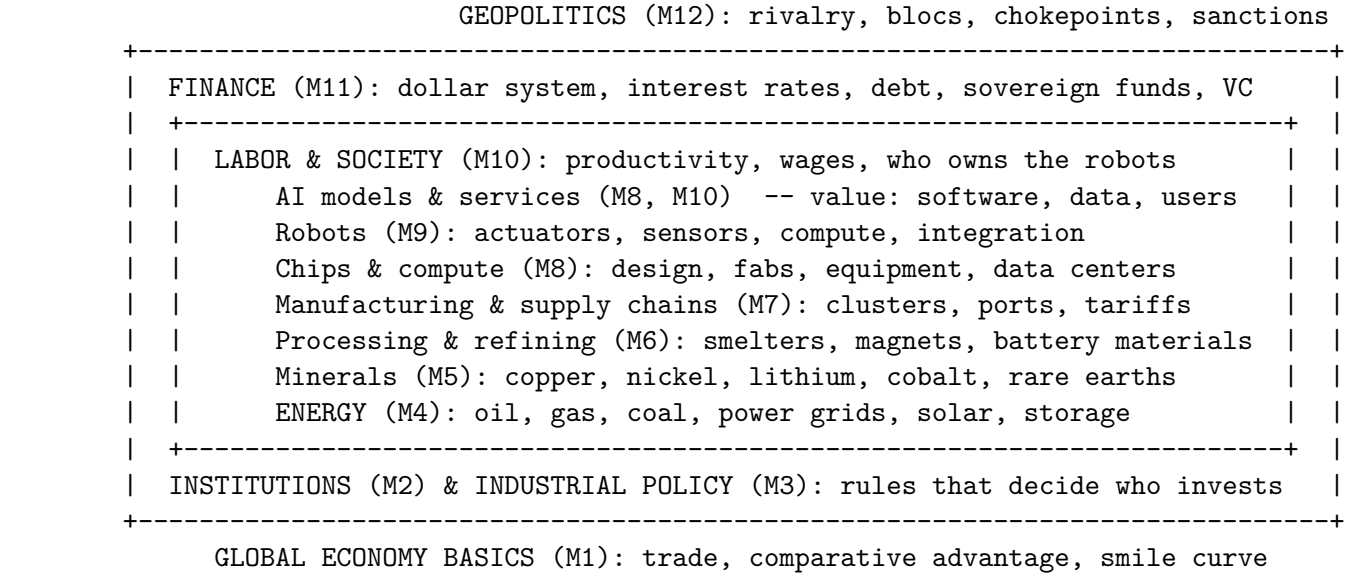
Figure	Value	Year	Source
BYD new energy vehicle sales	4,602,436 (+7.7%)	2025	BYD via CnEVPost
BYD overseas sales	about 1.05 million (more than double 2024)	2025	CnEVPost / Reuters
BYD net profit change	about -19%	2025	CnEVPost
BYD in-house share of components	about 75% (Seal teardown); nearly 80% of core components	2023; 2026	UBS; Rhodium Group via CNBC
Global venture funding to AI	about 50% of $425 billion	2025	Crunchbase (Month 11)
Southeast Asia startup funding	$5.37 billion	2025	DealStreetAsia (Month 11)
Global robot installations forecast	about 806,000 per year	2029	IFR
US-UAE AI campus planned capacity	5 gigawatts (Stargate UAE 1 GW within it)	2026	Middle East Institute
Humain planned data center capacity	about 6 gigawatts	by 2034	Humain via InTechnologies

Think like a founder

- **Your first product is a learning machine.** Choose a first business that teaches you the customers, the supply chain and the money of an industry you want to lead in ten years.
- **Build credibility as an asset.** Clean books, honest reporting and good governance from the first day lower your cost of capital and make you the partner that sovereign funds, development banks and multinationals choose.
- **The value capture lesson:** across the whole chain from ore to robot to AI, value is captured by those who control scarce capabilities, own customer relationships, and have access to cheap patient capital. A founder starts with the second, earns the third, and builds toward the first.

Month synthesis and course capstone: the whole system on one page

Here is the full twelve-month system as one diagram. Read it from the bottom up: each layer depends on the ones below it, and finance and geopolitics wrap around all of them.



And as a table: what each layer teaches about value capture, and one 2026 fact that shows it.

Layer (Month)	Who captures value	Typical chokepoint	A 2026 illustration
Global economy (M1)	Ends of the smile curve: design, brands, services	Market access	US tariffs reshaped trade routes; Vietnam's exports to the US hit about $153 billion in 2025
Institutions (M2)	Countries with credible rules and markets	Trust	MSCI proposed expelling Indonesia from its indexes over market-integrity concerns

Layer (Month)	Who captures value	Typical chokepoint	A 2026 illustration
Industrial policy (M3)	States that combine support with discipline	State capacity	China's decade-long robotics strategy preceded 59% of world robot installations
Energy (M4)	Low-cost producers and owners of routes	Straits, pipelines, grids	Hormuz disruption sent Brent futures from $72 to a $126.41 peak (April 30, 2026)
Minerals (M5)	Owners of low-cost deposits, in booms	Geology	Africa and Indonesia hold key deposits but capture little downstream value
Processing (M6)	Refiners with cheap energy and capital	Midstream refining	China refines about 91% of rare earths and makes 94% of magnets
Manufacturing (M7)	Dense supplier clusters	Agglomeration	Morocco's car exports reached about $17 billion; Gotion's Kenitra battery plant began commissioning in October 2026
Chips and compute (M8)	Designers, equipment makers, leading fabs	Advanced chips, lithography	Advanced US chips called "crown jewels", kept out of trade talks
Robots (M9)	Component makers and integrators	Magnets, reducers, scale	97% of humanoid shipments from Chinese vendors in H1 2026
AI and labor (M10)	Owners of models, data, and capital	Compute, talent	AI took about half of global venture funding in 2025
Finance (M11)	Issuers of reserve currency; cheap-capital owners	Dollar system, cost of capital	10-year US yield above 5%; rupiah over 18,000 per dollar
Geopolitics (M12)	Holders of chokepoints, until alternatives exist	All of the above	Rare earth controls won China leverage at the Busan truce

Re-reading the first robot

Month 1 opened with an invented industrial robot arm sold to a factory for $30,000, and split its price into the value added at each stage: the mine $200, the smelter and refinery $700, the component makers $11,100, the robot company $10,000, and the distributor and integrator $8,000. After twelve months it is a map of the whole course.

Stage (value added in the Month 1 example)	Why its share is small or large	Where the course explained it
Mine ($200)	Ore is abundant somewhere; prices swing with cycles; owners capture rents in booms only	Month 5 (geology, price formation); Month 2 (resource curse); Month 6 Week 3 (Africa's bargaining power)
Smelter and refinery ($700)	Margins depend on cheap power and cheap capital; China's scale sets the price	Month 4 (energy); Month 6 Week 1 (smelting economics, negative fees); Month 11 Week 2 (cost of capital)
Components ($11,100)	Precision reducers, magnets and chips are hard to copy and sit at chokepoints	Month 9 Week 2 (anatomy, the reducer race); Month 5 Week 3 (rare earths); Month 8 (chips)
Robot company ($10,000)	Design, software and scale on a home market; prices fall fast as volumes grow	Month 9 Weeks 3-4 (Unitree, cost curves); Month 7 Week 1 (clusters)
Distributor and integrator ($8,000)	Owns the customer and the know-how of deployment	Month 1 Week 3 (smile curve); this month, Week 4 (the robots-as-a-service example)

Now add the three forces that wrap around the chain. **Finance** (Month 11) charges every stage for its money: a refinery or component plant that borrows at 12% instead of 5% gives a large slice of its value added to lenders. **Geopolitics** (Month 7 Week 4 and this month) adds a toll at every border: a 10% or 12.5% Section 301 tariff, an export licence for magnets, a sanction on a bank. **Labour** (Month 10) decides how each stage's value added is divided between wages and profits, and whether the profits stay in the country or leave with a foreign owner.

The same table also answers the driving question for a country. Suppose a country in Southeast Asia or Africa mines the ore and imports everything else. Its GDP gains $200 from each robot, less than 1% of the price. If local firms also install and service the robot, the country captures $8,200, about 27%. If it also assembles robots for its region from imported components, it captures $18,200, about 61%. Refining adds only $700 more. The arithmetic is invented, but the pattern is the one we found again and again with real data: **the jumps in value come from the stages next to the customer and from hard-to-copy components, not from digging or first-stage processing.** That is why Month 6 favoured downstreaming as a bargain that raises local capture over time, and why this capstone points founders toward integration first.

The four loops that tie the system together

The layers feed back on each other. Four loops appeared in different months under different names.

1. **The cost-of-capital loop.** Stable institutions and stable money (Month 2, Month 11) lower the interest rate investors demand. A lower rate makes refineries, grids and factories profitable (Months 4, 6, 7). Profitable industry raises incomes and tax revenue, which strengthens institutions further. The loop also runs backward: Zambia's debt crisis (Month 11) and Indonesia's MSCI shock (Month 11) show how lost trust raises the cost of capital for everyone, including honest firms.

2. **The learning loop.** Firms get cheaper by producing more, and they produce more when they have a large home market. China's robot makers learned on the world's largest robot market before they became exporters (Month 9 Week 1); BYD learned on China's car market (this month). A country without a home market for robots, such as most of Africa today (519 industrial robots installed in 2025, Month 9), cannot easily start this loop, which is why using robots comes before making them.
3. **The chokepoint loop.** Control of a bottleneck gives power (rare earths, lithography, the dollar). Using that power gives others a reason to build alternatives, which slowly erodes the bottleneck. We saw it with rare earths (Months 5, 7 and 9 and Week 1 of this month), with frozen Russian reserves and the dollar (Month 11), and with export controls on chips (Month 8). The loop is slow, usually a decade, which is why chokepoints still work in the short run.
4. **The energy loop.** Cheap, reliable power makes refining, manufacturing, data centres and robots cheaper (Months 4, 6, 8). Those industries then demand more power and more minerals for grids and batteries (Month 5). The 2026 Hormuz shock showed the downside: when the base layer is shocked, the price rises travel all the way up to the interest rate on a loan in Semarang (Week 2 of this month, Month 11).

A good test of whether you understand the course is to pick any event in the news and ask which loop it pushes, and in which direction.

The course in seven sentences

1. Everything in the material economy rests on **energy**, and every energy shock travels through the entire chain.
2. **Owning resources is not the same as capturing value**; value goes to those who process, design, integrate, brand and finance.
3. **Power concentrates at chokepoints**, often in unglamorous midstream steps like refining, magnets, lithography and payment networks.
4. **The cost of capital decides who can build** capital-heavy layers, and it is set by institutions, currency stability and trust as much as by central banks.
5. **Institutions are the deepest cause** of whether a country climbs the chain or stays stuck exporting raw materials and cheap labor.
6. **Geopolitics turns interdependence into leverage**, splitting the strategic layers of the chain while the commercial layers stay more open.
7. **A founder captures value by owning the customer and the know-how of deployment first**, then earning cheap capital, then moving toward scarce capabilities.

Common misconceptions

1. **"The US and China are decoupling completely."** They still trade hundreds of billions of dollars of goods. What is happening is selective decoupling in strategic technologies, with truces and tariff bargaining in ordinary goods.
2. **"Whoever has the minerals controls the future."** Mining is necessary but not sufficient. The 2025 rare earth episode showed that power sat in refining and magnet-making, not in mining, where China's share was about 60% against 94% in magnets.
3. **"BRICS is building an alternative to the dollar."** BRICS members disagree on payments, currencies and even wars. The 2026 New Delhi declaration made no firm commitment on payment interoperability, and India said there is no BRICS currency proposal.

4. **"Automation will make developing countries irrelevant."** Automation changes the path, but cheaper robots and AI also lower the cost of building industry in developing countries. The winners will be those who adopt and integrate fastest, not only those with the cheapest labor.
5. **"A country must build the whole chain to benefit."** No country, including the United States, builds every layer. The goal is to capture valuable, defensible positions in the chain, not full independence.

Exercises

1. **(Quantitative)** Brent rose from about $72 a barrel (February 27) to about $113 (the January contract's October 30, 2026 settle of $113.43; the expiring December contract settled at $117.64 that day). A country imports 1.5 million barrels a day. Estimate the increase in its annual oil import bill at the higher price. What share of GDP is this if GDP is $1.4 trillion?
2. **(Quantitative)** Industrial robot installations were 603,307 in 2025, with the IFR forecasting 806,000 in 2029. What compound annual growth rate does this imply? If Southeast Asia's share rose from about 5% to 8% of installations, how many robots a year would it install in 2029?
3. **(Scenario analysis)** Using the 2x2 scenario grid in Week 3, describe what each scenario means for Indonesia's nickel-to-battery strategy. Which actions are robust across all four?
4. **(Analysis)** Choose one chokepoint (rare earth magnets, advanced lithography, the Strait of Malacca, or the dollar payment system). Explain who controls it, how it has been used, how long alternatives would take, and who would capture value if alternatives were built.
5. **(Essay, 1,000 words)** "Africa and Southeast Asia cannot build ore-to-robot supply chains; they should focus on services." Argue for or against, using evidence from at least four months of this course, and end with a "Judgment".
6. **(Founder exercise)** Pick one of the seven opportunity areas in Week 4. Apply the five questions. Write a one-page plan: customer, problem, first product, revenue model, capital needed, chokepoints you depend on, scenario risks, and how social impact is measured.
7. **(Capstone)** Draw your own version of the one-page system map. Add one layer or connection you think this course underweighted, and explain why it matters for value capture.

Reading list

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Glossary

- **AfCFTA**: African Continental Free Trade Area, a pact to create a single African market.
- **Alignment / hedging**: choosing one great power's side versus keeping ties with both.
- **BRICS**: a group of 11 emerging economies (2026) that coordinates on trade, finance and diplomacy.
- **Chokepoint**: a node in a network (a strait, a technology, a payment system) that others cannot easily avoid.
- **Economic statecraft**: using economic tools (tariffs, controls, sanctions, investment rules) for strategic goals.
- **Entity List**: a US list of foreign companies barred from receiving certain US goods and technologies.
- **Export controls**: government restrictions on selling certain goods or technology abroad.
- **IEEPA**: International Emergency Economic Powers Act, a US law; the Supreme Court ruled in 2026 that it does not authorize tariffs.
- **Integrator**: a company that selects, installs and programs equipment such as robots for customers.
- **Premature deindustrialization**: when countries see manufacturing's share of jobs peak at lower income levels than earlier industrializers.
- **Rules of origin**: rules deciding which country a product "comes from" for tariff purposes.
- **Scenario planning**: building several plausible futures to test strategies against uncertainty.
- **Section 301 / Section 122 / Section 232**: US trade laws allowing tariffs for unfair trade practices, balance-of-payments problems, and national security respectively.
- **Transshipment**: routing goods through a third country to disguise their origin.
- **Vertical integration**: owning several stages of a supply chain within one company.
- **Weaponized interdependence**: using control over network chokepoints to pressure others.

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Appendix A: Model Answers, Months 1-6

This appendix gives model answers to the exercises at the end of Months 1 to 6. For quantitative exercises it shows every step of the calculation, so you can check your own working line by line. For essay, policy and founder exercises it gives a model answer of about 200 to 400 words that takes a clear position, followed by a short note on what a strong answer must include. The model essays are deliberately shorter than the word counts the exercises ask for: they show the shape of a good argument, not the full length. Numbers come from the chapters themselves; where an exercise asks you to research something new, the model answer says which facts you would need to check.

How to use it: try each exercise first, then read the model answer. If your number differs, find the first line where your working and the model working part ways. That line is usually where the misunderstanding is.

Month 1: How the Global Economy Works

Exercise 1: Value added along a smartphone chain

Value added at each stage is the price that stage sells for minus the price of what it bought from the stage before it.

Step 1. Compute value added stage by stage.

- Mining: sells metals for 6 dollars and buys nothing in this simplified chain. Value added = 6 - 0 = 6 dollars.
- Refining: sells materials for 15 dollars, bought metals for 6. Value added = 15 - 6 = 9 dollars.
- Components: sells parts for 180 dollars, bought materials for 15. Value added = 180 - 15 = 165 dollars.
- Assembly: sells the phone for 210 dollars, bought parts for 180. Value added = 210 - 180 = 30 dollars.
- Brand owner: sells to retailers for 340 dollars, bought the phone for 210. Value added = 340 - 210 = 130 dollars.
- Retail: sells to consumers for 400 dollars, bought for 340. Value added = 400 - 340 = 60 dollars.

Step 2. Check the total.

6 + 9 + 165 + 30 + 130 + 60 = 400 dollars. The sum of value added equals the final price, as it must: each dollar of the final price is counted once, at the stage that created it.

Step 3. Compute each stage's share of 400 dollars.

Stage	Value added (dollars)	Share of final price
Mining	6	6 / 400 = 1.5%
Refining	9	9 / 400 = 2.25%
Components	165	165 / 400 = 41.25%
Assembly	30	30 / 400 = 7.5%
Brand owner	130	130 / 400 = 32.5%
Retail	60	60 / 400 = 15.0%
Total	400	100%

Step 4. Interpret.

The ends of the smile are the components stage (41.25 percent, which in a real phone means chips, screens and memory built on scarce know-how and intellectual property) on the upstream side, and the brand owner plus retail (32.5 + 15 = 47.5 percent) on the downstream side. Assembly, the bottom of the smile, keeps only 7.5 percent. Mining and refining together keep 3.75 percent: in this chain raw materials sit even lower than assembly, which matches the chapter's point that ore is usually low on the curve unless supply is concentrated or prices spike.

A caution: "components" lumps together very different things. A real teardown would split it into high-margin chips and low-margin plastic and glass parts, so the smile is sharper than this five-stage table shows.

Exercise 2: Absolute and comparative advantage

Data: Country X needs 6 hours per ton of copper and 3 hours per solar panel. Country Y needs 30 hours per ton of copper and 10 hours per panel.

(a) Absolute advantage. Absolute advantage means making a good with fewer hours. X needs 6 hours for copper against Y's 30, and 3 hours for a panel against Y's 10. X has the absolute advantage in both goods.

(b) Opportunity costs. The opportunity cost of one good is how much of the other good you give up by using those hours.

- X, one ton of copper: 6 hours, which could make 6 / 3 = 2 panels. So 1 ton of copper costs X 2 panels.
- X, one panel: 3 hours, which could make 3 / 6 = 0.5 tons of copper.
- Y, one ton of copper: 30 hours, which could make 30 / 10 = 3 panels.
- Y, one panel: 10 hours, which could make 10 / 30 = 0.333 tons of copper.

	Cost of 1 ton copper	Cost of 1 panel
Country X	2 panels	0.5 tons of copper
Country Y	3 panels	0.333 tons of copper

X gives up fewer panels per ton of copper (2 against 3), so X has the comparative advantage in copper. Y gives up less copper per panel (0.333 against 0.5), so Y has the comparative advantage in panels. Even though Y is worse at everything, it is "less worse" at panels.

(c) A trade price at which both gain. Any price between the two opportunity costs works: 1 ton of copper must trade for more than 2 panels (so X gains) and fewer than 3 panels (so Y gains). Pick 1 ton of copper = 2.5 panels.

Suppose X sells Y 1 ton of copper and receives 2.5 panels.

- X: making 1 ton of copper takes 6 hours. Making 2.5 panels itself would have taken 2.5 x 3 = 7.5 hours. X gets the panels for 6 hours of work instead of 7.5, saving 1.5 hours per ton traded.
- Y: making 2.5 panels takes 2.5 x 10 = 25 hours. Making 1 ton of copper itself would have taken 30 hours. Y gets the copper for 25 hours instead of 30, saving 5 hours per ton traded.

Both countries save hours, so both gain, even though X is more productive at everything. That is the core of comparative advantage.

Exercise 3: Doubling times and catch-up

Part 1. The rule of 70 says doubling time in years is roughly 70 divided by the growth rate in percent.

- At 2 percent: 70 / 2 = 35 years (exact: ln 2 / ln 1.02 = 35.0 years).
- At 5 percent: 70 / 5 = 14 years (exact: 14.2 years).
- At 8 percent: 70 / 8 = 8.75 years (exact: 9.0 years).

The rule is very accurate at low rates and slightly understates the doubling time at higher rates.

Part 2. Catch-up. Country A has 5,000 dollars per person and grows 6 percent a year. Country B has 40,000 dollars and grows 1.5 percent.

Step 1. Write the condition. A catches B after t years when 5,000 x 1.06^t = 40,000 x 1.015^t.

Step 2. Rearrange. (1.06 / 1.015)^t = 40,000 / 5,000 = 8.

Step 3. Take logarithms. t = ln 8 / ln(1.06 / 1.015).

- ln 8 = 2.0794.
- 1.06 / 1.015 = 1.04433, and ln 1.04433 = 0.04338.
- t = 2.0794 / 0.04338 = 47.9 years.

Step 4. Check with the rule of 70. The gap closes at about 6 - 1.5 = 4.5 percentage points a year. The ratio of 8 is three halvings (8 to 4 to 2 to 1). Each halving takes about 70 / 4.5 = 15.6 years, so three take about 46.7 years. That is close to the exact 47.9 years.

Interpretation: even very fast growth, held for nearly half a century, is needed to close an eightfold gap. That is why sustained growth, not one good decade, decides which countries get rich.

Exercise 4: Currency mismatch on a dollar loan

Step 1. Debt in local currency before the move: 50 million dollars x 15,000 = 750,000 million = 750 billion local units.

Step 2. Debt after the move: 50 million x 18,000 = 900 billion local units.

Step 3. Percentage rise: (900 - 750) / 750 = 150 / 750 = 20 percent. The dollar amount owed has not changed at all, but measured in the currency the firm earns, the debt is 20 percent bigger. (Equivalently, the dollar rose 18,000 / 15,000 - 1 = 20 percent against the local currency.)

Step 4. Annual interest in local currency.

- Before: 7% x 50 million dollars = 3.5 million dollars a year. At 15,000, that is 3.5 million x 15,000 = 52.5 billion local units.
- After: still 3.5 million dollars, but at 18,000 that is 63.0 billion local units.

Step 5. Interest cover (operating profit divided by interest).

- Before: 900 / 52.5 = 17.1 times.
- After: 900 / 63.0 = 14.3 times.

Interpretation: at this level of profit the firm is still safe; cover of 14 times is comfortable. But the example shows the mechanism of the 1997-1998 crisis: a depreciation raises debt and interest in local terms with no change in the firm's business. In 1997-1998 the rupiah lost about 81 percent of its value, which is far larger than this 20 percent move. A loss of 81 percent in the rupiah's dollar value means a dollar costs about 1 / (1 - 0.81) = 5.3 times as many rupiah as before, so debt and interest in local terms would rise more than fivefold, and cover of 17 times would collapse to about 3, even before profits fell in the recession. Firms with thinner cover went bankrupt.

Exercise 5: Essay, "Ireland's GDP is a fiction"

Model answer.

The statement is half right. Ireland's GDP is not a fiction in the accounting sense: it correctly measures the value added that is legally booked in Irish companies. It is a fiction if you read it as a measure of the income that Irish residents earn or can spend.

Three measures show the gap. GDP counts production booked inside Ireland. GNI (gross national income) subtracts income paid to foreign owners, such as profits sent to foreign parent companies; in 2025 Irish GNI was only about 71 percent of GDP, against 89 to 98 percent in mineral exporters such as Zambia, Chile and Botswana. GNI* goes further: it removes the profits of companies that moved their headquarters to Ireland and the depreciation of foreign-owned intellectual property and leased aircraft. In 2025 GDP was about 602 billion euros, but GNI* was about 334 billion euros, only 55.4 percent of GDP. Almost half of Ireland's measured output is, in practice, profit belonging to foreign shareholders.

The origin was the 2015 jump in real GDP of 26.3 percent as first published in July 2016 (the CSO has since revised it to 24.4 percent, still by far the largest in the European Union). No factories appeared; multinationals moved patents and other intellectual property onto Irish balance sheets to use Ireland's 12.5 percent tax rate, so worldwide profits from that property were recorded as Irish production.

What this teaches about value capture is the important part. In modern chains the most valuable thing is often intangible, such as design, software or brand, and intangibles can be moved on paper to wherever tax is lowest. So the place where value is recorded, the place where the physical work happens, and the place where the owners live can all be different. A country can look rich on GDP while its people capture much less. The same lens applies to a mining country: if most profit leaves through foreign owners, transfer pricing or trading companies, GDP overstates national gain.

Judgment: GDP is the right measure of production and the wrong measure of national welfare for any economy dominated by foreign-owned, mobile capital. For Ireland, GNI* is the better guide to what the country can afford.

What a strong answer must include: clear definitions of GDP, GNI and GNI*; the chapter's numbers (26.3 percent growth in 2015 as first published, since revised to 24.4 percent; GNI at about 71 percent of GDP, GNI* at 55.4 percent of 602 billion euros); the mechanism of moving intellectual property for tax reasons; a fair statement of what GDP does measure correctly; and a link to value capture beyond Ireland, such as mining countries where profits leave.

Exercise 6: Essay, one product's value chain

Model answer (product: a mid-range Android phone bought in Southeast Asia for about 250 dollars).

The chain runs: mines (copper, cobalt, lithium, rare earths, gold), refiners and chemical makers (largely in China), component makers (the processor from a chip designer such as Qualcomm or MediaTek, manufactured by TSMC in Taiwan; memory from Korea or China; the screen from Korea or China; camera sensors from Japan or Korea), the assembler (often in China, Vietnam or India), the brand owner, the operating system (Google's Android with its app store), distributors and retailers, and finally me.

Evidence first. The best published evidence is the 2010 iPhone 4 study: Apple kept 58.5 percent of the 549 dollar price, Korean firms 4.7 percent, and Chinese assembly labour about 1.8 percent, about 10 dollars. A 2017 follow-up found brands still capture most smartphone profits. These are Apple figures; my phone's brand almost certainly keeps less, because Android brands compete hard on price.

Now the reasons, using the three mechanisms. Contestability: assembly can be done by many firms in many countries, so assemblers earn thin margins; the brand can move production from China to Vietnam or India. Scarcity: only a few firms can design leading phone processors and only one or two can manufacture them at the leading edge, so the chip slice earns high margins; the same goes for the operating system, which almost no one can replace. Bargaining power: the brand chooses among assemblers and screen suppliers, so suppliers compete for its orders. But the brand itself has little power against Google or the chip makers, which is why Android brands have thinner margins than Apple.

Guesses, clearly labelled: I estimate (not measured) that for my phone, chips and memory take perhaps a quarter to a third of the factory cost, the retailer 10 to 15 percent of the shelf price, and the miner under 2 percent. These would need a teardown report to check.

Judgment: value in my phone is captured by whoever controls the scarce layers, chips and software, not by whoever ships the box.

What a strong answer must include: a chain with at least six stages and named countries; clear marking of where margins are high and low; each of the three mechanisms applied to a specific stage; published evidence (such as the Kraemer, Linden and Dedrick figures) separated from labelled estimates; and an honest note on what data would settle the guesses.

Exercise 7: Essay, the IMF and the Asian financial crisis

Model answer.

The IMF's response was partly wrong in its medicine but right in its basic aim, and it was not the main cause of the damage.

The case against is strong. The crisis began in private balance sheets, not government budgets.

Indonesia ran a budget surplus before 1997; its problem was companies that had borrowed short-term dollars without hedging and weak, politically connected banks. Yet the IMF's programmes, about 17 billion dollars for Thailand, 36 billion for Indonesia and 58 billion for South Korea, came with budget cuts and high interest rates. Critics led by Joseph Stiglitz argued that austerity deepened recessions that were already severe, and that closing Indonesian banks in late 1997 without a full deposit guarantee set off bank runs. The results were terrible: Indonesia's real GDP shrank about 13 percent in 1998, inflation reached about 70 percent and the rupiah lost about 81 percent of its value. Long lists of structural conditions also looked like outsiders using a crisis to impose reforms.

The case for is also real. When a currency is in free fall, a country must give investors a reason to stop selling. Higher interest rates were meant to do that. With dollar debts everywhere, a still weaker currency would have bankrupted even more firms. The rescue money, though not enough to stop the panic at first, gave governments time. South Korea and Thailand recovered within a few years, and the IMF later admitted that its early fiscal tightening was a mistake.

Judgment: the fiscal austerity was wrong for countries whose problem was private debt, and the Indonesian bank closures were badly handled. But the deeper cause was the mix of fixed exchange rates, open capital accounts, short-term foreign-currency borrowing and weak bank supervision. Those made a sudden stop likely whatever the IMF did afterwards. The lasting lesson for Asian governments, which they acted on, was to build large reserves, float currencies and supervise banks, so they would never need the IMF on those terms again.

What a strong answer must include: the chapter's numbers (package sizes, the 81 percent rupiah fall, GDP minus 13 percent, inflation about 70 percent); the distinction between private and public debt; the strongest version of both sides, including the confidence argument; the bank-closure episode; a clear judgment that separates the cause of the crisis from the quality of the response.

Exercise 8: Founder exercise, remittances

Step 1. Estimate total annual fees. Fees = 728.6 billion dollars x 6.36% = 728.6 x 0.0636 = about 46.3 billion dollars a year.

Step 2. Size the prize from lower costs. Digital services cost about 4.6 percent and non-digital ones about 7.3 percent. If a business moved a flow from 7.3 to, say, 3 percent, migrants would save 4.3 cents per dollar sent. On a corridor of 1 billion dollars a year, that is 43 million dollars saved, and the business could still earn 1 to 2 percent, or 10 to 20 million dollars of revenue.

Model answer (business sketch). Build a mobile app for one corridor, for example Indonesian workers in Malaysia, Taiwan or Hong Kong sending money home, that collects money in the sending country through local bank transfer or e-wallets, converts at the mid-market rate plus a small, clearly shown margin, and pays out instantly to bank accounts and e-wallets at home. Revenue comes from a fee of about 1.5 to 2.5 percent in total, plus later products built on payment history: savings accounts for families at home, small loans, and insurance.

Incumbents are banks, money-transfer firms with agent networks, and digital players that already offer cheaper transfers. The scarce things are not software: they are licences in both countries, banking partners willing to hold the money, cheap foreign-exchange liquidity, and above all migrants' trust, which spreads by word of mouth inside worker communities. Regulation is the main barrier: money-transfer licences, anti-money-laundering and know-your-customer rules (checking the identity of each customer), foreign-exchange rules, and data rules. A new firm usually starts by partnering with a licensed bank.

Judgment: the opportunity is real, but the winner in a corridor is decided by licences, partners and trust, so a founder should pick one corridor and own it deeply rather than launch everywhere.

What a strong answer must include: the fee calculation (about 46 billion dollars); the gap between digital and non-digital costs; a concrete corridor; a revenue model; named types of incumbents; the scarce assets (licences, liquidity, trust); and the specific regulations.

Month 2: Institutions and Why Nations Get Rich

Exercise 1: Dutch disease and a furniture exporter

(a) Profit before the boom.

- Revenue in local units: 400 dollars x 15,000 = 6,000,000 local units.
- Local costs: 5,400,000 local units.
- Profit: 6,000,000 - 5,400,000 = 600,000 local units.
- In dollars: 600,000 / 15,000 = 40 dollars per set (a 10 percent margin on revenue).

(b) Profit after the boom. The currency appreciates to 13,000 per dollar and local costs rise 8 percent.

- Revenue: 400 x 13,000 = 5,200,000 local units. The dollar price abroad has not changed, but each dollar now buys fewer local units.
- Costs: 5,400,000 x 1.08 = 5,832,000 local units.
- Profit: 5,200,000 - 5,832,000 = -632,000 local units.
- In dollars: -632,000 / 13,000 = about -48.6 dollars per set. The firm now loses money on every set.

(c) Cost cut needed to break even. To break even, costs must fall to the new revenue, 5,200,000.

- Required cut: (5,832,000 - 5,200,000) / 5,832,000 = 632,000 / 5,832,000 = 10.8 percent.
- Put another way, costs must end up 3.7 percent below their level before the boom (5,200,000 / 5,400,000 = 0.963), even though local wages and rents are rising.

Interpretation: a profitable exporter with a 10 percent margin is wiped out by a 13 percent appreciation plus 8 percent cost inflation, with no change in its own efficiency. That is Dutch disease: the resource boom makes every other export industry less competitive, which is why the chapter stresses saving resource revenue abroad.

Exercise 2: A sovereign fund fiscal rule

(a) Spending at 3 percent: 3% x 500 billion = 15 billion dollars a year.

(b) After a 20 percent crash, the fund is worth 500 x 0.8 = 400 billion dollars. Spending at 3 percent of the new value: 3% x 400 = 12 billion dollars. Spending must fall by 15 - 12 = 3 billion dollars, which is 20 percent, the same percentage as the fall in the fund.

(c) Why spend 2.7 rather than 3 percent. Four reasons:

- The 3 percent is an expected return, not a guaranteed one. Spending a bit less leaves a buffer for years when returns are below expectation, so the fund's real value is protected.
- As (b) shows, a rule tied to fund value makes spending swing with markets. A lower spending rate, and Norway's practice of smoothing, means a crash does not force sudden budget cuts.

- When a fund is very large relative to the economy (Norway's was about 2.3 trillion dollars in September 2026), spending the full 3 percent pumps a lot of money into the domestic economy, adding to inflation and Dutch disease pressure. Spending less keeps the economy balanced.
- Future generations also own the fund. Spending below the expected return is a deliberate choice to grow the fund's real value over time, which matters if oil income will fall as the world uses less oil.

Exercise 3: A rentier government's budget

All figures are shares of non-oil GDP.

Step 1. Non-oil primary deficit = non-oil revenue - primary spending = 15 - 38 = -23 percent of non-oil GDP. (This is almost exactly Saudi Arabia's 2025 figure of 23.3 percent in the chapter.)

Step 2. Overall primary balance = oil revenue + non-oil revenue - primary spending.

- Good year: 20 + 15 - 38 = -3 percent of non-oil GDP.
- Bad year: 12 + 15 - 38 = -11 percent of non-oil GDP.

Interpretation: a swing of 8 points in oil revenue moves the overall balance from a manageable deficit to a large one, while the non-oil deficit, the part the government actually controls, stays fixed at 23 percent. That is why the IMF watches the non-oil primary balance: it shows how dependent the state is on oil, whatever the oil price is doing that year.

Exercise 4: North and South Korea

Step 1. Ratio = 52.57 / 1.87 = 28.1. South Korea's income per person is about 28 times the North's (by the Bank of Korea estimate, which uses South Korean prices).

Step 2. The ratio shrinks at roughly the difference in growth rates: 3.5 - 1.5 = 2 percentage points a year. More exactly, each year the ratio is multiplied by 1.015 / 1.035.

Step 3. Halving the ratio from 28 to 14 means solving (1.035 / 1.015)ˆt = 2.

- t = ln 2 / ln(1.035 / 1.015) = 0.6931 / 0.01951 = 35.5 years.
- Rule of 70 check: 70 / 2 = 35 years.

Step 4. To close the gap completely (ratio of 28 to 1) at this pace would take t = ln 28.1 / 0.01951 = 3.336 / 0.01951 = about 171 years.

Interpretation: even if North Korea kept up its unusual 3.5 percent growth of 2025, which the chapter links to arms sales to Russia and trade with China rather than to reform, it would still be 14 times poorer than the South in the 2060s. Catch-up growth of the kind South Korea, Taiwan and China achieved, 6 to 10 percent a year, came with institutional changes that let people and firms keep the rewards of investment. Without such change, slow growth only narrows the gap over centuries.

Exercise 5: Essay, does China's rise disprove the institutions theory?

Model answer.

No. China's rise disproves a crude version of the theory, that only democracies can grow fast, but it supports the more careful version, and it leaves the hardest test still to come.

The case for "disproves" is strong. China has grown faster for longer than any large economy in history while keeping a one-party state with no independent courts, no free press and no open contest for power. By 2025 it led the world in electric vehicles, batteries, solar panels, shipbuilding and industrial robots. If inclusive institutions were necessary for prosperity, this should not have happened. Defenders of China's model add that its state was unusually capable: local officials were promoted for delivering growth, which created strong incentives without elections.

The case against is stronger. China's growth began only after it introduced more inclusive economic institutions: letting farmers sell surplus crops after 1978, letting private firms form, opening to foreign investment and trade. Growth tracked those reforms, not the one-party system. In the Mao era, with the same party and more extractive economic rules, China stayed poor. The Korean peninsula shows the same lesson: one people, two sets of institutions, a 28 to 1 gap in income. Acemoglu, Johnson and Robinson also predict that growth under extractive political institutions can be fast during catch-up, when a country copies known technologies, but becomes harder when it must innovate at the frontier, because frontier innovation needs people who feel safe challenging incumbents.

Judgment: China shows that a strong, development-focused state can deliver decades of catch-up growth without pluralist politics, so "inclusive political institutions are necessary for fast growth" is false as a universal rule. But the economic side of the theory survives: China grew when it opened markets and protected investors in practice. The open question is the 2030s. If China keeps leading in AI, chips and robots under tighter political control, the theory will need serious revision. Both confident predictions, "China will stall" and "China proves institutions do not matter", should be treated with suspicion.

What a strong answer must include: the distinction between economic and political institutions; the timing of China's reforms after 1978; the strongest pro-China argument (capable state, incentives for officials); the catch-up versus frontier distinction; at least one comparison case such as the two Koreas; and a judgment that says what evidence would change your mind.

Exercise 6: Essay, Botswana and the DRC

Model answer.

Botswana and the DRC differ because of who controlled the state and the mineral rents, and whether rules constrained that control, not because of geology or the fact of colonialism alone.

Botswana inherited institutions that colonial rule left largely intact. British rule over Bechuanaland was light; pre-colonial Tswana practices such as the kgotla, a public assembly where chiefs heard their people, limited chiefs' power. Its first leaders, led by Seretse Khama, were cattle owners who wanted secure property rights, and they built a merit-based civil service. Mineral rights were vested in the national state, not in any group's chiefs, so there was less reason to fight over them. The state bargained hard: from 1975 it owned 50 percent of Debswana and later 15 percent of De Beers. Its democracy has held, and in 2024 the ruling party lost power and handed over peacefully.

The DRC inherited the opposite. Leopold's rubber regime and then Belgian rule were built around extraction, with almost no preparation for self-government. Independence in 1960 brought army mutiny, the Katanga secession, Lumumba's killing and then Mobutu's kleptocracy, which looted Gecamines until copper and cobalt output collapsed. Two wars and continuing conflict in the east followed. Today the DRC mines about 73 percent of the world's cobalt, but China produced about 78 percent of refined cobalt in 2024 (IEA) and 79 percent in 2025 (Cobalt Institute), and the DRC

has no commercial plant making cobalt sulfate or cathode precursors.

Each can learn from the other in 2026. The DRC can learn from Botswana that bargaining power comes from credible, predictable rules: its 2026 export quota of 96,600 tonnes, about 42 percent of 2025 output, and its concentrate export ban will raise revenue only if investors believe the rules will not change again. Botswana can learn from the DRC's situation the cost of depending on one commodity: diamonds are roughly three-quarters of merchandise exports (72.7 percent in the third quarter of 2025; some estimates run up to 80 percent), and with prices down 37 percent in 2026 and the economy shrinking in 2024 and 2025, it urgently needs to diversify. Its 2025 deal with De Beers, which gives the state more control of sales, moves it toward the customer end of the chain, but it is still diamonds.

Judgment: institutions explain most of the gap; the right lesson for both is to turn mineral bargaining power into capabilities outside the mine.

What a strong answer must include: the specific mechanisms (who held mineral rights, constraints on rulers, civil service quality, bargaining with mining firms); honest treatment of colonial history in both; the chapter's numbers (73 percent of mined cobalt, 78-79 percent of refined cobalt, 96,600 tonnes, Debswana 50 percent, diamonds roughly three-quarters of exports, 2024-2025 contraction); and two-way lessons, not just "the DRC should copy Botswana".

Exercise 7: Essay, can the Gulf become an AI producer?

Model answer.

The Gulf can become a major owner and host of AI infrastructure, and possibly a producer of some AI services, but it is unlikely to become a major producer of the scarce layers of the AI chain, chips and frontier models, by the 2030s.

The case for is money and energy. Saudi Arabia's PIF held assets of about 3.4 trillion riyals, about 905 billion dollars, at the end of 2025, the UAE has its own large funds, and both have cheap energy and land, exactly what data centers need. The plans are huge: a 5-gigawatt UAE-US AI Campus in Abu Dhabi, whose first 200 megawatts of Stargate UAE were due in late 2026, and Saudi Arabia's Humain aiming for about 1.9 gigawatts by 2030 and over 6 gigawatts by 2034. Saudi data center capacity already rose from 68 megawatts in 2021 to 467 megawatts in early 2026. Capital can also buy stakes in chip and model companies.

The barriers are institutional and human. Rentier states rely on oil money rather than taxes, so the state has had little need to build a broad, skilled private economy. Private-sector jobs have mostly gone to foreign workers, often under the kafala system that ties a worker's legal status to an employer; non-Saudis were 41.6 percent of Saudi Arabia's population in 2022. Frontier research needs large numbers of top engineers who choose to stay and start firms, open debate, and firms willing to fail. Chip design and fabrication need decades of accumulated know-how that money alone cannot buy quickly; the key chips depend on US export licences, which gives Washington a veto.

The 2026 war changed the calculation in two ways. It added physical risk: Iranian drones targeted Amazon data centers in the UAE and Bahrain, and Iran's Revolutionary Guard declared regional tech firms legitimate targets, so the UAE was reported to be spreading its 5-gigawatt plan across many sites with underground construction and air defence. And it showed fiscal risk: Saudi oil revenue was down 23 percent year on year by the third quarter, and The Line was declared no longer a priority.

Judgment: the Gulf will be a big AI landlord and investor, with a real chance of building strong AI service firms in Arabic-language and government markets; becoming a chip or frontier-model producer would require a change in labour and research institutions that has barely begun.

What a strong answer must include: the distinction between hosting, owning and producing; the chapter's numbers on capacity and plans; the rentier-state mechanism and the labour market; the dependence on US chips; and both war effects, physical security and the fiscal hit.

Exercise 8: Founder exercise, an institutional assessment

Model answer (example country: Zambia; a student should check each claim against current sources such as the World Bank's Business Ready report, court statistics and news from the last two years).

Contract enforcement: medium to weak. Commercial courts exist and work in English, but cases can take years, so firms rely on arbitration, often abroad, for large contracts. Score: 2 out of 5.

Property security: mixed. Formal title is reasonably secure in cities, but much land is held under customary tenure, and mining licences have been changed or revoked under past governments, for example in the long dispute over Konkola Copper Mines. Score: 2 out of 5.

Ease of entry: fairly good for ordinary firms, which can register quickly, but harder in mining and electricity, where licences and access to the state power company decide who can operate. Score: 3 out of 5.

Predictability of rule changes: improving but fragile. Mining tax rules have changed repeatedly over two decades, and the 2020 debt default and the long restructuring that followed show how fiscal stress drives sudden changes. Score: 2 out of 5.

Proposed business: a digital mining-licence and land-rights registry and monitoring service, sold as software to the ministry of mines and as a data service to investors and lenders. It would map every licence, its status, payments and inspections, and make the record public. It profits from improving property security and predictability: when a licence's status is public and timestamped, it is harder to revoke quietly or to issue twice. Who pays: the government pays a licence fee, ideally funded by a development-finance grant at first (lenders such as the World Bank fund this kind of work); investors, banks and insurers pay subscriptions for verified data because it lowers their risk. The biggest risk is that officials who gain from opacity resist it, so the business needs a champion at a senior level and a revenue base from private users that survives a change of minister.

What a strong answer must include: one named country; evidence for each of the four questions, with sources and dates, separating evidence from judgment; a score or clear rating for each; a business tied to one specific institution; a credible payer; and an honest view of who loses from the improvement and how they might resist.

Month 3: The Industrialization Playbooks

Exercise 1: Allen's logic, when does a machine pay?

(a) Break-even wage. The machine costs 5,000 dollars a year and replaces 4 workers. It breaks even when the wages saved equal its cost: 4 x wage = 5,000, so wage = 5,000 / 4 = 1,250 dollars per worker per year. Above that wage the machine is cheaper than the workers; below it, the workers are cheaper.

(b) Adoption.

- Country A: wages of 1,500 dollars. Four workers cost 4 x 1,500 = 6,000 dollars a year, more than the machine's 5,000. The machine saves 1,000 dollars a year, so it is adopted.
- Country B: wages of 800 dollars. Four workers cost 4 x 800 = 3,200 dollars, less than 5,000. The machine would lose 1,800 dollars a year, so it is not adopted.

(c) Cheaper electricity. Reading the exercise as the machine's total annual cost falling from 5,000 to 3,000 dollars, the break-even wage becomes 3,000 / 4 = 750 dollars. Now Country B's four workers cost 3,200 dollars, more than 3,000, so the machine saves 200 dollars a year and is adopted in B as well. (If you read it as only the running part of the cost falling by some amount, the method is the same: divide the new total annual cost by 4 and compare with each country's wage.)

What it tells you: where automation happens depends on relative prices, the cost of machines and energy compared with the cost of labour, not on wages alone. Britain mechanized first because its labour was dear and its coal cheap. If solar power makes electricity very cheap in sunny low-wage countries (Month 4), the break-even wage falls and automation becomes profitable in places that were once "too cheap" to automate. Cheap energy can pull automation, and the industries built on it, toward the sun.

Exercise 2: South Korea's export growth

Step 1. Set up the equation, with values in millions of dollars: 35 x (1 + g)^15 = 10,000.

Step 2. Divide: (1 + g)^15 = 10,000 / 35 = 285.7.

Step 3. Take the 15th root: 1 + g = 285.7^(1/15). Using logarithms, ln 285.7 = 5.655, and 5.655 / 15 = 0.3770, so 1 + g = e^0.3770 = 1.458.

Step 4. g = 0.458, so exports grew about 45.8 percent a year on average for 15 years.

Step 5. Comparison. At 8 percent a year for 15 years, exports would grow by a factor of 1.08^15 = 3.17. Starting from 35 million dollars, that gives 35 x 3.17 = about 111 million dollars in 1977, against Korea's actual 10 billion dollars. Korea's exports multiplied about 286 times; the 8 percent country's multiplied about 3 times.

Check with the unrounded data the exercise gives: from 41 million dollars in 1961 to 10.05 billion dollars in 1977 is 16 years, and (10,050 / 41)^(1/16) = 1.410, or about 41 percent a year; from 33 million dollars in 1960 it is (10,050 / 33)^(1/17) = 1.400, about 40 percent a year. The stylised path gives a slightly higher rate because it squeezes the same growth into 15 years, but every version is around 40 to 46 percent a year.

Interpretation: these are dollar values, not adjusted for inflation, and they start from a tiny base, which flatters the growth rate. Even so, sustained growth of more than 40 percent a year shows what export discipline plus a fast-growing world market could do: firms that had to export grew into world-scale producers.

Exercise 3: The cost per job of the CHIPS Act

(a) Awarded cost per job: 31.5 billion / 25,000 = 1,260,000 dollars per job. Using the 13.1 billion actually disbursed by April 2026, the figure so far is 13.1 billion / 25,000 = 524,000 dollars per

job, but this is not a fair final figure, because the jobs assume all projects are finished while most of the money is not yet paid.

(b) Why "cost per job" is incomplete. Things it leaves out:

- Security value. The main goal was to reduce dependence on chips made almost entirely in Taiwan. A supply disruption could cost the economy far more than the subsidy.
- Private investment leveraged. Companies announced private investments many times larger than the grants. Each subsidy dollar may unlock several dollars of investment.
- Indirect jobs and linkages. Fabs need suppliers, construction workers and services, and chips feed downstream industries. Direct jobs are only part of the employment effect.
- Learning and capability. A working leading-edge fab builds skills that can be used for future generations of chips; this is the linkage logic of Week 1.
- Costs beyond the subsidy. Tax credits (the investment tax credit is separate from the 31.5 billion), the opportunity cost of public money, the risk of subsidy races with other countries, and the risk of supporting fabs that never become competitive.
- The Intel equity stake. Part of the money became shares that were worth about 44.6 billion dollars on paper in July 2026, so some "cost" may come back as a financial return.

A fair evaluation asks whether US fabs reach world-class yields and costs and win outside customers, not just how many people they employ.

Exercise 4: Essay, land reform and the Philippines

Model answer.

Studwell is broadly right, and the Philippines is the clearest Asian case of what happens when land reform is blocked: a country that looked set to industrialize in the 1970s shifted its workers from farms straight into low-productivity services instead.

The mechanism runs through the three channels in Week 1. Without thorough reform, large estates and tenancy survived, so small farmers did not get the gains that made Japanese, Korean and Taiwanese rice farms so productive. Rural incomes stayed low, so the home market for simple manufactured goods stayed thin. And landed families kept political power. In the Philippines many of the same families owned land, banks and protected businesses, and they had little reason to accept the export discipline that forced Korean firms to compete. Reform came late and slowly: the 1988 Comprehensive Agrarian Reform Program (CARP) targeted about 8 million hectares, and research summarised by the American Economic Association found that about 80 percent of the target area was eventually transferred, decades after East Asia's reforms. Some economists, such as Adamopoulos and Restuccia, argue CARP itself reduced productivity by capping farm size and restricting land sales.

The industrial result is visible in the numbers. According to the IBON Foundation's reading of official data, manufacturing's share of Philippine GDP peaked at 29.1 percent in 1974 and fell to a historic low of 17.3 percent in 2025. A University of the Philippines study found that manufacturing's share of employment has been stuck at about 10 percent for five decades. Compare Taiwan, which capped rents at 37.5 percent of the harvest in 1949 and pushed land to tillers in 1953, and then built an export manufacturing base that by 2025 made it richer per person than Japan.

Judgment: land reform is not sufficient; Philippine industry was also hurt by import-substitution policies, the debt crisis of the 1980s and weak state capacity. But the failure to break landlord power early is the root, because it shaped who controlled politics and banks. Whether it must come

"before" industrialization is less certain; Malaysia and parts of Latin America built some industry without it. The safer claim is that late or partial reform makes the climb much harder.

What a strong answer must include: the three mechanisms (yields, rural demand and savings, political power); at least two sourced numbers on the chosen country, with years (the figures above come from search excerpts of IBON, University of the Philippines and American Economic Association pages and should be checked against the originals before citing); a contrast with an East Asian reformer; other causes besides land; and a judgment on whether "before" is too strong.

Exercise 5: Essay, the US stake in Intel

Model answer.

As a financial trade the stake has so far been good; as industrial policy it was a poor use of public money, because it gave up the discipline that makes subsidies work.

The case for is real. In August 2025 the government converted 5.7 billion dollars of awarded but unpaid CHIPS grants, plus 3.2 billion dollars from the Secure Enclave defence programme, into 433.3 million Intel shares at 20.47 dollars, about 8.9 billion dollars for 9.9 percent of the company. By July 15, 2026, at 102.99 dollars a share, the Government Accountability Office valued the stake at about 44.6 billion dollars, a paper gain of about 35.7 billion dollars, five times the money committed. Supporters add that Intel is the only US-owned firm trying to make leading-edge logic chips, so its survival is a security matter; a shareholder has a lasting interest in its health; and the warrant to buy another 5 percent protects against the foundry business being sold off.

The case against rests on export discipline. Korea's Heavy and Chemical Industry drive worked because support was tied to measurable performance in world markets, and firms that failed lost it. In the Intel deal, Commerce removed Intel's unmet project milestones in exchange for equity, declared all six projects complete once Intel had spent 7.865 billion dollars, and stopped requiring progress reports on the original milestones. Support continued while performance conditions were dropped: the opposite of discipline. A state that owns part of Intel also has a reason to steer customers toward it, weakening the competition that disciplined Korean and Taiwanese firms. And the share price rose for many reasons, including the AI boom and expectations of government favour, so the gain does not prove capability was built.

Judgment: the stake was a good financial trade and a weak disciplinary one. The real test is whether US fabs reach world-class yields and costs by the early 2030s and win outside customers for leading-edge chips. Until then, public money has bought a share in a bet, not a capability.

What a strong answer must include: the deal terms and the GAO valuation; the removal of milestones; a clear definition of export discipline and the Korean comparison; the strongest argument for (security, upside, warrant); the distinction between financial return and industrial success; and a test for judging the outcome.

Exercise 6: Design exercise, a robot component industry by 2040

Model answer (for a Southeast Asian country with an existing auto and electronics base, such as Thailand or Malaysia; the same logic fits Morocco in Africa).

Point 1. Pick a narrow target: precision reducers and joint modules for industrial and service robots, building on existing machining and auto-parts firms. This uses the linkage feature: reducers sit

inside every robot arm and many machine tools. Performance measure: domestic content and unit cost of reducers, benchmarked each year against Chinese and Japanese prices.

Point 2. Support several firms, not one champion. Offer matched grants and cheap loans to three to five firms, local and foreign joint ventures, for machining centres, heat treatment and testing equipment. This is the competition feature. Performance measure: each firm's export sales and qualification with at least one foreign robot maker within five years; firms that miss lose further support.

Point 3. Make support temporary and dated. Tariff protection on imported reducers, if any, steps down to zero over ten years, announced in advance; grants end in 2035. This is the credible-horizon feature. Performance measure: the share of each firm's revenue that comes without subsidy, which must rise every year.

Point 4. Bring in know-how from abroad. Offer tax incentives to Japanese, Korean, European or Chinese reducer and motor makers that license technology to local partners or train local engineers, and fund scholarships with a return-home condition for diaspora engineers. This is the openness feature. Performance measure: number of local engineers certified and number of patents or licensed designs used in local production.

Point 5. Build a small, expert agency that talks to firms constantly but is shielded from capture: an industrial-robotics desk staffed by engineers, publishing every subsidy, recipient and result. This is the state-capacity feature (Rodrik's embedded autonomy). Performance measure: public annual scorecard of each supported firm's exports, costs and defect rates, and an independent audit every three years.

The sixth feature, discipline and exit, runs through every point: each measure above has a published threshold, and missing it ends support. The design does not try to build everything from ore to robot; it targets one component where tolerances are in micrometres, not nanometres, and where Chinese price falls show the chokepoint is breakable.

What a strong answer must include: five concrete policy points; each of the six design features used at least once and named; a measurable, time-bound performance test for every point; a clear exit rule; a narrow target justified by linkages and feasibility; and realism about competing with China.

Exercise 7: Founder exercise, one robot component

Model answer (component: precision gear reducers; figures from Month 9).

Who makes them. For three decades two Japanese firms dominated. Nabtesco says it has about 60 percent of the global market for precision reducers in medium and large industrial-robot joints (its own estimate), and Harmonic Drive Systems invented and long led strain wave gears. Chinese challengers are closing in: Leaderdrive of Suzhou had about 12 percent of the global harmonic reducer market in 2025 by one estimate, with revenue up 47.3 percent that year. World output is roughly 12 million strain wave gears, 3 million planetary gearboxes and 2 million cycloidal or RV reducers a year (Epoch AI, April 2026). So production sits mainly in Japan and China, with some in Germany and Korea.

Where on the smile curve. Medium to high, and sliding. Reducers need hard know-how, so leaders earned high margins, but Chinese entry is pushing prices down: Leaderdrive's average harmonic reducer price fell about 42 percent from 2017 to 2024, and its gross margin fell from 52.2 percent in 2021 to 36.9 percent in 2025. The chokepoint is splitting into a commodity tier and a premium tier;

Leaderdrive still earns about 15 percentage points more margin on exports than at home, because foreign buyers pay for proven reliability.

Could a new firm enter? Not at the premium tier at first. A new firm should start where qualification is easier and customers are close: repair, refurbishing and replacement reducers for robots already installed in local auto and electronics plants, where downtime is costly and imported spares are slow; then simple planetary gearboxes for local automated guided vehicles, conveyors and agricultural machinery. First customers would be local system integrators and factory maintenance teams. Only after years of reliability data would it approach robot makers.

Judgment: entry is possible as a service-led niche that builds precision machining skill, not as a head-on challenger to Nabtesco or Leaderdrive.

What a strong answer must include: named leading firms and countries, with sourced shares and dates; a smile-curve position justified by margins and contestability; an honest view of Chinese price pressure; a realistic first customer and entry product; and what the firm would need (machine tools, heat treatment, testing, qualification time).

Month 4: Energy, the Base Layer

Exercise 1: Energy per person

Step 1. Convert yearly energy to power. A watt is one joule per second. One year has 365.25 x 24 x 3,600 = 31,557,600 seconds, about 3.156 x 10^7 seconds. One exajoule (EJ) is 10^18 joules.

Step 2. World average power: 600 x 10^18 J / 3.156 x 10^7 s = 1.90 x 10^13 watts, which is 19 terawatts.

Step 3. Per person: 1.90 x 10^13 / 8.2 x 10^9 = about 2,320 watts. This confirms the figure of about 2,300 watts per person: on average, every person on Earth has the equivalent of 23 bright 100-watt bulbs burning day and night.

Step 4. Per person per year in gigajoules: 600 EJ / 8.2 billion = 73.2 GJ per person.

Step 5. Sub-Saharan Africa. At one-eighth of the world average, each person uses 73.2 / 8 = 9.15 GJ. Half the world average is 73.2 / 2 = 36.6 GJ. The increase per person is 36.6 - 9.15 = 27.4 GJ, which is three-eighths of the average (0.5 - 0.125 = 0.375).

Step 6. Total extra energy: 1.2 billion x 27.4 GJ = 32.9 x 10^9 GJ = 32.9 EJ.

Step 7. As a share of current world supply: 32.9 / 600 = 5.5 percent.

Interpretation: bringing 1.2 billion people from one-eighth to half of the world average would need only about 5 or 6 percent more world energy, roughly what world supply grows in a few years. The barrier is not the global quantity of energy but finance, grids and institutions in the countries that need it.

Exercise 2: An oil supply shock

(a) Net supply loss. Supply lost: 4 mb/d. Replaced by stocks: 1.5 mb/d. Net loss: 4 - 1.5 = 2.5 mb/d. As a share of the market: 2.5 / 104 = 2.4 percent.

(b) Required price rise. With an elasticity of -0.07, a 1 percent price rise cuts demand by 0.07 percent. To cut demand by 2.4 percent, the price must rise by about 2.4 / 0.07 = 34 percent.

(This simple ratio is a first approximation; for large shocks the true response is not exactly linear.)

For comparison, without the stock release the loss would be 4 / 104 = 3.85 percent, and the price would need to rise by 3.85 / 0.07 = 55 percent. The stock release cut the needed price rise by about 21 percentage points.

(c) Why actual 2026 prices did not follow the naive calculation.

- Prices rose less in some months because buffers did some of the work demand would otherwise have done. The IEA agreed a collective release of 400 million barrels on March 11, 2026. Commercial inventories were also drawn down: by the end of September global observed stocks were 509 million barrels below pre-war levels. Some exporters rerouted oil, such as Saudi Arabia through its East-West pipeline, and ceasefire hopes (for example the two-week ceasefire from about April 7) lowered expected future shortages.
- Prices rose more in other months because buffers ran down and expectations worsened. Spare capacity sat mostly in the Gulf, behind the same chokepoint, so it could not be used. As stocks fell to their lowest since at least 2016, each new attack, such as the September bombing of the East-West pipeline, hit a market with less cushion. Futures prices move on expected future shortages, not only today's barrels, so fear of a longer closure pushed prices up quickly. Diesel rose even more than crude because Gulf product exports recovered more slowly (1.9 mb/d in September against 4.7 mb/d before the war).
- The elasticity itself changes with time. In the first weeks demand barely responds; over months people and firms adjust, so the IEA forecast that 2026 demand would fall 2.6 mb/d. A longer shock needs less price rise per barrel lost.

So Brent went from about 72 dollars before the war to a peak of 126.41 dollars on April 30, a 76 percent rise, larger than the 34 to 55 percent from the simple model, at the moments when buffers and hope were weakest.

Exercise 3: The cost of solar power and the cost of capital

Step 1. Annual output. 100 MW x 8,760 hours x 0.19 = 166,440 MWh, which is 166.44 million kWh a year.

Step 2. Turn the up-front cost into an equal annual payment with the capital recovery factor (CRF): CRF = r x (1 + r)^n / ((1 + r)^n - 1), with n = 25 years.

(a) At 5 percent.

- 1.05^25 = 3.386. CRF = 0.05 x 3.386 / 2.386 = 0.07095.
- Annual capital cost: 0.07095 x 65 million = 4.61 million dollars.
- Add operations: 4.61 + 1.00 = 5.61 million dollars a year.
- Cost per kWh: 5.61 million / 166.44 million kWh = 3.37 US cents per kWh.

(b) At 9 percent.

- 1.09^25 = 8.623. CRF = 0.09 x 8.623 / 7.623 = 0.10181.
- Annual capital cost: 0.10181 x 65 million = 6.62 million dollars.
- Total: 6.62 + 1.00 = 7.62 million dollars.
- Cost per kWh: 7.62 / 166.44 = 4.58 cents per kWh.

(c) At 13 percent.

- 1.13^25 = 21.231. CRF = 0.13 x 21.231 / 20.231 = 0.13643.
- Annual capital cost: 0.13643 x 65 million = 8.87 million dollars.
- Total: 8.87 + 1.00 = 9.87 million dollars.
- Cost per kWh: 9.87 / 166.44 = 5.93 cents per kWh.

Cost of capital	Cost per kWh
5%	3.37 cents
9%	4.58 cents
13%	5.93 cents

The same farm, with the same sun and the same panels, costs 76 percent more per kWh at 13 percent than at 5 percent (5.93 / 3.37 = 1.76). Cutting the cost of capital from 13 to 9 percent lowers the cost by about 23 percent (from 5.93 to 4.58 cents). For solar, which has no fuel cost, finance is the largest single cost.

Policies that could lower the cost of capital from 13 to 9 percent in a country like Indonesia, Kenya or Nigeria:

- Make the buyer creditworthy. Most risk comes from the state utility failing to pay. Clearing utility arrears, setting tariffs that cover costs, and payment guarantees backed by the treasury or a development bank make the contract bankable.
- Reduce currency risk. Allow contracts in local currency with indexation, or use currency hedging facilities supported by development banks, so lenders do not price in a possible devaluation.
- Use standard, predictable contracts and auctions, rather than one-off negotiated deals that can be reopened by the next government. Honour existing contracts.
- Blend public and private money. Development finance institutions and climate funds can take the first loss or offer long, cheap loans, lowering the average cost.
- Build the local bond market and let pension funds invest in infrastructure, so long-term local savings fund long-term assets.
- Fix land, permitting and grid connection rules, so projects are not delayed for years, which raises risk and cost.

Exercise 4: Smelter economics

(a) Break-even electricity price.

- Margin available for power: 2,600 - 1,500 = 1,100 dollars per tonne.
- Electricity used: 14 MWh per tonne = 14,000 kWh.
- Maximum price: 1,100 / 14,000 = 0.0786 dollars per kWh = 7.86 cents per kWh (78.6 dollars per MWh).

Above about 7.9 cents the smelter loses money on every tonne.

(b) Comparison.

- Evening solar with storage, from Week 4: about 9 to 10 cents per kWh before financing costs and margins. That is above the break-even, and a smelter must run day and night, so solar plus storage alone does not work at these prices. Daytime solar at 3.4 to 5.9 cents (Exercise 3) is cheap enough, but only for part of the day.

- Gas power at 25 dollars per MMBtu: fuel alone costs 6.5 x 25 = 162.5 dollars per MWh, about 16 cents per kWh. That is double the break-even, so a smelter on imported LNG at 2026 Asian spot prices would lose about (16.25 - 7.86) x 14,000 / 100 = about 1,175 dollars per tonne.
- The US Aluminum Association says a competitive smelter needs about 4 cents per kWh on a 10 to 20 year contract. At 4 cents, power costs 14,000 x 0.04 = 560 dollars a tonne, leaving a margin of 2,600 - 1,500 - 560 = 540 dollars.

Where it could be profitable: places with long-term power well under 7.9 cents, ideally near 4 cents, around the clock. That means hydropower regions (Norway, Iceland, Quebec in Canada, and potentially hydro-rich parts of Africa such as the Congo river basin if grids and institutions allow), Gulf states with cheap domestic gas sold at regulated prices rather than LNG export prices, and China's coal-rich and hydro provinces. Solar plus storage could get there as battery costs keep falling, especially in sunny places with a low cost of capital, or in hybrid systems with hydro or gas backup.

Exercise 5: Essay, the 2026 war and renewables as security policy

Model answer.

The statement is mostly true, but it needs one correction: renewables are a security policy against fuel shocks, while creating a new dependence on Chinese equipment that is a different and smaller kind of risk.

The evidence for is strong. The 2026 war showed how exposed fuel importers are. Hormuz flows fell from about 20 mb/d before the war to about 7 mb/d in August. Brent rose from about 72 dollars to a peak of 126.41 dollars, and Asian LNG rose about 143 percent to above 25 dollars per MMBtu, which puts the fuel cost of gas power at about 16 cents per kWh. Poorer importers such as Pakistan and Bangladesh had LNG cargoes cancelled. A country that generates electricity from its own sun, wind, water or nuclear plants does not need to buy fuel through a strait. Pakistan is the case: it imported about 16.6 GW of Chinese panels in 2024, and when LNG cargoes were cut in 2026, every rooftop panel reduced exposure. Ember called this the second big fossil shock in four years.

The counter-argument is supply-chain dependence. Almost all panels and most batteries come from China, so a country swaps dependence on Gulf fuel for dependence on Chinese factories. But the two risks differ. A fuel cut stops power within weeks, because fuel is consumed. A panel embargo stops only new installations; panels already on roofs keep producing for 25 years. Stocks of panels and batteries are also cheap to hold. Renewables also do not cover everything: aviation, shipping, fertiliser and many industrial processes still need oil and gas, and variable power needs storage, grids and backup.

Judgment: the statement is right. For an energy importer, building domestic renewables is one of the cheapest security investments available. A good policy combines them with storage, grid upgrades, some strategic reserves of fuel and equipment, and more than one supplier of panels and batteries.

What a strong answer must include: the chapter's 2026 numbers (Hormuz flows, Brent peak, LNG prices, the 400 million barrel release); a mechanism linking imported fuel to vulnerability; the Pakistan case; the China dependence counter-argument, treated seriously; the difference between a flow risk (fuel) and a stock risk (equipment); limits of renewables; and a clear judgment.

Exercise 6: Essay, Norway and Nigeria

Model answer.

Oil made Norway richer and more resilient and Nigeria more fragile because Norway had inclusive institutions before oil and used them to separate oil money from day-to-day spending, while Nigeria found oil in a young, divided state where oil money became the prize of politics.

Norway struck oil in 1969, when it was already a democracy with a competent civil service, low corruption and a tax-based state. It saves most oil revenue in the Government Pension Fund Global, invested abroad and worth about 2.3 trillion dollars in September 2026, and the government spends only about the expected real return, set at 3 percent, with 2.7 percent proposed for 2027. Investing abroad means oil dollars do not flood the home economy, which limits Dutch disease: the currency does not rise so much that other exports die. Because spending is tied to the fund's return rather than this year's oil price, Norway's effective fiscal breakeven is low and its budget does not swing with oil.

Nigeria became independent in 1960, with borders drawn by colonial rule around many groups, and oil exports grew as civil war broke out. Oil revenue went straight into the budget and became the main thing politicians fought over. The result is the opposite of Norway on every channel in Month 2: rents concentrated in the state, weak accountability because government relied on oil rather than taxes, and public spending on wages and fuel subsidies that made the budget depend on high prices. Dutch disease hurt agriculture and manufacturing. Oil theft and failed refineries meant Nigeria imported most of its fuel for decades while exporting crude; its output was about 1.42 mb/d in September 2026. Many factories run on diesel generators, so the country with oil does not even have cheap industrial energy.

Judgment: the difference is institutional, not geological. The policies Norway uses, a fund, a spending rule, investment abroad, can be copied, but they only work where governments can commit to them and citizens can hold them to it.

What a strong answer must include: the timing of oil discovery relative to institutions; Norway's fund and spending rule with numbers; a clear explanation of Dutch disease and fiscal breakevens; Nigeria's specific failures (fuel imports, theft, subsidies, generators); a Month 2 mechanism such as rentier politics; and a judgment on what is transferable.

Exercise 7: Founder exercise, a business built on cheap daytime solar

Model answer (business: solar ice and cold storage for small-scale fishers in a coastal town; all equipment figures below are planning assumptions to be checked against supplier quotes).

The product. Fishers lose a large part of their catch value when fish spoils before sale. The business makes block ice during the day and runs a small insulated cold room, selling ice by the block and cold storage by the crate-day. Ice is a way of storing solar energy as cold, so the business needs little battery storage.

Energy needed. Assume the plant makes 2 tonnes of ice a day. Small ice machines use very roughly 80 to 100 kWh per tonne (assumption), so about 200 kWh, plus 40 kWh for the cold room during the day and 20 kWh at night for controls and lights: about 240 kWh a day in daylight and 20 kWh at night.

Solar and battery size. With about 4.5 peak sun hours a day (typical of sunny tropical coasts; assumption), daytime load needs 240 / 4.5 = 53 kWp; add 20 percent for losses and cloudy days, so about 65 kWp of panels. A 25 kWh battery covers the night load with margin.

Up-front cost (assumptions). Panels, inverters and mounting at about 0.50 dollars per watt installed

for a small system: 65,000 x 0.5 = 32,500 dollars. Battery at the chapter's 200 dollars per kWh: 25 x 200 = 5,000 dollars. Ice machines, cold room and building: about 40,000 dollars. Total: about 77,500 dollars.

Finance. A blend: a grant or concessional loan from a development or climate fund for part of the equipment, an equipment loan secured on the machines, and founder equity. Repayments can be matched to revenue by paying monthly.

Customers and payment. Fishers and fish traders pay per block of ice and per crate stored, by mobile money, with discounts for monthly subscriptions. Revenue check (assumption): ice at 40 dollars a tonne, 2 tonnes a day for 300 days, gives 24,000 dollars a year plus storage fees, with no fuel bill, unlike a diesel ice plant paying 25 cents per kWh or more.

Judgment: the business is attractive because it sells a product (preserved fish value) rather than electricity, and its main cost, power, is nearly free after the up-front investment. The risk is finance and demand: it needs a market close enough to sell all the ice.

What a strong answer must include: a product other than electricity; an energy estimate in kWh per day with stated assumptions; sizing of solar (kWp) and battery (kWh); an up-front cost; a financing plan; named customers and how they pay; and a comparison with the diesel or grid alternative.

Exercise 8: Data exercise, two estimates of data-centre electricity

Model answer (paragraph).

The IEA put data-centre electricity use at about 485 TWh in 2025, while the Energy Institute reported 788 TWh. The chapter's note explains most of the gap: the Energy Institute's figure, which comes from S&P Global, includes cryptocurrency mining, which the IEA counts separately. Other definitional choices also matter and should be checked in each source's methodology: whether telecom networks or small server rooms inside companies are included, and whether the figure is measured at the server or includes cooling and power losses. For planning, the definition decides the answer. A government sizing a grid for AI campuses needs the load of large, concentrated facilities that run all day and connect at high voltage; cryptocurrency mines can move or shut down when prices change, and office server rooms are spread across the ordinary grid. A founder pitching a data-centre power business who quotes the larger number without saying it includes crypto overstates the AI market by more than half (788 / 485 = 1.62). The rule: always quote the definition with the number, and compare like with like. A planner should also ask which estimate is a forecast and which is a measurement, because both sources revise their figures.

What a strong answer must include: both figures with sources; at least one confirmed definitional difference (cryptocurrency mining) and other items to check; why concentrated, constant AI load differs from other loads for grid planning; and the size of the error from mixing definitions.

Month 5: Critical Minerals from Geology to Market

Exercise 1: Grade arithmetic for a copper mine

(a) Ore per tonne of copper. Each tonne of ore contains 0.45% copper, of which 87% is recovered. Copper recovered per tonne of ore = 0.0045 x 0.87 = 0.003915 tonnes. Ore needed per tonne of copper = 1 / 0.003915 = 255.4 tonnes.

(b) Total rock moved. A strip ratio of 2.5 to 1 means 2.5 tonnes of waste rock for every tonne of ore. Waste per tonne of copper = 255.4 x 2.5 = 638.6 tonnes. Total rock = ore + waste = 255.4 + 638.6 = 894.0 tonnes (equivalently 255.4 x 3.5). To make one tonne of copper, the mine moves almost 900 tonnes of rock.

(c) Grade falls to 0.40%. Copper per tonne of ore = 0.0040 x 0.87 = 0.00348 tonnes. Ore per tonne of copper = 1 / 0.00348 = 287.4 tonnes. The increase in ore throughput = 287.4 / 255.4 - 1 = 12.5 percent. (Shortcut: with the same recovery, throughput scales with 0.45 / 0.40 = 1.125.)

Interpretation: a grade fall of about one-ninth means the mine must dig, crush and grind 12.5 percent more ore, with roughly 12.5 percent more energy, water and equipment wear, just to stand still. This is why falling grades in older mines push costs up over time.

Exercise 2: The lithium conversion margin

Step 1. Cost of feed per tonne of lithium carbonate equivalent (LCE): 7.7 tonnes of SC6 x 970 dollars = 7,469 dollars.

Step 2. Gross margin per tonne of LCE = carbonate price - feed cost = 10,300 - 7,469 = 2,831 dollars. This is what is left for conversion costs and profit.

Step 3. Converter's profit after conversion costs of 2,500 dollars = 2,831 - 2,500 = 331 dollars per tonne, about 3.2 percent of the carbonate price.

Step 4. Spodumene rises 30 percent and carbonate only 10 percent.

- New SC6 price: 970 x 1.3 = 1,261 dollars. Feed cost: 7.7 x 1,261 = 9,710 dollars.
- New carbonate price: 10,300 x 1.1 = 11,330 dollars.
- New gross margin: 11,330 - 9,710 = 1,620 dollars.
- New profit: 1,620 - 2,500 = -880 dollars per tonne. The converter now loses 880 dollars on each tonne.

Interpretation: the converter sits between two prices and is squeezed when the input price rises faster than the output price. A margin of a few hundred dollars per tonne can turn into a loss with modest price moves, because the feed is about three-quarters of the product's value. This is why converters try to own mines or sign contracts that link feed prices to product prices.

Exercise 3: The nickel cost curve

The illustrative cost curve from Week 4:

Producer group	Volume (Mt)	Cash cost (dollars per tonne)	Cumulative (Mt)
Indonesian NPI and HPAL	2.4	10,000	2.4
Other low-cost	0.5	13,000	2.9
Mid-cost	0.5	16,000	3.4
High-cost	0.4	20,000	3.8

Part 1. Demand of 3.7 Mt. The first three groups supply 3.4 Mt. The remaining 3.7 - 3.4 = 0.3 Mt

must come from the high-cost group, which can supply up to 0.4 Mt. The marginal producer is the high-cost group, so the market-clearing price is about 20,000 dollars per tonne.

Part 2. Add 0.3 Mt of Indonesian supply at 10,000 dollars. The low-cost Indonesian group becomes 2.7 Mt. Cumulative supply: 2.7 (Indonesia) + 0.5 = 3.2; + 0.5 = 3.7 Mt through the mid-cost group. That exactly meets demand of 3.7 Mt without any high-cost mine. The marginal producer is now the mid-cost group, so the price falls to about 16,000 dollars per tonne.

Part 3. Who gains and who loses.

- Buyers gain most. They save 20,000 - 16,000 = 4,000 dollars on every tonne: 4,000 x 3.7 million = about 14.8 billion dollars a year.
- High-cost producers (Australia and older sulphide mines) lose everything: they are no longer needed and close, as Australian output fell 54 percent in 2025.
- Other low-cost and mid-cost producers keep their volume but their margin per tonne falls by 4,000 dollars. The mid-cost group's margin falls to zero; it is now the marginal producer.
- Indonesian producers as a group gain volume but may lose total margin. Before: margin of 20,000 - 10,000 = 10,000 dollars on 2.4 Mt = 24.0 billion dollars. After: margin of 16,000 - 10,000 = 6,000 dollars on 2.7 Mt = 16.2 billion dollars. Their total margin falls by about 7.8 billion dollars. The new tonnes added 0.3 x 6,000 = 1.8 billion, but the lower price cost them 4,000 x 2.4 = 9.6 billion on existing tonnes.

The last point explains Indonesia's 2026 behaviour: cutting its ore quota (the RKAB) was an attempt to move back up the cost curve and raise the price. The 2025 quota approved 379 million wet tonnes of ore, of which about 320 million were actually mined; the 2026 quota was first set at 260-270 million wet tonnes and revised on October 29, 2026 to 290-300 million. It was a cut either way, because as the dominant low-cost supplier its own expansion was lowering its own profits. (The cost-curve figures above are illustrative round numbers chosen to show the mechanism; the quota figures are the real ones given in Month 5, Week 4.)

Exercise 4: Rare earths in a humanoid robot

Step 1. Rare earth content cost: 40% of 400 dollars = 160 dollars.

Step 2. If rare earth prices triple, that 160 dollars becomes 480 dollars, an increase of 480 - 160 = 320 dollars. The magnets now cost 400 + 320 = 720 dollars (assuming the rest of magnet cost is unchanged and the whole increase is passed on).

Step 3. Effect on the robot: 320 / 30,000 = 1.07 percent. The robot's cost rises from 30,000 to 30,320 dollars.

Interpretation: price risk is small; a 1 percent cost rise can be absorbed or passed on. Supply risk is large: without magnets, the actuators cannot be built and the robot cannot be shipped at all, so the loss is 100 percent of the revenue, not 1 percent. A robot maker should therefore manage magnet supply, not magnet price. That means qualifying more than one supplier (including at least one outside China), holding a few months of buffer stock, signing longer contracts even at a premium, and designing motors that can use less rare earth content or alternative magnets. Paying a 20 or 50 percent "security premium" for a non-Chinese magnet adds only a fraction of a percent to the robot's cost, which is the chapter's point: the premium is affordable; the hard part is finding someone willing to sign the long contract.

Exercise 5: Essay, are subsidies the main cause of China's refining dominance?

Model answer.

The claim is too narrow. State support mattered, but China's refining dominance is mainly the result of several reinforcing mechanisms, of which direct subsidy is only one, and it is now self-sustaining through clustering and demand.

State strategy and patience. From the 1980s China treated rare earths and other minerals as strategic, built state research institutes, consolidated producers into large state groups and set quotas. This is broader than subsidy: it is a long-term plan that private firms elsewhere, answerable to shareholders each quarter, did not match.

Low costs, including environmental costs. For decades Chinese separation plants operated with weak environmental enforcement, producing toxic and radioactive waste around Baotou and in the southern clay districts. Prices fell so low that Western producers, including Mountain Pass for a period, closed. This was a hidden subsidy paid by local people's health and land, not by the budget.

Learning curves and clustering. Once nearly all separation was in China, engineers, equipment makers, chemical suppliers and magnet makers gathered there, so costs kept falling and know-how deepened. The IEA notes that grain boundary diffusion in magnet making is highly patented, with only one equipment supplier outside China and equipment reportedly more than ten times as expensive. A newcomer faces a missing ecosystem, not just a missing plant.

Cost of capital. State banks lend cheaply and accept low returns, which matters for capital-heavy plants. This is the part closest to subsidy.

Domestic demand. China's EV, wind, electronics and motor industries became the world's largest buyers, so a Chinese refiner always had a customer nearby.

The results are visible: across key energy minerals other than rare earths, the top refiner's average market share rose to 72 percent in 2025, and since 2005 China accounted for over 90 percent of growth in copper smelting. The IEA estimates operating costs outside the dominant supplier are on average about 50 percent higher.

Judgment: subsidies and cheap state credit started the process, and low environmental standards accelerated it, but clustering, learning and home demand now keep it going. That is why matching China needs more than money: it needs guaranteed buyers, price floors and time to build an ecosystem.

What a strong answer must include: at least four named mechanisms, each with a clear causal story; the distinction between budget subsidy, hidden environmental subsidy and cheap capital; at least two numbers (72 percent, 90 percent of smelting growth, 50 percent higher costs outside China); and a judgment on which mechanisms matter now versus historically.

Exercise 6: Essay, Hormuz and the cost of metals far away

Model answer.

The 2026 closure of the Strait of Hormuz raised the cost of producing copper, nickel and lithium thousands of kilometres from the Gulf through two channels: energy and, less obviously, sulphuric acid.

Step 1: the sulphur source. Sulphur is mostly a by-product of refining oil and processing sour gas. The Middle East supplies about one-quarter of global sulphur, and about half of seaborne sulphur trade passes through Hormuz (IEA). When tankers and bulk carriers could not move freely, a large part of the world's traded sulphur was stuck.

Step 2: sulphur to acid. Most sulphuric acid is made by burning sulphur. With less sulphur available, acid makers paid more or cut output. China, a large acid producer, curbed sulphuric acid exports in May 2026 to protect its own users, tightening supply further for everyone else.

Step 3: acid to metals. Sulphuric acid is used to leach copper from oxide ores (common in Chile and the African Copperbelt), to dissolve laterite nickel ore in Indonesian HPAL plants that make battery-grade MHP, and to process lithium, cobalt and rare earths. An HPAL plant or a copper leaching operation consumes acid in large volumes, so its cost per tonne of metal rose directly. The IEA reports that in some cases acid costs overtook energy costs as the largest cost item in mineral processing.

Step 4: the energy channel. At the same time, Brent rose to a peak of 126.41 dollars and Asian LNG above 25 dollars per MMBtu. Mines run trucks on diesel, which rose even more than crude, and smelters and refineries need electricity and heat. Higher fuel prices raised costs everywhere, and more for producers relying on imported fuel.

Step 5: the result. Producers with their own acid plants (such as copper smelters, which make acid as a by-product) gained an advantage; leaching operations that buy acid lost. Higher costs for marginal producers push up the cost curve, so prices for the metals tend to rise.

Judgment: the chain shows that the critical-minerals system depends on chemicals and energy that are themselves concentrated. A country building processing should secure acid supply, for example by pairing leaching plants with sulphide smelters or sulphur-burning plants, as carefully as it secures ore.

What a strong answer must include: the IEA figures on sulphur (about a quarter of supply, half of seaborne trade through Hormuz); China's May 2026 acid export curbs; specific processes that use acid (copper leaching, HPAL nickel, lithium); the IEA finding that acid costs sometimes overtook energy; the parallel energy channel; and who gains and loses.

Exercise 7: Policy design, a rare earth magnet plant outside China

Model answer (support package).

Equity. The state, through a national investment fund, takes a minority stake, for example 10 to 20 percent through convertible preferred shares, as the US Department of Defense did with 400 million dollars in MP Materials. This shares the upside and gives the state a seat, without running the company.

Loans. A state development bank or export-credit agency lends for separation and magnet equipment at long tenor, as with MP's 150 million dollar loan for heavy rare earth separation, ideally co-financed with a development finance institution to lower the cost of capital.

Price floor. A ten-year floor of 110 dollars per kg of NdPr, structured as a contract for difference: when the market price is below 110, the state pays the difference; when it is above 110 after the plant reaches full capacity, the state takes a share of the upside (MP's deal gives the government 30 percent).

Offtake. Binding purchase agreements for most of the magnet output from defence buyers and from domestic carmakers, motor makers and robot makers, so the plant has customers at a price above the Chinese price. Without this, a floor protects the feedstock but not the magnet sales.

Stockpile. A strategic stockpile of NdPr oxide and finished magnets, a few months of national demand, bought partly from the plant. The IEA estimates stockpiling 11 high-risk materials would cost countries outside the dominant supplier less than 900 million dollars a year net in total.

Maximum annual taxpayer cost of the floor:

- Floor minus market price: 110 - 60 = 50 dollars per kg.
- Output: 1,000 tonnes = 1,000,000 kg.
- Cost: 50 x 1,000,000 = 50 million dollars a year.

Over ten years at the same prices, that is 500 million dollars, before any upside share. Compare the benefit: the IEA says rare earths are less than 1 percent of a vehicle's value and that tripling their price would raise a car's cost by just 0.1 percent, so a domestic supply secured for 50 million dollars a year is cheap insurance if a supply cut would stop car or robot factories.

Judgment: the package works only if all parts come together, as in the MP template; a price floor without offtake, or equity without technology, leaves the plant exposed to Chinese price cuts.

What a strong answer must include: each of the five tools with a reason; links to the MP Materials terms; the calculation of 50 million dollars a year; the upside-sharing clause; performance conditions (such as output targets and cost reduction milestones) to avoid an open-ended subsidy; and the cost compared with the security benefit.

Exercise 8: Minerals per kilowatt-hour

Base case from the Week 2 table (per kWh of NMC 811 cells):

- Lithium: about 0.55 kg LCE x about 16 dollars = about 8.80 dollars (the table rounds to about 9).
- Nickel: 0.7 kg x 15.50 dollars = 10.85 dollars (about 11).
- Cobalt: 0.09 kg x 38.60 dollars = 3.47 dollars (about 3.50).
- Manganese and other: under 1 dollar.
- Total: about 24 dollars, against about 10 dollars for LFP. The gap is about 14 dollars per kWh.

The table prices lithium at the end-September 2026 level (CNY 118,800-126,800 per tonne, about 16 to 17 dollars per kg, SMM). By October 30 Chinese lithium carbonate had fallen to about CNY 90,000 per tonne (about 12.60 dollars per kg), which cuts the lithium line from about 9 to about 7 dollars per kWh in both chemistries. Because lithium is in both columns, that move leaves the NMC-LFP gap, and every step below, unchanged.

Step 1. New prices: nickel 20 dollars per kg, cobalt 25 dollars per kg.

- Nickel: 0.7 x 20 = 14.00 dollars, up 3.15 dollars.
- Cobalt: 0.09 x 25 = 2.25 dollars, down 1.22 dollars.
- Lithium and other metals unchanged.

Step 2. New NMC 811 total: about 24 + 3.15 - 1.22 = about 25.9 dollars, call it about 26 dollars per kWh. (Using the unrounded figures, the change is +1.93 dollars.)

Step 3. Gap with LFP. LFP contains no nickel or cobalt, so its value stays about 10 dollars. The gap widens from about 14 to about 16 dollars per kWh, an increase of about 1.93 dollars. Higher nickel prices hurt NMC more than cheaper cobalt helps it, because each kWh holds almost eight times as much nickel as cobalt (0.7 kg against 0.09 kg).

Step 4. Scale to 1.5 TWh, which is 1.5 billion kWh. The exercise does not say how much of that is NMC 811, so first take the upper bound where all of it is:

- Nickel revenue swing: 1.5 billion x 3.15 = +4.73 billion dollars a year.
- Cobalt revenue swing: 1.5 billion x 1.22 = -1.84 billion dollars a year.
- Net swing in nickel plus cobalt revenue: about +2.89 billion dollars a year.

In reality much of the 1.5 TWh is LFP, which uses neither metal, so scale these figures by the NMC share. If, say, 40 percent of batteries were NMC 811 (an illustrative share, not a measured one), the swings would be +1.89 billion for nickel, -0.73 billion for cobalt and +1.16 billion net.

Interpretation: the price move shifts money toward nickel producers (mostly Indonesia) and away from cobalt producers (mostly the DRC), and it makes LFP more attractive still, which over time pushes more demand away from both metals.

Month 6: Processing, Downstreaming and Resource Nationalism

Exercise 1: Aluminium power economics

Step 1. Annual electricity use: 500,000 tonnes x 14 MWh = 7,000,000 MWh (7 TWh) a year, about the annual electricity use of a small country.

Step 2. Annual power bill and power cost per tonne.

Power price	Annual power bill	Power cost per tonne
25 dollars per MWh	7,000,000 x 25 = 175 million dollars	14 x 25 = 350 dollars
50 dollars per MWh	7,000,000 x 50 = 350 million dollars	14 x 50 = 700 dollars
90 dollars per MWh	7,000,000 x 90 = 630 million dollars	14 x 90 = 1,260 dollars

Step 3. Profit per tonne. Revenue minus non-power costs = 2,500 - 1,400 = 1,100 dollars per tonne available to pay for power.

- At 25 dollars: 1,100 - 350 = 750 dollars profit per tonne, or 375 million dollars a year.
- At 50 dollars: 1,100 - 700 = 400 dollars profit per tonne, or 200 million dollars a year.
- At 90 dollars: 1,100 - 1,260 = -160 dollars per tonne, a loss of 80 million dollars a year.

Step 4. Break-even power price: 1,100 / 14 = 78.6 dollars per MWh (7.86 cents per kWh). The smelter is profitable at 25 and 50 dollars, and loses money at 90 dollars.

Which countries have power that cheap, and why (from Month 4): Iceland, Norway and other hydropower regions, because old dams that are already paid off produce very cheap power around the clock; Gulf states, because they sell domestic gas to industry at low regulated prices; and China's coal-rich and hydro provinces, because of cheap local coal and large dams. The US Aluminum

Association's benchmark of about 40 dollars per MWh on a 10 to 20 year contract shows the industry standard: smelters need cheap, constant, long-term power, not just cheap power at some hours. Typical industrial grid power of about 90 dollars per MWh in many countries is too expensive.

Exercise 2: Value added versus export value

(a) The plant's value added. Value added = sales - purchased inputs.

- Sales: 1,000 million dollars.
- Purchased inputs: coal and coke 250 million dollars + electrodes and parts 150 million dollars + ore 300 million dollars = 700 million dollars.
- Value added = 1,000 - 700 = 300 million dollars.

Check: value added is paid out as wages, interest and profit: 50 + 100 + 150 = 300 million dollars. It matches. (The exercise mentions no taxes, so the model assumes taxes are zero; in reality taxes would be part of value added and would stay in the country.)

(b) The part of value added that stays in the country.

- Local wages: 50 million dollars stay.
- Interest to foreign banks: 100 million dollars leave.
- Profit to the foreign owner: 150 million dollars leave.
- Value added that stays = 50 million dollars, only one-sixth of the plant's value added.

(c) Total domestic gain including ore. The 300 million dollars paid for ore goes to local mines. Total domestic gain = 50 + 300 = 350 million dollars. (This assumes the mines are locally owned and use local inputs; if the mines are partly foreign-owned or import fuel and equipment, the true figure is lower.)

Comparison with the 1 billion dollars export figure. The headline export number is about 2.9 times the domestic gain (1,000 / 350). Only 35 percent of the export value stays in the country, and most of that is the ore, which the country could have sold even without a smelter. The smelter itself adds only 50 million dollars of local income. This is the chapter's lesson: export value is not national gain. To raise the local share, the country needs local ownership, local lenders, local suppliers of coke and parts, and taxes that capture part of the profit.

Exercise 3: Cobalt quota arithmetic

Step 1. Pre-ban export revenue: 200,000 tonnes x 25,000 dollars = 5.0 billion dollars.

Step 2. Price needed under the quota to match it: 5.0 billion dollars / 96,600 tonnes = 51,760 dollars per tonne. The price must more than double (51,760 / 25,000 = 2.07 times) to compensate for the 52 percent cut in volume.

Step 3. Revenue at the 2026 prices.

- August, about 55,600 dollars per tonne: 96,600 x 55,600 = 5.37 billion dollars a year, about 0.37 billion dollars more than before the ban. The DRC was better off.
- October, about 38,600 dollars per tonne: 96,600 x 38,600 = 3.73 billion dollars a year, about 1.27 billion dollars less than before. The DRC was worse off.

So the quota paid only while prices stayed above about 51,800 dollars. (These are annualized revenue rates at a single month's price, a simplification.)

Step 4. Other costs and benefits to include.

- Stockpiles. Cobalt not exported is stored or left "in solution" (Glencore's own-sourced output fell 52 percent in the first nine months of 2026). It has a future value if sold later at good prices, but it ties up working capital and may sell into a weak market.
- Royalties and taxes. The state receives a fiscal share of revenue, not all of it, so the state's gain or loss is smaller than the revenue change, but in the same direction. Cobalt's "strategic substance" status carries a higher royalty rate, so higher prices raise royalties per tonne.
- Copper. Most DRC cobalt is a by-product of copper mines, so mines keep running for copper; the quota mainly changes what happens to the cobalt.
- Relations with buyers and substitution. High and unpredictable prices push battery makers toward cobalt-free chemistries such as LFP, which shrinks long-run demand. Buyers also see the DRC as a less reliable supplier.
- Domestic processing. If the quota attracts investment in local refining, each tonne could later earn more; so far little new refining has come.
- Market power over time. Bunching of exports before quota deadlines and inventory draws by Chinese buyers show the limits of a monopoly producer when buyers hold stocks.

Exercise 4: Corridor economics

Step 1. Trucking cost to Durban: 300,000 tonnes x 350 dollars = 105 million dollars a year.

Step 2. Lobito is 30 percent cheaper: saving per tonne = 0.30 x 350 = 105 dollars, so the Lobito cost is 350 - 105 = 245 dollars per tonne.

Step 3. Annual saving: 300,000 x 105 = 31.5 million dollars a year.

Step 4. Offsetting a cost rise. If falling grade raises costs by 500 dollars per tonne, the corridor offsets 105 / 500 = 21 percent of the increase. The mine's net cost still rises by 500 - 105 = 395 dollars per tonne.

Interpretation: the corridor helps, especially for lower-grade deposits on thin margins, but it cannot by itself save a mine whose costs rise sharply. Transport is one cost among many; power, grade and recovery matter more over time.

Exercise 5: Scorecard essay, the Philippines and nickel

Model answer (scores 0 to 2 on each of the eight conditions; Philippine facts below come from search results on USGS, BusinessWorld and Mining Weekly pages and should be checked against the originals before citing).

1. Indispensability: 1. The Philippines was the world's second-largest nickel miner in 2025, about 270,000 tonnes or 6.9 percent of world output, but far behind Indonesia's 67 percent, and its reserves of about 4.8 million tonnes are much smaller than Indonesia's 62 million. Buyers can go elsewhere, mainly to Indonesia.

2. Cost competitiveness of processing: 0. Grid power is expensive by regional standards, many deposits are on remote islands, typhoons interrupt mining, and there is no coal-based captive power cluster like Indonesia's parks.

3. A capable partner: 1. Two HPAL plants operate, Coral Bay and Taganito, built with Japanese partners, which shows partners exist; but no Tsingshan-scale investor is building new parks.

4. Market access: 1. Ore goes mostly to China, and increasingly to Indonesia's smelters, so the country sells to a few buyers who also compete with it.
5. Policy credibility: 1. In 2025 the Senate passed a raw-ore export ban after a five-year transition, then lawmakers dropped it in June 2025 and passed a margin-based fiscal regime instead. The reversal shows how quickly rules can swing.
6. Fiscal and ownership capture: 1. Many mines are Filipino-owned, a plus, and the new margin-based royalty and windfall tax should raise the state's share; but ore exports capture little value per tonne.
7. Linkages and skills: 1. The HPAL plants train chemical engineers, but there is no stainless or battery industry downstream.
8. Environmental and social licence: 0 to 1. Laterite mining clears forest and silts coastal waters; provinces such as Palawan have resisted new mining.

Total: about 6 or 7 out of 16, low on the conditions (1 and 2) that the chapter says decide whether a ban can work.

Recommended policy: do not copy Indonesia's ban. The Philippines is not indispensable, and a ban would mostly send buyers to Indonesia. Instead, use the new fiscal regime to raise the state's share of margins, offer incentives for more HPAL capacity tied to cheaper, cleaner power, partner with Japanese, Korean, US and European buyers that want nickel outside Chinese-controlled chains, and set environmental standards that buyers facing carbon and traceability rules will accept.

Judgment: downstreaming in the Philippines should be a bargain with selected partners, built on lower processing costs and cleaner supply, not a ban.

What a strong answer must include: one country and mineral; a score for each of the eight conditions with a dated, sourced piece of evidence; a total and an interpretation linked to the chapter's rule (high on 1, 2 and 3 gives capacity; 6, 7 and 8 decide development); a specific policy; and a judgment that says whether a ban, a bargain or cost reduction fits.

Exercise 6: Essay, did Indonesia's nickel downstreaming mainly benefit China?

Model answer.

The claim is partly true: Chinese firms and Chinese industry captured the largest share of the profit and control. But it is wrong as a whole judgment, because Indonesia gained a great deal that it would never have had without the ban, and global buyers gained too.

Using the value-capture table actor by actor:

- The Indonesian state gained a far larger tax base, investment, exports and geopolitical weight. Ferronickel exports rose from 2.9 billion dollars in 2019 to 14.1 billion dollars in 2024, and total nickel-product exports reached 22.1 billion dollars in 2024. FDI rose 44.2 percent in 2021-2022 to 45.6 billion dollars. The cost: lower royalties per tonne as prices collapsed, environmental liabilities and dependence on China.
- Indonesian workers and regions gained tens of thousands of jobs; IMIP alone employed about 86,000 Indonesians in 2025. They also bore pollution, accidents such as the December 2023 explosion that killed 21 workers, rising local costs and layoffs.

- Chinese firms gained control of the world's lowest-cost nickel supply. Chinese firms control about 75 percent of Indonesia's nickel refining capacity, and in the worked RKEF example most of the operating profit and capital charges leave the country.
- Chinese battery and stainless industries gained cheap inputs and gave up little; China took 82 percent of Indonesia's nickel exports in 2024.
- Global buyers gained lower prices: LME nickel fell from over 48,000 dollars in early 2022 to about 15,000 dollars by mid-2025.
- Rival producers lost: Australia's output fell 54 percent in 2025.

So the biggest winners were Chinese firms and the world's nickel buyers, many of them Chinese. Indonesia captured a smaller share of profit than of output, and much of what stays is the ore value, which it would have earned anyway, though at lower volumes.

But the counterfactual matters. Without the ban, Indonesia would still be exporting ore for about 1 billion dollars a year, by the government's estimate, with no smelters and no stainless industry. Its absolute gain is large even if its share is modest.

Judgment: "mainly benefited China" is right about profit share and control, wrong about whether Indonesia was better off. The real test is the next stage: whether Indonesia turns an enclave into an ecosystem with Indonesian-owned firms, engineers, cleaner power and buyers beyond China.

What a strong answer must include: each actor in the table with evidence and numbers; the distinction between share of value and absolute gain; a counterfactual; the environmental and labour costs; and a judgment on what would change the balance.

Exercise 7: Policy memo, Zambia and the Lobito Corridor

Model answer (one-page memo).

To: Minister of Mines and Minerals Development, Zambia. Subject: Using the Lobito Corridor to capture more value from copper and cobalt. Recommendation: treat the corridor as a development corridor, not only a faster export route.

1. Power first. Processing fails without reliable power, the single biggest obstacle in the Copperbelt. Zambia relies heavily on hydropower, which drought can cut sharply, as in 2024. Contract new solar-plus-storage capacity at mine sites and processing hubs, expand trading through the Southern African Power Pool, and offer long-term power contracts to processors. Measure: hours of supply interruption at industrial sites.

2. Processing at a corridor hub. Zambia already refines about 270,000 tonnes of copper. Offer a special economic zone at a Copperbelt rail junction for cathode, copper rod and wire, and cobalt hydroxide refining, with stable tax terms for 15 years. Avoid export bans: Zambia has about 4 percent of world copper and is not indispensable. Use lower transport costs (up to 30 percent cheaper via Lobito) to make processing competitive rather than force it.

3. Local suppliers. Require mines and the railway to publish procurement plans and set rising, realistic targets for local purchases of services, maintenance, reagents and parts, with a supplier development fund. Measure: share of mine spending with Zambian-owned firms.

4. Debt discipline. Zambia defaulted in 2020 and is borrowing again, for example the African Development Bank's 255 million dollar loan and 10 million dollar grant, approved in August 2026, for its part of the corridor. Borrow only on concessional terms, prefer private concession

finance and equity for commercial parts, and publish every corridor loan. Never pledge copper revenues as collateral.

5. Two corridors, not one patron. Use the US- and EU-backed Lobito route and the Chinese-backed TAZARA rehabilitation (about 1.2 billion dollars) as competing exits. Competition between corridors lowers tariffs and strengthens Zambia's bargaining power. Negotiate open access so that any shipper, Western or Chinese, can use either line on published tariffs.

Risks: delays (the new line targets about 2030), weak rail operations and cross-border friction; price falls that close marginal mines. Ask: approve the power and processing-zone plan, and a debt ceiling for corridor borrowing.

What a strong answer must include: each of the five topics with a concrete action; the chapter's numbers (30 percent cost cut, financing amounts, TAZARA, Zambia's production and refining); a reason not to rely on a ban; debt safeguards; and a balanced approach to both great powers.

Exercise 8: Founder exercise, a business without a mine or smelter

Model answer (business: power-quality and predictive-maintenance services for processing plants and mines along the Lobito Corridor).

Customer. Copper and cobalt mines, smelters, refineries and the railway operator in the DRC and Zambian Copperbelt. Their plants depend on large motors, pumps, furnaces and conveyors, and on a power supply that is often unreliable.

Problem. Unplanned stoppages are very costly: a smelter or concentrator that stops loses output by the hour, and power dips damage equipment. Spare parts and specialist engineers often come from South Africa or abroad and take days to arrive. The chapter notes that power, maintenance and services in frontier regions are scarce, and scarcity is margin.

Why now. The Lobito Corridor is raising traffic and investment: about 753 million dollars in railway financing, a target of 4.6 million tonnes a year of capacity, first copper-cobalt shipments to US customers in 2026, and Western buyers who want traceable supply. New processing plants and the railway itself will need maintenance and monitoring. Cheap sensors and AI models for spotting equipment faults make the service possible for a small team.

How it makes money. Fit vibration, temperature and power-quality sensors to critical equipment; charge a monthly subscription per plant for monitoring and alerts, plus fees for on-site repair crews and a stock of common spare parts held locally. Later, sell power-management services, such as controlling on-site solar and batteries to ride through grid failures, and traceability data for buyers.

Biggest risk. Political and price risk combined: a fall in copper or cobalt prices (cobalt fell from about 55,600 dollars to 38,600 dollars per tonne between August and October 2026) leads clients to cut spending on services first, and policy changes such as quotas, export bans or new local-content rules can delay projects. Mitigation: serve both copper and cobalt clients, include the railway as a client, keep contracts short enough to adjust, and hire and train local technicians to meet local-content rules.

What a strong answer must include: a specific customer; a problem with evidence; a "why now" linked to the corridor or the nickel parks; a clear revenue model; the biggest political or price risk with a number; and how the business avoids owning the capital-heavy, low-margin steps.

Appendix B: Model Answers, Months 7-12

This appendix gives model answers to the exercises at the end of Months 7 to 12. For quantitative exercises, every step of the calculation is shown, using the numbers the chapters give. For essay, analysis and founder exercises, there is a model answer of roughly 200 to 400 words that takes a clear position, followed by a short note on what a strong answer must include. The model essays are shorter than the word counts the exercises ask for: treat them as the skeleton of a full answer, not as a replacement for writing your own.

A note on rounding: intermediate results are kept to two or three decimal places and rounded only at the end. Where an exercise leaves an assumption open, the assumption is stated in the answer.

Month 7: Manufacturing and Global Supply Chains

Exercise 7.1: Value added calculation

Given: the Mexican factory imports 70 dollars of Chinese parts and 10 dollars of US parts, and exports a finished server component to the US for 120 dollars.

(a) Mexico's value added is the value of what the factory sells minus the value of the inputs it bought from other countries.

- Imported inputs: 70 + 10 = 80 dollars.
- Mexico's value added: 120 - 80 = 40 dollars.
- As a share of the export price: 40 / 120 = 33.3 percent.

(b) US customs records the full invoice value of the good, by the country where it was last substantially transformed. So the US records 120 dollars of imports from Mexico, and 0 dollars from China, for this item.

(c) The Chinese value inside the US import is the 70 dollars of Chinese parts. That is 70 / 120 = 58.3 percent of the import value, more than Mexico's own 33.3 percent. (The remaining 10 dollars, 8.3 percent, is US value coming home.)

(d) The bilateral US-China deficit counts only goods shipped directly from China. When Chinese parts are assembled in Mexico, Vietnam or elsewhere, the Chinese content is recorded as an import from the assembling country. In this example, gross data say the US depends on Mexico for 120 dollars and on China for nothing, but in value-added terms China supplied 70 dollars and Mexico 40. Multiply this across many products and the US looks less dependent on China than it is. The chapter's evidence fits: China's direct share of US imports fell from 21 to 16

percent between 2017 and 2022, while China's share of the value added inside US imports barely moved, from 15.7 to 15.3 percent (Stanford SCCEI). Trade diversion moved the last assembly step, not the underlying dependence.

Exercise 7.2: Learning curve

A 15 percent learning rate means each doubling of cumulative output multiplies unit cost by 1 - 0.15 = 0.85.

(a) Doublings needed for the start-up to go from 120 to 40 dollars:

- Required cost ratio: 40 / 120 = 1/3 = 0.3333.
- We need 0.85^n = 0.3333.
- Take natural logarithms: n x ln(0.85) = ln(0.3333).
- ln(0.85) = -0.1625; ln(0.3333) = -1.0986.
- n = -1.0986 / -0.1625 = 6.76 doublings.

Check: 0.85^7 = 0.3206, so 7 doublings would take the cost to 120 x 0.3206 = 38.47 dollars; 6 doublings would leave it at 120 x 0.3771 = 45.25 dollars. So the answer lies between 6 and 7, at about 6.76.

(b) Cumulative volume needed:

- Volume multiplier: 2^6.76 = 108.4.
- Cumulative volume: 50,000 x 108.4 = about 5.42 million motors.

Notice that this is more than the Chinese firm's current cumulative output of 4 million. The start-up must make about 5.4 million motors in total, which means about 5.37 million more than it has made so far, just to reach the cost the Chinese firm has today.

(c) Strategy implications. The Chinese firm is not standing still. It is at 4 million units, or 80 times the start-up's volume (4,000,000 / 50,000 = 80, about 6.3 doublings ahead). If it doubles again to 8 million, its cost falls to 40 x 0.85 = 34 dollars. Chasing it head-on on the same product is almost hopeless, because the leader moves down the curve as fast as, or faster than, the follower can climb it, and the follower loses money on every unit along the way. Sensible strategies are:

- Do not compete on the commodity motor. Pick a niche where the Chinese giant has little volume (for example, a special motor for a local application, a sealed motor for wet or dusty environments, or a motor plus driver plus software bundle).
- Compete on something other than unit cost: speed of delivery, local service, design-in support, traceable non-Chinese content (which, as Week 4 showed, can earn a premium after the tariff and magnet shocks).
- Borrow volume: become a licensed second source or joint venture partner of an established maker, so that you inherit some of its learning.
- Use subsidies or protected procurement only as a bridge with a firm deadline, as East Asian industrial policy did (Month 3), not as a permanent crutch.

Exercise 7.3: Landed cost

The Week 3 table, rebuilt with a US tariff on China of 10 percent and freight from China of 12 dollars a unit:

Cost element	From China	From Mexico
Factory price	100 USD	115 USD
Freight per unit	12 USD (crisis)	2 USD
US tariff	10% of 100 = 10 USD	0 USD (USMCA-compliant)
Inventory and delay cost	3 USD	1 USD
Total landed cost	125 USD	118 USD

Step by step:

- China: 100 + 12 + 10 + 3 = 125 dollars.
- Mexico: 115 + 2 + 0 + 1 = 118 dollars.

Mexico wins by 7 dollars a unit, even with a tariff on China of only 10 percent, because crisis freight adds 8 dollars compared with normal times.

Break-even tariff rate. Let the tariff be T percent of the 100-dollar factory price, so the duty is T dollars.

- China's landed cost: 100 + 12 + T + 3 = 115 + T.
- Set equal to Mexico: 115 + T = 118, so T = 3.
- The two locations are equal at a tariff of 3 percent on China.

Comparison with normal freight (4 dollars): China's cost would be 100 + 4 + T + 3 = 107 + T, and break-even would be T = 11, an 11 percent tariff. So a freight crisis is worth roughly 8 percentage points of tariff protection to Mexico. The lesson is that transport risk and tariffs are substitutes in the location decision: when shipping is unreliable, nearshoring needs much less tariff help. (Strictly, a crisis would also raise inventory costs for the China route, which would push the break-even tariff even lower.)

Exercise 7.4: Shipping shock

The chapter's robot vacuum example compared normal freight of 1,500 dollars per container with crisis freight of 6,000 dollars, added the cost of 12 extra days in transit at a 10 percent annual cost of capital, and added 2 to 4 dollars of higher diesel and plastic costs on a 300-dollar product. We apply the same method to a 40-dollar fan, 2,000 units per container.

Step 1, freight per unit:

- Normal: 1,500 / 2,000 = 0.75 dollars, which is 0.75 / 40 = 1.9 percent of the price.
- Crisis: 6,000 / 2,000 = 3.00 dollars, which is 3.00 / 40 = 7.5 percent of the price.
- Increase: 3.00 - 0.75 = 2.25 dollars.

Step 2, inventory cost of 12 extra days:

- Value of goods in one container: 2,000 x 40 = 80,000 dollars.
- Cost: 80,000 x 10 percent x 12 / 365 = 263 dollars per container.
- Per unit: 263 / 2,000 = 0.13 dollars.

(Check against the chapter: for the vacuum, 300,000 x 10 percent x 12 / 365 = 986 dollars, the "about 1,000 dollars" the chapter gives.)

Step 3, materials. For the vacuum, 2 to 4 dollars on 300 is about 0.7 to 1.3 percent of the price. Assume the same share for the fan: 40 x 0.7 percent to 40 x 1.3 percent = about 0.27 to 0.53 dollars. (A fan is mostly plastic and copper, so this may understate it.)

Step 4, total shock compared with normal times:

- Low end: 2.25 + 0.13 + 0.27 = 2.65 dollars.
- High end: 2.25 + 0.13 + 0.53 = 2.91 dollars.
- As a share of the price: 2.65 / 40 = 6.6 percent to 2.91 / 40 = 7.3 percent, call it about 7 percent.

For comparison, the vacuum's shock was about 3 percent. The fan's shock is more than twice as large in percentage terms, mainly because freight is charged per container and a cheap product carries less value per cubic metre. If the fan maker's margin is 5 percent, a 7 percent shock wipes out all profit and more.

Who is most hurt by chokepoint closures:

- Makers of low-value, bulky goods (furniture, fans, plastic goods, building materials, cheap textiles), where freight is a large share of price.
- Low-margin contract manufacturers that cannot pass costs on to powerful brand-owning buyers.
- Firms on long routes that cross the chokepoints (Asia to Europe through the Red Sea, Gulf energy to Asia through Hormuz and Malacca).
- Firms with just-in-time production and little inventory, which stop when parts are late.
- Firms that depend on energy-intensive inputs (plastics, chemicals, fertilisers) priced off oil and gas.

High-value, light, high-margin goods (phones, chips, medicines) are the least hurt.

Exercise 7.5: Essay: "Vietnam is the next South Korea"

Model answer.

The claim is premature. Vietnam has copied the first half of South Korea's story, export growth through manufacturing, but not the second half, the creation of domestic firms that own technology and brands. Unless that changes, Vietnam is more likely to become the next Malaysia, a prosperous assembly and components platform, than the next Korea.

The growth is real. Vietnam's goods trade reached 930 billion dollars in 2025 and exports 475 billion, up 17 percent (Vietnam Customs and the National Statistics Office). Samsung's phone complexes, opened from 2009, turned the country into an electronics exporter in a decade. Disbursed foreign direct investment (FDI) hit an estimated 27.6 billion dollars in 2025, the highest in five years.

But look at who captures the value. Foreign-invested firms produced 367 billion dollars, or 77.3 percent, of 2025 exports, while domestic firms' exports fell 6.1 percent to about 108 billion. The domestic sector ran a trade deficit of about 29 billion dollars. Vietnam's surplus of about 134 billion dollars with the US is matched by a deficit of about 116 billion with China, the signature of an assembly platform that imports Chinese inputs. Vietnamese officials themselves say value retained "remained low, mainly concentrated in assembly".

Korea was different in exactly this respect. As Month 3 showed, Korea used export discipline, directed credit and protection to build national champions such as Samsung, Hyundai and POSCO,

which learned foreign technology and then competed with it. Korea limited foreign ownership; Vietnam's model depends on it.

To make the claim true, Vietnam would need to: tie incentives for foreign firms to measurable local sourcing and supplier development, as Thailand did in autos; build a domestic components sector (moulds, circuit boards, precision parts) with patient credit; invest heavily in engineering education and research; and grow a few domestic firms, perhaps from groups such as Vingroup or FPT, into exporters with their own designs. It must also manage political risk: its US surplus already earned it the higher 12.5 percent tariff group in July 2026.

Judgment: Vietnam is a success, but a Korea-style one only if it shifts from attracting factories to growing firms. That shift is a policy choice, not a natural next step.

What a strong answer must include.

- A clear definition of what made Korea Korea (domestic firms, technology learning, export discipline), not just high growth.
- Value-added evidence, not only gross exports: the 77.3 percent foreign share, the domestic-sector deficit, the surpluses and deficits with the US and China.
- Policy differences (FDI dependence versus national champions) and at least one risk (tariffs, transshipment accusations).
- Concrete, specific changes Vietnam would need to make, and a marked judgment.

Exercise 7.6: Essay: Morocco versus Ethiopia

Model answer.

Morocco built a durable cluster because it combined a trade position that did not depend on one country's goodwill with deep physical infrastructure and a stable, long-term strategy. Ethiopia built parks around a single trade preference and a single product, garments, and when the preference was withdrawn, the parks emptied.

Morocco's automotive exports were 159.2 billion dirhams in 2025, about 17 billion dollars and about a third of goods exports. The base is Tanger Med, which handled 11.1 million TEU and 526,862 vehicles in 2025. Morocco has had free trade with the EU since 2000 and with the US since 2006, so no single partner's decision can switch off its access. Its factories sit a truck and ferry ride from Spanish and French plants, which makes just-in-time supply possible. And it pushed suppliers of wiring harnesses, seats and stamping to locate nearby, so much of the export value is components rather than final assembly alone.

Ethiopia's Hawassa Industrial Park opened in 2016 with a target of 60,000 jobs and about 1 billion dollars of exports a year. It peaked at more than 35,000 workers. All of Ethiopia's park network produced net exports of about 141 million dollars in 2020/21. The model rested on duty-free access to the US under AGOA. When the US suspended Ethiopia on 1 January 2022 over the Tigray war, 18 foreign firms left the parks and about 11,500 jobs were lost. Exports to the US fell 31.7 percent in 2023, and laid-off Hawassa workers saw their labour income halve. Ethiopia is landlocked, its firms were mostly foreign cut-and-sew operations with few local suppliers, and its competitiveness without the preference was thin.

My advice to Kenya, Nigeria or Tanzania: first, secure several markets (the EU, the US, the African Continental Free Trade Area, the Gulf) so that no single preference is decisive. Second, invest in one excellent port-and-zone complex with fast customs before scattering parks. Third, choose products

where you are competitive even without a preference, and require anchor investors to develop local suppliers. Fourth, keep politics stable and rights respected, because trade preferences are political and can be lost.

Judgment: the decisive difference was not wages or subsidies but resilience: Morocco built a cluster that would survive a shock, and Ethiopia built one that only worked while conditions were perfect.

What a strong answer must include.

- Numbers for both cases (Morocco's exports and port volumes; Ethiopia's targets, actual jobs and losses).
- Mechanisms: diversified trade access, proximity, supplier development and stability versus single-preference, single-product dependence.
- Honesty about Morocco's limits (foreign brands, EV transition risk) so the comparison is not one-sided.
- Specific, practical advice for a named country, and a marked judgment.

Exercise 7.7: Essay with a judgment: Was 2025-2026 US tariff policy a success?

Model answer.

Success must be defined against stated goals. The administration promised three things: to reduce dependence on China, to shrink the trade deficit, and to raise revenue and bring investment home without hurting American consumers. Measured that way, the policy succeeded on the first goal, partly on the third, and failed on the second.

Dependence on Chinese final goods fell. China's share of US goods imports fell from about 22 percent in 2017 to 9 percent by the end of 2025, and real imports from China dropped 28 percent in 2025 (PIIE). The trade-weighted average tariff on Chinese goods stood at 35.2 percent in August 2026, far above 3.1 percent in January 2018.

The deficit did not shrink. The US goods deficit was about 1.23 trillion dollars in 2025, slightly larger than in 2024. Imports moved to Taiwan, Mexico and Vietnam instead of stopping, and China's global surplus hit a record 1.189 trillion dollars. As Month 1 explained, the deficit reflects the gap between US saving and investment, which tariffs barely touch. And value-added dependence on China barely moved, because Chinese parts kept arriving inside goods assembled elsewhere.

Revenue rose, but Americans paid. Tariff revenue rose by an estimated 214.7 billion dollars above the 2022-2024 average by February 2026, yet the New York Federal Reserve found US importers bore 86 to 94 percent of the burden. Then the Supreme Court struck down the IEEPA tariffs in February 2026, and about 126 billion dollars of the roughly 166 billion collected had been certified for refund by October. The effective rate actually collected in August 2026 was only 7.1 percent.

The magnet squeeze of 2025 exposed the deepest weakness: tariffs could not fix dependence on a chokepoint input, and China's export controls shut Ford factories within weeks. Finally, the legal basis of the tariffs changed four times in eighteen months, which raised uncertainty and delayed investment.

Judgment: the policy was a narrow success and a broad failure. It reduced direct reliance on Chinese final goods, but it did not cut the deficit, it taxed mainly Americans, much of its revenue is being refunded, and it left the critical-input dependence untouched.

What a strong answer must include.

- An explicit definition of success before the evidence.
- At least five numbers from the chapter, correctly attributed (the answer above uses about ten).
- The distinction between statutory and effective rates, and between gross and value-added dependence.
- Both the supporters' and critics' case, and a clearly marked judgment.

Exercise 7.8: Founder exercise: one robot component

Model answer (servo motor driver and controller board).

Choice: the servo drive, the circuit board that powers and controls each joint motor. Week 3's actuator breakdown puts the driver at 15 to 20 percent of actuator cost, about 70 dollars in a 400-dollar actuator.

Who makes it today: Japanese firms (Yaskawa, Panasonic, Mitsubishi Electric), German firms (Siemens, Beckhoff) and a fast-growing group of Chinese makers (Inovance and many smaller Shenzhen firms). Chokepoints: power semiconductors (MOSFETs and gate drivers, many from Infineon, onsemi, STMicro and Chinese makers), microcontrollers, and encoders, plus magnets in the motor it controls. Circuit board assembly is not a chokepoint; Malaysia, Thailand and Vietnam already do it at scale.

Where to start: Penang, Malaysia, which has about 13 percent of global chip assembly, packaging and testing, an electronics manufacturing services base and engineers who know power electronics.

Business concept: a compact, open-protocol joint drive for small robot arms and mobile robots, with firmware that accepts motors and encoders from several suppliers and microcontrollers from at least two sources. Target customer: Southeast Asian and Indian robot integrators and small robot makers building arms, carts and agricultural machines, who today import Chinese drives with patchy documentation. Source of advantage: (1) multi-source design, so customers can avoid single-country dependence and document non-Chinese content for US and EU buyers; (2) local engineering support in the customers' time zone; (3) software tools for tuning and diagnostics, which create recurring revenue and switching costs. Revenue: hardware at a 30 to 35 percent gross margin plus a yearly software and support licence.

Main risk: Chinese competitors at much higher volume (Exercise 7.2 shows why we must not compete on price alone). Response: stay in the niche where documentation, traceability and support matter more than the last dollar.

What a strong answer must include.

- A real component with its share of cost, current producers and specific chokepoint inputs.
- A plausible location justified by an existing cluster, not just low wages.
- A named target customer and a source of advantage other than "cheaper than China".
- Awareness of learning-curve and policy risk.

Month 8: Semiconductors, Compute and the AI Supply Chain

Exercise 8.1: Wafer economics

(a) One large die.

- Candidate dies per wafer: 120.

- Good dies at 70 percent yield: 120 x 0.70 = 84.
- Cost per good die: 25,000 / 84 = 297.62 dollars.

(b) Two chiplets instead.

- Candidate chiplets per wafer: 250.
- Good chiplets at 85 percent yield: 250 x 0.85 = 212.5, so about 212 good chiplets.
- Cost per good chiplet: 25,000 / 212.5 = 117.65 dollars.
- Cost of one pair (the equivalent of one large die): 2 x 117.65 = 235.29 dollars.
- Saving in silicon: 297.62 - 235.29 = 62.33 dollars, or 62.33 / 297.62 = 20.9 percent.

Notice where the gain comes from. Splitting the die raises the number of candidate dies slightly (250 half-size chiplets versus the 240 you would get from simply doubling 120, because smaller rectangles waste less of the round wafer's edge) and, more importantly, raises yield from 70 to 85 percent, because a single defect now ruins only half as much silicon.

(c) Why chiplets might still cost more overall:

- Advanced packaging. The two chiplets must be joined, often on a silicon interposer or bridge (for example TSMC's CoWoS), which is expensive and was itself a bottleneck in 2024-2026 (Week 3).
- Extra silicon area. Each chiplet needs its own input-output circuits and die-to-die interfaces, so two 200 square millimetre chiplets usually do less work than one 400 square millimetre die.
- Packaging yield. If the assembly step fails, two good chiplets can be lost together.
- Testing. Each chiplet must be tested before packaging ("known good die"), adding cost.
- Performance and power. Signals between chiplets travel further than within one die, costing some speed and energy.

So chiplets win when dies are large and yields are poor (as at the leading edge), and lose for smaller, mature chips where a monolithic die already yields well.

Exercise 8.2: Data center power

(a) Annual electricity use.

- Average IT power: 300 MW x 0.75 utilisation = 225 MW.
- Total facility power including cooling and overhead: 225 MW x 1.3 PUE = 292.5 MW.
- Hours in a year: 8,760.
- Annual energy: 292.5 MW x 8,760 h = 2,562,300 MWh.
- Convert to TWh (1 TWh = 1,000,000 MWh): 2.56 TWh.

(b) Share of Indonesia's consumption.

- 2.56 / 300 = 0.0085, or about 0.85 percent of national electricity use, from one site.

That sounds small, but it is concentrated. A 292.5 MW continuous load is about the output of a mid-sized power station running all year, delivered to one point on the grid. Using the chapter's 100 MW example as a guide (0.876 TWh a year, about 70 million dollars at 80 dollars per MWh), this site's bill would be about 2,562,300 x 80 = 205 million dollars a year. On water, AMRO's estimate of about 4.2 million litres a day per 100 MW suggests up to about 12.6 million litres a day if the site used similar evaporative cooling (4.2 x 3 for 300 MW), unless it uses liquid cooling or recycled water.

(c) What an energy minister should require:

- Additional clean power: the developer contracts new renewable generation (and storage) equal to its consumption, as one Johor operator did with a 500 MW deal with Tenaga Nasional, rather than drawing on existing coal-heavy supply.
- Full-cost tariffs and grid contributions: the developer pays for its own transmission connection and substation and pays a tariff that reflects true cost, so households do not subsidise it.
- Siting and phasing: build where the grid has spare capacity, and connect in phases tied to new generation coming online.
- Water limits: closed-loop or liquid cooling, recycled water, and no approvals for the most water-intensive designs (Johor banned these in late 2025).
- Flexibility: agree to curtail load during grid emergencies.
- Local value: training and hiring targets for local engineers, and access for local AI firms to some capacity on fair terms.

Exercise 8.3: Memory shock

(a) New cost.

- Memory cost rises 150 percent: 25 x 1.50 = 37.50 dollars more.
- New memory cost: 25 + 37.50 = 62.50 dollars.
- New total cost: 200 - 25 + 62.50 = 237.50 dollars.

(b) Percentage increase in total cost: 37.50 / 200 = 18.75 percent.

(c) Effect on profit if the price stays at 215 dollars.

A note first: the exercise's numbers do not quite agree. A cost of 200 dollars and a price of 215 dollars give a margin of 15 dollars, which is 15 / 215 = 7.0 percent, not 6 percent. Both readings are worked here; the conclusion is the same.

- Using the stated 6 percent margin: profit before = 0.06 x 215 = 12.90 dollars per phone; profit after = 12.90 - 37.50 = -24.60 dollars, a loss of 24.60 dollars per phone.
- Using the stated cost and price: profit before = 215 - 200 = 15.00 dollars; profit after = 215 - 237.50 = -22.50 dollars, a loss of 22.50 dollars per phone.

Either way the conclusion is the same: a component that was only 12.5 percent of cost (25 / 200) can turn a profitable product into a loss-maker when its price spikes, because the phone maker's margin is thinner than the cost increase. To break even it would need to raise its price by about 22.50 to 24.60 dollars, roughly 10.5 to 11.4 percent. This is what happened in 2026, when conventional DRAM contract prices rose about 93 to 98 percent in a single quarter (TrendForce) as memory makers diverted capacity to high-bandwidth memory for AI.

Exercise 8.4: Token economics

(a) Compute cost per student per month.

- Input: 2 million tokens x 0.10 dollars per million = 0.20 dollars.
- Output: 0.4 million tokens x 0.50 dollars per million = 0.20 dollars.
- Total: 0.40 dollars per student per month.

(b) Gross margin (counting compute as the only cost of revenue).

- Gross profit: 5.00 - 0.40 = 4.60 dollars.
- Gross margin: 4.60 / 5.00 = 92 percent.

In practice, payment fees, app store commissions, hosting and support also count as cost of revenue, so the real gross margin will be lower.

(c) Usage triples.

- Compute cost: 3 x 0.40 = 1.20 dollars.
- Gross margin: (5.00 - 1.20) / 5.00 = 76 percent.

Still healthy. The risk is the heavy user: a student who uses 10 times the average costs 4.00 dollars, leaving only 1.00 dollar of gross profit. Most subscription AI products protect themselves with usage caps or tiered plans.

(d) Model prices fall 13-fold.

- At original usage: 0.40 / 13 = 0.031 dollars; margin (5.00 - 0.031) / 5.00 = 99.4 percent.
- At tripled usage: 1.20 / 13 = 0.092 dollars; margin (5.00 - 0.092) / 5.00 = 98.2 percent.

The lesson is the reverse of what it first seems. Falling model prices do not make this business safe; they make it easy to copy. If compute costs only 3 to 9 cents a student, a competitor can charge 1 dollar a month and still make money, and large platforms can bundle tutoring for free. The 13-fold fall comes from Epoch AI's estimate that the cost of a given level of AI performance fell about 13 times a year from 2023 to 2026. So the start-up's durable advantage cannot be cheap compute. It must be things that do not fall in price: local curriculum fit (for example, Indonesian exam syllabi), relationships with schools, student data that improves the product, trust with parents, and distribution.

Exercise 8.5: Essay: "Taiwan's silicon shield makes war less likely"

Model answer.

The silicon shield is real but weakening, and it deters war less than its name suggests. It raises the cost of an attack for everyone, but it does not change China's core motive, and the world is deliberately building alternatives that will thin the shield over the next decade.

The argument for the shield is strong today. TSMC earned 122.42 billion dollars in 2025 with a 59.9 percent gross margin, and advanced processes of 7 nm and below made up 74 percent of its wafer revenue. Taiwan held about 69 percent of the world's leading-edge logic capacity in 2022 (SIA and BCG). North American customers produced 74 percent of TSMC's revenue in late 2025. A war or blockade would cut off Nvidia, Apple and much of China's own electronics industry, a shock larger than the 2026 Hormuz closure. That gives the US a strong reason to defend Taiwan and gives China a reason to hesitate.

But three facts weaken it. First, overseas fabs: TSMC's US commitment rose from 12 billion dollars in 2020 to about 265 billion by July 2026, plus plants in Japan and Germany, and Taiwan's share of leading-edge capacity is projected to fall to about 47 percent by 2032. The more chips are made elsewhere, the less the world needs Taiwan specifically. Second, China's motive is political, not economic. Beijing views unification as a matter of national legitimacy, and economic damage alone has rarely stopped a state pursuing a core territorial goal, as Russia's invasion of Ukraine in 2022 showed. Third, China is reducing its own dependence: Goldman Sachs projects China's self-supply of advanced chips rising from about 8 percent in 2025 to about 66 percent by 2035, so the cost of losing TSMC to China is falling.

There is also a dangerous twist. If dependence on TSMC is mutual, a blockade rather than an

invasion might be used as leverage, betting that the world would pressure Taiwan to negotiate rather than lose its chips.

Judgment: the shield makes war less likely in the next few years, mainly by strengthening American resolve, but it is a wasting asset. Deterrence in the 2030s will rest on military balance and diplomacy, not on chips. For founders, a Taiwan disruption remains a low-probability, very-high-impact risk that justifies avoiding single-sourced chips.

What a strong answer must include.

- The mechanism of the shield (mutual dependence raising the cost of war) stated clearly.
- Evidence on TSMC's scale and share, and on overseas fab construction and its limits (the most advanced processes still start in Taiwan).
- Separate analysis of US and Chinese interests, including that China's motive is not mainly economic.
- Recognition that the shield changes over time, and a marked judgment.

Exercise 8.6: Essay: Were US chip export controls, 2022-2026, a success?

Model answer.

If success means stopping China's AI progress, the controls failed. If it means slowing China's access to the best hardware and buying the US time, they largely succeeded, though the 2025-2026 policy of selling second-best chips for a fee muddled that success.

The controls bit hardest on hardware. Since 2019 ASML has not shipped EUV machines to China, and since 2023 Dutch and Japanese controls have covered some advanced DUV tools too. SMIC makes 7 nm-class chips only with DUV multi-patterning, which is slower and yields worse. The Council on Foreign Relations judged the best US AI chips about five times more powerful than the best Chinese ones in 2026, with the gap possibly widening to 17 times by late 2027. Epoch AI estimates Huawei's 2026 compute at under 4 percent of Nvidia's, with about 1.5 million Ascend units against 5.9 million Nvidia chips. Much of Huawei's earlier output relied on about 2.9 million TSMC-made dies obtained before controls tightened, a stockpile Epoch expects to run out in 2026.

But AI progress did not stop. Models such as DeepSeek showed that Chinese labs could compete through algorithmic efficiency. And the controls gave Beijing an overwhelming reason to build its own industry: Goldman Sachs projects China's advanced-chip self-supply rising from about 8 percent in 2025 to about 66 percent by 2035, and China's share of ASML's sales falling from 41 percent in 2024 to 33 percent in 2025, with about 20 percent expected for 2026, means Western toolmakers lose revenue that funds their research.

The Trump administration's approach weakened coherence. Licensing the H20 and then the H200 with a 25 percent government share treated controls as a revenue source and bargaining chip. China then discouraged purchases, and H200 shipments were under 1 percent of Nvidia's data center revenue by mid-2026. The policy collected little money and signalled that controls were negotiable.

Judgment: a qualified, time-limited success. The controls slowed China's frontier hardware by several years, mainly because allies controlled lithography, but they accelerated China's drive to self-sufficiency, which will erode their leverage in the 2030s. They worked best when consistent and coordinated with allies, and worst when used as a deal-making tool.

What a strong answer must include.

- A stated definition of success (slowing versus stopping; hardware versus AI capability).
- At least five chapter facts, such as the ASML ban, the CFR performance gap, Epoch's Huawei estimates, Goldman's self-supply projection, ASML's China share and the H200 arrangement.
- The counter-effect: controls as an accelerant of Chinese self-reliance.
- The role of allies (the Netherlands, Japan), and a marked judgment.

Exercise 8.7: Essay with a judgment: Should Indonesia or Malaysia welcome more AI data centers?

Model answer.

Yes, but only on conditions that make data centers pay their full cost and leave behind more than rent. Without conditions, a data center is closer to a mine than a design center: it uses local land, power and water to serve a value chain controlled elsewhere.

Johor shows both sides. It went from about 10 MW of capacity in early 2021 to roughly 825 MW to 1 GW by 2026, with about 2 GW under construction and an estimated 35 billion dollars of investment. That brought construction jobs, tax and better connectivity. But BMI estimates Johor's data centers will need about 30 TWh a year by 2031, about 18 percent of Peninsular Malaysia's electricity, up from 6 percent in 2026, and Wood Mackenzie expects about 40 percent of Johor's demand by 2035. The most water-intensive designs can use up to 50 million litres a day, which led Johor to ban them in late 2025, and 2026 brought Malaysia's first community protests.

The value-capture picture is sobering. Following one dollar down the AI stack, the fat margins sit with Nvidia (about 75 percent), TSMC (about 60 percent) and ASML (about 53 percent). Electricity is only about a tenth of a data center's annual hardware cost. The host earns like a utility: steady but small. Yet that same fact gives the host bargaining power: because chips dominate cost, operators will pay high prices for fast, reliable power. Speed to power is the scarce thing a host sells, so it should not sell it cheaply.

Three policy conditions:

1. Bring your own clean power and pay full cost: every new site contracts new renewable generation and storage equal to its use, pays for its own grid connection, and pays a tariff that does not shift costs onto households.
2. Water and siting rules: closed-loop or liquid cooling, recycled water, location where the grid has headroom, and phased connection tied to new supply.
3. Local capability: binding training and hiring targets, a share of capacity available to local AI firms and universities at fair prices, and links to local chip design and engineering, as Malaysia is attempting with Penang.

Judgment: welcome them selectively. A country that prices power, water and land properly and uses data centers to grow local AI businesses gains; one that subsidises them with cheap power gains little and pays in grid strain.

What a strong answer must include.

- Johor numbers on capacity growth, power share and water.
- The value-capture logic (where margins sit in the AI stack; power as a small share of cost but a scarce location factor).
- Three specific, enforceable policy conditions.
- A clear judgment, not "it depends".

Exercise 8.8: Founder exercise: an AI supply chain bottleneck

Model answer (local-language data).

Bottleneck: high-quality, legally licensed training and evaluation data in Southeast Asian languages, especially for speech, education and government services. Frontier models are trained mostly on English and Chinese text; languages such as Javanese, Sundanese, Tagalog, Khmer and regional varieties of Malay and Indonesian are thin in the data and weak in model quality.

Company: a Semarang-based data and evaluation company. Product 1: licensed datasets built with consent from schools, broadcasters, publishers and paid native-speaker contributors (speech recordings, transcribed dialogue, exam-style questions with verified answers). Product 2: evaluation benchmarks and testing services that tell model makers and local companies how well a model performs in each language and domain. Product 3: fine-tuning services for local firms (banks, telecoms, schools) that need a model to work in local languages.

Customers: global model labs and cloud providers that want to sell in Southeast Asia (the chapter shows labs earning at a combined rate of over 100 billion dollars a year and racing for new markets), plus regional companies and governments.

Economics: data licences at tens to hundreds of thousands of dollars per dataset with recurring updates; evaluation as a subscription; contributors paid fairly and given a revenue share, which is also the social-impact metric. Capital needed is modest compared with hardware: mainly people, legal work and quality control.

What if the bottleneck disappears? It may: as models improve with synthetic data and cross-language transfer, raw data becomes less scarce. That is why the business must shift over five years from selling data to selling evaluation, fine-tuning and trusted local deployment, where knowledge of local institutions and relationships, not data volume, is the advantage. The plan should test this every year: if labs stop buying data, revenue must already come mainly from services.

What a strong answer must include.

- A specific bottleneck tied to evidence in the chapter.
- A named customer who pays, and why they pay now.
- A revenue model with rough numbers.
- A credible answer to "what if the bottleneck disappears", showing a path to a more durable advantage.

Month 9: Robotics, Anatomy and Industrial Economics

Exercise 9.1: Robot density

Robot density is the number of operational industrial robots per 10,000 manufacturing employees.

Original data:

- Robots per employee: 120,000 / 9,500,000 = 0.012632.
- Per 10,000 employees: 0.012632 x 10,000 = 126.3, so a density of about 126.

After the revision to 14 million employees:

- 120,000 / 14,000,000 = 0.008571.
- Per 10,000 employees: 85.7, so a density of about 86.

The density falls by 126.3 - 85.7 = 40.6 points, or about 32 percent, with no change on the factory floor.

Three-sentence explanation: The revision changes only the denominator, the count of workers, because the robots were counted separately (by the industry federation from supplier shipments) and none were added or removed. What changes is our picture of how automated the average worker's job is: the same machines are now spread over 47 percent more workers (14 / 9.5 = 1.47), so the country looks less automated, has more room for robot adoption, and may be ranked lower internationally. This is exactly what happened to China in April 2026, when the IFR revised its 2024 density from 567 to 166 per 10,000 workers after using a much larger employment figure, while its stock of about two million robots stayed the same.

Exercise 9.2: Payback

Net annual saving = wages saved for three workers minus the robot's maintenance cost. Payback period = purchase cost / net annual saving.

(a) Worker cost 45,000 dollars a year.

- Wages saved: 3 x 45,000 = 135,000 dollars.
- Net saving: 135,000 - 4,000 = 131,000 dollars a year.
- Payback: 120,000 / 131,000 = 0.92 years, about 11 months.

(b) Worker cost 12,000 dollars a year.

- Wages saved: 3 x 12,000 = 36,000 dollars.
- Net saving: 36,000 - 4,000 = 32,000 dollars a year.
- Payback: 120,000 / 32,000 = 3.75 years.

(c) Worker cost 4,000 dollars a year.

- Wages saved: 3 x 4,000 = 12,000 dollars.
- Net saving: 12,000 - 4,000 = 8,000 dollars a year.
- Payback: 120,000 / 8,000 = 15 years, more than twice the robot's 7-year life. The robot never pays for itself.

Break-even labour cost (payback = 7 years). Let w be the annual cost per worker.

- We need 120,000 / (3w - 4,000) = 7.
- So 3w - 4,000 = 120,000 / 7 = 17,142.86.
- 3w = 21,142.86.
- w = 7,047.62 dollars a year per worker, about 587 dollars a month.

Interpretation: below about 7,000 dollars a year per worker, this robot cell is not worth buying even before considering the cost of capital (which would push the threshold higher). That threshold sits above the base wages of experienced factory operators in Indonesia (385 dollars a month, about 4,620 dollars a year) and Vietnam (302 dollars a month) in JETRO's 2025 survey (Month 7), though total employer cost including benefits is higher than base wages. This is the core reason robot adoption is fastest in high-wage economies and slowest in low-wage ones, unless robots are needed for quality, safety or labour shortages rather than to save wages.

Exercise 9.3: Learning curve for a humanoid BOM

A 15 percent learning rate means each doubling of cumulative production multiplies the bill of materials (BOM) by 0.85.

At 400,000 units:

- Ratio: 400,000 / 25,000 = 16 = 2^4, so 4 doublings.
- BOM: 40,000 x 0.85^4 = 40,000 x 0.5220 = 20,880 dollars.

At 6.4 million units:

- Ratio: 6,400,000 / 25,000 = 256 = 2^8, so 8 doublings.
- BOM: 40,000 x 0.85^8 = 40,000 x 0.2725 = 10,900 dollars.

Doublings to reach 15,000 dollars:

- Required ratio: 15,000 / 40,000 = 0.375.
- 0.85^n = 0.375, so n = ln(0.375) / ln(0.85) = -0.9808 / -0.1625 = 6.04 doublings.
- Cumulative volume: 25,000 x 2^6.04 = 25,000 x 65.6 = about 1.64 million units.

Context from the chapter: Epoch AI estimates about 16,000 humanoids were produced in 2025, and Morgan Stanley projects a China-built BOM of 16,300 dollars at about one million units a year by 2034. Our 15 percent curve says the BOM reaches 15,000 dollars at about 1.6 million cumulative units, which is consistent with that kind of forecast. The big uncertainty is not the arithmetic but whether demand will arrive to carry production to millions: learning curves describe what happens if volume comes; they do not create it.

Exercise 9.4: Supply ceiling

Step 1, reducers available to robots each year (30 percent of world output):

- Planetary: 3,000,000 x 0.30 = 900,000.
- Strain wave: 12,000,000 x 0.30 = 3,600,000.

Step 2, humanoids each component could support:

- Planetary: 900,000 / 8 per robot = 112,500 humanoids.
- Strain wave: 3,600,000 / 6 per robot = 600,000 humanoids.

Step 3, the binding constraint is the smaller number. Planetary reducers bind, capping output at 112,500 humanoids a year. At that level the design uses 112,500 x 6 = 675,000 strain wave gears, under a fifth of the 3.6 million available.

Interpretation: this is about seven times the roughly 16,000 humanoids produced in 2025 (112,500 / 16,000 = 7.0), so reducer supply is not yet stopping anyone, but it would bind quickly under the 1.2 million a year that Bank of America forecasts for 2030. Two levers loosen it: redesigning to use fewer planetary reducers (for example, more strain wave gears or direct-drive joints), and new reducer capacity, which Epoch AI expects to at least double by 2028 because reducers, unlike chips, need no clean rooms. A founder should read this as a signal that reducer capacity, and the machine tools that make reducers, are where money can be made in the next five years.

Exercise 9.5: Robot hour

Assumptions carried over from the Week 3 example: a 5-year life, 300 working days a year, and 6,000 dollars a year for maintenance, repairs, software and electricity. Changed: price 25,000 dollars, three 8-hour shifts, 70 percent of human productivity.

Step 1, hours of operation:

- Per year: 3 shifts x 8 hours x 300 days = 7,200 hours.
- Over the life: 7,200 x 5 = 36,000 hours.

Step 2, capital cost per hour: 25,000 / 36,000 = 0.69 dollars.

Step 3, running cost per hour: 6,000 / 7,200 = 0.83 dollars.

Step 4, total cost per robot hour: 0.69 + 0.83 = 1.53 dollars.

Step 5, cost per human-equivalent hour: 1.53 / 0.70 = 2.18 dollars.

Compare the Week 3 result of about 7.30 dollars: lower price, more shifts and higher productivity together cut the effective cost by about 70 percent. (If running costs rise with use, say to 9,000 dollars a year at three shifts, the figure becomes (0.69 + 1.25) / 0.70 = 2.78 dollars.)

Comparison with a worker (Indonesia). JETRO's FY2025 survey of Japanese-affiliated companies, cited in Month 7, gives a monthly base wage of about 385 dollars for an experienced factory operator in Indonesia (August 2025). With a 40-hour week, a month has about 40 x 52 / 12 = 173 hours, so the base wage is 385 / 173 = 2.22 dollars an hour. Employer costs on top of base pay (social insurance, allowances, the annual religious-holiday bonus) push the total higher, perhaps to around 2.70 to 3.00 dollars an hour, but that markup is an assumption, not a figure from the survey. For Vietnam, the same survey gives 302 dollars a month, about 1.74 dollars an hour in base wage.

Conclusion: at 2.18 dollars per human-equivalent hour, this robot is roughly at parity with an experienced Indonesian operator and slightly above a Vietnamese one. In the United States, Germany or Japan, where hourly labour costs are many times higher, it would be overwhelmingly cheaper. A student answer should find and cite its own wage source; good options are JETRO's annual survey, national statistics offices, or the ILO's wage database.

Exercise 9.6: Essay: Japan's eroding reducer chokepoint versus China's durable magnet chokepoint

Model answer.

The two chokepoints differ in what they are made of. Japan's is accumulated skill in a precision mechanical part; China's is the refining and processing of materials at scale, protected by geology, environmental cost and long lead times. Skill can be learned by a rival with enough customers. Processing capacity at scale takes a decade to build and a willingness to bear pollution costs.

Start with reducers. Nabtesco says it holds about 60 percent of precision reducers for medium and large industrial robot joints, and Harmonic Drive Systems long led strain wave gears. But Chinese firms are learning fast. Leaderdrive grew revenue 47.3 percent in 2025 and holds about 12 percent of the global harmonic reducer market. The mechanism is learning-by-doing on a home market: China installed 354,226 industrial robots in 2025, 59 percent of the world, and 55 percent came from Chinese brands. Chinese robot makers were willing to try domestic reducers, which gave suppliers millions of hours of field feedback. Policy pushed dual sourcing, and cheap equity capital

let challengers build capacity before profit. The result: Leaderdrive's average price fell about 42 percent from 2017 to 2024, its gross margin fell from 52.2 to 36.9 percent, and the scarcity rent is being competed away. Lead times are short because reducers need precision machining, not clean rooms.

Magnets are different. China holds about 91 percent of magnet rare earth refining and 94 percent of sintered magnet production. No rival has a home market advantage here, because the constraint is not customers but supply. New rare earth mines take about eight years on average, refining produces radioactive waste that most democracies are reluctant to accept, and Chinese producers' scale keeps prices low enough to bankrupt newcomers, which is why the US guaranteed MP Materials a 110-dollar floor price. When China imposed licences in April 2025, carmakers shut lines within weeks.

Judgment: the asymmetry is real and will last at least a decade. Reducer power will shift towards proven reliability and premium qualification rather than disappear, while magnet power will erode only as fast as state-backed projects outside China come online, which on current lead times means the 2030s.

What a strong answer must include.

- Both mechanisms explicitly: learning-by-doing and home markets for reducers; lead times, environmental cost and scale for magnets.
- Evidence: market shares, Chinese price and margin falls, installation numbers, refining shares, the eight-year lead time, the 2025 licence episode.
- An explanation of why a home market helps for one chokepoint but not the other.
- Nuance: the reducer chokepoint splits into tiers and moves down to machine tools, rather than vanishing.

Exercise 9.7: Essay: a 15-year robot chain strategy for one country

Model answer (Malaysia).

Malaysia should aim to own three of the five layers by 2041: refining and materials, mid-range components, and integration and software, while partnering for ores and for frontier AI models and leaving generic humanoid assembly to others.

Layer 1, ores: do not build strategy here. Malaysia has some rare earth resources, but its advantage is processing, not mining. Secure feedstock through long-term contracts with Australian, African and other Southeast Asian mines.

Layer 2, refining and materials: expand. Malaysia hosts the Lynas plant, which the IEA calls the "distant second" in rare earth separation after China. Over years 1 to 7, the goal is to add heavy rare earth separation and a magnet plant with a Japanese or Korean partner, backed by long-term offtake contracts with buyers outside China who want non-Chinese supply. Strict radioactive waste rules are a condition of social licence, not a cost to cut.

Layer 3, precision components: enter selectively. Malaysia has about 13 percent of global chip assembly, packaging and testing and a strong electronics base in Penang. That fits servo drives, encoders, sensors and motor assembly better than reducers. Years 3 to 10: build these, using Exercise 7.8's logic of multi-source designs and documented non-Chinese content. Reducers are a deliberate "no" for now: Chinese makers are already in a price war (Leaderdrive's margin fell from 52.2 to 36.9 percent), so entering the commodity tier would burn capital.

Layer 4, assembly: limited. Do not compete with Chinese humanoid makers on generic robots. Assemble specialised robots for Southeast Asia's own factories, such as semiconductor handling and palm oil and logistics automation, where proximity and service matter.

Layer 5, AI and software: invest heavily in integration, fine-tuning open robot models for local tasks, and service firms. This is the layer most open to local founders and needs talent more than capital.

Enablers: cheap clean power for refining, engineering education, and patient capital from Khazanah with clear performance targets.

Judgment: a full ore-to-robot chain is not the right goal. A Malaysia that is the region's trusted non-Chinese source of magnets and robot electronics, plus its best integrators, would capture far more value than one that tries to make everything.

What a strong answer must include.

- All five layers addressed, with an explicit list of layers not entered and the reasons.
- Evidence specific to the chosen country, using the chapter's layer assessment.
- Time phasing over 15 years, with partners named by type.
- A link to value capture: why the chosen layers earn more, and what would change the strategy.

Exercise 9.8: Founder exercise: a dangerous or dirty job

Model answer (cleaning industrial tanks and silos in Central Java).

Job: cleaning the inside of storage tanks, silos and drains in palm oil, food, chemical and fuel facilities. Workers enter confined spaces with toxic gas, low oxygen and slippery residues, a well-known cause of fatal accidents worldwide.

Estimate (assumptions, to be checked against local data): about 2,000 workers in Central Java doing this full or part time. Wage about 385 dollars a month (using the JETRO Indonesian operator figure as a proxy), about 4,620 dollars a year, plus safety equipment, standby crews and insurance that could double the cost per cleaning job. A typical tank cleaning takes a crew of four about three days, and the plant loses production while the tank is empty.

Robot: a tracked, remote-controlled cleaning robot with high-pressure nozzles, cameras and gas sensors, roughly 40,000 to 60,000 dollars if built from purchased motors, drives, sensors and an open-source control stack.

Sell, rent or service? Sell the outcome, not the robot. Using Exercise 9.2's logic, a 50,000-dollar robot replacing two workers at about 4,620 dollars each would take 50,000 / (9,240 - 4,000 maintenance) = about 9.5 years to pay back, longer than its likely life. On wages alone, selling the machine does not work in a low-wage market. But the value is not mainly wages: it is avoided deaths and injuries, shorter downtime, and compliance with safety rules. If the service cuts a three-day job to one day and a plant loses, say, 3,000 dollars a day of output, each job saves about 6,000 dollars plus risk. Charging 3,000 dollars per job, one robot doing 40 jobs a year earns 120,000 dollars, enough to pay back the robot in under a year after operator wages and transport.

Social impact metric: confined-space worker-hours eliminated and incidents avoided, while retraining the existing cleaners as robot operators.

What a strong answer must include.

- Explicit estimates of workers, wages and risk, with sources or clearly labelled assumptions.

- A robot cost and a payback or unit-economics calculation.
- A justified choice between selling, renting and service, recognising that in low-wage markets the case rests on safety, quality and downtime, not wage savings.
- A measurable social-impact indicator.

Month 10: AI, Robots and the Labor Economy

Exercise 10.1: Growth accounting

The growth accounting identity from Week 1 is: output growth = (capital share x capital growth) + (labour share x labour growth) + TFP growth. With a capital share of 35 percent, the labour share is 100 - 35 = 65 percent.

Step 1, contribution of capital: 0.35 x 3% = 1.05 percentage points.

Step 2, contribution of labour: 0.65 x 0.5% = 0.325 percentage points.

Step 3, TFP growth is the residual: 2.5% - 1.05% - 0.325% = 1.125%, about 1.1 percent a year.

Step 4, the effect of AI adding 0.3 percentage points to TFP growth for ten years, holding capital and labour growth unchanged.

- Without AI, output grows 2.5 percent a year. After 10 years the output index is 1.025^10 = 1.2801.
- With AI, output grows 2.8 percent a year. After 10 years the index is 1.028^10 = 1.3184.
- Ratio: 1.3184 / 1.2801 = 1.0297.

So output is about 3.0 percent higher after ten years than it would have been without AI. (A shortcut gives almost the same answer: 1.003^10 = 1.0304, or simply 10 x 0.3 = 3 percent.)

A refinement: in the Solow model, higher TFP also raises the return on capital, so firms invest more and capital grows faster. Over the long run this roughly multiplies the TFP effect by 1 / labour share, here 1 / 0.65 = 1.54, so the full output gain could approach 3.0 x 1.54 = about 4.6 percent. The simple calculation above, which holds capital growth fixed, is therefore a lower bound.

Exercise 10.2: Hulten's theorem

The Week 1 method: aggregate TFP gain = (share of tasks exposed) x (share of exposed tasks profitably automated) x (average cost saving on those tasks).

Step 1, share of all tasks affected: 30% x 40% = 12% of all tasks.

Step 2, TFP gain over 10 years: 12% x 25% = 3.0%.

Step 3, per year: about 3.0 / 10 = 0.3 percent a year (more precisely 1.030^(1/10) - 1 = 0.296 percent).

Comparison:

- Against Acemoglu's 0.66 percent over ten years: our 3.0 percent is about 4.5 times larger (3.0 / 0.66 = 4.5).
- Against Goldman Sachs's 7 percent rise in global GDP: our estimate is under half as large (3.0 / 7 = 0.43).

Which assumption explains most of the difference? Compare our inputs with the chapter's illustrative Acemoglu-style example (20 percent exposed, 25 percent profitable, 15 percent saving, giving 0.75 percent):

- Breadth (share of all tasks affected): ours is 12 percent against 5 percent, a factor of 2.4.
- Depth (cost saving): ours is 25 percent against 15 percent, a factor of 1.67.
- Combined: 2.4 x 1.67 = 4.0, which takes 0.75 percent to 3.0 percent.

So breadth, how many tasks AI is assumed to affect within a decade, explains most of the gap with Acemoglu. (In his own paper, about 4.6 percent of all tasks are affected, with an overall cost saving of about 14 percent; his 0.66 percent figure is roughly 4.6 x 14.4 percent.)

The gap with Goldman Sachs has two further causes that Hulten's theorem leaves out. First, Goldman measures GDP, not TFP: as in Exercise 10.1, higher TFP draws in more capital, so the GDP effect is larger than the TFP effect (3.0 / 0.65 = 4.6 percent on our numbers). Second, Goldman assumes much wider adoption, deeper automation of whole jobs, and new tasks and products, which Hulten's theorem does not count at all. The tenfold disagreement among forecasters is a disagreement about breadth and new tasks, not about arithmetic.

Exercise 10.3: The labour share

Week 3 starting point: revenue 10 million dollars; wages 6 million (150 workers); other costs 2 million; profit 2 million. After automation, wages are 3 million (75 workers) and robot costs 1.5 million. Now competition forces revenue down to 9 million.

Step 1, new profit: 9 - 3 (wages) - 1.5 (robots) - 2 (other costs) = 2.5 million dollars.

Step 2, value added (wages + profit + robot costs, counting robot costs as capital income to the robot maker and lenders, as the chapter does): 3 + 2.5 + 1.5 = 7 million dollars.

Step 3, labour share of value added: 3 / 7 = 42.9 percent.

Comparison:

	Before automation	Automation, revenue 10m	Automation, revenue 9m
Wages	6.0m	3.0m	3.0m
Robot costs	0	1.5m	1.5m
Profit	2.0m	3.5m	2.5m
Labour share	75.0%	37.5%	42.9%

Step 4, who captures the gain. Automation saves 3 million in wages and costs 1.5 million in robots, a net resource saving of 1.5 million a year. With the 1 million price cut:

- Consumers gain 1.0 million through lower prices.
- The firm's owners gain 0.5 million (profit rises from 2.0 to 2.5 million, instead of to 3.5 million).
- The robot maker's owners and lenders receive 1.5 million (payment for the robots, part of which is their profit).
- The 75 displaced workers lose 3.0 million in wages, unless they find other jobs.

Competition shifts most of the firm's gain to consumers, and the labour share is a little higher than without the price cut (42.9 against 37.5 percent) because profit is smaller. But workers are still

the big losers in the first round. The longer-run question is the productivity effect: if lower prices raise demand enough, the firm may hire again (see Exercise 10.4), and consumers' savings are spent elsewhere, creating jobs in other sectors.

Exercise 10.4: Demand elasticity

A 25 percent productivity boost means each worker handles 1.25 times as much. The workers needed per unit of service fall to 1 / 1.25 = 0.80 of the old level. Employment = demand x workers needed per unit.

Demand rises 10 percent:

- Employment index: 1.10 x 0.80 = 0.88.
- Employment falls 12 percent.

Demand rises 40 percent:

- Employment index: 1.40 x 0.80 = 1.12.
- Employment rises 12 percent.

Break-even: employment stays constant when demand x 0.80 = 1, so demand = 1 / 0.80 = 1.25. Demand must rise 25 percent, exactly matching the productivity gain.

Example with numbers: a call centre with 1,000 agents. After the boost it needs 800 agents for the old volume. At +10 percent demand it needs 880 (120 fewer); at +40 percent it needs 1,120 (120 more); at +25 percent it needs exactly 1,000.

The general rule: automation of a task raises employment in that activity only if demand grows faster than productivity. That depends on the price elasticity of demand and on how much of the productivity gain is passed on in lower prices. ATMs and bank tellers (Week 4) are an example of demand growing enough; many customer-service centres, where demand is limited, may not be.

Exercise 10.5: Essay: US versus German robot outcomes

Model answer.

The same technology produced different results because it landed in different institutions and different demand conditions. Of the mechanisms, I find export demand the most convincing primary cause, with worker voice deciding how the remaining adjustment was shared.

The evidence is stark. Acemoglu and Restrepo found that in US local labour markets from 1990 to 2007, one more robot per thousand workers reduced the employment-to-population ratio by about 0.18 to 0.34 percentage points and wages by 0.25 to 0.5 percent. Dauth and co-authors found that in Germany, which has more robots per worker (449 per 10,000 manufacturing workers in 2024 against 307 for the US), each robot replaced about two manufacturing jobs, about 275,000 in total from 1994 to 2014, but these losses were fully offset by new service jobs.

Mechanism one, worker voice and job protection. German works councils and unions made dismissals costly and negotiated, so incumbents were redeployed to new tasks in the same plant. The adjustment fell on young people, who found fewer factory openings, rather than on existing workers.

Mechanism two, training. Germany's apprenticeship system made it easier to move workers into new tasks and pushed young people toward university.

Mechanism three, export demand. German carmakers sold into expanding world markets, so robots raised output and kept plants growing. Many US robots arrived in regions already hit by import competition from China and Mexico, where demand was shrinking.

Mechanism four, where robots were used. In Germany robots clustered in the expanding car industry; in the US they were spread into declining sectors.

Why I rank demand first: Exercise 10.4 shows that whether automation destroys jobs in an activity depends on whether demand grows faster than productivity. Worker protection can redistribute who bears the loss, but it cannot create jobs if output is falling; Germany's protections worked because there was growth to redeploy workers into. Without export demand, works councils would have managed decline, not avoided it.

Judgment: demand explains whether jobs were lost in total; institutions explain who lost and how. For a developing country, this implies automation goes best when tied to growing export markets and paired with training and worker voice.

What a strong answer must include.

- The two studies with their key numbers.
- At least three distinct mechanisms, explained (not just listed).
- A reasoned ranking with an argument for why the chosen mechanism matters most.
- Awareness that Germany also had costs (fewer jobs for young people, a falling labour share).

Exercise 10.6: Essay: "The horse analogy is wrong because humans own capital and vote"

Model answer.

The claim is right in principle and potentially wrong in practice. Humans differ from horses because they can claim a share of machine income through ownership and politics. But that protection works only if ownership is broad and political institutions are inclusive. Where either fails, the horse analogy could come uncomfortably close.

Leontief's analogy is powerful. Horses were indispensable until the combustion engine, then the US horse population, which peaked in the 1910s, collapsed in a few decades. If machines become better than humans at nearly everything, labour income could collapse the same way. The rebuttal, as the chapter says, is that horses could not demand a share of tractor profits; humans can.

But look at the conditions. First, ownership. In the United States in 2026, the richest 1 percent of households held about 50.9 percent of corporate equities and the bottom half about 1 percent. In most developing countries ownership is even more concentrated in founding families, the state and foreign investors. If robots and AI raise the capital share, as the Week 3 logistics example shows (the labour share fell from 75 to 37.5 percent), most people own almost none of the machines.

Second, ownership across borders. Much of the value of robots and AI is captured by firms in the US, China, Japan and Taiwan. A worker in Semarang does not vote in the countries that tax Nvidia or the leading AI labs.

Third, voting must work. Month 2 showed that extractive institutions let elites capture resource rents. A society where income comes from capital rather than labour looks like a petrostate: ordinary people are less needed as workers and taxpayers, so their political leverage can weaken.

Fourth, it depends on whether labour remains a complement. In Allen's account of Engels' pause, wages eventually rose because capital and labour were complements. If AI becomes a near-perfect substitute, that safety valve closes.

The claim fails, then, when ownership is concentrated, when gains flow abroad, when institutions are extractive, and when machines substitute rather than complement. Alaska shows the remedy: a public fund that turned oil rent into citizen capital.

Judgment: humans are not horses only if they act like owners. That requires deliberate policy, such as broad ownership, sovereign funds and fair taxes, and it will not happen automatically.

What a strong answer must include.

- A fair statement of the analogy and of the rebuttal.
- Specific conditions under which the rebuttal fails, supported by evidence (ownership concentration, international capture, institutions, substitution versus complementarity).
- Links to Month 2 (institutions) and the Alaska case.
- A judgment that specifies what would make the claim true.

Exercise 10.7: Policy design: a robot dividend

Model answer (Indonesia, 2040).

Goal: by 2040, pay every Indonesian resident an annual dividend from the returns of automation, while keeping the fund safe from politicians.

Sources of money:

1. Public stakes. A share of the state's existing holdings, now consolidated under Danantara (Month 11), would be transferred into a separate Citizens' Automation Fund, with a rule that a fixed share of dividends from state-owned firms in telecoms, energy, mining and future robot and AI ventures flows in each year.
2. Resource rents. A fixed share of nickel and other mineral royalties, which already fund the industrial base on which robots depend.
3. Taxes on concentrated automation income: a modest levy on very large companies' profits above a normal return (an excess-profit tax), not a tax on robots themselves, which would slow productive investment.
4. Public data and spectrum: licence fees for spectrum and royalties for commercial access to government datasets used to train AI.

Design: like Alaska, the fund saves the money and invests it in a diversified, mainly global portfolio, so the dividend depends on investment returns, not on any one industry. Each year it pays out a fixed share, for example 3 to 4 percent of the fund's average value over five years, divided equally among residents.

Protection from political capture:

- A legal basis that is hard to change (Alaska's fund is in its constitution), with a supermajority needed to alter withdrawal rules.
- An independent board with fixed terms, professional managers and no politicians, following Norway's model.
- Full public reporting of holdings and returns, and an annual audit by the Supreme Audit Agency (BPK) and an international auditor.

- A ban on lending to the government or investing in politically chosen domestic projects, the failure that destroyed 1MDB in Malaysia.
- A universal, equal payment, which builds a large constituency that defends the fund, as in Alaska.

Evidence on work: Jones and Marinescu found the Alaska dividend did not reduce employment, so fears of idleness should not block the design.

Judgment: a robot dividend is less about taxing machines than about turning shared assets into citizen capital, under institutions strong enough to stop elites capturing the rent.

What a strong answer must include.

- Specific funding sources (taxes, public stakes, rents) with a reason for each, and awareness that taxing robots directly can slow productivity.
- Concrete anti-capture rules drawn from Alaska, Norway and Month 2's institutions framework, and an example of failure (1MDB).
- A payout rule that protects the capital.
- The labour-market evidence on dividends.

Exercise 10.8: Founder exercise: a job in your city

Model answer (pharmacy assistant in Semarang).

Tasks, with a classification:

Task	Likely outcome in 10 years
Reading prescriptions and checking doses	Augmented (AI checks; human confirms)
Picking medicines from shelves	Automated in large pharmacies (robot dispensers)
Checking for drug interactions	Augmented
Managing stock and re-ordering	Automated
Counselling patients on how to take medicine	Augmented (AI drafts, human explains)
Spotting a patient who needs a doctor urgently	Unaffected or augmented; human judgment
Handling payments and insurance claims (BPJS)	Mostly automated

Product: an AI assistant for small independent pharmacies (apotek) that reads handwritten or digital prescriptions, flags dose errors and interactions, suggests counselling points in Indonesian and Javanese, and manages stock automatically. It augments the assistant: the time saved on stock and paperwork is redirected to counselling patients, which raises the pharmacy's value to customers.

Who captures the value:

- The pharmacy owner gains through fewer errors, less wasted stock and more loyal customers.
- The assistant keeps the job and becomes more skilled (the Brynjolfsson, Li and Raymond study found a 34 percent gain for novices), which may support higher pay if the pharmacy shares the gain.
- Patients gain from fewer medication errors.
- My company captures a subscription, say 300,000 rupiah a month per pharmacy, while the AI model provider captures a cost per use, which Month 8 shows is falling rapidly.

Risk: large chains may build their own tools or replace staff with robot dispensers. My defence is focusing on independent pharmacies, local languages and integration with local insurance systems.

What a strong answer must include.

- A complete task list for a real job, with each task classified.
- A product that clearly augments rather than replaces, and explains how freed time is used.
- A value-capture analysis covering the worker, the employer, the customer, the founder and upstream suppliers.

Month 11: Money, Finance and Sovereign Capital

Exercise 11.1: Debt stability

The Week 3 debt equation: change in debt ratio = (r - g) / (1 + g) x d - pb, where d is debt as a percentage of GDP, r the average interest rate, g nominal GDP growth and pb the primary balance (positive means a surplus). To keep the debt ratio stable, set the change to zero, so the required primary balance is pb = (r - g) / (1 + g) x d.

Case 1: r = 9 percent, g = 6 percent, d = 80 percent.

- r - g = 0.09 - 0.06 = 0.03.
- Divide by 1 + g: 0.03 / 1.06 = 0.0283.
- Multiply by debt: 0.0283 x 80 = 2.26.
- Required primary balance: a surplus of about 2.3 percent of GDP.

Check: with no primary surplus, debt would rise from 80 to about 82.3 percent of GDP in one year, and keep rising.

Case 2: r = 5 percent, g = 6 percent, d = 80 percent.

- r - g = 0.05 - 0.06 = -0.01.
- Divide by 1.06: -0.00943.
- Multiply by 80: -0.75.
- Required primary balance: -0.75 percent of GDP, so the government can run a primary deficit of about 0.75 percent of GDP and still hold debt steady at 80 percent.

The swing between the two cases is 2.26 - (-0.75) = 3.0 percentage points of GDP a year, from a 4-point fall in the interest rate. For a country with GDP of, say, 1.4 trillion dollars, that is about 42 billion dollars a year of spending cuts or tax rises that a lower interest rate makes unnecessary. This is why credibility, which lowers r, is worth so much: when interest is above growth, debt grows on autopilot.

Exercise 11.2: The cost of capital for a battery-materials plant

The Week 2 annuity formula: annual payment = investment x r / (1 - (1 + r)^-n). Here the investment is 2,000 million dollars and n = 25 years.

At 4 percent:

- (1.04)^-25 = 0.3751.
- 1 - 0.3751 = 0.6249.
- Annuity factor: 0.04 / 0.6249 = 0.06401.

- Annual capital cost: 2,000 x 0.06401 = 128.0 million dollars.

At 11 percent:

- (1.11)^-25 = 0.0736.
- 1 - 0.0736 = 0.9264.
- Annuity factor: 0.11 / 0.9264 = 0.11874.
- Annual capital cost: 2,000 x 0.11874 = 237.5 million dollars.

Difference: 237.5 - 128.0 = 109.5 million dollars a year.

Per tonne, at 60,000 tonnes a year:

- At 4 percent: 128.0 million / 60,000 = about 2,134 dollars per tonne.
- At 11 percent: 237.5 million / 60,000 = about 3,958 dollars per tonne.
- Difference: 109.5 million / 60,000 = about 1,824 dollars per tonne.

Over 25 years the high-cost plant pays about 109.5 x 25 = 2.7 billion dollars more, more than the cost of the plant itself. In a commodity market where both plants sell at the same world price, the plant with cheap capital can undercut the other by about 1,800 dollars a tonne and still earn the same return. Compared with the chapter's 1 billion dollar, 20-year example (a gap of about 54 million dollars a year, or 1,070 dollars a tonne), the gap here is larger because the plant is bigger, the life is longer (which favours cheap money more, because interest compounds over more years) and the rate gap is 7 points. This is how finance, not geology, often decides who captures the value of downstream processing.

Exercise 11.3: Founder dilution

We follow the chapter's method: the option pool is created first and dilutes the founder before any money comes in; each round dilutes all existing holders in proportion.

Step 1, option pool:

- Founder: 100% x (1 - 0.12) = 88%.

Step 2, seed round: 500,000 dollars at a 2 million dollar pre-money valuation.

- Post-money valuation: 2.0 + 0.5 = 2.5 million dollars.
- Seed investors' stake: 0.5 / 2.5 = 20%.
- Founder: 88% x (1 - 0.20) = 70.4%.

Step 3, Series A: 4 million dollars at a 16 million dollar pre-money valuation.

- Post-money valuation: 16 + 4 = 20 million dollars.
- Series A investors' stake: 4 / 20 = 20%.
- Founder: 70.4% x (1 - 0.20) = 56.32%.

Step 4, paper value of her stake: 56.32% x 20 million = 11.26 million dollars.

Cap table after both rounds:

Holder	Stake	Paper value at 20m post-money
Founder	56.32%	11.26m
Option pool	12% x 0.8 x 0.8 = 7.68%	1.54m
Seed investors	20% x 0.8 = 16.00%	3.20m

Holder	Stake	Paper value at 20m post-money
Series A investors	20.00%	4.00m
Total	100.00%	20.00m

Two cautions. First, "paper value" is not money: it assumes every share is worth the Series A price, but investors usually hold preferred shares with a liquidation preference (they get their money back first), so in a modest sale the founder could receive far less than 11.26 million. Second, Series A investors often require the option pool to be topped up before their investment (the "option pool shuffle"), which would dilute the founder further. Her real wealth comes only when the company is sold or listed, which is why the MSCI episode in this month matters to founders: a weak local stock market makes exits harder.

Exercise 11.4: Bank Indonesia and the impossible trinity, October 2026

Model answer.

The impossible trinity says a country can have at most two of three things: a fixed exchange rate, free capital movement and an independent monetary policy. Indonesia keeps capital mostly open and sets its own rate, so it cannot also fix the rupiah. In practice it tries to get a little of all three: it lets the rupiah float but smooths sharp moves. In October 2026 that balancing act was under maximum strain.

The pressures were severe. Brent was still far above its pre-war level of about 72 dollars: on October 30 the front-month January contract settled at 113.43 dollars (the expiring December contract at 117.64), raising Indonesia's import bill as a net oil importer. The US 10-year Treasury yield had hit about 5.24 percent on October 1, its highest since 2002, pulling money toward dollars. And MSCI's October 15 proposal to reclassify Indonesia to standalone status sent the Jakarta Composite down 11.6 percent in one day, with UBS estimating about 4 billion dollars of forced passive selling. The rupiah weakened toward 18,161 per dollar by November 2.

Bank Indonesia's options were:

1. Raise the BI-Rate (for example from 5.75 to 6.00 percent, as about half of the economists Bloomberg surveyed expected), to defend the rupiah at the cost of slower growth.
2. Hold the rate and sell reserves, which stood at 146.3 billion dollars at the end of September, to smooth the rupiah.
3. Let the rupiah fall, accepting imported inflation.
4. Restrict capital outflows, which would break the "free capital" corner and deepen MSCI's doubts.

It chose a targeted middle path. Holding the policy rate avoided tightening credit for the whole economy after 100 basis points of hikes in May and June. Raising the deposit facility rate to 5.00 percent lifted the floor of short-term market rates, paying more for funds parked at the central bank, which supports the rupiah at the margin without raising every loan rate. Cheaper hedging for foreign investors attacked the specific problem: BI said these incentives brought in about 9.9 billion dollars between June and October 20. Cutting reserve requirements kept banks able to lend.

What I would have done: the same, plus a clear statement that further hikes would follow if the rupiah's fall started feeding into core inflation. The shock was partly about market governance, not interest rates, so a rate rise would have hurt borrowers without fixing MSCI's concerns. The best

defence of the rupiah in this case is credible market reform, which is the government's job, not the central bank's.

What a strong answer must include.

- A correct statement of the trinity and where Indonesia sits in it.
- The specific October 2026 pressures (oil, US yields, MSCI) with numbers.
- At least three options and their trade-offs.
- An explanation of why a deposit facility rate rise is a narrower tool than a policy rate rise, and a reasoned personal choice.

Exercise 11.5: Essay: Is the dollar's dominance good or bad for developing countries?

Model answer.

On balance, dollar dominance is a net benefit to most developing countries in normal times and a heavy cost in crises. The benefits are steady and spread out; the costs arrive in concentrated shocks. That makes the real question not whether to leave the dollar, which no developing country can do alone, but how to reduce exposure to its worst effects.

The benefits are real. The dollar is on one side of 89 percent of foreign exchange trades (BIS, April 2025), and roughly half of world trade is invoiced in dollars. One common currency lowers the cost of trading: an Indonesian exporter can sell to Kenya without needing a rupiah-shilling market. Dollar trade finance is deep and cheap, and US Treasury markets give central banks a safe, liquid place to hold their 13.22 trillion dollars of reserves.

The costs are large. First, original sin: most developing countries cannot borrow abroad in their own currency, so when their currency falls, their debts rise. Indonesia learned this in 1997-98, when the rupiah fell from about 2,400 to 16,000-17,000 per dollar, companies' dollar debts rose about six-fold in rupiah terms and the economy shrank about 13 percent. Second, US rate shocks: decisions in Washington are made for the US economy, not for Jakarta. In 2026, a 10-year Treasury yield above 5 percent pulled capital toward the US and forced Bank Indonesia to raise rates by 100 basis points. Third, sanctions risk: in 2022 about 300 billion dollars of Russia's reserves were frozen, showing that "safe" assets are only as safe as relations with the issuer.

The costs fall hardest on countries with large short-term dollar debts and weak institutions, and they can be reduced: hold large reserves, borrow in local currency, build domestic bond markets and let the currency float. Since 1998 Indonesia has done all four, which is why the rupiah's slide from about 15,000 to over 18,000 per dollar between 2023 and 2026 was painful but not catastrophic.

Judgment: dollar dominance is good for developing countries as traders and bad for them as borrowers. The right response is not de-dollarisation for its own sake, but reducing dollar debt, building local capital markets and diversifying reserves modestly, including into gold.

What a strong answer must include.

- All three costs named in the question (original sin, rate shocks, sanctions risk) and both benefits, each with evidence.
- A historical case (1997-98) and a current one (2026).
- A distinction between the country as trader and as borrower, or a similar insight into who bears the costs.
- A clearly marked judgment with policy implications.

Exercise 11.6: Essay: Governance rules for an African mineral fund

Model answer (a hypothetical "Zambian Minerals Development Fund").

A fund that both saves mineral wealth and finances a battery and motor supply chain has two jobs that pull against each other: savers want safety and diversification abroad; builders want risky domestic projects. The answer is to separate them by law, with strict rules for each.

Structure, borrowed from Chile and Norway: three compartments. (1) A stabilisation compartment, like Chile's Economic and Social Stabilization Fund, receives copper revenue above a long-run reference price set by an independent panel, and releases it only when prices fall below that reference. (2) A savings compartment, like Norway's fund, invests abroad in a diversified, mostly passive portfolio, to avoid the Dutch disease. Spending is limited to its expected real return (Norway uses 3 percent). (3) A development compartment, like Temasek, invests in domestic battery-precursor, cathode and motor projects, capped at, for example, 20 percent of total assets.

Rules for the development compartment:

- Commercial test: it invests only alongside private or foreign partners who put in at least as much money as the fund does, and only at a target return above its cost of capital. This prevents vanity projects.
- Kill rule: projects that miss milestones lose follow-on funding.
- No lending to the government, no rescues of state firms, and no budget transfers in. Danantara's new development arm, which may receive budget money as equity, shows how this line gets blurred.

Governance, borrowed from Norway and Temasek:

- Founding law with a supermajority required to change withdrawal rules.
- A professional board with fixed, staggered terms, appointed through a public process; no serving politicians or party officials.
- Published annual accounts audited by an international firm, a full list of holdings, and an annual hearing in parliament.
- Compliance with the Santiago Principles for sovereign funds.

Avoiding 1MDB: 1MDB failed because a fund close to the head of government borrowed heavily, invested secretly and lost billions to theft. So: no borrowing without parliamentary approval; all transactions above a threshold published within 30 days; conflict-of-interest declarations for staff and board; and an independent whistleblower channel.

Judgment: a mineral fund can help finance a supply chain only if it is first a trustworthy saver. Credibility lowers the country's cost of capital, which, as Week 2 showed, may be worth more to industry than any single investment the fund makes.

What a strong answer must include.

- Specific mechanisms borrowed from each model: Norway (overseas investment, spending rule, transparency), Temasek (commercial ownership of companies), Chile (counter-cyclical stabilisation).
- A way to reconcile saving and domestic industrial investment, such as separate compartments and caps.
- Concrete anti-capture rules tied to how 1MDB failed.
- A link to the cost-of-capital argument.

Exercise 11.7: Research: the MSCI decision on Indonesia

Model answer and method.

As of November 2026, when this book was written, MSCI has not decided. Students doing this exercise after June 30, 2027 should replace the outline below with the actual outcome. This model shows what to look for and how to judge it.

What was known in November 2026 (Week 2 case study):

- On October 15, 2026, MSCI proposed reclassifying Indonesia to standalone status, citing "continued suspected coordinated trading behavior and lack of evidence of meaningful enforcement". Consultation runs until April 30, 2027, with a decision on or before June 30, 2027. MSCI said it expects to go ahead unless there is "clear evidence of material improvements".
- Reforms announced: the minimum free float raised gradually toward 15 percent, disclosure of shareholders above 1 percent, fines of 2.1 trillion rupiah (about 116 million dollars) on entities linked to manipulation, and police investigations.
- Flows: the Jakarta Composite fell 11.6 percent on October 16 and 5 percent on October 19, wiping out about 65 billion dollars. UBS estimated about 4 billion dollars of passive selling if standalone status is confirmed, plus up to 15 billion from active funds; Goldman Sachs estimated 5.6 billion dollars of passive outflows. Indonesia's weight in the MSCI Emerging Markets index was 0.46 percent.

How to research the outcome:

1. Read MSCI's own announcement on its investor relations site (the primary source), and note the exact classification and effective date.
2. List reforms actually enforced, not only announced: check the Financial Services Authority (OJK) and the Indonesia Stock Exchange for implemented free-float rules, completed enforcement cases and court outcomes.
3. Measure flows: foreign net buying or selling from the Indonesia Stock Exchange's data, fund-flow reports, and the rupiah's movement against the dollar around the decision date. Compare actual outflows with the UBS and Goldman estimates.
4. Judge: did enforcement change behaviour (for example, fewer suspicious price spikes in thinly traded stocks)?

Judgment framework: if MSCI keeps Indonesia in Emerging Markets, the lesson is that enforcement, not rhetoric, moved the decision. If it confirms standalone status, the cost will show up as a higher cost of equity for Indonesian firms and harder exits for venture investors, which falls hardest on future founders.

What a strong answer must include.

- The actual decision and its date from MSCI's primary source (no guessing before it exists).
- A distinction between reforms announced and reforms enforced.
- Flow data compared with the pre-decision estimates.
- An assessment of what it means for Indonesia's cost of capital.

Month 12: Geopolitics, Scenarios and Strategy, 2026-2050

Exercise 12.1: The cost of an oil shock

The exercise uses Brent at about 72 dollars a barrel on February 27, 2026 and about 113 dollars on October 30, 2026, the settle of the front-month January contract (113.43 dollars). The expiring December contract settled at 117.64 dollars the same day; that version is worked at the end.

Step 1, price increase per barrel: 113 - 72 = 41 dollars.

Step 2, extra cost per day: 41 x 1,500,000 barrels = 61.5 million dollars a day.

Step 3, extra cost per year: 61.5 million x 365 = 22,447.5 million dollars, about 22.4 billion dollars.

Step 4, share of GDP: 22.4 / 1,400 = 0.0160, about 1.6 percent of GDP.

For context, the total annual import bill rises from 72 x 1.5 million x 365 = 39.4 billion dollars (2.8 percent of GDP) to 113 x 1.5 million x 365 = 61.9 billion dollars (4.4 percent of GDP), an increase of 57 percent.

With the unrounded January settle of 113.43 dollars, the increase is 41.43 dollars a barrel: 41.43 x 1.5 million x 365 = 22.7 billion dollars a year, still about 1.6 percent of GDP.

With the expiring December contract's 117.64 dollars instead: the increase is 45.64 dollars a barrel, so 45.64 x 1.5 million x 365 = 25.0 billion dollars a year, about 1.8 percent of GDP. Which contract you use moves the answer by about 2 billion dollars, a reminder to say which price series a number comes from.

Why it matters: an extra 1.6 percent of GDP spent on imported oil widens the current account deficit by roughly that amount, pressures the currency (Month 11) and, if fuel is subsidised, raises the budget deficit too. The numbers are illustrative, because a country does not buy all its oil at the spot price on one day: contracts, hedges, refined product imports and falling demand at high prices all change the real figure. Also, the price on one day is not the yearly average: Brent peaked at 126.41 dollars on April 30, 2026, fell back to the low 70s after a ceasefire and a June memorandum of understanding, rose above 100 dollars again when fighting resumed in July, and fell to 104.77 dollars on November 2 on hopes of peace talks (Month 11).

Exercise 12.2: Robot installation growth

Step 1, growth factor over four years (2025 to 2029): 806,000 / 603,307 = 1.336.

Step 2, compound annual growth rate (CAGR): 1.336^(1/4) - 1.

- The fourth root of 1.336: ln(1.336) = 0.2897; 0.2897 / 4 = 0.0724; e^0.0724 = 1.0751.
- CAGR = about 7.5 percent a year.

Check: 603,307 x 1.0751^4 = 603,307 x 1.336 = 806,000.

Step 3, Southeast Asia's installations.

- 2025 at 5 percent: 603,307 x 0.05 = about 30,165 robots.
- 2029 at 8 percent: 806,000 x 0.08 = 64,480 robots a year.
- Growth: 64,480 / 30,165 = 2.14 times, which implies a CAGR for Southeast Asia of 2.14^(1/4) - 1 = about 20.9 percent a year, nearly three times the world rate.

Interpretation: a rise in share from 5 to 8 percent sounds modest, but it means Southeast Asia's robot market more than doubles in four years. For a founder, that is the size of the integration, maintenance and training market (Week 4's first opportunity area): roughly 34,000 extra robot installations a year by 2029, each needing someone to install, program and service it.

Exercise 12.3: Scenario analysis: Indonesia's nickel-to-battery strategy

Model answer.

Indonesia mines about 67 percent of the world's nickel (2025, USGS) and wants to move through processing toward battery materials, cells and eventually motors. In the four scenarios:

A, "Open Automation" (managed rivalry, fast diffusion). Demand for batteries, EVs and robots grows fast and trade stays open. Indonesia's nickel is in demand, but it competes head-on with large, automated Chinese plants, many of which are the same Chinese firms operating in Indonesia. Its edge is cheap clean power and ore next door; its risk is that nickel-free lithium iron phosphate (LFP) batteries keep gaining share.

B, "Slow Convergence" (managed rivalry, slow diffusion). Chinese overcapacity keeps prices low, as in the 2023-2025 battery-materials glut. Smelter margins are thin and cathode or cell plants lose money for years.

C, "Two Stacks" (deep split, fast diffusion). Western carmakers pay a premium for battery materials not controlled by China, and rules of origin reward traceable local value. This is Indonesia's best box, but only if its plants' ownership, technology and power meet US and EU rules. Today most of Indonesia's nickel processing is Chinese-owned and coal-powered, which disqualifies much of it. Indonesia must hedge with some lower-carbon lines built with Japanese, Korean or Western partners.

D, "Fortress World" (deep split, slow diffusion). Blocs close and technology is expensive. Plants built on Chinese technology cannot sell to the West, so domestic and ASEAN markets become the main outlet and the industry is hostage to politics.

Robust actions (sensible in all four):

1. Decarbonise the power supply for industrial parks (solar, hydro, geothermal), which lowers costs in A, qualifies for premiums in C, and reduces oil and coal shock exposure in every box.
2. Build traceability and environmental and labour standards that buyers trust, which is required in C and D and helps sales in A and B.
3. Diversify partners and buyers across blocs, rather than relying on one country's capital and technology.
4. Lower the cost of capital through governance reform (Month 11's MSCI lesson).
5. Develop skills and local integration and service firms for batteries, EVs and robots, the robust business from Week 3.
6. Sign long offtake contracts before building new cathode or cell plants, so that new capacity is a hedged bet rather than a speculation.

Judgment: Indonesia's downstream plants are bets on scenario C and should be designed for it; the robust core of the strategy is clean power, traceability, governance and skills, not more smelters.

What a strong answer must include.

- A specific reading of each of the four scenarios for this strategy, not a generic description of the scenarios.
- Awareness of Indonesia's actual position (dominant nickel share, Chinese ownership, coal power, the LFP threat).
- A short list of actions that do acceptably in all four boxes, with the reason each is robust.
- The distinction between robust actions and bets on one scenario.

Exercise 12.4: Analysis: one chokepoint (rare earth magnets)

Model answer.

Who controls it: China. In 2024 China accounted for about 60 percent of mined production of magnet rare earths, 91 percent of refining and 94 percent of sintered permanent magnet production (IEA). Control rests not on geology but on midstream processing built over decades with state support, tolerance of toxic waste and low prices that drove competitors out.

How it has been used: in April 2025 China required export licences for seven rare earths and magnets; magnet exports to the US fell to nearly zero in May, and carmakers in the US, Europe, Japan and India cut or stopped production lines. In October 2025 China announced controls reaching products made abroad with more than 0.1 percent Chinese rare earth value. Weeks later, at Busan, it suspended them for a year in exchange for US tariff cuts and the suspension of the Affiliates Rule. The IEA estimated that fully implemented controls could put up to 6.5 trillion dollars of economic activity outside China at risk each year.

How long alternatives take: new rare earth mines take about eight years on average from discovery to production (IEA), and magnet plants are even scarcer. Even with existing and announced projects, by 2035 supply outside China would cover only about half of mining needs, a quarter of refining needs and under a fifth of magnet demand outside China. The IEA estimates about 60 billion dollars of investment over a decade would be needed. Realistically, a meaningful alternative supply chain takes 10 to 15 years.

Who would capture value if alternatives were built:

- Countries that offer reliable, environmentally responsible processing, such as Malaysia (Lynas) and possibly Vietnam, Brazil or African deposit holders that add refining, would gain a strategic role that buyers and governments pay a premium for.
- Firms with state-backed price floors and offtake contracts, such as MP Materials with its 110-dollar-per-kilogram floor, would capture a security premium.
- Magnet and motor makers outside China would capture margin, but the scarcity rent would shrink as supply grows, as happened with reducers in Month 9.

Judgment: the magnet chokepoint will remain effective through the early 2030s. Its value to China decays every time it is used, and the most durable winners of diversification will be the processors that combine cheap clean power, credible waste management and long-term contracts.

What a strong answer must include.

- All four parts of the question: control, use, time to alternatives, and value capture.
- Specific numbers and dates (shares, licence episodes, lead times, investment needs).
- The insight that chokepoints decay when used, but slowly.
- A view on which actors gain, including the risk that scarcity rents are competed away.

Exercise 12.5: Essay: "Africa and Southeast Asia cannot build ore-to-robot supply chains; they should focus on services"

Model answer.

I argue against the statement as written, but agree with half of it. A complete ore-to-robot chain inside either region by the 2040s is unlikely. But "focus on services" is the wrong conclusion. The right strategy is to own several layers, including some physical ones, and to start from services as the entry point, not the destination.

The case for the statement is strong. Month 7 showed that China makes about 27 to 32 percent of world manufacturing value added, and its clusters keep a cost lead through agglomeration and learning curves. Month 9 showed the precision-component layer (reducers, force sensors, magnets) takes decades to learn: Japan's reducer lead lasted thirty years. Month 5 and Month 12 showed China refines about 91 percent of magnet rare earths and makes 94 percent of sintered magnets. And Month 9 found all of Africa installed only 519 industrial robots in 2025, so there is no home market to learn on.

But the evidence also shows physical layers within reach. Month 6 showed Indonesia's nickel ban pulled processing into the country; Indonesia now dominates world nickel and has a processing base. Malaysia hosts the largest rare earth refinery outside China. Month 7 showed Morocco built an automotive cluster exporting about 17 billion dollars a year in twenty years, and in October 2026 Gotion began commissioning an LFP battery plant in Kenitra.

And services alone are a trap. Month 1's robot table, as Month 12 reads it, shows that a country that mines the ore and also installs and services the robot captures about 8,200 dollars of a 30,000-dollar robot (about 27 percent), and that adding regional assembly raises its share to about 18,200 dollars, about 61 percent. Month 10 showed that whoever owns the machines captures the gains from automation. A region that only services foreign-owned robots would keep a slice but cede the ownership question.

The right sequence: start with integration, software and service, where knowledge of local customers matters and capital needs are low; use that to build a home market for robots; then move backward into assembly, battery materials, motors and drives, with partners. Southeast Asia can reach components; Africa's realistic path is processing in a few places with cheap power, assembly in one or two hubs, and services everywhere.

Judgment: not ore-to-robot and not services only, but "several layers, sequenced". Services are the door, physical layers the house; a strategy that stops at the door leaves the value with whoever owns the machines.

What a strong answer must include.

- A clear position (for, against, or a defended middle) stated at the start.
- Evidence from at least four Months, accurately cited.
- Both regions treated separately, since their starting positions differ.
- A link to value capture and the ownership question, and a marked judgment.

Exercise 12.6: Founder exercise: one opportunity area

Model answer (traceability and compliance software).

The five questions:

1. Where is the scarce capability? Proving origin. After the 2025 magnet squeeze and the July 2026 Section 301 tariffs, US and EU buyers need to show where minerals and parts came from, how much carbon they contain and whether forced labour was used. Credible verified data is scarce; the software itself is not.
2. Who owns the customer? Mid-sized exporters of battery materials, motors and electronics in Southeast Asia, who answer to demanding foreign buyers but lack compliance teams.
3. Cost of capital? Low capital needs: software and auditors, not plants. Month 11 showed that a few percentage points of interest decide who wins in capital-heavy businesses; a light business like this one is not decided by its borrowing rate.
4. Robust across scenarios? Strongest in C and D (rules of origin and bloc rules require proof); still useful in A and B, where buyers' carbon and labour rules apply.
5. Social value? Better labour and environmental conditions are what we verify, so impact and revenue come from the same audit.

One-page plan:

- Customer: exporters of nickel products, battery precursors, motors and wiring in Indonesia, Vietnam and Malaysia selling to US, EU, Japanese and Korean buyers.
- Problem: buyers demand origin, carbon and labour documentation; suppliers lose orders or face tariffs (for example, the 12.5 percent Section 301 rate for some countries) when they cannot prove it.
- First product: a supplier-mapping tool that collects documents from each tier, reads customs and shipping records with AI to flag hidden dependencies (the Nexperia lesson from Month 7), and produces buyer-ready compliance reports.
- Revenue model: annual subscription per exporter (for example, 12,000 to 30,000 dollars depending on size) plus per-report fees; later, buyers pay to access verified supplier data.
- Capital needed: about 500,000 to 1 million dollars for 18 to 24 months (engineers, two auditors, sales), from a seed round, as in Month 11's dilution example.
- Chokepoints we depend on: cloud hosting and AI model providers (keep multi-provider), access to customs data, and above all credibility, so we partner with recognised certification bodies.
- Scenario risks: if the US-China split narrows (towards A), the trusted-supplier premium shrinks; if rules change often, we must update quickly, which is also why customers keep paying.
- Social impact metrics: number of worker-years covered by verified labour audits, tonnes of carbon emissions documented and reduced, and suppliers brought into compliance.

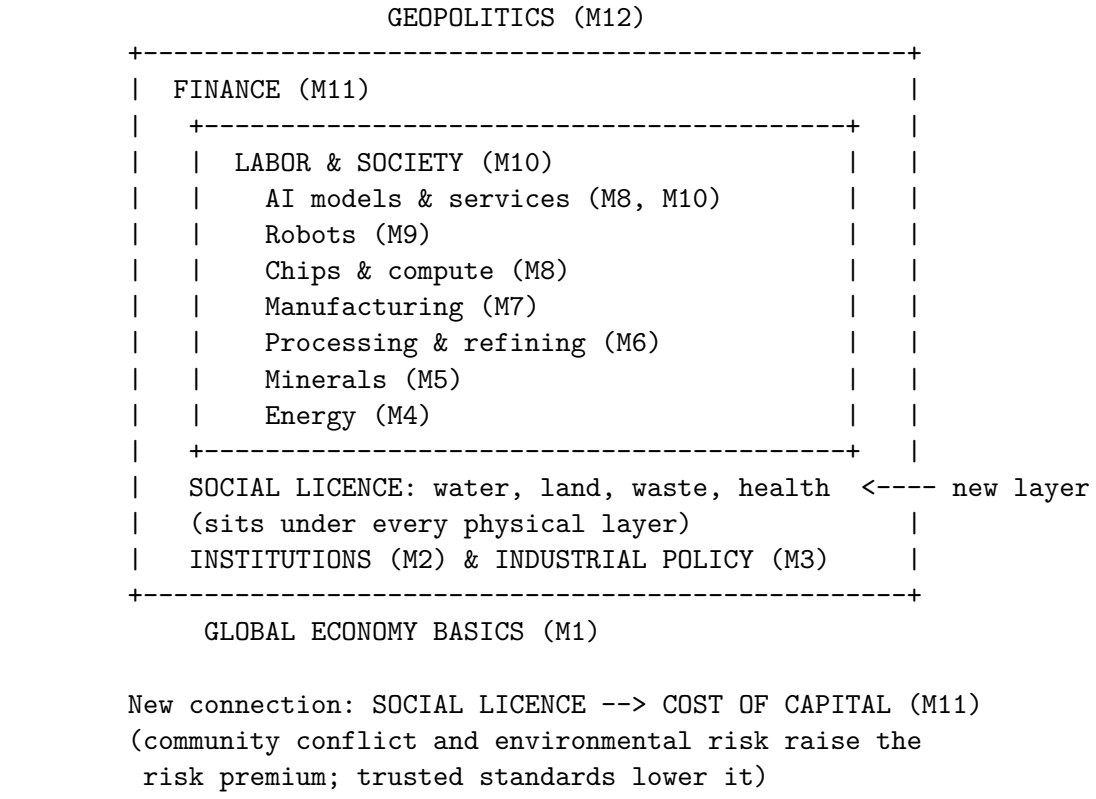
What a strong answer must include.

- All five questions answered for the chosen area, honestly, including weaknesses.
- Every element of the one-page plan the exercise lists.
- Numbers for revenue and capital, even if rough.
- A social-impact measure that is tied to the business model, not added on.

Exercise 12.7: Capstone: your own system map

Model answer.

The course map stacks energy, minerals, processing, manufacturing, chips, robots and AI, wrapped by labour, finance, institutions and geopolitics. I would add one layer that the course mentions in several months but never puts on the map: social licence, meaning the consent of the communities and environments that host the physical layers (water, land, air, health).



Why it matters for value capture. The course's evidence shows social licence acting as a real constraint again and again. In Month 8, Johor banned the most water-intensive data centers, which can use up to 50 million litres a day, and 2026 brought Malaysia's first community protests against data centers. In Months 5, 9 and 12, rare earth refining produces toxic and sometimes radioactive waste; China's dominance was built partly on tolerating that damage, and the lack of that tolerance is one reason alternatives take a decade. Month 12's Southeast Asia scenario lists environmental backlash against nickel processing as part of the bad path.

The value-capture mechanism has three parts. First, social licence is a hidden cost that decides where processing locates: places that price it at zero attract plants at first and then face shutdowns, protests or lawsuits. Second, it feeds the cost-of-capital loop: lenders and buyers increasingly price environmental and community risk, so credible standards lower the interest rate and win premium contracts in the "Two Stacks" scenario. Third, it is where host countries can bargain: a community and government that set clear, enforced terms (water pricing, waste rules, local benefit sharing) can capture more of the rent from the physical layers, as Johor is now trying to do with data centers.

Judgment: the course treats energy as the base layer, and it is. But beneath energy sits permission. The countries that build the physical layers of the robot chain in the 2030s will be those that can win and keep their citizens' consent, and that is an institutional capability, not a technical one.

What a strong answer must include.

- A clear map, in text if needed, showing where the new layer or connection sits.
- One addition only, argued in depth, rather than a list.
- Evidence from at least two Months showing the addition matters.
- A specific mechanism linking the addition to value capture (who gains, who pays, how it changes the cost of capital, bargaining power or location).

Appendix C: The Mathematical Toolkit of the Material Economy

This appendix collects, in one place, the mathematics that sits underneath the twelve Months of the course. Each Month uses these tools in passing: a learning curve in Month 4, a cost curve in Month 5, a net present value in Month 6, an input-output ripple in Month 9, a debt equation in Month 11, a scenario tree in Month 12. Here you get each tool on its own, built from the ground up, so you can use it on any problem you meet as a founder, an investor or a policy analyst.

Every section follows the same pattern:

- **Intuition**: what the tool is really measuring, in plain words.
- **The formula**: written in plain text so it survives any printer. We write powers as x^2, multiplication as x or *, square roots as sqrt(x), and natural logarithms as ln(x).
- **A worked example**: real-looking numbers tied to ore, robots or artificial intelligence (AI). Where a number is a real statistic it carries a year and a source; where it is an assumption made up for teaching, it is marked "illustrative". Do not quote an illustrative number as a fact.
- **A Python snippet**: short code you can paste into any Python 3 interpreter to reproduce the example and then change the inputs. Only the standard library is needed, except in the input-output section, which shows a pure-Python version and an optional numpy version.
- **Two practice problems with answers**: try them before you read the answers.

A note on notation. Money is written as "USD 5" or "5 dollars". Percentages in formulas are written as decimals (5% = 0.05). "Per year" is written "/yr". A growth rate is always "per period", and you must keep the period consistent: never mix a monthly rate with a yearly number of periods.

A note on attitude. Mathematics in economics is not there to look clever. It is there to stop you lying to yourself. Most bad investment decisions, failed industrial policies and silly headlines come from one of a small number of arithmetic mistakes: forgetting that money later is worth less than money now, confusing an average with a margin, reading a nominal number as a real one, or treating a single forecast as certain. The tools below are the antidotes.

C.1 Compounding, growth rates and the rule of 70

Intuition

Almost everything interesting in this course grows or shrinks by a percentage each year rather than by a fixed amount: gross domestic product (GDP), the number of robots installed, the cost of a battery, the computing power used to train AI models, the debt of a government. When something

grows by a constant percentage, it grows *exponentially*. Each year's growth is added to the base, so next year's growth is calculated on a bigger number. This is compounding.

Human intuition is linear, so it badly underestimates compounding. A country growing at 7% a year does not become 70% richer in ten years; it roughly doubles. A cost falling 10% a year does not reach zero in ten years; it falls to about a third. Getting this right is the first habit of a quantitative thinker.

The formula

If a quantity starts at X0 and grows at rate g per period for t periods:

```
X_t = X0 * (1 + g)^t
```

To find the growth rate between two observed values (the compound annual growth rate, CAGR):

```
CAGR = (X_end / X_start)^(1 / t) - 1
```

The doubling time T is the t that makes (1 + g)^t = 2:

```
T = ln(2) / ln(1 + g)
```

For small g, ln(1 + g) is close to g, and ln(2) is about 0.693, so:

```
T is roughly 70 / (growth rate in percent)
```

This is the **rule of 70**. At 7% growth, doubling takes about 70 / 7 = 10 years (the exact answer is 10.24). Some people use 72 because 72 divides neatly by 2, 3, 4, 6, 8, 9 and 12; both are approximations. The same rule works in reverse for decline: something falling at 10% a year halves in roughly 7 years (exactly 6.6 years, because ln(0.9) is about -0.105).

Economists also use *continuous* growth: X_t = X0 * e^(g*t). Then the log of X rises in a straight line, which is why growth charts are often drawn on a logarithmic (log) scale. On a log scale, a straight line means a constant percentage growth rate, and a steeper line means faster growth. Keep that in mind whenever you see a chart of robot installations or AI training compute.

Worked example: a nickel refinery, a country and a robot price

(a) Capacity growth. An (illustrative) nickel refining cluster produces 50,000 tonnes a year and its output grows 12% a year. Rule of 70: doubling in 70 / 12 = 5.8 years. Exact: ln(2) / ln(1.12) = 0.693 / 0.1133 = 6.1 years. The rule is slightly optimistic at high rates; above about 10% use the exact formula.

(b) A country's income. A country with GDP per person of USD 5,000 grows 5% a year in real terms for 30 years:

```
5,000 * 1.05^30 = 5,000 * 4.32 = about USD 21,600
```

Thirty years of 5% growth more than quadruples income. That is roughly the gap between a lower-middle-income country and an upper-middle-income one. At 3% instead, the result is 5,000 * 1.03^30 = about USD 12,100. Two percentage points per year, held for a generation, is the difference between two very different countries. This is why Month 3 spends so much time on what sustained high growth requires.

(c) A falling robot price. Suppose an industrial robot arm that cost USD 50,000 now costs USD 20,000 eight years later (illustrative). The CAGR is:

```
(20,000 / 50,000)^(1/8) - 1 = 0.4^(0.125) - 1 = 0.892 - 1 = -10.8% per year
```

So the price fell about 10.8% a year on average. Note that a total fall of 60% does *not* mean 60 / 8 = 7.5% a year; the compound rate is higher because each year's fall is taken from a smaller base.

Python snippet

```python
import math

def grow(x0, g, t):
    return x0 * (1 + g) ** t

def cagr(x_start, x_end, years):
    return (x_end / x_start) ** (1 / years) - 1

def doubling_time(g):
    return math.log(2) / math.log(1 + g)

print(round(grow(5000, 0.05, 30)))          # about 21610
print(round(cagr(50000, 20000, 8), 4))      # about -0.1082
print(round(doubling_time(0.12), 2))        # about 6.12 years
print(round(70 / 12, 2))                    # rule of 70: 5.83
```

Practice problems

1. AI training compute for frontier models has been reported to grow very fast. Suppose (illustrative) it grows 4 times per year. What is the annual growth rate in percent, and how many months does it take to double?
2. A battery factory's output rose from 2 gigawatt-hours (GWh) to 30 GWh in 6 years. What was its CAGR?

Answers.

1. Growing 4 times per year means g = 300% (since 1 + g = 4). Doubling time = ln(2) / ln(4) = 0.5 years, or 6 months. Do not use the rule of 70 here: 70 / 300 = 0.23 years is badly wrong, because the rule only works for small rates.
2. CAGR = (30 / 2)^(1/6) - 1 = 15^(0.1667) - 1 = 1.570 - 1 = about 57% per year.

C.2 Present value, net present value, internal rate of return and the cost of capital

Intuition

A dollar today is worth more than a dollar in five years, for three reasons: you could invest today's dollar and earn a return; prices may rise in the meantime (inflation); and the future dollar might never arrive (risk). Discounting turns future money into today's money so you can compare a project's costs, which come first, with its revenues, which come later.

This matters enormously in the material economy, because mines, refineries, chip fabs, power plants and robot factories are all *capital-intensive*: huge spending up front, cash coming back over decades. For such projects the discount rate, which is set by the cost of capital, often matters more than the engineering. A project that is profitable when money costs 6% a year can be a loss when money costs 14%. This is a central reason why the same solar farm or refinery is cheaper to build in Germany than in Zambia, even when the sunshine or the ore is better in Zambia (see Month 4, Week 4, and Month 11).

The formulas

Present value (PV) of an amount C received t years from now, at discount rate r:

```
PV = C / (1 + r)^t
```

Net present value (NPV) of a project with cash flows C0, C1, ..., Cn (costs negative, revenues positive):

```
NPV = C0 + C1/(1+r) + C2/(1+r)^2 + ... + Cn/(1+r)^n
```

Decision rule: accept a project if NPV > 0 at the right discount rate, because it creates more value than it costs, after paying investors the return they require.

Internal rate of return (IRR) is the discount rate that makes NPV exactly zero. It is the project's own rate of return. Decision rule: accept if IRR > cost of capital. There is no general algebraic formula for IRR; you find it by trial and error (the computer does this quickly).

Annuity shortcut. If a project pays the same amount C every year for n years, starting next year:

```
PV = C * [1 - (1 + r)^(-n)] / r
```

Cost of capital and WACC. Most firms fund projects with a mix of debt (loans, bonds) and equity (owners' money). The weighted average cost of capital (WACC) is:

```
WACC = (D / V) * r_d * (1 - tax rate) + (E / V) * r_e
```

where D is debt, E is equity, V = D + E, r_d is the interest rate on debt and r_e is the return equity investors demand. Debt gets a tax adjustment because interest is usually tax-deductible. Equity is more expensive than debt because shareholders are paid last and take more risk. In poorer or riskier countries both r_d and r_e are higher, which is called a *country risk premium*.

Worked example 1: a nickel refinery

An (illustrative) high-pressure acid leach (HPAL) nickel plant, of the type discussed in Month 6, costs USD 1.5 billion, spent as USD 500 million in each of years 0, 1 and 2 while it is built. From year 3 to year 22 (20 years) it earns a net cash flow of USD 250 million a year. All figures in millions of dollars.

At a discount rate of 8%:

- PV of construction: -500 - 500/1.08 - 500/1.08^2 = -500 - 463 - 429 = -1,392
- PV of the 20 years of cash: an annuity of 250 for 20 years is worth 250 * 9.818 = 2,454 as of year 2; discount that two more years: 2,454 / 1.08^2 = 2,104
- NPV = 2,104 - 1,392 = about +713

At a discount rate of 12%, the same project has NPV = about +144. Still positive, but the margin of safety has shrunk by four-fifths. The IRR, found by trial, is about 13.4%. So:

- A Chinese or Japanese sponsor with access to money at 8% sees a very attractive project.
- A sponsor whose investors demand 14% (common for a first-of-its-kind plant in a country with unstable rules) should reject it, because the IRR of 13.4% is below 14%.

Same rock, same technology, different verdict. This is the arithmetic behind the Month 6 observation that downstreaming is easiest for whoever has the cheapest capital.

Worked example 2: a robot factory and its WACC

A start-up wants to build a factory for robot gear reducers (the precision gearboxes in robot joints, discussed in Month 9). It costs USD 40 million today and is expected to earn USD 8 million a year for 10 years (illustrative).

First, the cost of capital. The firm plans 60% debt at 7% interest, 40% equity at a required 15% return, with a 22% corporate tax rate:

```
WACC = 0.6 * 0.07 * (1 - 0.22) + 0.4 * 0.15
     = 0.0328 + 0.0600 = 0.0928, about 9.3%
```

NPV at 8% is +13.7 million. At 12% it is +5.2 million. At 15% it is almost exactly zero (+0.15 million). So the IRR is about 15.1%. With a WACC of 9.3% the project clearly passes. But if the founders could only raise equity (say at 25%, typical for risky venture money), the project would fail the test. This is why founders in capital-intensive hardware look so hard for cheap debt, government guarantees, development-bank loans or strategic partners: the financing structure can decide whether a good factory gets built.

Python snippet

```python
def npv(r, cash_flows):
    return sum(c / (1 + r) ** t for t, c in enumerate(cash_flows))

def irr(cash_flows, lo=-0.9, hi=1.0):
    # bisection: assumes NPV falls as r rises (normal project)
    for _ in range(200):
        mid = (lo + hi) / 2
        if npv(mid, cash_flows) > 0:
            lo = mid
        else:
            hi = mid
    return mid

refinery = [-500, -500, -500] + [250] * 20
print(round(npv(0.08, refinery)), round(npv(0.12, refinery)))  # 713 144
print(round(irr(refinery), 4))                                 # about 0.1344

factory = [-40] + [8] * 10
print(round(npv(0.0928, factory), 1), round(irr(factory), 4))  # 10.7 0.151
```

Common traps

- **Mixing nominal and real.** If your cash flows include future inflation, discount with a nominal rate; if they are in today's prices, use a real rate. Mixing them is one of the commonest errors in feasibility studies.
- **Ignoring the timing of construction.** Delays are expensive mainly because they push revenue later, and later revenue is discounted more.
- **Trusting IRR alone.** IRR does not tell you the size of the value created, and for projects whose cash flows change sign more than once (for example, a mine with a big closure and clean-up cost at the end) there can be more than one IRR. Always look at NPV too.

Practice problems

1. A data center will pay you USD 100 million in 5 years. What is it worth today at 10%?
2. In worked example 2, what is the NPV at the WACC of 9.28%? Is the answer consistent with the IRR?

Answers.

1. PV = 100 / 1.1^5 = 100 / 1.6105 = about USD 62.1 million.
2. Using the annuity formula, PV of 8 per year for 10 years at 9.28% = 8 * [1 - 1.0928^(-10)] / 0.0928 = 8 * 6.34 = about 50.7, so NPV = 50.7 - 40 = about +10.7 million. Consistent: NPV is positive because the discount rate (9.28%) is below the IRR (15.1%).

C.3 The levelized cost of electricity

Intuition

Electricity runs the whole material economy: smelters, refineries, chip fabs, data centers and robot factories. When people argue about whether solar is "cheaper" than coal or gas, they usually mean the *levelized cost of electricity* (LCOE). LCOE answers one question: if you build this power plant and run it over its whole life, what constant price per megawatt-hour (MWh) would you need to receive just to cover all costs, including the return investors require?

LCOE is really just the NPV idea from section C.2 turned around. Instead of asking "is the NPV positive at a given price?", it asks "what price makes the NPV exactly zero?". That makes it a natural meeting point for engineering (capacity factor, lifetime) and finance (cost of capital).

The formula

The full definition:

```
LCOE = (sum over years of discounted costs) / (sum over years of discounted electricity output)

LCOE = [ sum_t (I_t + OM_t + F_t) / (1+r)^t ] / [ sum_t E_t / (1+r)^t ]
```

where I_t is investment, OM_t is operation and maintenance, F_t is fuel, E_t is electricity produced, and r is the discount rate.

Discounting the electricity, not only the money, can feel strange. The reason is that we are looking for a price p per MWh such that the discounted revenue (p * E_t each year) equals the discounted costs; solving for p gives exactly this ratio.

For a plant with all investment up front and steady output, there is a convenient shortcut using the *capital recovery factor* (CRF), which converts an up-front cost into an equal yearly payment over n years:

```
CRF = r * (1 + r)^n / ((1 + r)^n - 1)
```

```
LCOE = (Capex * CRF + fixed O&M per year) / (MWh per year) + fuel and variable cost per MWh
```

Annual output per kilowatt (kW) of capacity = 8,760 hours * capacity factor / 1,000, in MWh. The capacity factor is the share of the year's maximum possible output that the plant really produces: perhaps 15-25% for solar depending on location, 30-50% for wind, 50-90% for gas, coal and nuclear depending on how they are used.

Worked example: the same solar farm in two countries

An (illustrative) utility-scale solar farm costs USD 600 per kW of capacity, has fixed operation and maintenance of USD 12 per kW per year, a capacity factor of 18% and a life of 25 years. One kW produces 8,760 * 0.18 / 1,000 = 1.577 MWh per year.

Country A, cost of capital 6%.

- CRF(6%, 25 years) = 0.0782
- Annualized capital cost = 600 * 0.0782 = USD 46.9 per kW per year
- Total yearly cost = 46.9 + 12 = 58.9
- LCOE = 58.9 / 1.577 = about **USD 37 per MWh**

Country B, cost of capital 12%.

- CRF(12%, 25 years) = 0.1275
- Annualized capital cost = 600 * 0.1275 = 76.5
- LCOE = (76.5 + 12) / 1.577 = about **USD 56 per MWh**

The panels, the sunshine and the engineers are identical. Only the cost of money differs, and the electricity costs 50% more. (Month 4, Week 4 runs the same comparison with a farm costing USD 700 per kW at a 20% capacity factor and finds about 55% more; the gap is a little larger there because operations and maintenance are a smaller share of its cost. Both examples are illustrative, and both say the same thing.) For a country that wants to attract an aluminium smelter, a nickel refinery or a data center, all of which are electricity-hungry (Month 4 and Month 6), that 50% can decide the location. The International Energy Agency (IEA) has repeatedly made this point: in many emerging economies the cost of capital for clean-energy projects is two to three times higher than in advanced economies, and financing costs make up the largest share of the LCOE (IEA, Cost of Capital Observatory, 2023-2024; approximate). This is one of the most important numbers in the whole course, because it links Month 4 (energy) to Month 11 (finance).

Comparison with gas. An (illustrative) combined-cycle gas plant costs USD 1,000 per kW, fixed O&M USD 25 per kW per year, capacity factor 60%, life 25 years, discount rate 8%. CRF(8%, 25) = 0.0937, so capital plus O&M = (93.7 + 25) / (8,760 * 0.6 / 1,000) = 118.7 / 5.256 = about USD 22.6 per MWh. Fuel: using the heat rate of 6.5 million British thermal units (MMBtu) per MWh from Month 4 and gas at USD 10 per MMBtu, fuel adds 65 per MWh. Total LCOE: about USD 88 per MWh. Notice the structure: solar is almost all capital cost, so it is sensitive to interest rates; gas is mostly fuel, so it is sensitive to gas prices and geopolitics.

Limits of LCOE

LCOE is a good first comparison but it hides three things. First, *timing*: a MWh at 7 pm on a hot evening is worth more than a MWh at noon on a sunny day, and solar produces most at noon. Second, *firmness*: a smelter needs power 24 hours a day, so solar must be paired with batteries or backup, which adds cost. Analysts sometimes use "firmed" LCOE or system costs to deal with this. Third, *grid costs*: transmission lines to a remote solar farm are not in the plant's LCOE. Always ask "the LCOE of what, delivered when, to where?".

Python snippet

```
def crf(r, n):
    return r * (1 + r) ** n / ((1 + r) ** n - 1)

def lcoe(capex_per_kw, fixed_om, cap_factor, r, years, fuel_per_mwh=0.0):
    mwh_per_kw = 8760 * cap_factor / 1000
    return (capex_per_kw * crf(r, years) + fixed_om) / mwh_per_kw + fuel_per_mwh

print(round(lcoe(600, 12, 0.18, 0.06, 25), 1))        # about 37.4
print(round(lcoe(600, 12, 0.18, 0.12, 25), 1))        # about 56.1
print(round(lcoe(1000, 25, 0.60, 0.08, 25, 65), 1))   # about 87.6
```

Practice problems

1. In the solar example at 6%, what happens to LCOE if the site is sunnier, with a capacity factor of 24% instead of 18%?
2. Gas rises to USD 25 per MMBtu, as in parts of Asia in 2026 (Month 4). What is the new LCOE of the gas plant?

Answers.

1. Output becomes 8,760 * 0.24 / 1,000 = 2.102 MWh per kW. LCOE = 58.9 / 2.102 = about USD 28 per MWh. LCOE falls in proportion to the capacity factor: 18/24 = 0.75 of the original.
2. Fuel = 6.5 * 25 = 162.5 per MWh. LCOE = 22.6 + 162.5 = about USD 185 per MWh, more than double. This is why a gas price spike can make energy-intensive industry in gas-importing countries unprofitable overnight.

C.4 Learning curves and Wright's law

Intuition

In 1936 the aircraft engineer Theodore Wright published a paper showing that each time the cumulative number of airplanes built doubled, the labour hours needed per airplane fell by a roughly constant percentage, about 20% in his data ("Factors Affecting the Cost of Airplanes", Journal of the Aeronautical Sciences, 1936). The pattern has since been found in hundreds of technologies: solar modules, lithium-ion batteries, wind turbines, transistors and many chemicals. It is called **Wright's law** or the **experience curve**.

The key idea is that cost falls with *experience* (how many units the world has made in total), not with time. Time only matters because experience accumulates over time. If production stalls, the

cost decline stalls. If production explodes, as solar did in China after 2010, costs collapse faster than anyone predicted. Why does this happen? Workers and engineers learn; processes get automated; suppliers scale up; designs are simplified; waste falls; and bigger factories spread fixed costs over more units.

Wright's law is the single most important equation for anyone asking whether humanoid robots, or a robot supply chain in Africa or Southeast Asia, could become cheap by the 2040s (Month 9 and Month 12).

The formula

Let C1 be the cost of the first unit (or the cost at some reference level of cumulative output N1), and N the cumulative number made. Then:

```
C(N) = C1 * N^(-b)       or, relative to a reference point,   C(N) = C_ref * (N / N_ref)^(-b)
```

The *learning rate* LR is the percentage cost decline per doubling of cumulative output:

```
1 - LR = 2^(-b)    so    b = -log2(1 - LR)
```

A 20% learning rate gives b = 0.322; an 18% rate gives b = 0.286; a 10% rate gives b = 0.152.

Because ln C = ln C1 - b * ln N, the learning curve is a *straight line* on a log-log chart, with slope -b. That is how you estimate a learning rate from data: take logs of cost and cumulative output and fit a straight line (a linear regression).

A useful rule: the number of doublings needed to halve cost is ln(0.5) / ln(1 - LR). At LR = 20%, that is 3.1 doublings, meaning cumulative output must rise about 2^3.1 = 8.6 times.

What the real data say

- **Solar photovoltaic (PV) modules:** learning rates of roughly 20-25% per doubling over several decades are the usual range in the research literature (for example, Way, Ives, Mealy and Farmer, "Empirically grounded technology forecasts and the energy transition", Joule, 2022).
- **Lithium-ion battery packs:** BloombergNEF's annual survey put the average pack price at about USD 115 per kilowatt-hour (kWh) in 2024 and about USD 108 per kWh in 2025 (BNEF Battery Price Survey, December 2024 and December 2025, https://about.bnef.com/insights/clean-transport/lithium-ion-battery-pack-prices-fall-to-108-per-kilowatt-hour-despite-rising-metal-prices-bloombergnef), down from about USD 1,474 per kWh in 2010 in inflation-adjusted dollars, a fall of about 93%. Long-run learning-rate estimates for batteries are mostly in the high teens to low twenties of percent per doubling.
- **Robots:** there is no agreed public learning rate for humanoid robots yet, because cumulative production is still tiny. Anything you read is an assumption. Treat it as one.

Worked example 1: battery packs

Assume an 18% learning rate and a pack price of USD 115 per kWh today (the 2024 figure, used as a round reference). If the world's cumulative battery production grows tenfold:

```
cost factor = 10^(-0.286) = 0.517
new price = 115 * 0.517 = about USD 59 per kWh
```

A tenfold rise in cumulative output is about 3.3 doublings, and 0.82^3.3 is about 0.52. The two methods agree. Note an important caveat: raw material prices (lithium, nickel, cobalt) sit *outside* the learning curve. In 2022, when lithium prices spiked, average pack prices actually *rose* for the first time in over a decade (BNEF, 2022). Learning reduces processing and manufacturing costs; it does not repeal geology (Month 5).

Worked example 2: a humanoid robot cost path

Suppose (illustrative) a humanoid robot costs USD 100,000 to build when cumulative production is 10,000 units, and the learning rate is 20%.

- At 100,000 cumulative units (10 times more, 3.3 doublings): 100,000 * 10^(-0.322) = about USD 47,700.
- At 1,000,000 cumulative units (100 times more, 6.6 doublings): 100,000 * 100^(-0.322) = about USD 22,700.

So a million cumulative robots, on these assumptions, takes the cost to roughly the price of a small car. Whether the 20% assumption holds depends on how much of the robot's cost is "learnable" (assembly, motors, reducers, electronics) and how much is raw material or chips that follow their own curves. A sensible model splits the bill of materials into parts, each with its own learning rate and floor, which is exactly the exercise in Month 9, Week 3.

The founder lesson: whoever produces the most units learns fastest and gets cheapest, which lets them sell more, which makes them learn faster. This self-reinforcing loop is why early scale matters so much in hardware, and why China's large domestic market gives its robot makers an advantage that is hard to match from a small country without an export strategy.

Python snippet

```
import math

def wright_cost(c_ref, n_ref, n, learning_rate):
    b = -math.log2(1 - learning_rate)
    return c_ref * (n / n_ref) ** (-b)

def estimate_learning_rate(cum_outputs, costs):
    # least-squares slope of ln(cost) on ln(cumulative output)
    xs = [math.log(n) for n in cum_outputs]
    ys = [math.log(c) for c in costs]
    mx, my = sum(xs) / len(xs), sum(ys) / len(ys)
    slope = sum((x - mx) * (y - my) for x, y in zip(xs, ys)) / sum((x - mx) ** 2
↪   for x in xs)
    return 1 - 2 ** slope          # slope = -b

print(round(wright_cost(115, 1, 10, 0.18), 1))              # about 59.5
print(round(wright_cost(100000, 1e4, 1e6, 0.20)))           # about 22706
print(round(estimate_learning_rate([1, 2, 4, 8], [1000, 800, 640, 512]), 3))  #
↪   0.2
```

Practice problems

1. With a 15% learning rate, how many doublings of cumulative output are needed to cut cost by 75%?
2. A robot joint module cost USD 1,000 when 1,000 had been made and USD 640 when 4,000 had been made. What is the learning rate?

Answers.

1. Solve 0.85^n = 0.25: n = ln(0.25) / ln(0.85) = 8.5 doublings, which means cumulative output must rise about 2^8.5 = 370 times. Slow learning rates require enormous scale.
2. From 1,000 to 4,000 is two doublings. (1 - LR)^2 = 640 / 1,000 = 0.64, so 1 - LR = 0.8 and LR = 20%.

C.5 Production functions, growth accounting and the Solow residual

Intuition

A production function is a recipe that says how much output (Y) an economy or a firm gets from its inputs: capital (K, meaning machines, buildings, robots, data centers) and labour (L, meaning hours of work, often adjusted for skill). The most common recipe in economics is the **Cobb-Douglas** function, named after the mathematician Charles Cobb and the economist (later US senator) Paul Douglas, who fitted it to US manufacturing data in 1928.

Growth accounting uses the production function to split a country's growth into three parts: more capital, more labour, and everything else. "Everything else" is called **total factor productivity (TFP)**, or the **Solow residual**, after Robert Solow. It is "residual" because it is what is left over after you subtract what inputs can explain. It captures better technology, better organization, better institutions and better allocation of resources, but also measurement errors. Economists like to say it is "a measure of our ignorance".

This matters for the driving question because it separates two very different growth stories. A country can grow by piling up capital (building factories, smelters, roads) or by getting more output from the same inputs (learning, innovation). The first kind slows down because of diminishing returns; the second kind can continue.

The formula

Cobb-Douglas:

```
Y = A * K^alpha * L^(1 - alpha)
```

where A is TFP and alpha is between 0 and 1. With exponents adding to 1, there are *constant returns to scale*: double both K and L, and Y doubles. But each input alone has *diminishing returns*: double K alone and Y rises by only 2^alpha times.

Under competitive markets, alpha equals the share of national income paid to capital (profits, rent, interest) and 1 - alpha equals the share paid to labour (wages). Historically alpha has been about 0.3 to 0.4 in many countries.

Taking logs and differentiating over time gives the **growth accounting equation**:

```
g_Y = g_A + alpha * g_K + (1 - alpha) * g_L
```

so

`g_A (the Solow residual) = g_Y - alpha * g_K - (1 - alpha) * g_L`

The marginal product of capital (the extra output from one more unit of K) is:

`MPK = alpha * Y / K = alpha * A * (K / L)^(alpha - 1)`

As K per worker rises, MPK falls. That is diminishing returns in one line.

Worked example 1: diminishing returns to robots

A factory has A = 2, K = 100 (units of capital, think robot-equivalents), L = 50 workers and alpha = 0.35.

- Y = 2 * 100^0.35 * 50^0.65 = about 127.5
- Double robots only (K = 200): Y = about 162.5, a rise of 27.5% (because 2^0.35 = 1.275)
- Double both (K = 200, L = 100): Y = about 254.9, exactly double

MPK at the start = 0.35 * 127.5 / 100 = 0.446 units of output per extra unit of capital.

Worked example 2: growth accounting for an industrializing economy

An (illustrative) economy grows 4% a year. Its capital stock grows 6% a year, its labour input 1.5% a year, and alpha = 0.35.

- Contribution of capital = 0.35 * 6 = 2.1 percentage points
- Contribution of labour = 0.65 * 1.5 = 0.975 points
- Solow residual = 4 - 2.1 - 0.975 = **0.925 points**

So about three-quarters of growth comes from adding inputs and about a quarter from TFP. That pattern, heavy on capital, is typical of fast-industrializing economies. Solow's own 1957 study ("Technical Change and the Aggregate Production Function", Review of Economics and Statistics) found the opposite for the United States from 1909 to 1949: about seven-eighths of the growth in output per hour of work came from the residual, not from more capital per worker. Later, Alwyn Young ("The Tyranny of Numbers", Quarterly Journal of Economics, 1995) argued that much of the East Asian miracle came from input accumulation (more capital, more workers, more education) rather than unusually fast TFP growth, which started a long debate covered in Month 3.

How AI and robots fit

Robots and AI are capital, but unusual capital. In the simple Cobb-Douglas world, more robots raise wages, because each worker has more capital to work with. But newer models (Acemoglu and Restrepo's task framework, Month 10) let machines *replace* labour in specific tasks. That can shift income from labour to capital, which shows up as a rising alpha. If AI also raises A, the whole economy gets more productive. The open question of Month 10 is how big each effect is.

Python snippet

```
def cobb_douglas(A, K, L, alpha):
    return A * K ** alpha * L ** (1 - alpha)

def solow_residual(gY, gK, gL, alpha):
```

```
    return gY - alpha * gK - (1 - alpha) * gL

print(round(cobb_douglas(2, 100, 50, 0.35), 1))   # 127.5
print(round(cobb_douglas(2, 200, 50, 0.35), 1))   # 162.5
print(round(solow_residual(4, 6, 1.5, 0.35), 3))  # 0.925
```

Practice problems

1. An economy grows 6%, capital grows 10%, labour grows 1%, and alpha = 0.4. What is TFP growth?
2. With alpha = 0.35, by what percentage does output rise if capital rises 10% and labour is unchanged (TFP unchanged)?

Answers.

1. g_A = 6 - 0.4 * 10 - 0.6 * 1 = 6 - 4 - 0.6 = 1.4% per year.
2. 1.1^0.35 = 1.034, so output rises about 3.4%. (The linear approximation 0.35 * 10 = 3.5% is close.)

C.6 Input-output tables and the Leontief inverse

Intuition

When a customer buys a robot, the robot maker needs motors, gearboxes, sensors and chips. The motor maker needs copper wire and magnets. The magnet maker needs refined rare earths. The refinery needs ore, electricity and chemicals. And the electricity plant needs steel, which needs iron ore… Every purchase sends ripples back up the chain, and each ripple creates smaller ripples of its own.

The economist Wassily Leontief built a way to add up all those ripples at once, for which he received the Nobel Prize in economics in 1973. His tool, the **input-output table**, records how much each industry buys from every other industry per unit of its own output. A bit of matrix algebra, the **Leontief inverse**, then tells you the *total* output needed from every industry, direct and indirect, to deliver one unit of final demand. National statistics offices and the Organisation for Economic Co-operation and Development (OECD) publish such tables; the OECD's Inter-Country Input-Output (ICIO) tables and its Trade in Value Added (TiVA) database are built on them, and they are how Month 1's global value chain statistics are calculated.

For the driving question, input-output analysis answers questions like: if the world buys a million more robots, how much more copper, nickel and rare earth is needed, and in which countries does the extra income land?

The formula

Let x be the vector of total outputs of each sector, f the vector of final demand (what consumers, governments, investors and foreigners buy), and A the matrix of *technical coefficients*: A[i][j] is the value of input from sector i needed to make one dollar of output in sector j. Total output must cover both intermediate use and final use:

```
x = A x + f
```

Solving for x:

```
x = (I - A)^(-1) f
```

where I is the identity matrix (ones on the diagonal, zeros elsewhere). L = (I - A)^(-1) is the **Leontief inverse**. Each entry L[i][j] tells you how much output sector i must produce, in total, for each dollar of final demand for sector j's product.

There is a beautiful way to see why this works. The ripples form a series:

```
x = f + A f + A^2 f + A^3 f + ...
```

Round 0 is the final demand itself. Round 1 (A f) is what the robot maker buys directly. Round 2 (A^2 f) is what its suppliers buy, and so on. As long as each sector uses less than one dollar of inputs per dollar of output, the ripples shrink and the series adds up to (I - A)^(-1) f, exactly as the geometric series 1 + a + a^2 + ... adds up to 1 / (1 - a) when a is less than 1.

The value added of each sector per dollar of output is 1 minus its column sum in A. Multiply by x and you get the income (wages plus profits) created in each sector. Total value added always equals total final demand: this is the identity behind GDP (Month 1).

Worked example: how robot demand ripples to ore

Take a three-sector economy (illustrative coefficients). Sector 1 is *metals* (mining plus refining), sector 2 is *components* (motors, reducers, sensors, batteries, chips), sector 3 is *robots* (assembly and integration). Each column says what one dollar of that sector's output needs:

Input from per dollar of	Metals	Components	Robots
Metals	0.05	0.20	0.10
Components	0.02	0.10	0.30
Robots	0.00	0.01	0.05
Value added (1 - column sum)	0.93	0.69	0.55

So a dollar of robots uses 10 cents of metals directly (frames, casings), 30 cents of components and 5 cents of other robots (robots used in robot factories). Now suppose final demand for robots is USD 1,000 million, with no final demand for the other two sectors.

The ripples, in millions of dollars:

Round	Metals	Components	Robots
0 (final demand)	0	0	1,000
1 (direct purchases)	100.0	300.0	50.0
2	70.0	47.0	5.5
3	13.5	7.8	0.7
4	2.3	1.3	0.1
Total (all rounds)	about 186	about 356	about 1,056

Look at metals. The *direct* metal content of the robots is only USD 100 million, but the *total* demand for metals is about USD 186 million, because components are themselves metal-heavy (copper windings, magnets, steel gears, battery materials). Most of the extra metal demand arrives

in round 2, hidden one step back in the chain. Someone who only reads the robot maker's bill of materials underestimates the pull on ore by almost half.

Value added: metals 0.93 * 186.2 = 173.2; components 0.69 * 356.3 = 245.8; robots 0.55 * 1,056.4 = 581.0. Total = 1,000, exactly the final demand, as it must be.

Now the geography. If a country builds the robot assembly sector but imports all its components and metals, the components and metals rows become imports, and its domestic value added from the 1,000 million of robots is only the 581 million from assembly (in reality less, because the assembly sector's value added is mostly wages and profits, some of which may go to foreign owners). If it also builds components, it captures another 246 million. This is the input-output way of saying what Month 6 and Month 9 say in words: owning the middle of the chain (components) matters at least as much as owning the ore.

Python snippet

```
A = [[0.05, 0.20, 0.10],
     [0.02, 0.10, 0.30],
     [0.00, 0.01, 0.05]]
f = [0.0, 0.0, 1000.0]

def matvec(M, v):
    return [sum(M[i][j] * v[j] for j in range(len(v))) for i in range(len(M))]

# Method 1: add up the ripples  x = f + Af + A^2 f + ...
total, ripple = f[:], f[:]
for _ in range(50):
    ripple = matvec(A, ripple)
    total = [t + r for t, r in zip(total, ripple)]
print([round(v, 1) for v in total])          # about [186.2, 356.3, 1056.4]

# Method 2 (optional, needs numpy): the Leontief inverse directly
# import numpy as np
# L = np.linalg.inv(np.eye(3) - np.array(A))
# print(L @ np.array(f))
```

Limits

Input-output models assume fixed recipes (no substitution when prices change), constant returns to scale and no capacity limits. They are excellent for "how big is the ripple?" in the short run, and weaker for the long run, when engineers redesign products to use less of whatever has become expensive (for example, robot motors that use fewer rare-earth magnets).

Practice problems

1. Using the same A matrix, final demand is now USD 1,000 million of *components* only (for example, motors exported to robot makers abroad). The Leontief inverse column for components is about (0.237, 1.120, 0.012). What total metals output is required?
2. If final demand for robots is USD 500 million, what are total outputs? (Hint: the model is linear.)

Answers.

1. Metals = 0.237 * 1,000 = about USD 237 million. Per dollar of final demand, components pull more metal than finished robots do (0.237 versus 0.186), because a robot dollar includes lots of assembly and integration value that uses little metal.
2. Exactly half of the 1,000 case: about 93 (metals), 178 (components) and 528 (robots), in millions of dollars.

C.7 Value-added accounting and the smile curve

Intuition

The price of a finished product is not "earned" by the company that sells it. It is the sum of the value added at every stage. Value added at a stage = what the stage sells its output for minus what it pays for inputs bought from other firms. Value added pays the stage's workers (wages), its lenders (interest), its owners (profits) and the government (some taxes). Summing value added across all stages gives the final price; summing across all firms in a country gives its GDP.

This accounting reveals two things that ordinary trade statistics hide. First, a country that assembles expensive products can export huge *gross* values while keeping only a thin slice of *value added*. Second, when you line up the stages from research and design through manufacturing to brand and service, value added per unit often forms a U shape: high at the two ends, low in the middle. That is the **smile curve** of Stan Shih, introduced in Month 1, Week 3.

The formula

For each stage s:

```
VA_s = Sales_s - Purchased inputs_s

Sum over all stages of VA_s = final price paid by the end customer

Share of stage s = VA_s / final price

Domestic value added in a country's exports = sum of VA_s over stages located in that country
```

A margin is different from value added. The gross margin of a firm is (Sales - cost of goods sold) / Sales, and the operating margin subtracts wages and overheads too. A stage can have large value added (many well-paid workers) but a thin profit margin, or small value added but a fat margin (a few engineers licensing a design).

Worked example: a humanoid robot sold for USD 30,000

All numbers are illustrative, chosen to be roughly consistent with the cost structures discussed in Month 9.

Stage	Sells for	Buys inputs	Value added	Share
Mining (ore, concentrate)	300	0	300	1.0%

Stage	Sells for	Buys inputs	Value added	Share
Refining and materials	800	300	500	1.7%
Components (motors, reducers, sensors, battery, chips)	13,000	800	12,200	40.7%
Assembly (contract manufacturer)	15,000	13,000	2,000	6.7%
Brand, design, software, sales and service	30,000	15,000	15,000	50.0%
Total			30,000	100%

(For simplicity the mining stage's own purchases, such as fuel and explosives, are folded into its value added.)

Three lessons.

- **The ore is about 1% of the value.** Even with refining, the material stages capture under 3% here. That does not mean ore does not matter: without it there is no robot, and in a shortage its price can jump. But in normal times the value is elsewhere.
- **The middle has a fork.** "Components" is not one stage. Inside it, a precision reducer or an advanced chip may earn high margins (scarce know-how), while a simple wiring harness earns little. The smile curve is a tendency, not a law; a hard-to-copy middle can sit high, as TSMC shows (Month 8).
- **Gross exports mislead.** A country that hosts only assembly exports a robot worth 15,000 on the customs form, but its domestic value added is 2,000. If it also hosts mining, it keeps 2,300 out of 30,000, about 7.7%. If it hosts components too, it keeps 14,500, about 48%.

Drawing the smile in text

A quick text chart of value added per stage (each # is about 1,000 dollars):

```
Mining      | (0.3)
Refining    | (0.5)
Components  | ############
Assembly    | ##
Brand/soft  | ###############
```

Strictly, Shih's smile places research and design at the far left, before components, and splits brand and service at the right; our five stages compress that. The point survives: the bottom of the smile is the commodity middle (refining, simple assembly), where many firms can do the job and competition squeezes margins.

Python snippet

```
stages = [("Mining", 300, 0), ("Refining", 800, 300),
```

```
          ("Components", 13000, 800), ("Assembly", 15000, 13000),
          ("Brand/software", 30000, 15000)]
final_price = stages[-1][1]
for name, sales, inputs in stages:
    va = sales - inputs
    print(f"{name:15s} VA={va:6d}  share={va / final_price:6.1%}  " + "#" *
    ↪  round(va / 1000))

def domestic_share(located_here):
    va = sum(s - i for n, s, i in stages if n in located_here)
    return va / final_price

print(round(domestic_share({"Mining", "Assembly"}), 3))   # 0.077
```

Practice problems

1. A country mines and refines the materials and does the assembly, but imports components and does not own the brand. What share of the 30,000 robot's value added does it capture?
2. A competitor moves assembly to a cheaper country and cuts the assembler's price to the brand from 15,000 to 14,500, with the component price unchanged. If the retail price stays at 30,000, how does value added shift?

Answers.

1. 300 + 500 + 2,000 = 2,800, which is about 9.3%.
2. Assembly value added falls from 2,000 to 1,500, and the brand's rises from 15,000 to 15,500. Cheaper assembly does not lower the retail price unless competition forces it; it transfers value to whoever has pricing power, here the brand owner. This is the smile curve in motion.

C.8 Market concentration: HHI and concentration ratios

Intuition

How much market power does a seller have? A rough but powerful answer comes from how concentrated the market is. If a hundred small mines each sell 1% of the world's copper, none can move the price. If one country produces three-quarters of the world's cobalt, its decisions (an export ban, a quota) move the global price, as the Democratic Republic of the Congo (DRC) showed in 2025 (Month 6).

Two measures dominate. The **concentration ratio** CRk is the combined market share of the k largest players: CR3 for the top three, CR4 for the top four. The **Herfindahl-Hirschman Index** (HHI) squares every player's share (in percent) and adds them up. Squaring gives big players much more weight, so HHI captures both how few the players are and how unequal they are.

The formulas

```
CRk = s1 + s2 + ... + sk        (shares of the k largest, in percent)

HHI = s1^2 + s2^2 + ... + sn^2   (all players, shares in percent)
```

HHI ranges from near 0 (perfect competition) to 10,000 (a monopoly with 100%). With n equal players, HHI = 10,000 / n, so 1 / HHI (in share terms) gives the "equivalent number of equal-sized firms":

```
equivalent number of firms = 10,000 / HHI
```

Competition authorities use HHI thresholds. The US Department of Justice and Federal Trade Commission's 2023 Merger Guidelines treat a market with HHI above 1,800 as highly concentrated, and presume that a merger raising HHI by more than 100 points in such a market may substantially lessen competition. A merger of two firms with shares a and b raises HHI by exactly 2ab, since (a + b)^2 - a^2 - b^2 = 2ab.

Worked example 1: cobalt by country

The US Geological Survey (USGS) reports that the DRC produced about 73% of the world's mined cobalt in 2025 and Indonesia about 14% (USGS, Mineral Commodity Summaries 2026; the figures Months 2, 5 and 6 use). Using those two shares plus rounded, illustrative shares of 3% for Russia, 2% for Australia, 2% for the Philippines, and the remaining 6% spread across many small producers:

```
HHI (top five) = 73^2 + 14^2 + 3^2 + 2^2 + 2^2 = 5,329 + 196 + 9 + 4 + 4 = 5,542
```

The small producers add very little (six producers of 1% each would add only 6). So HHI is about 5,550, equivalent to fewer than two equal-sized producers (10,000 / 5,550 = 1.8). CR3 = 90%.

Two warnings. Country concentration is not the same as firm concentration: many DRC mines are owned by Chinese firms, some by Swiss-based Glencore, some partly by the Congolese state company Gecamines. And mining concentration is not refining concentration: China refined about 78% of the world's cobalt in 2024 (IEA, Global Critical Minerals Outlook 2025) and 79% in 2025 (Cobalt Institute), as Month 2 reports, a far higher share than its share of mining. For value capture, the refining HHI often matters more than the mining HHI (Month 5, Week 3).

Worked example 2: a merger among chip-equipment makers

An (illustrative) segment of chip equipment has eight firms with shares 30, 20, 15, 10, 10, 5, 5 and 5 percent.

- CR3 = 30 + 20 + 15 = 65%
- HHI = 900 + 400 + 225 + 100 + 100 + 25 + 25 + 25 = 1,800

The second and third firms propose to merge. The change is 2 * 20 * 15 = 600, so HHI becomes 2,400. Under the 2023 US guidelines this market moves from the threshold into "highly concentrated" with a large increase, and the merger would face a strong presumption against it. In practice, many chip-equipment segments are far more concentrated than this; extreme ultraviolet (EUV) lithography has a single supplier, ASML, an HHI of 10,000 (Month 8).

Python snippet

```python
def hhi(shares_percent):
    return sum(s ** 2 for s in shares_percent)

def crk(shares_percent, k):
    return sum(sorted(shares_percent, reverse=True)[:k])
```

```
cobalt = [73, 14, 3, 2, 2] + [1] * 6
print(hhi(cobalt), crk(cobalt, 3), round(10000 / hhi(cobalt), 2))  # 5548 90 1.8

equipment = [30, 20, 15, 10, 10, 5, 5, 5]
merged = [30, 35, 10, 10, 5, 5, 5]
print(hhi(equipment), hhi(merged), hhi(merged) - hhi(equipment))  # 1800 2400 600
```

Practice problems

1. Four firms each have 25% of a market. What are the HHI, CR3 and the equivalent number of firms?
2. A rare-earth refining market has one player with 90% and ten players with 1% each. Compute HHI. Then a new refinery outside the dominant country takes 10 points of share from the leader. What is the new HHI?

Answers.

1. HHI = 4 * 625 = 2,500; CR3 = 75%; equivalent firms = 10,000 / 2,500 = 4.
2. Before: 8,100 + 10 = 8,110. After: 80^2 + 10^2 + 10 * 1^2 = 6,400 + 100 + 10 = 6,510. One meaningful new entrant cuts HHI by 1,600 points, which is why governments pay so much to fund a single non-Chinese refinery: the first entrant changes concentration far more than the fifth.

C.9 Price elasticity, supply curves, cost curves and the commodity cycle

Intuition

Why did lithium prices rise roughly tenfold between early 2021 and late 2022 and then fall by more than 80% by 2024 (Month 5; based on Chinese spot lithium carbonate prices, approximate), and then swing again in 2026, from a high of CNY 200,500 per tonne on May 13 to about CNY 90,000 by October 30 (SMM, Month 5)? Why can a single mine closure move the copper price, while a new chip design barely moves the price of sand? The answer lies in *elasticity*: how strongly quantities respond to prices. In the short run, both demand and supply of most minerals are very *inelastic*. Buyers cannot quickly redesign their products, and producers cannot open a mine in less than several years. When neither side can adjust quantities, a small shock must be absorbed almost entirely by price.

The second idea is the **cost curve** (or supply curve built from mine costs). Line up every mine in the world from cheapest to most expensive and plot each one's cost against cumulative capacity. The price settles near the cost of the **marginal mine**, the most expensive one still needed to meet demand. Every cheaper mine earns a surplus (economists call it a *rent*) equal to the price minus its own cost.

The formulas

Price elasticity of demand:

```
e_d = (% change in quantity demanded) / (% change in price)
```

It is negative (higher price, less demand). |e_d| < 1 is inelastic, |e_d| > 1 is elastic. Price elasticity of supply e_s is defined the same way and is positive.

If supply suddenly falls by s percent (at the old price), the price must rise until demand falls and supply recovers enough to close the gap. A good approximation:

```
% change in price = s / (e_s - e_d) = s / (e_s + |e_d|)
```

Rent of mine i (per tonne) = Price - Cost_i. Total rent = sum over producing mines of (Price - Cost_i) * Output_i.

The cobweb model of the commodity cycle: producers decide capacity based on the price when they invest, but the capacity arrives years later.

```
Supply this period:  Q_t = c + d * P_(t-1)
Demand this period:  P_t = a - b * Q_t
```

If b * d < 1, prices oscillate and settle down; if b * d > 1, the swings grow. Real cycles are messier, but the core mechanism, investment lags plus inelastic demand, is real.

Worked example 1: a supply shock

Short-run demand elasticity for a battery metal is -0.2 (illustrative but typical in size for short-run mineral demand). A mine accident and an export quota remove 5% of world supply.

- If supply cannot respond at all in the short run (e_s = 0): price rises 5 / 0.2 = 25%.
- If other producers can lift output a little (e_s = 0.1): price rises 5 / (0.1 + 0.2) = about 17%.

A 5% shortage can cause a 17-25% price jump. Run the arithmetic backwards: when the DRC suspended cobalt exports in February 2025 and then moved to export quotas, a large price reaction was exactly what the elasticities predicted.

Worked example 2: the cost curve and the marginal mine

Six (illustrative) copper mines, sorted by cost:

Mine	Capacity (thousand tonnes)	Cost (USD per tonne)	Cumulative capacity
A	100	4,000	100
B	150	5,500	250
C	200	6,500	450
D	120	7,500	570
E	80	9,000	650
F	60	11,000	710

If demand is 600 thousand tonnes, mines A to D (570) are not enough, so mine E must run. The price must be at least about USD 9,000, the cost of the marginal mine. Mine A earns a rent of 9,000 - 4,000 = 5,000 per tonne. Mine F stays closed.

If demand falls to 500 thousand tonnes (a recession), mine E is no longer needed, D becomes the marginal mine and the price falls toward USD 7,500. Mine A's rent falls from 5,000 to 3,500 per tonne, a 30% drop, while the price fell only 17%. Low-cost producers have profits that move more

than proportionally with price; high-cost producers swing between profit and loss. That is why the cheapest mines (and the countries that host them) survive the downturns, and why governments seeking resource revenue should care where their deposits sit on the cost curve, not only how big they are. In reality, prices can briefly fall *below* the marginal mine's full cost, because a mine that has already been built keeps running as long as the price covers its cash (operating) cost.

Worked example 3: a cobweb cycle

Demand: P = 20,000 - 20 * Q (Q in thousand tonnes). Supply: Q_t = 200 + 0.03 * P_(t-1). The equilibrium is P = 10,000, Q = 500. Start from a boom price of 14,000.

Period	Quantity	Price
1	620	7,600
2	428	11,440
3	543	9,136
4	474	10,518
5	516	9,689
6	491	10,187

The high price triggers too much investment, the glut crashes the price, the crash kills investment, and the shortage pushes the price back up. Here b * d = 20 * 0.03 = 0.6, so the swings shrink. In real mining, a "period" is five to fifteen years, which is why one cycle can last a whole career.

Python snippet

```
def price_change(supply_cut_pct, e_s, e_d):
    return supply_cut_pct / (e_s + abs(e_d))

print(price_change(5, 0.0, -0.2), round(price_change(5, 0.1, -0.2), 1))  # 25.0
↪  16.7

mines = [("A", 100, 4000), ("B", 150, 5500), ("C", 200, 6500),
         ("D", 120, 7500), ("E", 80, 9000), ("F", 60, 11000)]

def clearing_price(demand):
    cum = 0
    for name, cap, cost in mines:
        cum += cap
        if cum >= demand:
            return name, cost
    return None, float("inf")

print(clearing_price(600), clearing_price(500))   # ('E', 9000) ('D', 7500)

P = 14000
for t in range(1, 7):
    Q = 200 + 0.03 * P
    P = 20000 - 20 * Q
```

```
    print(t, round(Q), round(P))
```

Practice problems

1. Demand elasticity is -0.3 and supply elasticity is 0.2. A new technology adds 4% to supply. Roughly what happens to the price?
2. In the cost curve example, demand rises to 700 thousand tonnes. Which mine is marginal, what is the price, and what is the total rent of mine B?

Answers.

1. A supply increase of 4% lowers the price by about 4 / (0.2 + 0.3) = 8%.
2. Cumulative capacity reaches 710 only with mine F, so F is marginal and the price is about USD 11,000. Mine B's rent = (11,000 - 5,500) * 150,000 tonnes = USD 825 million per year.

C.10 Debt dynamics: r minus g

Intuition

Is a government (or a company, or a country as a whole) with debt equal to 80% of its income in trouble? It depends less on the level than on the *direction*, and the direction depends on a race between two numbers: r, the interest rate paid on the debt, and g, the growth rate of the income that services it. If r is greater than g, interest piles up faster than income grows, so debt rises as a share of income unless the borrower runs a surplus before interest (a *primary surplus*). If r is less than g, growth shrinks the debt ratio on its own.

This one comparison explains a lot of Month 11: why low interest rates in the 2010s let rich countries carry large debts, why the rise in rates after 2022 worried finance ministers everywhere, and why so many developing countries, which borrow at high interest and often in foreign currency, slid into debt distress in 2022-2024. The economist Olivier Blanchard's 2019 presidential address to the American Economic Association ("Public Debt and Low Interest Rates") made r minus g the centre of the modern debate.

The formula

Let d be debt as a share of GDP, r the nominal interest rate on debt, g nominal GDP growth, and pb the primary balance (revenue minus spending excluding interest) as a share of GDP. Then:

```
d_(t+1) = d_t * (1 + r) / (1 + g) - pb
```

The debt ratio is stable when d_(t+1) = d_t, which needs a primary balance of:

```
pb* = d * (r - g) / (1 + g)    (approximately d * (r - g))
```

Use r and g in the same terms: both nominal (the usual case) or both real.

Worked example: two countries with debt at 60% of GDP

Country H (high interest): r = 7%, g = 4% nominal, primary balance zero. After one year: 0.60 * 1.07 / 1.04 = 0.617. After ten years: about **79.7%** of GDP. To keep debt stable at 60%, it needs a primary surplus of 0.60 * 0.03 / 1.04 = **1.7% of GDP** every year. With a surplus of only 1% of GDP, debt still rises to about 68% in ten years.

Country L (low interest): r = 4%, g = 7%. After ten years with zero primary balance: about **45.1%** of GDP. It could even run a primary *deficit* of about 1.7% of GDP and keep debt stable.

The same starting debt, two opposite futures. This is why credit markets care so much about a country's growth prospects and why a sudden rise in r (a global rate shock, or a downgrade) can turn a manageable debt into a crisis. Foreign-currency debt adds another channel: if the local currency falls, debt measured against local GDP jumps immediately (section C.11).

Applied to a company

The same arithmetic applies to a robot start-up financed with debt. If revenue grows faster than the interest rate on the loan, leverage falls by itself; if growth stalls below the interest rate, leverage creeps up even with no new borrowing, until a refinancing crisis arrives.

Python snippet

```
def debt_path(d0, r, g, pb, years):
    d, path = d0, [d0]
    for _ in range(years):
        d = d * (1 + r) / (1 + g) - pb
        path.append(d)
    return path

print(round(debt_path(0.60, 0.07, 0.04, 0.00, 10)[-1], 3))  # 0.797
print(round(debt_path(0.60, 0.07, 0.04, 0.01, 10)[-1], 3))  # 0.683
print(round(debt_path(0.60, 0.04, 0.07, 0.00, 10)[-1], 3))  # 0.451
print(round(0.60 * (0.07 - 0.04) / 1.04, 4))                # 0.0173
```

Practice problems

1. Debt is 90% of GDP, r = 5%, g = 3%. What primary balance stabilizes debt?
2. A country with debt of 50% of GDP, half of it in US dollars, sees its currency lose 30% of its value against the dollar (it takes 1 / 0.7 = 1.43 times as much local currency to buy a dollar). Ignoring everything else, what is the new debt ratio?

Answers.

1. pb* = 0.90 * 0.02 / 1.03 = about 1.75% of GDP surplus.
2. The dollar half (25% of GDP) is now worth 25 * 1.43 = 35.7% of GDP in local currency; the local half stays at 25%. New ratio is about 60.7% of GDP, a jump of more than 10 points overnight, without a single new loan.

C.11 Exchange rates and terms-of-trade shocks

Intuition

Most of the material economy is priced in US dollars: oil, copper, nickel, lithium, chips, many robots. Most workers, however, are paid in local currencies. So the exchange rate decides how much of a dollar price ends up as local income and how competitive local costs look to foreign buyers.

The **terms of trade** (ToT) compare the prices of what a country sells to the prices of what it buys. When copper prices fall and the price of imported machines does not, a copper-exporting country is poorer, even if it produces exactly the same tonnes. Commodity exporters live with large terms-of-trade swings; Month 2's resource curse and Month 6's downstreaming debate are partly about escaping them.

The formulas

Exchange rate quoted as local currency (LC) per US dollar: e. A rise in e means the local currency has *depreciated* (weakened).

```
Local-currency price = dollar price * e
Dollar price of a local cost = local cost / e
```

Real exchange rate (a measure of competitiveness):

```
RER = e * P_foreign / P_domestic
```

A higher RER means foreign goods are relatively more expensive, so local producers are more competitive.

Terms-of-trade index:

```
ToT = (export price index / import price index) * 100
```

Approximate income effect of a terms-of-trade shock:

```
change in national income (% of GDP) is roughly (exports / GDP) * (% change in export prices) -
```

Worked example 1: a copper miner when the currency falls

A mine sells copper at USD 100 per unit of output (scaled for simplicity). Its local costs (wages, local power, local services) are 3,000 LC per unit; its dollar costs (imported equipment, fuel, foreign debt service) are USD 25. The exchange rate is 50 LC per dollar.

- Local costs in dollars: 3,000 / 50 = 60
- Profit: 100 - 60 - 25 = USD 15 per unit

The local currency depreciates by 20%: one LC now buys 20% fewer dollars, so e rises to 50 / 0.8 = 62.5.

- Local costs in dollars: 3,000 / 62.5 = 48
- Profit: 100 - 48 - 25 = USD 27 per unit, an increase of 80%

Depreciation is a gift to exporters with local costs, and a tax on local consumers of imported goods. That is why commodity currencies often fall in commodity downturns: the falling currency cushions local producers. But a firm or government with dollar *debt* and local *revenue* is hurt (see the debt practice problem in C.10).

Worked example 2: a terms-of-trade shock

A copper-exporting economy has exports equal to 30% of GDP, mostly copper. Copper prices fall 25%; import prices are unchanged. Roughly:

```
income loss = 0.30 * 25% = 7.5% of GDP
```

The ToT index falls from 100 to 75. Even if the economy produces exactly as many tonnes, its people can buy 7.5% of GDP less from abroad. Few recessions in rich countries are that large. This is the mechanical reason resource-dependent economies are volatile, and why diversification and stabilization funds (Month 11) matter.

Python snippet

```
def miner_profit(price_usd, local_cost_lc, usd_cost, e):
    return price_usd - local_cost_lc / e - usd_cost

print(miner_profit(100, 3000, 25, 50), miner_profit(100, 3000, 25, 62.5))  # 15.0
↪  27.0

def tot_income_effect(x_share, dpx, m_share, dpm):
    return x_share * dpx - m_share * dpm

print(tot_income_effect(0.30, -25, 0.30, 0))   # -7.5 (% of GDP)
```

Practice problems

1. A factory in Thailand imports a robot priced at USD 20,000. The baht weakens from 33 to 36 per dollar. By what percentage does the robot's cost in baht rise?
2. An oil exporter's export prices rise 40% and its import prices rise 10%. Compute the new ToT index (base 100) and, if exports and imports are each 25% of GDP, the approximate income effect.

Answers.

1. From 660,000 to 720,000 baht, a rise of 9.1% (36 / 33 - 1). Automation is more expensive for firms whose currency weakens, which slows robot adoption in emerging markets after a currency shock.
2. ToT = 140 / 110 * 100 = 127.3. Income effect = 0.25 * 40 - 0.25 * 10 = +7.5% of GDP.

C.12 Scenario and expected-value reasoning

Intuition

Nobody knows whether humanoid robots will be cheap by 2035, whether a Taiwan crisis will cut chip supply, or whether a new export ban will double the price of a mineral. Good decisions under uncertainty do not need a correct forecast. They need three things: a small set of distinct *scenarios*, honest *probabilities* (even rough ones), and the *payoff* of each choice in each scenario. Then you compare choices by their **expected value** (EV), the probability-weighted average payoff, while also looking at the downside you cannot survive.

Month 12 uses scenarios to think about the world to 2050. Here we make the method quantitative.

The formulas

```
EV = p1 * V1 + p2 * V2 + ... + pn * Vn       (probabilities sum to 1)

Variance = sum of p_i * (V_i - EV)^2,   standard deviation = sqrt(variance)
```

Value of information. If you can learn something before committing (a pilot plant, a drill programme, a prototype), the value of that information is:

```
EV(decide after learning) - EV(decide now)
```

It is worth paying for information whenever its cost is below this value.

Worked example: should a founder build a robot-integration factory now?

A founder can invest USD 2 million now. Three scenarios, with payoffs net of the investment (illustrative):

Scenario	Probability	Net payoff (USD million)
Demand never comes	0.5	-2
Modest success	0.3	+5
Big success	0.2	+30

```
EV = 0.5 * (-2) + 0.3 * 5 + 0.2 * 30 = -1 + 1.5 + 6 = +6.5 million
```

Positive, so worth doing, but notice the *median* outcome is a loss of 2 million. Most such bets lose; the average is carried by the rare big success. This is the mathematics of venture capital (Month 11): investors hold many such bets so that the averages work out, while a single founder lives through only one draw. That difference is why diversification belongs to investors and focus to founders, and why a founder should never bet money they cannot afford to lose.

Adding a pilot. Suppose a USD 0.3 million pilot (a single customer installation) reveals whether the "demand never comes" scenario is true. If it is, the founder stops and loses only the 0.3 million. Otherwise she builds, paying the pilot cost on top:

```
EV(with pilot) = 0.5 * (-0.3) + 0.3 * (5 - 0.3) + 0.2 * (30 - 0.3)
               = -0.15 + 1.41 + 5.94 = +7.2 million
```

The pilot raises EV from 6.5 to 7.2 million, so the information is worth about 0.7 million more than it costs (value of information about 1.0 million, minus its 0.3 million cost). Mining companies do the same with exploration drilling, and governments should do the same with industrial policy: test small, then scale what works (Month 3).

Scenarios are not forecasts

Shell's planners popularised scenario planning in the 1970s by building a few internally consistent stories rather than one forecast. The point is not to guess the right one but to find choices that do acceptably in all of them ("robust" strategies) and to spot early signals that tell you which world you are in. Two common mistakes: giving one scenario 90% probability and then calling it a scenario analysis, and ignoring "tail" scenarios with small probabilities but huge consequences (an export ban, a war at a chokepoint).

Python snippet

```python
def ev(probs, values):
    return sum(p * v for p, v in zip(probs, values))
```

```
def stdev(probs, values):
    m = ev(probs, values)
    return sum(p * (v - m) ** 2 for p, v in zip(probs, values)) ** 0.5

p = [0.5, 0.3, 0.2]
now = [-2, 5, 30]
pilot = [-0.3, 5 - 0.3, 30 - 0.3]
print(ev(p, now), round(ev(p, pilot), 2), round(stdev(p, now), 1))  # 6.5 7.2
↪ 12.1
```

Practice problems

1. A planned refinery is worth +400 million (net present value) if exports of its feedstock remain open (probability 0.7) and -300 million if the supplying country bans exports of the feedstock (probability 0.3). What is the EV? What if a long-term supply contract reduces the loss in the ban scenario to -100 million but costs 50 million in all scenarios?
2. In the founder example, suppose the modest scenario pays only +1 million instead of +5 million (keeping its probability at 0.3). If the big-success probability is q and the "never comes" probability is 0.7 - q, what is the smallest q that makes building now worth it (EV above zero)?

Answers.

1. EV = 0.7 * 400 + 0.3 * (-300) = 280 - 90 = +190 million. With the contract: 0.7 * 350 + 0.3 * (-150) = 245 - 45 = +200 million. The contract adds 10 million in EV and, more importantly, shrinks the worst case, which can matter more than EV for a firm that cannot survive a 300 million loss.
2. EV = (0.7 - q) * (-2) + 0.3 * 1 + 30q = -1.4 + 2q + 0.3 + 30q = 32q - 1.1. EV $>$ 0 requires q $>$ 1.1 / 32 = about 0.034. A 3.4% chance of a big success is enough to make the bet positive in expectation, which shows how strongly EV depends on the size of the rare upside. Check that payoff assumption as hard as the probabilities, and remember the median outcome is still a loss.

C.13 Basic statistics pitfalls: base effects, real versus nominal, purchasing power parity and friends

Intuition

Most misleading claims about the global economy use true numbers. The trick is in how they are compared. A short list of traps covers most cases. Learn to spot them and you will read the news, company reports and even government statistics with a sharper eye.

Pitfall 1: base effects

A growth rate depends on the starting point. If a country installs 100 industrial robots one year and 300 the next, it has "tripled robot installations, +200%", while a country going from 50,000 to 55,000 grew only 10% but added 50 times as many robots. Year-on-year inflation or growth after a collapse (2021 after the 2020 pandemic slump) also looks spectacular because the base was low.

`Always ask: growth from what? And what is the absolute change?`

Pitfall 2: real versus nominal

Nominal values are in money of the day; real values remove inflation. The conversion:

```
real growth = (1 + nominal growth) / (1 + inflation) - 1, roughly nominal growth minus inflatio

real value in base-year prices = nominal value * (price index in base year / price index this y
```

Example: a mining region's wages rise 10% while local inflation is 6%. Real wages rise 1.10 / 1.06 - 1 = 3.8%, not 10%. A commodity price "at a record high" in nominal dollars may be far below its peak in real terms; always check whether a long-run price chart is inflation-adjusted.

Pitfall 3: market exchange rates versus purchasing power parity

GDP converted at market exchange rates tells you what an economy can buy on world markets: chips, machines, oil. GDP at **purchasing power parity** (PPP) adjusts for the fact that local goods and services (haircuts, rent, food) are cheaper in poorer countries, so it better measures living standards. The World Bank's International Comparison Program produces the PPP conversion factors.

Example (illustrative): a country's GDP per person is USD 5,000 at market rates, and its overall price level is 40% of the US level. Its GDP per person at PPP is roughly 5,000 / 0.40 = USD 12,500. Both numbers are correct; they answer different questions. For the material economy, market rates usually matter more: a robot, a lithography machine or a barrel of oil is bought at world prices, not PPP prices.

Pitfall 4: averaging percentages

A price that rises 50% and then falls 50% is down 25% (1.5 * 0.5 = 0.75), not back where it started. The right average of yearly growth rates is the geometric mean (the CAGR of section C.1), not the arithmetic mean. Likewise, "percent" and "percentage points" differ: a tax rate rising from 10% to 12% is up 2 percentage points but 20 percent.

Pitfall 5: shares, totals and per person

China's share of something can fall while its quantity rises (the pie grew). A country's total GDP can rise while GDP per person falls (population grew faster). Before drawing a conclusion, ask whether the right denominator is the world total, the population, the number of workers or the value of output.

Pitfall 6: survivorship and selection

Studies of "successful industrial policies" look only at the countries we remember; studies of "successful founders" look only at firms that survived. The failures, which are needed to estimate the odds, have disappeared from the sample. Section C.12's founder example is the cure: always ask what fraction of attempts end in each outcome.

Pitfall 7: correlation is not causation

Countries with more robots per worker are richer, but richness also causes robots (rich countries have high wages, so automation pays). Before saying X causes Y, ask whether Y could cause X, whether

a third factor (education, institutions) causes both, and whether there is a natural experiment that separates them. This is the core of the institutions debate in Month 2.

Python snippet

```
def real_growth(nominal, inflation):
    return (1 + nominal) / (1 + inflation) - 1

def ppp_gdp(gdp_market, price_level_vs_us):
    return gdp_market / price_level_vs_us

def geometric_mean_growth(rates):
    prod = 1.0
    for r in rates:
        prod *= 1 + r
    return prod ** (1 / len(rates)) - 1

print(round(real_growth(0.10, 0.06), 4))              # 0.0377
print(ppp_gdp(5000, 0.40))                            # 12500.0
print(round(geometric_mean_growth([0.5, -0.5]), 4))   # -0.134 per year, not 0
```

Practice problems

1. A lithium price index rose from 100 to 900 over two years and then fell to 180 the following year. Over the three years, what were the total change and the CAGR? What was the percentage fall in the last year?
2. A country's nominal GDP grew 8% last year, inflation was 3% and population grew 1.5%. Roughly how fast did real GDP per person grow?

Answers.

1. Total change: 180 / 100 = 1.8, so +80% over three years; CAGR = 1.8^(1/3) - 1 = about 21.6% a year. The last-year fall was 1 - 180 / 900 = 80%. The same series can be described as "up 80%" or "crashed 80%"; both are true, and only the base differs.
2. Real growth = 1.08 / 1.03 - 1 = 4.85%; per person = 1.0485 / 1.015 - 1 = about 3.3%.

C.14 One-page formula sheet

Tool	Formula (plain text)	Use it for
Compounding	X_t = X0 * (1 + g)^t	GDP, robot stock, compute
CAGR	(X_end / X_start)^(1/t) - 1	average growth between two dates
Rule of 70	doubling time is roughly 70 / g%	quick mental checks (small g only)
NPV	sum of C_t / (1 + r)^t	refinery, factory, mine decisions
IRR	r that makes NPV = 0	compare with cost of capital
WACC	(D/V) * r_d * (1 - tax) + (E/V) * r_e	discount rate for a firm

Tool	Formula (plain text)	Use it for
LCOE	(Capex * CRF + O&M) / MWh + fuel	comparing power sources
CRF	r(1 + r)^n / ((1 + r)^n - 1)	annualizing a capital cost
Wright's law	C = C_ref * (N / N_ref)^(-b), b = -log2(1 - LR)	battery, solar, robot cost paths
Cobb-Douglas	Y = A * K^alpha * L^(1 - alpha)	output from capital and labour
Solow residual	g_A = g_Y - alpha * g_K - (1 - alpha) * g_L	how much growth is productivity
Leontief	x = (I - A)^(-1) f	how demand ripples to ore
Value added	VA = sales - purchased inputs	who captures value in a chain
HHI	sum of shares^2 (shares in %)	market concentration
Elasticity	%dQ / %dP	how prices react to shocks
Price shock	%dP is roughly s / (e_s + abs(e_d))	supply cuts, export bans
Debt dynamics	d' = d(1 + r)/(1 + g) - pb	sovereign and corporate debt
Terms of trade	(Px / Pm) * 100	commodity exporter income
Expected value	sum of p_i * V_i	scenario decisions
Real growth	(1 + nominal)/(1 + inflation) - 1	removing inflation

How the tools fit together

Read the formula sheet as a chain, the same chain the course follows from ore to robot to AI. Geology and the cost curve (C.9) set the price of a mineral and who earns rent from it; concentration (C.8) tells you who can move that price. Energy costs (C.3) and the cost of capital (C.2) decide where processing can be profitable. Learning curves (C.4) decide how fast the costs of batteries, solar and robots fall, and therefore when new markets open. Input-output analysis (C.6) traces how demand for robots pulls on ore, and value-added accounting (C.7) says which countries and firms keep the money. Production functions (C.5) turn all of this into national growth, while exchange rates, terms of trade and debt (C.10 and C.11) decide whether that growth survives a shock. Scenarios and expected value (C.12) let you act without knowing the future, and the statistics pitfalls (C.13) keep you honest about the numbers you used along the way.

Judgment: if you master only three of these tools, choose NPV with the cost of capital, Wright's law and the value-added table. Together they answer the founder's three core questions: is this worth building, will it get cheaper, and who will keep the money.

Appendix D: Country Atlas of the Material Economy

This appendix is a reference atlas. It takes 22 economies that matter for the chain from ore to robot to artificial intelligence (AI) and gives each one a short profile in the same format: a key-facts table, its role in the chain, its strengths and bottlenecks, what changed in 2025-2026, and a 2040 outlook. Use it alongside the Months.

How to read the tables:

- **GDP 2025** is nominal gross domestic product (GDP) at market exchange rates, in US dollars (USD), from the International Monetary Fund (IMF) World Economic Outlook (WEO) database of October 2026. Population and GDP per capita come from the same database. 2025 values are IMF estimates, not final national accounts, and the IMF revises them.
- **Electricity mix** gives the main sources of electricity generation as a share of the total, from the Our World in Data compilation of Ember and Energy Institute data. The latest year is 2025 for most countries and 2024 for Indonesia, Saudi Arabia, the United Arab Emirates (UAE), Zambia and the Democratic Republic of the Congo (DRC). Shares are rounded, so they may not add to 100%. "Geothermal and other" is the Our World in Data category "other renewables excluding bioenergy", which for Indonesia and the Philippines is mostly geothermal (an inference from those countries' known geothermal fleets).
- **Robots** means industrial robots, measured by the International Federation of Robotics (IFR). "Installations" is the number of new robots put into service in a year. "Density" is the number of operating robots per 10,000 manufacturing employees, from the IFR's April 2026 release (data for 2024). Where the IFR's public releases give no number for a country, the table says so rather than guessing.
- **Minerals** names the main mined or refined minerals and, where the Months already use them, shares of world production from the US Geological Survey (USGS) Mineral Commodity Summaries 2026.
- **Top exports** names the leading export groups. Where a 2025 official figure was found, it is given.

Every profile ends with a **Sources** line. [V] means the page was fetched and read for this appendix or for a Month that this appendix draws on; [E] means the number comes from a search-result snippet and should be checked before you rely on it. The 2040 outlooks are marked **Judgment:** because they are reasoned forecasts, not facts.

Three sources are used for every country, so they are listed once here and not repeated in each Sources line:

- [V] IMF, World Economic Outlook database, October 2026, via the IMF DataMapper API (indicators NGDPD, LP and NGDPDPC; series run to 2031): https://www.imf.org/external/datamapper/api/v1/N
- [V] Our World in Data, "Share of electricity production by source" (Ember and Energy Institute data), CSV download: https://ourworldindata.org/grapher/share-elec-by-source
- [V] IFR, "Five Million Robots now Operate in Factories Globally", press release, 24 September 2026, and World Robotics 2026 Executive Summary: https://ifr.org/ifr-press-releases/news/five-million-robots-now-operate-in-factories-globally and https://ifr.org/img/worldrobotics/Exec _Industrial_Robots.pdf ; density figures from IFR, "Robot Density Surges in Europe, Asia, and Americas", 8 April 2026 (as used in Month 9).

1. United States

Indicator	Value
GDP 2025	about USD 30,861 billion (IMF, Oct 2026)
Population 2025	about 341.9 million (IMF)
GDP per capita 2025	about USD 90,270 (IMF)
Top exports	aircraft and parts, refined oil and crude, liquefied natural gas (LNG), machinery, chips and medical products; plus the world's largest services exports (software, finance, intellectual property)
Main minerals	copper, gold, molybdenum, lithium (small), rare earths at Mountain Pass (about 13% of world rare earth mining, 2025, USGS MCS 2026)
Robots	38,428 installations in 2025, up 12%, now the world's second-largest market; density 307 (2024)
Electricity mix 2025	gas 40%, nuclear 17%, coal 16%, wind 10%, solar 9%, hydro 5%

Role in the chain. The United States sits at both high ends of the smile curve (Month 1). It designs most of the world's advanced chips (Nvidia, AMD, Qualcomm, Apple), builds the leading AI models, owns the biggest cloud companies and supplies the deepest capital markets in the world. Its big technology firms guided 2026 capital spending of roughly USD 720-740 billion (Month 8). But the middle of the chain, from refining minerals to making precision robot parts, is thin. Month 9 cites Morgan Stanley's estimate that a humanoid robot built without Chinese suppliers costs nearly three times as much as one built with them.

Strengths and bottlenecks. Its strengths are technology, capital, universities, a huge home market and cheap natural gas. Its bottlenecks are physical:

- refining and magnet making, where China dominates (Month 5);
- electricity grids and permitting, which slow new data centres and factories;
- skilled factory labour and the speed of building plants, which Epoch AI puts at "2+ years in most Western countries" against 6 to 9 months at best in China (Month 9);
- policy volatility, especially tariffs, which changed many times in 2025-2026 (Month 7).

2025-2026 developments. Washington moved from subsidies to direct ownership in the mineral chain. In July 2025 the Department of Defense (since renamed the Department of War) agreed to

buy USD 400 million of preferred stock in MP Materials, set a floor price of USD 110 per kilogram for neodymium-praseodymium (NdPr) oxide for ten years, and agreed to buy all the output of a planned "10X" magnet plant, which MP sited in Northlake, Texas, in February 2026 (about 10,000 tonnes a year of capacity, targeted for about 2028). In February 2026 the government launched Project Vault, a strategic minerals reserve for civilian industry backed by a loan of up to USD 10 billion from the Export-Import Bank (EXIM) plus about USD 2 billion of private money. In June 2026 the CHIPS office agreed up to USD 1.6 billion of support for USA Rare Earth.

2040 outlook. Judgment: the United States will still capture the largest share of value in AI in 2040, because it controls models, chip design, cloud platforms and capital. It will rebuild part of the middle of the chain (magnets, some refining, advanced packaging), but at higher cost than China, so its robots will be more expensive unless it allows allied supply from Japan, Korea, Mexico and Southeast Asia.

Sources: IFR (see above); USGS MCS 2026 via Month 5; [E] CSIS, "Industrial Policy Is Back, And Minerals Are at the Center", Feb 2026: https://www.csis.org/analysis/industrial-policy-back-and-minerals-are-center ; [E] MP Materials, "MP Materials Selects Northlake, Texas. . . ", 26 Feb 2026: https://mpmaterials.com/news/mp-materials-selects-northlake-texas-as-the-site-of-10X-a-new-us-rare-earth-magnet-manufacturing-campus ; [E] EXIM, "EXIM Approves Project Vault Loan", 2 Feb 2026: https://www.exim.gov/news/project-vault ; [E] NIST, CHIPS agreement with USA Rare Earth, 3 Jun 2026: https://www.nist.gov/news-events/news/2026/06/department-commerces-chips-program-finalizes-definitive-agreement-usa-rare ; Month 8 for capital spending and Month 9 for Epoch AI and Morgan Stanley.

2. China

Indicator	Value
GDP 2025	about USD 19,626 billion (IMF, Oct 2026)
Population 2025	about 1,404.9 million (IMF)
GDP per capita 2025	about USD 13,970 (IMF)
Top exports	electrical machinery and electronics, machinery, electric vehicles, batteries, solar panels, steel, ships; goods exports USD 3.77 trillion (2025, WTO via Month 7)
Main minerals	rare earths (69% of world mining, 2025), natural graphite (82%), plus large coal, aluminium, tungsten and gallium output; dominant refiner of copper (about 48% of refined output), lithium chemicals, cobalt and rare earths (USGS MCS 2026; IEA, via Month 5)
Robots	354,226 installations in 2025, up 20%, 59% of the world; 55% supplied by Chinese brands; density 166 (2024, after IFR revision)
Electricity mix 2025	coal 54%, hydro 13%, solar 11%, wind 11%, nuclear 5%, gas 3%

Role in the chain. China is the only economy that is strong at almost every link: mining of some

minerals, refining of most, components (magnets, batteries, reducers, motors), final assembly of electronics, cars and robots, and a growing share of AI models. Month 9 shows that China installs more than half of the world's new industrial robots and, according to J.P. Morgan Private Bank, makes about 70% of the world's robot actuators.

Strengths and bottlenecks. China's strengths are scale, supplier density (Month 7), fast construction, cheap capital and a huge engineering workforce. Its bottlenecks are:

- frontier AI chips and the machines to make them, limited by US and Dutch export controls (Month 8);
- dependence on imported ores (iron ore, copper concentrate, nickel from Indonesia, cobalt from the DRC);
- a shrinking property sector, weak household demand and deflationary pressure;
- trade resistance, as other countries put tariffs on Chinese goods to protect their own industries.

2025-2026 developments. China turned its refining dominance into a policy tool. It imposed export controls on several medium and heavy rare earths in April 2025 and broader controls in October 2025, then suspended parts of them under the US-China trade arrangement of late October 2025 (see Month 9 for the latest dates). Robot installations hit a record, Chinese brands held more than half of their home market, and humanoid firms such as Unitree listed on the stock market (Month 9).

2040 outlook. Judgment: China will still be the world's cheapest place to build robots in 2040, and probably the largest producer of humanoids, because its supplier base and home market reinforce each other. Its share of world refining will fall from today's extreme levels as Western, Gulf and Southeast Asian plants open, but it will remain the price-setter for most processed minerals. The key uncertainty is whether it can close the gap in leading-edge chips; if it does, the world splits into two complete AI stacks.

Sources: IFR (see above); USGS MCS 2026 and IEA via Month 5; Month 7 (WTO, China customs via Reuters) and Month 9 (J.P. Morgan Private Bank, IFR) for exports, surplus and actuators.

3. Japan

Indicator	Value
GDP 2025	about USD 4,453 billion (IMF, Oct 2026)
Population 2025	about 123.3 million (IMF)
GDP per capita 2025	about USD 36,120 (IMF)
Top exports	cars (motor vehicles 21.9 trillion yen, 2025, JAMA), general machinery (19.7 trillion yen), electrical machinery including chips and parts (18.5 trillion yen); total exports a record 110.4 trillion yen in 2025
Main minerals	almost none mined; large smelting and refining of copper, nickel and zinc from imported ores
Robots	36,219 installations in 2025, down 19%; density 446 (2024); the world's largest exporter of robots (IFR)

Indicator	Value
Electricity mix 2025	gas 33%, coal 32%, solar 10%, nuclear 9%, hydro 7%, bioenergy 5%

Role in the chain. Japan is the precision supplier of the world. It makes industrial robot arms (Fanuc, Yaskawa, Kawasaki, Nachi, Denso, Epson), most of the world's precision reducers for robot joints (Nabtesco and Harmonic Drive Systems, Month 9), chip-making equipment (Tokyo Electron, Screen, Lasertec), key chip materials (silicon wafers, photoresists, specialty gases) and image sensors (Sony). Japanese trading houses and smelters also finance mines abroad and refine copper and nickel at home.

Strengths and bottlenecks. Its strengths are deep manufacturing know-how, long supplier relationships, strong balance sheets and a large stock of overseas investment (Japan's current account surplus reached a record 31.87 trillion yen in 2025, mostly from investment income). Its bottlenecks are:

- a shrinking and ageing workforce, which raises the value of automation but also limits engineers;
- weak home demand for robots (installations fell 19% in 2025, and the IFR sees no major recovery soon);
- imported energy, which made the 2026 oil and LNG shock painful;
- price competition from Chinese makers of reducers and robots, which are cheaper and improving fast (Month 9).

2025-2026 developments. Exports set a record in 2025 despite US tariffs, as AI-related demand for chips and electronic parts offset weaker car shipments to the United States. Japan's position in robot parts came under pressure: Chinese harmonic reducer makers expanded, and Harmonic Drive Systems began shifting toward integrated actuators (Month 9). The government continued to fund Rapidus, the venture aiming to make 2-nanometre-class chips in Hokkaido (Month 8).

2040 outlook. Judgment: Japan will remain a top supplier of precision parts, equipment and materials in 2040, but its share of the robot market will shrink as China takes the low and middle segments. Its best path is to sell premium components and equipment to everyone, including Chinese, Indian and Southeast Asian robot makers, and to use its own robots to offset labour shortages.

Sources: IFR (see above); [E] Nippon.com, "Japan's Trade Deficit Halves to 2.7 Trillion Yen in 2025", Feb 2026: https://www.nippon.com/en/japan-data/h02684 ; [E] JAMA, Motor Industry of Japan 2026: https://www.jama.or.jp/english/reports/docs/MIoJ2026_e.pdf ; [E] moto-no.jp export breakdown from Ministry of Finance data: https://moto-no.jp/area/ex_jp_2025/ ; [E] Japan Times, "Japan's 2025 current account surplus hits record", 9 Feb 2026: https://www.japantimes.co.jp/business/2026/02/09/economy/japan-record-current-account-surplus ; Month 8 and Month 9 for Rapidus, Nabtesco and Harmonic Drive.

4. South Korea

Indicator	Value
GDP 2025	about USD 1,882 billion (IMF, Oct 2026)
Population 2025	about 51.7 million (IMF)
GDP per capita 2025	about USD 36,410 (IMF)

Indicator	Value
Top exports	semiconductors USD 173.4 billion (24% of exports), cars USD 72.0 billion, ships USD 32.0 billion, petrochemicals, steel; total exports a record USD 709.7 billion in 2025 (Ministry of Trade, Industry and Resources)
Main minerals	almost none mined; Korea Zinc is a major refiner of zinc, lead and minor metals
Robots	about 30,000 installations in 2025, down 1%; world's highest density, 1,220 per 10,000 workers (2024)
Electricity mix 2025	coal 31%, nuclear 30%, gas 28%, solar 6%, bioenergy 3%

Role in the chain. Korea is the memory chip and battery power of the chain. SK hynix and Samsung together held about 83% of the market for high-bandwidth memory (HBM), the memory stacked beside AI chips (Counterpoint, second quarter of 2026, Month 8). LG Energy Solution, Samsung SDI and SK On are among the largest battery makers outside China. Hyundai owns Boston Dynamics, maker of the Atlas humanoid, and Korean factories are the most automated in the world.

Strengths and bottlenecks. Its strengths are world-class manufacturing in memory, displays, batteries, shipbuilding and cars, strong engineering talent and fast execution by large firms. Its bottlenecks are:

- almost total dependence on imported energy and minerals;
- high concentration (the top ten exporters made 39% of exports in 2025, a record, according to Korean statistics reported by Kyunghyang Shinmun);
- the world's lowest fertility rate, which will shrink the workforce;
- exposure to both US tariffs and Chinese demand, and to US-China rivalry in chips.

2025-2026 developments. The AI memory boom drove record exports: semiconductor exports rose 22.2% in 2025, and the Ministry reported record monthly exports of USD 120.9 billion in September 2026. Memory prices roughly doubled in early 2026 (Month 8). Robot installations stayed flat at about 30,000, but the IFR expects new car-industry investment to lift demand from 2027. Hyundai planned to deploy Atlas humanoids in its US factories from 2028 (Month 9).

2040 outlook. Judgment: Korea will remain a top-three supplier of AI memory and a leader in robot adoption, and could become a major humanoid maker by combining Hyundai's robots, LG's batteries and Samsung's chips. But demographics and energy imports are serious constraints. Its best strategy is to keep its technological lead in memory and to move robot and battery assembly closer to customers (the United States, Europe, Southeast Asia) while keeping the most valuable know-how at home.

Sources: IFR (see above); [V] Yonhap, "S. Korea's exports hit all-time high of USD 709.7 bln in 2025", 1 Jan 2026: https://en.yna.co.kr/view/AEN20260101000852320 ; [E] Kyunghyang Shinmun, "Exports surpassed 700 billion dollars, but the top 10 companies accounted for 40%", 10 Feb 2026: https://www.khan.co.kr/en/article/202602101202007 ; [E] MOTIR, September 2026

exports press release: https://english.motir.go.kr/eng/article/EATCLdfa319ada/2743/view ; Month 8 (Counterpoint, TrendForce) and Month 9 (McKinsey on Atlas).

5. Taiwan

Indicator	Value
GDP 2025	about USD 921 billion (IMF, Oct 2026)
Population 2025	about 23.3 million (IMF)
GDP per capita 2025	about USD 39,510 (IMF)
Top exports	information and communication technology (ICT) products including AI servers USD 251.2 billion, electronic components (mainly chips) USD 222.9 billion; total exports a record USD 640.8 billion in 2025, up 34.9% (Ministry of Finance)
Main minerals	none of significance; imports almost all energy
Robots	installations not given in the IFR press release; density 302 (2024)
Electricity mix 2025	gas 49%, coal 36%, solar 5%, wind 4%, hydro 2%, nuclear 1%

Role in the chain. Taiwan is the factory floor of advanced computing. TSMC makes most of the world's leading-edge logic chips, including Nvidia's AI processors, and runs the most important advanced packaging lines (CoWoS, the method that joins AI chips to HBM memory). Taiwanese firms such as Foxconn, Quanta and Wistron assemble most AI servers. In 2025, for the first time since 1999, the United States (30.9% of exports) took more of Taiwan's exports than mainland China and Hong Kong combined (26.6%). TSMC's 2025 revenue was USD 122.4 billion at a 59.9% gross margin (Month 8), a sign that this manufacturing step captures high value, unlike ordinary assembly.

Strengths and bottlenecks. Its strengths are unmatched process know-how, a dense supplier cluster around Hsinchu and Tainan, and deep trust with US chip designers. Its bottlenecks are:

- geopolitical risk from China's claim over the island, the single largest risk in the global AI supply chain;
- energy: about 85% of its power comes from imported gas and coal, and it closed its last nuclear reactor in 2025, which is visible in the drop of nuclear to about 1% of generation;
- water, land and engineers for new fabs;
- extreme concentration: ICT and electronic components made about 74% of exports in 2025.

2025-2026 developments. The AI boom produced Taiwan's fastest growth in 15 years: the government estimated 2025 GDP growth at 7.37%, and Taiwan became the world's 12th-largest goods exporter, its best ranking since 1994.

2040 outlook. Judgment: Taiwan will still be the most important place for leading-edge chip production in 2040, but its share will fall (the SIA/BCG projection in Month 8 is 47% of sub-10-nanometre capacity by 2032) as TSMC, Samsung, Intel and Rapidus build in the United States, Japan and Korea. Taiwan's value capture will stay high as long as it keeps the newest process node

and packaging at home. The main risk is political and military, not economic; any blockade would be a global economic shock larger than the 2026 Hormuz crisis.

Sources: IFR density (see above); [E] Invest Taiwan, "Taiwan reports record high foreign trade in 2025", 13 Jan 2026: https://investtaiwan.nat.gov.tw/newsPage62553eng?lang=eng&search=62553 ; [E] OCAC/CNA, "Taiwan's exports, imports, trade surplus smash records in 2025", Jan 2026: https://ocacnews.net/article/417350 ; [E] Kaohsiung Times, US share of exports, May 2026: https://www.kaohsiungtimes.com/the-golden-age-of-taiwan-u-s-economic-and-trade-relations-driven-by-ai/ ; [E] Focus Taiwan, "Taiwan ranks 12th globally in exports in 2025", 25 Mar 2026: https://focustaiwan.tw/business/202603250007 ; [E] Taiwan Business Topics via TIMTOS, 7.37% growth estimate: https://www.timtos.com.tw/en/news/4DC3B0EF1600FD46/info.html?cr=6&lt=data ; Month 8 for TSMC and SIA/BCG.

6. Germany

Indicator	Value
GDP 2025	about USD 5,116 billion (IMF, Oct 2026)
Population 2025	about 83.5 million (IMF)
GDP per capita 2025	about USD 61,260 (IMF)
Top exports	cars and car parts, machinery, chemicals, pharmaceuticals, electrical equipment; goods export surplus 200.5 billion euros in 2025 (Destatis)
Main minerals	lignite (brown coal), potash, salt; refines copper (Aurubis) from imported concentrate
Robots	fewer than 25,000 installations in 2025, down 8%, fifth-largest market; density 449 (2024)
Electricity mix 2025	wind 27%, coal 21%, solar 18%, gas 17%, bioenergy 10%, hydro 4%, oil and other 4%

Role in the chain. Germany is the machine builder of Europe. It makes the tools that make things: machine tools, industrial automation (Siemens), robot arms (KUKA, now Chinese-owned by Midea), drives and gearboxes, optics (Zeiss, which supplies the lenses inside ASML's lithography machines), lasers (Trumpf) and chemicals (BASF). It is the largest robot user in Europe, with 41% of European Union installations in 2025.

Strengths and bottlenecks. Germany's strengths are its "Mittelstand" of specialised mid-sized firms, vocational training, high-quality engineering and strong brands. Its bottlenecks are:

- high and volatile energy costs since the loss of cheap Russian gas in 2022, which hit chemicals and metals hardest;
- an industrial model built on combustion-engine cars, now losing share to Chinese electric vehicles;
- slow digital adoption and weak software and AI platforms;
- an ageing workforce and slow permitting.

2025-2026 developments. GDP grew only 0.2% in 2025 after two years of recession, and manufacturing output fell for the third year in a row (-1.3%). China again became Germany's largest

trading partner (251.8 billion euros of trade), but German exports to China fell almost 10% while imports from China rose almost 9%, leaving a record deficit of about 90 billion euros (IW Koeln).

2040 outlook. Judgment: Germany will remain a leader in high-end machinery, automation and industrial software for factories, but it will lose mass car production to China and to lower-cost European neighbours. Its robot future lies in being the integrator and the premium supplier of precision parts and factory software, not the cheapest maker. The decisive question is energy: if Germany can build cheap renewable power and grids, its industry can adapt; if not, energy-intensive steps will move to the Gulf, the United States and Asia.

Sources: IFR (see above); [V] Destatis, "Gross domestic product up 0.2% in 2025", press release 017, 15 Jan 2026: https://www.destatis.de/EN/Press/2026/01/PE26_017_811.html ; [E] Destatis, "China is Germany's most important trading partner once again in 2025", 20 Feb 2026: https://www.destatis.de/EN/Press/2026/02/PE26_056_51.html ; [E] IW Koeln, "China-Schock trifft vor allem die Autoindustrie", 24 Feb 2026: https://www.iwkoeln.de/studien/juergen-matthes-china-schock-trifft-vor-allem-die-autoindustrie.html .

7. Netherlands

Indicator	Value
GDP 2025	about USD 1,322 billion (IMF, Oct 2026)
Population 2025	about 18.0 million (IMF)
GDP per capita 2025	about USD 73,260 (IMF)
Top exports	chip-making machines (ASML net sales 32.7 billion euros, 2025), refined petroleum, chemicals, food and flowers, plus large re-exports through Rotterdam
Main minerals	natural gas (Groningen field now closed); none of significance for the chain
Robots	not given in IFR press releases
Electricity mix 2025	gas 35%, wind 25%, solar 21%, coal 7%, bioenergy 5%, nuclear 3%

Role in the chain. The Netherlands owns one of the world's tightest chokepoints: ASML is the only company that makes extreme ultraviolet (EUV) lithography machines, without which no leading-edge chip can be made. ASML recognised 48 EUV systems worth 11.6 billion euros in 2025 (Month 8). The country also hosts NXP (chips for cars), ASM International (deposition equipment), and Nexperia (simple chips for cars and electronics). The port of Rotterdam is Europe's largest gateway for oil, chemicals, containers and metals.

Strengths and bottlenecks. Strengths: monopoly-level technology in lithography, a cluster of suppliers around Eindhoven (the "Brainport" region), excellent logistics and finance. Bottlenecks:

- ASML depends on suppliers in Germany (Zeiss optics) and the United States (light sources), so it is part of a multi-country system rather than a stand-alone national champion;
- export controls make the Netherlands a front line in the US-China chip war, and China's share of ASML sales was expected to fall to about 20% in 2026 from 41% in 2024 (Month 8).

2025-2026 developments. The Nexperia affair showed how small links can stop big chains. On 30 September 2025 the Dutch government used the Goods Availability Act to intervene in Nexperia, owned by China's Wingtech, and a Dutch court suspended its chief executive. China then blocked exports from Nexperia's Chinese plants, causing chip shortages at carmakers, before mostly lifting the ban in November 2025 as part of a wider US-China understanding; the Dutch order was suspended. The dispute continued in courts: a Dutch court ordered an investigation in February 2026, Nexperia's Chinese unit declared independence, a Chinese court froze about USD 300 million of Nexperia assets in August 2026, and in November 2026 Nexperia sued Wingtech for damages.

2040 outlook. Judgment: ASML will almost certainly still be the sole supplier of the most advanced lithography in 2040, because the know-how and supplier network took 30 years to build; China's attempts to copy it will produce older-generation machines first. The Netherlands will keep capturing very high value from a tiny number of machines. The Nexperia case suggests that Europe will screen foreign ownership of chip firms much more tightly.

Sources: Month 8 for ASML figures (ASML results; SCMP citing ASML); [E] Reuters, "Chinese court freezes USD 300 million of Nexperia assets in Wingtech case", 31 Aug 2026: https://www.reuters.com/world/asia-pacific/chinese-court-freezes-dutch-chipmaker-nexperia-bvs-stakes-four-china-units-2026-08-31/ ; [E] NL Times, "Nexperia sues Chinese parent company Wingtech", 3 Nov 2026: https://nltimes.nl/2026/11/03/nijmegen-based-nexperia-sues-chinese-parent-company-wingtech-2-billion ; [E] Nexperia, Sustainability Report 2025 (timeline of the intervention): https://www.nexperia.com/dam/jcr:5ca32ae2-52cd-4d8e-a514-d13dd11285e7/NEXPERIA_Sustainability_Report_2025.pdf ; [E] Prime Law, account of the Enterprise Chamber case: https://www.primelaw.nl/en/corporate-litigation/dutch-nexperia-inquiry-proceedings/ .

8. India

Indicator	Value
GDP 2025	about USD 3,909 billion (IMF, Oct 2026)
Population 2025	about 1,463.9 million (IMF)
GDP per capita 2025	about USD 2,670 (IMF)
Top exports	IT and business services, refined petroleum, pharmaceuticals, smartphones (USD 30.1 billion in 2025, about 76% from Apple, Month 7), gems and jewellery, machinery
Main minerals	coal, iron ore, bauxite, rare earths in beach sands (limited production), some copper and zinc
Robots	almost 10,500 installations in 2025, up 15%, sixth-largest market; 27% average annual growth 2020-2025
Electricity mix 2025	coal 71%, solar 9%, hydro 9%, wind 5%, nuclear 3%, gas 2%

Role in the chain. India is the world's largest supplier of software and engineering services and a fast-growing assembler of electronics. About 55 million iPhones, roughly a quarter of world output, were assembled in India in 2025 (Month 7). It has a vast pool of engineers who design chips for

global companies, but until 2026 it had no commercial chip factories. Its home market for cars, phones and, soon, robots is huge and young.

Strengths and bottlenecks. Strengths: scale, a young workforce, strong engineering education, a large domestic market and good relations with the United States, Japan and the Gulf. Bottlenecks:

- low manufacturing share of GDP and a thin supplier base, so many components are still imported from China;
- slow land acquisition, complex regulation and tariffs on imported inputs;
- a coal-heavy grid and patchy power quality for precision manufacturing.

2025-2026 developments. India's chip programme started to produce. Micron opened India's first semiconductor assembly and test plant at Sanand, Gujarat, on 28 February 2026; Kaynes Semicon reached commercial production on 31 March 2026 and CG Semi (with Renesas) on 4 July 2026. Tata Electronics and Taiwan's PSMC are building India's first commercial fab at Dholera (28-110 nanometres, about 50,000 wafers a month), due to be commissioned in 2028. In July 2026 the cabinet approved "Semicon 2.0" with 1.275 lakh crore rupees (about USD 13.4 billion). A 7,280 crore rupee scheme aims to build 6,000 tonnes a year of rare earth magnet capacity.

2040 outlook. Judgment: India is the most likely second centre of large-scale electronics and robot assembly after China by 2040, because of its market size and the strategic support of the United States and Japan. It will be strong in robot software, AI applications and services, and in assembly; it will probably still import advanced chips and many precision components. Success depends on infrastructure and on lowering tariffs on inputs, the lessons of Month 3's East Asian playbooks.

Sources: IFR (see above); Month 7 (Bloomberg via Mobile World Live; Financial Express); [E] Micron, "Micron Celebrates Opening of India's First Semiconductor Assembly and Test Facility", 28 Feb 2026: https://investors.micron.com/news/press-release/2026/Micron-Celebrates-Opening-of-Indias-First-Semiconductor-Assembly-and-Test-Facility-02-28-2026/default.aspx ; [E] Hindustan Times, explainer on plants in production, 15 Aug 2026: https://www.hindustantimes.com/ht-explainers/india-independence-day-pm-narendra-modi-semiconductors-chips-critical-minerals-technology-production-in-reality-red-fort-101786792244730.html ; [E] DSCI, State of India's Semiconductor Industry 2026: https://www.dsci.in/files/content/knowledge-centre/2026/India-semiconductor-report.pdf ; [E] TNF India, rare earth magnet scheme, Sept 2026: https://www.tnf-india.com/intelligence/signal/manufacturing-and-engineering-india/2026-w39 .

9. Indonesia

Indicator	Value
GDP 2025	about USD 1,446 billion (IMF, Oct 2026)
Population 2025	about 284.7 million (IMF)
GDP per capita 2025	about USD 5,080 (IMF)
Top exports	coal, palm oil, iron and steel including stainless steel and ferronickel (ferronickel exports USD 14.1 billion in 2024), nickel intermediates, electronics, garments and footwear

Indicator	Value
Main minerals	nickel (about 2.6 million tonnes, 67% of world mining, 2025), cobalt as a nickel by-product (14% of world), coal, bauxite, copper and gold (Grasberg), tin
Robots	not given in IFR press releases; small relative to Thailand
Electricity mix 2024	coal 61%, gas 19%, hydro 7%, bioenergy 6%, geothermal and other 5%, oil 2%

Role in the chain. Indonesia is the world's nickel processing hub. Its 2020 ore export ban (Month 6) pulled in mostly Chinese investment: smelters grew from 2 in 2014 to about 54 by 2025, and Indonesia now makes nickel pig iron, ferronickel, stainless steel and battery-grade intermediates such as mixed hydroxide precipitate (MHP). It is therefore one of the few developing countries that has moved one or two steps up from raw ore. It is also a large producer of copper, tin and bauxite.

Strengths and bottlenecks. Strengths: huge reserves (62 million tonnes of nickel), a large domestic market and young workforce, and a proven willingness to use policy to force processing. Bottlenecks:

- most processing capacity is Chinese-controlled (about 75% of nickel refining, CSIS) and most nickel exports go to China (82% in 2024), so much of the value and know-how leaves;
- processing runs on captive coal power, which raises carbon and pollution concerns and can exclude Indonesian nickel from some Western supply chains;
- serious safety and environmental problems at industrial parks, including the December 2023 furnace explosion at IMIP that killed 21 workers;
- weak manufacturing beyond metals, with electronics and machinery lagging Vietnam and Thailand (Month 7).

2025-2026 developments. The government cut nickel ore mining quotas for 2026 to support prices (initially 260-270 million wet tonnes, revised to about 290-300 million, S&P Global via Month 6), which pushed smelters to import more ore from the Philippines. Danantara, the new sovereign wealth fund launched in February 2025, took control of state companies (Month 11).

2040 outlook. Judgment: Indonesia will remain the centre of world nickel and a major stainless steel and battery-material producer in 2040. The question is whether it moves further, into cathodes, cells, motors and robot assembly for its own market, and whether more of the ownership and engineering becomes Indonesian. That requires cleaner power, better safety enforcement and much deeper technical education. Indonesia shows that downstreaming can create industrial capacity quickly, but capturing value takes longer than building plants.

Sources: USGS MCS 2026 (nickel, cobalt) [V] via Months 5, 6 and 9: https://pubs.usgs.gov/periodicals/mcs2026/mcs2
nickel.pdf ; Month 6 (NEXT Indonesia Center; CSIS 2025; AP; S&P Global); Month 11 (Danantara).

10. Vietnam

Indicator	Value
GDP 2025	about USD 494 billion (IMF, Oct 2026); Vietnam's own statistics office reported about USD 514 billion
Population 2025	about 102.3 million (IMF)
GDP per capita 2025	about USD 4,830 (IMF)
Top exports	electronics, computers and components (USD 107.7 billion in 2025, up 48.4%), phones, textiles and footwear, machinery, furniture; total goods exports about USD 475 billion, up 17%
Main minerals	rare earths (large reserves, little production), bauxite, coal, tungsten, titanium
Robots	not given in IFR press releases
Electricity mix 2025	coal 48%, hydro 33%, solar 7%, gas 6%, wind 5%

Role in the chain. Vietnam is the leading "China plus one" assembly base. Samsung makes roughly half of its smartphones there, and Apple sources much of its US-bound electronics from Vietnam. Foreign-invested firms make about three-quarters of its exports (77.3% in 2025, Month 7). It is mainly a final-assembly step: the parts come largely from China, which gave Vietnam a trade deficit with China of about USD 116 billion in 2025 while it ran a surplus of about USD 134 billion with the United States (Month 7).

Strengths and bottlenecks. Strengths: competitive wages (a base wage for experienced operators of about USD 302 a month, JETRO 2025, Month 7), proximity to Chinese suppliers, many free-trade agreements and political stability. Bottlenecks:

- a thin domestic supplier base and low local value added;
- power shortages in the north in some recent years and a grid still heavy on coal;
- exposure to US tariffs and to US rules against goods that are mostly Chinese inside;
- a shortage of engineers and managers for higher-value work.

2025-2026 developments. GDP grew 8.02% in 2025, the strongest in Southeast Asia, and about 8.2% in the first half of 2026, driven by AI-related electronics exports; computer and electronics exports grew about 50% year on year in the first seven months of 2026 (VinaCapital). A December 2025 amendment to the geology and minerals law classified rare earths as a "special strategic" mineral and banned raw rare earth exports; in October 2026 a draft rule proposed allowing exports only of separated oxides, metals, alloys and magnets.

2040 outlook. Judgment: Vietnam will be one of Asia's largest electronics exporters in 2040, and it could become a significant assembler of robots and drones for export. Whether it captures more value depends on building local suppliers and engineering skills, as Korea and Taiwan did, rather than staying an assembly platform. A domestic rare earth magnet industry before the 2030s is unlikely.

Sources: [E] Reuters, "Vietnam's annual growth reaches 8% as trade surplus with US hits record", 5 Jan 2026 (page opened but only the headline loaded, so treated as a snippet): https://www.reuters.com/world/asia-pacific/vietnams-annual-growth-reaches-8-trade-surplus-with-us-hits-record-despite-2026-01-05/ ; [E] General Statistics Office of Vietnam, 2025 socio-economic report: https://www.nso.gov.vn/du-lieu-va-so-lieu-thong-

ke/2026/01/bao-cao-tinh-hinh-kinh-te-xa-hoi-quy-iv-va-nam-2025/ ; [E] VCCI, electronics exports 2025: https://vccinews.com/news/62336/exports-2026-standardization-pressure-amid-rising-protectionism.html ; [E] VinaCapital, Macro Commentary July 2026: https://vinacapital.com/wp-content/uploads/2026/08/20260811-Monthly-Macro-Summary-July-2026.pdf ; [E] The Diplomat, "Vietnam, Rare Earths, and the Evolving China Plus One Strategy", Aug 2026: https://thediplomat.com/2026/08/vietnam-rare-earths-and-the-evolving-china-plus-one-strategy/ ; [E] TNGlobal, draft rare earth export rules, 5 Oct 2026: https://technode.global/2026/10/05/vietnam-can-ease-rare-earth-export-ban-can-export-deep-processed-products ; [E] VnEconomy (NSO figures for GDP size): https://en.vneconomy.vn/solutions-for-advancing-science-and-technology.htm ; Month 7 (JETRO, NSO).

11. Thailand

Indicator	Value
GDP 2025	about USD 577 billion (IMF, Oct 2026)
Population 2025	about 71.6 million (IMF)
GDP per capita 2025	about USD 8,060 (IMF)
Top exports	computers and parts, cars and car parts, gems and jewellery, rubber products, machinery, food; total exports a record USD 339.6 billion in 2025, up 12.9% (Ministry of Commerce)
Main minerals	limited: some gypsum, tin and gold; imports most energy
Robots	about 3,600 installations in 2023 (IFR data via Statbase); no 2025 figure in IFR press releases
Electricity mix 2025	gas 66%, coal 18%, solar 5%, bioenergy 5%, hydro 4%, wind 2%

Role in the chain. Thailand is Southeast Asia's car and hard-disk-drive hub, the "Detroit of Asia", with a deep base of Japanese-owned suppliers. It is now becoming a hub for electric vehicles (EVs) built by Chinese carmakers, for printed circuit boards and chip assembly, and for data centres. Chinese brands rose from 3.2% of its car market in 2020 to 21.2% in 2025, while Japanese brands fell from about 88% to 69.3% (Month 7).

Strengths and bottlenecks. Strengths: a mature industrial base and supplier network, good infrastructure, an investment agency (the Board of Investment, BOI) that moves fast, and the highest factory wages among its middle-income neighbours, which pushes automation. Bottlenecks:

- an ageing population, like a rich country's but at a middle-income level;
- heavy dependence on imported gas for power, painful in the 2026 energy shock;
- a car industry built on combustion engines, disrupted by Chinese EVs;
- political instability and relatively weak domestic research and development.

2025-2026 developments. BOI investment applications reached a record of about USD 60.2 billion in 2025, up 67%; the digital sector led with 746 billion baht, mostly data centres, and Sunwoda announced Thailand's first upstream battery cell factory. The government published a National Semiconductor and Advanced Electronics Strategy for 2026-2050 focused on power, sensor and analog chips, and in August 2026 set up a new oversight framework for data centres.

2040 outlook. Judgment: Thailand is well placed to become Southeast Asia's main hub for EV and robot assembly and for industrial automation services by 2040, because it already has the suppliers, the skills and the high labour costs that make robots pay. Its weakness is upstream: it has few minerals and depends on imported energy, so it will be an assembler and integrator, not an ore-to-robot chain on its own.

Sources: [E] Thailand Public Relations Department, "Thailand's Exports Grow 12.9% to Record High in 2025", 26 Jan 2026: https://thailand.prd.go.th/en/content/category/detail/id/52/iid/469585 ; [E] BOI, "Thailand's Investment Applications Reach New Highs in 2025", press release 15/2569, 26 Jan 2026: https://www.boi.go.th/upload/content/PR15_2569EN.pdf ; [E] Thailand.go.th, Sunwoda battery cell plant: https://www.thailand.go.th/public/issue-focus-detail/record-breaking-influx-thailand-becomes-a-magnet-for-global-high-tech-investment ; [E] BOI, National Semiconductor and Advanced Electronics Strategy 2026-2050: https://www.boi.go.th/upload/content/Thailand%20National%20Semiconductor%2 ; [E] Bangkok Post, data centre oversight, 19 Aug 2026: https://bangkokpost.pressreader.com/article/28195602462087 ; [E] Statbase (IFR data): https://statbase.org/data/tha-industrial-robots-installations/ ; Month 7 (MarkLines via China Daily).

12. Malaysia

Indicator	Value
GDP 2025	about USD 473 billion (IMF, Oct 2026)
Population 2025	about 33.9 million (IMF)
GDP per capita 2025	about USD 13,960 (IMF)
Top exports	electrical and electronic (E&E) products, which led record exports in 2025; palm oil; LNG and petroleum products; optical and scientific equipment
Main minerals	tin, bauxite, rare earths (ionic clay deposits estimated at 16.1 million tonnes of feedstock, CSIS), plus oil and gas
Robots	not given in IFR press releases
Electricity mix 2025	coal 45%, gas 34%, hydro 18%, solar 2%

Role in the chain. Malaysia is the world's back-end chip specialist. It handles about 13% of global chip assembly, testing and packaging (Bank Negara Malaysia and the Malaysia Semiconductor Industry Association, via Month 7), with plants run by Intel, Infineon, AMD and many others in Penang and Kulim. It hosts Lynas Malaysia at Gebeng, near Kuantan, now described as the world's largest single rare earth processing plant and the main producer of separated heavy rare earths outside China. And it has become a major data centre location: 925.6 megawatts (MW) were live in October 2026, with 2.0 gigawatts (GW) under construction (BMI via Month 8).

Strengths and bottlenecks. Strengths: decades of electronics experience, good infrastructure, English-speaking engineers and a neutral position between the United States and China. Bottlenecks:

- packaging and testing capture only a small share of chip value (about 6% of semiconductor value added, Month 8), so Malaysia needs to move into design and advanced packaging;
- data centres compete for electricity and water; Johor's data centres are projected to use about 18% of Peninsular Malaysia's electricity by 2031 (BMI via Month 8);

- rare earth mining from ionic clays carries environmental risk and has been limited to one mine in Perak;
- a coal-heavy grid that some foreign customers want to see decarbonised.

2025-2026 developments. In 2025 Lynas produced dysprosium and terbium oxides, becoming the first company outside China to make commercial quantities of dysprosium oxide; it added samarium oxide in March 2026 and announced an expansion of heavy rare earth separation. Malaysia's moratorium on exports of unprocessed rare earths, in force since 1 January 2024, was eased in October 2026: new guidelines allow raw exports for research or processing, on condition that the exporter commits to processing in Malaysia.

2040 outlook. Judgment: Malaysia is the Southeast Asian economy best placed to own a piece of the magnet chain outside China by 2040, because it already separates heavy rare earths at scale. If it adds metal and magnet making, it could supply motors for robots and EVs across the region. The risk is that data centres take the power and land that higher-value manufacturing needs.

Sources: [E] MITI, "Malaysia's 2025 Trade Performance", 20 Jan 2026: https://www.miti.gov.my/miti/resources/Mec ; [E] CSIS, "The Race for Malaysia's Rare Earths", 20 Oct 2026: https://www.csis.org/analysis/race-malaysias-rare-earths-indo-pacific-strategy ; [E] CSIS, "Rare Earth Export Restrictions One Year Later", Apr 2026: https://www.csis.org/analysis/rare-earth-export-restrictions-one-year-later ; [E] Lynas, Appendix 4E annual results, Aug 2026: https://announcements.asx.com.au/asxpdf/20260826/pdf/0737nx07s3 ; [E] The Star, "Malaysia allows first export of raw rare earths under new rules", 22 Oct 2026: https://www.thestar.com.my/business/business-news/2026/10/22/malaysia-allows-first-export-of-raw-rare-earths-under-new-rules ; [E] Straits Times, Lynas Gebeng plant, Apr 2026: https://www.straitstimes.com/asia/se-asia/delicate-extraction-malaysia-offers-rare-earths-alternative-to-china ; Months 7 and 8 (Bank Negara, BMI).

13. Philippines

Indicator	Value
GDP 2025	about USD 487 billion (IMF, Oct 2026)
Population 2025	about 113.9 million (IMF)
GDP per capita 2025	about USD 4,280 (IMF)
Top exports	electronic products, especially chip assembly and testing; nickel ore; copper concentrate and gold; coconut products and bananas; plus large services exports (business process outsourcing) and remittances
Main minerals	nickel ore (about 270,000 tonnes of nickel content in 2025, the world's second-largest producer, USGS MCS 2026), copper, gold, chromite
Robots	not given in IFR press releases
Electricity mix 2025	coal 59%, gas 17%, hydro 10%, geothermal and other 8%, solar 2%

Role in the chain. The Philippines plays two roles. It is the world's largest exporter of nickel ore, shipping lower-grade ore mainly to China and, increasingly, to Indonesian smelters that need it

for blending. And it has a long-established chip assembly and testing industry (Texas Instruments, Analog Devices, onsemi and others), plus one of the world's largest outsourced services industries, which is now being reshaped by AI.

Strengths and bottlenecks. Strengths: a young, English-speaking workforce, large mineral resources and close security ties with the United States and Japan. Bottlenecks:

- almost no processing: only two high-pressure acid leaching (HPAL) plants, Coral Bay and Taganito, refine nickel at home;
- the highest electricity prices in Southeast Asia, a coal-heavy grid and weak transmission across thousands of islands;
- weak infrastructure, slow permitting and local opposition to mining (Palawan has a reported long moratorium on new permits);
- exposure of its call-centre industry to automation by AI (Month 10).

2025-2026 developments. The Philippines decided not to copy Indonesia's ore ban. In June 2025 lawmakers dropped a proposed raw-ore export ban from the mining fiscal reform, and Republic Act 12253, signed on 5 September 2025, instead set margin-based royalties and a windfall profits tax. Then Indonesia cut its 2026 mining quotas, and Philippine ore exports to Indonesia jumped: up about 50% in 2025 to 15.3 million tonnes and up 82% year on year in January-July 2026. In August 2026 President Marcos signed Executive Order 122, a framework to promote domestic processing and new critical minerals reserves while still allowing ore exports.

2040 outlook. Judgment: the Philippines will most likely remain a major ore supplier with a modestly larger processing industry by 2040. It could become a partner in a Southeast Asian battery chain, feeding ore and intermediates to Indonesian and regional refiners, rather than building a full chain alone. High power costs are the main obstacle to heavy processing.

Sources: [V] USGS MCS 2026, Nickel (via Month 9): https://pubs.usgs.gov/periodicals/mcs2026/mcs2026-nickel.pdf ; [E] BusinessWorld, "Lawmakers drop ore export ban", 12 Jun 2025: https://bworldonline.com/top-stories/2025/06/12/678715/lawmakers-drop-ore-export-ban/ ; [E] Discovery Alert, "Philippines Nickel Industry: Why Ore Ships Out Unprocessed", Oct 2026 (RA 12253, HPAL plants): https://discoveryalert.com/education/philippines-nickel-industry-ore-exports-processing/ ; [E] Mining.com.au, ore exports to Indonesia, 28 Oct 2026: https://mining.com.au/regulatory-gaps-could-derail-the-philippines-critical-minerals-ambitions-oecd-warns/ ; [E] Lawin News, Executive Order 122, Aug 2026: https://lawin.news/marcos-approves-framework-to-develop-philippine-critical-minerals-industry/ .

14. Saudi Arabia

Indicator	Value
GDP 2025	about USD 1,274 billion (IMF, Oct 2026)
Population 2025	about 36.0 million (IMF)
GDP per capita 2025	about USD 35,370 (IMF)
Top exports	crude oil and refined products (the large majority of goods exports), petrochemicals and plastics, fertilisers, aluminium

Indicator	Value
Main minerals	oil and gas; phosphate, bauxite and aluminium, gold, copper (state miner Ma'aden); claimed large untapped deposits including rare earths
Robots	not given in IFR press releases
Electricity mix 2024	gas 63%, oil 34%, solar 2%

Role in the chain. Saudi Arabia is the base layer: the world's largest crude oil exporter and the swing producer inside OPEC+ (Month 4). It is trying to use oil money and cheap land and energy to move up the chain into metals (Ma'aden's aluminium and phosphate), AI computing and, eventually, manufacturing. The Public Investment Fund (PIF), with about 3.396 trillion riyals (about USD 905 billion) under management at end-2025 (Month 11), is the main tool.

Strengths and bottlenecks. Strengths: very cheap energy and land, huge capital, strong state capacity to build fast, and space for solar power. Bottlenecks:

- the budget still swings with oil prices: the 2025 deficit was 276.6 billion riyals, about 5.5% of GDP (Month 4);
- exposure to Gulf conflict: in August-September 2026 Saudi crude output fell to between about 6 and 8.3 million barrels a day, far below its target of about 10.5, because of the Hormuz disruption (IEA via Month 4);
- AI chips depend on US export licences.

2025-2026 developments. PIF launched Humain, a state AI company, in May 2025. In November 2025 the US Commerce Department authorised exports of up to the equivalent of 35,000 Nvidia GB300 Blackwell chips each to Humain and to the UAE's G42. Saudi data centre capacity rose from 68 MW in 2021 to more than 467 MW in the first quarter of 2026 (Saudi Press Agency via SAB); Humain announced partnerships with Nvidia, AMD, AWS and others and targets of roughly 2-3 GW by 2030, with most large sites due between 2027 and 2029. In August 2026 Aramco and Ma'aden formed an exploration joint venture for copper and other critical minerals.

2040 outlook. Judgment: Saudi Arabia will still be a top oil exporter in 2040, but it has a real chance to become a large exporter of energy-intensive products: aluminium, green or low-carbon steel, fertilisers and AI computing ("exporting electricity as tokens"). It is unlikely to become a robot manufacturing centre, because it lacks the supplier base and workforce, but it will be a large buyer and a funder of robot companies abroad.

Sources: Month 4 (IEA Oil Market Report; Ministry of Finance via Bloomberg; IMF breakeven via AGBI) and Month 11 (PIF Annual Report 2025); [E] US Department of Commerce, "Statement on UAE and Saudi Chip Exports", 19 Nov 2025: https://www.commerce.gov/news/press-releases/2025/11/statement-uae-and-saudi-chip-exports ; [E] SAB/Zawya newsletter, data centre capacity, Aug 2026: https://www.sab.com/content/dam/sabpws/corporate/business-insight/2026/august/2026-08-ZAWYA_SAB-August-Newsletter-2.pdf ; [E] Data Centre Magazine, "HUMAIN and NVIDIA Expand Saudi AI Factory Pipeline by 84MW", 22 Oct 2026: https://datacentremagazine.com/news/humain-and-nvidia-expand-saudi-ai-factory-pipeline-by-84mw ; [E] Arab News, Saudi data centre targets, Sep 2026: https://www.arabnews.com/business/saudi-arabias-data-center-race-can-the-kingdom-become-the-middle-easts-cloud-hub-3000478 ; [E] The National, Aramco-Ma'aden venture, 18 Aug 2026: https://www.thenationalnews.com/business/energy/2026/08/18/aram and-maaden-form-venture-to-tap-saudi-arabias-25-trillion-mineral-wealth/ .

15. United Arab Emirates

Indicator	Value
GDP 2025	about USD 614 billion (IMF, Oct 2026)
Population 2025	about 11.3 million (IMF)
GDP per capita 2025	about USD 54,150 (IMF)
Top exports	crude oil and refined products, re-exports through Dubai (gold, phones, machinery), aluminium (Emirates Global Aluminium), plus large trade, finance, aviation and logistics services
Main minerals	oil and gas; no significant metal mining at home, but large investments in mines abroad
Robots	not given in IFR press releases
Electricity mix 2024	gas 68%, nuclear 23% (Barakah), solar 9%

Role in the chain. The UAE is a capital and logistics hub that is trying to become an AI hub. Its sovereign funds (Mubadala reported about 1.4 trillion dirhams, about USD 385 billion, under management at end-2025, and the Abu Dhabi Investment Authority is estimated at about USD 1.13 trillion, Month 11) invest in chips, AI and mines worldwide. G42, the Abu Dhabi AI company, is building Stargate UAE with OpenAI, Oracle, Nvidia, Cisco and SoftBank: a 1 GW cluster inside a planned 5 GW UAE-US AI Campus, announced in May 2025.

Strengths and bottlenecks. Strengths: capital, fast decisions, cheap energy, nuclear power, a business-friendly legal system in its free zones and a close strategic bargain with Washington on AI chips. Bottlenecks:

- a very small citizen population and dependence on foreign workers;
- exposure to Gulf war risks, which became real in 2026;
- reliance on US export approvals for advanced chips, which come with security conditions;
- little domestic manufacturing depth beyond aluminium, petrochemicals and defence.

2025-2026 developments. In November 2025 the United States authorised G42 to buy the equivalent of up to 35,000 Nvidia GB300 chips, and in July 2026 the US Commerce Department gave the UAE "enhanced favorable treatment", including licence-free access for approved government and commercial entities. The UAE ambassador said in May 2026 that the first advanced chips had arrived. But after Iranian drone strikes in the 2026 war, Reuters reported in September 2026 that the UAE was revising the plan, possibly from a single 5 GW campus to a network of protected sites. As of late September 2026, no official confirmation was found that the first 200 MW of Stargate UAE was in service.

2040 outlook. Judgment: the UAE will be one of the world's main AI computing locations outside the United States and China in 2040, serving the Middle East, Africa and South Asia, and a major investor in African and Asian mines and infrastructure. It is not a candidate for a robot supply chain of its own, but it is a likely financier and customer of one, for example in North Africa or South Asia.

Sources: Month 11 (Mubadala; Global SWF); [E] US Department of Commerce, 19 Nov 2025: https://www.commerce.gov/news/press-releases/2025/11/statement-uae-and-saudi-chip-exports ; [E]

OpenAI, "Introducing Stargate UAE", 22 May 2025: https://openai.com/index/introducing-stargate-uae ; [E] Federal Register, "Enhanced Favorable Treatment for the United Arab Emirates Under the Export Administration Regulations", 14 Jul 2026: https://www.govinfo.gov/app/details/FR-2026-07-14/2026-14132 ; [E] Khaleej Times, first chips delivered, 9 May 2026: https://www.khaleejtimes.com/business/tech/5g uae-us-ai-campus-under-construction ; [E] Reuters, "UAE revises AI data center plan after Iranian attacks, sources say", 11 Sep 2026: https://www.reuters.com/world/middle-east/uae-revises-ai-data-center-plan-after-iranian-attacks-sources-say-2026-09-11/ ; [E] Intelligence Age Gulf, status of first 200 MW, 28 Sep 2026: https://iagulf.com/en/articles/engineering-benefits .

16. Chile

Indicator	Value
GDP 2025	about USD 357 billion (IMF, Oct 2026)
Population 2025	about 20.0 million (IMF)
GDP per capita 2025	about USD 17,840 (IMF)
Top exports	copper (50.3% of all exports in 2025; copper exports about USD 54.4 billion), lithium carbonate, iron ore, molybdenum, fruit, salmon, wood pulp; mining as a whole 61.2% of exports (Cochilco)
Main minerals	copper (about 5.4 million tonnes in 2025, the largest producer, about 23% of world mining; 180 million tonnes of reserves), lithium (about 19% of world mining; 9.2 million tonnes of reserves), molybdenum, iodine
Robots	not given in IFR press releases
Electricity mix 2025	solar 25%, hydro 22%, coal 18%, gas 15%, wind 13%, bioenergy 6%

Role in the chain. Chile is the world's copper mine and a top lithium producer. Copper is in every motor, cable, battery and data centre, so Chile sits at the very start of the ore-to-robot chain. It also processes more at home than most mining countries: it smelts part of its copper and turns brine from the Atacama salt flat into lithium carbonate. But most copper still leaves as concentrate, and the highest-value steps (cathodes for batteries, cells, motors) happen in China, Korea and Japan.

Strengths and bottlenecks. Strengths: world-class deposits, strong institutions by regional standards, experienced mining companies (state-owned Codelco, BHP's Escondida, Antofagasta) and excellent solar resources in the north. Bottlenecks:

- falling ore grades and ageing mines: 2025 copper output was the lowest since 2016 and about 7.6% below the 2018 peak;
- water scarcity in the Atacama, which forces desalination;
- accidents and delays at Codelco: an accident at El Teniente in July 2025 cut that division's output by 13%;
- domestic political conflict over how much of the mining rent the state should take.

2025-2026 developments. Codelco and SQM formed Novandino Litio, a joint venture for lithium in the Salar de Atacama until 2060, in which the state's share of operating margin is about 70% in

2025-2030 and about 85% from 2031 (Month 6); Codelco booked a gain of about USD 2.0 billion from it. High copper prices lifted mining's fiscal contribution, even as volumes fell.

2040 outlook. Judgment: Chile will still be the world's largest copper producer in 2040, though its share will drift down as the DRC, Peru and Zambia grow. It has a good chance to add one or two processing steps, such as more copper smelting and refining powered by solar, and lithium chemicals beyond carbonate, but a full battery or robot chain is unlikely given its small market and distance from Asian factories. Its success model is rent capture plus good governance, not industrialisation.

Sources: USGS MCS 2026 via Month 5; Month 6 (Government of Chile; Nueva Mineria); [V] Codelco, "Codelco closed 2025 with an EBITDA of USD 6.67 billion…", 27 Mar 2026: https://www.codelco.com/en/prensa/2026/codelco-cerro-2025-con-un-ebitda-de-us-6-670-millones-una-utilidad ; [E] Diario Financiero, Chile's 2025 copper output (Cochilco), 10 Feb 2026: https://www.df.cl/empresas/produccion-de-cobre-de-chile-cae-casi-2-en-2025-y-escondida-por-si-sola ; [E] CNN Chile, Cochilco yearbook (export shares, fiscal contribution), 11 Sep 2026: https://www.cnnchile.com/negocios/produccion-cobre-2025-cayo-a-nivel-mas-bajo-desde-2016-pero-aportes-fiscales-se-dispararon-mas-de-40 .

17. Brazil

Indicator	Value
GDP 2025	about USD 2,280 billion (IMF, Oct 2026)
Population 2025	about 213.4 million (IMF)
GDP per capita 2025	about USD 10,680 (IMF)
Top exports	soybeans, crude oil, iron ore, meat, sugar, coffee, wood pulp, aircraft (Embraer)
Main minerals	iron ore (Vale), niobium (the dominant world supplier), bauxite, manganese, graphite, lithium (Minas Gerais), and rare earths, with some of the largest reserves outside China (USGS)
Robots	almost 4,300 installations in 2025, up 38%; the car industry's purchases rose 212%
Electricity mix 2025	hydro 52%, wind 16%, solar 12%, bioenergy 7%, gas 7%, coal 2%, nuclear 2%

Role in the chain. Brazil is a resource giant with a real industrial base. It supplies iron ore, niobium (used in high-strength steel), and growing amounts of lithium, graphite and rare earths, and it has one of the cleanest large power systems in the world, which is an advantage for low-carbon smelting and data centres. It also makes cars, aircraft and farm machinery. Chinese carmakers such as BYD are now building plants there, which explains the jump in robot purchases (IFR, Month 9).

Strengths and bottlenecks. Strengths: minerals, clean and cheap hydro-wind-solar power, a large home market and strong agricultural technology. Bottlenecks:

- the "Brazil cost": complex taxes, high interest rates, poor logistics and bureaucracy;
- a long history of protecting industry without forcing it to export, the Latin American failure case of Month 3;
- low productivity growth and weak technical education;

- political swings between more and less market-friendly policies.

2025-2026 developments. Brazil became a front line in the race for rare earths outside China. Serra Verde, which runs the Pela Ema ionic clay mine in Goias state, the only scaled producer of all four magnet rare earths outside Asia, secured a USD 565 million loan from the US International Development Finance Corporation (DFC) in February 2026, agreed a 15-year offtake with floor prices for a US-government-backed vehicle, and was bought by USA Rare Earth in a deal of about USD 2.8 billion completed in September 2026. In the same month President Lula signed a National Policy on Critical and Strategic Minerals, and officials warned against selling raw minerals abroad "for a song".

2040 outlook. Judgment: Brazil has the ingredients for a real mine-to-magnet chain (rare earths, clean power, a car industry and a home market), more than any other Latin American country. Whether it builds one depends on fixing its cost of capital and logistics and on negotiating processing commitments into its mineral deals. A realistic 2040 outcome is that Brazil separates rare earths and makes some magnets and EV motors for its own and regional markets, while still exporting most minerals in raw or semi-processed form.

Sources: IFR (see above); [E] Serra Verde, "Serra Verde Secures USD 565 million Financing from US DFC", 5 Feb 2026: https://svpm.com.br/en/financingdfc/ ; [E] USA Rare Earth, "USA Rare Earth Completes Combination with Serra Verde Group", 4 Sep 2026: https://investors.usare.com/news-releases/news-release-details/usa-rare-earth-completes-combination-serra-verde-group ; [E] Agencia Brasil, "US company acquires Brazilian rare earth miner", 23 Apr 2026: https://agenciabrasil.ebc.com.br/en/economia/noticia/2026-04/us-company-acquires-brazilian-rare-earth-miner ; [E] Agencia Brasil, "Brazil creates policy to boost production of critical minerals", 17 Sep 2026: https://agenciabrasil.ebc.com.br/en/economia/noticia/2026-09/brazil-creates-policy-boost-production-critical-minerals ; [E] Mining.com, "Brazil targets bigger role in critical minerals rush", Sep 2026: https://www.mining.com/brazil-targets-bigger-role-in-critical-minerals-rush .

18. Mexico

Indicator	Value
GDP 2025	about USD 1,835 billion (IMF, Oct 2026)
Population 2025	about 133.4 million (IMF)
GDP per capita 2025	about USD 13,760 (IMF)
Top exports	cars and car parts, computers and electronics, electrical machinery, crude oil, medical devices, agricultural products; Mexico supplied 15.6% of US goods imports in 2025, about USD 534 billion (Month 7)
Main minerals	silver (the world's largest producer), copper, zinc, gold, fluorspar; lithium (state-controlled since 2022, little production)
Robots	fewer than 5,200 installations in 2025, down 7%, a third year of decline; density 62 (2024)
Electricity mix 2025	gas 62%, oil 9%, hydro 8%, solar 7%, wind 6%, coal 3%, nuclear 3%

Role in the chain. Mexico is North America's factory. Under the United States-Mexico-Canada Agreement (USMCA) it assembles cars, trucks, televisions, computers and, increasingly, AI servers for the US market. US-Mexico goods trade was about USD 872.8 billion in 2025 (Month 7), and record foreign direct investment of USD 40.87 billion arrived the same year. Mexico mainly does assembly and parts manufacturing; the chips, many components and the design come from the United States, Asia and Europe.

Strengths and bottlenecks. Strengths: a land border with the world's richest market, a trade agreement that gives tariff-free access (88% of US imports from Mexico were USMCA-compliant in January-May 2026), a large and skilled manufacturing workforce in the north and centre, and cheap US natural gas by pipeline. Bottlenecks:

- electricity: state control of the power sector has slowed new private generation and grid investment, which limits new factories and data centres;
- water shortages in the industrial north;
- insecurity, organised crime and weak rule of law in some states;
- dependence on US trade policy, which made the USMCA review a constant risk.

2025-2026 developments. The USMCA was not renewed at its 1 July 2026 review, so it remains in force with annual reviews (Month 7). Mexico aligned more closely with US trade policy: from 1 January 2026 it raised tariffs, by up to 50% on cars, on 1,463 tariff lines from countries without a trade agreement with it, including China, South Korea, India, Thailand and Indonesia.

2040 outlook. Judgment: Mexico will be the main nearshoring base for the United States in 2040, and a large assembler of robots, drones and AI hardware for the North American market, because geography and the trade agreement are hard to replace. Its value capture will rise only if it invests in power, water, security and engineering education.

Sources: IFR (see above); Month 7 (CRS; US Census via CargoFax; Secretaria de Economia via Rio Times; USTR); [E] Reuters, "Mexico to hike tariffs on China starting Thursday", 30 Dec 2025: https://www.reuters.com/business/autos-transportation/mexico-hike-tariffs-china-starting-thursday-2025-12-30/ ; [E] El Pais English, "Mexico completes its trade shift…", 1 Jan 2026: https://english.elpais.com/economy-and-business/2026-01-01/mexico-completes-its-trade-shift-with-the-entry-into-force-of-tariffs-on-china-and-countries-without-trade-agreements.html .

19. Morocco

Indicator	Value
GDP 2025	about USD 184 billion (IMF, Oct 2026)
Population 2025	about 37.7 million (IMF)
GDP per capita 2025	about USD 4,880 (IMF)
Top exports	cars and car parts (automotive exports 159.2 billion dirhams, about USD 17 billion, in 2025), phosphates and fertilisers (OCP revenue 113.9 billion dirhams, about USD 12.3 billion, in 2025), aircraft parts, textiles, food

Indicator	Value
Main minerals	phosphate rock (50 billion of the world's 73 billion tonnes of reserves, 68%, USGS MCS 2026), cobalt (small, from Bou Azzer), silver, barite
Robots	Africa as a whole installed only 519 industrial robots in 2025 (IFR); no separate figure for Morocco
Electricity mix 2025	coal 62%, wind 16%, gas 11%, solar 6%, oil 4%, hydro 2%

Role in the chain. Morocco is Africa's clearest success in moving from raw materials into manufacturing. Its state champion OCP turns phosphate rock into fertilisers and phosphoric acid at home rather than exporting rock (Month 6). It built a car industry from almost nothing around Renault and Stellantis plants and the Tanger Med port, which handled 11.1 million containers (TEU) in 2025 (Month 7). Now it is linking the two: phosphate is a key input to lithium iron phosphate (LFP) batteries, and Chinese battery firms are building there.

Strengths and bottlenecks. Strengths: proximity to Europe, free-trade agreements with the European Union and the United States, political stability, a world-class port, phosphate and strong wind and sun. Bottlenecks:

- a coal-heavy grid today, which matters for exports to a Europe that taxes carbon at the border;
- water stress and drought;
- many factories are foreign-owned, and local suppliers and research are still thin;
- youth unemployment and uneven education.

2025-2026 developments. Gotion High-Tech began commissioning production equipment at its LFP battery gigafactory in Kenitra in the week of 20 October 2026, described as the first of its kind in Africa and the Middle East; the first phase cost about 13 billion dirhams (about USD 1.3 billion) and is reported as 20 GWh by the company and 10 GWh by the African Development Bank, which approved a 100 million euro loan in July 2026. In September 2026 Volkswagen's PowerCo and Gotion agreed a Kenitra joint venture to make 100,000 tonnes a year of LFP cathode material (about 480 million euros) mainly for cell plants in Spain and Slovakia.

2040 outlook. Judgment: Morocco is the African country most likely to host a complete battery chain from mineral (phosphate, some cobalt and manganese) to cell by 2040, serving Europe. It could add electric motors and robot assembly for European carmakers if it keeps its stability, cleans its grid with wind and solar, and builds local engineering firms. It will be an industrial partner of Europe and China rather than a fully independent chain.

Sources: USGS MCS 2026 via Month 6; Month 6 (OCP 2026) and Month 7 (Office des Changes; Tanger Med); IFR via Month 9 for Africa's 519 units ([V] https://ifr.org/post/exciting-times-for-robot-suppliers); [V] Morocco World News, "Gotion to Launch Morocco's First Battery Gigafactory in Kenitra the Week of October 20", 12 Oct 2026: https://www.moroccoworldnews.com/2026/10/341754/gotion-to-launch-moroccos-first-battery-gigafactory-in-kenitra-the-week-of-october-20/ ; [E] AfDB, "African Development Bank Approves EUR 100 Million to Gotion Power Morocco", 24 Jul 2026: https://www.afdb.org/en/news-and-events/african-development-bank-approves-eu100-million-gotion-power-morocco-develop-africas-first-lithium-ion-phosphate-battery-gigafactory-95727 ; [E]

Volkswagen Group, "Volkswagen Group, PowerCo and Gotion deepen strategic partnership", 28 Sep 2026: https://www.volkswagen-group.com/en/press-releases/volkswagen-group-powerco-and-gotion-deepen-strategic-partnership-20710 .

20. Democratic Republic of the Congo

Indicator	Value
GDP 2025	about USD 93 billion (IMF, Oct 2026)
Population 2025	about 106.6 million (IMF)
GDP per capita 2025	about USD 870 (IMF)
Top exports	refined copper and copper concentrate, cobalt hydroxide, gold, diamonds, crude oil, tin and coltan (tantalum ore)
Main minerals	cobalt (about 230,000 tonnes, 73% of world mining, 2025; 6.0 of 12 million tonnes of world reserves), copper (about 14% of world mining), tin, tantalum, gold, lithium (Manono, undeveloped) (USGS MCS 2026)
Robots	negligible; Africa as a whole installed 519 in 2025
Electricity mix 2024	hydro 84%, solar 16% of reported grid generation; most people have no grid electricity, and mines often rely on diesel or imported power

Role in the chain. The DRC is the single most important source of cobalt and the world's second-largest copper miner after Chile. Its Copperbelt in Katanga supplies the metals inside batteries, motors and cables. Most mines are owned by Chinese firms (CMOC is the largest cobalt producer) or Glencore, and most processing beyond copper cathode and cobalt hydroxide happens in China, so the DRC captures royalties and taxes but little industrial value.

Strengths and bottlenecks. Strength: geology of world importance, which gives bargaining power. Bottlenecks:

- conflict in the east, where the Rwanda-backed M23 group seized mineral-rich areas, including the Rubaya coltan zone, and the city of Uvira after the December 2025 peace accord;
- weak state capacity, corruption and the human cost of artisanal mining, including child labour (Month 6);
- shortages of electricity and transport: exports travel thousands of kilometres by road and rail to ports in Tanzania, South Africa and, increasingly, Angola via the Lobito Corridor;
- extreme poverty, which limits the domestic market and skills.

2025-2026 developments. The DRC used its market power like a cartel of one. It banned cobalt exports in February 2025, replaced the ban with quotas in October 2025 (96,600 tonnes a year for 2026 and 2027, less than half of 2024 output, S&P Global via CES Intelligence), and in April 2026 created a strategic minerals reserve. Cobalt prices rose sharply (Month 6). On 4 December 2025 it signed a Strategic Partnership Agreement with the United States, giving US-aligned firms preferred access to some projects and setting five-year targets to export 50% of copper and 30% of cobalt through the Lobito Corridor, on the same day as the DRC-Rwanda peace accords in Washington.

US-linked investors followed, including Virtus Minerals' purchase of Chemaf for about USD 700 million. Fighting continued, and the United States sanctioned Rwanda's defence ministry in October 2026.

2040 outlook. Judgment: the DRC will still dominate cobalt and be one of the top two copper producers in 2040, and it will probably capture more rent through quotas and partnerships. Building refineries and battery plants at home is possible only with reliable power (for example new hydro on the Congo River) and peace in the east; without them, the processing will happen in Zambia, Angola, Morocco, the Gulf or China. The DRC shows that market power over a mineral is not the same as industrial development.

Sources: [V] USGS MCS 2026, Cobalt (via Month 9): https://pubs.usgs.gov/periodicals/mcs2026/mcs2026-cobalt.pdf ; USGS copper share via Month 5; [E] IEA policy database, "DRC ARECOMS Decision No. 004/2025, Cobalt Quota System": https://www.iea.org/policies/29138-drc-arecoms-decision-no-0042025-cobalt-quota-system ; [E] IEA policy database, "DRC-US Strategic Partnership Agreement": https://www.iea.org/policies/29541-drc-us-strategic-partnership-agreement ; [E] CES Intelligence, DR Congo 2026 risk assessment, Jul 2026: https://www.ces-intelligence.com/analysis/dr-congo-2026-geopolitical-risk-assessment ; [E] Responsible Statecraft, Orion and Virtus deals, May 2026: https://responsiblestatecraft.org/trump-congo-drc/ ; [E] US State Department, sanctions on Rwandan violators of the Washington Accords, 2 Mar 2026: https://www.state.gov/releases/office-of-the-spokesperson/2026/03/sanctioning-rwandan-violators-of-the-washington-accords-for-peace-and-prosperity ; [E] BBC, "US sanctions Rwanda's defence ministry over Congo war", 29 Oct 2026: https://www.bbc.com/news/articles/cm87zw51qw0jo .

21. Zambia

Indicator	Value
GDP 2025	about USD 29 billion (IMF, Oct 2026)
Population 2025	about 21.9 million (IMF)
GDP per capita 2025	about USD 1,310 (IMF)
Top exports	copper (about 70% of export earnings, and more than 10% of GDP), cobalt, gold, emeralds, sugar and maize
Main minerals	copper (890,346 tonnes in 2025 by the mines ministry, a record, or about 940,000 tonnes by the USGS estimate; about 4% of world supply), cobalt, manganese, gemstones
Robots	negligible; Africa as a whole installed 519 in 2025
Electricity mix 2024	hydro 86%, coal 11%, solar and other about 3% (2024 was a severe drought year)

Role in the chain. Zambia is the southern half of the Central African Copperbelt. It mines and smelts copper and sells it as cathode or concentrate, mainly to China and to traders in Switzerland and Singapore. Investors include First Quantum, Barrick, Vedanta and Chinese firms. It is landlocked, so railways to Angola (Lobito) and Tanzania (TAZARA) decide how cheaply it can export (Month 6).

Strengths and bottlenecks. Strengths: large copper reserves (about 6% of known world reserves, US State Department), relative political stability and peaceful transfers of power, and a government committed to investor-friendly reform. Bottlenecks:

- electricity: almost all power is hydro, so droughts like 2024 cause blackouts that hit mines; industry estimates Zambia needs at least 2,000 MW of new capacity for its copper plans;
- debt: Zambia defaulted on its Eurobonds in 2020 and only completed most of its restructuring (about USD 12.4 billion of external debt) by 2025-2026;
- landlocked logistics and high transport costs;
- little manufacturing beyond copper smelting and refining.

2025-2026 developments. Copper output reached a record in 2025, but first-half 2026 output was nearly flat at 447,182 tonnes, far below the path to the government's target of 3 million tonnes a year by 2031. Zambia completed its IMF programme in January 2026 (about USD 1.7 billion disbursed) and bought back most of its defaulted Eurobonds in June 2026. President Hakainde Hichilema won a second term in August 2026 with about 60% of the vote and plans a new, growth-focused IMF programme. The US and African Development Bank financed the Zambia side of the Lobito rail link (USD 255 million AfDB loan approved in August 2026, Month 6).

2040 outlook. Judgment: Zambia will probably produce 1.5 to 2 million tonnes of copper a year by the 2030s, not 3 million by 2031, because power, rail and new mines take longer than plans assume. It can capture more value by refining all its copper at home and by making copper products (wire, cable, rods) for the region, using new solar and regional power trade.

Sources: USGS via Month 6; [E] bne IntelliNews, "Hichilema wins second term as Zambia bets on copper-led growth", 18 Aug 2026: https://www.intellinews.com/hichilema-wins-second-term-as-zambia-bets-on-copper-led-growth-461723/ ; [E] US State Department, 2026 Investment Climate Statement: Zambia: https://www.state.gov/reports/2026-investment-climate-statements/zambia ; [E] Zambian Economist, "Zambia Economic Outlook 2026: After the Vote, the Hard Part", 21 Aug 2026: https://zambianeconomist.com/zambia-economic-outlook-after-2026-election/ ; [E] News Diggers, "Copper output rises to 447,181 in H1 2026", 1 Aug 2026: https://diggers.news/business/2026/08/01/copper-output-rises-to-447181-in-h1-2026-govt/ ; [E] IMF, sixth review of the Extended Credit Facility with Zambia, 27 Jan 2026: https://www.imf.org/en/news/articles/2026/01/27/pr-26024-zambia-imf-executive-board-completes-sixth-review-under-the-extended-credit-facility ; [E] Al Jazeera, copper share of exports, 12 Aug 2026: https://www.aljazeera.com/news/2026/8/12/zambia-to-vote-as-economic-hardship-tests-government-reforms .

22. South Africa

Indicator	Value
GDP 2025	about USD 427 billion (IMF, Oct 2026)
Population 2025	about 63.1 million (IMF)
GDP per capita 2025	about USD 6,770 (IMF)
Top exports	platinum-group metals (PGMs), gold, coal, iron ore, manganese and chrome ores, cars (BMW, Mercedes-Benz, Toyota, Volkswagen plants), citrus and wine

Indicator	Value
Main minerals	platinum (about 71% of world mining), palladium (about 37%), manganese (about 38%), chrome, gold, coal, iron ore, vanadium (USGS MCS 2026 via Month 5)
Robots	Africa as a whole installed only 519 in 2025 (IFR); no separate public figure for South Africa
Electricity mix 2025	coal 81%, solar 8%, wind 5%, nuclear 4%, oil 1%

Role in the chain. South Africa is the continent's most industrialised economy and the world's main source of platinum, manganese and chrome. Platinum and palladium are used in catalytic converters, hydrogen fuel cells and electrolysers; manganese goes into steel and some battery chemistries. It has deep mining finance and engineering firms, the Johannesburg Stock Exchange, a car export industry and universities that train mining and electrical engineers for all of Africa. Yet it captures relatively little value downstream, because high power costs make smelting and refining expensive (BloombergNEF).

Strengths and bottlenecks. Strengths: minerals, financial markets, institutions such as independent courts and a free press, and engineering skills. Bottlenecks:

- electricity: years of rolling blackouts ("load shedding") until May 2025, and tariffs for energy-intensive users that rose more than 900% since 2008 (Minerals Council via ESI-Africa);
- freight rail and ports run by the state company Transnet, whose decline cut mineral exports, now being opened to private operators;
- very high unemployment and inequality, crime and corruption;
- ageing deep mines: PGM output fell 4.1% in 2025 and remains below pre-pandemic levels.

2025-2026 developments. The power system recovered. By 30 October 2026 South Africa had gone 532 consecutive days without load shedding since 16 May 2025, Eskom's energy availability factor was the highest in six years at 68.15%, and Eskom reported about 7,235 MW of available capacity above current demand. South Africa was removed from the Financial Action Task Force "grey list" and received a credit rating upgrade from S&P in late 2025. But high electricity prices kept smelters under pressure, and some PGM operations, such as Bokoni, suspended mining in 2026.

2040 outlook. Judgment: South Africa will remain Africa's main mining finance and engineering hub and the world's main PGM supplier in 2040. If cheaper renewable power and private rail arrive, it could become the processing centre for southern Africa: refining copper and cobalt from Zambia and the DRC, making manganese and vanadium battery materials, and assembling cars, EVs and eventually robots for the African market.

Sources: USGS MCS 2026 via Month 5; [V] Eskom, "Generation Recovery Plan sustains Eskom reaching a 68.15% EAF…", 30 Oct 2026: https://www.eskom.co.za/generation-recovery-plan-sustains-eskom-reaching-a-68-15-eaf-and-diesel-cost-reduction-of-r4-81-billion-year-on-year/ ; [E] ESI-Africa, "Mining reforms progress, but bottlenecks persist", 10 Feb 2026: https://www.esi-africa.com/business-and-markets/esi-africa-at-mining-indaba-reforms-progress-but-bottlenecks-persist/ ; [E] BloombergNEF, South Africa Transition Factbook 2026: https://assets.bbhub.io/professional/sites/44/South-Africa-Transition-Factbook.pdf ; [E] BusinessDay, PGMs and Bokoni, 19 Aug 2026: https://www.businessday.co.za/opinion/2026-08-19-ashley-nyiko-mabasa-pgms-could-be-the-basis-of-a-new-industrial-strategy/ .

Comparison of the 22 economies

The table below puts the atlas on one page. GDP and GDP per capita are 2025 IMF estimates (October 2026 WEO). "Main link" is the step of the ore-to-robot-to-AI chain where each economy is strongest today. The last column is the author's judgment, summarising the outlooks above.

Economy	GDP 2025 (USD bn)	GDP per capita (USD)	Main link in the chain today	Likely 2040 role (Judgment)
United States	30,861	90,270	AI models, chip design, cloud, capital	Still captures most AI value; rebuilds some mid-chain
China	19,626	13,970	Refining, components, assembly, robots	Cheapest robot maker; refining share falls but stays dominant
Japan	4,453	36,120	Precision parts, robot arms, chip tools	Premium component and equipment supplier
South Korea	1,882	36,410	Memory chips, batteries, robot use	Memory leader; possible humanoid maker
Taiwan	921	39,510	Leading-edge chips, AI servers	Still the leading-edge centre, smaller share
Germany	5,116	61,260	Machinery, automation, factory software	Premium automation and integration
Netherlands	1,322	73,260	EUV lithography (ASML), logistics	Keeps lithography monopoly
India	3,909	2,670	Software, electronics assembly	Second assembly centre after China
Indonesia	1,446	5,080	Nickel processing	Nickel and battery-material hub
Vietnam	494	4,830	Electronics assembly	Large electronics and drone assembler
Thailand	577	8,060	Cars, electronics, data centres	Regional EV and robot assembly hub
Malaysia	473	13,960	Chip packaging, rare earth separation	Possible non-China magnet link

Economy	GDP 2025 (USD bn)	GDP per capita (USD)	Main link in the chain today	Likely 2040 role (Judgment)
Philippines	487	4,280	Nickel ore, chip testing, services	Ore supplier to a regional chain
Saudi Arabia	1,274	35,370	Oil, capital, aluminium	Energy-intensive exports, AI compute
UAE	614	54,150	Capital, logistics, AI compute	Gulf AI hub and global investor
Chile	357	17,840	Copper and lithium mining	Top copper producer, more refining
Brazil	2,280	10,680	Iron ore, niobium, rare earths	Possible rare earth separation and magnets
Mexico	1,835	13,760	Assembly for the US market	North American robot and hardware assembler
Morocco	184	4,880	Phosphate, cars, batteries	Africa's most complete battery chain
DRC	93	870	Cobalt and copper mining	Dominant cobalt supplier; processing elsewhere
Zambia	29	1,310	Copper mining and smelting	Larger copper refiner for the region
South Africa	427	6,770	PGMs, manganese, mining finance	Southern Africa's processing and finance hub

Three patterns stand out. First, value per person follows position in the chain, not mineral wealth: the economies that make chips, tools and software (the United States, the Netherlands, Germany, Taiwan) are far richer per person than the economies that supply ore (the DRC, Zambia, the Philippines). Second, the fastest climbers (China, Korea, Taiwan, and now Vietnam, India and Morocco) rose by building manufacturing, not by owning minerals. Third, no single developing country in Southeast Asia or Africa has every link. A cheap robot supply chain from ore to finished robot by the 2040s is most plausible as a regional network: in Southeast Asia, Indonesian and Philippine nickel, Malaysian rare earth separation and chip packaging, and Thai and Vietnamese assembly; in Africa, Congolese and Zambian copper and cobalt, Moroccan phosphate and battery plants, and South African finance and engineering. Judgment: Southeast Asia is closer to this goal than Africa, because it already has manufacturing clusters and a robot-using industry, while Africa installed only 519 industrial robots in 2025. Month 12 returns to these scenarios.

Appendix E: Master Glossary and Reading List

This appendix gathers the reference material scattered across the twelve Months into one place. Part 1 is a single glossary that merges every term defined in the Months' glossary sections, alphabetized and deduplicated. Where several Months defined the same idea, the definitions are combined into one; the label in brackets gives the Month that first introduces the term, plus any later Months that define it again. Abbreviations are written out in full, with the short form in brackets. Part 2 lists, Month by Month, the books, reports and articles each chapter recommends, with titles and authors only. A work recommended by several Months is listed in full once and noted briefly in later Months. Part 3 names ten data sources worth returning to long after you finish the course.

Use the glossary in two ways. While reading, look up a term the moment it stops you. After each Month, cover the definitions and try to explain every term introduced that Month in your own words; any term you cannot explain clearly is a sign to reread that part of the chapter.

Glossary

Entries are in alphabetical order. Each gives the term, the Month where it first appears in brackets, and a definition.

A

- **Accelerator (artificial intelligence accelerator)** (Month 8): A chip specialised for the calculations used to train and run artificial intelligence (AI) models, such as a graphics processing unit or Google's tensor processing unit.
- **Actuator** (Month 9): The complete joint module that makes a robot move: an electric motor, a transmission (reducer or screw), a motor driver and position and force sensors packaged together.
- **Advanced packaging** (Month 8): Techniques that place several chips, for example a processor and stacks of memory, very close together in one package with dense connections between them. Taiwan Semiconductor Manufacturing Company's Chip on Wafer on Substrate is the leading example, and packaging capacity became a bottleneck of the AI boom.
- **Advantages of backwardness** (Month 3): Alexander Gerschenkron's idea that a late-developing country can copy the newest technology instead of inventing it, but needs strong institutions, such as large banks or the state, to gather the large amounts of capital that modern industry requires.
- **African Continental Free Trade Area (AfCFTA)** (Month 12): An agreement among African countries to create a single continental market by cutting tariffs and other barriers on

trade between them.

- **African Growth and Opportunity Act (AGOA)** (Month 7): A United States law giving eligible sub-Saharan African countries duty-free access to the US market for many products, especially clothing.
- **Agglomeration economies** (Month 7): The cost advantages firms gain from locating near many similar firms: a shared pool of skilled workers, nearby suppliers, faster repair and the spread of know-how.
- **Alignment and hedging** (Month 12): Two strategies for a middle power facing a great-power rivalry. Alignment means choosing one side, such as the United States or China; hedging means keeping good relations with both, avoiding hard choices and extracting benefits from each.
- **Anode and cathode** (Month 5): The negative and positive electrodes of a battery. In a lithium-ion battery the anode is usually graphite, and the cathode is a metal oxide or phosphate whose chemistry (for example lithium iron phosphate) decides which metals the battery needs.
- **Assembly, testing and packaging (ATP), also outsourced semiconductor assembly and test (OSAT)** (Month 7): The back end of chipmaking, where finished silicon wafers are cut into individual chips, packaged and tested.
- **Augmentation** (Month 10): Using a technology to make workers more productive at a task rather than to replace them, for example an AI tool that helps a call-centre agent answer faster.
- **Automation** (Month 10): Using a technology, such as a robot or software, to perform a task that a person used to do.
- **Autonomous mobile robot (AMR)** (Month 9): A robot that moves around a factory or warehouse on its own, using sensors and maps to navigate, for example to carry parts or shelves.

B

- **Backward and forward linkages** (Month 3; also Month 6): The connections an industry creates in the local economy. Backward linkages are the demand it creates for local inputs (a smelter buying power, lime and services); forward linkages are the supplies it provides to later industries (refined metal for battery makers).
- **Backward and forward participation** (Month 1): Two measures of how deeply a country is involved in global value chains. Backward participation is the foreign value added contained in its exports; forward participation is its own value added that ends up in other countries' exports.
- **Backwardation** (Month 4; also Month 5): A state of a futures market in which near-term prices are higher than prices for later delivery.
- **Barrel and million barrels per day** (Month 4): The standard units of the oil market. A barrel (bbl) is 159 litres of oil; million barrels per day (mb/d) measures flows of production, consumption or trade.
- **Basis point** (Month 11): One-hundredth of a percentage point (0.01 percentage points). A rise in a bond yield from 6.00 percent to 6.25 percent is a rise of 25 basis points.
- **Baumol's cost disease** (Month 10): The tendency for the prices of services with slow productivity growth, such as teaching, nursing or live music, to rise relative to goods whose productivity grows fast. It means sectors AI and robots cannot speed up take a growing share of spending and limit economy-wide growth.
- **Beneficiation and downstreaming (hilirisasi)** (Month 6): Processing raw minerals inside the producing country before export, for example smelting nickel ore into nickel pig iron

or battery intermediates. Hilirisasi is the Indonesian word for this policy, made famous by Indonesia's nickel ore export ban.

- **Bill of materials (BOM)** (Month 9): The list of a product's parts and their costs, excluding assembly labour, overhead and profit.
- **Blended finance** (Month 11): Using public or philanthropic money, for example a guarantee or a first-loss tranche, to reduce the risk of a project so that private investors will fund the rest.
- **Brent, West Texas Intermediate, Dubai and Murban** (Month 4): Benchmark crude oils whose prices are used to price most of the world's oil. Brent comes from the North Sea, West Texas Intermediate (WTI) from the United States, and Dubai and Murban from the Gulf.
- **BRICS** (Month 12): A group of large emerging economies named after its first members, Brazil, Russia, India, China and South Africa. It expanded in 2024 and 2025 and counted 11 members by 2026, including Indonesia; it coordinates on trade, finance and diplomacy but is not a unified bloc.
- **By-product** (Month 5): A metal recovered as a secondary output of mining or refining another metal, such as cobalt from copper mines in the Congo or gallium from aluminium refining.

C

- **Capacity factor** (Month 4): The actual output of a power plant over a period divided by the maximum it could have produced running at full power the whole time.
- **Capital outflow and capital inflow** (Month 11): Money leaving a country (outflow) or entering it (inflow) to buy assets such as bonds, shares, property or companies. Sudden outflows can weaken a currency and push up interest rates.
- **Capital recovery factor (CRF)** (Month 4): A formula that converts an up-front investment into an equal annual payment over a project's life at a given interest rate.
- **Captive power** (Month 6): A power plant built for and owned by an industrial site rather than connected to the public grid.
- **Carried interest (carry)** (Month 11): The share of a venture or private equity fund's profits, often 20 percent, kept by the fund managers (the general partners) as their reward for performance, on top of an annual management fee.
- **Carry trade** (Month 11): Borrowing in a low-interest-rate currency to invest in a higher-interest-rate currency and pocket the difference.
- **Chaebol** (Month 3): Large family-controlled South Korean business groups, such as Samsung, Hyundai and LG, that grew with state credit and export targets during Korea's industrialization and still dominate its economy.
- **China plus one** (Month 7): A strategy of keeping production in China while adding at least one other country, such as Vietnam, India or Mexico, to reduce the risk of tariffs, lockdowns or political shocks.
- **Chip on Wafer on Substrate (CoWoS)** (Month 8): Taiwan Semiconductor Manufacturing Company's advanced packaging technology, which places a processor and high-bandwidth memory side by side on a silicon layer. It is used for most leading AI accelerators, and its limited capacity rationed AI chip supply.
- **Chokepoint** (Month 7; also Month 12): A node in a network that others cannot easily avoid: a narrow strait or canal, a single critical input, a unique machine or a payment system. Whoever controls a chokepoint gains leverage over everything that passes through it, though not always profit.
- **Class 1 and Class 2 nickel** (Month 5): Class 1 nickel is high-purity nickel (at least 99.8 percent) suitable for batteries and for delivery on the London Metal Exchange; Class 2 nickel is lower-purity product, such as nickel pig iron and ferronickel, used mainly for stainless steel.

- **Clearing house and margin** (Month 5): A clearing house is the body that stands between buyers and sellers on an exchange and guarantees their trades; margin is the collateral it demands from traders to cover possible losses.
- **Cobot (collaborative robot)** (Month 9): A robot arm designed to work safely next to people without a protective fence, usually lighter, slower and easier to program than a traditional industrial robot.
- **Codified knowledge** (Month 10): Formal knowledge written down in documents, manuals and textbooks. Because it exists as text, it is relatively easy for AI to learn and reproduce, unlike tacit knowledge.
- **Commodity dependence** (Month 2): The condition of a country whose merchandise exports are more than 60 percent commodities (raw materials, fuels and farm products), as defined by the United Nations Conference on Trade and Development (UNCTAD).
- **Common Framework** (Month 11): A process launched by the Group of Twenty (G20) major economies in 2020 to coordinate debt restructurings for poor countries among all creditors, including China and private lenders.
- **Comparative advantage** (Month 1): The ability to produce a good at a lower opportunity cost than another producer, meaning giving up less of other goods to make it. David Ricardo showed that countries gain from trade by specialising in their comparative advantage, even if one is better at everything.
- **Concentrate** (Month 5): An upgraded product made at or near a mine by crushing ore and separating out much of the waste rock.
- **Contango** (Month 5): A state of a futures market in which prices for later delivery are higher than the current price, usually signalling a surplus. The opposite of backwardation.
- **Contract for difference and price floor** (Month 5): Arrangements in which a government or buyer tops up a producer's revenue when the market price falls below an agreed level (and, in a contract for difference, receives money back when it rises above it).
- **Corridor (development or trade corridor)** (Month 6): A coordinated transport route, such as a railway, road and port, that links a mining or industrial region to markets. The Lobito Corridor, running from the Congo and Zambia to Angola's Atlantic coast, is the main example in the course.
- **Cost curve** (Month 5; also Month 9): A chart that ranks producers from lowest to highest cost per unit. In commodity markets the cost of the marginal (highest-cost) producer still needed tends to set the long-run price; in Month 9 the term also describes how a product's cost falls with volume.
- **Cost of capital** (Month 4; also Month 11): The return investors require to fund a project, which is the price of money for that project. For capital-intensive projects such as solar farms, smelters and chip fabs it largely determines the final cost, so countries with high risk premiums are at a disadvantage.
- **Creative destruction** (Month 2): The process, named by Joseph Schumpeter, by which new firms and technologies replace old ones. Inclusive institutions allow it; elites who would lose from it often try to block it.
- **Credible commitment** (Month 2): A promise that others can believe because the person or government making it is constrained from breaking it, for example by courts, a parliament or a treaty.
- **Cultivation System (Cultuurstelsel)** (Month 2): The Dutch colonial system in Java from the 1830s to the 1870s that forced villagers to grow export crops such as coffee and sugar for the colonial government.
- **Currency mismatch** (Month 1): Having debts in one currency, often US dollars, while

earning income in another, often the local currency. If the local currency falls, the debt becomes heavier, which was a central cause of the 1997-1998 Asian financial crisis.

- **Cycloidal reducer (rotate vector or RV reducer)** (Month 9): A stiff, strong gearbox that uses lobed discs rolling inside a ring of pins to reduce speed and raise torque. It is used in the heavy-load joints of industrial robots, and Japan's Nabtesco long dominated its supply.

D

- **Degrees of freedom** (Month 9): The number of independent ways a robot, arm or limb can move.
- **Developmental state** (Month 3): A state with a capable, relatively insulated bureaucracy that actively steers private firms toward national industrial goals using credit, licences, trade protection and performance targets, as in Japan, South Korea and Taiwan.
- **Dexterous hand** (Month 9): A robot hand with many independently controlled fingers, often with touch sensing, able to grasp and manipulate varied objects.
- **Dilution** (Month 11): The fall in an existing owner's percentage of a company when it issues new shares, for example to venture investors.
- **Directed credit and financial repression** (Month 3): Government control of bank lending and interest rates to channel savings into chosen sectors, often by paying savers below-market interest.
- **Displacement effect** (Month 10): The fall in demand for labour when machines take over tasks that workers used to do.
- **Domestic processing requirement** (Month 6): A rule that minerals must be processed inside the country before they can be exported. A softer form of downstreaming policy than an outright ore export ban.
- **Dutch disease** (Month 2; also Month 4): The decline of manufacturing and other tradable sectors when a resource boom raises the value of a country's currency and pushes up local wages and costs.
- **Dynamic comparative advantage** (Month 1): Comparative advantage that changes over time as a country builds skills, capital, infrastructure and capabilities. South Korea moving from wigs to ships to chips is the classic example.
- **Dynamic random-access memory (DRAM)** (Month 8): The fast working memory used in computers, phones and servers, made mainly by Samsung, SK hynix and Micron.

E

- **Economic statecraft** (Month 12): Using economic tools, such as tariffs, export controls, sanctions and investment screening, to pursue strategic or security goals rather than purely economic ones.
- **Effective tariff rate** (Month 1; also Month 7): The total customs duties collected (or estimated) divided by the total value of imports. It measures the real weight of tariffs better than headline rates, because exemptions and shifts in trade change what is actually paid.
- **Electronic design automation (EDA)** (Month 8): The software used to design chips, dominated by a few mainly American firms such as Synopsys and Cadence. A quiet chokepoint, because no advanced chip can be designed without it.
- **Electrowinning** (Month 6): Recovering metal from a solution by passing an electric current through it so that pure metal plates onto a cathode.
- **Enclave** (Month 6): An industrial zone or project with few links to the rest of the domestic economy: it imports its inputs, exports its output and employs relatively few local suppliers.
- **Encoder** (Month 9): A sensor that measures the exact angle or position of a robot joint or

motor shaft, so the controller knows where each part of the robot is.

- **Energy intensity** (Month 4): The amount of energy used to produce one unit of gross domestic product. It falls as economies become more efficient or shift from heavy industry toward services.
- **Engels' pause** (Month 10): The period in early industrial Britain, roughly 1790 to 1840, when output per worker rose but real wages stagnated, so the gains went mainly to owners of capital.
- **Entity List** (Month 12): A United States government list of foreign companies and organisations that may not receive certain US goods, software and technologies without a licence.
- **Exajoule (EJ)** (Month 4): A unit of energy equal to 10 to the power 18 joules, used for national and global energy totals. The world uses roughly 600 exajoules of primary energy a year.
- **Exorbitant privilege** (Month 11): The advantages the United States gains from issuing the world's main reserve currency, such as borrowing cheaply in its own currency and the power to impose financial sanctions.
- **Export controls** (Month 5; also Months 7, 8 and 12): Government licensing requirements or bans on exporting certain goods, technologies or materials, usually for security reasons. Examples include US limits on advanced chips and chipmaking tools sold to China, and China's 2025 controls on rare earths and magnets.
- **Export discipline** (Month 3): Tying government support, such as cheap credit, subsidies or protection, to firms' success in export markets, and cutting support to firms that fail.
- **Export processing zone (EPZ) and special economic zone (SEZ)** (Month 3): Designated areas where firms can import inputs duty-free and face simpler rules and taxes, usually on condition that they export.
- **Export quota** (Month 6): A limit on the quantity of a good that may be exported in a period.
- **Exposure (to artificial intelligence)** (Month 10): The share of a job's tasks that AI could perform or speed up significantly.
- **Extractive economic institutions** (Month 2): Rules designed to transfer income and wealth from the many to a few, for example forced labour, insecure property rights for ordinary people, or monopolies granted to an elite.
- **Extractive Industries Transparency Initiative (EITI)** (Month 2): A global standard under which member countries and companies publish payments, revenues and, increasingly, contracts in oil, gas and mining, so citizens can see where resource money goes.
- **Extractive political institutions** (Month 2): Political systems that concentrate power in a narrow elite with few checks on how it is used. They usually support extractive economic institutions that benefit that elite.
- **Extreme ultraviolet (EUV) lithography** (Month 8): Chip-printing technology that uses light with a wavelength of 13.5 nanometres to draw the finest circuit patterns. The machines are made only by the Dutch company ASML, making them a single point of failure for leading-edge chips.

F

- **Fab (fabrication plant)** (Month 8): A factory that processes silicon wafers into chips. A leading-edge fab costs 20 billion US dollars or more, which is why only a handful of companies can build them.
- **Fabless** (Month 8): A chip company that designs chips but outsources their manufacturing to a foundry. Nvidia, Apple's chip unit, AMD and Qualcomm are fabless.

- **Fiscal breakeven oil price** (Month 4): The oil price at which a petrostate's government budget balances. When the market price falls below it, the government must borrow, cut spending or draw down savings.
- **Fiscal-military state** (Month 2): A state organised around raising taxes and borrowing to fight wars, as in eighteenth-century Britain. Charles Tilly and others argue that war-driven taxation built the capable states of Europe.
- **Force majeure** (Month 4): A contract clause that excuses a party from its obligations when extraordinary events beyond its control, such as a war or blockade, prevent performance.
- **Foreign direct investment (FDI)** (Month 1; also Month 7): Cross-border investment that gives the investor lasting control or significant influence over a business in another country, conventionally at least 10 percent of voting power, such as building or buying a factory.
- **Foreign exchange reserves** (Month 1): Foreign assets, mostly government bonds of major currencies such as the US dollar, held by a central bank.
- **Foundry** (Month 3; also Month 8): A company that manufactures chips designed by other firms. Taiwan Semiconductor Manufacturing Company (TSMC) invented the pure-play foundry model in 1987 and is by far the largest.
- **Free float** (Month 11): The share of a listed company's stock that is available for public trading, excluding shares held by founders, governments and other long-term holders.

G

- **General-purpose technology** (Month 10): A technology that spreads across the whole economy, keeps improving and spawns complementary innovations, such as the steam engine, electricity and computers. Many economists argue AI is one.
- **Global value chain (GVC)** (Month 1): A production process for one product whose stages, from design and raw materials to assembly, marketing and service, are spread across two or more countries.
- **Grade** (Month 5): The share of a useful metal in ore, for example 0.6 percent copper.
- **Graphics processing unit (GPU)** (Month 8): A processor with thousands of small parallel cores, originally designed for graphics and now the main chip for training and running AI models. Nvidia dominates the market.
- **Gross domestic product (GDP)** (Month 1): The market value of all final goods and services produced within a country in a period, usually a year. It equals the sum of value added by all producers in the country.
- **Gross national income (GNI)** (Month 1): Gross domestic product plus income that residents earn abroad, minus income that foreigners earn in the country. For countries with large foreign-owned mines or factories, it can be much lower than GDP.
- **Gross trade versus value added** (Month 7): Gross trade counts the full value of goods each time they cross a border; trade in value added counts only the value actually created in each country.

H

- **Heckscher-Ohlin model** (Month 1): A theory that countries export goods that use intensively the factors of production (labour, capital or land) that they have in abundance, and import goods that use their scarce factors.
- **High-bandwidth memory (HBM)** (Month 8): Stacks of dynamic random-access memory chips placed next to a processor to move data at very high speed. Essential for AI accelerators, and made by only three firms, led by SK hynix.
- **High-pressure acid leaching (HPAL)** (Month 5): A process that dissolves laterite nickel

ore in hot, pressurised sulphuric acid to produce battery-grade intermediates such as mixed hydroxide precipitate. Central to Indonesia's move into battery nickel.
- **Household responsibility system** (Month 3): China's reform from 1978 that gave farm families control over production on their plots and let them sell surplus output, sharply raising farm productivity and freeing labour for industry.
- **Hulten's theorem** (Month 10): The result that the economy-wide productivity gain from a cost-saving technology roughly equals the share of economic activity it affects multiplied by the cost saving on that activity.
- **Hydrometallurgy and pyrometallurgy** (Month 6): The two families of metal processing. Hydrometallurgy uses water-based chemistry, such as leaching and electrowinning; pyrometallurgy uses heat, such as smelting in furnaces.
- **Hyperscaler** (Month 8): A very large cloud computing company, such as Amazon, Microsoft, Google, Meta or Oracle, that builds and operates huge data centres. Hyperscalers are the main buyers of AI chips.

I

- **Import substitution industrialization (ISI)** (Month 3): A strategy of building domestic industries to replace imports, protected by tariffs and quotas.
- **Impossible trinity (trilemma)** (Month 1; also Month 11): The principle that a country cannot have a fixed exchange rate, free movement of capital and an independent monetary policy all at once; it must give up one of the three.
- **Inclusive economic institutions** (Month 2): Rules that secure property rights for broad groups of people, enforce contracts fairly, provide public services and allow new firms to enter markets freely.
- **Inclusive political institutions** (Month 2): Political systems that are both sufficiently centralised to keep order and pluralistic enough to spread and constrain power among many groups.
- **Indirect rule** (Month 2): A form of colonial rule exercised through local chiefs or rulers who answered to the colonial power, used widely by Britain in Africa. It often strengthened unaccountable local authority.
- **Industrial policy** (Month 3): Government action aimed at changing the structure of the economy toward particular sectors, technologies or activities, using tools such as subsidies, credit, tariffs, public investment and procurement.
- **Industrial robot** (Month 9): Under the International Organization for Standardization standard ISO 8373, an automatically controlled, reprogrammable, multipurpose manipulator programmable in three or more axes.
- **Infant-industry argument** (Month 3): The case for temporarily protecting a new domestic industry from foreign competition until it gains experience and scale and can compete on its own.
- **Inference** (Month 8): Running a trained AI model to produce outputs, such as answering a question.
- **Institutions** (Month 2): The formal and informal rules that structure economic, political and social interaction: laws, property rights, courts, norms and customs.
- **International Emergency Economic Powers Act (IEEPA)** (Month 7; also Month 12): A 1977 United States law giving the president powers in a declared national emergency. It was used in 2025 to impose broad tariffs, but the Supreme Court ruled in February 2026 that it does not authorise tariffs.

J

- **Jevons paradox** (Month 8): The observation, first made by William Jevons about coal, that when greater efficiency lowers the cost of using a resource, total use of it can rise.

K

- **Kafala system** (Month 2): A sponsorship system in Gulf states that ties a migrant worker's legal status to a particular employer, limiting the worker's ability to change jobs or leave the country.
- **Keiretsu and zaibatsu** (Month 3): Japanese business groups. The prewar zaibatsu were family-controlled holding companies; the postwar keiretsu are groups of firms linked by cross-shareholdings, long-term supply ties and often a main bank.

L

- **Labour share** (Month 10): The share of national income paid to workers as wages, salaries and benefits, with the rest going to owners of capital as profits, interest and rent.
- **Land reform** (Month 3): Redistribution of farmland from large landlords to the families who farm it.
- **Landed cost** (Month 7): The total cost of getting a product to the buyer: factory price plus freight, insurance, duties, inventory cost and other charges.
- **Lead firm** (Month 1): The firm that governs a value chain by setting product standards, choosing suppliers and controlling the brand or design, such as Apple in smartphones.
- **Learning curve (Wright's law; learning rate)** (Month 4; also Months 7 and 9): The pattern that a product's unit cost falls by a roughly constant percentage, the learning rate, each time cumulative production doubles. It explains the collapse in solar and battery costs and is used to forecast robot costs.
- **Levelized cost of electricity (LCOE)** (Month 4): The average cost per kilowatt-hour of building, financing and operating a power plant over its whole lifetime. It allows solar, wind, gas and coal plants to be compared on one basis.
- **Limited partner (LP) and general partner (GP)** (Month 11): In a venture capital fund, limited partners are the investors who provide the money, such as pension funds and sovereign funds; general partners are the managers who choose investments and earn fees and carried interest.
- **Liquefied natural gas (LNG)** (Month 4): Natural gas cooled to about minus 162 degrees Celsius so that it becomes liquid and can be shipped by tanker across oceans.
- **Lithium carbonate equivalent (LCE)** (Month 5): The standard unit for measuring lithium quantities across different chemical forms. One tonne of lithium metal is about 5.32 tonnes of lithium carbonate equivalent.
- **Lithium iron phosphate (LFP) and nickel manganese cobalt (NMC)** (Month 5): The two main lithium-ion battery cathode chemistries. Lithium iron phosphate is cheaper and safer and needs no nickel or cobalt; nickel manganese cobalt stores more energy per kilogram.
- **Lithium sulphate** (Month 6): An intermediate lithium chemical produced between mineral concentrate and battery-grade lithium carbonate or hydroxide. Making it locally is one way a lithium producer can capture some processing value.
- **Lithography** (Month 8): Printing circuit patterns onto a silicon wafer using light shone through a pattern called a mask. The most advanced machines use extreme ultraviolet light.
- **London Metal Exchange (LME)** (Month 5): The main global exchange for trading base metals such as copper, aluminium, nickel and zinc, and for setting their reference prices.

M

- **Manufacturing value added (MVA)** (Month 7): The value created by a country's manufacturing sector: its output minus purchased inputs.
- **Merit order** (Month 4): The ranking of power plants from lowest to highest marginal cost, used by grid operators to decide which plants run first.
- **Mita** (Month 2): The Spanish colonial system of forced labour in the Andes, used from 1573 to 1812 to supply workers to silver and mercury mines.
- **Mixed hydroxide precipitate (MHP)** (Month 5): An intermediate product containing nickel and cobalt, made from laterite ore by high-pressure acid leaching and later refined into battery chemicals.
- **Modified gross national income (GNI*)** (Month 1): Ireland's adjusted measure of national income, which removes the effects of re-domiciled multinational companies, depreciation of foreign-owned intellectual property and leased aircraft.
- **Monopsony** (Month 6): A market with one dominant buyer, which can push down the price it pays. A mining country selling to a single refining country faces monopsony power.
- **Moore's law** (Month 8): The observation by Intel co-founder Gordon Moore that the number of transistors on a chip doubles roughly every two years.

N

- **Nearshoring** (Month 7): Moving production to a country close to the final market, such as Mexico for the United States or Morocco for Europe, to cut transport time and political risk.
- **Neodymium-iron-boron (NdFeB) magnet** (Month 5; also Month 9): The strongest common type of permanent magnet, made from neodymium, iron and boron, often with dysprosium or terbium added. It is used in electric vehicle and robot motors, and China makes the great majority of the world's supply, commonly estimated at above 90 percent.
- **New trade theory** (Month 1): An explanation of trade, associated with Paul Krugman, based on economies of scale and consumers' love of product variety. It explains why similar countries trade similar goods with each other.
- **Node (process node)** (Month 8): A generation of chipmaking technology, labelled in nanometres, for example 3 nanometres or 2 nanometres.
- **Nominal GDP and real GDP** (Month 1): Nominal gross domestic product is measured at current prices; real gross domestic product is measured at constant prices from a base year, removing the effect of inflation so that growth in actual output can be seen.
- **Non-oil primary deficit** (Month 2): The gap between a government's non-oil revenue and its spending excluding interest payments. It measures how dependent a petrostate's budget is on oil money.

O

- **Obsolescing bargain** (Month 6): Raymond Vernon's idea that bargaining power shifts from a foreign investor to the host government once the investor's capital is sunk in a mine or plant, so the government can later demand better terms.
- **Organization of the Petroleum Exporting Countries (OPEC) and OPEC+** (Month 4): OPEC is a group of major oil-exporting countries, founded in 1960, that coordinates production to influence prices; OPEC+ is its wider alliance since 2016 with Russia and other producers.
- **Original sin** (Month 1; also Month 11): The inability of many developing countries to borrow abroad in their own currency, which forces them to borrow in dollars or other foreign currencies

and creates currency mismatch.

P

- **Petrostate** (Month 4): A state whose government budget and economy depend heavily on oil and gas revenues, such as Saudi Arabia, Iraq or Nigeria. Closely related to the rentier state.
- **Planetary gearbox** (Month 9): A compact, relatively cheap gearbox in which several small gears circle a central gear inside a ring. Common in lower-cost robot joints, including many Chinese humanoids.
- **Planetary roller screw** (Month 9): A high-load screw mechanism that turns rotation into straight-line motion using threaded rollers. Used in the linear actuators of humanoid robot legs and arms; precision grinding makes it costly.
- **Power law** (Month 11): A distribution in which a few outcomes account for most of the total.
- **Power usage effectiveness (PUE)** (Month 8): Total data centre electricity use divided by the electricity used by its computing equipment. A value of 1.0 is ideal, meaning no power is spent on cooling or other overheads.
- **Premature deindustrialization** (Month 3; also Months 10 and 12): The pattern, documented by Dani Rodrik, in which manufacturing's share of jobs and output peaks and starts to fall at much lower income levels in today's developing countries than in earlier industrializers.
- **Price cannibalization** (Month 4): The fall in the market value of solar or wind power as more of it is built, because many plants produce at the same time and push down prices in those hours.
- **Price reporting agency** (Month 5): A company, such as Fastmarkets or Benchmark Mineral Intelligence, that surveys trades and publishes price assessments for opaque markets with no liquid exchange, such as lithium and rare earths.
- **Primary balance** (Month 11): Government revenue minus spending excluding interest payments. A primary surplus is needed to stabilise debt when interest rates are higher than economic growth.
- **Primary energy and total energy supply (TES)** (Month 4): Energy in its original form, such as coal, oil, gas, uranium, sunlight or wind, before conversion into electricity or fuels and the losses that conversion involves. Total energy supply is a country's primary energy production plus imports minus exports and stock changes.
- **Production Linked Incentive (PLI)** (Month 3; also Month 7): India's scheme, launched in 2020, that pays manufacturers a percentage of their additional sales in targeted sectors such as smartphones, electronics and batteries.
- **Productivity effect** (Month 10): The rise in demand for labour that follows automation because lower costs and higher incomes increase demand for goods and services across the economy.
- **Productivity J-curve** (Month 10): The pattern in which measured productivity dips during the early years of investment in a new technology, while firms reorganise and build intangible assets, and rises later.
- **Purchasing power parity (PPP)** (Month 1): An exchange rate that makes the cost of a common basket of goods equal across countries. Used to compare living standards, because market exchange rates understate incomes in poorer countries.

R

- **Reducer** (Month 9): A transmission between a robot's motor and its joint that lowers speed and raises torque. The main types are strain wave gears, cycloidal reducers and planetary gearboxes; precision reducers are a key chokepoint in robot supply.

- **Reinstatement effect** (Month 10): The rise in demand for labour from new tasks and jobs that a technology creates, such as software developers after computers.
- **Rent (economic rent and resource rent)** (Month 1; also Month 6): Income above the minimum needed to keep a resource in its current use, usually earned from scarcity or control. Resource rent is the surplus profit from a scarce natural deposit above normal returns on the capital and labour used to extract it.
- **Rentier state** (Month 2): A state that lives mainly on external rents, especially from oil or gas exports, rather than on taxing its own citizens.
- **Reserve and resource** (Month 5): A reserve is the part of a mineral deposit that can be mined economically with today's prices and technology; a resource is the larger amount that might be mined someday.
- **Resource curse** (Month 2): The tendency of countries that depend on natural resource exports to experience slower growth, weaker institutions, more corruption or more conflict than similar countries without such resources.
- **Resource nationalism** (Month 2; also Month 6): Government action to increase national control over natural resources or the nation's share of the value they generate, through higher royalties, state ownership, export bans or local processing rules.
- **Reversal of fortune** (Month 2): The finding by Acemoglu, Johnson and Robinson that former colonies that were relatively rich in 1500 tend to be relatively poor today, and vice versa, which they explain by the institutions colonisers set up.
- **Robot density** (Month 9): The number of operational industrial robots per 10,000 manufacturing employees, the International Federation of Robotics' standard measure of how automated a country's factories are. South Korea leads the world.
- **Robot tax** (Month 10): Any tax, or reduction in subsidy, aimed at the use of robots or automation, proposed to slow displacement or fund support for workers.
- **Rotary kiln electric furnace (RKEF)** (Month 5): A process that dries and pre-heats laterite nickel ore in a rotary kiln and then smelts it in an electric furnace to make nickel pig iron or ferronickel, mostly for stainless steel.
- **Rule of 70** (Month 1): A shortcut for growth: the time it takes something to double is roughly 70 divided by its annual percentage growth rate. An economy growing 7 percent a year doubles in about 10 years.
- **Rules of origin** (Month 7; also Month 12): The rules that decide which country a product legally comes from for tariff purposes, often based on how much value was added locally or which processing steps happened where.

S

- **Santiago Principles** (Month 11): A set of voluntary standards for the governance, transparency and investment practices of sovereign wealth funds, agreed in 2008 by funds and their host governments.
- **Scenario planning** (Month 12): Building several plausible, internally consistent futures to test strategies against uncertainty, rather than betting on a single forecast. Popularised by planners at Shell.
- **Section 122, Section 232 and Section 301** (Month 7; also Month 12): Sections of United States trade law that allow tariffs: Section 122 for temporary tariffs to address balance-of-payments problems, Section 232 for national-security tariffs, and Section 301 for tariffs against unfair foreign trade practices.
- **Short squeeze** (Month 5): A rapid price rise forced when traders who bet on falling prices (short sellers) must buy back their positions to meet margin calls, as in the London Metal

Exchange nickel market in March 2022.

- **Simple Agreement for Future Equity (SAFE)** (Month 11): A common early-stage investment contract in which an investor gives a start-up money now in exchange for the right to receive shares later, usually at the next priced funding round.
- **Six-axis force/torque sensor** (Month 9): A sensor that measures force and twisting in all three directions at a robot's wrist or ankle, letting it feel contact, balance and handle objects gently.
- **Smile curve** (Month 1; also Month 7): A chart showing that value added and margins are high at the start of a value chain (research, design, key equipment) and at the end (brands, software, services, customers), and low in the middle (assembly and basic manufacturing).
- **Social license to operate** (Month 2): The ongoing acceptance of a mine or project by local communities and other stakeholders. Unlike a government permit, it can be lost at any time, stopping a project.
- **So-so automation** (Month 10): Automation that displaces workers but raises productivity only slightly, such as self-checkout machines, so it lowers labour demand without creating much new income.
- **Sovereign wealth fund (SWF)** (Month 2; also Month 11): A government-owned investment fund, often financed by resource revenues or foreign exchange reserves, such as Norway's Government Pension Fund Global or Indonesia's Danantara.
- **Spare capacity** (Month 4): Unused oil production capacity that can be brought online within about 90 days and sustained.
- **Stablecoin** (Month 11): A digital token designed to hold a fixed value, usually one US dollar, backed by reserves such as cash and US Treasury bills. Growing stablecoin use extends the dollar's reach.
- **State capacity** (Month 2): A state's ability to collect taxes, enforce laws and contracts and provide public goods. Fiscal capacity (taxing) and legal capacity (enforcing) are its main parts.
- **Strain wave gear (harmonic drive)** (Month 9): A compact, zero-backlash reducer that uses a flexible toothed ring deformed by an oval wave generator inside a rigid ring. Used in lighter robot joints such as wrists; Japan's Harmonic Drive Systems is a leading maker.
- **Strategic minor minerals** (Month 5): Small-market minerals such as gallium, germanium, indium, antimony and tungsten, with critical uses in chips, defence and batteries, and supply often concentrated in China.
- **Strategic Petroleum Reserve and emergency stocks** (Month 4): Government-held oil stocks that can be released in a supply crisis. Members of the International Energy Agency must hold emergency stocks equal to at least 90 days of net imports.
- **Sudden stop** (Month 1): An abrupt halt or reversal of foreign capital inflows into a country, usually forcing a sharp currency fall, higher interest rates and a recession.
- **Superstar firm** (Month 10): A highly profitable, dominant firm that captures a large share of its market with relatively few workers, and therefore a low labour share.
- **System integrator (integrator)** (Month 9; also Month 12): A firm that selects, designs, installs and programs complete robot or automation systems for a customer. The course sees integration as a realistic entry layer for founders in Africa and Southeast Asia.

T

- **Tacit knowledge** (Month 10): Knowledge learned through experience and practice that is hard to write down, such as how to run a furnace well or calm an angry customer.
- **Tax-to-GDP ratio** (Month 2): Total tax revenue as a share of gross domestic product, a basic measure of state capacity.

- **Terawatt-hour (TWh)** (Month 8): One billion kilowatt-hours, a unit for large amounts of electricity. Global data centres used about 485 terawatt-hours in 2025, according to the International Energy Agency.
- **Terms of trade** (Month 1): The ratio of a country's export prices to its import prices.
- **Token** (Month 8): A small chunk of text, about three-quarters of an English word on average, that AI language models read and write. Model prices and usage are measured in tokens.
- **Torque** (Month 9): Rotational force: what a robot joint needs to lift, push or hold a load. Reducers trade motor speed for higher torque.
- **Total factor productivity (TFP)** (Month 10): The part of output growth not explained by more labour or capital, capturing better technology, organisation and efficiency. It is the main long-run source of rising living standards.
- **Trade in value added (TiVA)** (Month 1): Trade measured by where value is actually added rather than by the gross value of shipments. The Organisation for Economic Co-operation and Development (OECD) TiVA database is the standard source.
- **Training** (Month 8): Building an AI model by repeatedly adjusting its parameters on large amounts of data. Frontier training runs use tens of thousands of accelerators for months.
- **Transistor** (Month 8): A tiny electronic switch, the basic building block of every chip. Leading chips contain tens of billions of them.
- **Transshipment** (Month 7; also Month 12): Routing goods through a third country, sometimes with minimal processing, to change or disguise their apparent origin, often to avoid tariffs. Rules of origin and enforcement aim to stop it.
- **Treatment and refining charges (TC/RC)** (Month 5): The fees miners pay smelters to process their concentrate into metal. When smelting capacity is excessive, as in copper in 2024-2026, these fees can fall to near zero or below.
- **Triffin dilemma** (Month 11): The conflict, identified by the economist Robert Triffin, between a reserve-currency country supplying the world with enough of its currency, which requires running deficits, and keeping confidence in that currency.
- **Twenty-foot equivalent unit (TEU)** (Month 7): The standard unit of container shipping capacity, equal to one 20-foot container; a 40-foot container counts as 2 TEU.

U

- **United States-Mexico-Canada Agreement (USMCA)** (Month 7): The 2020 trade agreement among the United States, Mexico and Canada that replaced the North American Free Trade Agreement (NAFTA).
- **Universal bank** (Month 3): A bank that both lends to and owns shares in industrial firms, often with seats on their boards. Typical of German industrialization, it concentrated capital for heavy industry.
- **Universal basic capital** (Month 10): Giving every citizen ownership of a share of productive capital, such as a stake in a national fund, so that income from machines and AI is widely shared.
- **Universal basic income (UBI)** (Month 10): A regular, unconditional cash payment to every citizen, regardless of work or income. Alaska's oil dividend is the closest real example discussed in the course.
- **Upgrading** (Month 1): Moving to higher-value activities in or across value chains. Process upgrading means producing more efficiently; product upgrading, making better products; functional upgrading, taking on new functions such as design; chain upgrading, moving into a new industry.
- **Utility death spiral** (Month 4): A cycle in which customers leaving the grid, for example by

installing rooftop solar, raise costs for those who remain, pushing more of them to leave.

V

- **Value added** (Month 1; also Month 6): The value of a firm's output minus the value of the inputs it buys from other firms. It is each producer's contribution to gross domestic product, and the sum of all value added in a country equals its GDP.
- **Vertical integration** (Month 9; also Month 12): A company owning several stages of a supply chain, such as making its own key components instead of buying them. BYD, which makes its own batteries and chips, is the course's leading example.
- **Virtual power plant** (Month 4): Software that coordinates many small energy devices, such as home batteries, electric vehicle chargers and flexible loads, so they act together like one power plant.

W

- **Wafer** (Month 8): A thin disc of very pure silicon, usually 300 millimetres across, on which hundreds of chips are made at once.
- **Weaponized interdependence** (Month 12): Henry Farrell and Abraham Newman's term for using control over the chokepoints of a global network, such as the dollar payment system or a unique technology, to pressure other countries.

Y

- **Yield** (Month 8): The share of chips on a wafer that work correctly. High yields on a new node are a large part of what makes Taiwan Semiconductor Manufacturing Company hard to match.

Reading list by month

Each Month's recommended reading, by author and title. A work already listed under an earlier Month appears in short form with a pointer back.

Month 1: How the Global Economy Works

- David Ricardo, "On the Principles of Political Economy and Taxation" (chapter 7)
- Paul Krugman, Maurice Obstfeld and Marc Melitz, "International Economics: Theory and Policy"
- Richard Baldwin, "The Great Convergence: Information Technology and the New Globalization"
- World Bank, "World Development Report 2020: Trading for Development in the Age of Global Value Chains"
- Kenneth Kraemer, Greg Linden and Jason Dedrick, "Capturing Value in Global Networks: Apple's iPad and iPhone"
- Gary Gereffi, "Global Value Chains and Development: Redefining the Contours of 21st Century Capitalism"
- Diane Coyle, "GDP: A Brief but Affectionate History"
- Matthew Klein and Michael Pettis, "Trade Wars Are Class Wars"
- Dani Rodrik, "The Globalization Paradox"
- Paul Blustein, "The Chastening: Inside the Crisis That Rocked the Global Financial System and Humbled the IMF (International Monetary Fund)"

Month 2: Institutions and Why Nations Get Rich

- Daron Acemoglu and James A. Robinson, "Why Nations Fail: The Origins of Power, Prosperity, and Poverty"
- Daron Acemoglu and James A. Robinson, "The Narrow Corridor: States, Societies, and the Fate of Liberty"
- Douglass C. North, "Institutions, Institutional Change and Economic Performance"
- Michael L. Ross, "The Oil Curse: How Petroleum Wealth Shapes the Development of Nations"
- Paul Collier, "The Bottom Billion"
- Charles Tilly, "Coercion, Capital, and European States, AD 990-1992"
- Jeffrey Herbst, "States and Power in Africa: Comparative Lessons in Authority and Control"
- Steffen Hertog, "Princes, Brokers, and Bureaucrats: Oil and the State in Saudi Arabia"
- Adam Hochschild, "King Leopold's Ghost"
- Daron Acemoglu, Simon Johnson and James A. Robinson, "The Colonial Origins of Comparative Development: An Empirical Investigation" (article)
- David Albouy, "The Colonial Origins of Comparative Development: An Empirical Investigation: Comment" (article, a critique to read alongside the paper above)

Month 3: The Industrialization Playbooks

- Joe Studwell, "How Asia Works: Success and Failure in the World's Most Dynamic Region"
- Alice Amsden, "Asia's Next Giant: South Korea and Late Industrialization"
- Robert Wade, "Governing the Market: Economic Theory and the Role of Government in East Asian Industrialization"
- Chalmers Johnson, "MITI and the Japanese Miracle" (MITI is Japan's Ministry of International Trade and Industry)
- Ha-Joon Chang, "Kicking Away the Ladder: Development Strategy in Historical Perspective"
- Robert C. Allen, "The British Industrial Revolution in Global Perspective"
- Alexander Gerschenkron, "Economic Backwardness in Historical Perspective"
- Reka Juhasz, Nathan Lane and Dani Rodrik, "The New Economics of Industrial Policy" (article)
- Nathan Lane, "Manufacturing Revolutions: Industrial Policy and Industrialization in South Korea" (article)
- Chris Miller, "Chip War: The Fight for the World's Most Critical Technology"

Month 4: Energy, the Base Layer

- Vaclav Smil, "Energy and Civilization: A History"
- Daniel Yergin, "The Prize: The Epic Quest for Oil, Money, and Power"
- Daniel Yergin, "The New Map: Energy, Climate, and the Clash of Nations"
- Energy Institute, "Statistical Review of World Energy 2026"
- International Energy Agency, "Oil Market Report" (monthly) and "World Energy Outlook 2025"
- International Energy Agency, "Reducing the Cost of Capital" and the Cost of Capital Observatory
- International Renewable Energy Agency, "Renewable Power Generation Costs in 2024"
- Ember, "Global Electricity Review 2026"
- International Energy Agency, "Energy and AI" and "Key Questions on Energy and AI"

Month 5: Critical Minerals from Geology to Market

- U.S. Geological Survey, "Mineral Commodity Summaries 2026"
- International Energy Agency, "Global Critical Minerals Outlook 2026"
- Ed Conway, "Material World: The Six Raw Materials That Shape Modern Civilisation" (published in the United States as "Material World: A Substantial Story of Our Past and Future")
- Javier Blas and Jack Farchy, "The World for Sale: Money, Power and the Traders Who Barter the Earth's Resources"
- Siddharth Kara, "Cobalt Red: How the Blood of the Congo Powers Our Lives"
- Henry Sanderson, "Volt Rush: The Winners and Losers in the Race to Go Green"
- Guillaume Pitron, "The Rare Metals War"
- Ernest Scheyder, "The War Below: Lithium, Copper, and the Global Battle to Power Our Lives"
- Oliver Wyman for the London Metal Exchange, "Independent Review of Events in the Nickel Market in March 2022"

Month 6: Processing, Downstreaming and Resource Nationalism

- International Energy Agency, "Global Critical Minerals Outlook 2026" (see Month 5)
- Richard Gray and Duc Minh Nguyet Nguyen (Center for Strategic and International Studies), "Indonesian Industrialization: Downstreaming Up the Value Chain"
- SOAS University of London, Department of Development Studies, "Processing Power: How Indonesia Used Industrial Policy to Move up the Global Nickel Value Chain" (case study)
- World Trade Organization, "Indonesia: Measures Relating to Raw Materials" (dispute DS592), Panel Report
- Raymond Vernon, "Sovereignty at Bay"
- Paul Collier, "The Plundered Planet"
- Tom Burgis, "The Looting Machine"
- Natural Resource Governance Institute, "Natural Resource Charter" (2nd edition)
- Organisation for Economic Co-operation and Development, "Due Diligence Guidance for Responsible Supply Chains of Minerals from Conflict-Affected and High-Risk Areas"

Month 7: Manufacturing and Global Supply Chains

- Marc Levinson, "The Box: How the Shipping Container Made the World Smaller and the World Economy Bigger"
- Richard Baldwin, "The Great Convergence" (see Month 1)
- Patrick McGee, "Apple in China: The Capture of the World's Greatest Company"
- Dan Wang, "Breakneck: China's Quest to Engineer the Future"
- Joe Studwell, "How Asia Works" (see Month 3)
- World Bank, "World Development Report 2020" (see Month 1)
- Kraemer, Linden and Dedrick, "Capturing Value in Global Networks" (see Month 1)
- United Nations Conference on Trade and Development, "Review of Maritime Transport 2025: Staying the Course in Turbulent Waters"
- Chad P. Bown (Peterson Institute for International Economics), "The Trump-China trade wars: Five takeaways from US imports in 2025"
- Soumaya Keynes and Chad P. Bown, "How to Win a Trade War: An Optimistic Guide to an Anxious Global Economy"

Month 8: Semiconductors, Compute and the AI Supply Chain

- Chris Miller, "Chip War" (see Month 3)

- Stephen Witt, "The Thinking Machine: Jensen Huang, Nvidia, and the World's Most Coveted Microchip"
- Semiconductor Industry Association and Boston Consulting Group, "Emerging Resilience in the Semiconductor Supply Chain"
- Saif M. Khan, Alexander Mann and Dahlia Peterson (Center for Security and Emerging Technology), "The Semiconductor Supply Chain: Assessing National Competitiveness"
- International Energy Agency, "Energy and AI" and "Key Questions on Energy and AI" (see Month 4)
- Luke Emberson and David Roodman (Epoch AI), "The plunging price of thought"
- World Trade Organization and partners, "Global Value Chain Development Report 2023" (chapter 4)
- Taiwan Semiconductor Manufacturing Company, "Annual Report 2025"

Month 9: Robotics, Anatomy and Industrial Economics

- International Federation of Robotics, "World Robotics 2026: Industrial Robots" (Executive Summary)
- Jean-Stanislas Denain and Yann Riviere (Epoch AI), "How fast could robot production scale up?"
- McKinsey and Company, "Turning humanoid supply chain constraints into billion-dollar wins"
- Bank of America Institute, "Physical AI, part 2: Humanoid robots"
- International Energy Agency, "Global Critical Minerals Outlook 2026" (see Month 5)
- Kevin M. Lynch and Frank C. Park, "Modern Robotics: Mechanics, Planning, and Control"
- Ed Conway, "Material World" (see Month 5)
- Dan Wang, "Breakneck" (see Month 7)
- Richard Baldwin, "The Great Convergence" (see Month 1)

Month 10: AI, Robots and the Labor Economy

- Daron Acemoglu and Simon Johnson, "Power and Progress: Our Thousand-Year Struggle over Technology and Prosperity"
- Daron Acemoglu, "The Simple Macroeconomics of AI" (National Bureau of Economic Research Working Paper 32487)
- David Autor, Caroline Chin, Anna Salomons and Bryan Seegmiller, "New Frontiers: The Origins and Content of New Work, 1940-2018" (article)
- Erik Brynjolfsson, Danielle Li and Lindsey Raymond, "Generative AI at Work" (article)
- Erik Brynjolfsson, Bharat Chandar and Ruyu Chen, "Canaries in the Coal Mine? Six Facts About the Recent Employment Effects of Artificial Intelligence"
- Carl Benedikt Frey, "The Technology Trap: Capital, Labor, and Power in the Age of Automation"
- James Bessen, "Learning by Doing: The Real Connection between Innovation, Wages, and Wealth"
- International Labour Organization, "Generative AI and Jobs: A Refined Global Index of Occupational Exposure" (ILO Working Paper 140)
- John Maynard Keynes, "Economic Possibilities for our Grandchildren" (essay)
- Thomas Piketty, "Capital in the Twenty-First Century"

Month 11: Money, Finance and Sovereign Capital

- Barry Eichengreen, "Exorbitant Privilege: The Rise and Fall of the Dollar"
- Adam Tooze, "Crashed: How a Decade of Financial Crises Changed the World"

- Paul Blustein, "The Chastening" (see Month 1)
- Carmen Reinhart and Kenneth Rogoff, "This Time Is Different: Eight Centuries of Financial Folly"
- Deborah Brautigam, "The Dragon's Gift: The Real Story of China in Africa"
- World Bank, "International Debt Report 2025"
- Sebastian Mallaby, "The Power Law: Venture Capital and the Making of the New Future"
- Brad Feld and Jason Mendelson, "Venture Deals"
- Norges Bank Investment Management, half-year and annual reports

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- Henry Farrell and Abraham Newman, "Underground Empire: How America Weaponized the World Economy"
- Chris Miller, "Chip War" (see Month 3)
- Ed Conway, "Material World" (see Month 5)
- Joe Studwell, "How Asia Works" (see Month 3)
- Daron Acemoglu and James A. Robinson, "Why Nations Fail" (see Month 2)
- Dani Rodrik, "Premature Deindustrialization" (article)
- Peter Schwartz, "The Art of the Long View: Planning for the Future in an Uncertain World"
- International Energy Agency, "Rare Earth Elements: Pathways to Secure and Diversified Supply Chains"
- International Energy Agency, "Global Critical Minerals Outlook 2026" (see Month 5)
- International Federation of Robotics, "World Robotics 2026" (see Month 9)
- International Monetary Fund, "World Economic Outlook, October 2026: Resilience Under Strain, Urgent Choices"

Ten core data sources to keep using

Numbers go out of date; sources do not. Most of these ten are free (the full World Robotics report is paid, but its summaries are not), all are updated regularly, and between them cover every layer of the chain from ore to robot to artificial intelligence, plus the money that finances it. When a figure in this book matters to you, check it against the latest release from one of these.

1. **International Monetary Fund, World Economic Outlook (IMF WEO)**: The twice-yearly baseline for growth, inflation, debt and current accounts in every country, with forecasts; use it to check any macroeconomic number and to see how the outlook is shifting.
2. **World Bank, World Development Indicators (WDI)**: The broadest free database of country statistics over decades, from gross domestic product per person and electricity access to education and poverty; use it for long-run comparisons between countries.
3. **United Nations Comtrade (Commodity Trade Statistics Database)**: Detailed records of what every country exports and imports, product by product and partner by partner; use it to trace where ores, magnets, chips and robots actually flow.
4. **U.S. Geological Survey, Mineral Commodity Summaries (USGS MCS)**: The annual two-page summaries of world mine production and reserves for over 90 minerals; use it as the first stop for who digs what and how much is left in the ground.
5. **International Energy Agency (IEA)**: The leading source on energy and critical minerals, from oil market reports to the Global Critical Minerals Outlook and Energy and AI; use it for supply, demand, refining concentration and scenarios.
6. **Ember**: An independent energy think tank with free, frequently updated electricity data by

country and fuel; use it to track how fast solar, wind, coal and gas are changing each country's power mix.

7. **International Federation of Robotics, World Robotics (IFR)**: The annual count of industrial and service robots installed and in operation by country and industry, including robot density; use it to measure who is automating and how fast.
8. **World Semiconductor Trade Statistics (WSTS)**: The industry's monthly and annual chip sales figures by region and product type, such as memory and logic, with forecasts; use it to read the semiconductor cycle and the AI boom's effect on it.
9. **Bank for International Settlements (BIS)**: The central banks' bank, publishing data on cross-border lending, debt securities, credit, exchange rates and dollar funding; use it to follow global liquidity and the dollar system.
10. **Organisation for Economic Co-operation and Development, Trade in Value Added (OECD TiVA)**: Estimates of where value is actually created inside traded goods, by country and industry; use it to see how much of a country's exports is its own value added rather than imported parts.

A good habit: each time you finish a Month, pick one of these sources, download one table relevant to that Month, and update one number in your notebook. By the end of the course you will know where every important number in the material economy lives.